Which hand?

Detecting left- vs right-hand nose touches from smartphone motion sensors

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Abstract

A person holds a smartphone at arm's length and brings it to the nose with the left or the right hand. This report predicts which hand did it from the phone's motion sensors (a web browser, about 60 Hz), as a calibrated probability $P(\text{right})$, for persons the model has never seen. The data are 598 instructed trials of 20 persons; every evaluation leaves one person out. The mirror symmetry of the body explains the signal: the phone's early rotation about gravity changes sign with the hand. Among 17 candidates, a pre-specified rule chose a one-dimensional convolutional network on gravity-frame and heading-following body-frame channels. The nested estimate of this selection, with the candidates fixed, is an accuracy of 88.5 % [84.1, 92.3] (area under the ROC curve 0.945); the shipped model's own numbers are exploratory. With its in-fold Platt calibrator, its probabilities are consistent with calibration (expected calibration error 0.026, null band up to 0.042; slope 0.97 [0.69, 1.50]; intercept +0.07 [−0.25, +0.39]). Three attempts raise the accuracy to 96.4 %. Without a start cue, a simpler reference classifier changes by −5.1 points (exploratory, the best of 22 pipelines): it shows a hand for 76.4 % of the recordings, right on 79.9 % of them (74.6 % forced-choice). The main gaps: two left-handed persons, right-hand free-choice trials only, and a production simulation without the final model.

Key results

Unseen persons	Nested estimate of the selection (candidates fixed): accuracy 88.5 % [84.1, 92.3], AUC 0.945 [0.906, 0.975]. The shipped body-frame 1D CNN alone: 89.3 % [85.0, 93.1] (exploratory: best of 16 configurations in its lane, then of 17 candidates). Section 5.2.
The percentage	With its in-fold Platt calibrator, consistent with calibration: ECE 0.026 (perfect-calibration null band up to 0.042), slope 0.97 [0.69, 1.50], intercept +0.07 [−0.25, +0.39]; no further remedy helps. Section 5.4.
Several attempts	$k = 3$: accuracy 96.4 % [93.2, 98.9] (mean $P(\text{right})$). The displayed $\sigma(\sum_i z_i/\tau_3)$, right in 96.3 %, is neither over- nor under-confident on average (mean confidence minus accuracy −0.008 [−0.024, +0.016]; ECE, slope and intercept not computed). Section 5.5.
Natural use	Right-hand recall on the free-choice trials 85.0 % (17 of 20); all are right-hand so far, so a constant right answer would score 100 %. Section 5.7.
No start cue	Lane m4's pipeline with its reference classifier yaw8, *not* the final model (exploratory, best of 22 pipelines): 72.8 % [68.9, 76.6] with half credit for abstentions (outputs of exactly 0.5 included); a hand shown for 76.4 % of the recordings, right on 79.9 % of those; forced choice 74.6 %; yaw8 on the stored windows 77.9 %. Section 6.
Data	20 persons (12 Android, 8 iOS; 2 left-handed), 598 instructed trials, 20 free-choice trials. Section 2.

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1. Introduction

The question. A person holds a smartphone at arm's length and brings it to touch the nose, with the left or the right hand. The task, set by the principal investigator (PI), is to predict which hand from the phone's motion sensors alone, as the browser's DeviceMotion interface delivers them to a web page at about 60 Hz [1]. The answer is a percentage, $P(\text{right})$: the probability that the right hand moved the phone. It should be calibrated (of the touches given 80 %, about 80 % should be right-hand touches), and it must hold for persons the model has never seen, because the PI will test it on new persons.

Why it is hard. Five things make the task hard.

1. Persons differ in arm length, speed and style, and the model sees no movement of the test person.
2. Phones differ in sensors, sampling and browser [2]: iOS reports the accelerometer with the opposite sign to Android, and the browser delivers only about 60 Hz.
3. The device axes turn with the grip, so the same arm movement gives different device-frame signals when the phone is held upright, sideways or with the screen away.
4. The two hands make mirror-image movements, so most simple descriptors (speed, duration, path length) are the same for both hands.
5. In production there is no start cue: the page must find the approach in a continuous stream, reject other movements such as the return of the arm, and decide when it has seen enough.

The approach. The analysis rests on the left–right mirror symmetry of the body (Section 3). Mirroring a movement flips the sign of a few chiral (mirror-odd) quantities, chiefly the rotation about gravity, and leaves the rest unchanged, so only chiral quantities can carry the hand. Every trial is standardised into a gravity frame that does not depend on the grip and a body frame that follows the phone's heading (Section 4). Four model families, from a one-line rule to deep sequence models, are evaluated by leave-one-person-out (LOSO) cross-validation, with all tuning nested inside the training persons. A pre-specified cross-lane rule selects the final model among the lanes' candidates, and a nested run of this selection gives its selection-aware estimate (Section 5). Calibration, several attempts, natural free-choice use and robustness to an unseen grip are tested on the same folds. A separate simulation removes the start cue and adds detection, online logic and a stopping rule (Section 6). Secondary analyses ask what else the same data reveal: speed, grip, handedness and the value of demographics (Section 7). The appendices state every decision with its options, evidence and rationale.

Status of this version. This is a preliminary version, written for the researcher's review. The analysis was organised in lanes, and the appendices keep their names. The lanes on interpretable models (m1, Appendix C), classical machine learning (m2, Appendix D), time-frequency and deep models (m3, Appendix E), the cue-free production pipeline (m4, Appendix G), preprocessing (p1, Appendix B) and the case study (x1, Appendix I) each passed two independent audits, one for rigour and one for reproducibility, and were fixed after each. The final-model lane (selection, calibration and export; Appendix F) is a first pass that has not yet been independently audited. The secondary analyses of speed, grip and handedness passed two audits with fixes; the demographics test is a quick, unaudited first pass. The final model has not yet been run inside the cue-free production simulation, so the production numbers of Section 6 use that simulation's own reference classifier. No review of the report's completeness has run yet.

Related work. Table 1 lists prior work on detecting the hand or the handedness from phone or wrist sensors. Most studies infer the hand from touch traces or grip during ordinary phone use [3–6], and several split the data by trial or at random, so that the same person can appear in training and test. Such splits overstate the accuracy for new persons [7–9], and with few persons a cross-validated estimate has wide error bars [10]. Closest to this task, Mohammadi-Ghazi et al. saw opposite rotation senses of the two hands in a clinical finger-to-nose test with wrist sensors, but built no classifier [11]. The studies in Table 1 report accuracies; a calibrated probability is a separate requirement, and modern classifiers, deep networks in particular, are often over-confident [12]. This study therefore evaluates person-wise throughout and checks calibration for every model it ships.

Table 1: Prior work on detecting the hand or the handedness from phone or wrist sensors. Accuracies are as reported; most evaluations are not person-wise.

Study	Signal	N	Result	Caveat
GripSense [3]	gyroscope, touch, vibration	10	84.26 % hand posture	controlled study
Löchtefeld et al. [13]	sensors during the unlock gesture, k-NN	12	98.51 %	all right-handed
Guo et al. [4]	touch traces; acc + gyro for hand changes	14	≈94 % precision	random 10-fold CV
Nelavelli & Ploetz [5]	sign of the swipe-arc curvature	12	99.53 % consistent	no held-out users
Fernández et al. [6]	touch features	174	up to 99.92 % TPR	oversampling before random splits
Chen et al. [14]	motion sensors, LSTM	n/r	89.2–97.4 %	unclear if test users unseen
Singh & Singh [15]	acc, gyro, orientation; XGBoost	n/r	85.04 % (3 classes)	protocol not reported
Mohammadi-Ghazi et al. [11]	wrist IMU, finger-to-nose	17	opposite rotation sense	clinical, no classifier

2. Data

The app. The data were recorded with NoseTouch, a web app that runs in the phone's browser on iOS and Android with nothing to install. It guides the participant hands-free with spoken and visual prompts (which hand, which grip, which speed), records the motion streams, segments each attempt online and uploads every attempt at once. Appendix A.1 lists what is recorded and which streams feed the models.

The protocol. A visit follows the street protocol of Table 2. Tier 0 is one free-choice trial: the participant chooses the hand, moves naturally and reports the hand afterwards. These free-choice trials form the natural-use set, which is scored but never used for training. Tiers 1–4 are instructed: the app prompts the hand, the grip (upright, top left or top right) and the speed (natural, fast or slow) or a movement variant, in blocks of one hand. A visit was planned to take about 5 minutes; the median completed visit took 9.7 min.

Participants. The participants were recruited among Bar-Ilan University students, most of them master's students, and each was offered candy, chocolates and gummies for their time. Figure 1 shows a data-collection session[1]. One additional participant, PD01, is analysed separately as an exploratory case study and never enters the modelling (Appendix I).

Curation. The data snapshot holds 878 attempts. A pre-specified exclusion cascade, applied in order, keeps 598 primary trials and 20 natural-use trials (Figure 2). The largest rule, E1, removes the 172 attempts the app itself did not accept; rule E4d removes 30 trials with a

[1] A video of a data-collection session: https://drive.google.com/file/d/1V9PF623HkJ8MjR6loJhUHnDkCx60W-Jl/view?usp=sharing.

Table 2: The street protocol (protocol 2.0.0). *A* is the first hand, *B* the other; within a block the order of trials is random. 29 planned trials, 39 with the optional bonus; the app estimates 4.5 minutes without the bonus.

Tier	Blocks	Grip (facing the person)	Trials
0 choice	one	free	1 natural; the participant picks the hand and reports it afterwards
1 basics	*A*, *B*	upright	4 per hand: natural, natural, fast, slow
2 sideways	*B*, *A*; *A*, *B*	top left, top right	3 per block: natural, natural, fast
3 variety	*B*, *A*	upright	4 per hand: tilted touch + 3 drawn from start shoulder or centre, curved, pause, twist, elbow high, bent start, lean in, jerky
4 bonus	*A*, *B*, *B*, *A*	upright	optional: natural, fast, natural per hand + 2 more variety cells



Figure 1: A data-collection session (cropped so that no face is visible). Participants follow the app's prompts, here **LEFT** at normal speed, with the phone at arm's length; the candy on the table was the compensation.

sampling gap near the movement. Quality flags (for example possible clipping or a rotated screen) and the predictor's input gates are not exclusions: they are kept and scored apart, so every model sees the data production will see. Labels are never changed, although 2 blocks of one person show the signature of a swapped hand (Appendix A.8). Appendix A.4 gives every rule and its count.

The dataset in numbers. Table 3 describes the 20 persons of the analysis: 12 women and 8 men, aged 19 to 29 years (mean 25.1); 18 are right-handed and 2 left-handed. Of these persons, 12 recorded on Android and 8 on iOS. The primary set has 302 left-hand and 296 right-hand trials, a median of 28 per person (range 8 to 45). The natural-use set has one free-choice trial per person, and all 20 of them are right-hand trials.

Gaps. Three gaps limit what these data can show. Only 2 persons are left-handed, so no effect of handedness can be estimated. Only 8 persons used an iPhone, so platform comparisons have little power. All free-choice trials are right-hand trials, so natural use measures right-hand recall only: a constant right answer would score 100 %.

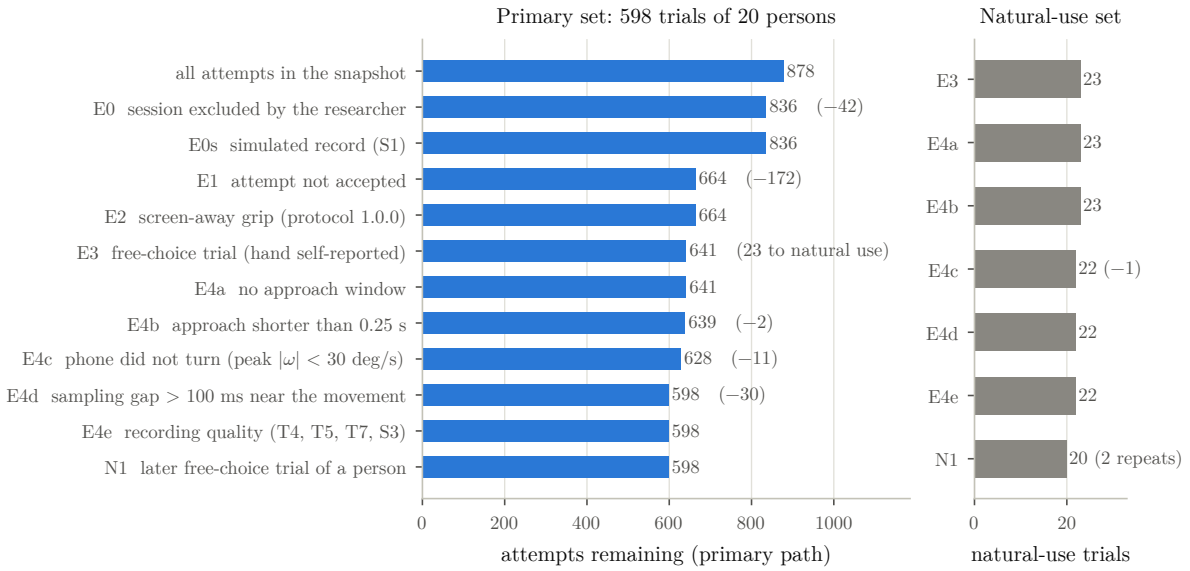


Figure 2: The exclusion cascade, applied in order; an attempt is charged to the first rule it fails. Rule E3 moves the free-choice trials, whose hand is self-reported, to the natural-use set, which then passes the same quality rules; N1 keeps one free-choice trial per person (the earliest). The result is the primary set used for training and evaluation, and the natural-use set used for scoring only.

Table 3: Participants of the analysis.

Quantity	Value
Persons	20
Women / men	12 / 8
Age (years), mean ± SD (range)	25.1 ± 2.9 (19–29)
Height (cm), mean ± SD (range; n)	168.8 ± 8.8 (152–187; 17)
Dominant hand right / left	18 / 2
Usual phone hand right / both / left	18 / 2 / 0
Persons on Android / iOS	12 / 8
Included sessions (Android / iOS)	24 (15 / 9)
Device models	14
Persons with several sessions	3
Primary trials (left / right)	598 (302 / 296)
Primary trials (Android / iOS)	598 (383 / 215)

3. Why the hand is detectable

3.1. The mirror

Let $\mathbf{R}(t)$ map device coordinates to body coordinates, with $\mathbf{f}$ the specific force (the browser's acceleration including gravity), $\mathbf{a}$ the acceleration and $\boldsymbol{\omega}$ the angular velocity, all in device coordinates. Mirroring the body across its sagittal plane, $\mathbf{p}' = \mathbf{M}\mathbf{p}$, and relabelling the device's x axis, $\mathbf{R}' = \mathbf{M}\mathbf{R}\mathbf{S}$ with $\mathbf{M} = \mathbf{S} = \mathrm{diag}(-1, 1, 1)$, gives for every grip and every movement

$$\mathbf{f}' = \mathbf{S}\mathbf{f}, \qquad \mathbf{a}' = \mathbf{S}\mathbf{a}, \qquad \boldsymbol{\omega}' = -\mathbf{S}\boldsymbol{\omega}. \tag{1}$$

Acceleration is a polar vector; angular velocity is a pseudovector and picks up $\det \mathbf{S} = -1$. With $\hat{\mathbf{g}}$ the unit vector pointing up, the yaw rate about gravity, $\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$, is a pseudoscalar: a scalar that changes sign under the mirror and is unchanged by any rotation. This report calls such quantities chiral (mirror-odd), and quantities that keep their sign mirror-even:

$$(-\mathbf{S}\boldsymbol{\omega}) \cdot (\mathbf{S}\hat{\mathbf{g}}) = -\,\boldsymbol{\omega} \cdot \hat{\mathbf{g}} \quad \text{(mirror)}, \qquad (\mathbf{Q}\boldsymbol{\omega}) \cdot (\mathbf{Q}\hat{\mathbf{g}}) = \boldsymbol{\omega} \cdot \hat{\mathbf{g}} \quad \text{(any grip } \mathbf{Q} \in SO(3)\text{)}. \tag{2}$$

Holding the phone with the screen facing away is the proper rotation $\mathbf{Q} = \mathbf{R}_y(180^\circ) = \mathrm{diag}(-1, 1, -1)$, so it leaves every chiral feature unchanged (Table 4). An antisymmetric model, with logit $z(\mathbf{x}) = g(\mathbf{x}) - g(m(\mathbf{x}))$, where $m(\cdot)$ is the mirror, then satisfies $P(\text{right})(m(\mathbf{x})) = 1 - P(\text{right})(\mathbf{x})$ exactly. Only chiral features can carry the hand: a mirror-even feature takes the same distribution for both hands whenever the two hands move as mirror images.

Table 4: The mirror and the screen-away flip in device coordinates.

Quantity	Mirror m (left ↔ right)	Screen away, $\mathbf{Q} = \mathbf{R}_y(180^\circ)$
$\mathbf{f}$, $\mathbf{a}$, $\hat{\mathbf{g}}$ (polar vectors)	$(-x, y, z)$	$(-x, y, -z)$
$\boldsymbol{\omega}$ (pseudovector)	$(x, -y, -z)$	$(-x, y, -z)$
$\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$, $\boldsymbol{\omega} \cdot \mathbf{a}$, $\det[\mathbf{a}(t), \mathbf{a}(t+\tau), \hat{\mathbf{g}}]$	sign flips	unchanged
$\lVert\boldsymbol{\omega}\rVert$, $\mathbf{a} \cdot \hat{\mathbf{g}}$, duration (mirror-even)	unchanged	unchanged
Hand label	swapped	unchanged
Grip label	top left ↔ top right	toward ↔ away

3.2. The mechanism

Early in the approach the forearm swings toward the body's midline. Seen from above, the phone therefore turns counter-clockwise in the right hand and clockwise in the left hand. Near the nose the wrist turns the phone partly back to face the person, so the net rotation about gravity over the whole approach is a weaker and more variable signal than the early turn. The early turn carries the clearest signal.

3.3. The evidence on 20 persons

The yaw about gravity accumulated over the first half of the approach separates the hands with a pooled area under the receiver operating characteristic curve (AUC) of 0.85. The net yaw over the whole approach has a lower pooled AUC, 0.77; the difference was not tested in a

paired comparison (Figure 3). Within persons the median AUC is 0.90, and it is above 0.5 for 20 of 20 persons. The sign of the first-half yaw alone, with no training, classifies 79 % of the trials, and it separates the hands with the same sign on both platforms (AUC 0.86 on Android, 0.84 on iOS).

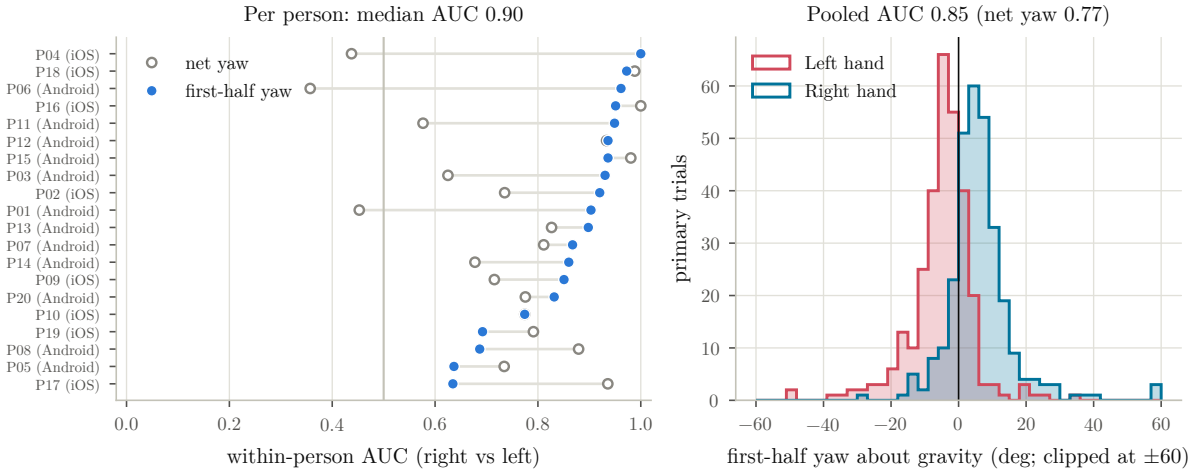


Figure 3: The yaw about gravity over the first half of the approach has a higher pooled AUC than the net yaw over the whole approach (0.85 against 0.77; no paired test). Left: AUC within each of the 20 persons; right: the pooled distributions. The early turn carries the clearest signal, so the models get time-resolved inputs.

The class means tell the same story (Figure 4). The yaw rate about gravity has opposite signs for the two hands over most of the first half of the approach, and the cumulative yaw keeps most of that early turn. The mirrored left-hand mean has the shape of the right-hand mean but a smaller early peak, and the mirror-even channels differ slightly between the hands: the two hands are close to, but not exactly, mirror images. Mirror augmentation is therefore a plausible, not an exact, way to share data between the hands. In the feature catalogue of Appendix C.2, chiral features reach a LOSO AUC of up to 0.87, while every mirror-even feature stays between 0.46 and 0.53. The sensor conventions behind the mirror are verified on the data: of the 64 possible sign patterns, the mirror predicted by the conventions maps right-hand onto left-hand waveforms best, on both platforms (Appendix A.2).

3.4. Consequences for modelling

Three consequences shape the models. First, they get chiral, time-resolved inputs in the gravity frame, not only whole-movement integrals. Second, a body frame that follows the phone's heading gives chiral channels that do not depend on the grip (Section 4), which matters when the test grip was never seen in training (Section 5.3). Third, the mirror doubles the training data: a mirrored trial is a valid trial of the other hand (mirror augmentation), or the model can be built antisymmetric. The return of the arm is a time-reversed approach, so its chiral signs are reversed: on the 288 trials with a clean return window, the first-half yaw rule is right on 76 % of the approaches and on only 17 % of the returns. A cue-free detector that takes the return for the approach therefore answers the wrong hand (Appendix G.3).

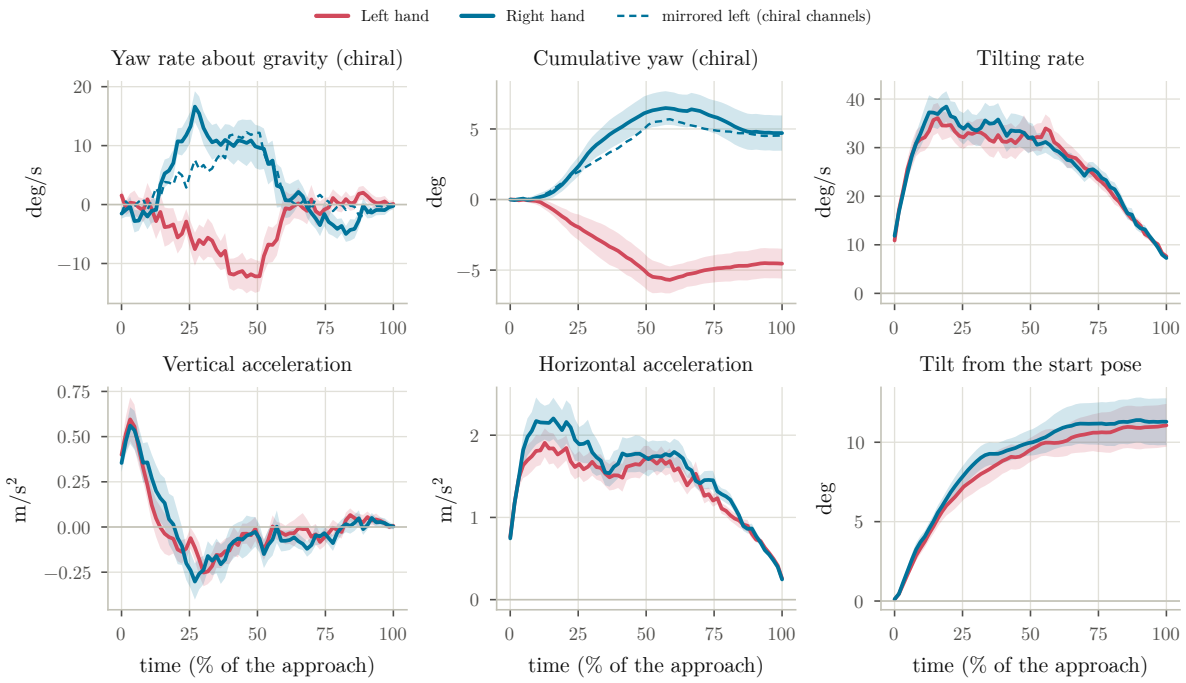


Figure 4: Mean time course of the gravity-frame channels over the time-normalised approach, left against right hand (band: ± standard error across the 20 persons). Dashed: the left-hand mean mirrored, which should lie on the right-hand mean if the movement is mirror-symmetric. The chiral channels carry the hand early in the approach; the mirror-even channels do not.

3.5. Limits

The mirror of the accelerometer is approximate, because the sensor sits off the phone's centre line. The two arms need not move as exact mirror images either: the dominant and the non-dominant arm differ in how they are controlled [16, 17]. Section 7 finds no detectable dominant-hand signal in mirror-invariant kinematics.

4. Methods

4.1. Standardisation

Every trial passes through the same standardisation, on oracle bounds in training and on detected bounds in production (Appendices A.2 and A.3). The streams are resampled linearly onto a uniform 60 Hz grid; gaps up to 100 ms are interpolated. The accelerometer sign of iOS is normalised to the W3C device frame, whose x axis points to the right edge, y to the top edge and z out of the screen. A gravity frame is built without the magnetometer: the gravity direction at rest before and after the movement is propagated through the movement with the gyroscope, which gives the unit up vector $\hat{\mathbf{g}}$ at every sample, the vertical and horizontal parts of $\mathbf{a}$ and $\boldsymbol{\omega}$, the yaw rate $\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$ and its integral. A body frame follows the phone's heading: its axes are the horizontal part of the screen normal and the horizontal direction perpendicular to it (Appendix E.1, equation (6)). A rotation of the phone in the hand leaves

the body-frame channels unchanged, so they do not depend on the grip. All channels pass a zero-phase, fourth-order Butterworth low-pass filter at 15 Hz. The primary analysis uses the oracle window: the approach bounds that the app stored online (Appendix A.6). A fixed panel of three pre-specified models judged each of these choices against its alternatives, and all 11 defaults were kept (Appendix B).

4.2. Evaluation design

Every evaluation is leave-one-person-out (LOSO): a model is trained on all other persons and tested on the held-out person, once for each of the 20 persons. There is no separate test set. Results pool the held-out predictions of all folds. Hyperparameters, calibrators and aggregation scales are fitted on the training persons only, with inner person-wise splits (nested tuning). In every fold, a leak check re-runs the fold on a random half of the test trials in a new order and requires identical predictions. A training person contributes at most 40 trials, so that no person dominates a model.

All 95 % confidence intervals (CIs) in this report are person-cluster bootstrap intervals: 2000 resamples of persons, not of trials [18]. Two models are compared on the same trials through their per-person differences, with a paired person bootstrap for the CI and an exact sign-flip test for the p-value. P-values are adjusted within a named family of comparisons, by Holm's method or, where stated, by the Benjamini–Hochberg (BH) method. Overlapping CIs are not taken as a test.

A configuration chosen by its LOSO result among several is exploratory, and its number of configurations tried is reported with it. The headline of a selection is the nested estimate: the whole selection is repeated inside each outer fold, on the training persons only, so the held-out person never influences the choice.

A pipeline may abstain by returning $P(\text{right}) = 0.5$ exactly. An abstention is never a right-hand answer: it earns half credit, and wherever a pipeline can abstain the report gives its coverage (the share answered), its accuracy on the answered trials and its forced-choice accuracy. Natural use is scored as right-hand recall on the free-choice trials; all of them are right-hand trials so far, so a constant right answer would score 100 %.

4.3. Metrics

Accuracy ($P(\text{right})$ above 0.5 read as right), AUC, the log loss and the Brier score [19] measure discrimination and the quality of the probabilities. Calibration is judged by three numbers together: the expected calibration error (ECE; ten equal-mass bins) against its perfect-calibration null band, that is, the range of ECE values that a perfectly calibrated model with the same predicted probabilities shows by chance; and the slope and intercept of a logistic recalibration of the outcome on the logit, each with its CI. A calibrated model has a slope near 1 and an intercept near 0. Several attempts $i = 1, \ldots, k$ of one person and hand are combined as

$$P(\text{right})_k = \sigma\left(\frac{1}{\tau_k}\sum_{i=1}^{k} z_i\right), \tag{3}$$

where z_i is the logit of attempt i, σ the logistic function and τ_k a scale fitted inside each fold (Appendix F.9). The k-attempt curves use the 19 persons with at least 5 trials of each hand.

4.4. Model families

Four families of models were developed in parallel, each with its own appendix.

- Rules and interpretable models: the sign of the first-half yaw; antisymmetric logistic models on six physics features (PHYS6); a temporal filter; a shallow tree; a generalised additive model (Appendix C).
- Classical machine learning on 691 engineered features: logistic regression (LR), support vector machines, LightGBM gradient-boosted trees and stacked ensembles (Appendix D).
- Time-frequency and deep sequence models: MiniRocket [20], logistic models on time-frequency planes and one-dimensional convolutional networks (1D CNN) (Appendix E).
- The reference floor, yaw8: a mirror-augmented logistic regression on the cumulative yaw at 8 instants, fixed before any model was built (Appendix A.9).

4.5. The cross-lane selection

Lanes m1, m2, m3 and p1 delivered their finalists. With the reference floor, body-frame variants of engineered models and a pre-specified cross-lane ensemble, 17 candidates entered the selection. A rule fixed before the comparison chose the final model (Appendix F.2): (1) score every candidate in its production form, with the calibration it would ship; (2) find the lowest log loss and the leaders that paired comparisons cannot separate from it (Holm); (3) drop the leaders with a Holm-significant accuracy drop on a grip absent from training, a natural-use shortfall or a screen-away failure the rest-pose check cannot repair; (4) among the remaining leaders, take the most portable. The same rule, rerun inside every outer fold on the training persons' inner predictions, gives the nested, selection-aware estimate. It treats the candidates as fixed: it does not repeat the lanes' own choices made on the outer folds.

5. Results

5.1. Model comparison

Figure 5 shows the 17 candidates as their lanes deliver them. Most of them beat the reference floor, yaw8 (77.9 % [73.1, 82.9]): 11 of 16 candidates are more accurate after Holm adjustment. The differences among the best candidates are small: the candidate with the lowest log loss and the others that paired comparisons cannot separate from it form 6 leaders (11 of the 16 other candidates are separated from it, Holm over these 16 comparisons). Of the candidates, 7 were chosen on the outer folds in their own lane, so the forest is exploratory. Appendices C to E give each family's own results and decisions.

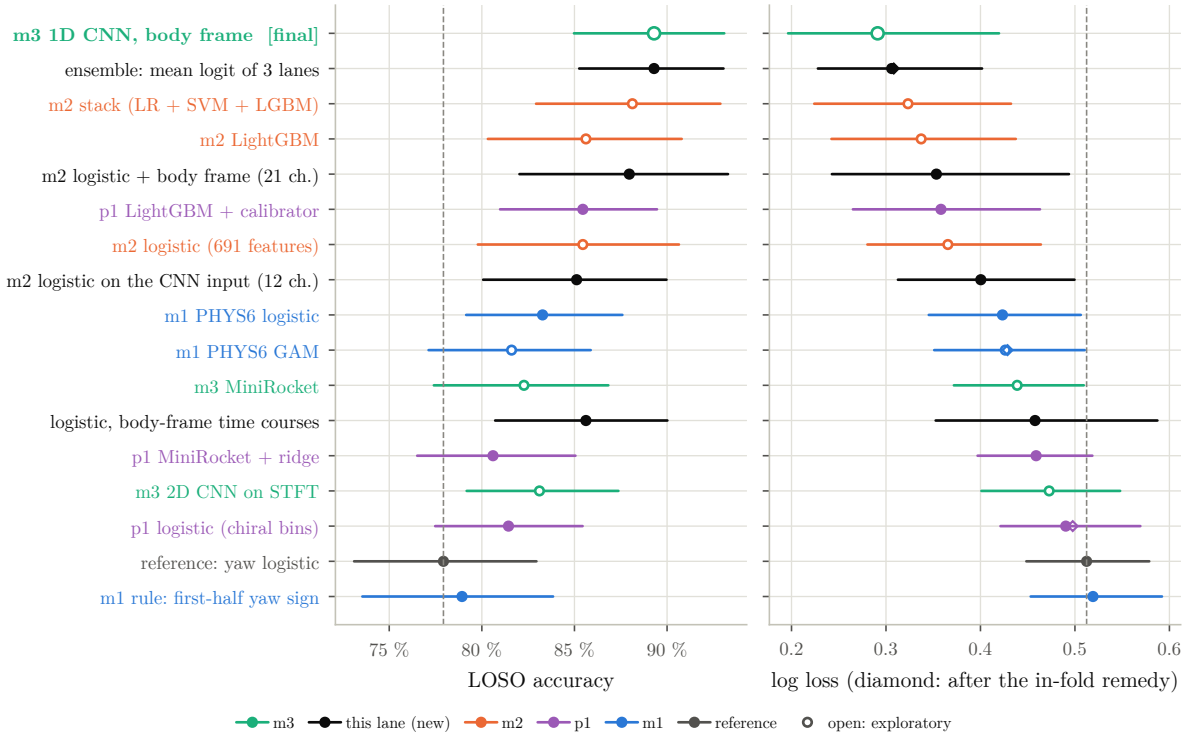


Figure 5: The 17 candidates of the cross-lane selection as their lanes deliver them (LOSO accuracy and log loss), coloured by lane. Open markers: chosen on the outer folds in their lane (exploratory); diamonds: log loss after the calibration remedy chosen inside each fold; dashed: the reference floor. The forest is exploratory; the nested estimate (Table 5) is the honest number.

5.2. The final model and its selection-aware estimate

The final model is the body-frame 1D CNN: a one-dimensional convolutional network on 12 channels in the gravity frame and the body frame (Appendix E). It has the lowest log loss in production form. The headline is the nested, selection-aware estimate of the cross-lane selection, which reruns the rule inside every outer fold and treats the 17 candidates as fixed (Table 5, Figure 6): accuracy 88.5 % [84.1, 92.3], AUC 0.945 [0.906, 0.975] and log loss 0.307 [0.212, 0.432]. Inside the outer folds the selection chose the CNN in 16 of 20 folds. It does not repeat the lanes' own outer-fold choices: lane m3's own nested estimates, which repeat its configuration choice in every fold, are 89.1 % among its grip-invariant configurations and 87.8 % among all of them (both post hoc; Appendix E.2).

The shipped model's own LOSO numbers are exploratory: it was the best of 16 configurations in its lane, then chosen among 17 candidates. They are accuracy 89.3 % [85.0, 93.1], AUC 0.951 [0.912, 0.980] and log loss 0.291 [0.196, 0.420], an optimism of +0.8 [−0.2, +2.3] points of accuracy over the nested estimate. Its body-frame input was also designed after outer-fold results had been seen, an optimism that no nested estimate removes (Appendix E.2). The rest of this section describes the shipped model, and its numbers share this exploratory status. The CNN is +11.7 [+8.0, +15.6] points more accurate than the reference floor (Holm $p < 0.001$ over the 16 comparisons with the reference).

Table 5: Selection-aware estimate of the cross-lane selection, the naive best-of-candidates number and its optimism (paired person bootstrap).

estimate	accuracy	AUC	log loss
nested cross-lane selection (selection-aware)	88.5 [84.1, 92.3]	0.945 [0.906, 0.975]	0.307 [0.212, 0.432]
best on the outer folds: m3 1D CNN, body frame (exploratory)	89.3 [85.0, 93.1]	0.951	0.291 [0.196, 0.420]
optimism (naive - nested)	+0.8 [−0.2, 2.3]	+0.006 [0.001, 0.013]	−0.015 [−0.034, −0.003]

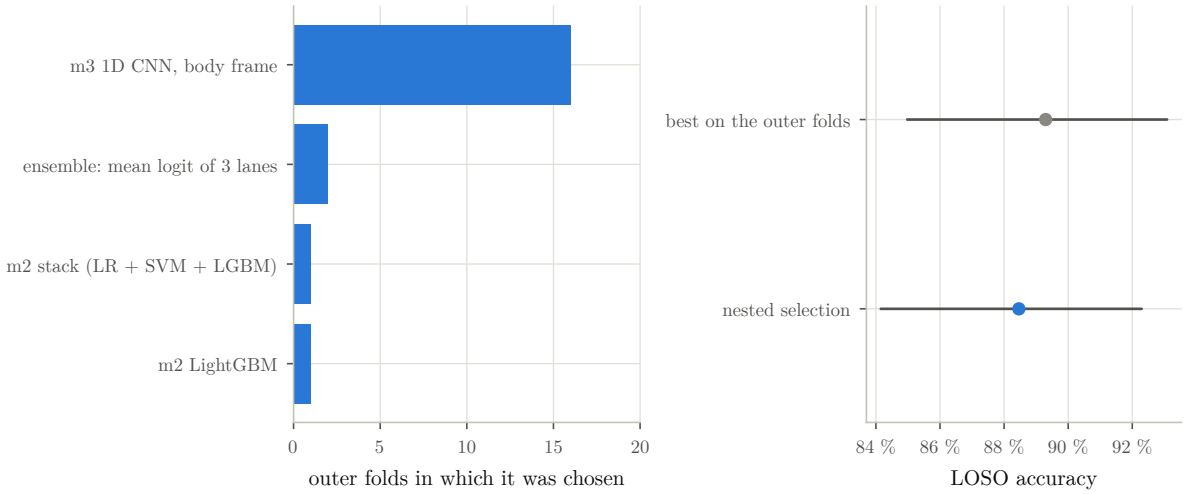


Figure 6: Nested cross-lane selection. Left: how often each candidate was chosen, by the lowest inner log loss on the training persons, over the 20 outer folds. Right: the selection-aware estimate against the best candidate picked on the outer folds. The naive pick is optimistic by +0.8 [−0.2, +2.3] points of accuracy.

The input and the model class were compared separately. On the same body-frame channels, the CNN is +4.3 [+1.7, +7.6] points more accurate than lane m2's engineered features with logistic regression (Holm $p = 0.016$ over six input-against-class comparisons), and on the gravity frame alone it does not beat lane m1's linear models (Appendices F.5 and E.11). With all 21 channels, lane m2's logistic regression is not detectably less accurate than the CNN (−1.4 [−4.3, +1.4] points, Holm $p = 0.738$) but has a higher log loss (+0.064 [+0.024, +0.108], Holm $p = 0.023$).

5.3. Robustness to an unseen grip

The robustness tests decided among the leaders (Figure 7). Trained on upright grips only and tested on the sideways grips, the leaders that read device-frame channels collapse: lane m2's logistic regression changes by −31.0 points, its stack by −25.3, the preprocessing panel's gradient-boosted trees by −21.6 and lane m2's LightGBM by −19.8 (each Holm $p \leq 0.014$ over the 17 grip tests). The body-frame CNN changes by −4.1 [−7.0, −1.4] points (unadjusted $p = 0.016$; Holm $p = 0.188$ over the 17 grip tests) and reaches 85.3 % on the unseen grips. It is one of the two leaders without a Holm-significant drop; the other, an ensemble, changes by −10.5 points (Holm $p = 0.312$). Of these two, the rule took the more portable CNN. The documented fallback is lane m1's PHYS6 logistic: −5.9 points less accurate (Holm $p = 0.008$

over the comparisons with the best candidate), but it changes by only −1.1 points on the unseen grips (87.5 %; Appendix F.13).

A phone held with the screen facing away defeats every model that reads the screen-relative frame: on synthesised screen-away twins of the recorded trials the CNN falls to 28.6 %. A model-free check of the rest pose before the movement, exported with the model, detects 95.7 % of the twins; with the check and an un-flip the CNN recovers 86.6 %. The check also fires on 4.3 % of the recorded trials, which costs −2.7 [−7.4, +0.1] points of accuracy on them (Appendix F.12).

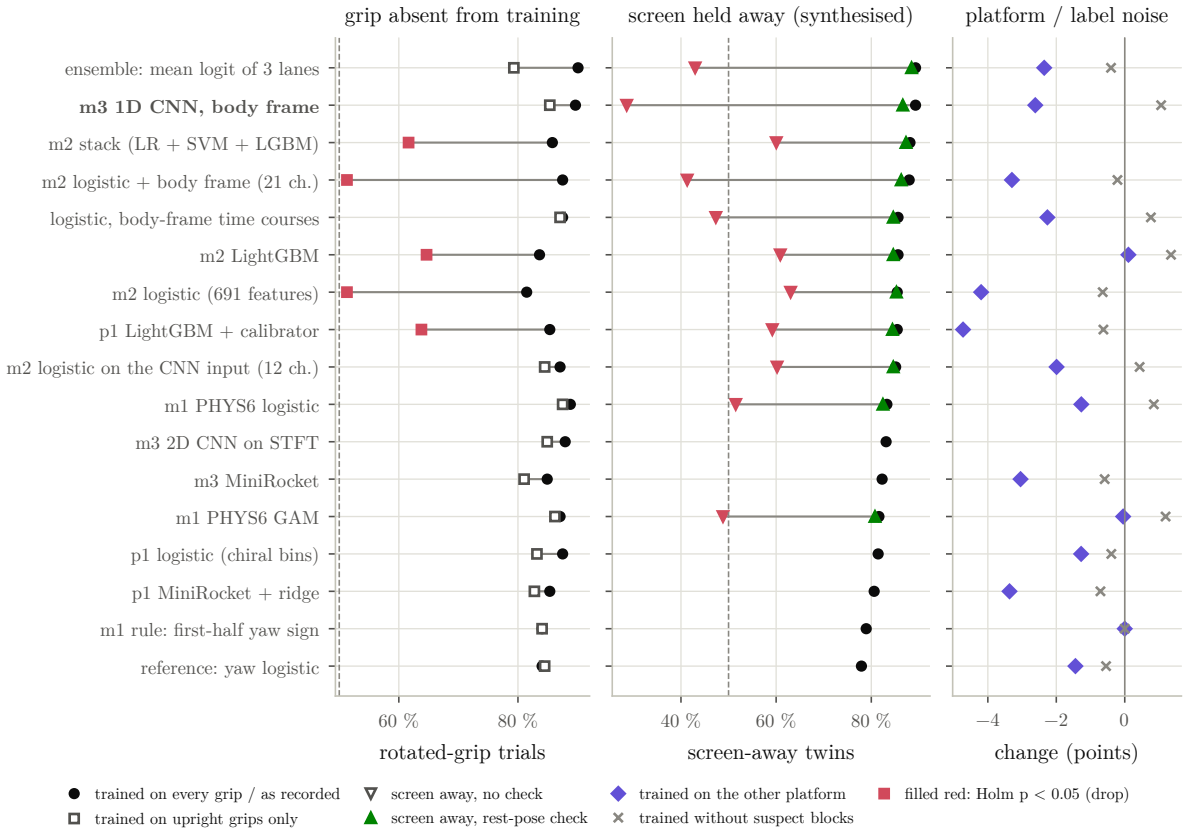


Figure 7: Robustness of every candidate (bold: the final model). Left: accuracy on the sideways grips when trained on every grip (dot) or on upright grips only (square). Middle: accuracy as recorded and on synthesised screen-away twins, without and with the rest-pose check. Right: accuracy change when trained on the other platform only and without suspect label blocks. Device-frame models fail an unseen grip, which removes them from the selection.

5.4. Calibration

The final model ends with a Platt calibrator fitted inside each fold (lane m3's choice, Appendix E). With it, its probabilities are consistent with calibration (Figure 8): ECE 0.026, inside its perfect-calibration null band (up to 0.042); slope 0.97 [0.69, 1.50]; intercept +0.07 [−0.25, +0.39]. Four further remedies fitted inside the folds on top of it (Platt, temperature, isotonic and beta calibration) each raise the log loss slightly, none detectably (Holm $p = 1.000$ for each, over

the four remedies), so none is added. No difference between the platforms is detectable (slope difference $p = 0.446$, calibration in the large $p = 0.554$, unadjusted), but with 8 iOS persons this comparison has little power, and per-platform intercepts were not saved (Appendix F.7).

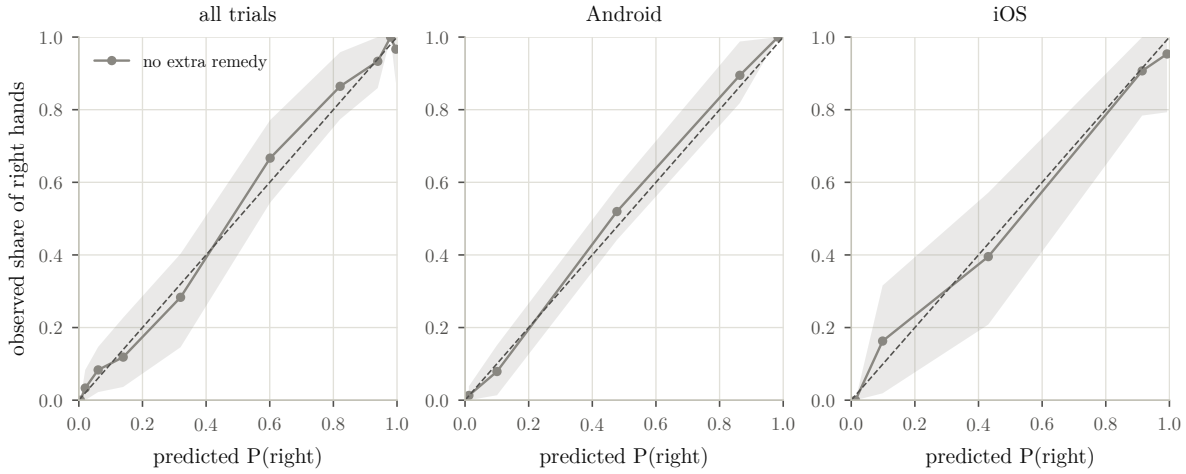


Figure 8: Reliability of the final model, overall and per platform (equal-mass bins, person-bootstrap 95 % bands; its in-fold Platt calibrator, no further remedy). ECE 0.026 (null band up to 0.042), slope 0.97 [0.69, 1.50], intercept +0.07 [−0.25, +0.39]: the percentage shown is consistent with calibration.

5.5. Several attempts

Combining attempts helps (Figure 9). With $k = 3$ attempts of one person and hand, the mean $P(\text{right})$ of the evaluation harness's k-curve reaches 96.4 % [93.2, 98.9]. The aggregates of Figure 9 give nearly the same answers (each is right in 96.3 %) and differ in the percentage they show. The displayed aggregate of equation (3), with τ_3 fitted in-fold (median 1.54), is neither over- nor under-confident on average: its mean confidence minus its accuracy is −0.008 [−0.024, +0.016]. This is calibration in the large only; its ECE with a null band, slope and intercept were not computed. The mean $P(\text{right})$ is under-confident (−0.122 [−0.138, −0.105]) and has a higher log loss (difference −0.089 in favour of the scaled sum, Holm $p = 0.010$ over the paired comparisons of the aggregates; Appendix F.9).

5.6. Who and where it fails

Figure 10 breaks the accuracy down. It is 90.6 % [86.6, 94.1] on Android and 87.0 % [77.0, 94.1] on iOS; the platforms were not compared by a test. The accuracy is below 80 % for 3 persons, the lowest being 61.8 %; the two left-handed persons score 96.4 % and 88.0 % (two persons: no CI). The weakest grip-by-speed cell still reaches 86.5 % [76.7, 94.6]; the weakest movement variant reaches 60.0 % [31.2, 87.5] on 15 trials of 8 persons. Of the 64 errors, 24 are confident errors (a probability below 0.2 for the true hand), and they cluster in a few persons (7 in P09, 5 in P20; Appendix F.10).

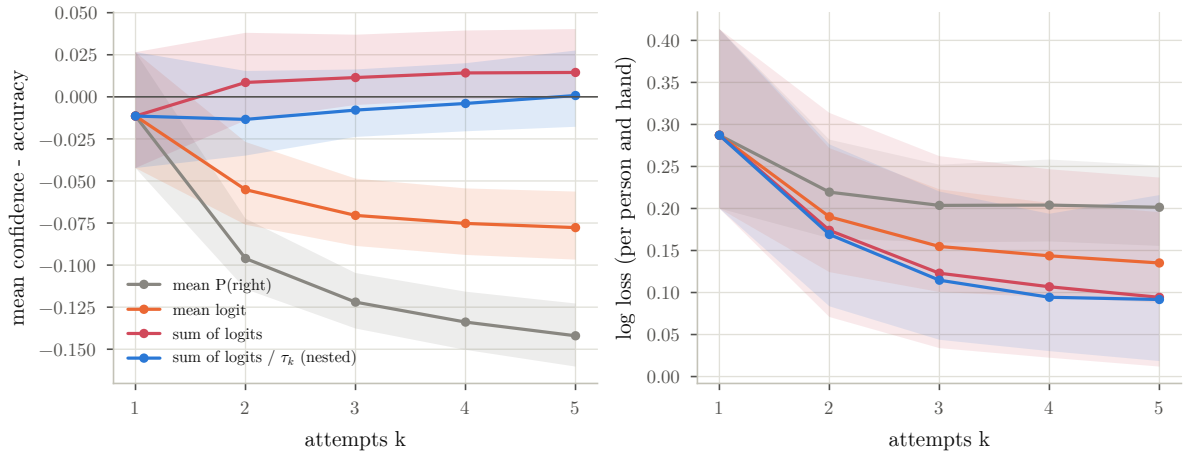


Figure 9: Several attempts of one person and hand (19 persons with at least 5 trials of each hand). Left: over-confidence (mean confidence minus accuracy) of four aggregates (above 0: over-confident); right: their log loss. The scaled sum of logits is neither over- nor under-confident on average; the mean $P(\text{right})$ is under-confident and is not shown to the user.

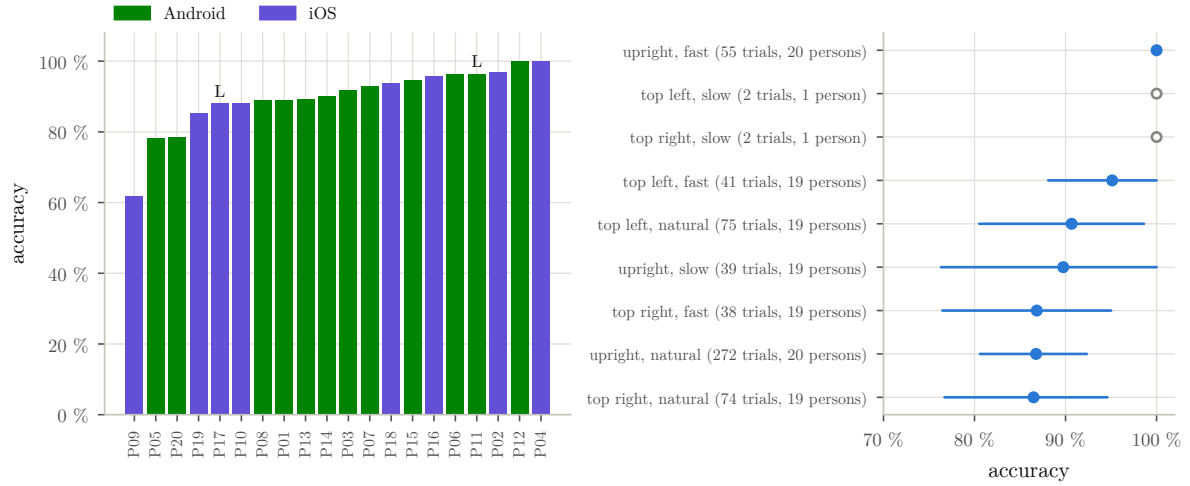


Figure 10: The final model by person (left; sorted; colour: platform; L: the two left-handed persons) and by grip × speed cell (right; CI only for cells with at least 5 persons, open markers otherwise).

5.7. Natural use

On the free-choice trials the final model's right-hand recall is 85.0 % (17 of 20), against 17.7 correct trials expected from the same persons' instructed trials: no detectable shortfall (Holm $p = 1.000$ over the 17 candidates). All free-choice trials are right-hand trials so far, so a constant right answer would score 100 % and this recall says nothing about the left hand. The mean probability of the reported hand is lower on free choice than on the same persons' instructed right-hand trials (74.1 % against 82.8 %), but the difference is not detectable ($p = 0.158$, unadjusted; Appendix F.6).

5.8. More persons?

The learning curve flattens (Figure 11). With 19 training persons the CNN reaches 89.3 %. An inverse power law fitted to random subsets of training persons extrapolates to 89.6 % [88.8, 90.8] at 40 persons, a gain along the fitted curve of +0.5 [+0.1, +1.3] points from 19 persons. The band resamples the repetitions at fixed persons, so it understates the uncertainty, and the curve is fitted to few sizes. The extrapolation suggests that more persons of the same kind would add little average accuracy; new persons are needed above all for the groups the data lack: left-handers and left-hand free choice.

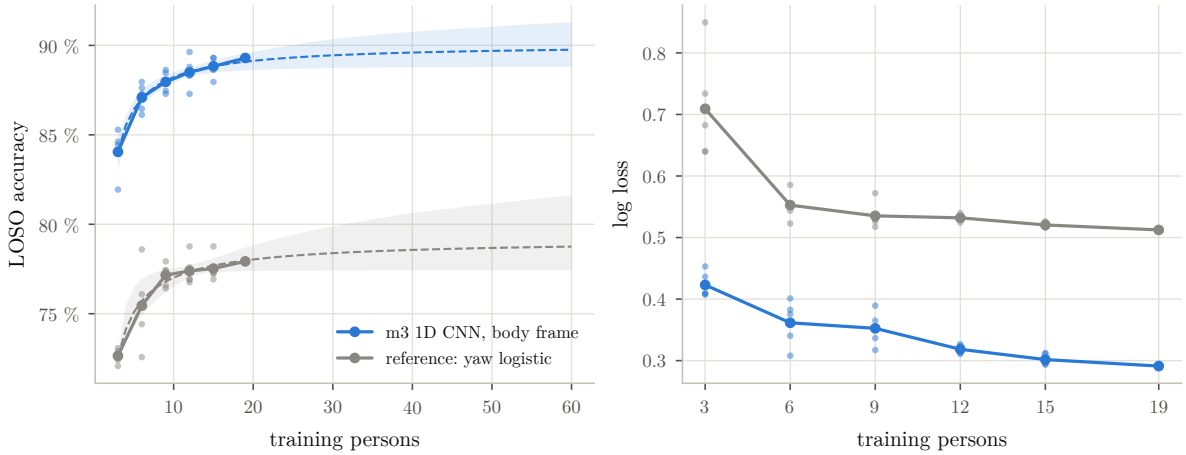


Figure 11: Learning curves: training persons drawn at random inside every outer fold (dots: repetitions; line: their mean; dashed: an inverse power law with a 95 % bootstrap band over the repetitions, which understates the uncertainty). The extrapolation to 40 persons gives 89.6 % for the CNN; by this extrapolation, more persons of the same kind would add little average accuracy.

6. Towards production: cue-free detection and stopping

In production there is no start cue and no stored bound. The page receives a continuous stream; it must find the approach, reject other movements, decide when to answer and show a calibrated $P(\text{right})$. A simulation of this pipeline ran on the full recordings (Appendix G). Its numbers use the simulation's own reference hand classifier, yaw8 (a mirror-augmented logistic regression on the cumulative yaw at 8 instants), *not* the final CNN. They measure what detection and stopping cost, not how the final model performs without a cue; running the final model inside this simulation is the next step (Section 8).

6.1. Finding the approach

Figure 12 shows what each detector hands to the classifier. The baseline detector, v0, takes the first movement after a fixed stillness: its first detection is the approach in 53 % of recordings, part of the approach in 12 %, an earlier movement in 28 %, the return in 2 % and nothing in 5 %. The production detector cuts the stream into candidate movement segments and scores

each with a segment classifier that does not use the hand, from features such as the stillness before the segment, its displacement and the change of the up vector. It accepts the most likely candidate when its probability of being an approach is at least 0.50. Its scored segment is the approach in 67 % of recordings, part of the approach in 10 %, an earlier movement in 5 % and the return in 2 %; it rejects 17 % (Appendix G.4).

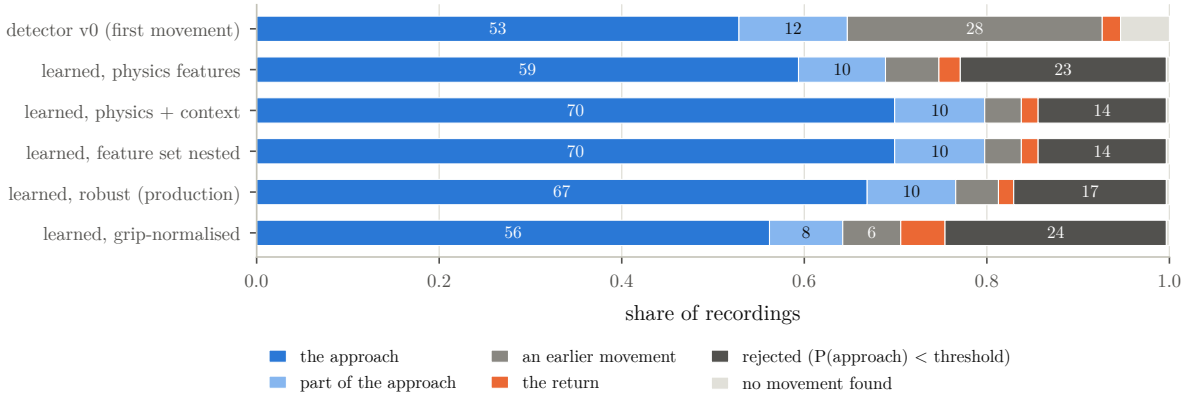


Figure 12: What each cue-free detector hands to the hand classifier on the 598 full recordings (rest, approach, hold, return), scored against the app's stored events (percentages). v0 takes the first movement after a fixed stillness; the learned detectors pick the candidate with the highest P(approach) and reject it below the threshold. The production detector scores far fewer earlier movements than v0.

6.2. What detection costs

Figure 13 gives the accuracy ladder of the yaw8 classifier: 77.9 % [73.1, 82.9] on the oracle windows, 67.7 % [63.4, 72.1] behind v0 and 72.8 % [68.9, 76.6] in the production pipeline, where an abstention earns half credit. An abstention is a rejected recording or a calibrated $P(\text{right})$ of exactly 0.5. The production pipeline accepts an approach in 82.9 % of recordings, but 39 of these 496 get exactly 0.5 from the calibrator (Section 6.4). It therefore shows a hand for 76.4 % of recordings (its coverage), is right on 79.9 % of those, and scores 74.6 % when forced to answer every recording. Detection changes the accuracy by −5.1 [−7.6, −2.7] points against the oracle windows (Holm $p = 0.047$, family of 26 comparisons). The pipeline was chosen among 22 cue-free pipelines on the outer folds, so its number is exploratory.

The detector relies on the stillness the app enforced before each trial. When the stillness before the approach is cut, the production pipeline falls from 73.1 % with 1 s of stillness kept to 56.4 % with none, when it answers only 33 % of recordings (Appendix G.2). A model-free check of the rest pose catches a phone held with the screen away: on synthesised screen-away twins it stops 95.7 % of recordings and cuts the share answered wrongly from 38.8 % to 2.0 % (Holm $p < 0.001$, family of 11), while it stops 4.3 % of the recordings held toward the person (Appendix G.10).

6.3. Online logic and the stopping rule

Multi-attempt sessions were synthesised by splicing a person's cued recordings with 1 s of stillness at each junction, and the pipeline was replayed on them sample by sample (Figure 14). With up

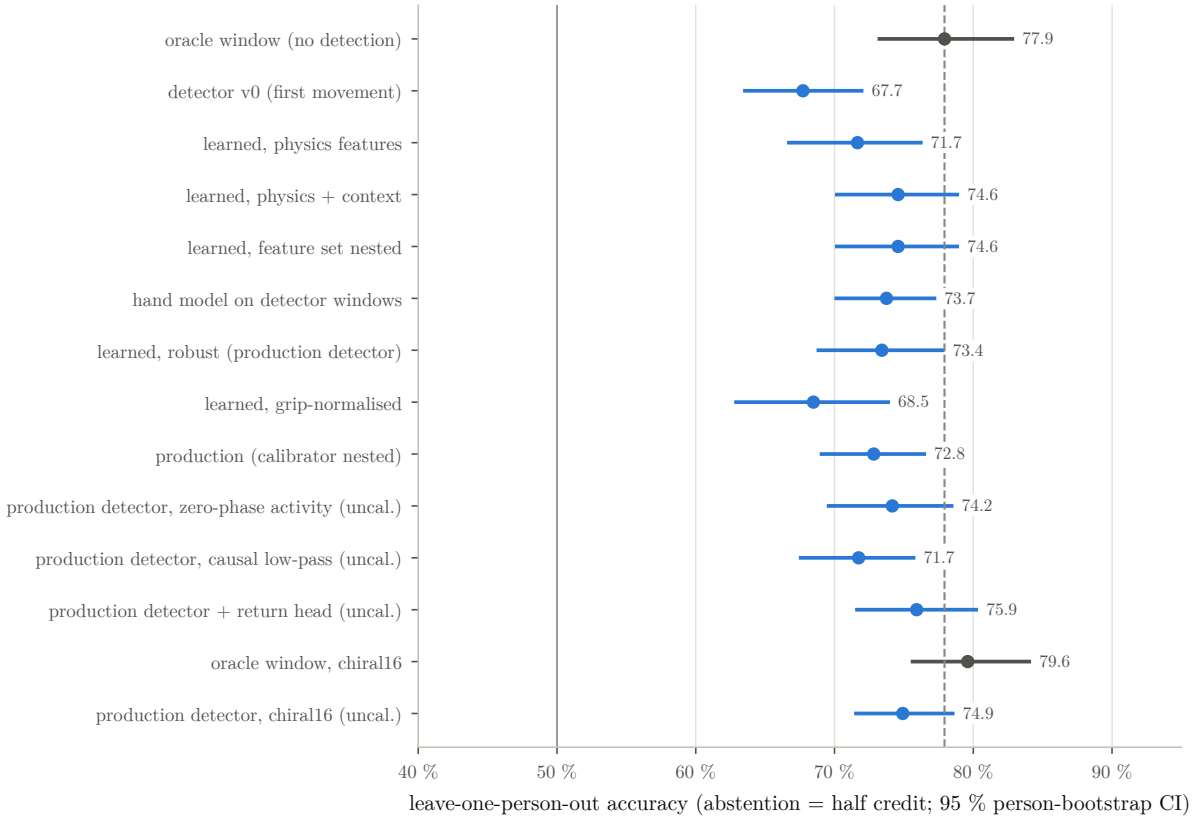


Figure 13: Hand accuracy of the cue-free pipelines on the full recordings (no start cue, no stored bounds), against the same yaw8 classifier on the oracle windows (dashed). An abstention earns half credit. The gap between the oracle windows and the production pipeline is the cost of detection.

to three accepted attempts the session accuracy is 76.5 % [69.2, 83.7], against 88.4 % [82.0, 94.5] when only the true approaches are kept. Only 77 % of the accepted attempts are approaches; the others, earlier movements and returns, cancel most of the gain of combining (Appendix G.8). The stopping rule answers once the aggregated $P(\text{right})$ leaves the band between $1 - p_{\text{stop}}$ and $p_{\text{stop}} = 0.90$, or after 5 accepted attempts. It reaches 83.2 % [74.6, 90.8] after 3.66 accepted attempts on average, and it answers 100 % of the sessions. Its aggregation scale is fitted in each fold on the training persons' sessions, but this scale was added after the held-out sessions had shown the problem, so the rule is post hoc and exploratory (6 configurations; Appendix G.9).

6.4. The calibrated output

For one touch, a calibrator chosen inside each fold among antisymmetric ones keeps the mirror property, $\mathrm{P}(\text{mirror}) = 1 - P$; symmetrised isotonic regression won in 20 of 20 folds. The calibrated one-touch output has ECE 0.048 (null band 0.020 to 0.060), slope 0.94 [0.75, 1.18] and intercept −0.13 [−0.29, 0.03], consistent with calibration (Appendix G.6). At the stop, the session-fitted scale gives ECE 0.107, at the upper edge of its null band (up to 0.108), slope 1.03 [0.73, 1.81] and intercept −0.70, whose CI was not saved: a borderline result that needs a

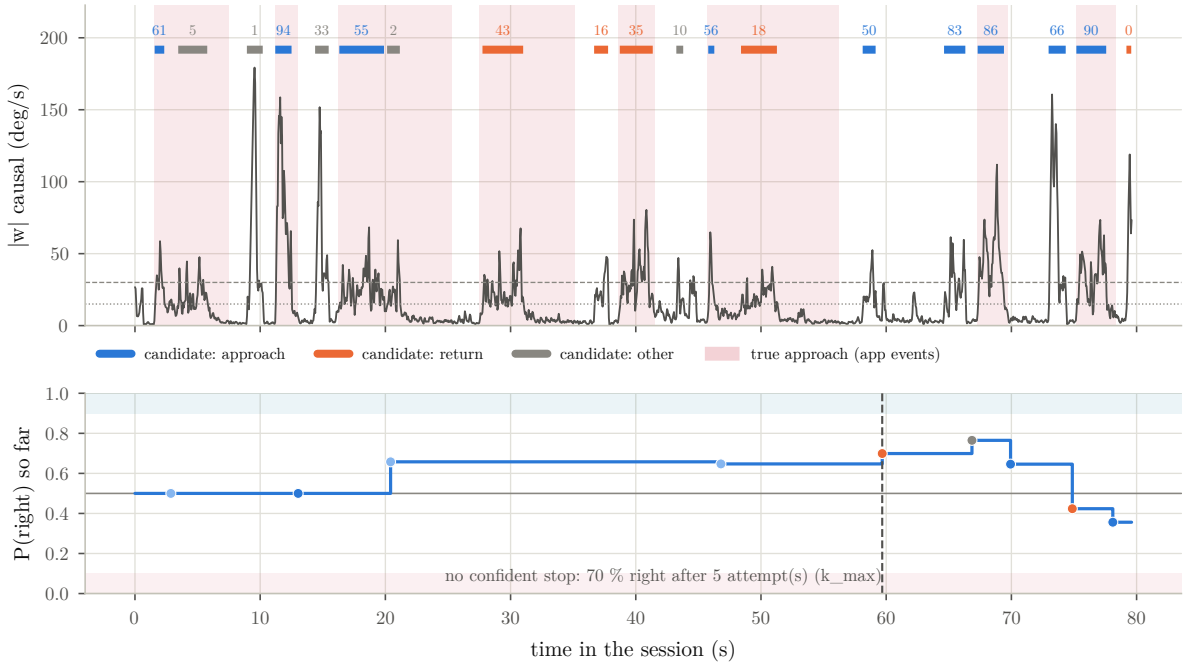


Figure 14: Online decision on one synthetic session of the left hand, replayed sample by sample. Top: causal activity and the candidate segments with their P(approach). Bottom: the aggregated $P(\text{right})$ after each accepted attempt (blue: a true approach; orange: a return; grey: another movement) and the stopping band. The session was not selected for success.

prospective test.

6.5. Natural use: the main failure mode

The detector fails more often on free choice (Table 6). It rejects 9 of 20 natural-use recordings (45 %, against 17 % on the primary set; 3.2 rejections expected from each person's own rate, Poisson-binomial $p = 0.0015$, unadjusted, a single pre-planned test). The production pipeline accepts an approach in 11 of them, 1 of which gets exactly 0.5, so it shows a hand for 10 (50.0 %). Its right-hand recall is 90.0 % on the trials with a hand shown, 72 % forced-choice and 70.0 % [57.5, 82.5] with half credit. All these trials are right-hand trials, so a constant right answer would score 100 %. Free-choice approaches look less like the instructed approaches the detector learned from, and the detector must learn them before production (Appendix G.12).

6.6. Instructions for the user

Three instructions follow from these results: hold the phone with the screen facing you, tilted slightly up; keep still for 1 s before the movement; and make several attempts. The decision comes a median 754 ms after the first screen touch (Appendix G.7).

Table 6: Primary and natural-use recordings (one first free-choice trial per person, all right hands): share answered (an approach accepted), accuracy on the answered recordings, forced-choice accuracy (the best candidate always scored) and half-credit accuracy with a person-bootstrap 95 % CI.

Pipeline	Set	Trials	Answered	Acc. answered	Forced choice	Half credit	CI
oracle window (no detection)	primary	598	100%	77.9%	77.9%	77.9%	[73.1, 82.9]
oracle window (no detection)	natural use	20	100%	75.0%	75.0%	75.0%	[55.0, 90.0]
detector v0 (first movement)	primary	598	95%	68.7%	67.7%	67.7%	[63.4, 72.1]
detector v0 (first movement)	natural use	20	80%	68.8%	65.0%	65.0%	[45.0, 82.5]
learned, robust (production detector)	primary	598	83%	78.2%	74.9%	73.4%	[68.7, 77.9]
learned, robust (production detector)	natural use	20	55%	81.8%	70.0%	67.5%	[52.5, 82.5]
learned, grip-normalised	primary	598	75%	74.5%	70.6%	68.5%	[62.8, 74.0]
learned, grip-normalised	natural use	20	55%	81.8%	70.0%	67.5%	[52.5, 82.5]
production (calibrator nested)	primary	598	83%	77.5%	74.6%	72.8%	[68.9, 76.6]
production (calibrator nested)	natural use	20	55%	86.4%	72.5%	70.0%	[57.5, 82.5]

6.7. The exported final model

The final model is exported for the inference page as a numpy-only reference implementation (Appendix F.14): 46084 fitted numbers in 1585 kB of JSON, about 20.05 ms per prediction. It reproduces the fitted pipeline to 2.2e-16 on all 618 trials. The export carries the rest-pose check and the scale τ_k of equation (3). It has not yet run behind the cue-free detector.

7. Secondary analyses: speed, grip, handedness and demographics

The same recordings carry more than the hand. These secondary models feed nothing that the page displays, so the report makes no calibration claim for them. All use LOSO with nested tuning and balanced accuracy, the mean recall over the classes, because the classes are unbalanced (Appendix H.1).

Instructed speed. The app ended a slow trial with stricter criteria than the other trials, so the oracle windows carry the speed label in their length. Re-running the app's detector with one parameter set for every trial gives label-blind windows (Appendix H.2). On these, the pre-specified multinomial LR on 16 mirror-invariant kinematic features reaches a balanced accuracy of 50.1 % [42.9, 56.4] (chance 33.3 %; Figure 15). On the oracle windows it reaches 67.8 % [60.7, 73.7], a label-informed and optimistic figure (paired difference −17.8 [−22.7, −12.5] points, Holm $p < 0.001$ over the balanced-accuracy comparisons of Appendix H.4). LightGBM is not better (difference −4.7 [−9.9, +0.6] points, Holm $p = 1.000$). Two caveats limit the result. The protocol confounds speed with grip: the grip alone predicts the instructed speed at 48.0 % [43.8, 52.9], and the headline does not beat it in balanced accuracy (Holm $p = 1.000$). And the fast instruction was weak: it raised the peak angular speed for 18 of 20 persons (Holm $p < 0.001$ over the manipulation contrasts), but its effect on the duration is uncertain (natural/fast duration ratio 1.12 [1.00, 1.23], Holm $p = 0.195$; Appendix H.3).

Grip. The grip is recognised at a balanced accuracy of 95.2 % [89.7, 100.0] from the up vector in device coordinates while the phone rests before the movement; a 12-feature model gives

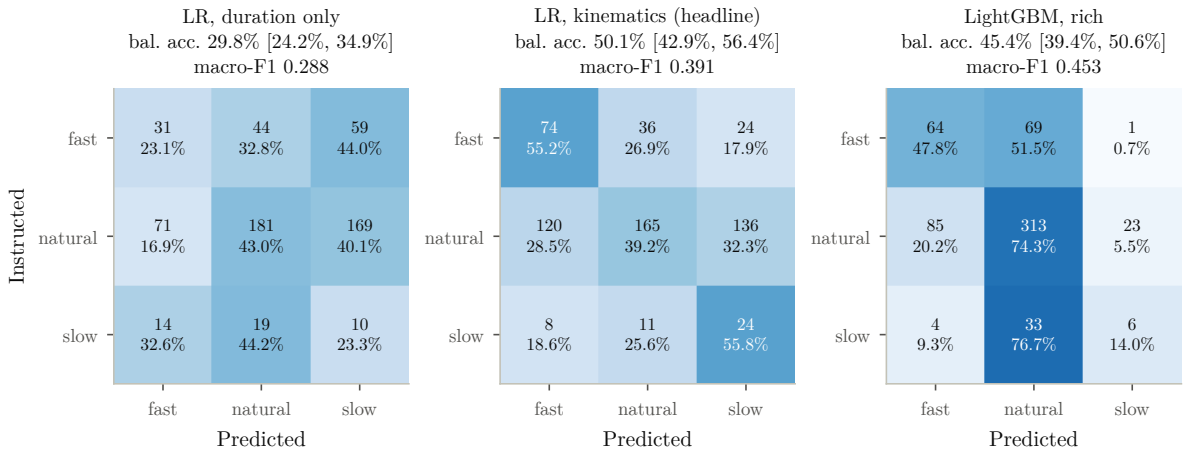


Figure 15: Confusion matrices of the instructed speed on label-blind windows (LOSO; counts and row shares; rows: instructed speed) for the duration-only baseline, the headline kinematic LR and LightGBM, with balanced accuracy and macro-F1. From the sensors alone the speed is recognised only moderately above chance.

identical predictions. The 18 errors come from 3 persons, and all 18 lie in the up-vector cluster of the predicted grip. These trials were probably held in another grip than instructed, although no independent record confirms it (Appendix H.8).

Dominant hand. Mirror-invariant kinematics show no detectable dominant-hand signal (Figure 16): the pooled AUC is 0.491 [0.453, 0.533], against a within-person permutation null with mean 0.495 ($p = 0.582$, unadjusted, a single pre-specified test). With only 2 left-handed persons, no person-level handedness model was fitted; they are shown individually (Appendix H.9).

Demographics. Production has no demographics. A quick, unaudited first-pass test asked whether they would help if it had (Appendix H.10). A second-stage logistic regression on the final model's logit, fitted inside each fold, was given the persons' age, sex, height, handedness and usual phone hand. Against the same second stage without covariates, adding them changes the accuracy by -3.2 [-6.3, -0.7] points (Holm $p = 0.062$ over the two add-on variants), and adding their interactions with the logit raises the log loss by $+0.056$ [$+0.014$, $+0.105$] (Holm $p = 0.031$; Figure 104 in Appendix H.10). The base model is the shipped final model, whose selection is exploratory. Demographics are not added. Natural-use recall cannot judge such an add-on, because a rule that always answers right scores 100.0 % on the all-right free-choice trials.

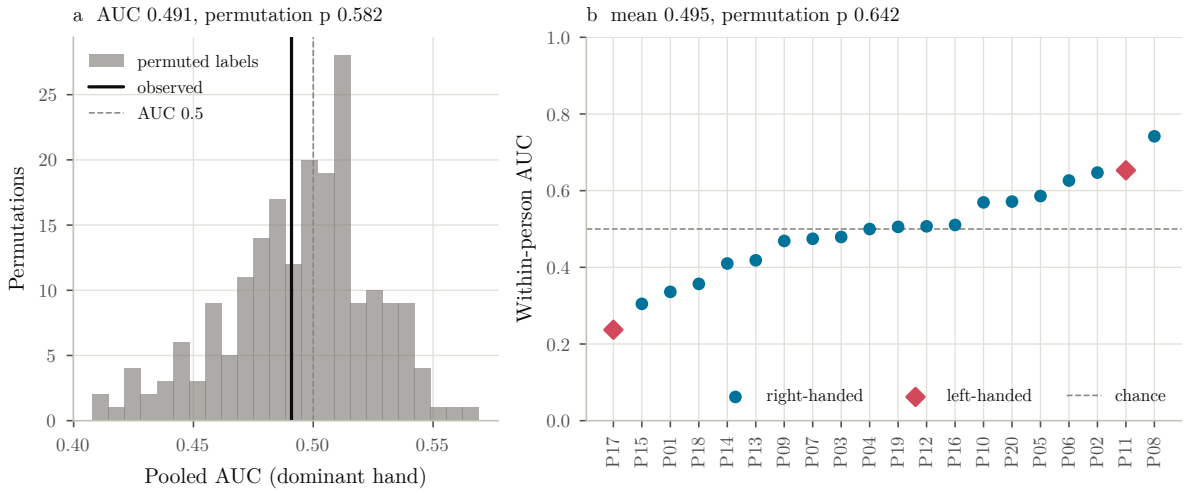


Figure 16: Can mirror-invariant kinematics tell a trial of the dominant hand? (a) The pooled AUC against its within-person label-permutation null (200 permutations, the whole nested pipeline re-run). (b) The AUC within each person; diamonds: the two left-handed persons, shown individually. No dominant-hand signal is detected.

8. Discussion

What the results mean. The hand that brings a phone to the nose can be read from the phone's motion sensors for persons the model has never seen. The signal is physical: early in the approach the phone turns about gravity in opposite senses for the two hands, and only chiral quantities carry the hand. A 1D CNN on gravity- and body-frame channels had the lowest log loss, although paired comparisons cannot separate it from the other leaders (6 in all); the cross-lane selection that chose it in 16 of 20 folds has a selection-aware accuracy of 88.5 % [84.1, 92.3]. Its body-frame input limits the change on a grip absent from training to −4.1 points, against −19.8 to −31.0 points for the leaders on device-frame channels. With its in-fold Platt calibrator its percentage is consistent with calibration, and combining three attempts with the scaled sum of logits raises the accuracy without making the percentage over- or under-confident on average. The main risk lies before the classifier: finding the approach without a cue costs accuracy, and natural free-choice movements are often rejected.

Limitations.

- Only 20 persons, mostly students in their twenties; 2 left-handed and 8 on iOS. Rare groups are barely represented.
- One free-choice trial per person, all right-hand: natural-use recall cannot show whether left-hand free choice is recognised as well, and a constant right answer would score 100 %.
- The main analysis uses the oracle windows stored by the app; the cue-free numbers come from lane m4's yaw8 classifier, not the final model.
- Screen-away holds and multi-attempt sessions are synthesised from the recordings, not recorded.
- The final model's own LOSO numbers are optimistic: it was the best of 16 configurations in its lane and was then chosen among 17 candidates on the outer folds, and its body-frame

input was designed after outer-fold results had been seen. The nested estimate removes the optimism of choosing among the 17 candidates, not that of the lanes' own outer-fold choices (7 candidates; the CNN the best of 16) nor that of the post hoc body-frame design.
- The browser delivers about 60 Hz, and the app segmented slow trials with criteria that depend on the instructed speed, which affects every analysis stratified by speed (Appendix H.2).

Next steps.

1. Run the final model inside the cue-free production simulation, and teach the detector free-choice approaches.
2. Build the Stage 3 inference page from the exported model, with the rest-pose check, the stillness check and the displayed aggregate of equation (3).
3. Record more participants, especially left-handed persons, and left-hand free-choice trials.
4. Close the open reviews listed in the status note (Section 1).
5. Let the PI evaluate the page prospectively on new persons.

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A. Data collection, curation and reproducibility

This appendix documents the data decisions behind every analysis: what the app records, how the two phone platforms are brought to one convention, which attempts become trials, what is flagged but kept, how the labels were screened, the reference floor that every model must beat, and how the report is rebuilt. Each subsection states the options, the evidence and the decision. Table 7 lists the decisions at a glance; Table 17 gives the notation.

Evaluation is always leave-one-person-out (LOSO): a model is trained on all other persons and tested on the held-out person, one fold per person [7, 8]. All 95 % confidence intervals (CIs) of this report are person-cluster bootstrap intervals: 2000 resamples of the persons with replacement, each resampled person bringing all of their trials [18]. Other abbreviations: area under the receiver operating characteristic curve (AUC), standard deviation (SD), standard error (SE), interquartile range (IQR), root mean square (RMS).

Table 7: The data decisions of this appendix at a glance. Each row is argued in the subsection named in the last column, with its figure or table.

Decision	Options considered	Choice	Where
Model inputs	every recorded stream; motion streams only	motion streams and segmentation bounds; no heading, magnetometer or Euler angles	A.1
Accelerometer sign	engine; calibration; trial votes	engine sign confirmed by the trial votes	A.2
Gyroscope bias	subtract the pre-GO mean; none	none	A.2
Sampling grid	native timestamps; a finer grid; 60 Hz	linear resampling to 60 Hz	A.3
Sampling gaps	interpolate all; drop every attempt with a gap; a threshold	interpolate up to 100 ms; exclude a longer gap near the movement (E4d)	A.3
Value precision	round iOS to Chrome's grid; native	native on both platforms	A.3
Platforms	a model per platform; platform as a feature; one model	one model; platform is a reporting stratum	A.3
Statistical unit	app code; session; person	the curated person	A.4
Exclusions	gates only; exclude every gate failure; exclude segments without a valid representation	E4b–E4d exclude; the gates are scored apart	A.4, A.7
Partial sessions	drop; keep	keep	A.5
Approach window	offline segmentation; stored online segmentation; screen contact	stored online segmentation (the oracle window)	A.6
Natural-use set	every free-choice trial; the first per person	the first per person (N1)	A.4
Suspect labels	correct them; keep them	keep them; sensitivity analyses	A.8
Reference floor	none; a pre-specified simple model	mirror-augmented yaw logistic regression	A.9

A.1. The app and what it records

The collector is a web app opened from a link or a QR code in the participant's own phone browser; nothing is installed. On an iPhone it runs in Safari (every iOS browser uses the WebKit engine); on Android in Chrome or Samsung Internet (the Chromium engine). The street protocol (Table 2) is hands-free: spoken and on-screen prompts announce the hand, the grip and the speed, the app waits for the phone to be still, gives a GO cue, detects the movement while it happens and signals its end. Figure 17 shows the gesture as the app's instructions draw it. Recruitment, compensation and a photo of a session (Figure 1) are in Section 2.

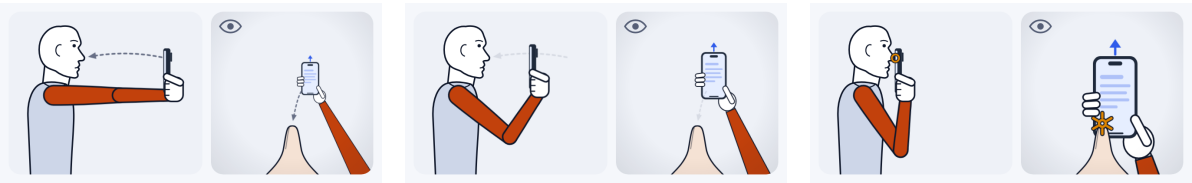
Every attempt is uploaded as one record as soon as it ends (Figure 18). A record holds:

- the motion streams of the browser's `devicemotion` event at about 60 Hz, the browser maximum: the linear acceleration $\mathbf{a}$, the specific force $\mathbf{f}$ (acceleration including gravity) and the rotation rate $\boldsymbol{\omega}$;
- orientation streams, all best effort and metadata only: Euler angles, relative and absolute (magnetometer-fused) quaternions, the magnetometer where a browser exposes it, and on iOS the compass heading;
- screen touches (time, position, contact size, force) and screen rotations;
- the app's online segmentation and the cue times: GO, onset, end, end cue (Appendix A.6);

- the return of the arm after the end cue, with its own onset and end (data schema 1.2);
- a short profile (age, sex, dominant hand, usual phone hand, optional height) and device details.

The app collects no camera, microphone or location data.

Decision: what feeds the models. The options were to use every stream or the motion streams only. Only the motion streams and the segmentation bounds reach a model. The heading streams (compass, absolute orientation, magnetometer) are excluded for two reasons. Earth's magnetic field is not mirrored with the movement, so a heading cannot be mirrored (Equation (1)), and a heading partly reveals the place where the phone was used. The Euler angles are excluded because they hit gimbal lock at the upright start pose, carry an arbitrary heading and, on Chrome, update only when an angle changes by at least 0.1°. Touches serve only to describe the contact (Appendix A.6). The return is never pooled with approaches: it reverses the approach in time, and its yaw points to the other hand (Appendix A.6; Appendix G). The profile never enters a model: it lives in a person table, apart from the trial table, and only stratifies results.



(a) Arm extended, still **(b)** Mid-reach **(c)** Nose touch and hold

Figure 17: The gesture as the app's instructions show it (right hand, upright grip, natural speed).

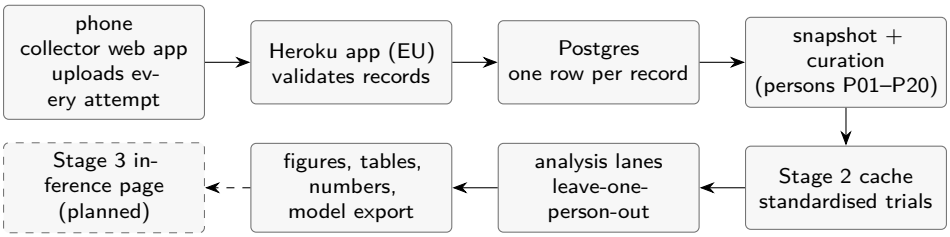


Figure 18: Data flow from the phone to this report. Every figure, table and number of the report is regenerated from the saved analysis outputs by one command (Appendix A.11).

A.2. Sensor conventions and platform normalisation

Table 8 summarises how the two browser engines deliver the sensors [1]. Both use the device frame of the World Wide Web Consortium (W3C) specification: x toward the right edge, y toward the top edge, z out of the screen. Two differences matter for the analysis. WebKit inverts the sign of every accelerometer axis, and Chrome rounds every value to 0.1 unit while WebKit does not. Figure 19 checks the resulting conventions end to end on real movements.

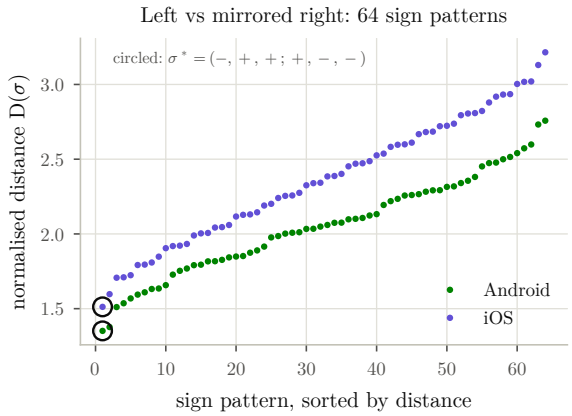
Table 8: What a phone browser delivers (condensed from the project's sensor notes and the W3C specification [1]).

	iOS: WebKit (every iOS browser)	Android: Chromium (Chrome, Samsung Internet)
Motion rate	≈60 Hz from a 1/60 s timer; duplicated or skipped samples	60 Hz pump
Value precision	unrounded doubles	rounded to 0.1 m/s^2 and 0.1 deg/s
Accelerometer sign	all axes inverted (face up: $z \approx -9.81$)	W3C (face up: $z \approx +9.81$)
`rotationRate`	deg/s; alpha, beta, gamma about x, y, z; not inverted	the same
`event.interval`	≈0.0167, in seconds	16, in ms, constant
Permission	`requestPermission()` inside a tap; a Cancel sticks	Chrome 151+ also has it; granted by default

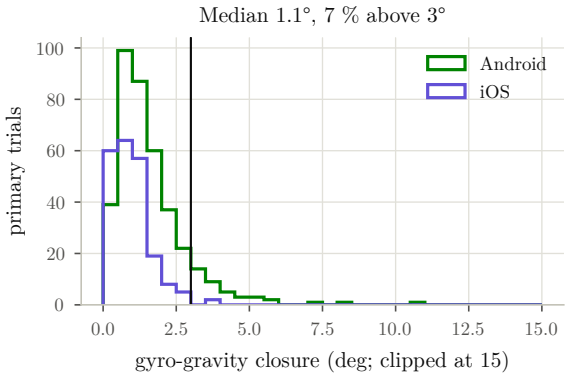
Decision: one versioned normalisation. Raw records are never modified. One versioned step (`norm_v2`) maps every session to the W3C convention before any feature is computed. The accelerometer sign could come from the engine (WebKit −1, Chromium +1), from a calibration, or from the data. Street sessions have no guided calibration, so the step takes the engine's expected sign and requires the trial votes to confirm it. In each trial with a fixed instructed grip, the rest gravity before the onset votes for the sign that makes it point up. Every included session's sign is supported: the street sessions by the engine sign that the votes confirm, the pilot session by its guided calibration. Android sessions resolve to +1 and iOS sessions to −1, as the engines predict. The `rotationRate` mapping (alpha, beta, gamma about x, y, z) is chosen by the gravity closure defined below: the W3C mapping closes with a median of 1.0°, against 13.4° for the runner-up mapping. Sessions too short to settle it keep the W3C default.

Evidence: the 64 sign patterns. An end-to-end check uses the movements themselves. The mirror of Equation (1) turns a right-hand movement into a left-hand one. In device coordinates it predicts the sign pattern $\sigma^* = (-, +, +; +, -, -)$ on $(a_x, a_y, a_z; \omega_x, \omega_y, \omega_z)$, applied together with the mirrored grip. For each of the $2^6 = 64$ sign patterns, Figure 19a measures the distance between every person's mean left-hand waveform and the sign-flipped mean right-hand waveform. The predicted pattern σ^* ranks 1 of 64 on Android and 1 on iOS. A wrong axis, a wrong sign or a failed iOS normalisation would put another pattern first.

Decision: gravity frame without a gyroscope-bias correction. The heading-free channels need the up vector $\hat{\mathbf{g}}$ during the movement. The gravity estimate (`gravity_v1`) takes the rest gravity before the onset and after the end, carries each through the movement with the gyroscope and blends the two. Its closure is the angle between the rest gravity carried to the nose and the rest gravity measured there; it tests the axis mapping and the propagation together. The options for the gyroscope bias were to subtract the mean rate before GO when it is below 3 deg/s (the analysis plan: the repository's Stage 2/3 plan, docs/09, whose decisions are cited as D1, D2, . . .) or to subtract nothing. Nothing is subtracted. Street sessions have no still calibration window, production has no GO, and the mean rate of a hand-held phone is tremor rather than bias. Without the correction the closure has a median of 1.1°, and 7 % of the primary trials exceed 3° (Figure 19b). The median is 1.3° on Android and 0.8° on iOS.



(a) The 64 sign patterns of the device-frame channels, sorted by the distance between left-hand and sign-flipped right-hand mean waveforms (channels scaled to unit RMS). Circled: the conventions' mirror σ^*, first on both platforms.



(b) Gyro–gravity closure of the primary trials: the angle between the rest gravity carried through the movement by the gyroscope and the rest gravity at the nose, with no bias correction. Line: 3°.

Figure 19: Two end-to-end checks of the sensor conventions on real movements. They support the sign normalisation, the `rotationRate` mapping and the gravity frame without a gyroscope-bias correction.

A.3. Sampling, gaps, precision and platforms

Decision: a uniform 60 Hz grid. Both platforms deliver 60 Hz. The median sample interval is 16.7 ms on Android and 17.0 ms on iOS, and the 99th percentile is 18.5 ms and 18.0 ms (Figure 20, left; Table 9). Isolated duplicated samples are dropped (flag T11, Appendix A.7), and the streams are resampled linearly onto a uniform 60 Hz grid. A finer grid would add no information, and the inference page sees the same browser rate. The low-pass filter and the analysis windows are preprocessing decisions (Appendix B).

Evidence: two kinds of gap. The intervals have a tail of gaps (Figure 20, right). Android gaps longer than 50 ms are short, with a median of 64 ms. iOS gaps are several times longer, with a median of 314 ms. In the 66 iOS attempts with a gap longer than 200 ms, these gaps last a median of 318 ms and start a median of 4.2 s after the recording starts; only 25 % of them follow an audio cue within 0.4 s. This pattern points to a recurring stall of the collection app's main thread, not to the sensor. The inference page must keep its main thread free while it records.

Decision: interpolate short gaps, exclude long gaps near the movement. The options were to interpolate every gap, to drop every attempt with a gap, or to set a threshold. Gaps up to 100 ms (at most 5 samples) are interpolated linearly. A longer gap stays missing in the stream. When it falls within $[\text{onset} - 0.5\,\text{s},\ \text{end} + 0.5\,\text{s}]$ it removes part of the movement, which interpolation cannot restore, and the trial is excluded (rule E4d, Appendix A.4). Elsewhere in the recording it does no harm and stays missing. The long gaps are an iOS phenomenon, so E4d removes mostly iOS trials (Table 9, attempts with a gap longer than 100 ms).

Decision: native precision. Chrome's rounding to a 0.1-unit grid adds noise with an SD of 0.029 unit ($0.1/\sqrt{12}$). The noise of a hand-held phone in the still window before GO is far larger (Figure 21; medians over trials): the SD of the specific force is 0.154 m/s^2 on Android and 0.147 m/s^2 on iOS, and the SD of the angular velocity 2.5 deg/s and 2.8 deg/s. The options were to round iOS values to Chrome's grid, so that the platforms match, or to keep the native values. Both platforms keep their native precision; rounding iOS is an ablation (A3, Appendix B).

Decision: one model for both platforms. After normalisation the platforms agree where it matters (Figure 22; Table 10). The class-mean yaw rate about gravity, $\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$, separates the hands with the same sign on both. The first-half yaw (the yaw about gravity accumulated over the first half of the approach) separates them with a pooled AUC of 0.86 on Android and 0.84 on iOS. iOS approaches are shorter (median 1.42 s against 1.80 s) and turn slightly faster (peak $|\boldsymbol{\omega}|$ 100 against 94 deg/s). The options were a model per platform, platform as an input feature, or one model for both. One model serves both platforms, and platform is a reporting stratum with a leave-one-platform-out check. Phones differ in their sensors [2]; here the normalisation removes the sign differences, and what remains (the iOS stalls, the rounding, the durations) is reported per platform. With 8 iOS persons, a per-platform model would also be trained on very few people.

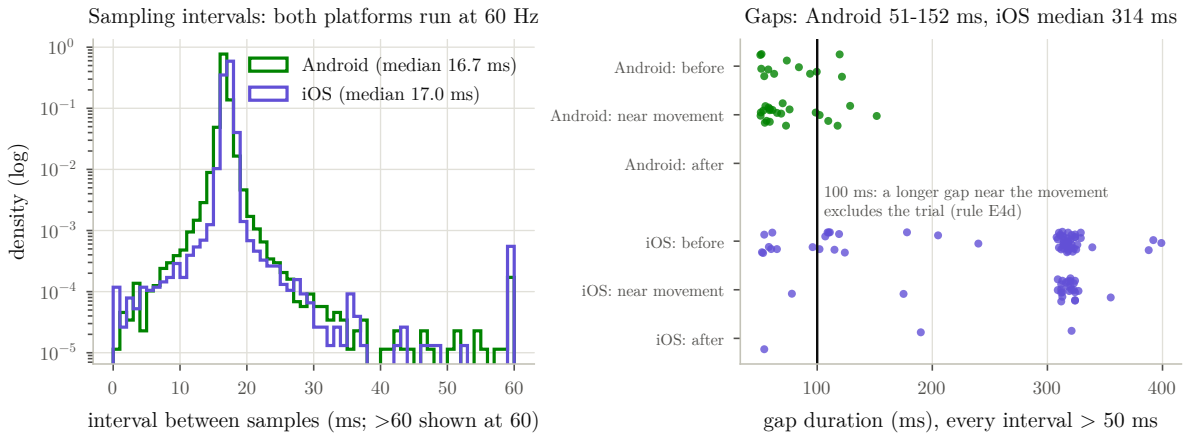


Figure 20: Sampling of the complete attempts. Left: the distribution of sample intervals on each platform (both run at 60 Hz). Right: every interval longer than 50 ms, by platform and by where it falls relative to the movement ("near": within 0.5 s of the approach). The line at 100 ms is the threshold of rule E4d. Supports the 60 Hz grid, the interpolation of short gaps and the exclusion of long gaps near the movement.

A.4. Curation and the exclusion cascade

The analysis reads one frozen snapshot of the upload database: 27 sessions with 878 attempts, each validated against the data schema, with no duplicate and no invalid record. The researcher's curation file maps every session to a person (P01–P20) and marks 24 sessions as included. The file's free-text reasons are replaced by fixed codes, and the private file that links persons to names is never opened (Appendix A.11).

Table 9: Sampling of the complete attempts per platform (medians over attempts): the rate, the sample intervals, the gaps and the hand-held noise in the still window before GO.

Quantity	Android	iOS
Complete attempts	418	257
Sampling rate, median (Hz)	60.0	60.0
Sample interval, median (ms)	16.7	17.0
Sample interval, 99th percentile (ms)	18.5	18.0
Gaps > 50 ms	34	89
Gap duration, median (ms)	64	314
Attempts with a gap > 100 ms	7	70
Rest SD of the specific force, median (m/s^2)	0.154	0.147
Rest SD of the angular velocity, median (deg/s)	2.52	2.83

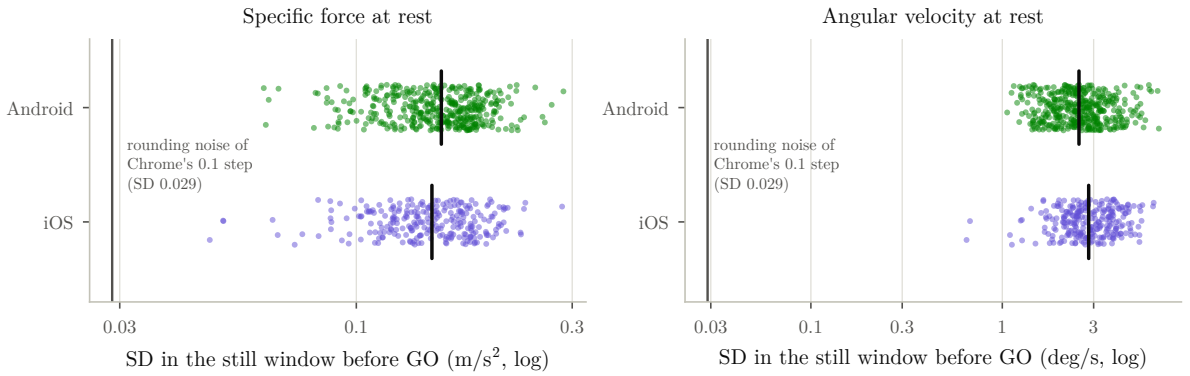


Figure 21: Hand-held noise in the still window before GO, per trial and platform (bar: median), against the noise of Chrome's 0.1-unit rounding (SD 0.029, vertical line). Left: specific force; right: angular velocity. The rounding is far below the physiological noise floor, so native precision is kept on both platforms.

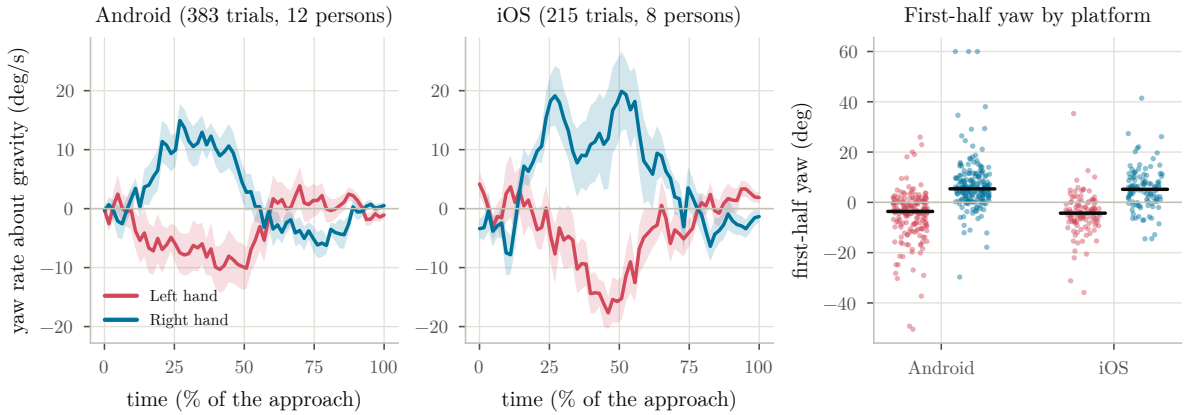


Figure 22: Android and iOS after standardisation: the class-mean yaw rate about gravity over the time-normalised approach (band: ±1 SE across persons) and the first-half yaw per trial (bar: median). The hands separate with the same sign on both platforms, which confirms the sign normalisation and supports one model for both.

Table 10: Android and iOS after standardisation (medians over the primary trials unless stated).

Quantity	Android	iOS
Persons	12	8
Trials	383	215
Approach duration (s)	1.80	1.42
Peak $\|\omega\|$ (deg/s)	94	100
Gyro-gravity closure (deg)	1.3	0.8
First-half yaw, right hand (deg)	5.4	5.2
First-half yaw, left hand (deg)	−3.6	−4.3
AUC of the first-half yaw	0.86	0.84
Trials with a clean return window	40%	63%

Decision: the person is the statistical unit. The options were the app's participant code, the session, or the curated person. The analysis groups by the curated person. One app code held two persons who shared a phone, and one person held several codes; grouping by code would put one person on both sides of a fold. The evaluation harness refuses any grouping that splits a person. 3 persons contributed several sessions; one of these sessions is the pilot of the earlier protocol (1.0.0). The demographics are read from each session's own record, agree across each person's sessions, and are stored in the person table only, so they cannot reach a model.

The exclusion cascade. Rules apply in order, and an attempt is charged to the first rule it fails (Table 11). Rule E3 diverts rather than removes.

- **E0** removes the 42 attempts of the sessions the researcher excluded: an incomplete run superseded by the person's later run, and runs that did not work.
- **E0s** (simulated records) and **E2** (the screen-away grip, collected only by the pilot protocol) remove nothing. Street visits never collect screen-away trials; the analysis synthesises them by a screen flip of the arrays (Appendix B).
- **E1** removes the 172 attempts the app did not accept (Table 12). Most showed no movement within the time limit after GO, or a phone still moving at the limit; the few others were redone, rejected for a wrong grip, aborted, flagged or interrupted. Overall, 1.26 attempts were needed per accepted trial.
- **E3** moves the 23 free-choice trials to the natural-use set. Their hand was chosen and reported by the person, not instructed, so they are never trained on and are scored apart.
- **E4a** (no approach window) removes nothing.
- **E4b** removes 2 approaches shorter than 0.25 s: a reach from arm's length cannot be that short, so a fragment was segmented (Figure 25).
- **E4c** removes 11 attempts whose phone did not turn (peak $|\boldsymbol{\omega}|$ below 30 deg/s, the app's own onset floor): every reach rotates the forearm. The natural-use set also loses a trial here.
- **E4d** removes 30 attempts with a gap longer than 100 ms near the movement (Appendix A.3).
- **E4e** (page hidden, rate below 50 Hz, no gyroscope, unsupported gravity sign) removes nothing.
- **N1** keeps one natural-use trial per person: the earliest free-choice trial that passed every rule, the person's first unpractised movement with a hand of their choice. The 2 free-choice

trials of repeat visits are reported apart.

The analysis sets. The primary set holds 598 trials of 20 persons in 24 sessions: 302 left-hand and 296 right-hand trials, 383 on Android and 215 on iOS. Each person contributes both hands, with 8 to 45 trials (median 28). The natural-use set holds 20 trials, one per person, and all are right-hand trials (20 right, 0 left). Natural-use accuracy is therefore right-hand recall on the free-choice trials (all right-hand so far; a constant right answer would score 100 %). The set supports an accuracy with a CI and nothing that needs both hands, such as an AUC or a calibration.

Decision: exclude only what has no valid representation. The analysis plan (docs/09, D5) foresaw no new exclusions: movement checks would act as predictor gates at evaluation. The options were to keep that plan, to exclude every gate failure, or to exclude only segments without a valid representation. Rules E4b, E4c and E4d exclude, because a fragment, a phone that does not turn and a hole in the movement have no valid representation. The remaining gate failures are valid movements and stay in training and in the headline (Appendix A.7). The cache keeps the arrays of every excluded complete attempt, so a sensitivity analysis that re-admits them needs no rebuild.

Table 11: Exclusion cascade: rules apply in order and an attempt is charged to the first rule it fails. E3 moves the free-choice trials (hand self-reported) to the natural-use set, which then passes the same quality rules; N1 keeps one natural-use trial per person (the earliest).

Step	Rule	Removed	Remaining	Persons	Sessions	Natural use
start	all attempts in the snapshot	0	878	21	27	0
E0	session excluded by the researcher	42	836	20	24	0
E0s	simulated record (S1)	0	836	20	24	0
E1	attempt not accepted	172	664	20	24	0
E2	screen-away grip (protocol 1.0.0)	0	664	20	24	0
E3	free-choice trial (hand self-reported)	0	641	20	24	23
E4a	no approach window	0	641	20	24	23
E4b	approach shorter than 0.25 s	2	639	20	24	23
E4c	phone did not turn (peak $\|\omega\|$ < 30 deg/s)	11	628	20	24	22
E4d	sampling gap > 100 ms near the movement	30	598	20	24	22
E4e	recording quality (T4, T5, T7, S3)	0	598	20	24	22
N1	later free-choice trial of a person	0	598	20	24	20

Table 12: Why attempts were not accepted (rule E1).

Reason (first that applies)	Attempts
failed: no movement within the time limit after GO	79
failed: still moving at the time limit	79
revoked: "redo last movement" on the pause sheet	4
not confirmed at tap review (protocol 1.0.0)	3
auto-rejected: grip detected as another grip	3
failed: aborted by the participant	2
flagged by the participant at review	1
failed: interrupted (page hidden or sensors stalled)	1

A.5. Completion, conditions and durations

Decision: keep partial sessions. Of the 23 street visits, 19 were completed and 4 ended early (Figure 23, left). The optional bonus tier was taken in 7 of the completed visits. The options were to drop partial sessions or to keep them. They are kept: the plan balances the hands within every tier, so even a visit that stops early contributes both hands. Dropping partial sessions would lose trials, and a whole person whose only visit ended early.

Evidence: visits run long. A completed visit took a median of 9.7 min (IQR 7.8–11.0 min), not the planned 5 min (Figure 23, right). The tiers took medians of 2.4, 3.1 and 1.8 min (tiers 1 to 3) and 2.0 min for the bonus tier. The next collection should plan for the measured duration, not the app's estimate.

Evidence: the conditions are balanced between hands. The primary trials span 19 instructed conditions of grip, speed and movement variant (Figure 24). Every condition has trials of both hands in similar numbers, so a condition cannot stand in for the label. Upright standard reaches at natural speed dominate, and the rare variants hold too few trials for a per-condition accuracy; the secondary analyses pool them (Appendix H).

Evidence: durations and the thresholds of E4b and E4c. The oracle approach (defined in Appendix A.6) lasts a median of 1.65 s, and 73 % of approaches last at most 2.5 s (Figure 25; Table 13). The instructed speed separates durations only loosely: the medians are 1.42 s for fast, 1.63 s for natural and 5.52 s for slow movements. Duration is therefore a poor proxy for the instructed speed. The reaction time from GO to the onset has a median of 575 ms. The fixed analysis window (−0.5 to +2.5 s around the onset) and the time normalisation follow from these durations (Appendix B).

A.6. Segmentation: the oracle window, online and offline

Two segmentations exist for every recording. The *online* segmentation is computed while the movement happens, from the samples seen so far: the app's `online_v1` detector drives the end cue and stores its bounds with the attempt. Its onset is back-dated to the end of the stillness

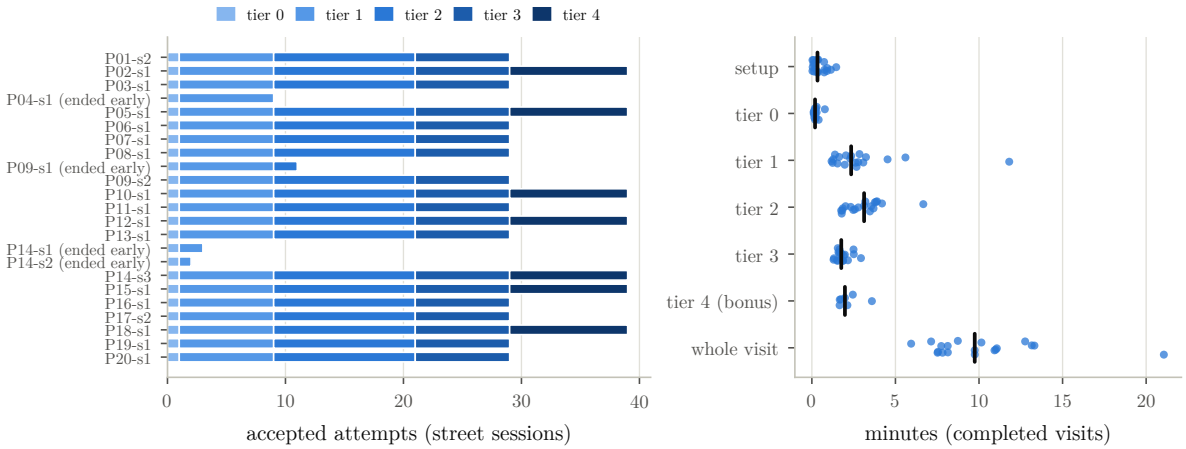


Figure 23: Completion. Left: accepted attempts per tier and street session (pseudonymous session labels; 4 visits ended early). Right: wall-clock durations of the setup, of each tier and of the whole visit, for the completed visits (bar: median). Partial sessions still hold both hands in every tier, so they are kept.

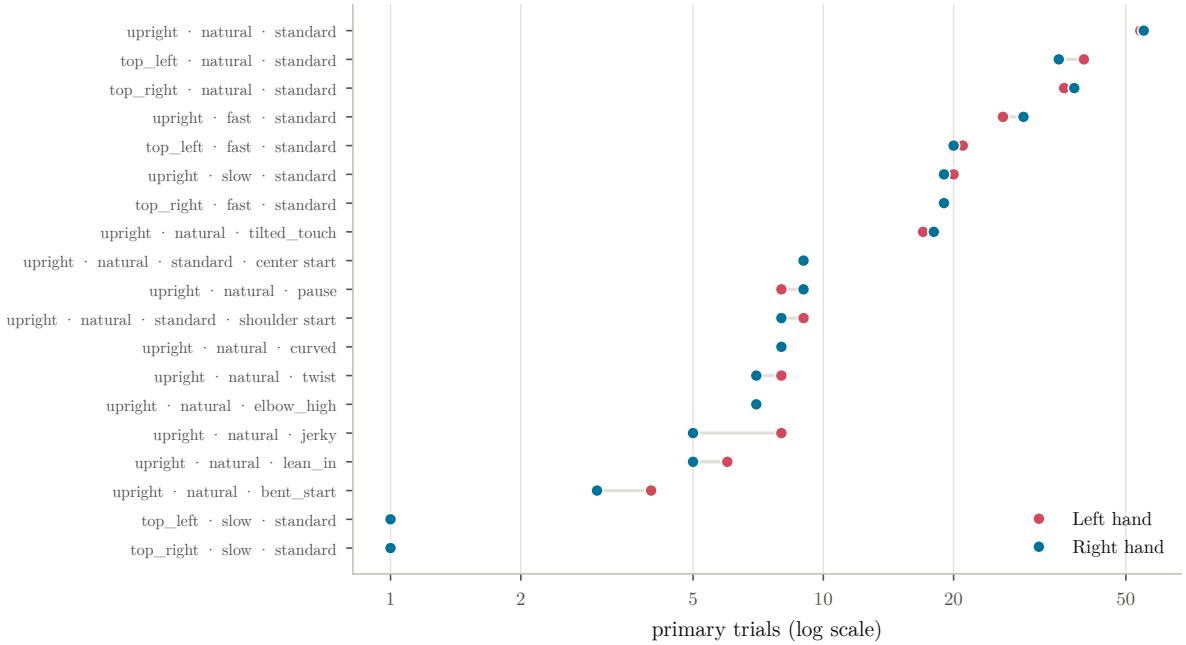


Figure 24: Primary trials per instructed condition (grip · speed · movement variant · start position), left against right hand, on a log scale. Every condition has both hands in similar numbers, so no condition can stand in for the label.

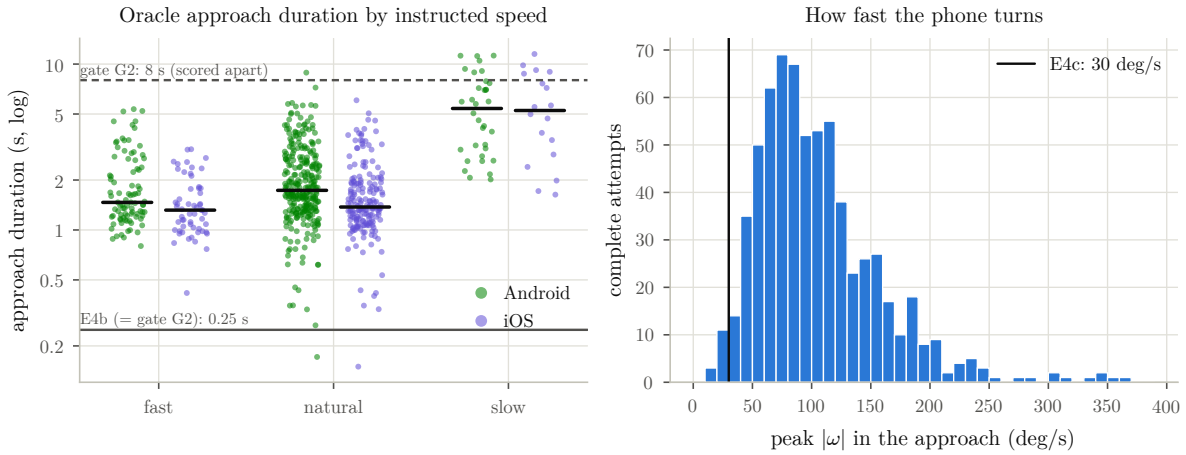


Figure 25: Left: oracle approach duration by instructed speed and platform (bar: median). Solid line: the 0.25 s threshold of E4b (a lower bound of gate G2); dashed line: the 8 s upper bound of gate G2 (longer approaches stay in the primary set and are scored apart). Right: peak angular speed within the approach, with the 30 deg/s threshold of E4c. Supports excluding fragments and phones that do not turn.

Table 13: Oracle approach duration of the primary trials by instructed speed and platform.

Speed	Platform	Trials	Median (s)	Q1 (s)	Q3 (s)	Peak |w| (deg/s)
fast	Android	85	1.47	1.20	2.63	116
fast	iOS	49	1.33	1.02	1.78	128
natural	Android	268	1.74	1.34	2.65	89
natural	iOS	153	1.37	1.13	1.90	94
slow	Android	30	5.84	3.06	7.83	80
slow	iOS	13	4.98	2.85	8.98	98

before the movement, and its end to the start of the final stillness at the nose; the end cue follows 250 ms after the end. The *offline* segmentation is computed after the recording, on the whole stream: `offline_v1` re-segments each recording in the analysis and needs some stillness after the end to find it. The *oracle window* is the approach between the stored online bounds. In production nothing marks the movement, and a cue-free detector must find it (Appendix G).

Decision: the oracle window is the stored online segmentation. The options were the offline segmentation (the analysis plan, docs/09 D10), the stored online segmentation, or the first screen contact as the end of the approach. The evidence is in Figure 26. The offline segmentation finds no end in 19 % of street attempts, because the arm returns soon after the end cue and leaves no long stillness at the nose. Where both exist they agree closely at the onset and less closely at the end; 115 primary trials have onsets more than 100 ms apart (flag T9). The stored online bounds exist for every complete attempt and can be reproduced exactly from the record. Screen contact is unreliable as a bound: only 72 % of the screen-toward attempts record a touch, and the first touch falls at a median 73 % of the oracle window, 333 ms before its end; the last part of the window is the phone settling against the nose. The offline bounds stay in the cache for an ablation (A5); the sensitivity of the models to the window bounds is

tested by jittering them (Appendix B).

Evidence: what a trial looks like. Figure 27 shows one left-hand and one right-hand trial of the same person, phone, grip and speed, after standardisation. The yaw rate about gravity takes opposite signs early in the two approaches. In both trials the return of the arm starts less than a second after the end cue and is as large as the approach. A detector that hands the return to an approach model gets a reversed movement and a confidently wrong answer; the return is therefore labelled separately and never pooled with approaches (Appendix G). A clean return window exists for 288 of the primary trials.

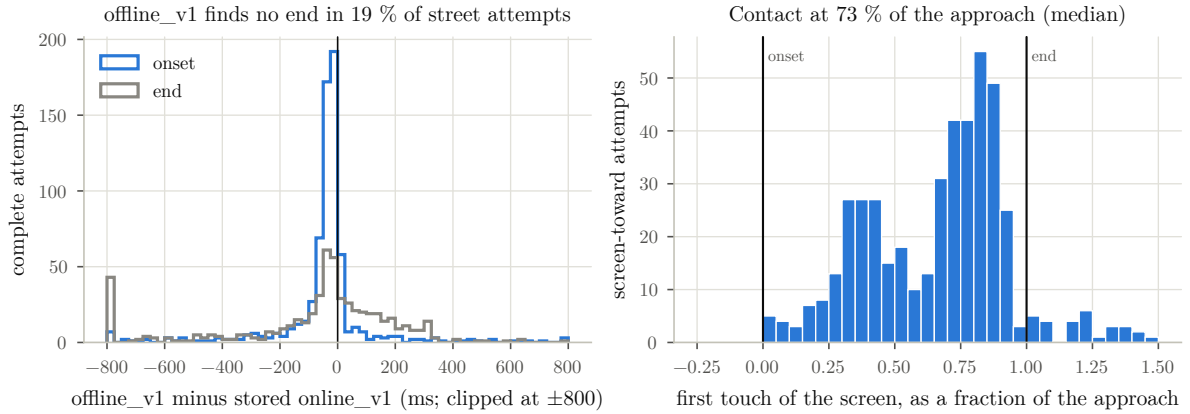


Figure 26: Left: offline minus stored online segmentation bounds, at the onset and at the end (clipped at ±800 ms; the offline segmentation finds no end in 19 % of street attempts). Right: the first screen touch as a fraction of the oracle window (0: onset, 1: end) in the screen-toward attempts. Supports the stored online bounds as the oracle window, with contact not used as a bound.

A.7. Flags and predictor gates

Two kinds of trial-level column are computed for every trial and never remove one (Figure 28; Table 14).

Flags (the trial checks T8–T12 of the data schema, docs/02, and a screen-rotation warning) describe the recording and serve only to stratify results:

- T8, grip not as instructed (23 trials): the grip before the onset differs from the instructed one, by the app's check or by the standardised up vector. The hand label is kept, because the hand instruction concerns the hand, not the grip.
- T9, online and offline onsets more than 100 ms apart (115 trials).
- T10, possible clipping (8 trials): $|\mathbf{f}|$ above 3.5 g or $|\boldsymbol{\omega}|$ above 950 deg/s somewhere in the raw recording.
- T11, repeated samples (126 trials): isolated duplicated samples were dropped before resampling.
- T12, a false start before GO (128 trials): the hold restarted before GO.

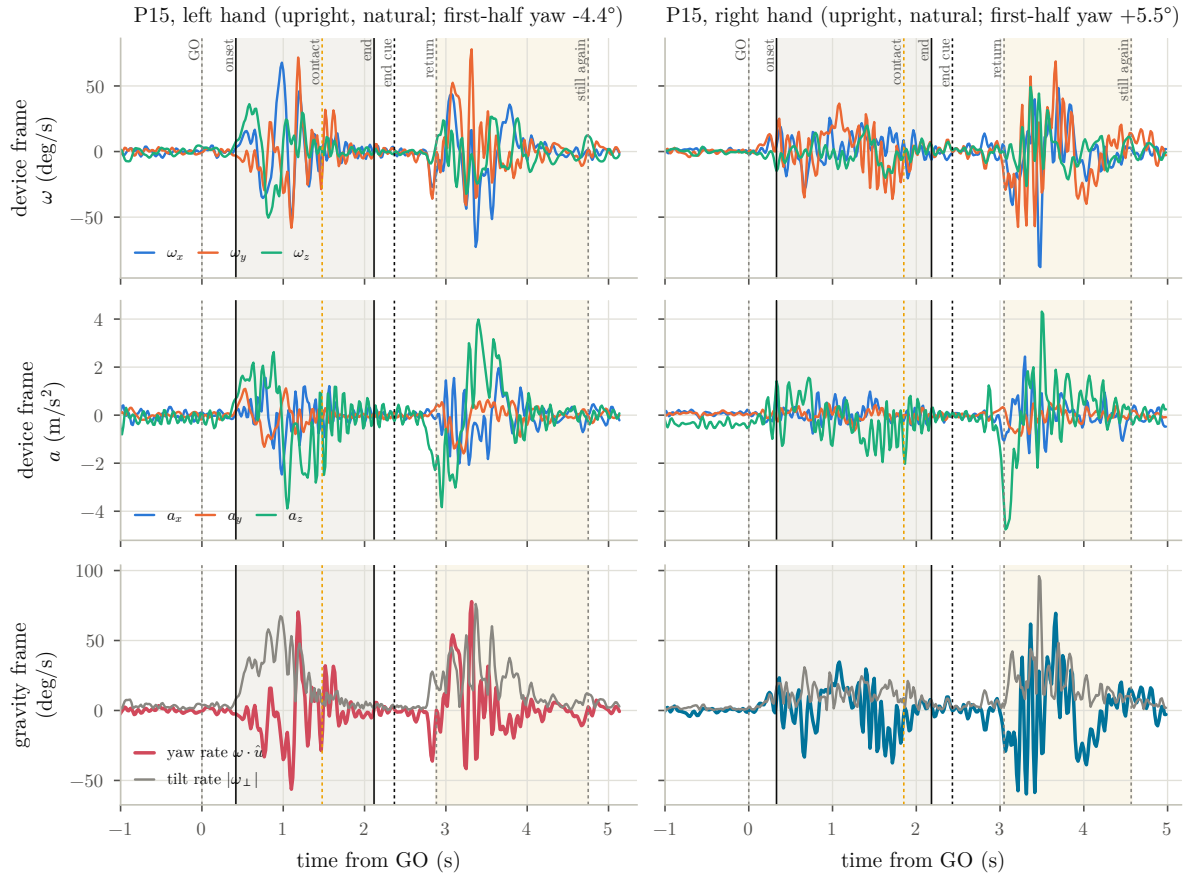


Figure 27: One left-hand and one right-hand trial of P15 (same phone, upright grip, natural speed), standardised: device-frame angular velocity and linear acceleration (top rows) and the heading-free gravity-frame yaw rate and tilt rate (bottom). Grey band: the oracle approach; beige band: the return of the arm. The yaw rate has opposite signs early in the two approaches; the return is a large separate movement that must be kept apart.

- Screen rotated during the movement (2 trials).

Predictor gates are the checks the inference page will apply before it answers (docs/09 D5). Gate G1 requires a rate of at least 50 Hz, no gap longer than 50 ms within 0.3 s of the approach, a visible page and a gyroscope; 16 primary trials fail it, all on Android, with gaps between 50 and 100 ms that were interpolated. Gate G2 requires a duration between 0.25 and 8 s and a peak $|\boldsymbol{\omega}|$ of at least 30 deg/s; 13 trials fail it, nearly all instructed slow reaches longer than 8 s. In all, 28 primary trials fail G1 or G2.

Decision: gate failures are kept and scored apart. The plan would not score gate failures. The options were to drop them from training and evaluation, or to keep them and report the gated metrics as well. They are kept in training and in the headline metrics, and every result also reports its metrics on the trials that pass both gates, as the inference page would see them.

The headline is then the harder, conservative number. For the reference floor (Appendix A.9) the accuracy is 77.9 % on all primary trials and 78.4 % on the trials that pass both gates.

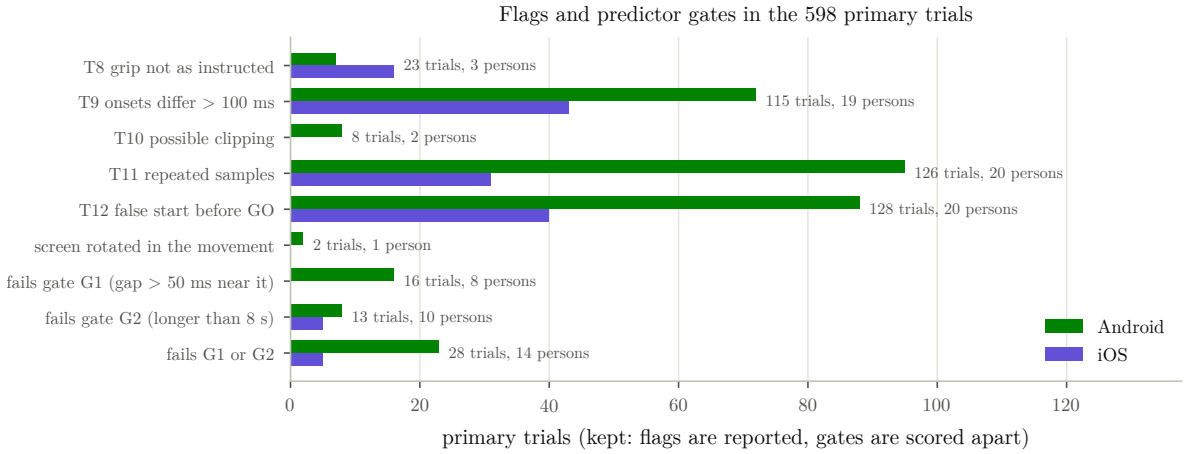


Figure 28: Flags (T8–T12, screen rotation) and predictor gates (G1, G2) in the primary trials, by platform. Flags are reported and used for stratification; trials failing a gate stay in training and in the headline, and every result also reports the metrics on the trials that pass both gates.

Table 14: Flags of docs/02 T8–T12 (kept in every analysis) and the predictor gates G1, G2 of docs/09 D5 (failing trials stay in training and in the headline metrics, and every result also reports the metrics on the trials that pass both gates): primary trials flagged or failing, by platform and by hand, and the persons concerned.

Check	Trials	Android	iOS	Left	Right	Persons
T8 grip not as instructed	23	7	16	8	15	3
T9 onsets differ > 100 ms	115	72	43	56	59	19
T10 possible clipping	8	8	0	4	4	2
T11 repeated samples	126	95	31	59	67	20
T12 false start before GO	128	88	40	72	56	20
screen rotated in the movement	2	2	0	1	1	1
fails gate G1 (gap > 50 ms near it)	16	16	0	4	12	8
fails gate G2 (longer than 8 s)	13	8	5	7	6	10
fails G1 or G2	28	23	5	10	18	14

A.8. Label consistency and the swap signature

Hands-free collection has a blind spot. An accepted attempt only means “completed and not redone”, so a person who used the wrong hand for a whole block leaves no trace in the record. The screen in Figure 29 checks each of the 193 planned blocks of the primary trials: the share of its trials whose first-half yaw sign agrees with the instructed hand.

The swap signature. A hand swap at a block change produces a specific pattern: two adjacent blocks of opposite instructed hands, each of at least 3 trials, in which every trial contradicts its label. A person whose hand moves differently produces another pattern, in which one hand disagrees and the other agrees. The signature is found from the data, without a typed person or block. It appears in 2 blocks: blocks 1 and 2 of P08, in which no trial agrees with its label. Low agreement confined to one hand of a person is not the signature. A post hoc example is P17, a left-hander, whose left-hand blocks mostly disagree with the rule while the right-hand blocks agree. Handedness is examined in Appendix H.

Decision: the labels are never changed. The options were to correct suspected labels or to keep the instructed labels and test their influence. The labels stay as instructed, and the ringed blocks are flagged for review. Two safeguards follow. A sensitivity analysis drops the ringed blocks from training, or from training and test (Appendix A.9). The wrong-hand screen of the analysis plan (docs/09 D6) runs inside each training fold: an antisymmetric logistic model on the chiral features, scored by inner LOSO over the training persons, flags training trials with $P(\text{label}) < 0.05$ and runs of at least 3 consecutive trials with $P(\text{label}) < 0.25$. Flagged trials leave training only, never evaluation, so the cleaning cannot inflate accuracy. Applied once to the whole primary set, with LOSO scores, these rules flag 10 trials.

A.9. The reference floor and label-noise sensitivity

Decision: a pre-specified reference floor. Every model is compared against one simple, pre-specified model, defined with the analysis foundation before any modelling lane ran. It is a logistic regression on the cumulative yaw about gravity at 8 instants of the time-normalised approach, trained with mirror augmentation (each training trial is added again mirrored, with the opposite label). It is a sanity floor, not a model choice; lane m4 uses the same model, under the name yaw8, as the hand classifier of its production simulation (Appendix G). Its LOSO accuracy is 77.9 % [73.1, 82.9], with an AUC of 0.855 and a log loss of 0.512 (Table 15). On the natural-use set it scores 75 %: this is right-hand recall on the free-choice trials (all right-hand so far; a constant right answer would score 100 %).

Evidence: several attempts help. When the $P(\text{right})$ of k attempts of one person and hand are averaged, the per-person balanced accuracy rises from 77.6 % at $k = 1$ to 87.2 % at $k = 3$ and 90.9 % at $k = 5$ (Figure 30, left; Table 16). Every k uses the same 19 persons, those with at least 5 trials of each hand. The inference page should therefore ask for several movements.

Evidence: label noise does not move the floor. Three variants test the labels (Figure 30, right; Table 15). Each is paired against the reference on the same trials, with an exact sign-flip test over persons; the family is these three comparisons. Without the swap-signature blocks in training the accuracy changes by +0.3 points (unadjusted $p = 0.62$). Without them in training and test it changes by +0.3 points ($p = 0.62$). With the D6 screen in every fold it changes by −0.3 points ($p = 0.59$). None differs even before the Holm adjustment over the three, which can only raise these p-values. The labels therefore stay as instructed, and the D6 screen stays an optional sensitivity analysis.



Figure 29: Label consistency of every planned block of every session (dot size: trials; colour: instructed hand; fill: share of trials whose first-half yaw sign agrees with the instructed hand; text: agreeing/total). Ringed: the swap signature, two adjacent blocks of opposite hands in which every trial contradicts its label. The labels are kept; the ringed blocks enter a sensitivity analysis.

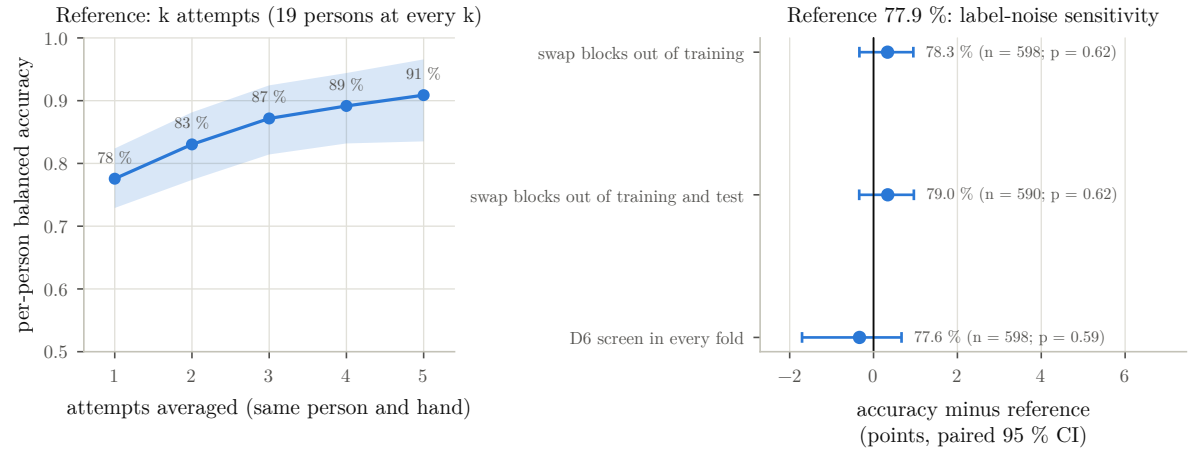


Figure 30: Reference floor. Left: per-person balanced accuracy when the $P(\text{right})$ of k attempts of one person and hand are averaged (the same 19 persons at every k). Right: the label-noise variants, paired against the reference on the same trials (95 % CI of the paired difference). Supports asking for several attempts and keeping the instructed labels.

Table 15: Reference floor and label-noise sensitivity (leave-one-person-out; 95 % person-cluster bootstrap CI; differences paired against the reference).

Model	Trials	Persons	Accuracy	95% CI	AUC	Log loss	Diff. vs ref.	p (sign flip)	Natural use
reference (yaw logreg)	598	20	77.9%	[73.1, 82.9]	0.855	0.512	–	–	75%
swap blocks out of training	598	20	78.3%	[73.7, 83.0]	0.856	0.513	+0.3%	0.62	75%
swap blocks out of training and test	590	20	79.0%	[74.2, 83.8]	0.861	0.507	+0.3%	0.62	75%
D6 screen in every fold	598	20	77.6%	[72.4, 82.7]	0.856	0.537	−0.3%	0.59	75%

Table 16: Reference floor: per-person balanced accuracy when the P(right) of k attempts of one person and hand are averaged (95 % person-cluster bootstrap CI).

Attempts averaged	Accuracy	CI low	CI high	Persons
1	77.6%	72.9%	82.4%	19
2	83.0%	77.4%	88.1%	19
3	87.2%	81.4%	92.4%	19
4	89.2%	83.2%	94.4%	19
5	90.9%	83.5%	96.6%	19

A.10. Demographics

The 20 persons are young adults (Figure 31): age 25.1 ± 2.9 years (range 19 to 29), 12 women and 8 men, and height 168.8 ± 8.8 cm (17 persons gave it). 18 are right-handed and 2 left-handed (P11 and P17, self-report). The usual phone hand is the right for 18 persons and both for 2. 12 persons used Android phones (15 sessions, 14 device models across all included sessions) and 8 used iPhones (9 sessions).

Consequences. With 2 left-handers, every handedness effect is exploratory, and the left-handers are shown individually (Appendix H). The narrow age range limits any claim about older users. Demographics never enter a model; they only stratify results, and the demographics add-on test is in Appendix H.

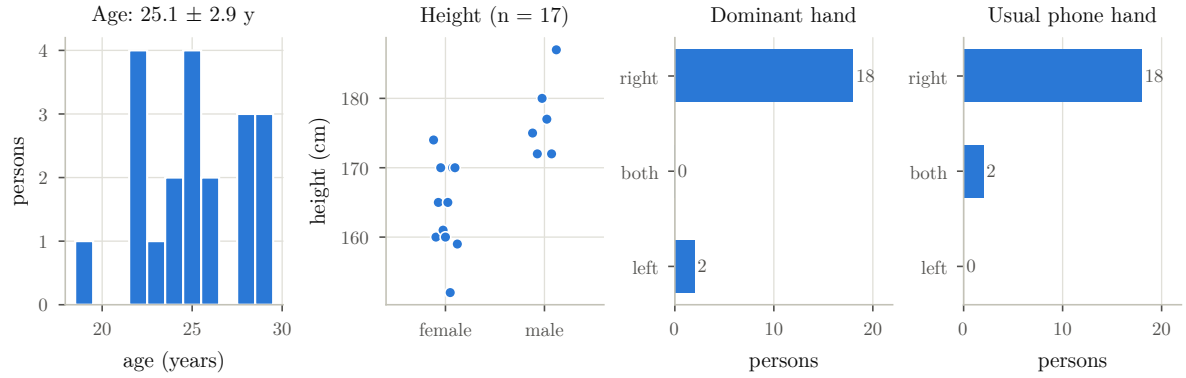


Figure 31: The 20 persons of the analysis: age, height by sex (17 persons gave a height), dominant hand and usual phone hand (self-report). With 2 left-handers, handedness effects can only be exploratory.

A.11. Reproducibility, provenance and privacy

One command rebuilds everything. One orchestrator regenerates every result, figure, table and number of this report from the frozen snapshot (`analysis/stage2/pipeline.py`):

```
cd analysis      # python = the frozen environment, .venv/Scripts/python.exe
python -m stage2.pipeline all --force   # cache, every unit, numbers, report
python -m stage2.pipeline all           # only what is missing or stale
python -m stage2.pipeline figures       # redraw every figure (no fitting)
python -m stage2.pipeline tables        # booktabs tables and CSV files
python -m stage2.pipeline numbers       # numbers files -> numbers.tex
python -m stage2.pipeline report        # asset step, then latexmk
```

Each target takes the options `--lanes` (units) and `--only` (names). The Python environment is frozen in a requirements file (`requirements-stage2.txt`), and every random choice draws from one fixed seed.

Compute is separate from presentation. A compute step runs the experiments and saves everything the presentation needs: predictions, per-fold information, curves and the data behind each figure. The figures, tables and numbers steps only read those files and never fit a model, so redrawing a figure after a change of an axis takes seconds. To change a figure, edit the presentation defaults of its figures module, run `python -m stage2.pipeline figures --lanes <unit> --only <NAME>` and rebuild the report; the asset step copies the new file.

Every number is generated. Each unit writes its numbers to a file of the form `{key: {value, format, source, description}}`, where `source` names the saved file the value was read from. The asset step turns them into the LaTeX macros that print every result in this report; no result is typed by hand. A missing key prints in red, and a final build stops on it.

Staleness and provenance. The cache records its data identity: a hash over the snapshot, the curation file, the cache parameters and the code that builds the cache. Every result records the data identity of the cache it was made from, with the git commit and a fingerprint of the analysis code. A changed snapshot or curation makes every result stale, and a plain `all` recomputes them; an edit of presentation code makes nothing stale. The footer of every page prints the provenance of the build: git 406405c · data 11b6c9b5 · 20 persons, 598 trials · leave-one-person-out.

Adding a participant. Five steps add a person:

1. put the new session in the snapshot, or export a new snapshot and point the configuration at it;
2. add its row to the curation file with the next person id (e.g. P21) and `include: true`;
3. run `python -m stage2.pipeline all`: the data identity has changed, so the cache is rebuilt and every result recomputed;

4. check that `python -m stage2.scripts.expected_counts --without P21` reproduces the pinned counts exactly, so every change is explained by the new person, then pin the new counts with `--write`;
5. run the test suite (`python -m pytest tests stage2/tests`).

The last person added, P20, passed this check: without P20 every count of the previous snapshot is reproduced exactly.

Privacy. The repository is public, so no name reaches any output. Persons appear only as P01–P20, and sessions only as pseudonymous labels such as `P14-s3`. The private file that links persons to names is never opened, and the curation's free-text reasons are replaced by fixed codes. Report-facing tables refuse app codes, backend ids, timestamps, operating-system and browser versions and screen geometry. The tests scan every table, JSON file, LaTeX file, caption, figure text, PNG and PDF, the cache tables and every source file for the name tokens of the raw records, case-insensitively. Exploratory participants such as PD01 live outside the snapshot and the curation; only the case-study lane reads them, no model is trained, tuned or scored on them, and a guard test checks that none of their ids reaches the main data (Appendix I).

Table 17: Notation.

Symbol	Meaning
$\mathbf{f}$, $\mathbf{a}$	specific force (acceleration including gravity) and linear acceleration, device frame, $\mathrm{m/s^2}$
$\boldsymbol{\omega}$	angular velocity (`rotationRate` as x, y, z), deg/s
$\hat{\mathbf{g}}$	unit vector pointing up (gravity frame), estimated by gyroscope propagation from the rest pose
$\mathbf{S}$, $\mathbf{M}$	$\mathrm{diag}(-1, 1, 1)$ on the device and on the body side
$\mathbf{Q}$	a grip: a constant proper rotation of the device frame
$m(\cdot)$	the left–right mirror of a trial or of a feature vector
$\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$	yaw rate about gravity (chiral, or mirror-odd: it changes sign under the mirror)
$P(\text{right})$	the model's probability that the right hand made the movement
τ_k	the scale of the summed logits z_i of k attempts, $P(\text{right}) = \sigma(\sum_i z_i/\tau_k)$, σ the logistic sigmoid

B. Preprocessing and data-treatment decisions

This appendix justifies every signal-processing and data-treatment choice of the pipeline (lane p1). Each choice was tested by changing one setting at a time and re-running the full leave-one-person-out (LOSO) evaluation with a fixed panel of three models. Every default survives this test. The appendix also covers the calibration of the panel's $P(\text{right})$ and the combination of several attempts. Abbreviations: confidence interval (CI), logistic regression (LR), gradient-boosted trees (GBM), MiniRocket with a ridge classifier (MR), area under the ROC curve (AUC), expected calibration error (ECE), Benjamini–Hochberg (BH). A difference in accuracy is given in percentage points (points).

B.1. How each choice was judged

The panel. Three models of different kinds were fixed before any option was compared, each with one configuration (Table 18). *LR*: an L2 logistic regression ($C = 1$) on the means of 8

time bins of the 2 chiral gravity-frame channels (the yaw rate $\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$ and its cumulative integral), trained with mirror augmentation. With mirror augmentation the fitted LR is antisymmetric, $p(m(x)) = 1 - p(x)$, so mirror-even inputs get no weight (Section 3). *GBM*: LightGBM with fixed parameters (100 trees, learning rate 0.1, 15 leaves, row and column subsampling 0.8 and 0.5) on 14 summary features of each of the 17 channels, with mirror augmentation. *MR*: MiniRocket [20] (2000 features requested) on the 8 gravity-frame channels, with a ridge classifier whose penalty and Platt scaling [21] were fitted by an inner 3-fold split over the training persons. The panel spans a linear model on two physically motivated channels, a tree ensemble on engineered features and a random-kernel time-series model. A treatment that helps or hurts all three is a property of the data, not of one model.

Options. An *option* is an alternative to one default. Options on the recorded stream (cut-off, filter order, causal filter, gap bridging, interpolation, sampling rate, rounding, timestamp jitter, gravity estimate, window, bound errors) were built by re-standardising the raw records with one setting changed. This produced 28 variant caches, including the default one. With every setting at its default, the rebuilt arrays equal the analysis cache exactly (largest difference 0 over 661 trials). Options on the training set (channel sets, normalisation, winsorising, augmentation, training-row filters, platform) were applied inside each training fold. In all, the lane ran 123 LOSO experiments on the 598 primary trials of 20 persons.

Paired comparison. Each option was compared with the same model under the current pipeline, on the same trials and folds. The comparison gives the pooled accuracy difference with its CI (2000 person resamples), the mean per-person log-loss difference with its CI, and an exact sign-flip test over persons. The comparisons form three families, each adjusted by BH and by Holm: the options (96 option-by-model comparisons), the robustness costs (8 test-time segmentation errors) and the pairs (16 remedies and size-matched controls).

Verdicts. *Worse* or *better*: the paired CI excludes 0 and the BH q is below 0.05. *Nominally worse* or *nominally better*: only the CI excludes 0; this is suggestive, not significant. *Identical*: every prediction is within 1e-06 of the default (solver precision). *No difference*: the CI includes 0. This is absence of evidence, not equivalence. The lower end of its CI is the largest loss the data allow, and the upper end is the largest gain forgone.

Decision rule. The rule was fixed in advance: change a default only if one option is better for at least two panel models and worse for none. An option run on one model cannot meet the rule; this applies to one option of the frame choice and one of the augmentation choice.

Other roles. A *robustness* run perturbs the test input of an unchanged model; it sets a requirement for production and is never an option. A *remedy* is judged only against the problem it addresses. A *scoring* run scores the trials an exclusion rule removes with the base model. A *control* removes as many training trials or persons at random as an option does. The leak check ran on every fold for the base and nested-calibrated runs, and passed; for the other variants it ran on 3 seeded folds.

Table 18: The fixed panel under the current pipeline, and with the calibrator chosen inside each training fold (leave-one-person-out, 20 persons; 95% person-cluster bootstrap CIs). ECE: 10 equal-mass bins, with the 95th percentile of the ECE of perfectly calibrated predictions on the same bins (no CI: the percentile bootstrap of the ECE is biased); slope: calibration slope (1 = calibrated, below 1 = over-confident); k = 3: mean P(right) of three attempts of one person and hand. Natural use: one free-choice trial per person, scored apart; a P(right) of exactly 0.5 is not an answer; every natural-use trial is right-handed, so the answered accuracy is right-hand recall.

Model	Accuracy % [95% CI]	AUC [95% CI]	Log loss [95% CI]	ECE (null 95th pct.)	Slope [95% CI]	k = 3 %	Natural use
LR	81.4 [77.5, 85.4]	0.868 [0.825, 0.906]	0.490 [0.421, 0.569]	0.069 (0.054)	0.82 [0.61, 1.18]	89.7 [85.2, 93.8]	85% correct (100% answered); half credit 85%
GBM	85.6 [81.0, 89.8]	0.922 [0.881, 0.959]	0.369 [0.260, 0.486]	0.049 (0.039)	0.76 [0.61, 0.99]	92.3 [86.7, 97.1]	90% correct (100% answered); half credit 90%
MR	80.6 [76.5, 85.1]	0.872 [0.835, 0.910]	0.459 [0.397, 0.518]	0.055 (0.053)	0.93 [0.78, 1.16]	88.4 [83.2, 93.3]	80% correct (100% answered); half credit 80%
GBM + nested calibrator	85.5 [81.0, 89.5]	0.921 [0.877, 0.958]	0.358 [0.265, 0.463]	0.032 (0.044)	0.92 [0.73, 1.21]	92.2 [86.6, 97.0]	90% correct (100% answered); half credit 90%
LR + nested calibrator	81.6 [77.7, 85.6]	0.866 [0.823, 0.905]	0.498 [0.430, 0.577]	0.085 (0.056)	0.87 [0.62, 1.33]	89.8 [85.4, 93.8]	80% correct (100% answered); half credit 80%

The panel's natural-use result is right-hand recall on the free-choice trials (all right-hand so far; a constant right answer would score 100 %): LR 85 %, GBM 90 % and MR 80 % of 20 trials (Table 18). No panel model returned a P(right) of exactly 0.5 on these trials (coverage 100 %, 100 % and 100 %), so the forced-choice accuracy equals the answered accuracy. GBM has the highest base accuracy, 85.6 % [81.0, 89.8]. Calling it the best panel model is a choice made on the outer folds among 3 models, so it is exploratory.

B.2. Overview: every default is kept

Figure 32 shows every option against the current pipeline for each panel model. Table 19 summarises the 11 choices and Table 20 gives the same choices per model. The rule keeps the default in 11 of 11 choices and changes none (0). Unless stated otherwise, q below is the BH-adjusted value within the options family.

No option is better than the current pipeline, even nominally (0 better, 0 nominally better). Of the 96 option comparisons, 8 have a CI that excludes 0; about 5 would do so by chance at the 95 % level. BH keeps 2, both worse, and Holm keeps 1:

- device-frame channels instead of the chiral channels in LR: −12.5 [−17.4, −7.9] points, BH $q = 0.028$, Holm $p = 0.028$;
- gravity-frame channels only in GBM: −6.0 [−8.8, −3.2] points, BH $q = 0.045$, Holm $p = 0.09$.

The other 6 CIs that exclude 0 are nominally worse and do not survive either adjustment. Of the remaining comparisons, 4 are identical to the default and 84 show no difference.

The per-person log loss points the same way. Its paired CI lies above 0 for the device-frame LR (+0.174 [+0.098, +0.248]), the gravity-frame GBM (+0.106 [+0.068, +0.147]) and the GBM without mirror augmentation (+0.057 [+0.027, +0.089]), among others. A few options lower the log loss nominally without changing accuracy: the fixed 3-s window for MR (−0.055 [−0.106, −0.003]), the approach plus hold for GBM (−0.022 [−0.044, −0.000]) and time warping for GBM (−0.020 [−0.037, −0.003]). The rule decides on accuracy, so these remain leads for future work, not changes. Figure 32 shows the accuracy differences of all 96 option-by-model comparisons. The full numbers (accuracy, accuracy difference with its CI, BH q, log-loss difference with its CI and verdict) are in `tables/p1/p1_options_a` and `p1_options_b` of the analysis outputs; at more than a page each, they are too long to reproduce here.

B.3. Low-pass filter: cut-off, order and phase

Options. The default is a zero-phase Butterworth low-pass at 15 Hz, order 4, applied after resampling to 60 Hz. The options were no low-pass, cut-offs of 5, 10 and 20 Hz, orders 2 and 8, and a causal (one-pass) filter of the same design.

Evidence. The movement lives below the cut-off: 99.2 % of the gyroscope power and 98.4 % of the accelerometer power of the approach lie below 15 Hz, on both platforms (Figure 33a). No filter option changes accuracy beyond noise (Figure 34). Only the 5-Hz cut-off in LR is nominally worse, −1.0 [−2.0, −0.2] points (BH $q = 0.61$). The largest loss compatible with the data is −2.2 points (no low-pass, MR), and the largest gain forgone is +2.9 points (5 Hz, GBM).

Phase. A causal filter delays the signal. At the current design it delays a 1-Hz movement by 22 ms; a 10-Hz cut-off would cost 38 ms, a 20-Hz cut-off 13 ms, and order 8 at 15 Hz 43 ms (Figure 33b). Its accuracy cost is within noise: LR −0.3 [−1.8, +1.2] and GBM −0.2 [−2.1, +1.6] points.

Decision. Keep the zero-phase 15-Hz filter of order 4. The predictor processes a recorded buffer, so a zero-phase filter is also available in production. If a later online stage needs a causal filter, the data bound its accuracy cost at −2.1 points.

B.4. Gaps, sampling rate, timestamps and gravity

Gaps and imputation. The default bridges gaps of up to 100 ms by linear interpolation. Trials with a longer gap near the movement are excluded by rule E4d (Section B.8). Bridging gaps of up to 250 ms changes no primary trial (0 trials differ), so its predictions are identical. Bridging only up to 50 ms changes 19 trials and shows no difference. Cubic-spline and sample-and-hold resampling show no difference either; the largest loss compatible with the data is −3.1 points (sample-and-hold, GBM). Decision: keep linear bridging up to 100 ms.

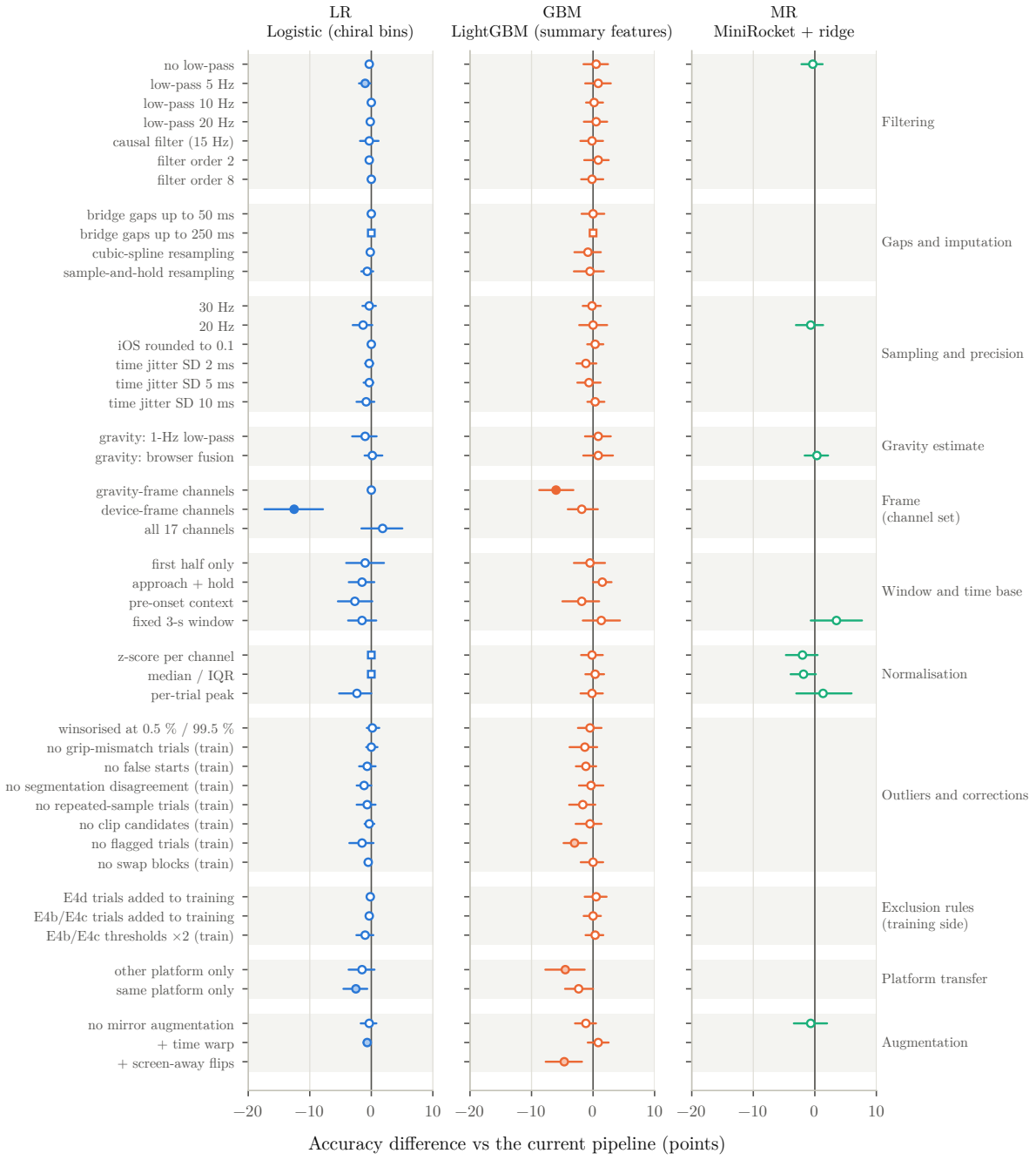


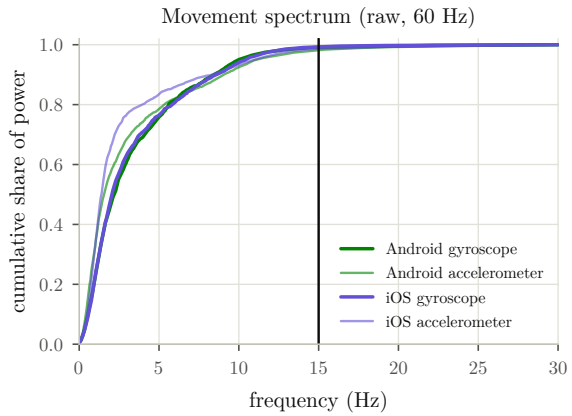
Figure 32: Every data-treatment option against the current pipeline, per panel model: paired difference in LOSO accuracy (points) with its 95 % CI (20 persons; 96 comparisons in one multiplicity family). Filled: CI excludes 0 and BH $q < 0.05$; light fill: CI alone excludes 0; open circle: no difference; open square: identical predictions. The exclusion-rule rows add the excluded trials to training and score the primary trials only. No option beats the default for two models, so every default is kept.

Table 19: Data-treatment decisions of the pipeline, judged by a fixed panel of three pre-specified models (LR: logistic on chiral bins; GBM: LightGBM on summary features; MR: MiniRocket + ridge). Options: how many alternatives were tested, on which models. Evidence: paired leave-one-person-out accuracy differences against the current default (20 persons; 95% person-bootstrap CIs; q: Benjamini-Hochberg over the 96 option comparisons); nominal: the CI excludes 0 but q does not pass. Gain forgone at most: the largest upper CI bound of any option, i.e. the most the default could be beaten by. Rule (pre-specified): change a default only if one option is better for at least two panel models and worse for none; an option run on one model cannot meet it.

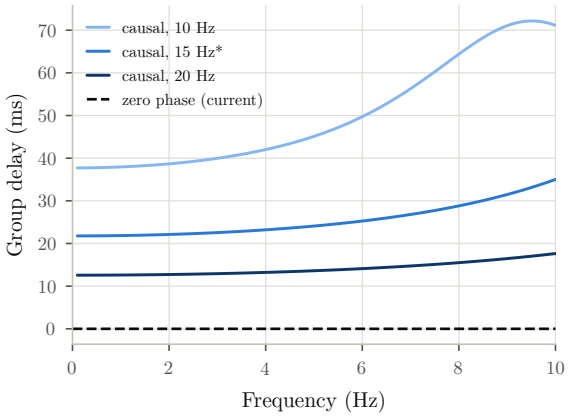
Choice	Current default	Options (models)	Evidence (accuracy, points [95% CI], q)	Gain forgone at most	Decision
Filtering	zero-phase Butterworth, 15 Hz, order 4	7 (LR, GBM, MR)	nominal only (1): low-pass 5 Hz (LR)	+2.9 (low-pass 5 Hz, GBM)	keep
Gaps and imputation	gaps up to 100 ms bridged linearly; E4d excluded	4 (LR, GBM)	identical: bridge gaps up to 250 ms (LR), bridge gaps up to 250 ms (GBM)	+1.8 (bridge gaps up to 50 ms, GBM)	keep
Sampling and precision	60 Hz, native precision	6 (LR, GBM, MR)	no difference for any option	+2.3 (20 Hz, GBM)	keep
Gravity estimate	gyro-propagated gravity (gravity_v1)	2 (LR, GBM, MR)	no difference for any option	+3.2 (gravity: browser fusion, GBM)	keep
Frame (channel set)	model default (LR chiral, GBM all 17, MR gravity frame)	3 (LR, GBM)	worse: device-frame channels (LR −12.5 [−17.4, −7.9], q 0.028); gravity-frame channels (GBM −6.0 [−8.8, −3.2], q 0.045)	+5.0 (all 17 channels, LR)	keep
Window and time base	approach, time-normalised (64 samples)	4 (LR, GBM, MR)	no difference for any option	+7.7 (fixed 3-s window, MR)	keep
Normalisation	none (physical units)	3 (LR, GBM, MR)	identical: z-score per channel (LR), median / IQR (LR)	+6.0 (per-trial peak, MR)	keep
Outliers and corrections	no winsorising; flagged trials and swap-signature blocks kept in training	8 (LR, GBM)	nominal only (1): no flagged trials (train) (GBM)	+1.7 (no segmentation disagreement (train), GBM)	keep
Exclusion rules (training side)	E4b (< 0.25 s), E4c (peak < 30 deg/s) and E4d (gap > 100 ms near the movement) trials excluded	3 (LR, GBM)	no difference for any option	+2.2 (E4d trials added to training, GBM)	keep
Platform transfer	pooled over platforms	2 (LR, GBM)	nominal only (2): other platform only (GBM); same platform only (LR)	+0.5 (other platform only, LR)	keep
Augmentation	mirror augmentation	3 (LR, GBM, MR)	nominal only (2): + screen-away flips (GBM); + time warp (LR)	+2.5 (+ time warp, GBM)	keep

Table 20: Data-treatment choices per panel model: range of the paired accuracy differences against the current default over the options tested (points), the number of options, and how many are worse or better after the Benjamini-Hochberg correction (*), only nominally, or identical.

Choice	LR	GBM	MR	Decision
Filtering	−1.0 to +0.0 (7 opt.; 1 nominal)	−0.2 to +0.8 (7 opt.)	−0.3 to −0.3 (1 opt.)	keep
Gaps and imputation	−0.7 to +0.0 (4 opt.; 1 identical)	−0.8 to +0.0 (4 opt.; 1 identical)	–	keep
Sampling and precision	−1.3 to +0.0 (6 opt.)	−1.2 to +0.3 (6 opt.)	−0.7 to −0.7 (1 opt.)	keep
Gravity estimate	−1.0 to +0.2 (2 opt.)	+0.8 to +0.8 (2 opt.)	+0.3 to +0.3 (1 opt.)	keep
Frame (channel set)	−12.5 to +1.8 (3 opt.; 1 worse*)	−6.0 to −1.8 (2 opt.; 1 worse*)	–	keep
Window and time base	−2.7 to −1.0 (4 opt.)	−1.8 to +1.5 (4 opt.)	+3.5 to +3.5 (1 opt.)	keep
Normalisation	−2.3 to +0.0 (3 opt.; 2 identical)	−0.2 to +0.3 (3 opt.)	−2.0 to +1.3 (3 opt.)	keep
Outliers and corrections	−1.5 to +0.2 (8 opt.)	−3.0 to +0.0 (8 opt.; 1 nominal)	–	keep
Exclusion rules (training side)	−1.0 to −0.2 (3 opt.)	+0.0 to +0.5 (3 opt.)	–	keep
Platform transfer	−2.5 to −1.5 (2 opt.; 1 nominal)	−4.5 to −2.3 (2 opt.; 1 nominal)	–	keep
Augmentation	−0.7 to −0.3 (2 opt.; 1 nominal)	−4.7 to +0.8 (3 opt.; 1 nominal)	−0.7 to −0.7 (1 opt.)	keep



(a) Cumulative power spectrum of the approach (median over 598 primary trials); the vertical line is the 15-Hz cut-off.



(b) Group delay of causal Butterworth low-pass filters (order 4) at 60 Hz; the zero-phase filter of the pipeline has none.

Figure 33: Why the cut-off is 15 Hz and why the filter is zero-phase. (a) Almost all movement power lies below the cut-off on both platforms. (b) A causal filter at the current design would delay a 1-Hz movement by 22 ms.

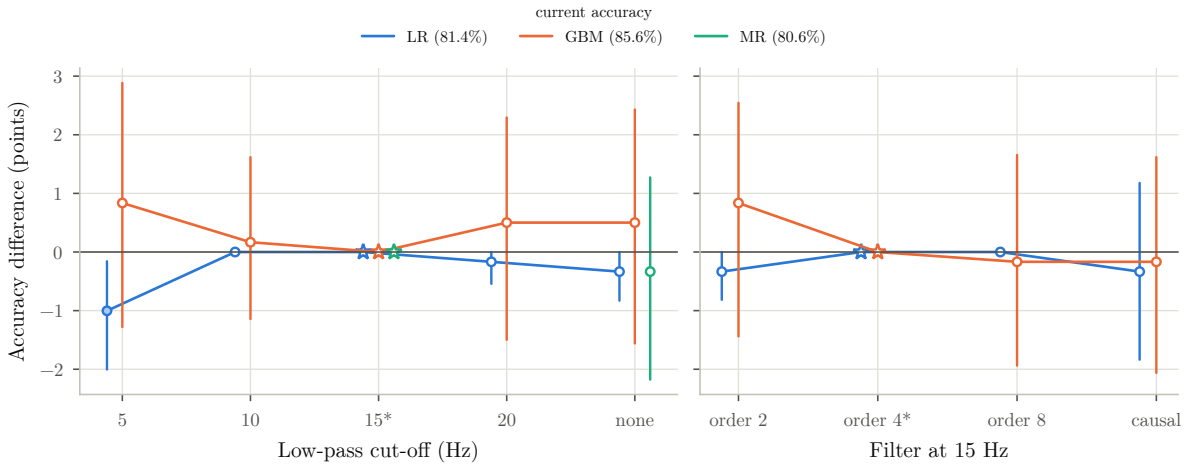


Figure 34: Low-pass filter options. Left: cut-off of the zero-phase Butterworth filter (order 4); right: its order at 15 Hz, and the causal filter of the same design. Paired accuracy difference against the current pipeline (star), per panel model (legend: current accuracy), with 95 % CIs; markers as in Figure 32. The default is kept.

Sampling rate and timestamps. The predictor must tolerate slower browsers and imprecise timestamps, because phones differ in rate and timing [2]. Keeping every second or third raw sample (30 and 20 Hz), rounding the iOS values to 0.1, or adding Gaussian noise of SD 2, 5 or 10 ms to the sample times shows no difference for any model (Figure 35). The largest drop is LR at 20 Hz, −1.3 [−3.0, +0.2] points. The largest loss compatible with the data is −3.1 points (20 Hz, MR). Decision: keep the native rate and precision, resampled to 60 Hz. This is a bound, not a proof of robustness.

Gravity estimate. The chiral channels are defined relative to gravity, so the gravity estimate matters in principle. The default propagates gravity with the gyroscope. A 1-Hz low-pass of the accelerometer and the browser's own fusion [1] show no difference (Table 19). The largest loss compatible with the data is −3.1 points (1-Hz low-pass, LR); the largest gain forgone is +3.2 points (browser fusion, GBM). Decision: keep the gyroscope-propagated estimate, because the browser fusion differs between platforms and is outside our control.

B.5. Representation: frame and window

Frame (channel set). Each panel model has its own default: LR uses the 2 chiral gravity-frame channels, MR the 8 gravity-frame channels and GBM all 17 channels. These are the only two options that are worse after the BH adjustment (Section B.2). Device-frame channels cost LR −12.5 [−17.4, −7.9] points, and restricting GBM to the gravity frame costs −6.0 [−8.8, −3.2] points. All 17 channels in LR show no difference (+1.8 [−1.6, +5.0] points). The device-frame channels cannot act as a grip-to-hand shortcut here, because the instructed grips are balanced across hands (χ^2 test, $p = 0.86$). They can, however, fail on grips that differ from training: in the final-model selection the device-frame candidates lose accuracy on the sideways grips and are removed by its grip test (Appendix F).

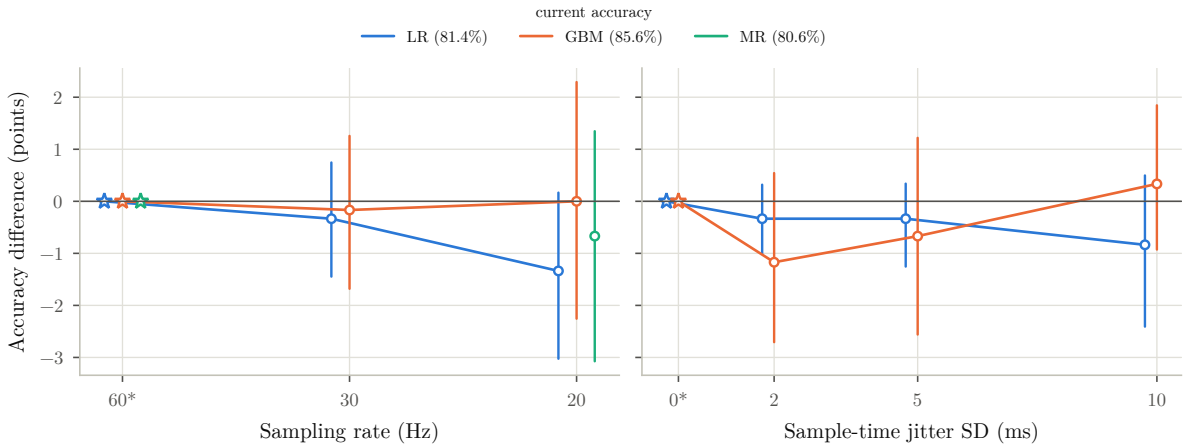


Figure 35: Robustness to the recording. Paired accuracy difference against the current pipeline (star) when every second or third raw sample is kept (30 and 20 Hz; the low-pass is capped at 0.4 times the rate), and when Gaussian noise is added to the raw sample times. 95 % CIs; markers as in Figure 32. A CI that includes 0 bounds the cost by its lower end; it does not show equivalence.

Window and time base. The default window is the approach, time-normalised to 64 samples. The options were the first half of the approach only, the approach plus the hold, the approach with pre-onset context, and a fixed 3-s window. None changes accuracy beyond noise (Table 19). The largest gain forgone is +7.7 points (fixed window, MR) and the largest loss compatible is −5.4 points (pre-onset context, LR). Decision: keep the approach. Production must segment the movement without a cue, and a window that needs the hold or more context adds latency and depends more on the detector.

B.6. Robustness to segmentation errors

Production has no GO cue, so the detector of Appendix G finds the approach with errors of tens to hundreds of milliseconds. To set a requirement for that detector, models trained on the oracle bounds (the approach bounds the app stored) were tested on bounds moved by independent uniform errors of up to ±50, ±100 and ±200 ms, or with the onset 100 ms late (Figure 36, Table 21).

No perturbation survives the BH adjustment in the robustness family. At ±100 ms the differences are LR −1.5 [−3.2, +0.0] and GBM −0.5 [−2.3, +1.2] points. At ±200 ms GBM is nominally worse, −2.0 [−4.2, −0.1] points (BH $q = 0.26$). The largest loss compatible with the data is therefore −4.2 points.

Training with jittered copies of the bounds (up to ±100 ms) was tried as a remedy. Against oracle-only training at the same jittered test bounds it shows no difference: LR +0.0 [−1.4, +1.4] and GBM +0.7 [−1.0, +2.5] points (BH $q = 0.88$ and 0.88 in the pairs family; Table 22). Decision: keep the oracle bounds in training. Treat bound errors as a requirement on the detector: at errors up to ±100 ms no cost is detectable with 20 persons, and the data allow a loss of at most −3.2 points (LR). The errors here are synthetic and uniform; the real detector's errors are evaluated in Appendix G.

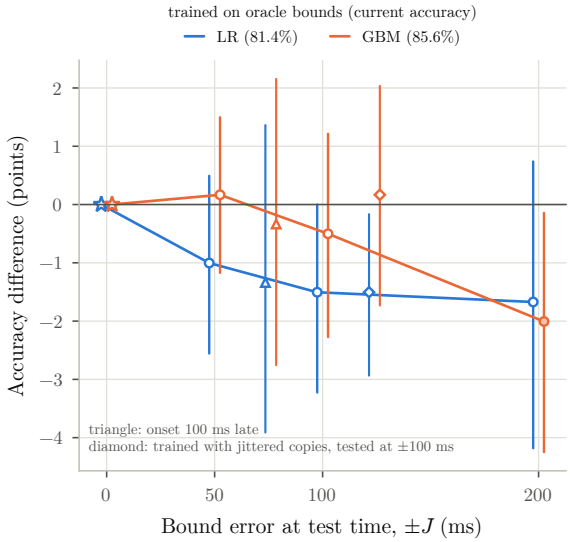


Figure 36: Robustness to segmentation errors: paired accuracy difference of a model trained on the oracle bounds when its test bounds are moved by independent uniform errors of up to $\pm J$ (circles) or the onset is 100 ms late (triangles). Diamonds: trained with jittered copies too, tested at ±100 ms and judged against the oracle-trained model at the same bounds. 95 % CIs. Jitter training does not help; the cost of bound errors is a requirement for the detector.

Table 21: Robustness to segmentation errors: models trained on the oracle bounds, tested on bounds moved by independent uniform errors (or an onset 100 ms late), against the same models on the oracle bounds; q: Benjamini-Hochberg over the 8 robustness comparisons. Last rows: training with jittered copies judged against oracle-only training, both tested on the same jittered bounds (q in the family of pairs). Verdict: * = the paired 95% CI excludes 0 and the Benjamini-Hochberg q of the exact sign-flip test (within the family) is below 0.05; nom. = the CI alone excludes 0; identical = predictions within 1e-06 of each other.

Model	Test-time perturbation	Acc. %	Diff. [95% CI]	q (BH)	Diff. LL [95% CI]	Verdict
GBM	bounds ±50 ms (test)	85.8	+0.2 [−1.2, +1.5]	0.86	+0.006 [−0.005, +0.018]	–
GBM	bounds ±100 ms (test)	85.1	−0.5 [−2.3, +1.2]	0.80	+0.010 [−0.016, +0.039]	–
GBM	bounds ±200 ms (test)	83.6	−2.0 [−4.2, −0.1]	0.26	+0.042 [+0.011, +0.072]	worse (nom.)
GBM	onset 100 ms late (test)	85.3	−0.3 [−2.8, +2.2]	0.95	−0.000 [−0.033, +0.030]	–
LR	bounds ±50 ms (test)	80.4	−1.0 [−2.6, +0.5]	0.42	+0.004 [−0.003, +0.011]	–
LR	bounds ±100 ms (test)	79.9	−1.5 [−3.2, +0.0]	0.26	+0.033 [+0.017, +0.051]	–
LR	bounds ±200 ms (test)	79.8	−1.7 [−4.2, +0.7]	0.26	+0.026 [+0.006, +0.049]	–
LR	onset 100 ms late (test)	80.1	−1.3 [−3.9, +1.4]	0.79	+0.036 [−0.008, +0.097]	–
LR	jitter-trained vs oracle-trained, both at ±100 ms	79.9	+0.0 [−1.4, +1.4]	0.88	−0.001 [−0.011, +0.008]	–
GBM	jitter-trained vs oracle-trained, both at ±100 ms	85.8	+0.7 [−1.0, +2.5]	0.88	−0.005 [−0.027, +0.017]	–

B.7. Normalisation

The default feeds the channels in physical units. The options were a z-score per channel, a median and interquartile-range scaling (both fitted on the mirror-augmented training rows, which keeps the mirror symmetry) and a per-trial peak scaling (Figure 37). For LR the two per-channel scalings are identical to the default, because LR standardises its own features (largest change of P(right): 2.1e-08). No option differs from the default for any model. The widest CIs belong to the per-trial peak scaling: LR −2.3 [−5.2, +0.0] and MR +1.3 [−3.0, +6.0] points. Decision: no normalisation. Physical units keep the speed and amplitude of the movement; a per-trial peak scaling discards the amplitude.

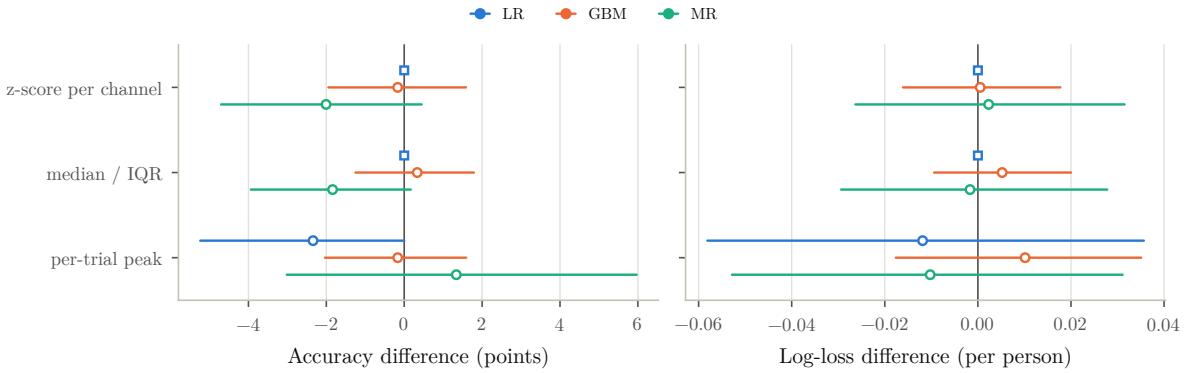


Figure 37: Normalisation of the input channels against none (physical units), per panel model: paired accuracy (left) and mean per-person log-loss (right) differences with 95 % CIs. Open squares: identical predictions (LR standardises its own features). No option improves on physical units.

B.8. Outliers, flags and exclusion rules

Two principles held throughout: evaluation trials are never dropped or relabelled, and any cleaning of the training set must beat a size-matched control.

Training-side cleaning. The options were winsorising at 0.5 % and 100 − 0.5 %, removing the trials of each quality flag (grip mismatch, false start, segmenter disagreement above 100 ms, repeated samples, possible clipping) or of all flags from training, and removing the swap-signature blocks (trials whose yaw sign suggests swapped hands). None helps (Figure 32). Removing all flagged trials is nominally worse for GBM, −3.0 [−4.8, −1.1] points (BH $q = 0.4$). Against a control that removes as many random trials per person and hand, it shows no difference: GBM −1.3 [−4.0, +1.2] and LR +1.3 [−0.5, +3.4] points (pairs family, Table 22). The control alone costs LR −2.8 [−5.3, −0.5] points (BH $q = 0.19$). So the loss comes from the smaller training set, not from the value of the flagged trials. Decision: keep flagged trials and swap blocks in training.

Scoring the removed trials. An exclusion that removes scorable trials makes the headline optimistic and the production gate too strict. Figure 38 scores the removed and flagged trials with the base models. The full table (`tables/p1/p1_subsets` of the analysis outputs) is too long to reproduce; the main rows follow.

- *E4d* (an unbridged gap of more than 100 ms within 0.5 s of the movement) removes 30 trials of 10 persons (4 Android, 26 iOS). The gap falls inside the approach in 16 of them (8 persons; median longest gap 317 ms) and only in the margins in 14. With linear imputation, the in-approach trials score 81.2 % [60.0, 95.0] (LR) and 75.0 % [49.9, 94.7] (GBM): compatible with the primary accuracy for LR and inconclusive for GBM. The margin-only trials score 100.0 % [100.0, 100.0] and 92.9 % [85.0, 100.0].
- *E4b/E4c* (shorter than 250 ms, or peak rotation below 30 deg/s) removes 13 trials of 7 persons. They score 53.8 % [14.3, 81.9] (LR, inconclusive) and 61.5 % [40.0, 84.6] (GBM, below the primary accuracy).
- *Thresholds twice as strict* would remove 91 more trials of 16 persons, which score 90.1 % [83.3, 94.4] (GBM): compatible with the primary accuracy, so stricter thresholds would discard scorable trials.
- *Subsets with fewer than 5 persons* get no CI; the values below are pooled over the trials of these few persons and are descriptive only. Grip-mismatch trials (23 trials, 3 persons): LR 82.6 %, GBM 87.0 %, MR 91.3 %. Possible clipping (8 trials, 2 persons): LR 37.5 %, GBM 87.5 %, MR 50.0 %. Swap-signature blocks (8 trials of 1 person): LR 25.0 %, GBM 75.0 %, MR 62.5 %. The swap blocks are a sensitivity analysis only; their labels were never changed.

Adding the excluded trials to training does not change the accuracy on the primary trials (E4d: LR −0.2 [−0.8, +0.3], GBM +0.5 [−1.4, +2.2] points; E4b/E4c: LR −0.3 [−0.9, +0.0], GBM +0.0 [−1.5, +1.3] points).

Decision. Keep E4b/E4c. They are justified a priori: a reach from arm's length lasts longer and rotates the forearm, so these trials are fragments rather than reaches (Appendix A); the small scored sample does not decide it. Keep E4d for comparability. There is no evidence that the imputed in-approach trials score worse, but the sample is small, so production should flag an attempt with a gap inside the movement rather than trust it.

B.9. Platform transfer

Android and iOS differ in sensors, rates, gaps and sensor fusion. A model that transfers only within a platform would need training data from each. The options trained on the same platform only or on the other platform only (Figure 39). Both also shrink the training set, so each was compared with a control that removes as many training persons at random.

Against the pooled pipeline, GBM trained on the other platform is nominally worse, −4.5 [−7.8, −1.4] points (BH $q = 0.21$), and LR trained on the same platform is nominally worse, −2.5 [−4.5, −0.7] points (BH $q = 0.5$). Against the size-matched controls, only GBM cross-platform training remains nominally worse, −3.2 [−5.8, −0.5] points (BH $q = 0.17$ in the pairs family); the other three pairs show no difference (Table 22). Decision: pool both platforms. Platform is a stratum, never a model input. The suggestive GBM loss is a reason to keep recruiting on both platforms.

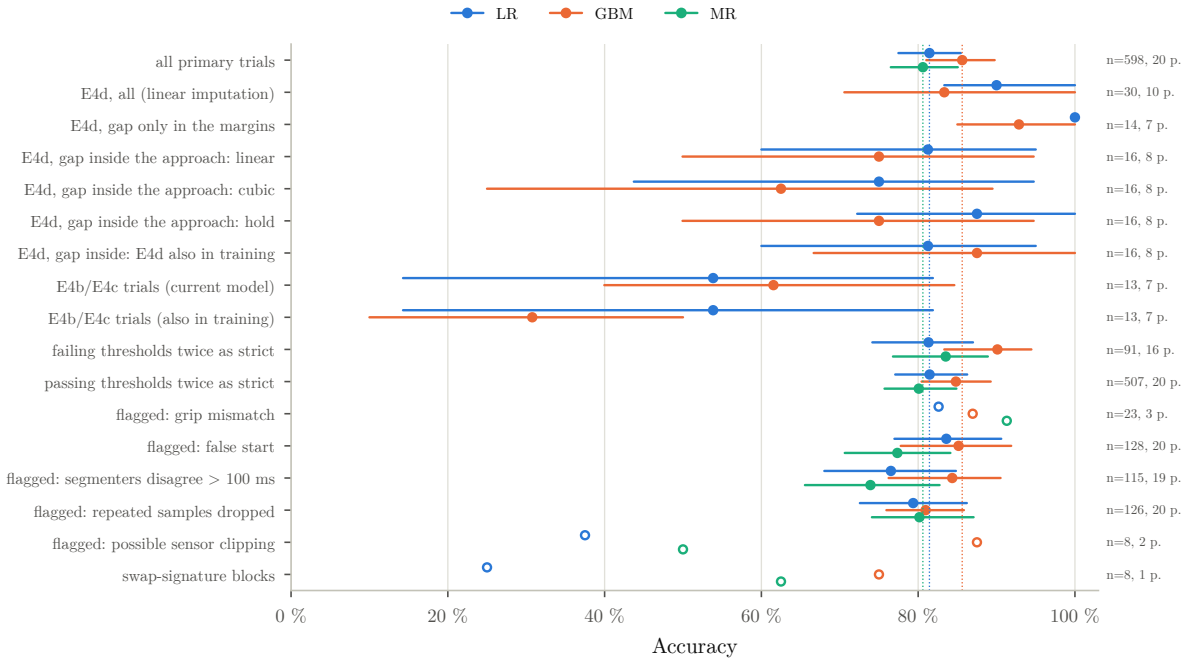


Figure 38: Accuracy on the trials that an exclusion rule removes or a flag marks, scored by the base models unless stated: E4d split by where the gap falls, with three imputations for gaps inside the approach; E4b/E4c; trials that thresholds twice as strict would remove; flagged trials; the swap-signature blocks. Dotted lines: all primary trials. 95 % CIs only for subsets of at least 5 persons (open markers: fewer, no CI); n = trials, p. = persons. The removed trials are not clearly harder to score, except the E4b/E4c fragments.

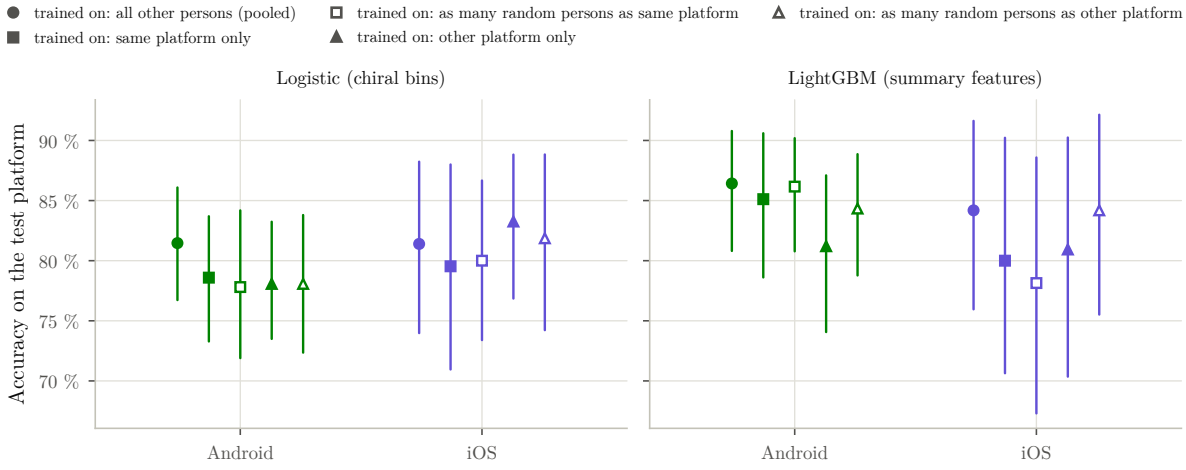


Figure 39: Platform transfer: accuracy on the persons of each platform (12 Android, 8 iOS) when the model is trained on all other persons (pooled), on the same platform only, or on the other platform only, each next to its size-matched control (open markers: as many training persons drawn at random). LOSO; 95 % CIs. Only the difference from the control isolates the platform; pooling is kept.

Table 22: The pairs family: remedies against the problem they address, options against their size-matched controls, and the controls against the current pipeline. Paired accuracy difference A minus B (points) with its 95 % CI; q: BH within the 16 comparisons of the family. The arm accuracies and log-loss differences are in `tables/p1/p1_pairs` of the analysis outputs.

A against B	Model	Difference [CI]	q	Verdict
Remedies				
Jitter-trained vs oracle-trained, both tested at ±100 ms	LR	+0.0 [−1.4, +1.4]	0.88	no difference
	GBM	+0.7 [−1.0, +2.5]	0.88	no difference
E4d trials in training vs primary-only training, scored on primary and E4d trials	LR	−0.2 [−0.7, +0.3]	1	no difference
	GBM	+0.8 [−0.8, +2.3]	0.68	no difference
Options against their size-matched controls				
All flagged trials out of training vs as many random trials out	LR	+1.3 [−0.5, +3.4]	0.32	no difference
	GBM	−1.3 [−4.0, +1.2]	0.88	no difference
Same platform only vs as many random persons	LR	+0.3 [−2.9, +3.3]	0.97	no difference
	GBM	+0.0 [−3.4, +3.3]	0.88	no difference
Other platform only vs as many random persons	LR	+0.5 [−2.2, +3.2]	0.88	no difference
	GBM	−3.2 [−5.8, −0.5]	0.17	nominally worse
Controls against the current pipeline				
As many random trials out as the flagged ones	LR	−2.8 [−5.3, −0.5]	0.19	nominally worse
	GBM	−1.7 [−3.9, +0.3]	0.28	no difference
As many random persons as the same platform	LR	−2.8 [−5.4, −0.3]	0.32	nominally worse
	GBM	−2.3 [−5.0, +0.4]	0.28	no difference
As many random persons as the other platform	LR	−2.0 [−4.0, +0.0]	0.28	no difference
	GBM	−1.3 [−3.5, +0.7]	0.66	no difference

B.10. Augmentation

Mirror augmentation adds the mirrored copy of every training trial with the other hand's label (Equation (1)); it encodes the left-right symmetry of the task. The options were no mirror augmentation, an added time warp (amplitude up to 0.15) [22], and added screen-away flips (GBM only). Without mirror augmentation, GBM accuracy shows no difference (−1.2 [−2.9, +0.5] points) but its log loss rises by +0.057 [+0.027, +0.089]. Screen-away flips are nominally worse for GBM, −4.7 [−7.7, −1.8] points (BH $q = 0.4$). Time warping is nominally worse for LR, −0.7 [−1.3, −0.2] points (BH $q = 0.75$), and nominally lowers the GBM log loss. Decision: keep mirror augmentation alone.

B.11. Calibration of the panel's P(right)

The product is a probability, so the panel's P(right) was checked for calibration (Figure 40). Table 23 gives every calibrator, Table 24 its paired difference from the uncalibrated model, and Table 26 the result per platform. Calibration was summarised by three numbers. The calibration slope is the coefficient of a logistic regression of the hand on logit P(right) (1 = calibrated, below 1 = over-confident). The intercept (calibration-in-the-large) is its offset. The ECE was computed over 10 equal-mass bins and compared with the ECE of perfectly calibrated predictions on the same bins (4000 simulations). This null treats trials as independent, so it is supporting evidence only.

Remedies. Four fixed calibrators (Platt [21], temperature [12], isotonic and beta) were fitted inside every outer training fold on cross-fitted predictions of the training persons (5 grouped inner folds) and applied to the held-out person. Comparing these 4 on the outer folds is exploratory. The headline is the *nested* choice: inside each training fold, grouped cross-validation chose among 4 candidates (none, temperature, Platt, beta). These maps are strictly increasing, so they keep the ranking within a fold. Isotonic is not a candidate because it ties predictions: it leaves 163 distinct values of 598 for GBM and 152 for LR, and it costs LR ranking (AUC −0.018 [−0.025, −0.011]).

GBM is over-confident and the nested calibrator corrects it. Uncalibrated, its slope is 0.76 [0.61, 0.99] and its intercept −0.18 [−0.70, 0.32]. Its ECE is 0.049 against a null 95th percentile of 0.039 ($p = 0.0027$; BH $q = 0.0037$ and Holm $p = 0.0067$ in the calibration test family of 4 tests, Table 25). Using training persons only, the fitted temperature exceeds 1 in 20 of 20 folds (median 1.34, range 1.28 to 1.45). With the nested calibrator (temperature in 13 folds, none in 5, beta in 2), the slope is 0.92 [0.73, 1.21], the intercept −0.13 [−0.54, 0.25], and the ECE 0.032 against a null 95th percentile of 0.044 ($p = 0.4$). The paired slope change is +0.17 [+0.10, +0.25] ($p < 0.001$; BH $q = 0.002$, Holm $p = 0.002$). Its effect on ranking and accuracy is negligible: AUC −0.001 [−0.007, +0.004], accuracy −0.2 [−0.6, +0.0] points. The log-loss gain, −0.011 [−0.031, +0.008], is within noise at 20 persons. Decision: when a GBM gives the product P(right), calibrate it with the nested calibrator.

The LR's calibrator hurts. Uncalibrated, the LR slope is 0.82 [0.61, 1.18] (its CI covers 1) and its intercept −0.11 [−0.34, 0.12]; its ECE, 0.069, exceeds the null 95th percentile of 0.054 ($p = 0.0022$; BH $q = 0.0037$, Holm $p = 0.0067$). The nested calibrator (none in 10 folds, temperature in 8, Platt in 2) gives a slope of 0.87 [0.62, 1.33] and an intercept of −0.10 [−0.31, 0.11], but a larger ECE, 0.085, against a null 95th percentile of 0.056 ($p < 0.001$). The slope moves by only +0.05 [+0.01, +0.16] (BH $q = 0.013$, Holm $p = 0.013$ in the calibration test family), while the paired log loss rises by +0.007 [+0.001, +0.016] and the Brier score by +0.003 [+0.001, +0.006]. Decision: leave the LR uncalibrated. Its deviations are not a monotone distortion that a rank-preserving map can remove. The fixed isotonic map (exploratory) lowers the ECE to 0.048 (null 95th percentile 0.052; slope 0.86 [0.68, 1.09], intercept −0.00 [−0.26, 0.25]), but it ties predictions and costs ranking.

MR carries its own inner Platt scaling. Its slope is 0.93 [0.78, 1.16], its intercept −0.03 [−0.31, 0.25], and its ECE 0.055 against a null 95th percentile of 0.053 ($p = 0.034$, unadjusted: MR was not part of the calibration test family).

Per platform. The pooled calibrator does not fully fix each platform (Table 26). Uncalibrated GBM is most over-confident on iOS (slope 0.62 [0.46, 0.96], intercept −0.34 [−1.64, 0.53]; ECE 0.078, null 95th percentile 0.058) and less so on Android (slope 0.87 [0.66, 1.19], intercept −0.11 [−0.61, 0.49]; ECE 0.050, null 0.050). After the nested calibrator, the Android slope is 1.03 [0.74, 1.46] (intercept −0.08 [−0.51, 0.38]; ECE 0.064, null 0.057) and the iOS slope 0.79 [0.60, 1.15] (intercept −0.25 [−1.19, 0.43]; ECE 0.074, null 0.068). On each platform the ECE stays slightly above its null band. With 8 iOS persons this is a caveat, not a finding.

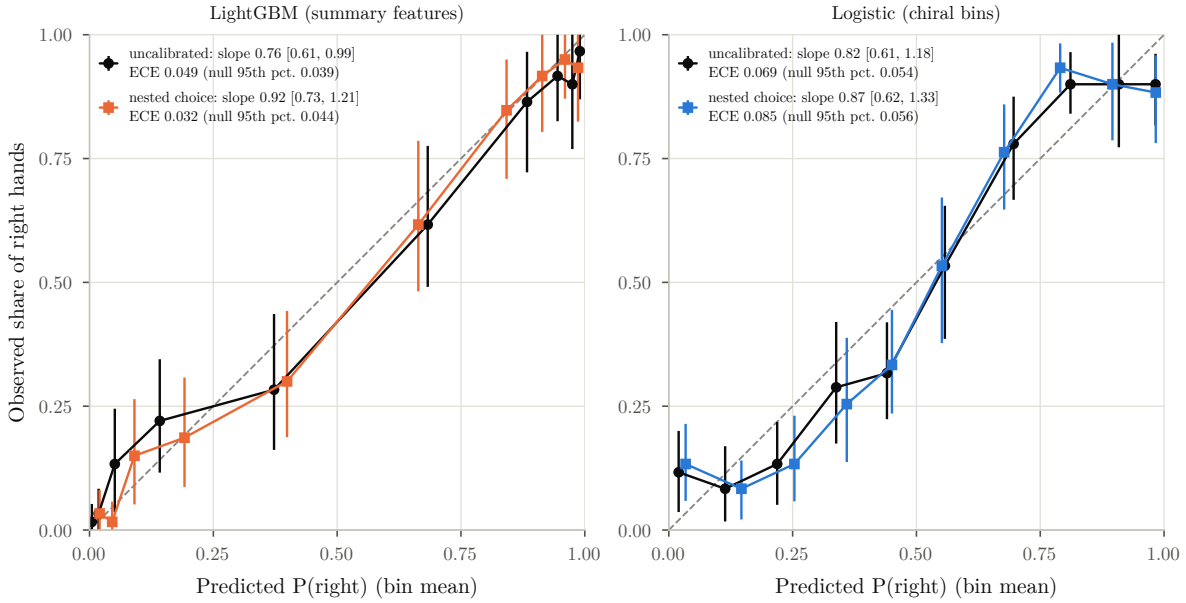


Figure 40: Reliability of $P(\text{right})$ for GBM (left) and LR (right), before calibration and after the calibrator chosen inside every training fold (10 equal-mass bins; bars: 95 % CI of the observed share of right hands; LOSO predictions). Legend: calibration slope with its CI, and the ECE with the 95th percentile of the ECE of perfectly calibrated predictions on the same bins. The nested calibrator brings the GBM slope to about 1; for LR it does not help.

B.12. Combining several attempts

The app may combine k attempts of one person. Attempts of one person are correlated, so summing their logits as if they were independent overstates the certainty, and averaging their probabilities understates it. Four rules were compared on random k-subsets of one person and hand, for k up to 5: the mean of $P(\text{right})$, the mean of the logits, the sum of the logits, and the sum divided by a scale τ_k fitted on the training persons of each fold, $P(\text{right}) = \sigma(\sum_i z_i/\tau_k)$, where z_i is the logit of attempt i and σ the logistic function (written T_k in the figure and tables). The same 19 persons, those with at least 5 trials of each hand, enter at every k (Figure 41, Tables 27 and 28).

Table 23: Calibrators fitted inside every training fold on inner cross-fitted predictions (GroupKFold over the training persons) and applied to the held-out person. Nested: the calibrator (or none) chosen inside each fold by grouped cross-validation among the rank-preserving candidates; the fixed calibrators are compared on the outer folds and are exploratory (4 configurations). ECE with the 95th percentile and the p-value of the ECE under perfect calibration (trials treated as independent). 95% person-cluster bootstrap CIs.

Model	Calibrator	Log loss [95% CI]	Brier [95% CI]	AUC	ECE (null 95th pct.; p)	Slope [95% CI]
GBM	none	0.369 [0.260, 0.486]	0.109 [0.077, 0.144]	0.922	0.049 (0.039; 0.003)	0.76 [0.61, 0.99]
GBM	Platt	0.359 [0.276, 0.449]	0.109 [0.080, 0.140]	0.920	0.039 (0.046; 0.20)	1.00 [0.81, 1.31]
GBM	temperature	0.356 [0.272, 0.447]	0.108 [0.079, 0.140]	0.921	0.033 (0.047; 0.47)	1.02 [0.82, 1.34]
GBM	isotonic	0.374 [0.275, 0.471]	0.108 [0.078, 0.137]	0.918	0.034 (0.042; 0.25)	0.85 [0.66, 1.17]
GBM	beta	0.363 [0.281, 0.452]	0.111 [0.081, 0.142]	0.918	0.041 (0.046; 0.14)	0.98 [0.80, 1.28]
GBM	nested	0.358 [0.265, 0.463]	0.108 [0.078, 0.141]	0.921	0.032 (0.044; 0.40)	0.92 [0.73, 1.21]
LR	none	0.490 [0.421, 0.569]	0.146 [0.123, 0.170]	0.868	0.069 (0.054; 0.002)	0.82 [0.61, 1.18]
LR	Platt	0.488 [0.432, 0.553]	0.150 [0.129, 0.172]	0.866	0.092 (0.057; <0.001)	1.02 [0.75, 1.48]
LR	temperature	0.488 [0.431, 0.552]	0.150 [0.129, 0.172]	0.868	0.089 (0.058; <0.001)	1.03 [0.75, 1.50]
LR	isotonic	0.475 [0.408, 0.551]	0.145 [0.121, 0.172]	0.850	0.048 (0.052; 0.12)	0.86 [0.68, 1.09]
LR	beta	0.497 [0.436, 0.567]	0.151 [0.130, 0.173]	0.865	0.102 (0.058; <0.001)	0.98 [0.69, 1.46]
LR	nested	0.498 [0.430, 0.577]	0.149 [0.127, 0.172]	0.866	0.085 (0.056; <0.001)	0.87 [0.62, 1.33]

For GBM at $k = 5$, the calibration slope is 0.40 [0.33, 0.91] for the sum of the logits (over-confident), 2.82 [2.15, 8.20] for the mean of $P(\text{right})$ and 1.89 [1.56, 4.55] for the mean of the logits (both under-confident). The scaled sum gives 1.13 [0.90, 2.85], the closest to 1. Its median scale is $\tau_5 = 3.15$, between 1 (the plain sum) and 5 (the mean of the logits). For LR the scaled sum gives 0.98 [0.63, 2.76] ($\tau_5 = 2.00$). By log loss, the GBM rules cannot be told apart with 19 persons: the scaled sum minus the sum is -0.140 [-0.388, $+0.032$], minus the mean of the logits -0.018 [-0.057, $+0.039$], and minus the mean of $P(\text{right})$ -0.071 [-0.131, $+0.014$]. The aggregate analysis has no perfect-calibration null band for its ECE and no intercept, so this verdict rests on the slope and the log loss only.

Decision: report the summed logits divided by τ_k, with τ_k refitted whenever the training data change. It is the principled rule, not a demonstrated improvement over the mean of the logits.

B.13. Accuracy by instructed condition

Table 29 stratifies the GBM predictions by the instructed condition. Conditions were never model inputs, no test was run, and the GBM designation is exploratory. Accuracy is 89.6 % [83.1, 96.0] at fast speed and 81.4 % [67.6, 94.6] at slow speed, and 78.6 % [69.5, 87.2] in protocol tier 3, the lowest of the four tiers. Strata with fewer than 5 persons have no CI. The secondary analyses of speed and grip are in Appendix H.

Table 24: Each calibrator minus the uncalibrated model on the same trials: pooled log loss, Brier score, AUC, calibration slope and accuracy, with paired person-bootstrap 95% CIs (both arms on the same resampled persons). A negative log-loss or Brier difference is an improvement; a slope difference above 0 corrects over-confidence; a strictly increasing calibrator cannot change the ranking within a fold.

Model	Calibrator	Diff. log loss	Diff. Brier	Diff. AUC	Diff. slope	Diff. acc. (points)
GBM	Platt	−0.010 [−0.038, +0.015]	+0.000 [−0.005, +0.004]	−0.002 [−0.004, −0.001]	+0.24 [+0.19, +0.33]	+0.0 [−1.1, +1.2]
GBM	temperature	−0.012 [−0.038, +0.012]	−0.001 [−0.005, +0.003]	−0.001 [−0.002, +0.000]	+0.26 [+0.21, +0.35]	+0.0 [+0.0, +0.0]
GBM	isotonic	+0.005 [−0.030, +0.030]	−0.001 [−0.009, +0.005]	−0.003 [−0.008, +0.002]	+0.09 [+0.02, +0.20]	+0.3 [−1.4, +2.0]
GBM	beta	−0.005 [−0.034, +0.020]	+0.002 [−0.003, +0.006]	−0.003 [−0.005, −0.001]	+0.22 [+0.18, +0.29]	−0.3 [−1.3, +0.8]
GBM	nested	−0.011 [−0.031, +0.008]	−0.001 [−0.005, +0.002]	−0.001 [−0.007, +0.004]	+0.17 [+0.10, +0.25]	−0.2 [−0.6, +0.0]
LR	Platt	−0.002 [−0.020, +0.014]	+0.004 [+0.001, +0.008]	−0.001 [−0.003, −0.000]	+0.20 [+0.14, +0.30]	−1.0 [−2.6, +0.3]
LR	temperature	−0.003 [−0.020, +0.013]	+0.004 [+0.001, +0.007]	−0.000 [−0.001, +0.001]	+0.21 [+0.15, +0.32]	+0.0 [+0.0, +0.0]
LR	isotonic	−0.015 [−0.039, +0.011]	−0.001 [−0.007, +0.006]	−0.018 [−0.025, −0.011]	+0.04 [−0.12, +0.13]	−0.5 [−2.2, +0.9]
LR	beta	+0.006 [−0.008, +0.019]	+0.005 [+0.002, +0.008]	−0.003 [−0.005, −0.001]	+0.16 [+0.07, +0.28]	−1.0 [−2.6, +0.3]
LR	nested	+0.007 [+0.001, +0.016]	+0.003 [+0.001, +0.006]	−0.002 [−0.004, +0.001]	+0.05 [+0.01, +0.16]	+0.2 [+0.0, +0.6]

Table 25: The calibration test family: the ECE of each uncalibrated panel model against perfect calibration, and the change of the calibration slope by its nested calibrator (two-sided bootstrap p); Benjamini-Hochberg q and Holm-adjusted p within the family.

Test	p	q (BH)	p (Holm)
gbm: ECE of the uncalibrated model against its perfect-calibration null	0.003	0.004	0.007
gbm: calibration-slope change, nested calibrator vs uncalibrated	<0.001	0.002	0.002
lr: ECE of the uncalibrated model against its perfect-calibration null	0.002	0.004	0.007
lr: calibration-slope change, nested calibrator vs uncalibrated	0.013	0.013	0.013

Table 26: Calibration per test platform (95% person-cluster bootstrap CIs within the platform; none below 5 persons).

Model	Calibrator	Platform	Trials	Pers.	Log loss [95% CI]	Brier [95% CI]	ECE (null 95th pct.)	Slope [95% CI]
GBM	none	android	383	12	0.335 [0.233, 0.474]	0.099 [0.068, 0.140]	0.050 (0.050)	0.87 [0.66, 1.19]
GBM	none	ios	215	8	0.429 [0.237, 0.638]	0.126 [0.068, 0.187]	0.078 (0.058)	0.62 [0.46, 0.96]
GBM	Platt	android	383	12	0.340 [0.257, 0.448]	0.102 [0.073, 0.140]	0.044 (0.060)	1.14 [0.86, 1.58]
GBM	Platt	ios	215	8	0.393 [0.247, 0.557]	0.121 [0.069, 0.175]	0.053 (0.073)	0.83 [0.61, 1.27]
GBM	temperature	android	383	12	0.336 [0.253, 0.446]	0.101 [0.072, 0.139]	0.047 (0.060)	1.17 [0.88, 1.61]
GBM	temperature	ios	215	8	0.393 [0.245, 0.559]	0.121 [0.068, 0.176]	0.058 (0.074)	0.84 [0.62, 1.28]
GBM	isotonic	android	383	12	0.338 [0.252, 0.435]	0.100 [0.071, 0.133]	0.037 (0.055)	1.00 [0.77, 1.37]
GBM	isotonic	ios	215	8	0.437 [0.249, 0.646]	0.122 [0.067, 0.176]	0.050 (0.068)	0.67 [0.42, 1.17]
GBM	beta	android	383	12	0.345 [0.261, 0.454]	0.105 [0.074, 0.144]	0.053 (0.059)	1.11 [0.85, 1.51]
GBM	beta	ios	215	8	0.396 [0.249, 0.560]	0.122 [0.069, 0.176]	0.053 (0.073)	0.82 [0.60, 1.26]
GBM	nested	android	383	12	0.342 [0.251, 0.477]	0.101 [0.072, 0.141]	0.064 (0.057)	1.03 [0.74, 1.46]
GBM	nested	ios	215	8	0.386 [0.232, 0.558]	0.120 [0.066, 0.176]	0.074 (0.068)	0.79 [0.60, 1.15]
LR	none	android	383	12	0.502 [0.432, 0.598]	0.150 [0.124, 0.177]	0.085 (0.069)	0.86 [0.57, 1.34]
LR	none	ios	215	8	0.469 [0.334, 0.634]	0.139 [0.097, 0.191]	0.083 (0.083)	0.77 [0.49, 1.53]
LR	Platt	android	383	12	0.501 [0.439, 0.577]	0.155 [0.130, 0.180]	0.112 (0.074)	1.07 [0.71, 1.67]
LR	Platt	ios	215	8	0.465 [0.358, 0.602]	0.142 [0.105, 0.189]	0.095 (0.091)	0.95 [0.60, 1.93]
LR	temperature	android	383	12	0.500 [0.438, 0.577]	0.154 [0.131, 0.179]	0.107 (0.074)	1.09 [0.71, 1.69]
LR	temperature	ios	215	8	0.465 [0.359, 0.598]	0.142 [0.105, 0.190]	0.100 (0.091)	0.96 [0.61, 1.99]
LR	isotonic	android	383	12	0.492 [0.421, 0.579]	0.150 [0.120, 0.180]	0.079 (0.068)	0.85 [0.63, 1.16]
LR	isotonic	ios	215	8	0.445 [0.323, 0.599]	0.137 [0.093, 0.191]	0.073 (0.084)	0.88 [0.59, 1.36]
LR	beta	android	383	12	0.512 [0.447, 0.602]	0.155 [0.131, 0.181]	0.117 (0.073)	1.00 [0.61, 1.65]
LR	beta	ios	215	8	0.469 [0.359, 0.605]	0.143 [0.105, 0.190]	0.086 (0.091)	0.94 [0.59, 1.90]
LR	nested	android	383	12	0.507 [0.436, 0.600]	0.152 [0.128, 0.178]	0.110 (0.072)	0.95 [0.59, 1.58]
LR	nested	ios	215	8	0.481 [0.355, 0.642]	0.142 [0.103, 0.193]	0.088 (0.086)	0.78 [0.49, 1.64]

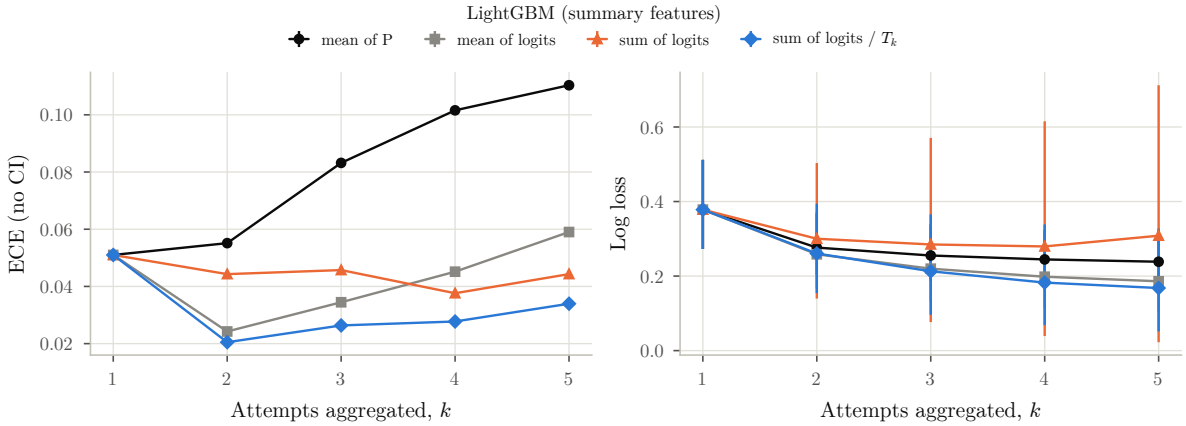


Figure 41: Combining k attempts of one person and hand (GBM; the same 19 persons at every k; every person and hand weighs the same). Left: ECE (no CI and no null band); right: log loss with 95 % CIs. Rules: mean of the probabilities, mean of the logits, sum of the logits, and that sum divided by the training-fold scale τ_k (T_k). The scaled sum has the calibration slope closest to 1 (Table 27).

Table 27: Calibration of k-attempt aggregates (19 persons, the same at every k; every person and hand weighs the same): calibration slope at $k = 2$, 3 and 5 (CI only at $k = 5$) and log loss at $k = 5$ with 95 % CIs. At $k = 1$ the rules coincide (slope 0.73 for GBM, 0.80 for LR). A slope above 1 is under-confident, below 1 over-confident. The full table, with accuracy, Brier score and ECE at every k, is `tables/p1/p1_aggregates` of the analysis outputs.

Model	Rule	Slope, $k = 2$	Slope, $k = 3$	Slope, $k = 5$ [CI]	Log loss, $k = 5$ [CI]
GBM	mean of P(right)	1.45	2.06	2.82 [2.15, 8.20]	0.238 [0.175, 0.318]
	mean of logits	1.17	1.46	1.89 [1.56, 4.55]	0.186 [0.105, 0.292]
	sum of logits	0.59	0.49	0.40 [0.33, 0.91]	0.308 [0.023, 0.712]
	sum of logits / τ_k	0.99	1.02	1.13 [0.90, 2.85]	0.168 [0.052, 0.327]
LR	mean of P(right)	2.05	3.07	4.59 [3.39, 8.94]	0.363 [0.314, 0.414]
	mean of logits	1.42	1.88	2.66 [1.81, 6.82]	0.318 [0.255, 0.385]
	sum of logits	0.71	0.63	0.53 [0.37, 1.36]	0.285 [0.089, 0.546]
	sum of logits / τ_k	0.98	0.95	0.98 [0.63, 2.76]	0.250 [0.128, 0.409]

Table 28: Paired differences between aggregation rules (sum of the logits divided by the training-fold T_k minus each other rule), with person-bootstrap 95% CIs and the two-sided bootstrap p of the log-loss difference; negative = the T_k rule is better.

Model	k	Sum of logits / T_k minus	Diff. log loss [95% CI]	Diff. Brier [95% CI]	p
GBM	2	mean of P	−0.016 [−0.049, +0.028]	−0.005 [−0.014, +0.006]	0.37
GBM	2	mean of logits	+0.002 [−0.013, +0.023]	+0.001 [−0.002, +0.004]	0.96
GBM	2	sum of logits	−0.040 [−0.112, +0.016]	−0.003 [−0.010, +0.003]	0.20
GBM	3	mean of P	−0.042 [−0.095, +0.028]	−0.010 [−0.023, +0.008]	0.18
GBM	3	mean of logits	−0.007 [−0.038, +0.034]	−0.000 [−0.006, +0.008]	0.59
GBM	3	sum of logits	−0.072 [−0.202, +0.023]	−0.005 [−0.016, +0.003]	0.18
GBM	4	mean of P	−0.062 [−0.118, +0.016]	−0.013 [−0.027, +0.007]	0.099
GBM	4	mean of logits	−0.016 [−0.051, +0.035]	−0.001 [−0.009, +0.009]	0.44
GBM	4	sum of logits	−0.097 [−0.277, +0.029]	−0.006 [−0.020, +0.004]	0.25
GBM	5	mean of P	−0.071 [−0.131, +0.014]	−0.013 [−0.029, +0.009]	0.093
GBM	5	mean of logits	−0.018 [−0.057, +0.039]	−0.001 [−0.010, +0.012]	0.43
GBM	5	sum of logits	−0.140 [−0.388, +0.032]	−0.007 [−0.023, +0.004]	0.25
LR	2	mean of P	−0.015 [−0.058, +0.033]	−0.008 [−0.018, +0.002]	0.51
LR	2	mean of logits	−0.011 [−0.039, +0.019]	−0.007 [−0.012, −0.002]	0.44
LR	2	sum of logits	−0.015 [−0.050, +0.016]	+0.002 [−0.002, +0.006]	0.39
LR	3	mean of P	−0.056 [−0.128, +0.035]	−0.020 [−0.038, −0.000]	0.20
LR	3	mean of logits	−0.034 [−0.089, +0.032]	−0.014 [−0.026, −0.003]	0.28
LR	3	sum of logits	−0.026 [−0.088, +0.026]	+0.002 [−0.004, +0.008]	0.40
LR	4	mean of P	−0.088 [−0.171, +0.027]	−0.029 [−0.049, −0.006]	0.12
LR	4	mean of logits	−0.054 [−0.124, +0.040]	−0.020 [−0.035, −0.005]	0.21
LR	4	sum of logits	−0.030 [−0.109, +0.034]	+0.002 [−0.004, +0.009]	0.44
LR	5	mean of P	−0.113 [−0.205, +0.019]	−0.036 [−0.058, −0.009]	0.097
LR	5	mean of logits	−0.068 [−0.146, +0.043]	−0.024 [−0.041, −0.004]	0.19
LR	5	sum of logits	−0.035 [−0.131, +0.044]	+0.001 [−0.006, +0.009]	0.49

Table 29: LightGBM (summary features) (the best panel model by outer-fold accuracy: an exploratory designation among 3) per instructed condition, stratified after prediction (conditions are never inputs; 95% person-cluster bootstrap CIs).

Condition	Level	Trials	Pers.	Accuracy % [95% CI]
platform	android	383	12	86.4 [80.8, 90.8]
platform	ios	215	8	84.2 [76.0, 91.6]
speed	fast	134	20	89.6 [83.1, 96.0]
speed	natural	421	20	84.8 [79.2, 90.2]
speed	slow	43	19	81.4 [67.6, 94.6]
orientation	top_left_toward	118	19	89.8 [82.5, 96.0]
orientation	top_right_toward	114	19	80.7 [72.4, 88.9]
orientation	upright_toward	366	20	85.8 [81.1, 89.7]
variant	bent_start	7	4	71.4
variant	curved	16	8	68.8 [43.8, 93.8]
variant	elbow_high	14	7	71.4 [57.1, 92.9]
variant	jerky	13	8	84.6 [54.5, 100.0]
variant	lean_in	11	6	72.7 [40.0, 100.0]
variant	pause	17	9	88.2 [72.2, 100.0]
variant	standard	470	20	87.4 [83.5, 91.5]
variant	tilted_touch	35	19	88.6 [77.8, 97.3]
variant	twist	15	8	66.7 [40.0, 92.9]
posture	sitting	438	15	84.5 [78.5, 89.4]
posture	standing	160	5	88.8 [84.4, 93.9]
start_position	center	18	10	77.8 [63.2, 94.1]
start_position	natural	563	20	86.0 [81.1, 90.6]
start_position	shoulder	17	9	82.4 [64.7, 100.0]
tier	1	159	20	89.3 [83.7, 94.2]
tier	2	221	19	85.5 [78.3, 91.9]
tier	3	140	19	78.6 [69.5, 87.2]
tier	4	61	7	91.8 [85.5, 96.7]

B.14. Caveats

- The panel is small and fixed. Not every option ran on every model; MR ran only on the most informative options, to save compute.
- With 20 persons, the resolution is a few points. "No difference" is absence of evidence; the lower CI bound is the largest loss the data allow.
- Every choice was judged on the oracle bounds. The cue-free detector could change the ranking of the window options.
- The segmentation errors are synthetic and uniform, not those of a real detector.
- The ECE null treats trials as independent. The fixed calibrators are compared on the outer folds (exploratory, 4 configurations).
- The GBM panel model uses device-frame channels. In the final-model selection such models lose accuracy on the sideways grips and are removed by the grip test (Appendix F), so the GBM verdicts transfer to the final model only where they do not depend on the device frame.
- The variants ran the leak check on 3 seeded folds only; the base and nested-calibrated runs ran it on every fold.

C. Interpretable models and the feature catalogue

Lane m1 asks two questions: which physical quantity carries the hand, and how far simple, explainable models go. This appendix states every decision of the lane with its options, its evidence and its rationale. The main text uses its results in Section 3 and Section 5.

C.1. Evaluation contract and designs

Every experiment ran through the shared harness of Appendix A: LOSO over 20 folds on the 598 primary trials of 20 persons. The lane ran 54 experiments; 54 of them passed the harness leak check on every fold. Every fitted quantity is fitted on the training persons only: hyperparameters, in-fold feature and treatment choices, thresholds, scalers, winsorising limits, spline knots, calibrators and the aggregation exponent. Choices use an inner split of the training persons into groups (GroupKFold over persons). The free-choice trials are scored and never trained on.

Each model carries one of six design labels. *Pre-specified*: fixed before the data were seen. *Specified after the pooled description*: PHYS6 (defined below), chosen after a pooled look at the components of all persons. *Nested*: a choice made inside every fold, so its estimate is selection-aware. *Exploratory*: a member of a family of configurations compared on the outer folds; its number of configurations is given. *Sensitivity run* and *window analysis*: never candidates for selection.

Paired differences come with a paired person-bootstrap CI (2000 resamples) and an exact sign-flip test over persons. Each family of comparisons is Holm-adjusted and named where it is used: the 5 contrasts against the rule, the 54 models against the reference, the 27 options against PHYS6, and the four post-hoc remedies of one model. The ECE uses 10 equal-mass bins. It is reported with the 95th percentile and the p value of its perfect-calibration null, in

which the labels are redrawn from the model's own $P(\text{right})$. The calibration slope is the slope of a logistic regression of the outcome on $\text{logit}(P(\text{right}))$, with a person-bootstrap CI; a slope above 1 means under-confidence. The intercept of the same regression is reported in Table 40 as a point estimate only: the lane computed no CI for it. On the free-choice trials, a $P(\text{right})$ of exactly 0.5 counts one half. All 20 free-choice trials are right-hand choices, so the natural-use score of this appendix is right-hand recall on the free-choice trials (a constant right answer would score 100 %), with a Wilson 95 % interval. The column $k = 3$ is the per-person balanced accuracy when the $P(\text{right})$ of three attempts of one person and hand are averaged.

C.2. The body frame, the catalogue and the physics

A heading-free body frame. From the gyro-propagated up vector $\hat{\mathbf{g}}$ and the screen normal, lane m1 builds two horizontal unit vectors: $\hat{\mathbf{h}}$, out of the screen towards the face, and $\hat{\mathbf{s}} = \hat{\mathbf{g}} \times \hat{\mathbf{h}}$, towards the person's right. The yaw rate $\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$, the roll rate $\boldsymbol{\omega} \cdot \hat{\mathbf{h}}$ and the lateral acceleration $\mathbf{a} \cdot \hat{\mathbf{s}}$ are chiral: they change sign under the mirror of eq. (1). Pitch, forward and vertical acceleration are mirror-even. The three screen-toward grips differ by a rotation about the screen normal, which leaves every feature unchanged. Under the screen-away flip, yaw is unchanged but roll and lateral change sign, because $\hat{\mathbf{h}}$ follows the screen normal. Unit tests of the lane check the three properties. A feature name gives the component and the fraction of the approach over which it is accumulated: `yaw_q50` is the yaw rotation over the first half (the first-half yaw), `yaw_q100` the net yaw, `roll_q50` the roll rotation and `latvel_q50` the lateral velocity change (the integral of $\mathbf{a} \cdot \hat{\mathbf{s}}$) up to the half of the approach.

Options for the univariate score. Each feature was scored alone by its LOSO AUC [23]. A univariate logistic model with an intercept was rejected: in LOSO, the fitted intercept follows the class balance of the training persons, which is anticorrelated with the balance of the held-out person, and it drags the AUC of an uninformative feature below 0.5. The catalogue therefore uses a direction-only rule, $P(\text{right}) = \sigma(s\,x/\text{RMS}(x))$, whose direction s and scale are fitted on the training persons, with no intercept.

Only chiral features carry the hand. The catalogue holds 71 features, 57 odd and 14 even (Figure 42, Table 30). Odd features reach a univariate AUC of up to 0.87: `latvel_q75` reaches 0.868 [0.820, 0.916] and orders the trials of 100 % of the persons correctly more often than not. Every even feature stays between 0.46 and 0.53. For 12 features the learned direction changes across folds. For such a feature, an AUC below 0.5 means that the direction learned on the other persons fails on the held-out person; it is not evidence for the opposite direction. *Decision:* models are built from chiral features; even descriptors enter at most as modulators of the chiral slopes.

The yaw turns early and turns back. The first-half yaw has a higher AUC than the net yaw (the CIs overlap and no paired test was run): `yaw_q50` reaches an AUC of 0.854 [0.803, 0.899] and separates 100 % of the persons; the net yaw reaches 0.764 [0.682, 0.840] (85 % of the persons) and the second-half yaw 0.511 [0.384, 0.628] (50 %). The describe step finds the same ordering with pooled AUCs of 0.85, 0.77 and 0.49 (Table 31). The phone turns towards

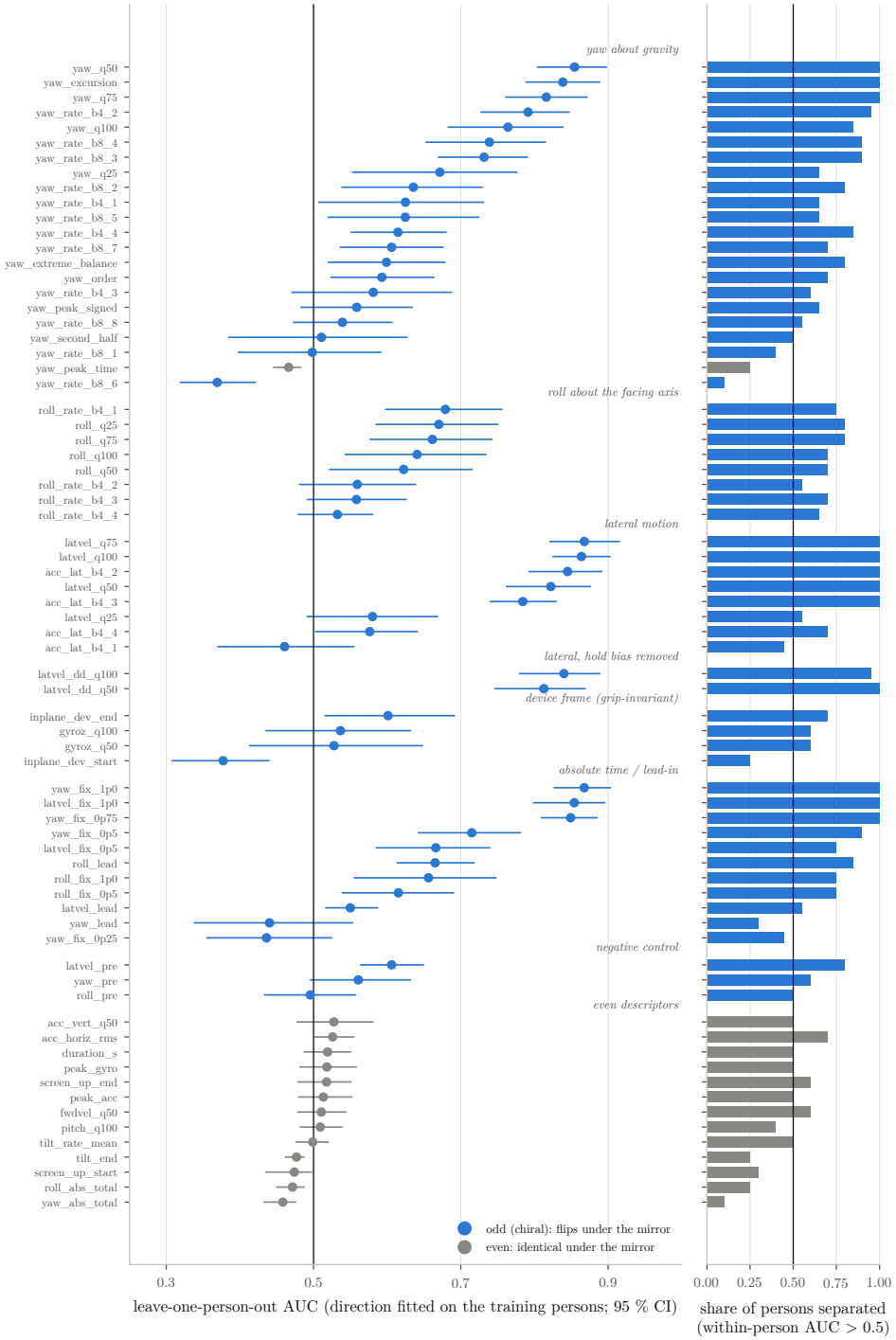


Figure 42: Feature catalogue: univariate LOSO AUC of each of the 71 features (direction and scale fitted on the training persons, no intercept; 95 % CI) and the share of persons whose own trials the feature orders correctly more often than not (598 trials, 20 persons). Odd (chiral) features reach 0.87; even features stay between 0.46 and 0.53. It supports building the models from chiral features only.

Table 30: Feature catalogue of lane m1: univariate leave-one-person-out AUC (direction-only rule sigmoid(s x / RMS(x)): direction s and scale fitted on the training persons, no intercept; 95 % person-bootstrap CI), share of persons whose own trials the feature orders correctly more often than not, median within-person AUC, accuracy of the sign rule (odd features; direction from the training persons) and per-hand medians. Screen flip: whether the feature changes sign when the phone is turned screen-away (every feature is unchanged by the three screen-toward grips). Formulas, frames and meanings: m1_feature_catalogue_full.csv.

Feature	Family	Parity	Unit	Screen flip	AUC	95% CI	Persons >0.5	Within median	Sign rule	Median L	Median R
yaw_q25	yaw	odd	deg	none	0.672	[0.552, 0.777]	65%	0.69	63.2%	−0.963	1.48
yaw_q50	yaw	odd	deg	none	0.854	[0.803, 0.899]	100%	0.90	78.9%	−3.88	5.33
yaw_q75	yaw	odd	deg	none	0.816	[0.760, 0.872]	100%	0.83	73.6%	−4.26	4.61
yaw_q100	yaw	odd	deg	none	0.764	[0.682, 0.840]	85%	0.77	66.6%	−3.81	2.87
yaw_second_half	yaw	odd	deg	none	0.511	[0.384, 0.628]	50%	0.52	48.0%	−0.821	−0.292
yaw_rate_b4_1	yaw	odd	deg/s	none	0.625	[0.506, 0.732]	65%	0.66	61.5%	−1.78	2.8
yaw_rate_b4_2	yaw	odd	deg/s	none	0.791	[0.727, 0.848]	95%	0.86	71.7%	−5.47	6.54
yaw_rate_b4_3	yaw	odd	deg/s	none	0.581	[0.470, 0.688]	60%	0.54	54.2%	−1.3	1.12
yaw_rate_b4_4	yaw	odd	deg/s	none	0.615	[0.550, 0.681]	85%	0.66	58.0%	0.627	−1.64
yaw_rate_b8_1	yaw	odd	deg/s	none	0.498	[0.397, 0.592]	40%	0.46	48.3%	−1.04	−0.00124
yaw_rate_b8_2	yaw	odd	deg/s	none	0.636	[0.538, 0.730]	80%	0.58	60.5%	−2.27	5.26
yaw_rate_b8_3	yaw	odd	deg/s	none	0.732	[0.669, 0.791]	90%	0.80	66.7%	−4.46	7.96
yaw_rate_b8_4	yaw	odd	deg/s	none	0.739	[0.652, 0.816]	90%	0.76	65.4%	−8.64	5.52
yaw_rate_b8_5	yaw	odd	deg/s	none	0.625	[0.519, 0.725]	65%	0.62	56.7%	−3.43	2.59
yaw_rate_b8_6	yaw	odd	deg/s	none	0.369	[0.318, 0.422]	10%	0.40	43.0%	−0.33	−0.576
yaw_rate_b8_7	yaw	odd	deg/s	none	0.606	[0.535, 0.677]	70%	0.59	57.7%	0.949	−2.28
yaw_rate_b8_8	yaw	odd	deg/s	none	0.539	[0.472, 0.607]	55%	0.53	53.0%	0.24	−0.245
yaw_peak_signed	yaw	odd	deg/s	none	0.559	[0.482, 0.635]	65%	0.56	56.7%	−36.3	39.4
yaw_peak_time	yaw	even	fraction	none	0.466	[0.445, 0.483]	25%	0.43	–	0.444	0.452
yaw_extreme_balance	yaw	odd	deg/s	none	0.599	[0.519, 0.679]	80%	0.62	56.7%	−3.22	5.55
yaw_order	yaw	odd	fraction	none	0.593	[0.523, 0.664]	70%	0.58	53.7%	−0.0317	0.0952
yaw_excursion	yaw	odd	deg	none	0.838	[0.788, 0.889]	100%	0.87	77.9%	−9.51	10.4
roll_q25	roll	odd	deg	sign	0.670	[0.584, 0.751]	80%	0.63	62.0%	0.753	−0.981
roll_q50	roll	odd	deg	sign	0.622	[0.521, 0.716]	70%	0.66	59.2%	0.957	−0.248
roll_q75	roll	odd	deg	sign	0.661	[0.576, 0.743]	80%	0.67	63.7%	1.42	−0.553
roll_q100	roll	odd	deg	sign	0.641	[0.542, 0.735]	70%	0.66	61.4%	1.17	−0.402
roll_rate_b4_1	roll	odd	deg/s	sign	0.679	[0.597, 0.756]	75%	0.62	63.5%	1.41	−2.07
roll_rate_b4_2	roll	odd	deg/s	sign	0.559	[0.480, 0.639]	55%	0.57	50.8%	0.307	0.745
roll_rate_b4_3	roll	odd	deg/s	sign	0.558	[0.491, 0.627]	70%	0.60	53.7%	0.525	−0.533
roll_rate_b4_4	roll	odd	deg/s	sign	0.532	[0.478, 0.581]	65%	0.55	51.0%	−0.265	−0.019
latvel_q25	lateral	odd	m/s	sign	0.580	[0.491, 0.669]	55%	0.56	53.3%	−0.013	−0.000997
latvel_q50	lateral	odd	m/s	sign	0.822	[0.761, 0.877]	100%	0.86	76.6%	−0.0773	0.068
latvel_q75	lateral	odd	m/s	sign	0.868	[0.820, 0.916]	100%	0.92	81.6%	−0.117	0.132
latvel_q100	lateral	odd	m/s	sign	0.864	[0.824, 0.903]	100%	0.92	80.3%	−0.133	0.134
acc_lat_b4_1	lateral	odd	m/s^2	sign	0.461	[0.369, 0.556]	45%	0.49	47.8%	−0.0283	−0.00133
acc_lat_b4_2	lateral	odd	m/s^2	sign	0.845	[0.792, 0.892]	100%	0.89	77.1%	−0.125	0.157
acc_lat_b4_3	lateral	odd	m/s^2	sign	0.784	[0.739, 0.830]	100%	0.83	70.6%	−0.0741	0.111
acc_lat_b4_4	lateral	odd	m/s^2	sign	0.576	[0.502, 0.642]	70%	0.58	56.9%	−0.0112	0.0113
latvel_dd_q50	dedrift	odd	m/s	sign	0.813	[0.746, 0.870]	100%	0.80	75.8%	−0.0538	0.0622
latvel_dd_q100	dedrift	odd	m/s	sign	0.840	[0.779, 0.890]	95%	0.89	77.3%	−0.101	0.13
gyroz_q50	device	odd	deg	sign	0.528	[0.412, 0.649]	60%	0.59	51.8%	0.389	0.717
gyroz_q100	device	odd	deg	sign	0.537	[0.435, 0.632]	60%	0.58	52.2%	0.38	0.676
inplane_dev_start	device	odd	deg	sign	0.377	[0.307, 0.440]	25%	0.35	42.1%	−0.247	0.168
inplane_dev_end	device	odd	deg	sign	0.601	[0.515, 0.692]	70%	0.63	56.4%	−0.749	0.029
yaw_fix_0p25	fixed	odd	deg	none	0.436	[0.355, 0.525]	45%	0.40	45.2%	−0.0232	0.142
yaw_fix_0p5	fixed	odd	deg	none	0.715	[0.642, 0.782]	90%	0.74	64.9%	−1.19	2.23
yaw_fix_0p75	fixed	odd	deg	none	0.849	[0.809, 0.886]	100%	0.85	77.3%	−3.31	5.11
yaw_fix_1p0	fixed	odd	deg	none	0.868	[0.826, 0.904]	100%	0.88	79.3%	−5.24	6.78
roll_fix_0p5	fixed	odd	deg	sign	0.615	[0.538, 0.691]	75%	0.62	59.2%	0.725	−0.537
latvel_fix_0p5	fixed	odd	m/s	sign	0.666	[0.584, 0.740]	75%	0.66	63.0%	−0.027	0.0132
roll_fix_1p0	fixed	odd	deg	sign	0.656	[0.555, 0.748]	75%	0.66	62.2%	1.1	−0.672
latvel_fix_1p0	fixed	odd	m/s	sign	0.854	[0.798, 0.897]	100%	0.89	77.8%	−0.0753	0.0891
yaw_lead	fixed	odd	deg	none	0.440	[0.337, 0.554]	30%	0.39	44.3%	−0.0109	−0.0761
roll_lead	fixed	odd	deg	sign	0.665	[0.613, 0.719]	85%	0.63	63.5%	0.107	−0.201
latvel_lead	fixed	odd	m/s	sign	0.550	[0.516, 0.588]	55%	0.53	54.3%	0.000996	−0.000343
yaw_pre	control	odd	deg	none	0.561	[0.495, 0.632]	60%	0.54	52.7%	−0.0401	−0.00128
roll_pre	control	odd	deg	sign	0.495	[0.432, 0.558]	50%	0.49	47.7%	−0.0426	−0.00112
latvel_pre	control	odd	m/s	sign	0.606	[0.563, 0.650]	80%	0.65	58.0%	−0.00636	0.00195
duration_s	even	even	s	none	0.519	[0.487, 0.551]	50%	0.50	–	1.66	1.63
peak_gyro	even	even	deg/s	none	0.518	[0.481, 0.559]	50%	0.51	–	88.8	93.9
peak_acc	even	even	m/s^2	none	0.513	[0.479, 0.553]	50%	0.51	–	4.34	4.48
tilt_end	even	even	deg	none	0.477	[0.461, 0.488]	25%	0.45	–	8.47	8.16
tilt_rate_mean	even	even	deg/s	none	0.499	[0.475, 0.521]	50%	0.50	–	24.9	25.3
pitch_q100	even	even	deg	sign	0.509	[0.481, 0.539]	40%	0.49	–	6.16	5.34
screen_up_start	even	even	1	sign	0.474	[0.434, 0.498]	30%	0.37	–	0.185	0.18
screen_up_end	even	even	1	sign	0.518	[0.478, 0.551]	60%	0.51	–	0.0806	0.0987
yaw_abs_total	even	even	deg	none	0.458	[0.432, 0.476]	10%	0.41	–	31.7	31.7
roll_abs_total	even	even	deg	none	0.471	[0.449, 0.488]	25%	0.42	–	19.8	20
acc_horiz_rms	even	even	m/s^2	none	0.526	[0.501, 0.555]	70%	0.54	–	1.54	1.56
acc_vert_q50	even	even	m/s	none	0.528	[0.477, 0.582]	50%	0.50	–	0.00838	0.0246
fwdvel_q50	even	even	m/s	sign	0.511	[0.478, 0.545]	60%	0.55	–	0.265	0.26

the side of the hand during the first part of the reach and partly back during the second. Whole-movement integrals lose part of the evidence.

Table 31: How well the yaw about gravity accumulated over parts of the approach separates the hands: AUC of right vs left pooled over trials, and within each person (median, lowest, and the persons above 0.5).

Feature	Pooled	Person median	Person min	Persons > 0.5	Persons
First-half yaw	0.85	0.90	0.63	20	20
Net yaw	0.77	0.77	0.36	17	20
Second-half yaw	0.49	0.48	0.00	10	20

The mirror maps one hand onto the other. After the left-hand mean is mirrored, the right-hand mean of the yaw rate is closer to it than to the unmirrored left mean for 18 of 20 persons (Table 32). The RMS difference between the two mean yaw-rate curves falls from 12.2 to 3.4 deg/s. The mirror does not change the even channels, so their small left-right differences stay as they are. This supports the antisymmetric models below: one set of weights serves both hands.

Table 32: Mirror symmetry of the gravity-frame channels (sign: how the channel changes under the mirror; chiral channels change sign): RMS of the right-hand mean, RMS difference between the right-hand (R) and the left-hand (L) mean, unmirrored and mirrored, and the persons whose right-hand mean is closer to their mirrored left-hand mean (closer).

Channel	Sign	RMS	L vs R	Mirrored L vs R	Closer	Persons
yaw_rate	−1	7.06	12.16	3.40	18	20
yaw_cum	−1	4.67	8.65	0.77	18	20
gyro_horiz	1	28.47	1.75	1.75	–	20
acc_vert	1	0.19	0.07	0.07	–	20
acc_horiz	1	1.61	0.19	0.19	–	20
tilt_from_start	1	9.37	0.57	0.57	–	20

Three chiral components with different timing. Figure 43 shows the mean time course of the three components per hand, with the left-hand mean mirrored. The right-minus-left difference of the mean rate peaks first for roll, then for yaw, then for lateral acceleration: at 16, 27 and 49 per cent of the approach. The mirrored left mean follows the right mean in all three components. The PHYS6 feature set takes each component at the half and at the end of the approach: `yaw_q50`, `yaw_q100`, `roll_q50`, `roll_q100`, `latvel_q50` and `latvel_q100`.

Even descriptors do not tell the hand. Figure 44 shows the per-hand distributions of six key features. The odd features shift in opposite directions for the two hands. The even ones overlap: the duration reaches an AUC of 0.52 and the peak angular speed 0.52. They were tested only as modulators of the chiral slopes (Appendix C.5).

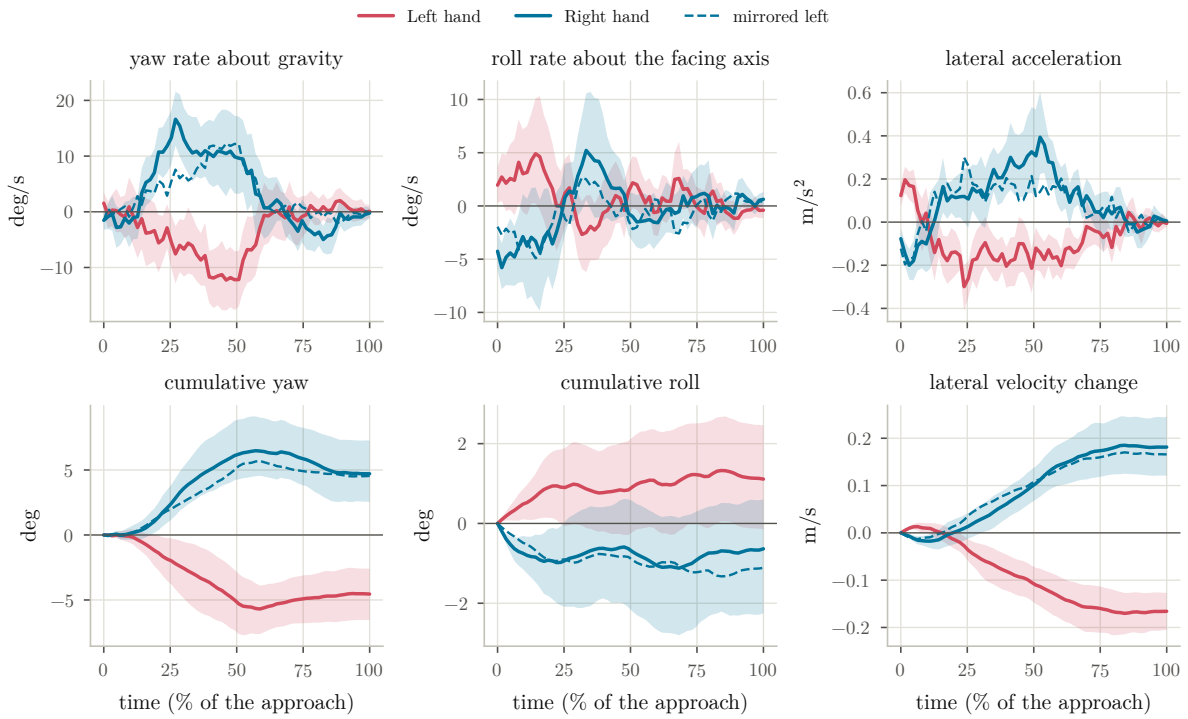


Figure 43: Mean time course of the three chiral components per hand (mean over the 20 persons of each person's mean; bands: person-bootstrap 95 % CI). Top: rates; bottom: their integrals over the approach. Dashed: the left-hand mean mirrored. The hand shows in yaw, roll and lateral motion, at different times of the approach.

Why the lateral feature behaves like yaw. This is a derivation, not a fitted result. The pair $(\hat{\mathbf{h}}, \hat{\mathbf{s}})$ turns with the heading of the screen normal at a rate W, which is the yaw rate for a turn about the vertical. Then $\mathrm{d}\hat{\mathbf{s}}/\mathrm{d}t = -W\hat{\mathbf{h}}$ and, with $\mathbf{v}$ the velocity,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{v}\cdot\hat{\mathbf{s}}\right) = \mathbf{a}\cdot\hat{\mathbf{s}} - W\left(\mathbf{v}\cdot\hat{\mathbf{h}}\right). \tag{4}$$

For a movement that starts and ends at rest, the integral of $\mathbf{a}\cdot\hat{\mathbf{s}}$ over the approach equals the integral of $W(\mathbf{v}\cdot\hat{\mathbf{h}})$: the turn rate weighted by the speed towards the face, plus sensor drift. The lateral feature is thus a speed-weighted yaw. Its univariate AUC (`latvel_q100`: 0.864) lies above that of the net yaw (0.764) and close to that of the first-half yaw (0.854).

Integration drift of the lateral feature. *Detection:* over the still hold, the lateral acceleration has a median absolute bias of 0.043 m/s^2 (90th percentile 0.120). Carried through the approach, this bias amounts to 52 % of |`latvel_q100`| in the median trial, and to more than half of it in 51 % of the trials. The bias itself differs between the hands (AUC 0.605 [0.564, 0.648]). *Why it matters:* a drift of that size could carry sensor or protocol artefacts instead of the movement. *Option tested:* PHYS6 with the lateral velocities integrated after removing the hold bias (exploratory, 33 configurations): 82.6 % [77.2, 87.8]; against PHYS6 +0.7 [−2.4, +3.2] points and log loss +0.008 [−0.013, +0.031], Holm $p = 1$ over the 27 options. *Decision:* the raw lateral integral is kept; its drift is a stated limitation.

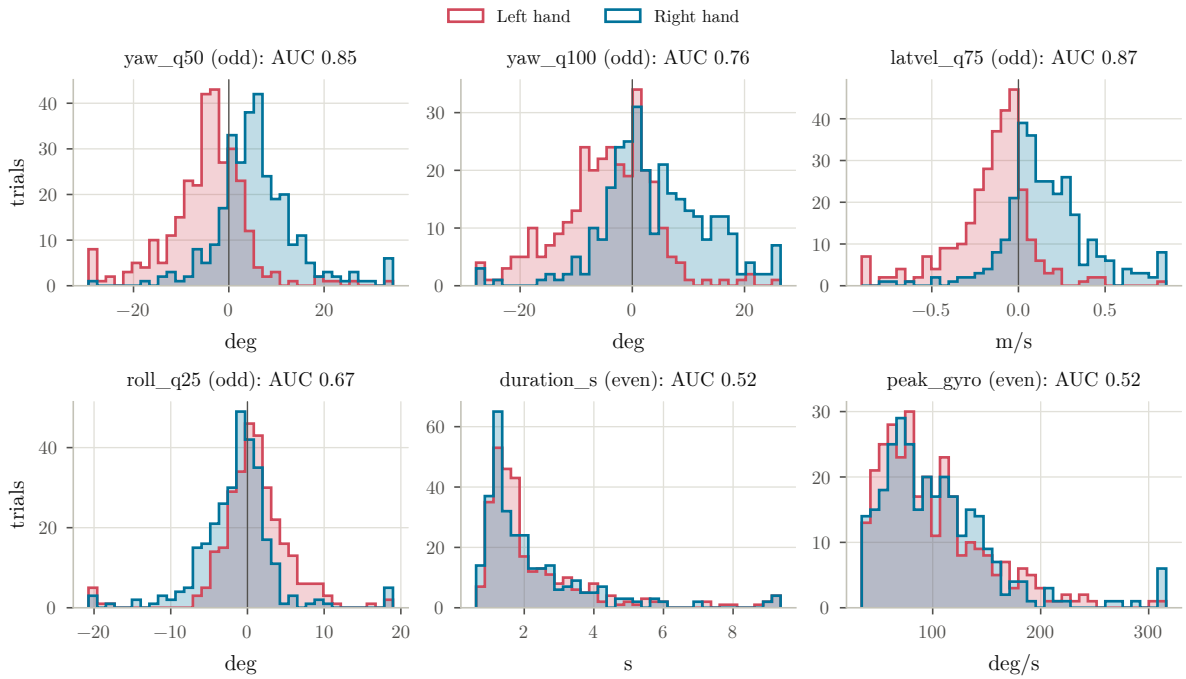


Figure 44: Per-hand distributions of four chiral and two even features (clipped at the 1st and 99th percentiles for display); titles give the univariate LOSO AUC. Even descriptors cannot tell the hand alone.

The still hold and the lead-in. A negative control was planned on the still hold 0.5 s to 0.2 s before the onset, where chance was expected. It is not at chance. The single features reach AUCs of 0.561 [0.495, 0.632] (yaw), 0.495 [0.432, 0.558] (roll) and 0.606 [0.563, 0.650] (lateral); the three together reach 58.9 % [54.5, 63.5], AUC 0.627 [0.576, 0.679]. Two explanations fit and cannot be separated here: a hand-specific drift of the held arm, and the settling of the arm after the previous trial, which used the same hand because hands are blocked. The v0 cue-free detector finds such an earlier movement first in 28 % of the recordings. In production, the hold is done with the hand being judged, so the hold is legitimate evidence, but its size here may be inflated by the blocked protocol. In the last 0.2 s before the onset (the lead-in), roll carries the most evidence (AUC 0.665 [0.613, 0.719]); the three components together reach 63.5 % [59.7, 67.5]. The movement starts slightly before the detected onset, so a cue-free detector should open its window earlier, not later (Appendix G).

C.3. Which components to keep

Options: antisymmetric logistic models on one component (two features each), on the three pairs, on all three (PHYS6), with the set chosen inside every fold, on every chiral approach feature, and with up to four features chosen in-fold by forward selection (Figure 45). Each is paired against PHYS6 with the Holm adjustment over the 27 options of that family.

Evidence: yaw alone reaches 78.9 % [74.0, 84.1] and lateral alone 80.9 % [76.9, 85.1]. Roll alone reaches only 59.4 % [51.6, 67.4] (AUC 0.584); it is worse than PHYS6 after the correction (log

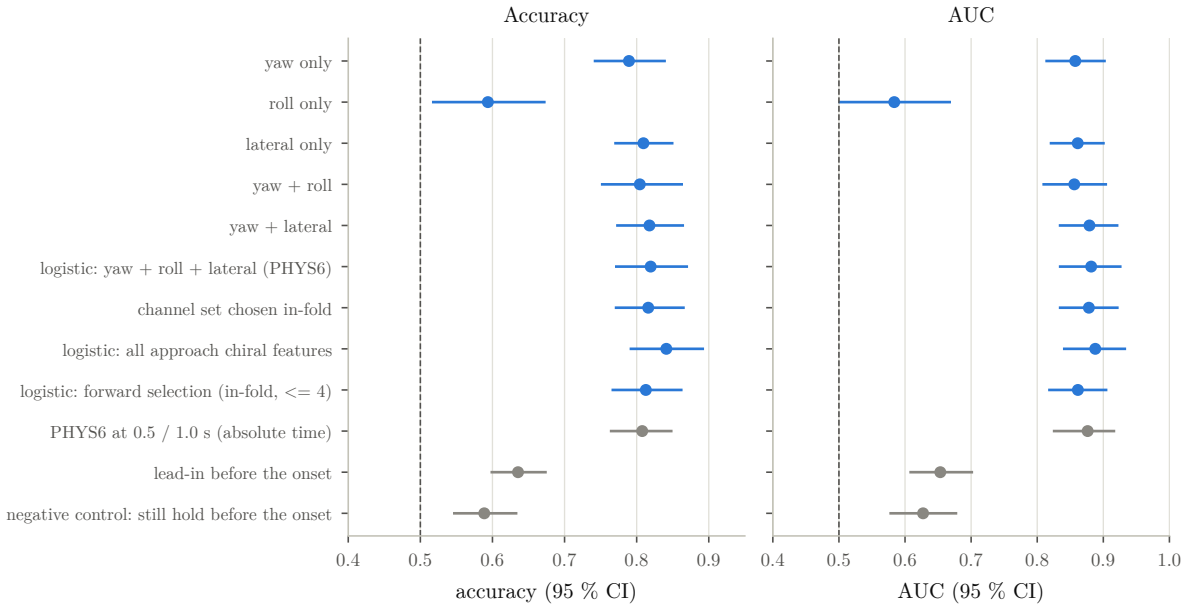


Figure 45: Accuracy and AUC (95 % CI) of antisymmetric logistic models on one, two or three chiral components, with the set or the features chosen in-fold, and on three windows (grey: absolute time after the onset, the lead-in and the still hold before the onset). Yaw and lateral carry the hand; the still hold is not at chance.

loss +0.251 [+0.166, +0.330], Holm $p = 0.002$). Yaw and lateral together reach 81.8 % [77.2, 86.6]; adding roll changes nothing measurable (yaw + lateral against PHYS6: −0.2 [−1.9, +1.7] points, log loss +0.005 [−0.017, +0.030], Holm $p = 1$). The three components taken at fixed times (0.5 s and 1.0 s after the onset, a window analysis independent of the end bound) reach 80.8 % [76.3, 85.0], against PHYS6 −1.2 [−4.5, +2.0] points (Holm $p = 1$).

Guards against the optimism of PHYS6. PHYS6 was specified after a pooled look at all persons, not pre-registered. Two nested guards test whether that look made it optimistic. The set chosen inside every fold among yaw, roll, lateral, the three pairs and PHYS6 is PHYS6 in 18 of 20 folds and reaches 81.6 % [77.0, 86.7] (against PHYS6 −0.3 [−1.4, +0.5] points, log loss +0.005 [−0.010, +0.026], Holm $p = 1$). The in-fold forward selection over every chiral approach feature reaches 81.3 % [76.5, 86.4], with a higher log loss than PHYS6 (+0.053 [+0.012, +0.101]; unadjusted CI above 0, Holm $p = 0.62$). Every chiral approach feature with an L2 penalty only (exploratory, 33 configurations) reaches 84.1 % [79.0, 89.3], against PHYS6 +2.2 [+0.0, +4.7] points (Holm $p = 1$).

Decision: PHYS6 is the baseline of the lane. Yaw about gravity and lateral motion carry the hand; roll is a weak, largely redundant early signal. Choosing the set inside the folds costs nothing measurable, and the wider in-fold search does no better (its higher log loss passes the unadjusted CI only): the small physical set is not an artefact of the pooled look.

Where in the movement. Antisymmetric logistic "temporal filters" on the binned chiral rates (16 bins of the time-normalised approach) show where the evidence lies (Figure 46). The yaw-rate weights average +0.28 in the first half of the approach and +0.02 in the second, and

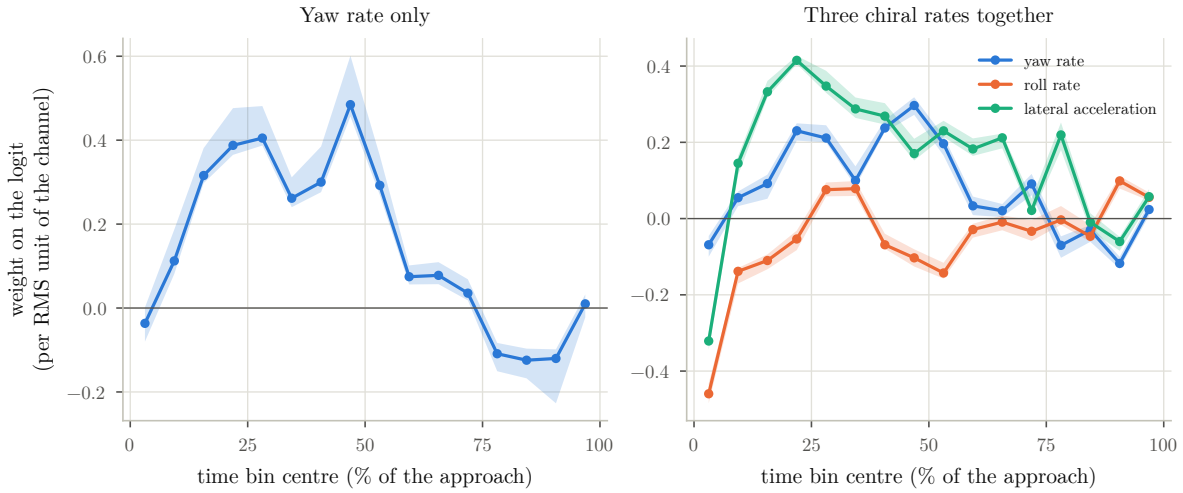


Figure 46: Weight profiles of the antisymmetric temporal filters on the binned chiral rates (median over the 20 LOSO folds; band: 10th to 90th percentile across folds). A positive weight means that a positive rate in that bin argues for the right hand. The yaw evidence sits in the first half of the approach and reverses near its end.

turn negative from 78 per cent of the approach onwards: the turn back. In the three-channel filter, the lateral weights are largest in the first third, and the first bin of roll and of lateral carries a negative weight. *Decision:* features must be time-resolved, not whole-movement integrals; PHYS6 samples each component at the half and at the end of the approach.

C.4. The model ladder

The ladder climbs from rules to fitted models (Figure 49). Every model of the lane, with its design and number of configurations, is in Table 37.

Rules. Four rules were compared (Table 33; per person in Table 34). The pilot rule says right if and only if the first-half yaw is positive. It has no fitted parameter and reaches 78.9 % [73.5, 83.8]. Its direction came from the pilot session of P01, a person of the evaluation set, and its $P(\text{right})$ is 0 or 1, so its log loss is not finite. The calibrated sign rule learns its direction on the training persons and outputs its training hit rate q (or $1-q$) as $P(\text{right})$. It makes the same decisions (78.9 % [73.5, 83.8]; median person 82.7 %). A fitted threshold (exploratory, 3 configurations) does not help: 77.6 % [72.2, 82.5]. The best single-feature rule, with feature, direction and threshold chosen inside every fold (exploratory, 3 configurations), reaches 80.8 % [76.1, 85.7]; it picks `latvel_q75` in 19 of 20 folds.

PHYS6: an antisymmetric logistic model. PHYS6 is a logistic model on the six chiral features, scaled by their RMS on the training persons, with no intercept and an L2 penalty whose strength is chosen on inner splits (half-decade grid from 0.0003 to 10). With no intercept and odd features only, the logit is odd: $P(\text{right})(m(\mathbf{x})) = 1 - P(\text{right})(\mathbf{x})$ for every trial, as eq. (2) requires. The model cannot learn a prior for one hand. It reaches 81.9 % [77.0, 87.1],

Table 33: Zero- and one-parameter rules, leave-one-person-out (95 % person-bootstrap CI). Parameters: fitted numbers per fold (the calibrated sign rule learns a direction and its hit rate; the threshold rule a direction, a threshold and the two side probabilities; the nested rule also chooses its feature inside every fold).

Rule	Accuracy	95% CI	Android	iOS	Median person	Natural use	k = 3	Parameters
first-half yaw > 0 (no training)	78.9%	[73.5, 83.8]	80.4%	76.3%	82.7%	85%	86.4%	0
first-half yaw sign, hit rate as P	78.9%	[73.5, 83.8]	80.4%	76.3%	82.7%	85%	86.4%	1 + 1
first-half yaw, fitted threshold	77.6%	[72.2, 82.5]	78.1%	76.7%	78.9%	85%	84.5%	2 + 2
best single-feature rule (in-fold choice)	80.8%	[76.1, 85.7]	79.1%	83.7%	80.7%	50%	88.1%	in-fold choice

Table 34: Per-person leave-one-person-out accuracy of the rules and of the small-model finalist.

Person	first-half yaw > 0 (no training)	first-half yaw sign, hit rate as P	first-half yaw, fitted threshold	best single-feature rule (in-fold choice)	PHYS6, data treatment chosen in-fold	Trials
P01	84%	84%	84%	76%	76%	45
P02	90%	90%	90%	87%	87%	31
P03	79%	79%	79%	96%	92%	24
P04	88%	88%	88%	100%	100%	8
P05	59%	59%	59%	59%	66%	32
P06	89%	89%	89%	96%	85%	27
P07	75%	75%	79%	75%	82%	28
P08	59%	59%	59%	81%	78%	27
P09	74%	74%	76%	62%	68%	34
P10	76%	76%	76%	94%	88%	34
P11	89%	89%	89%	89%	86%	28
P12	89%	89%	89%	89%	97%	38
P13	86%	86%	68%	64%	86%	28
P14	80%	80%	80%	78%	85%	40
P15	87%	87%	76%	76%	89%	38
P16	83%	83%	83%	96%	96%	24
P17	48%	48%	48%	80%	68%	25
P18	91%	91%	91%	88%	94%	32
P19	63%	63%	63%	78%	78%	27
P20	82%	82%	79%	75%	79%	28

AUC 0.882 [0.833, 0.928] and log loss 0.452 [0.374, 0.537]. Its coefficients are stable across folds (Table 35): per RMS unit, `yaw_q50` +0.85, `yaw_q100` +0.72, `latvel_q50` +0.45, `latvel_q100` +0.69, `roll_q50` −0.14 and `roll_q100` −0.33, each with the same sign in every fold. The small finalist, the same model with asinh-compressed features (Appendix C.5), has coefficients of the same signs (Table 36).

Table 35: Coefficients of the PHYS6 antisymmetric logistic model across the leave-one-person-out folds (per scaled unit of the feature: median, min, max; raw: per unit of the feature) and the share of folds that agree with the median sign.

feature	coef_std_median	coef_std_min	coef_std_max	coef_raw_median	unit	sign_agreement
yaw_q50	+0.85	+0.74	+1.01	+0.0752	deg	100%
yaw_q100	+0.72	+0.65	+0.84	+0.0608	deg	100%
roll_q50	−0.14	−0.27	−0.04	−0.0178	deg	100%
roll_q100	−0.33	−0.42	−0.20	−0.0379	deg	100%
latvel_q50	+0.45	+0.32	+0.62	+1.93	m/s	100%
latvel_q100	+0.69	+0.60	+0.90	+2.13	m/s	100%

Augmentation or antisymmetry. The alternative to the antisymmetric constraint is mirror augmentation [22]: train with an intercept on every trial and on its mirror image with the hand swapped. On odd features the two are the same model, with the L2 penalty halved because the rows are doubled; a unit test pins the equivalence at $C \to 2C$. In practice they share 99.8 % of the decisions (largest $|\Delta P(\text{right})|$ 0.061). The antisymmetric form was kept because it holds the symmetry exactly by construction, in the spirit of symmetry-constrained models [24].

Table 36: Coefficients of the small-model finalist (folds of its modal treatment) across the leave-one-person-out folds (per scaled unit of the feature: median, min, max; raw: per unit of the feature) and the share of folds that agree with the median sign.

feature	coef_std_median	coef_std_min	coef_std_max	coef_raw_median	unit	sign_agreement
yaw_q50	+0.82	+0.72	+0.93	+0.134	deg	100%
yaw_q100	+0.46	+0.40	+0.59	+0.0714	deg	100%
roll_q50	−0.11	−0.22	−0.01	−0.039	deg	100%
roll_q100	−0.36	−0.45	−0.27	−0.138	deg	100%
latvel_q50	+0.44	+0.34	+0.56	+3.8	m/s	100%
latvel_q100	+0.63	+0.58	+0.74	+3.38	m/s	100%

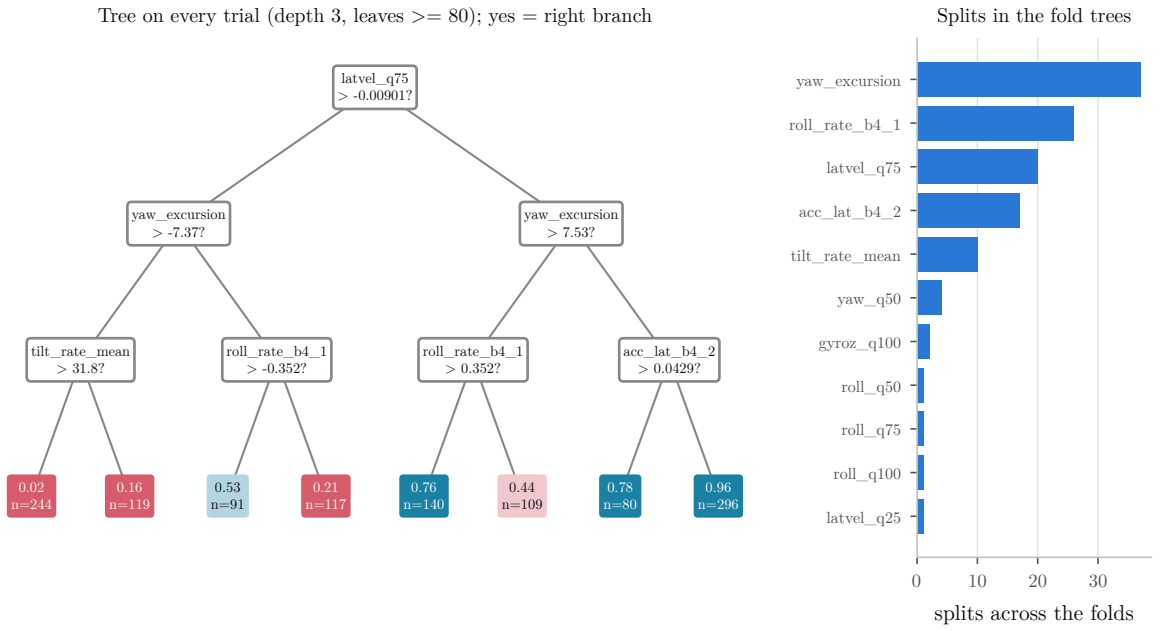


Figure 47: Left: the shallow tree with the hyperparameters chosen most often in the folds (10 of 20), refitted on every trial and its mirror image for illustration; a leaf shows P(right) and its number of (augmented) training rows. Right: how often each feature is split on across the fold trees. The first question is the same in every fold.

Temporal filter. The three-channel temporal filter of Figure 46 is, with its post-hoc calibrated variants, the most accurate model of the lane, but it is exploratory (33 configurations): 86.1 % [81.3, 90.9], AUC 0.907 [0.864, 0.945], log loss 0.431. Against PHYS6 it gains +4.2 [+0.7, +7.9] points, Holm $p = 1$ over the 27 options, and its log loss does not differ (−0.017 [−0.072, +0.039]). Its probabilities are under-confident (Appendix C.6).

Shallow tree. A tree of depth at most 3, mirror-augmented and symmetrised (exploratory, 6 configurations), reaches 79.7 % [75.4, 84.1], log loss 0.437 (Figure 47). Its first question is `latvel_q75` in 20 of 20 folds, and it splits almost only on chiral features; the one even feature it uses is the mean tilt rate. Its depth sits at the cap of its grid in 16 folds; the cap is a design choice that keeps the tree readable. Against PHYS6: −2.3 [−5.6, +1.1] points, log loss −0.027 [−0.078, +0.023], Holm $p = 1$.

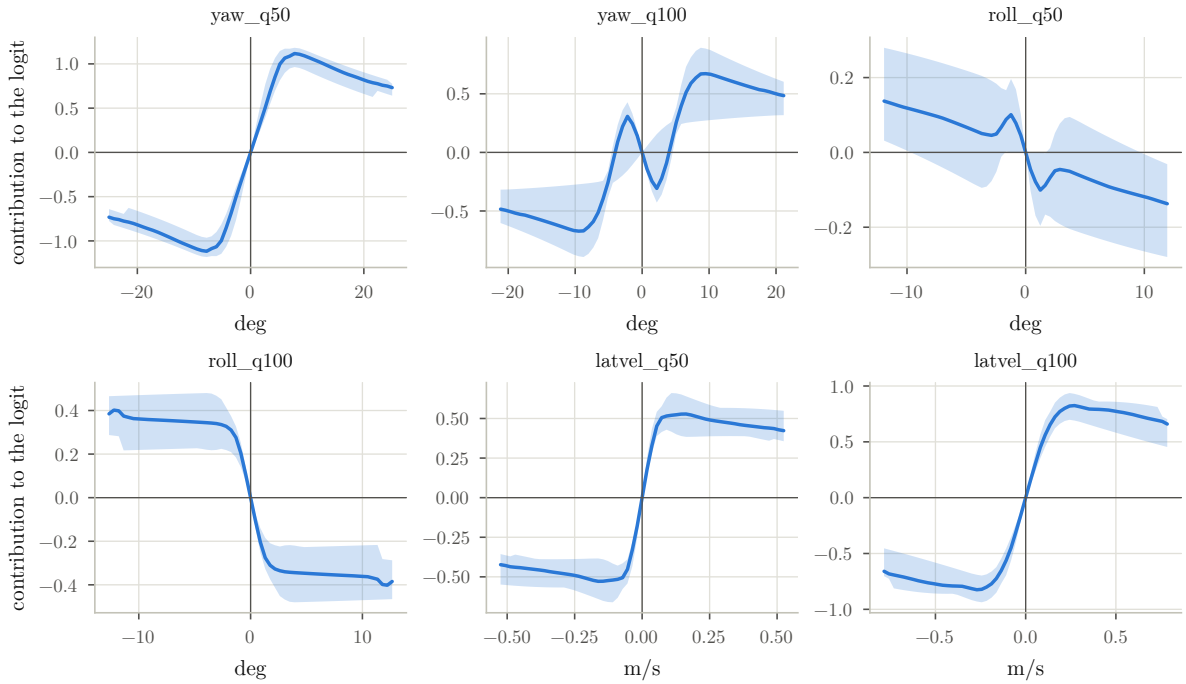


Figure 48: Shape functions of the GAM on PHYS6: median over the 20 folds and range across folds, in the units of each feature; each shape is odd. The evidence saturates beyond the bulk of the trials.

Generalised additive model. A generalised additive model (GAM) on PHYS6 replaces each linear term by a cubic spline (mirror-augmented and symmetrised; number of knots and L2 strength chosen in-fold: 7 knots in 7 folds, 9 in 13). It is exploratory (6 configurations): 81.6 % [77.1, 85.9], log loss 0.426. Its shapes are steepest around zero (Figure 48): the slope over the outer fifths of the range is smaller than the central slope for 6 of 6 features and reverses its sign for 4. Beyond the bulk of the trials, a larger value adds no further evidence. This is the saturation that the asinh treatment imposes on the linear model. Against PHYS6 the GAM's log loss does not differ after the correction (−0.030 [−0.089, +0.019], Holm $p = 1$).

What fitted numbers add to the rule. 5 fitted models were paired against the calibrated sign rule, with the Holm adjustment over these contrasts. PHYS6 gains +3.0 [−0.6, +7.1] points (Holm $p = 0.12$) and lowers the log loss by −0.078 [−0.148, +0.002] (Holm $p = 0.11$): neither survives. Three gains survive the correction. The small finalist lowers the log loss by −0.108 [−0.173, −0.042] (Holm $p = 0.024$); its accuracy gain of +4.3 [+1.0, +8.1] points does not survive (Holm $p = 0.05$, just above the 0.05 level). The GAM lowers the log loss by −0.108 [−0.165, −0.045] (Holm $p = 0.019$). The temporal filter gains +7.2 [+2.6, +12.1] points of accuracy (Holm $p = 0.018$). Against the pre-specified reference, a logistic model on the yaw at 8 instants (77.9 % [73.1, 82.9]), 8 of the 16 models of Figure 49 are more accurate beyond the unadjusted paired CI, and 1 after the Holm correction over the 54 lane models paired with it. The small finalist gains +5.4 [+2.6, +8.3] points over the reference (Holm $p = 0.09$).

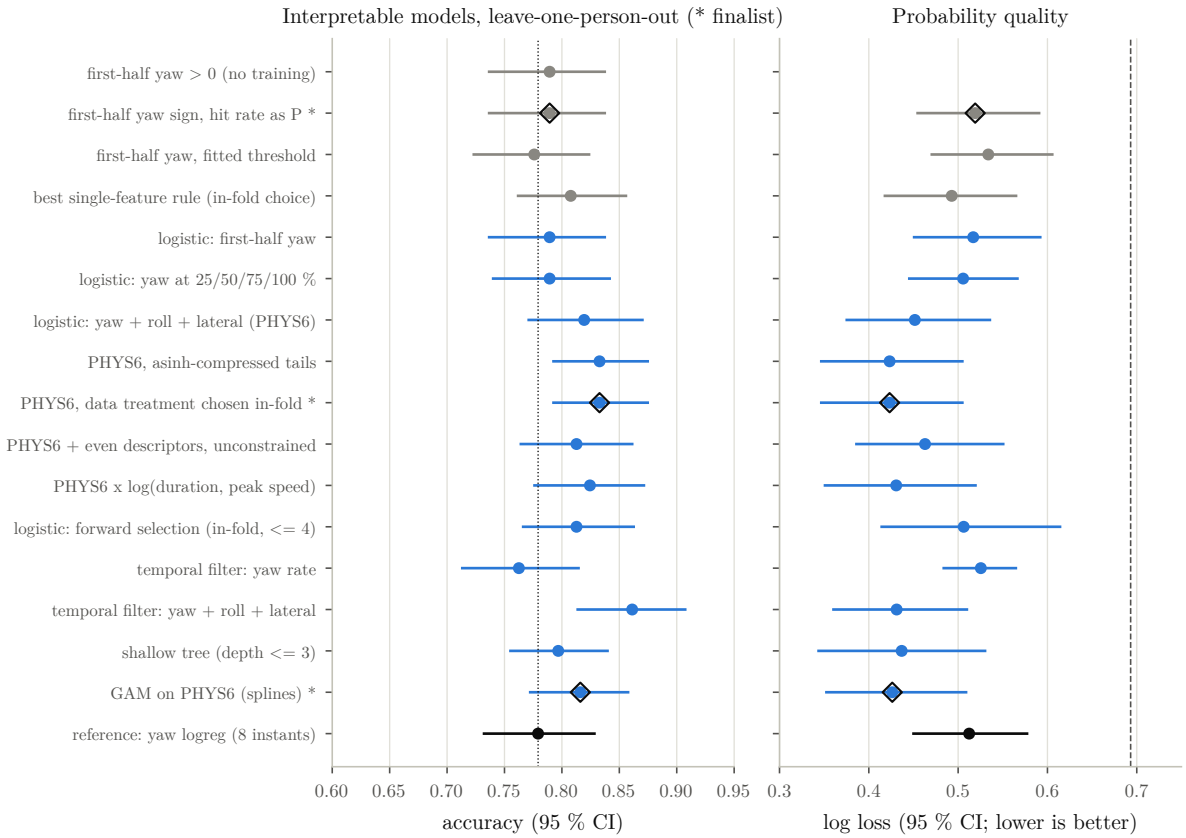


Figure 49: Accuracy and log loss (95 % CI) of the interpretable models, LOSO (grey: rules; blue: fitted models; black: the pre-specified reference, dotted line). Diamonds mark the finalists; dashed: the log loss of a constant 0.5. The zero-parameter rule has no finite log loss.

Table 37: Every lane m1 model, leave-one-person-out over the primary set (95 % person-bootstrap CI). ECE with the 95th percentile of its perfect-calibration null; Slope: calibration slope (1 = calibrated). Natural use: accuracy on the free-choice trials (never trained on; a P of exactly 0.5 counts one half); k = 3: per-person balanced accuracy when three attempts are averaged; vs ref.: paired accuracy difference to the reference floor in percentage points with the Holm-adjusted sign-flip p over every m1 model; Configs: configurations of its family compared on the outer folds (1 = not chosen on the outer folds). Design: pre-specified, specified after the pooled description, nested (choice inside every fold), outer-fold family member (exploratory) or sensitivity run.

Model	Design	Accuracy	95% CI	AUC	Log loss	ECE	ECE null 95%	Slope	Natural use	k = 3	vs ref. (pts)	Holm p	Configs
first-half yaw > 0 (no training)	pre-specified	78.9%	[73.5, 83.8]	0.790	–	–	–	–	85%	86.4%	+1.0	1	1
first-half yaw sign, hit rate as P	pre-specified	78.9%	[73.5, 83.8]	0.729	0.519	0.097	0.059	1.01	85%	86.4%	+1.0	1	1
first-half yaw, fitted threshold	family	77.6%	[72.2, 82.5]	0.737	0.534	0.081	0.060	0.93	85%	84.5%	−0.3	1	3
best single-feature rule (in-fold choice)	family	80.8%	[76.1, 85.7]	0.754	0.493	0.098	0.056	0.93	50%	88.1%	+2.8	1	3
logistic: first-half yaw	family	78.9%	[73.5, 83.8]	0.852	0.517	0.092	0.060	1.05	85%	86.5%	+1.0	1	33
logistic: yaw at 25/50/75/100 %	family	78.9%	[73.9, 84.3]	0.859	0.506	0.083	0.059	1.06	75%	87.2%	+1.0	1	33
logistic: yaw + roll + lateral (PHYS6)	post-pooled	81.9%	[77.0, 87.1]	0.882	0.452	0.073	0.056	1.07	80%	88.4%	+4.0	0.47	1
PHYS6, intercept + mirror augmentation	family	82.1%	[77.3, 87.3]	0.881	0.457	0.079	0.056	1.07	80%	88.6%	+4.2	0.28	33
PHYS6 + even descriptors, unconstrained	family	81.3%	[76.3, 86.2]	0.877	0.463	0.072	0.055	1.03	75%	87.6%	+3.3	0.97	33
PHYS6 x log(duration, peak speed)	family	82.4%	[77.5, 87.3]	0.890	0.431	0.065	0.053	1.07	75%	90.1%	+4.5	0.28	33
logistic: all approach chiral features	family	84.1%	[79.0, 89.3]	0.888	0.451	0.096	0.055	1.04	80%	90.6%	+6.2	0.029	33
logistic: forward selection (in-fold, <= 4)	family	81.3%	[76.5, 86.4]	0.862	0.506	0.071	0.053	0.78	80%	88.6%	+3.3	1	33
yaw only	family	78.9%	[74.0, 84.1]	0.858	0.509	0.085	0.059	1.04	80%	87.3%	+1.0	1	33
roll only	family	59.4%	[51.6, 67.4]	0.584	0.697	0.080	0.072	−0.27	60%	62.0%	−18.6	0.017	33
lateral only	family	80.9%	[76.9, 85.1]	0.861	0.517	0.109	0.059	1.01	55%	88.7%	+3.0	1	33
yaw + roll	family	80.4%	[75.0, 86.4]	0.856	0.499	0.084	0.059	1.02	75%	86.6%	+2.5	1	33
yaw + lateral	family	81.8%	[77.2, 86.6]	0.879	0.459	0.067	0.056	1.08	75%	88.7%	+3.8	0.61	33
channel set chosen in-fold	nested	81.6%	[77.0, 86.7]	0.878	0.458	0.066	0.055	1.05	80%	88.4%	+3.7	0.5	1
PHYS6 at 0.5 / 1.0 s (absolute time)	analysis	80.8%	[76.3, 85.0]	0.876	0.475	0.082	0.057	1.03	70%	90.2%	+2.8	1	1
lead-in before the onset	analysis	63.5%	[59.7, 67.5]	0.653	0.669	0.079	0.071	1.01	55%	65.9%	−14.4	0.0011	1
negative control: still hold before the onset	analysis	58.9%	[54.5, 63.5]	0.627	0.681	0.046	0.072	0.88	60%	61.5%	−19.1	0.0015	1
no scaling (raw units)	family	81.8%	[76.6, 87.3]	0.877	0.462	0.068	0.053	0.91	80%	88.5%	+3.8	0.5	33
robust scaling (median |x|)	family	82.1%	[77.6, 86.8]	0.881	0.453	0.080	0.056	1.10	80%	88.6%	+4.2	0.22	33
winsorised at the training 99th percentile	family	82.6%	[77.9, 87.6]	0.881	0.452	0.073	0.055	1.04	80%	88.7%	+4.7	0.11	33
asinh-compressed tails	family	83.3%	[79.2, 87.6]	0.890	0.423	0.047	0.053	1.04	75%	90.4%	+5.4	0.09	33
per-trial amplitude-normalised	family	82.3%	[78.0, 86.9]	0.886	0.432	0.055	0.053	1.06	75%	90.0%	+4.3	0.92	33
lateral de-drifted (hold bias removed)	family	82.6%	[77.2, 87.8]	0.879	0.456	0.085	0.055	1.06	80%	88.5%	+4.7	0.84	33
PHYS6, data treatment chosen in-fold	nested	83.3%	[79.2, 87.6]	0.890	0.423	0.047	0.053	1.04	75%	90.4%	+5.4	0.09	1
temporal filter: yaw rate	family	76.3%	[71.2, 81.6]	0.838	0.525	0.073	0.061	1.08	75%	85.6%	−1.7	1	33
temporal filter: yaw + roll + lateral	family	86.1%	[81.3, 90.9]	0.907	0.431	0.110	0.055	1.14	70%	93.5%	+8.2	0.024	33
shallow tree (depth <= 3)	family	79.7%	[75.4, 84.1]	0.878	0.437	0.034	0.049	0.88	68%	89.7%	+1.8	1	6
GAM on PHYS6 (splines)	family	81.6%	[77.1, 85.9]	0.886	0.426	0.028	0.052	1.00	75%	89.8%	+3.7	1	6
PHYS6 logistic + Platt	family	81.4%	[76.4, 86.5]	0.880	0.456	0.063	0.054	0.94	80%	88.2%	+3.5	0.84	33
PHYS6 logistic + temperature	family	81.9%	[77.0, 87.1]	0.881	0.454	0.066	0.053	0.95	80%	88.5%	+4.0	0.47	33
PHYS6 logistic + isotonic	family	81.8%	[77.1, 86.5]	0.868	0.460	0.047	0.049	0.82	78%	88.2%	+3.8	0.35	33
PHYS6 logistic + beta	family	80.9%	[75.7, 86.2]	0.878	0.461	0.067	0.053	0.93	80%	88.1%	+3.0	1	33
PHYS6 nested treatment + Platt	family	82.9%	[78.7, 87.3]	0.888	0.427	0.038	0.052	0.95	75%	90.0%	+5.0	0.11	33
PHYS6 nested treatment + temperature	family	83.3%	[79.2, 87.6]	0.889	0.425	0.045	0.052	0.96	75%	90.3%	+5.4	0.09	33
PHYS6 nested treatment + isotonic	family	81.9%	[77.9, 86.3]	0.878	0.458	0.038	0.049	0.80	75%	89.0%	+4.0	0.62	33
PHYS6 nested treatment + beta	family	82.4%	[78.1, 86.8]	0.886	0.432	0.043	0.051	0.93	75%	89.9%	+4.5	0.22	33
temporal filter + Platt	family	86.5%	[81.7, 90.9]	0.905	0.438	0.103	0.051	0.92	70%	93.7%	+8.5	0.012	33
temporal filter + temperature	family	86.1%	[81.3, 90.9]	0.906	0.437	0.099	0.052	0.93	70%	93.8%	+8.2	0.024	33
temporal filter + isotonic	family	84.9%	[80.6, 89.4]	0.895	0.383	0.047	0.046	0.90	70%	93.5%	+7.0	0.06	33
temporal filter + beta	family	86.5%	[81.8, 91.0]	0.905	0.457	0.098	0.051	0.85	70%	93.6%	+8.5	0.0063	33
GAM + Platt	family	81.1%	[76.6, 85.5]	0.884	0.431	0.024	0.051	0.92	75%	89.5%	+3.2	1	6
GAM + temperature	family	81.6%	[77.1, 85.9]	0.886	0.428	0.024	0.050	0.93	75%	89.7%	+3.7	1	6
GAM + isotonic	family	81.5%	[77.0, 86.1]	0.875	0.450	0.040	0.049	0.83	70%	89.4%	+3.6	1	6
GAM + beta	family	81.3%	[76.9, 85.5]	0.883	0.434	0.037	0.051	0.91	75%	89.3%	+3.3	1	6
first-half yaw sign, hit rate as P, swap-signature blocks out of training	sensitivity	78.9%	[73.5, 83.8]	0.735	0.519	0.083	0.059	0.96	85%	86.4%	+1.0	1	1
first-half yaw sign, hit rate as P, D6 screen in every fold	sensitivity	78.9%	[73.5, 83.8]	0.731	0.520	0.093	0.057	0.94	85%	86.4%	+1.0	1	1
PHYS6, data treatment chosen in-fold, swap-signature blocks out of training	sensitivity	83.4%	[79.3, 87.8]	0.890	0.423	0.046	0.053	1.01	75%	90.4%	+5.5	0.035	1
PHYS6, data treatment chosen in-fold, D6 screen in every fold	sensitivity	81.9%	[77.4, 86.7]	0.888	0.426	0.025	0.051	0.94	80%	89.6%	+4.0	0.21	1
GAM on PHYS6 (splines), swap-signature blocks out of training	sensitivity	81.1%	[76.7, 85.3]	0.880	0.439	0.025	0.051	0.91	75%	89.4%	+3.2	1	1
GAM on PHYS6 (splines), D6 screen in every fold	sensitivity	82.4%	[77.8, 87.0]	0.879	0.448	0.040	0.051	0.87	80%	89.4%	+4.5	0.47	1
reference: yaw logreg (8 instants)	pre-specified	77.9%	[73.1, 82.9]	0.855	0.512	0.072	0.056	0.88	75%	87.2%	–	–	1

C.5. Data treatment and model form

Options: 27 variants of PHYS6, each paired against it on the same trials (Figure 50, Table 38). Scaling: none, robust (median $|x|$) or RMS. Tails: winsorised at the training 99th percentile, or compressed by asinh. Per-trial amplitude normalisation. The de-drifted lateral velocity. The treatment chosen inside every fold. Model form: mirror augmentation with an intercept; an unconstrained model with an intercept and even descriptors; odd-by-even interactions with log duration and log peak speed. More features, in-fold selections, splines and the temporal filters complete the family.

Evidence: no option lowers the log loss of PHYS6 after the Holm correction over the 27 options. Asinh compression lowers it beyond the unadjusted paired CI (-0.030 [-0.058, -0.005], $p = 0.034$, Holm $p = 0.71$). Chosen inside every fold, the treatment is asinh in 20 of 20 folds, so the nested model and the asinh model make identical predictions. Worse after the correction: roll only, the lead-in and the still hold. The unconstrained model has a higher log loss than PHYS6 ($+0.011$ [$+0.003$, $+0.021$], unadjusted CI above 0, Holm $p = 0.52$).

Decision: the small model keeps the antisymmetric form, and its data treatment is chosen inside every fold (in practice asinh). The rationale is calibration rather than accuracy: the tail compression is the one remedy that removes the miscalibration of PHYS6 (Appendix C.6). The unconstrained form was rejected because it may learn asymmetries of this mostly right-handed sample that would not transfer to a left-handed user.

Hyperparameter grids. 11 pairs of an experiment and a hyperparameter chose an edge of their grid in at least one fold. Only the tree's depth (16 folds) and its minimum leaf size (13 folds) sit at an edge in most folds.

Table 38: Data treatment, representation and model form, each paired against the PHYS6 antisymmetric logistic model on the same trials (pooled accuracy difference in percentage points and mean per-person log-loss difference, 95 % paired person-bootstrap CI, exact sign-flip p; Holm over the 27 options of this family).

Option	Design	Accuracy	dAcc (pts)	dAcc CI	Holm p (acc)	Log loss	dLL	dLL CI	p (LL)	Holm p (LL)
logistic: first-half yaw	family	78.9%	−3.0	[−7.1, 0.6]	1	0.517	+0.065	[−0.010, 0.130]	0.098	1
logistic: yaw at 25/50/75/100 %	family	78.9%	−3.0	[−5.5, −0.6]	0.62	0.506	+0.054	[−0.014, 0.108]	0.12	1
PHYS6, intercept + mirror augmentation	family	82.1%	+0.2	[0.0, 0.5]	1	0.457	+0.006	[−0.000, 0.014]	0.089	1
PHYS6 + even descriptors, unconstrained	family	81.3%	−0.7	[−2.2, 0.7]	1	0.463	+0.011	[0.003, 0.021]	0.023	0.52
PHYS6 x log(duration, peak speed)	family	82.4%	+0.5	[−1.4, 2.3]	1	0.431	−0.022	[−0.043, 0.001]	0.079	1
logistic: all approach chiral features	family	84.1%	+2.2	[0.0, 4.7]	1	0.451	−0.002	[−0.029, 0.025]	0.91	1
logistic: forward selection (in-fold, <= 4)	family	81.3%	−0.7	[−3.5, 2.3]	1	0.506	+0.053	[0.012, 0.101]	0.028	0.62
yaw only	family	78.9%	−3.0	[−5.8, −0.7]	0.5	0.509	+0.059	[−0.009, 0.115]	0.094	1
roll only	family	59.4%	−22.6	[−31.0, −14.1]	0.0017	0.697	+0.251	[0.166, 0.330]	7.4e-05	0.002
lateral only	family	80.9%	−1.0	[−4.1, 1.7]	1	0.517	+0.063	[0.027, 0.102]	0.0027	0.065
yaw + roll	family	80.4%	−1.5	[−4.6, 1.5]	1	0.499	+0.052	[0.002, 0.093]	0.044	0.79
yaw + lateral	family	81.8%	−0.2	[−1.9, 1.7]	1	0.459	+0.005	[−0.017, 0.030]	0.7	1
channel set chosen in-fold	nested	81.6%	−0.3	[−1.4, 0.5]	1	0.458	+0.005	[−0.010, 0.026]	1	1
PHYS6 at 0.5 / 1.0 s (absolute time)	analysis	80.8%	−1.2	[−4.5, 2.0]	1	0.475	+0.024	[−0.035, 0.086]	0.46	1
lead-in before the onset	analysis	63.5%	−18.4	[−23.4, −14.0]	0.0002	0.669	+0.226	[0.142, 0.304]	0.00016	0.0039
negative control: still hold before the onset	analysis	58.9%	−23.1	[−28.4, −18.2]	5.1e-05	0.681	+0.234	[0.149, 0.311]	0.00011	0.0028
no scaling (raw units)	family	81.8%	−0.2	[−1.4, 1.1]	1	0.462	+0.011	[−0.001, 0.024]	0.11	1
robust scaling (median |x|)	family	82.1%	+0.2	[−0.5, 0.9]	1	0.453	+0.002	[−0.006, 0.010]	0.6	1
winsorised at the training 99th percentile	family	82.6%	+0.7	[0.0, 1.4]	1	0.452	−0.000	[−0.004, 0.004]	1	1
asinh-compressed tails	family	83.3%	+1.3	[−0.4, 3.2]	1	0.423	−0.030	[−0.058, −0.005]	0.034	0.71
per-trial amplitude-normalised	family	82.3%	+0.3	[−1.7, 2.8]	1	0.432	−0.017	[−0.063, 0.021]	0.47	1
lateral de-drifted (hold bias removed)	family	82.6%	+0.7	[−2.4, 3.2]	1	0.456	+0.008	[−0.013, 0.031]	0.5	1
PHYS6, data treatment chosen in-fold	nested	83.3%	+1.3	[−0.4, 3.2]	1	0.423	−0.030	[−0.058, −0.005]	0.034	0.71
temporal filter: yaw rate	family	76.3%	−5.7	[−9.4, −1.9]	0.16	0.525	+0.079	[0.012, 0.143]	0.034	0.71
temporal filter: yaw + roll + lateral	family	86.1%	+4.2	[0.7, 7.9]	1	0.431	−0.017	[−0.072, 0.039]	0.57	1
shallow tree (depth <= 3)	family	79.7%	−2.3	[−5.6, 1.1]	1	0.437	−0.027	[−0.078, 0.023]	0.32	1
GAM on PHYS6 (splines)	family	81.6%	−0.3	[−2.7, 2.2]	1	0.426	−0.030	[−0.089, 0.019]	0.31	1

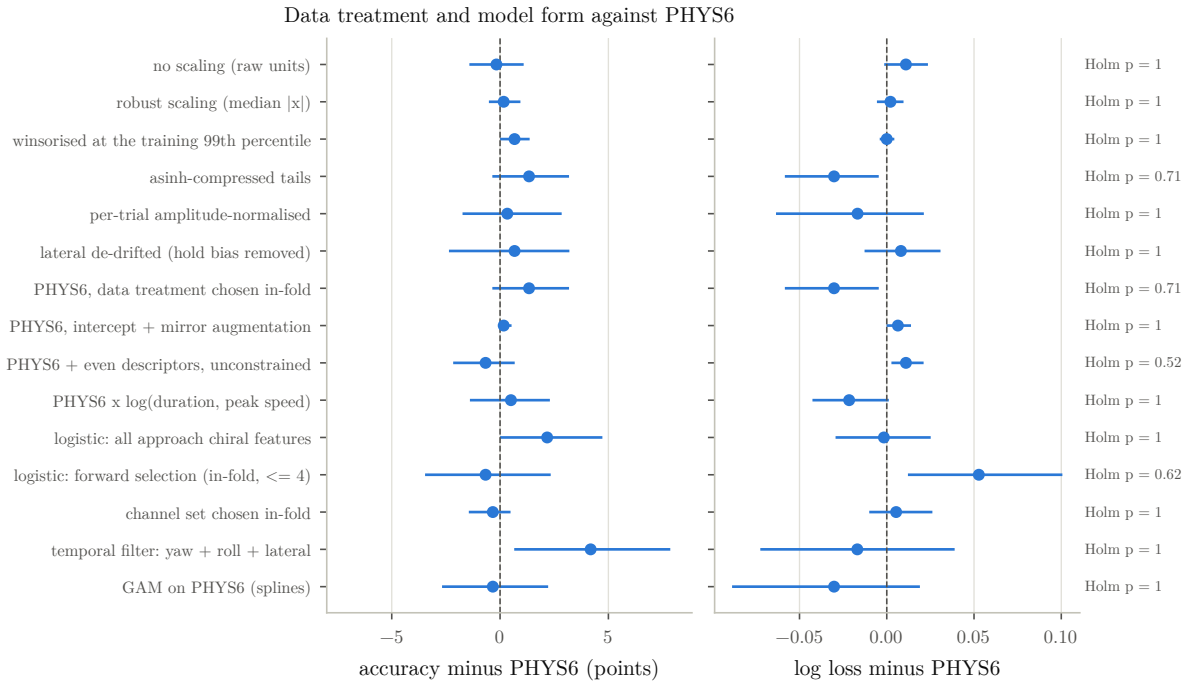


Figure 50: Data treatments and model forms against PHYS6 on the same trials: accuracy and log-loss differences (paired person-bootstrap 95 % CI); text: Holm-adjusted p of the per-person log-loss difference over the 27 options of the family. The component sets and windows of the family are in Figure 45; all options are in Table 38. No option lowers the log loss after the correction.

C.6. Finalists and their calibration

Selection. The lane hands three finalists to the synthesis, one per slot, under fixed criteria. A slot is judged by the log loss, because the product is a probability, and a replacement of the slot's anchor must survive the Holm correction over its family. *Rule:* the calibrated sign rule (pre-specified, 1 configuration). *Small model:* PHYS6 with its data treatment chosen in-fold (nested, 1 configuration); no member of the small-model family beats it in log loss after the correction. *Tree or GAM:* the GAM, which has the lowest LOSO log loss of the 6 tree and GAM configurations; this is an outer-fold choice, so the GAM is exploratory. The selection-aware result of the lane is the small finalist: 83.3 % [79.2, 87.6], AUC 0.890 [0.843, 0.934], log loss 0.423 [0.345, 0.506], Brier score 0.131 [19]. The synthesis re-selects across lanes in a nested loop (Appendix F).

Detection: PHYS6 is miscalibrated. PHYS6 has an ECE of 0.073, above the 95th percentile of its perfect-calibration null (0.056; $p = 0.0015$). 5 of its 10 reliability bins lie outside their person-bootstrap band (4 under-confident, 1 over-confident; binomial $p < 0.001$), while its calibration slope, 1.07 [0.76, 1.57], is compatible with 1 (Figure 51, left). This is an S-shaped error: the bins miss the diagonal on both sides, so no single global map of the logit can fix it. The temperature fitted on inner predictions is below 1 in 20 of 20 folds. *Why it matters:* the

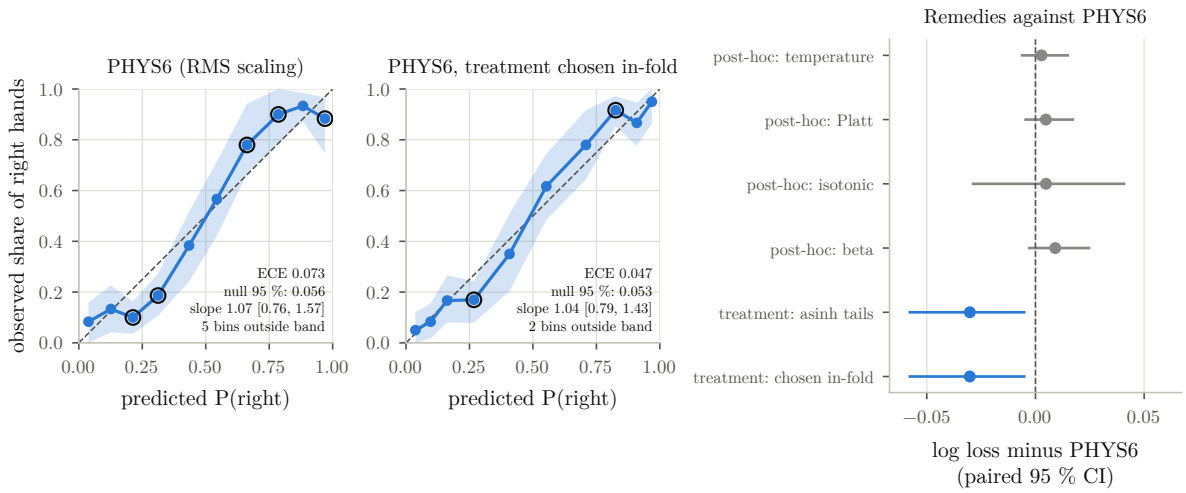


Figure 51: Calibration of PHYS6 and its remedies. Left, middle: reliability diagrams (10 equal-mass bins; band: person-bootstrap 95 % CI of the observed share; circled: bins whose band misses the diagonal) of PHYS6 and of the small finalist. Right: the log-loss change of each remedy against PHYS6 (paired 95 % CI). Post-hoc maps do not help; the asinh treatment does.

product shows $P(\text{right})$ as a percentage; a miscalibrated model misleads the user even when its decision is right.

Remedies. Four post-hoc maps were fitted on the inner cross-fitted predictions of the training persons (Table 39): Platt scaling [21], temperature scaling [12], isotonic regression and beta calibration. None lowers the log loss of PHYS6: Platt +0.005 [−0.005, +0.018], temperature +0.003 [−0.007, +0.015], isotonic +0.005 [−0.029, +0.041] and beta +0.009 [−0.004, +0.025], each with Holm $p = 1$ over the four. The asinh treatment is the effective remedy. It lowers the log loss by −0.030 [−0.058, −0.005] (beyond the unadjusted CI; Holm $p = 0.71$ over the 27 options) and leaves no miscalibration beyond chance: ECE 0.047 (null 95th percentile 0.053, $p = 0.14$), slope 1.04 [0.79, 1.43], 2 bins outside their band (binomial $p = 0.086$; Figure 51, middle). On the small finalist itself no post-hoc map lowers the log loss: Platt +0.003 [−0.002, +0.010] (Holm $p = 0.62$), temperature +0.001 [−0.004, +0.008] (Holm $p = 0.71$), isotonic +0.033 [+0.001, +0.077] (Holm $p = 0.27$) and beta +0.008 [−0.001, +0.019] (Holm $p = 0.4$). *Decision:* no calibration step is shipped on top of the small finalist. Every calibration result of the lane, with the calibration intercepts, is in Table 40.

The calibration of each finalist. Figure 52 shows the three finalists. The small finalist has an ECE of 0.047 (null 95th percentile 0.053, $p = 0.14$) and a slope of 1.04 [0.79, 1.43]: consistent with calibration. Per platform, its ECE also stays below the 95th percentile of its null: Android 0.053 (null 0.066, $p = 0.27$), iOS 0.070 (null 0.085, $p = 0.24$); the lane computed no per-platform slope. The GAM has an ECE of 0.028 (null 0.052, $p = 0.84$), a slope of 1.00 [0.77, 1.30] and 0 bins outside their band. The rule's ECE of 0.097 (null 0.059, $p < 0.001$) and its AUC of 0.729 are distorted and do not describe it. Its $P(\text{right})$ takes two values per fold (q or $1 - q$), so equal-mass bins and the ranking of the trials mostly sort the folds by q; in LOSO, q is lower when the held-out person is easy. Judged by its two decisions, the rule is consistent with

Table 39: Calibration remedies and treatments, paired against the uncalibrated model on the same trials: per-person log-loss and Brier differences with paired person-bootstrap 95 % CIs and sign-flip p (Holm within the remedies of one model), and the pooled ECE difference (point only).

Model	Against	Kind	dLog loss	dLL CI	p	Holm p	dBrier	dBrier CI	dECE
PHYS6 logistic + Platt	logistic: yaw + roll + lateral (PHYS6)	remedy	+0.005	[−0.005, 0.018]	0.53	1	−0.0007	[−0.002, 0.001]	−0.010
PHYS6 logistic + temperature	logistic: yaw + roll + lateral (PHYS6)	remedy	+0.003	[−0.007, 0.015]	0.76	1	−0.0015	[−0.003, −0.000]	−0.007
PHYS6 logistic + isotonic	logistic: yaw + roll + lateral (PHYS6)	remedy	+0.005	[−0.029, 0.041]	0.81	1	−0.0032	[−0.011, 0.004]	−0.026
PHYS6 logistic + beta	logistic: yaw + roll + lateral (PHYS6)	remedy	+0.009	[−0.004, 0.025]	0.28	1	+0.0005	[−0.002, 0.003]	−0.007
PHYS6 nested treatment + Platt	PHYS6, data treatment chosen in-fold	remedy	+0.003	[−0.002, 0.010]	0.31	0.62	+0.0005	[−0.001, 0.002]	−0.009
PHYS6 nested treatment + temperature	PHYS6, data treatment chosen in-fold	remedy	+0.001	[−0.004, 0.008]	0.71	0.71	−0.0004	[−0.001, 0.000]	−0.002
PHYS6 nested treatment + isotonic	PHYS6, data treatment chosen in-fold	remedy	+0.033	[0.001, 0.077]	0.068	0.27	+0.0040	[−0.001, 0.009]	−0.009
PHYS6 nested treatment + beta	PHYS6, data treatment chosen in-fold	remedy	+0.008	[−0.001, 0.019]	0.13	0.4	+0.0015	[−0.000, 0.004]	−0.004
temporal filter + Platt	temporal filter: yaw + roll + lateral	remedy	+0.006	[−0.011, 0.028]	0.61	1	−0.0037	[−0.006, −0.001]	−0.006
temporal filter + temperature	temporal filter: yaw + roll + lateral	remedy	+0.005	[−0.012, 0.026]	0.7	1	−0.0041	[−0.006, −0.002]	−0.010
temporal filter + isotonic	temporal filter: yaw + roll + lateral	remedy	−0.053	[−0.084, −0.021]	0.0055	0.022	−0.0154	[−0.028, −0.003]	−0.062
temporal filter + beta	temporal filter: yaw + roll + lateral	remedy	+0.024	[−0.003, 0.056]	0.14	0.42	−0.0038	[−0.006, −0.002]	−0.012
GAM + Platt	GAM on PHYS6 (splines)	remedy	+0.004	[−0.001, 0.010]	0.18	0.37	+0.0008	[−0.000, 0.002]	−0.004
GAM + temperature	GAM on PHYS6 (splines)	remedy	+0.001	[−0.003, 0.007]	0.69	0.69	−0.0001	[−0.001, 0.001]	−0.004
GAM + isotonic	GAM on PHYS6 (splines)	remedy	+0.022	[0.002, 0.046]	0.056	0.18	+0.0019	[−0.001, 0.005]	+0.013
GAM + beta	GAM on PHYS6 (splines)	remedy	+0.008	[0.001, 0.015]	0.044	0.18	+0.0015	[0.000, 0.003]	+0.009
asinh-compressed tails	logistic: yaw + roll + lateral (PHYS6)	treatment	−0.030	[−0.058, −0.005]	0.034	0.068	−0.0083	[−0.015, −0.002]	−0.026
PHYS6, data treatment chosen in-fold	logistic: yaw + roll + lateral (PHYS6)	treatment	−0.030	[−0.058, −0.005]	0.034	0.068	−0.0083	[−0.015, −0.002]	−0.026

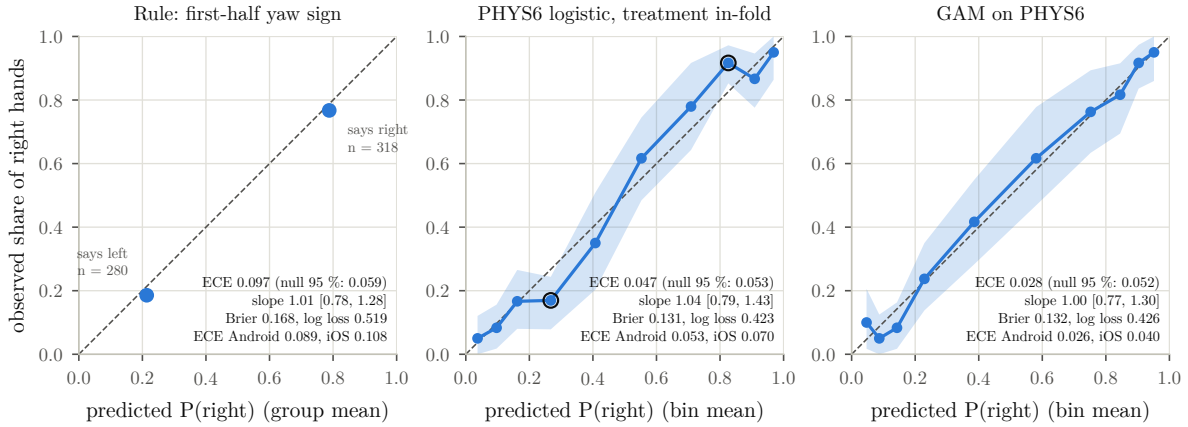


Figure 52: Reliability of the three finalists (LOSO probabilities; band: person-bootstrap 95 % CI of the observed share of right hands; dashed: perfect calibration). The rule's panel shows its two decision groups instead of bins. The small finalist and the GAM are consistent with calibration.

calibration (slope 1.01 [0.78, 1.28]; the intercept CI was not saved): when it says right, its mean $P(\text{right})$ is 0.79 against an observed share of right hands of 0.77 (318 trials); when it says left, 0.21 against 0.19 (280 trials). Its slope is 1.01 [0.78, 1.28].

The temporal filter. The uncalibrated temporal filter is under-confident: ECE 0.110 (null 0.055, $p < 0.001$), slope 1.14 [0.79, 1.83], 4 of 10 bins outside their band, all under-confident. Isotonic regression fitted on inner predictions lowers its log loss by −0.053 [−0.084, −0.021] (Holm $p = 0.022$ over its four remedies), to 0.383, with an ECE of 0.047, still just above its null band (null 0.046, $p = 0.04$), and a slope of 0.90 [0.69, 1.19]. The filter itself is exploratory (33 configurations) and does not beat the small finalist after the correction (+2.8 [+0.0, +5.7] points, Holm $p = 1$; log loss +0.013 [−0.039, +0.066]). It is reported as an exploratory alternative for the synthesis.

Table 40: Calibration of the lane m1 models and remedies (leave-one-person-out probabilities). ECE: 10 equal-mass bins, with the 95th percentile and the p value of its perfect-calibration null (labels redrawn from the model's own P); Slope / Intercept: the logistic recalibration of the outcome on logit(P) (slope > 1: under-confident), slope with a 95 % person-bootstrap CI; Bins out: reliability bins whose person-bootstrap band excludes the diagonal; Bins p: binomial tail of that count against the bands' nominal 5 % miss rate. Every calibrator is fitted on the inner cross-fitted predictions of the training persons.

Model	ECE	Null 95%	p	Slope	Slope CI	Intercept	Bins out	Bins p	Brier	Log loss	ECE Android	ECE iOS
logistic: yaw + roll + lateral (PHYS6)	0.073	0.056	0.0015	1.07	[0.757, 1.570]	−0.01	5	6.4e-05	0.139	0.452	0.080	0.070
PHYS6, data treatment chosen in-fold	0.047	0.053	0.14	1.04	[0.785, 1.435]	+0.01	2	0.086	0.131	0.423	0.053	0.070
temporal filter: yaw + roll + lateral	0.110	0.055	0.0005	1.14	[0.786, 1.827]	+0.00	4	0.001	0.126	0.431	0.120	0.099
GAM on PHYS6 (splines)	0.028	0.052	0.84	1.00	[0.769, 1.297]	+0.02	0	1	0.132	0.426	0.026	0.040
PHYS6 logistic + Platt	0.063	0.054	0.0055	0.94	[0.657, 1.401]	+0.00	2	0.086	0.138	0.456	0.072	0.063
PHYS6 logistic + temperature	0.066	0.053	0.0025	0.95	[0.663, 1.426]	−0.01	2	0.086	0.137	0.454	0.072	0.073
PHYS6 logistic + isotonic	0.047	0.049	0.076	0.82	[0.588, 1.159]	−0.00	0	1	0.137	0.460	0.048	0.074
PHYS6 logistic + beta	0.067	0.053	0.005	0.93	[0.641, 1.392]	+0.01	3	0.012	0.139	0.461	0.074	0.069
PHYS6 nested treatment + Platt	0.038	0.052	0.38	0.95	[0.718, 1.321]	+0.00	1	0.4	0.131	0.427	0.047	0.067
PHYS6 nested treatment + temperature	0.045	0.052	0.15	0.96	[0.725, 1.338]	+0.01	1	0.4	0.130	0.425	0.048	0.067
PHYS6 nested treatment + isotonic	0.038	0.049	0.31	0.80	[0.582, 1.126]	+0.01	1	0.4	0.136	0.458	0.046	0.061
PHYS6 nested treatment + beta	0.043	0.051	0.2	0.93	[0.696, 1.303]	+0.00	1	0.4	0.132	0.432	0.046	0.070
temporal filter + Platt	0.103	0.051	0.0005	0.92	[0.620, 1.520]	+0.01	5	6.4e-05	0.122	0.438	0.113	0.098
temporal filter + temperature	0.099	0.052	0.0005	0.93	[0.622, 1.531]	+0.00	4	0.001	0.122	0.437	0.113	0.099
temporal filter + isotonic	0.047	0.046	0.04	0.90	[0.693, 1.189]	−0.01	3	0.012	0.112	0.383	0.049	0.080
temporal filter + beta	0.098	0.051	0.0005	0.85	[0.551, 1.433]	+0.01	5	6.4e-05	0.122	0.457	0.112	0.101
GAM + Platt	0.024	0.051	0.92	0.92	[0.715, 1.206]	+0.01	0	1	0.133	0.431	0.016	0.058
GAM + temperature	0.024	0.050	0.93	0.93	[0.721, 1.220]	+0.02	0	1	0.132	0.428	0.021	0.049
GAM + isotonic	0.040	0.049	0.22	0.83	[0.624, 1.088]	+0.01	0	1	0.134	0.450	0.038	0.061
GAM + beta	0.037	0.051	0.39	0.91	[0.705, 1.192]	+0.01	1	0.4	0.134	0.434	0.032	0.059
asinh-compressed tails	0.047	0.053	0.14	1.04	[0.785, 1.435]	+0.01	2	0.086	0.131	0.423	0.053	0.070
first-half yaw sign, hit rate as P	0.097	0.059	0.0005	1.01	[0.783, 1.276]	−0.14	7	8.2e-08	0.168	0.519	0.089	0.108
shallow tree (depth <= 3)	0.034	0.049	0.5	0.88	[0.668, 1.224]	−0.09	0	1	0.138	0.437	0.047	0.081
logistic: forward selection (in-fold, <= 4)	0.071	0.053	0.0015	0.78	[0.533, 1.174]	−0.04	2	0.086	0.147	0.506	0.092	0.116
reference: yaw logreg (8 instants)	0.072	0.056	0.0025	0.88	[0.678, 1.198]	−0.11	5	6.4e-05	0.156	0.512	0.090	0.078

C.7. Combining k attempts

The inference app combines several attempts, and the probability it shows must stay honest for the aggregate. Let z_i be the logit of attempt i and σ the logistic function. Four aggregates of the small finalist were compared (Figure 53, Table 41): the mean of $P(\text{right})$; the mean of the logits; and two scaled sums, $P(\text{right}) = \sigma(\sum_i z_i/\tau_k)$, with $\tau_k = 1$ (the plain sum of the logits) or $\tau_k = k^\alpha$, where α is fitted on the training persons of each fold (median 0.25). The draws take k random trials of one person and hand (200 draws per cell, 19 persons with at least five trials per hand).

At $k = 3$ the four aggregates have about the same accuracy (sum of logits 89.7 %). They differ in log loss: mean of $P(\text{right})$ 0.348 [0.290, 0.408], mean of logits 0.320 [0.257, 0.389], sum of logits 0.260 [0.151, 0.393] and fitted α 0.252 [0.159, 0.362]. The ECE and its signed lean, shown without a null band, say why: averaging $P(\text{right})$ (ECE 0.152, lean +0.152) or logits (ECE 0.115, lean +0.115) keeps the confidence of one attempt while the accuracy rises, so the aggregate is under-confident. The sum of logits leans slightly over-confident (ECE 0.029, lean −0.029), as expected for correlated attempts treated as independent evidence. The fitted exponent and the plain sum cannot be told apart: log loss −0.008 [−0.031, +0.010], ECE −0.014 [−0.022, +0.017]. *Decision:* for this lane, the parameter-free sum of logits ($\tau_k = 1$). The final model fits its own τ_k on the training persons (Appendix F).

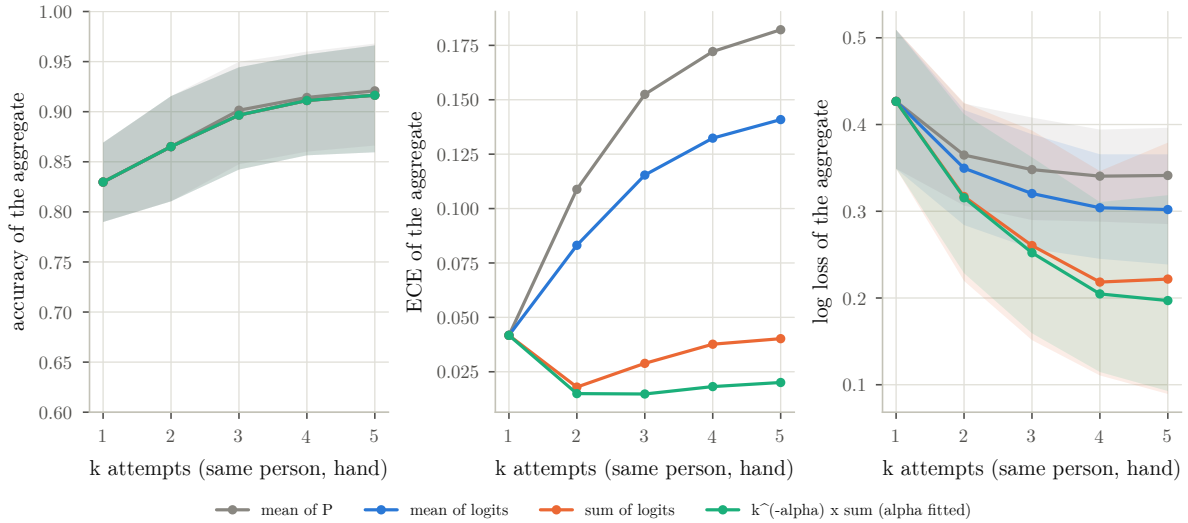


Figure 53: Combining k attempts of one person and hand with the small finalist (19 persons; 200 random draws per cell; bands: person-bootstrap 95 % CI; the ECE is shown without a band). Averages of P(right) or of logits become under-confident; the sums of logits stay close to calibrated.

Table 41: Calibration of k-attempt aggregates of the small-model finalist (random draws of k trials of one person and hand): mean of P, mean of logits, sum of logits, and $k^{-\alpha}$ times the sum, with α fitted on the training persons of each fold. Lean: mean of (observed share - P) towards the predicted side (> 0 under-confident, < 0 over-confident).

k	Aggregation	Accuracy	ECE	Lean	Log loss	LL 95% CI	Persons
1	mean_p	83.0%	0.042	+0.025	0.427	[0.349, 0.509]	19
1	mean_logit	83.0%	0.042	+0.025	0.427	[0.349, 0.509]	19
1	sum_logit	83.0%	0.042	+0.025	0.427	[0.349, 0.509]	19
1	fitted_alpha	83.0%	0.042	+0.025	0.427	[0.349, 0.509]	19
2	mean_p	86.5%	0.109	+0.109	0.365	[0.307, 0.424]	19
2	mean_logit	86.5%	0.083	+0.083	0.350	[0.284, 0.417]	19
2	sum_logit	86.5%	0.018	−0.018	0.317	[0.220, 0.425]	19
2	fitted_alpha	86.5%	0.015	+0.002	0.316	[0.229, 0.412]	19
3	mean_p	90.1%	0.152	+0.152	0.348	[0.290, 0.408]	19
3	mean_logit	89.7%	0.115	+0.115	0.320	[0.257, 0.389]	19
3	sum_logit	89.7%	0.029	−0.029	0.260	[0.151, 0.393]	19
3	fitted_alpha	89.7%	0.015	−0.005	0.252	[0.159, 0.362]	19
4	mean_p	91.4%	0.172	+0.172	0.340	[0.288, 0.394]	19
4	mean_logit	91.1%	0.132	+0.132	0.304	[0.245, 0.366]	19
4	sum_logit	91.1%	0.038	−0.038	0.218	[0.111, 0.346]	19
4	fitted_alpha	91.1%	0.018	−0.014	0.205	[0.115, 0.311]	19
5	mean_p	92.1%	0.182	+0.182	0.341	[0.285, 0.396]	19
5	mean_logit	91.6%	0.141	+0.141	0.302	[0.239, 0.366]	19
5	sum_logit	91.6%	0.040	−0.040	0.222	[0.089, 0.379]	19
5	fitted_alpha	91.6%	0.020	−0.019	0.197	[0.093, 0.319]	19

C.8. Failures and conditions

Persons. The small finalist's accuracy ranges from 65.6 % (P05) to 100.0 % (Figure 54, Table 42). 3 persons are below 70 %: P05 (65.6 %), P09 (67.6 %) and P17 (68.0 %). In each of them, one hand is at or below chance (per-hand ticks in Figure 54). The 2 left-handed persons are shown individually: P17 ranks 3 from the lowest and P11 (85.7 %) ranks 11. The model has no intercept and no even feature, so it cannot learn a right-hand prior from this mostly right-handed sample. With 2 left-handed persons, a gap cannot be told apart from the variation between persons.

Table 42: Per-person leave-one-person-out accuracy of the small-model finalist.

person_id	n	accuracy	acc_left	acc_right	log_loss	platform	handedness	n_confident_errors	rank_lowest
P01	45	75.6%	87.0%	63.6%	0.445	android	right	1	4
P02	31	87.1%	88.2%	85.7%	0.306	ios	right	0	13
P03	24	91.7%	83.3%	100.0%	0.354	android	right	0	16
P04	8	100.0%	100.0%	100.0%	0.188	ios	right	0	20
P05	32	65.6%	50.0%	81.2%	0.670	android	right	3	1
P06	27	85.2%	69.2%	100.0%	0.257	android	right	2	10
P07	28	82.1%	85.7%	78.6%	0.441	android	right	1	8
P08	27	77.8%	92.3%	64.3%	0.567	android	right	3	5
P09	34	67.6%	83.3%	50.0%	0.813	ios	right	8	2
P10	34	88.2%	83.3%	93.8%	0.298	ios	right	1	14
P11	28	85.7%	85.7%	85.7%	0.287	android	left	0	11
P12	38	97.4%	94.7%	100.0%	0.336	android	right	1	19
P13	28	85.7%	100.0%	71.4%	0.374	android	right	2	11
P14	40	85.0%	80.0%	90.0%	0.398	android	right	2	9
P15	38	89.5%	100.0%	78.9%	0.290	android	right	0	15
P16	24	95.8%	100.0%	90.9%	0.186	ios	right	0	18
P17	25	68.0%	38.5%	100.0%	0.567	ios	left	0	3
P18	32	93.8%	100.0%	88.2%	0.182	ios	right	0	17
P19	27	77.8%	92.3%	64.3%	0.591	ios	right	2	5
P20	28	78.6%	78.6%	78.6%	0.755	android	right	5	7

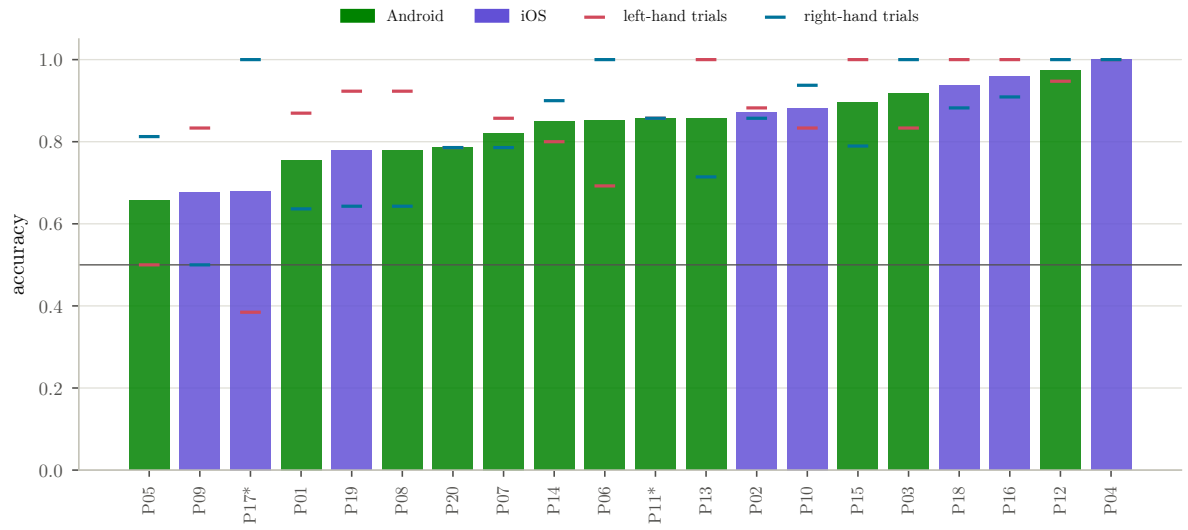


Figure 54: LOSO accuracy of each of the 20 persons with the small finalist (bars, coloured by platform) and per hand (ticks); * marks the two left-handed persons. Persons differ; the weakest have one hand near chance.

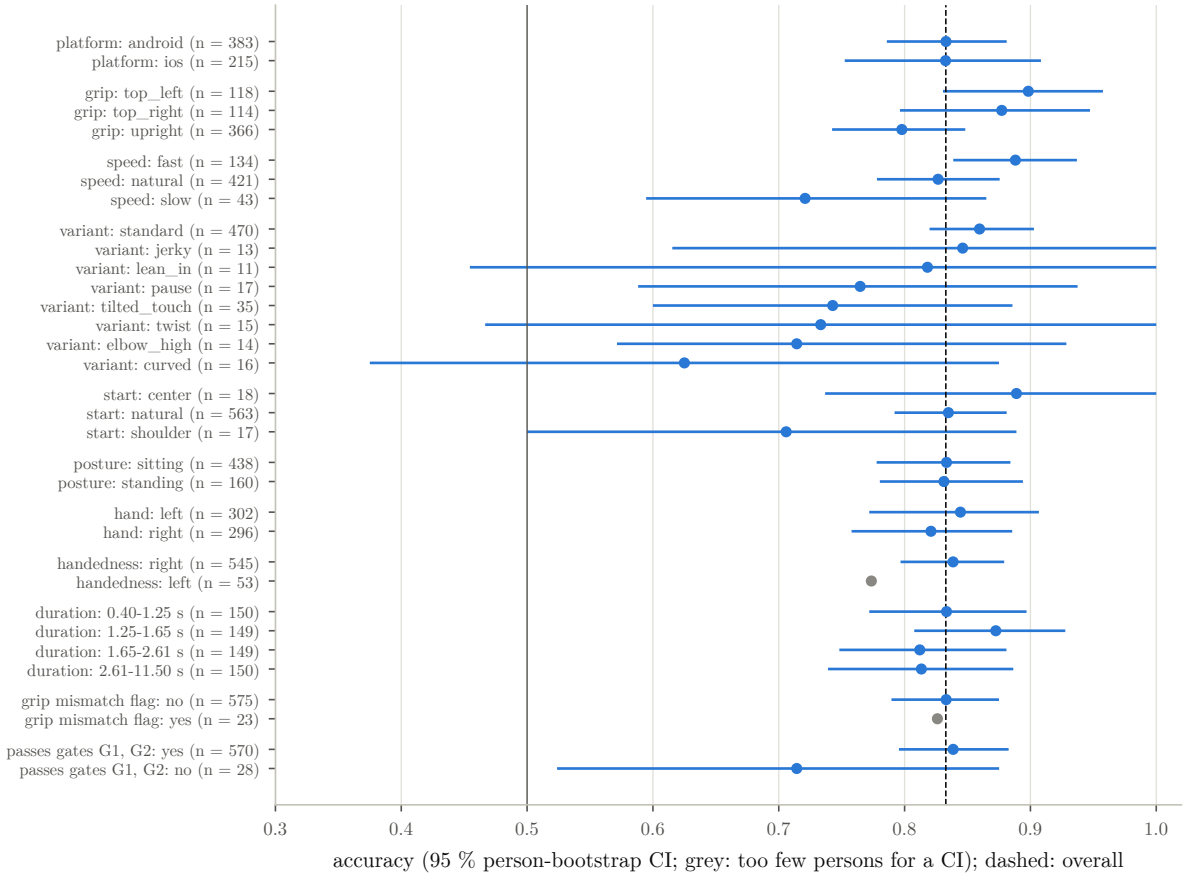


Figure 55: Accuracy of the small finalist by condition (strata after prediction; levels with at least 8 trials; person-bootstrap 95 % CIs only for levels with at least 5 persons, grey otherwise; dashed: overall). Descriptive: where the model is weak.

Conditions. Figure 55 and Table 43 stratify the small finalist's accuracy after prediction; conditions are never model inputs. The strata are descriptive: no test was run across levels, and overlapping CIs are not a test. Android reaches 83.3 % [78.6, 88.1] and iOS 83.3 % [75.2, 90.8]. The sideways grips reach 89.8 % [83.1, 95.8] (top left) and 87.7 % [79.6, 94.7] (top right); the upright grip 79.8 % [74.2, 84.8]. Fast movements reach 88.8 % [83.9, 93.7] and slow ones 72.1 % [59.5, 86.5] (43 trials). Among the movement variants with enough persons, the curved approach is the weakest (62.5 % [37.5, 87.5], 16 trials), against 86.0 % for the standard movement. The bent start covers only 4 persons and gets no CI (57.1 % of 7 trials). The 2 left-handed persons pooled reach 77.4 % (no CI).

Confident errors. A confident error has $P(\text{right})$ of the true hand below 0.2. The small finalist makes 31 of them among 100 errors on 598 trials (Figure 56, Table 44). Their mean yaw time course has the other hand's sign by construction, because they are selected on that evidence; it does not explain them. Slow movements are the most over-represented: 6 of the 31 confident errors (Wilson 95 % interval 9 % to 36 %) against 7 % of all trials, 2.7 times; this rests on a handful of trials. The non-circular evidence is block clustering. The confident errors fall in 19 of 193 blocks, against 27.5 expected when each person's confident errors are spread at random

Table 43: Accuracy of the small-model finalist by condition, platform, handedness, duration quartile and quality flag (strata after prediction; person-bootstrap CIs only for levels with at least 5 persons).

Stratum	Level	Trials	Persons	Accuracy	95% CI	Log loss
platform	android	383	12	83.3%	[78.6, 88.1]	0.428
platform	ios	215	8	83.3%	[75.2, 90.8]	0.415
instructed_rotation	top_left	118	19	89.8%	[83.1, 95.8]	0.343
instructed_rotation	top_right	114	19	87.7%	[79.6, 94.7]	0.383
instructed_rotation	upright	366	20	79.8%	[74.2, 84.8]	0.462
speed	fast	134	20	88.8%	[83.9, 93.7]	0.320
speed	natural	421	20	82.7%	[77.8, 87.6]	0.433
speed	slow	43	19	72.1%	[59.5, 86.5]	0.650
variant	bent_start	7	4	57.1%	–	0.661
variant	curved	16	8	62.5%	[37.5, 87.5]	0.781
variant	elbow_high	14	7	71.4%	[57.1, 92.9]	0.484
variant	jerky	13	8	84.6%	[61.5, 100.0]	0.366
variant	lean_in	11	6	81.8%	[45.5, 100.0]	0.418
variant	pause	17	9	76.5%	[58.8, 93.8]	0.440
variant	standard	470	20	86.0%	[82.0, 90.3]	0.401
variant	tilted_touch	35	19	74.3%	[60.0, 88.6]	0.462
variant	twist	15	8	73.3%	[46.7, 100.0]	0.524
start_position	center	18	10	88.9%	[73.7, 100.0]	0.453
start_position	natural	563	20	83.5%	[79.2, 88.1]	0.416
start_position	shoulder	17	9	70.6%	[50.0, 88.9]	0.628
posture	sitting	438	15	83.3%	[77.8, 88.4]	0.436
posture	standing	160	5	83.1%	[78.0, 89.4]	0.387
tier	1	159	20	81.8%	[74.2, 88.2]	0.445
tier	2	221	19	90.0%	[82.9, 96.3]	0.354
tier	3	140	19	74.3%	[66.2, 81.4]	0.514
tier	4	61	7	86.9%	[75.8, 96.7]	0.392
hand	left	302	20	84.4%	[77.2, 90.7]	0.415
hand	right	296	20	82.1%	[75.8, 88.6]	0.432
handedness	left	53	2	77.4%	–	0.419
handedness	right	545	18	83.9%	[79.7, 87.9]	0.424
duration_bin	0.40-1.25 s	150	18	83.3%	[77.2, 89.7]	0.426
duration_bin	1.25-1.65 s	149	18	87.2%	[80.8, 92.8]	0.370
duration_bin	1.65-2.61 s	149	20	81.2%	[74.8, 88.1]	0.478
duration_bin	2.61-11.50 s	150	17	81.3%	[73.9, 88.6]	0.419
grip_mismatch	no	575	20	83.3%	[79.0, 87.5]	0.419
grip_mismatch	yes	23	3	82.6%	–	0.524
false_start	no	470	20	83.6%	[78.8, 88.7]	0.414
false_start	yes	128	20	82.0%	[76.5, 87.9]	0.459
clip_candidate	no	590	20	83.2%	[79.0, 87.6]	0.426
clip_candidate	yes	8	2	87.5%	–	0.186
seg_disagree_100ms	no	483	20	83.9%	[79.5, 88.2]	0.415
seg_disagree_100ms	yes	115	19	80.9%	[71.4, 88.6]	0.457
repeated_samples	no	472	20	83.9%	[79.0, 88.6]	0.423
repeated_samples	yes	126	20	81.0%	[73.6, 89.1]	0.423
gate_pass	no	28	14	71.4%	[52.4, 87.5]	0.584
gate_pass	yes	570	20	83.9%	[79.6, 88.3]	0.415

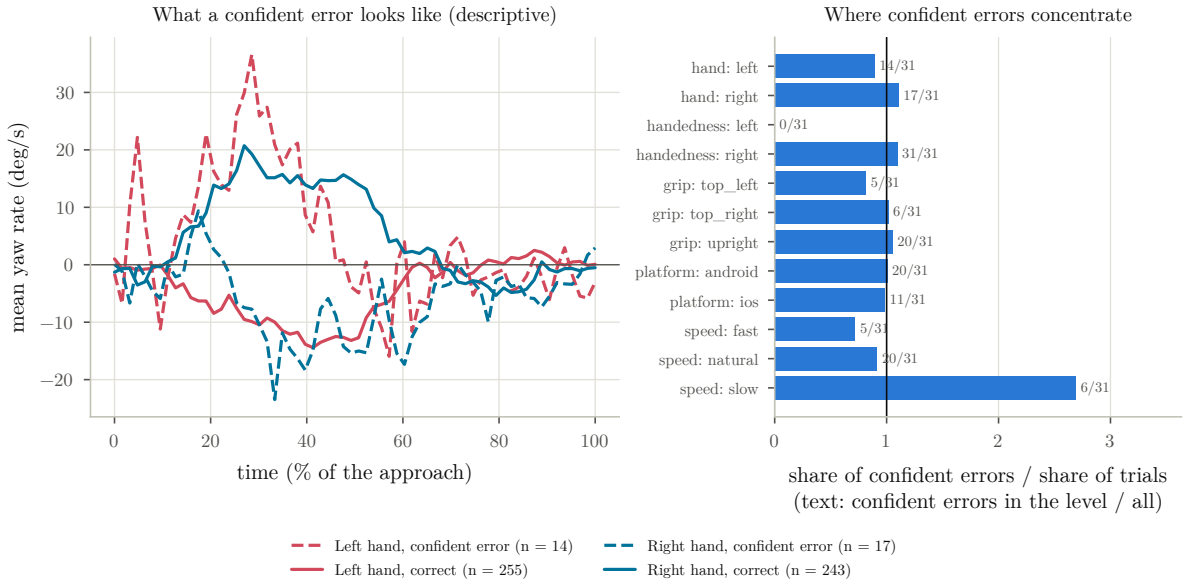


Figure 56: Left: mean yaw rate of correctly classified trials and of confident errors, by instructed hand (descriptive only: the errors are selected on this evidence). Right: share of confident errors over share of trials per condition, with the count of confident errors. Slow movements are over-represented.

over that person's trials (one-sided $p < 0.001$, 2000 permutations). They cluster in blocks, as a hand swap for a whole block would make them.

Table 44: Confident errors of the small-model finalist (P(true hand) < 0.2), with their conditions and flags.

trial_key	person_id	platform	hand	instructed_rotation	speed	variant	start_position	duration_s	p_label	grip_mismatch	false_start	clip_candidate	swap_block
P20-s1-t09a2	P20	android	left	top_left	natural	standard	natural	2.48	0.012	no	yes	no	0.00
P20-s1-t17a1	P20	android	right	top_right	natural	standard	natural	5.67	0.023	no	no	no	0.00
P09-s2-t13a1	P09	ios	right	top_right	natural	standard	natural	1.08	0.026	no	no	no	0.00
P09-s2-t17a1	P09	ios	right	top_left	natural	standard	natural	1.13	0.033	yes	no	no	0.00
P10-s1-t27a1	P10	ios	left	upright	natural	curved	natural	2.07	0.039	no	no	no	0.00
P20-s1-t11a1	P20	android	left	top_left	natural	standard	natural	2.17	0.044	no	yes	no	0.00
P20-s1-t10a3	P20	android	left	top_left	fast	standard	natural	4.52	0.059	no	no	no	0.00
P09-s2-t16a2	P09	ios	right	top_left	fast	standard	natural	1.78	0.071	yes	no	no	0.00
P09-s2-t14a1	P09	ios	right	top_right	natural	standard	natural	1.07	0.075	no	no	no	0.00
P12-s1-t36a1	P12	android	left	upright	natural	standard	shoulder	1.58	0.078	no	yes	no	0.00
P19-s1-t10a1	P19	ios	right	top_right	natural	standard	natural	3.11	0.081	no	no	no	0.00
P14-s3-t03a9	P14	android	left	upright	slow	standard	natural	9.35	0.084	no	no	no	0.00
P05-s1-t37a1	P05	android	left	upright	natural	curved	natural	1.80	0.086	no	yes	no	0.00
P05-s1-t38a1	P05	android	left	upright	natural	elbow_high	natural	1.75	0.092	no	no	no	0.00
P09-s2-t22a1	P09	ios	left	upright	natural	standard	center	1.42	0.093	no	yes	no	0.00
P08-s1-t07a2	P08	android	right	upright	slow	standard	natural	2.02	0.102	no	no	no	1.00
P09-s2-t24a1	P09	ios	left	upright	natural	twist	natural	1.65	0.106	no	no	no	0.00
P13-s1-t23a1	P13	android	right	upright	natural	tilted_touch	natural	1.08	0.116	no	yes	no	0.00
P06-s1-t05a4	P06	android	left	upright	slow	standard	natural	11.18	0.121	no	no	no	0.00
P09-s2-t23a1	P09	ios	left	upright	natural	bent_start	natural	1.50	0.137	no	no	no	0.00
P05-s1-t32a1	P05	android	right	upright	natural	standard	natural	1.35	0.138	yes	no	no	0.00
P06-s1-t21a1	P06	android	left	upright	natural	standard	shoulder	1.00	0.140	no	no	no	0.00
P20-s1-t03a1	P20	android	right	upright	slow	standard	natural	11.25	0.144	no	no	no	0.00
P09-s2-t12a3	P09	ios	right	top_right	fast	standard	natural	1.33	0.145	no	no	no	0.00
P01-s2-t03a1	P01	android	right	upright	slow	standard	natural	5.75	0.146	no	no	no	0.00
P07-s1-t11a1	P07	android	right	top_right	natural	standard	natural	1.65	0.148	no	no	no	0.00
P14-s3-t02a1	P14	android	left	upright	fast	standard	natural	3.42	0.156	no	no	no	0.00
P19-s1-t08a1	P19	ios	right	upright	slow	standard	natural	4.98	0.167	no	no	no	0.00
P08-s1-t08a3	P08	android	right	upright	fast	standard	natural	1.13	0.176	no	no	no	1.00
P08-s1-t06a1	P08	android	right	upright	natural	standard	natural	1.17	0.191	no	no	no	1.00
P13-s1-t21a1	P13	android	right	upright	natural	twist	natural	0.78	0.200	no	no	no	0.00

C.9. Natural-use transfer per component

All 20 free-choice trials, one per person, are right-hand choices. Every natural-use number in this section is therefore right-hand recall on the free-choice trials; a constant right answer

would score 100 %. Model-level recall rests on 20 trials and differs little across models: the rule answers 17 of 20 correctly (85 % [64, 95]), PHYS6 16, the small finalist 15 and the GAM 15. Exact McNemar tests of the rule against 5 other models find no difference after the Holm correction over them; the closest is the rule against the best single-feature rule (7 trials right only for the rule, 0 only for the other; Holm $p = 0.078$).

Abstentions. Two exploratory models abstain on a free-choice trial: they give $P(\text{right})$ exactly 0.5, which is never a right-hand answer. The tree abstains on 1 trial (coverage 95 %) and answers 13 of the remaining trials correctly. The isotonic-calibrated PHYS6 abstains on 1 trial (coverage 95 %) and answers 15 of the remaining trials correctly. Table 37 scores each abstention with half credit (tree 68 %, isotonic PHYS6 78 %). No forced choice was run: depending on how the tie is broken, a forced choice would answer the same number of trials correctly or one more.

A person-matched test per feature. A sharper question is whether each chiral component transfers from the instructed protocol to free choice (Figure 57, Table 45). For every free-choice trial, the sign of a feature (direction learned on the other persons) is compared with the reported hand. The expected count is the Poisson-binomial count from the same persons' instructed trials of that hand, so differences between persons cannot explain a shortfall. Two lateral features transfer worse after the Holm correction over the 9 features: `latvel_q75` points to the reported hand in 10 of 20 trials against 16.4 expected (Holm $p = 0.0031$), and `latvel_q100` in 11 against 16.3 (Holm $p = 0.018$). Yaw shows no detectable drop: `yaw_q50` 17 against 16.3, `yaw_q100` 11 against 12.8. Nor does roll. This explains the low recall of the best single-feature rule: it picks `latvel_q75` in 19 of 20 folds, and scores 10 of 20.

Logit contributions. In the small finalist, the mean contribution of each component towards the reported hand falls from the instructed to the free-choice trials (Table 46): yaw from +0.72 to +0.45 (difference −0.27 [−0.56, +0.01], Holm $p = 0.25$ over the three components) and lateral from +0.64 to +0.31 (difference −0.33 [−0.69, +0.00], Holm $p = 0.25$). Neither drop survives the correction; with one trial per person, the person-matched sign test above is the sharper evidence.

Decision: the lateral evidence depends on the instructed protocol; the yaw evidence shows no detectable dependence. The yaw-only rule is the robust candidate for natural use, even at a lower instructed accuracy. This rests on 20 right-hand trials and must be checked on free-choice trials of both hands.

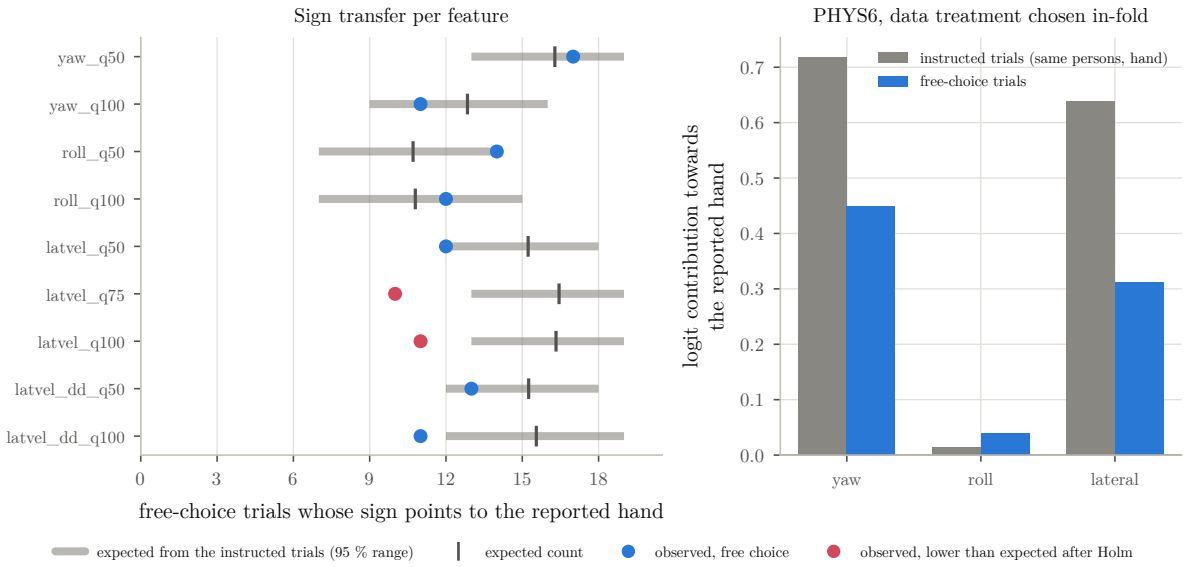


Figure 57: Natural use, per component. Left: free-choice trials whose feature sign points to the reported hand (dot) against the count expected from the same persons' instructed trials (bar: 95 % range; red: lower than expected after the Holm correction). Right: mean logit contribution of each component of the small finalist towards the reported hand. All 20 free-choice trials are right-hand choices, so these counts measure right-hand recall only (a constant right answer would score every trial). The lateral features transfer worse than yaw.

Table 45: Natural-use transfer of each chiral feature: free-choice trials whose sign points to the reported hand (direction learned on the other persons), against the count expected from the same persons' instructed trials of that hand (Poisson-binomial; two-sided p, Holm over the features).

Feature	Component	Trials	Observed	Expected	95% range	p	Holm p
yaw_q50	yaw	20	17	16.3	[13, 19]	1	1
yaw_q100	yaw	20	11	12.8	[9, 16]	0.43	1
roll_q50	roll	20	14	10.7	[7, 14]	0.13	0.63
roll_q100	roll	20	12	10.8	[7, 15]	0.62	1
latvel_q50	lateral	20	12	15.2	[12, 18]	0.088	0.53
latvel_q75	lateral	20	10	16.4	[13, 19]	0.00034	0.0031
latvel_q100	lateral	20	11	16.3	[13, 19]	0.0022	0.018
latvel_dd_q50	lateral	20	13	15.3	[12, 18]	0.26	1
latvel_dd_q100	lateral	20	11	15.6	[12, 19]	0.014	0.099

Table 46: Mean logit contribution of each component towards the reported hand on the free-choice trials and on the same persons' instructed trials of that hand; per-person difference with a person-bootstrap 95 % CI, sign-flip p and Holm over the components.

model	component	n	natural_mean	primary_mean	diff	diff_lo	diff_hi	p	natural_share_positive	primary_share_positive	p_holm
m1_lr_phys6	yaw	20	+0.32	+0.75	−0.43	−0.81	−0.11	0.019	75%	78%	0.056
m1_lr_phys6	roll	20	+0.02	+0.01	+0.01	−0.06	+0.07	0.79	60%	54%	0.79
m1_lr_phys6	lateral	20	+0.29	+0.58	−0.29	−0.60	−0.01	0.066	55%	80%	0.13
m1_lr_phys6_win_asinh	yaw	20	+0.45	+0.72	−0.27	−0.56	+0.01	0.085	80%	79%	0.25
m1_lr_phys6_win_asinh	roll	20	+0.04	+0.01	+0.02	−0.10	+0.15	0.73	60%	54%	0.73
m1_lr_phys6_win_asinh	lateral	20	+0.31	+0.64	−0.33	−0.69	+0.00	0.083	55%	80%	0.25
m1_lr_phys6_nested	yaw	20	+0.45	+0.72	−0.27	−0.56	+0.01	0.085	80%	79%	0.25
m1_lr_phys6_nested	roll	20	+0.04	+0.01	+0.02	−0.10	+0.15	0.73	60%	54%	0.73
m1_lr_phys6_nested	lateral	20	+0.31	+0.64	−0.33	−0.69	+0.00	0.083	55%	80%	0.25

C.10. Suspected wrong-hand blocks and sensitivity

Hands are blocked in the protocol, so a person who used the wrong hand for a whole block would leave a block of reversed trials. Labels are never changed; the analysis only flags and tests. *Detection 1:* a data-driven swap-signature screen flags blocks 1 and 2 of P08's first session (8 trials; Table 47). The small finalist is right on 37.5 % of them, against 83.9 % on every other trial. *Detection 2:* candidate wrong-hand blocks are blocks of at least two trials whose mean $P(\text{right})$ of the instructed hand is below 0.2 (Table 48). They are block 12 of P05's first session (2 trials, mean 0.09, neighbouring blocks 0.77), block 4 of P09's second session (3 trials, 0.08 against 0.58) and block 3 of P20's first session (3 trials, 0.04 against 0.91). Without them the accuracy would be 84.4 %, an upper bound if they are label errors.

Sensitivity: each finalist was retrained without the swap-signature blocks, and with the pre-registered wrong-hand screen inside every fold, which removes from training the trials with a $P(\text{right})$ of their label below 0.05 and runs of at least three confidently reversed trials (Table 49). The evaluation is unchanged; Holm is applied over the two runs of each finalist. Without the swap blocks, the small finalist changes by +0.2 [+0.0, +0.5] points (log loss −0.000 [−0.003, +0.002]). With the screen, which drops a median of 10 training trials per fold, it changes by −1.3 [−2.7, −0.2] points (Holm $p = 0.19$; log loss +0.002 [−0.011, +0.017]). The rule does not change. *Decision:* keep the instructed labels and train on every trial. The open question, whether to ask the participants about the candidate blocks, is left to the researcher.

Table 47: Blocks of the persons with swap-signature blocks (screen.block_consistency, data-driven): accuracy and mean P(instructed hand) of the small-model finalist per block. Labels are never changed.

person_id	session_label	block_index	hand	grip	n	n_correct	accuracy	mean_p_label	n_confident	swap_block	candidate
P08	P08-s1	1	left	upright	4	3	75.0%	0.50	0	1.00	0.00
P08	P08-s1	2	right	upright	4	0	0.0%	0.24	3	1.00	0.00
P08	P08-s1	3	right	top_right	3	3	100.0%	0.72	0	0.00	0.00
P08	P08-s1	4	left	top_right	2	2	100.0%	0.94	0	0.00	0.00
P08	P08-s1	5	left	top_left	3	3	100.0%	0.69	0	0.00	0.00
P08	P08-s1	6	right	top_left	3	3	100.0%	0.87	0	0.00	0.00
P08	P08-s1	7	right	upright	4	3	75.0%	0.61	0	0.00	0.00
P08	P08-s1	8	left	upright	4	4	100.0%	0.82	0	0.00	0.00

Table 48: Candidate wrong-hand blocks: blocks of at least two trials whose mean P(instructed hand) under the small-model finalist is below 0.2, with the mean P(instructed hand) of the neighbouring blocks of the same session. Labels are never changed; the D6-screen sensitivity run removes such trials inside each fold.

person_id	session_label	block_index	hand	grip	n	n_correct	accuracy	mean_p_label	n_confident	swap_block	candidate	neighbour_mean_p_label	neighbour_hands
P05	P05-s1	12	left	upright	2	0	0.0%	0.09	2	0.00	1.00	0.77	right
P09	P09-s2	4	right	top_right	3	0	0.0%	0.08	3	0.00	1.00	0.58	left;right
P20	P20-s1	3	left	top_left	3	0	0.0%	0.04	3	0.00	1.00	0.91	left;right

Table 49: Sensitivity of the finalist anchors to suspected wrong-hand training rows: trained without the swap-signature blocks, or with the docs/09 D6 screen inside every fold; evaluation unchanged (paired against the anchor; Dropped: training rows removed per fold).

Run	Anchor	Dropped (median)	dAcc (pts)	dAcc CI	dLL	dLL CI
first-half yaw sign, hit rate as P, swap-signature blocks out of training	m1_rule_fhy_sign_cal	8	+0.0	[0.0, 0.0]	−0.000	[−0.004, 0.004]
first-half yaw sign, hit rate as P, D6 screen in every fold	m1_rule_fhy_sign_cal	10	+0.0	[0.0, 0.0]	+0.001	[−0.004, 0.006]
PHYS6, data treatment chosen in-fold, swap-signature blocks out of training	m1_lr_phys6_nested	8	+0.2	[0.0, 0.5]	−0.000	[−0.003, 0.002]
PHYS6, data treatment chosen in-fold, D6 screen in every fold	m1_lr_phys6_nested	10	−1.3	[−2.7, −0.2]	+0.002	[−0.011, 0.017]
GAM on PHYS6 (splines), swap-signature blocks out of training	m1_gam_phys6	8	−0.5	[−1.3, 0.2]	+0.014	[−0.001, 0.037]
GAM on PHYS6 (splines), D6 screen in every fold	m1_gam_phys6	10	+0.8	[−1.4, 3.1]	+0.021	[−0.016, 0.070]

C.11. Caveats of the lane

- 20 persons: the CI of every model spans several points, and the per-person values are uncertain.
- PHYS6 was specified after a pooled look at all persons; its nested guards are reported in Appendix C.3. The nested treatment selection was added after a first run had shown the outer-fold result of the asinh compression.
- The ECE null treats trials as independent given $P(\text{right})$. Person effects make the real sampling variation larger, so its p value is optimistic. The reliability bands and the calibration slope resample persons. The calibration intercepts are point estimates without a CI.
- Roll and lateral acceleration assume the screen faces the person, as in every recorded grip. Under a screen-away grip they change sign; yaw does not. A production model must check the grip or rely on yaw for screen-away use.
- The natural-use set is one free-choice trial per person, all right-hand choices: it measures right-hand recall only, with a wide interval.
- The pilot rule's direction comes from a pilot session of P01, which is part of the evaluation set; the rule finalist learns its direction in-fold instead and makes the same decisions.
- The approach bounds are the oracle window. A cue-free pipeline finds its own window (Appendix G).

D. Classical machine learning on engineered features

Lane m2 asks how far classical (non-deep) models get on a rich set of hand-crafted per-trial features, and which data treatment and which features matter. Every result is leave-one-person-out over the 20 persons and 598 primary trials. Every hyperparameter is tuned inside the outer fold, on 3 inner person folds of the training persons. A paired difference compares two pipelines on the same trials, with a paired person-bootstrap CI. Its p-value is an exact sign-flip test over persons, Holm-adjusted within the comparison family named in the text (level 0.05). Two pipelines whose $P(\text{right})$ differ by less than 1e-03 on every trial are called identical (solver precision); they are not tested. The inputs are the oracle approach windows of the cache (Appendix B); production sees detected windows instead (Appendix G). Abbreviations used in this appendix: logistic regression (LR), support vector machine (SVM), radial basis

function (RBF), k nearest neighbours (k-NN), LightGBM gradient-boosted trees (LGBM), calibration-in-the-large (CITL).

D.1. Features and their mirror parity

Representation. One trial is described by 691 features computed from the 17 channels of the cache. Nine are in the device frame: the three axes of the linear acceleration, of the rotation rate and of the unit up vector. Eight are in the gravity frame: magnitudes, vertical and horizontal components, the yaw rate about gravity $\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$, its running integral (the cumulative yaw) and the tilt from the start. The approach window is time-normalised to 64 samples from onset to end. Seven feature families describe it (Table 50): means in $k = 8$ equal time bins (`bins`), summary statistics (`stats`), log band powers and the lowest Fourier coefficients (`fft`), wavelet sub-band energies (`wavelet`), correlations of every channel pair within a frame group (`xcorr`), cumulative integrals at the quarters of the window (`cumint`) and the log approach duration (`global`).

Mirror parity. Every feature is even (unchanged), odd (sign flip) or one of a swapped pair under the left-right mirror $m(\cdot)$ of Eqs. (1) and (2). The feature registry gives the exact mirror map of every feature (unit-tested): the features of a mirrored trial are a signed permutation of the original ones. Of the default set, 168 features are odd or swapped and 523 are even. The `wavelet` and `global` families are entirely even (0 and 0 odd features). This parity matters for the design: with mirror augmentation each even feature value enters the training set once with each label, so a linear model gives every even feature (and the intercept) zero weight. The augmented LR therefore uses only the mirror-sensitive features.

Table 50: Lane m2: feature families of the default set (approach window, 17 channels, k = 8 bins). Parity under the left/right mirror: even (unchanged), odd (sign flip) or swapped pairs (min/max and arg-min/arg-max of chiral channels); the mirror map is exact (unit test). With mirror augmentation a linear model gives the even features zero weight.

Family	Content	Features	Even	Odd	Swap pairs
bins	mean in k equal time bins of the window	136	88	48	0
stats	mean, SD, min, max, arg-min/max, skewness, kurtosis, energy, zero crossings	170	134	12	24
fft	log band powers and real/imaginary parts of the lowest Fourier coefficients	187	151	36	0
wavelet	log energy of every db2 sub-band	85	85	0	0
xcorr	correlation of every channel pair within a frame group	64	32	32	0
cumint	cumulative integral at quarters of the window (rates, accelerations)	48	32	16	0
global	log approach duration	1	1	0	0

D.2. Models, tuning and the headline estimate

Options. The lane compares 12 pipelines on the same features, all mirror-augmented: the pre-specified default, an L2-penalised LR; L1 and elastic-net LR; a linear SVM, an RBF SVM and a k-NN classifier, each with a Platt map [21]; a random forest and extra trees with one fixed configuration (300 trees, $\sqrt{p}$ of the p features per split, leaves of at least 3 rows; not tuned); HistGradientBoosting; LGBM; and a soft vote and a stack of the L2 LR, the RBF SVM and LGBM. The models are those of scikit-learn [25] and LightGBM. The L2 and L1 LR

and the linear SVM are tuned on a grid of their penalty. The other tuned models use a small seeded Bayesian search (Optuna) per outer fold: 8 trials for the RBF SVM, 8 for the k-NN, 8 for LGBM, 6 for the elastic net and 6 for HistGradientBoosting.

Three estimates with three roles. The *nested selection* (`m2_select`) chooses, inside every outer fold, among the L2 LR, the RBF SVM, LGBM and their soft vote by inner cross-validated log loss, then refits the winner on all training persons. It is one procedure, so its leave-one-person-out estimate is selection-aware: it is the lane's headline. The *default* (`m2_lr_full`) was fixed before any outer result. The best of the 12 pipelines by outer log loss (the stack) and the best tree pipeline (LGBM) are *exploratory*: each is the best of 12 (stack) and 12 (LGBM) configurations compared on the outer folds. The selection chose the soft vote in 12 and LGBM in 8 of the 20 outer folds. The median inner log-loss margin over the runner-up is 0.006, so the candidates are close. Every candidate's inner predictions are out of sample; the SVM's Platt map is cross-fitted across the inner folds, which corrects an in-sample map found by the audit.

Results. Figure 58 and Table 51 show every pipeline. The nested selection reaches an accuracy of 86.5 % [80.9, 91.7], an AUC of 0.932 [0.885, 0.973], a log loss of 0.334 [0.239, 0.440] and a Brier score of 0.099 [0.066, 0.135]. The default LR reaches 85.5 % [79.8, 90.6] with a log loss of 0.365 [0.280, 0.464]. The reference floor, a mirror-augmented LR on the cumulative yaw at 8 instants, reaches 77.9 % [73.1, 82.9]. The selection beats it by +8.5 [+4.4, +12.6] points (Holm $p = 0.006$) and the default by +7.5 [+3.5, +11.5] points (Holm $p = 0.010$). Both are adjusted over the 13 comparisons of m2 pipelines with the reference floor. Every pipeline of this family beats the floor after the adjustment. The smallest gain is that of the L1 LR (+6.0 [+1.8, +10.5] points, Holm $p = 0.019$), the largest that of the stack (+10.2 [+5.8, +14.6] points, Holm $p = 0.003$).

Against the default. The family is the 12 comparisons of the other pipelines and the selection with the default. No pipeline differs from the default in accuracy after the Holm adjustment. The stack, the best pipeline by outer log loss, gains +2.7 [+1.3, +4.3] points of accuracy (Holm $p = 0.073$) and changes the log loss by −0.042 [−0.071, −0.015] (Holm $p = 0.080$): not significant. The soft vote is the only pipeline with a lower log loss after the adjustment (−0.032 [−0.052, −0.015], Holm $p = 0.021$), and no pipeline has a higher one. The nested selection changes the log loss by −0.031 [−0.054, −0.007] (Holm $p = 0.202$). With 20 persons, the spread of a few points between pipelines is not resolved.

Natural use and abstentions. The free-choice trials are all right-hand trials so far (20 right, 0 left). Their score is therefore right-hand recall, not an accuracy over both hands, and a constant 'right' answer would score 100 %. The nested selection recognises 18 of 20 free-choice trials (right-hand recall 90 % [75, 100]), against a right-hand recall of 89.9 % on the primary trials. The k-NN and the RBF SVM recognise 16 (Table 51). None of the four reported pipelines abstains: none returns $P(\text{right}) = 0.5$ exactly on any primary or free-choice trial (free-choice coverage 100 % for the selection, 100 % for the default, 100 % for the stack and 100 % for LGBM). Their answered and forced-choice accuracies therefore equal the accuracies shown. The only abstaining configuration of the lane is an ablation (Appendix D.4).

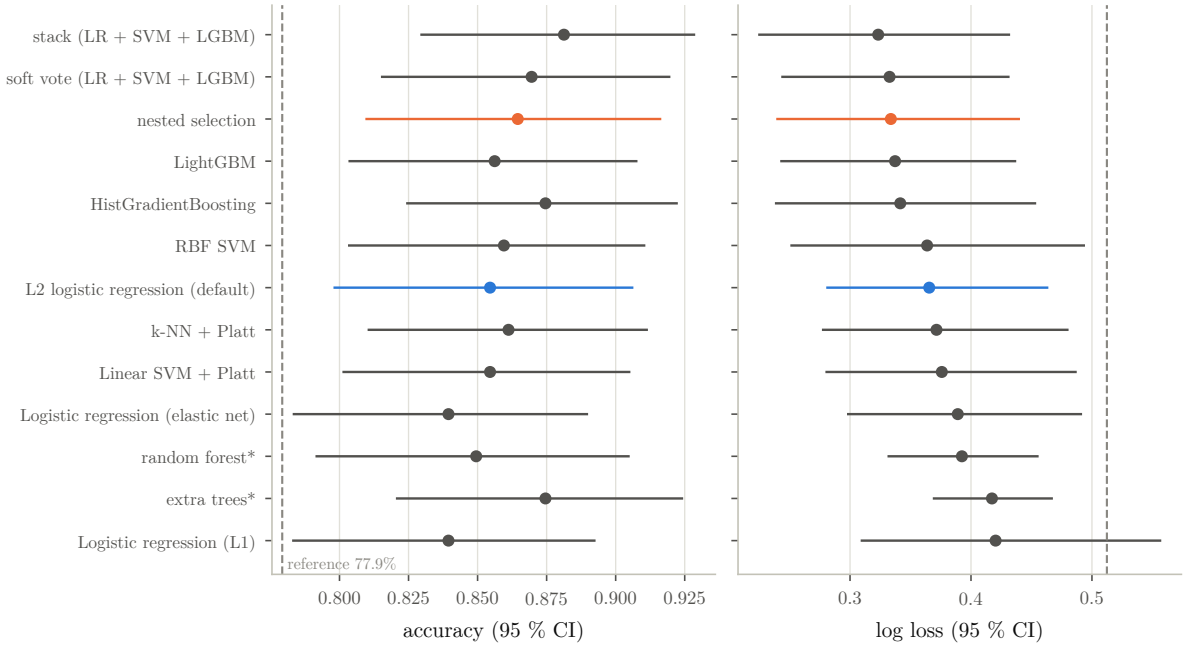


Figure 58: Classical pipelines on the same 691 features, leave-one-person-out over 20 persons, sorted by log loss (best at the top). Blue: the pre-specified default (L2 LR); orange: the nested selection, the selection-aware headline; dashed: the reference floor; * fixed hyperparameters. The best pipelines are each the best of 12 compared on the outer folds: exploratory.

Finalists. Lane m2 hands three finalists to the cross-lane nested selection of lane final (Table 52; Appendix F): the best linear, tree and ensemble pipelines by outer log loss. This choice is exploratory. For `m2_lr_full` the two records disagree: the finalists file counts 12 configurations compared on the outer folds, the saved result 1. Lane final uses the finalists file's count, the role in which the model became a candidate, so in the cross-lane selection it counts as exploratory. Inside lane m2 it remains the pre-specified default.

Table 51: Lane m2: classical models on the default per-trial features (leave-one-person-out, every model tuned inside the outer fold on inner person folds, mirror-augmented; 95 % person-cluster bootstrap CI). Free choice: right-hand recall on the free-choice trials (correct / scored; all scored trials are right-handed, so this is not an accuracy over both hands); primary: right-hand recall on the instructed trials as its comparator. k = 3: per-person balanced accuracy with 3 attempts averaged. p vs default: sign-flip test of the per-person log-loss difference against the pre-specified default, Holm-adjusted over the pipelines compared. "Configs compared": the size of the outer-fold comparison (1 = pre-specified or nested).

Model	Accuracy	95% CI	AUC	Log loss	ECE	Brier	Free choice: right recall	Primary: right recall	k = 3	p vs default (Holm)	Configs compared
Stack (LR + SVM + LGBM)	88.1%	[82.9, 92.9]	0.936	0.323	0.025	0.095	18/20	90.9%	93.7%	0.080	12
Soft vote (LR + SVM + LGBM)	87.0%	[81.5, 92.0]	0.934	0.333	0.042	0.098	18/20	89.2%	93.6%	0.021	12
Nested selection (LR / SVM / LGBM / soft vote, by inner log loss)	86.5%	[80.9, 91.7]	0.932	0.334	0.031	0.099	18/20	89.9%	93.1%	0.202	1
LightGBM	85.6%	[80.3, 90.8]	0.930	0.337	0.022	0.101	18/20	88.2%	93.2%	0.949	12
HistGradientBoosting	87.5%	[82.4, 92.3]	0.927	0.342	0.030	0.100	18/20	89.9%	93.5%	0.949	12
RBF SVM + Platt	86.0%	[80.3, 91.1]	0.924	0.364	0.036	0.108	16/20	86.5%	93.0%	1.000	12
L2 logistic regression, default features (pre-specified)	85.5%	[79.8, 90.6]	0.920	0.365	0.048	0.110	18/20	86.1%	92.8%	–	1
k-NN + Platt	86.1%	[81.0, 91.2]	0.916	0.371	0.043	0.110	16/20	84.8%	93.5%	1.000	12
Linear SVM + Platt	85.5%	[80.1, 90.5]	0.915	0.376	0.040	0.112	18/20	85.5%	92.6%	0.930	12
Logistic regression (elastic net)	83.9%	[78.3, 89.0]	0.908	0.389	0.034	0.117	18/20	83.4%	92.3%	0.930	12
Random forest (fixed hyperparameters)	84.9%	[79.1, 90.5]	0.925	0.392	0.102	0.119	18/20	86.8%	92.3%	0.949	12
Extra trees (fixed hyperparameters)	87.5%	[82.0, 92.4]	0.932	0.417	0.154	0.125	18/20	87.2%	93.0%	0.273	12
Logistic regression (L1)	83.9%	[78.3, 89.3]	0.898	0.420	0.034	0.125	18/20	82.4%	91.8%	0.202	12

Table 52: Lane m2 finalists handed to the cross-lane nested selection (results/finalists_m2.json): the best linear, tree and ensemble pipelines by outer log loss, an exploratory choice among all the model pipelines compared on the outer folds.

Finalist	Pipeline	Accuracy	95% CI	Log loss	ECE	Pipelines compared for the finalist choice
m2_lr_full	L2 logistic regression, default features (pre-specified)	85.5%	[79.8, 90.6]	0.365	0.048	12
m2_lgbm	LightGBM	85.6%	[80.3, 90.8]	0.337	0.022	12
m2_ens_stack	Stack (LR + SVM + LGBM)	88.1%	[82.9, 92.9]	0.323	0.025	12

D.3. Data treatment

Design. The default treatment was pre-specified: the time-normalised approach window, all 17 channels, the seven families with $k = 8$ bins, a z-score fitted on the training rows, mirror augmentation and the L2 LR. Each of 12 alternatives is paired against it on the same trials and Holm-adjusted within this family of 12 comparisons. The rule: adopt an alternative only if its held-out log loss is lower after the adjustment. No alternative has a lower log loss after the adjustment (Figure 59; Table 53, upper rows), so the default stays. The mirror options are treated in Appendix D.5.

Window. The alternatives are a fixed 3-s window in absolute time (from 0.5 s before to 2.5 s after the onset) and both windows side by side. Neither differs from the approach window (fixed: log loss -0.022 [-0.055, $+0.008$], Holm $p = 1.000$; both: -0.002 [-0.023, $+0.017$], Holm $p = 1.000$). The fixed window is also not onset-only. The cache freezes every channel from the app's end cue on, which happens in 402 of 598 trials (median 36 % of the post-onset window frozen; Spearman correlation of the freeze point with the approach duration 0.99). Its gravity reference also uses the rest after the end. Production has no end cue. Decision: keep the approach window.

Channel frame. Keeping only the device-frame channels worsens the log loss by $+0.126$ [$+0.077$, $+0.176$] (Holm $p < 0.001$) and the accuracy by -8.0 [-11.0, -4.9] points (Holm $p = 0.003$). Keeping only the gravity-frame channels worsens the log loss by $+0.107$ [$+0.067$, $+0.146$] (Holm $p = 0.001$); its accuracy does not differ (Holm $p = 0.706$). Decision: keep both frames. The device-frame features do not transfer to unseen grips, however (Appendix D.7).

Time bins. With $k = 4$ or $k = 16$ bins instead of 8, the log loss does not differ (Holm $p = 1.000$ and 1.000). Decision: keep $k = 8$.

Scaling. Without scaling, the log loss worsens by $+0.119$ [$+0.085$, $+0.155$] (Holm $p < 0.001$). The unscaled fit does not converge: the feature standard deviations span 5.6 orders of magnitude, and with the solver capped at 300 iterations 540 of 540 inner fits and 20 of 20 outer-fold fits stopped at the cap. Robust scaling (median and interquartile range) worsens the log loss by $+0.027$ [$+0.013$, $+0.041$] (Holm $p = 0.010$). Winsorising at 3 standard deviations and per-trial amplitude normalisation do not differ (Holm $p = 1.000$ and 1.000). Decision: keep the z-score. Trees are scale-free; the linear, kernel and k-NN models need the scaling.

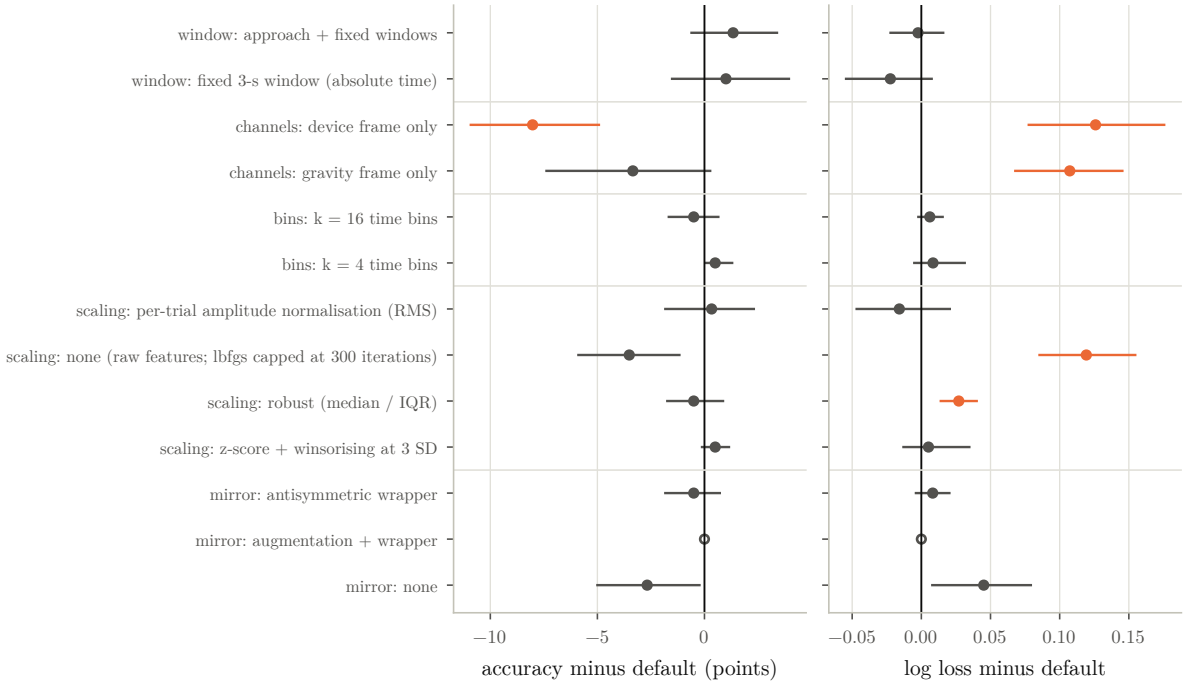


Figure 59: Data-treatment alternatives of the default L2 LR, each paired against the default on the same trials: accuracy (left) and log loss (right) differences with paired 95 % CIs. Orange: Holm-adjusted $p < 0.05$ within the family of 12 comparisons; open marker: identical within solver precision (not a test). No alternative improves the log loss, so the pre-specified default stays.

Inherited treatment. Filtering, resampling and gap handling are the cache's: a 15 Hz zero-phase low-pass filter, a 60 Hz grid, interpolation of gaps up to 100 ms and exclusion of longer gaps near the movement (Appendix B). This lane inherits them unchanged. Every feature is checked finite before a fit.

D.4. Feature-family ablation

Design. Each family is dropped in turn, or kept alone, in two designs (Figure 60; Table 53, lower rows). With mirror augmentation the family is the 11 comparisons with all families. Augmentation zeroes the even features, so this design tests only the mirror-sensitive part of each family. Without mirroring every feature can carry weight; this family has 14 comparisons with all families without mirroring.

Results. The `xcorr` family is the only one whose removal worsens the log loss after the adjustment, both with augmentation (+0.102 [+0.044, +0.146], Holm $p = 0.014$) and without mirroring (+0.084 [+0.049, +0.113], Holm $p = 0.004$). Alone, it almost matches all families: 84.9 % against 85.5 %, log loss +0.004 [−0.032, +0.041] (Holm $p = 1.000$). Every other single family has a worse log loss than all families after the adjustment, in both designs. Dropping `wavelet` or `global` with augmentation is identical to the default, because both carry zero weight.

Table 53: Lane m2: data-treatment and representation choices, each paired against its baseline on the same trials (the default L2 logistic regression; the ablations without mirroring against the default without mirroring): difference with paired person-bootstrap 95 % CI; p: exact sign-flip test over persons, Holm-adjusted within the comparison family (data treatment; families with augmentation; families without mirroring). "identical": predictions within solver precision of the baseline (no trial changes hand), not a test.

Choice	Variant	Features	Accuracy	Diff. vs baseline (pts)	p acc. (Holm)	Log loss	Diff. log loss	p log loss (Holm)	Log loss differs
window	window: approach + fixed windows	1398	86.8%	+1.3 [−0.7, +3.4]	1.000	0.366	−0.002 [−0.023, +0.017]	1.000	no
window	window: fixed 3-s window (absolute time)	708	86.5%	+1.0 [−1.6, +4.0]	1.000	0.344	−0.022 [−0.055, +0.008]	1.000	no
frame	channels: device frame only	367	77.4%	−8.0 [−11.0, −4.9]	0.003	0.499	+0.126 [+0.077, +0.176]	<0.001	yes
frame	channels: gravity frame only	325	82.1%	−3.3 [−7.4, +0.3]	0.706	0.466	+0.107 [+0.067, +0.146]	0.001	yes
bins	bins: k = 16 time bins	827	84.9%	−0.5 [−1.7, +0.7]	1.000	0.371	+0.006 [−0.003, +0.016]	1.000	no
bins	bins: k = 4 time bins	623	86.0%	+0.5 [+0.0, +1.3]	1.000	0.376	+0.008 [−0.006, +0.032]	1.000	no
scaling	scaling: per-trial amplitude normalisation (RMS)	691	85.8%	+0.3 [−1.9, +2.4]	1.000	0.342	−0.016 [−0.048, +0.021]	1.000	no
scaling	scaling: none (raw features; lbfgs capped at 300 iterations)	691	81.9%	−3.5 [−5.9, −1.1]	0.115	0.476	+0.119 [+0.085, +0.155]	<0.001	yes
scaling	scaling: robust (median / IQR)	691	84.9%	−0.5 [−1.8, +0.9]	1.000	0.394	+0.027 [+0.013, +0.041]	0.010	yes
scaling	scaling: z-score + winsorising at 3 SD	691	86.0%	+0.5 [−0.2, +1.2]	1.000	0.371	+0.005 [−0.014, +0.036]	1.000	no
mirror	LightGBM, mirror: none	691	86.8%	+1.2 [−0.6, +2.9]	0.599	0.382	+0.046 [+0.021, +0.075]	0.002	yes
mirror	Logistic regression (L2), mirror: antisymmetric wrapper	691	84.9%	−0.5 [−1.9, +0.8]	1.000	0.374	+0.008 [−0.005, +0.021]	1.000	no
mirror	Logistic regression (L2), mirror: augmentation + wrapper	691	85.5%	+0.0 [+0.0, +0.0]	–	0.365	+0.000 [−0.000, +0.000]	–	identical
mirror	Logistic regression (L2), mirror: none	691	82.8%	−2.7 [−5.1, −0.2]	0.653	0.410	+0.045 [+0.007, +0.080]	0.242	no
family drop	without bins	555	85.8%	+0.3 [−0.8, +1.4]	1.000	0.373	+0.008 [−0.012, +0.035]	1.000	no
family drop	without cumint	643	85.6%	+0.2 [−0.5, +0.9]	1.000	0.379	+0.013 [−0.007, +0.046]	1.000	no
family drop	without fft	504	85.5%	+0.0 [−1.3, +1.3]	1.000	0.375	+0.007 [−0.006, +0.027]	1.000	no
family drop	without global	690	85.5%	+0.0 [+0.0, +0.0]	–	0.365	−0.000 [−0.000, −0.000]	–	identical
family drop	without stats	521	85.3%	−0.2 [−1.2, +1.2]	1.000	0.380	+0.013 [−0.004, +0.037]	1.000	no
family drop	without wavelet	606	85.5%	+0.0 [+0.0, +0.0]	–	0.365	−0.000 [−0.000, +0.000]	–	identical
family drop	without xcorr	627	82.4%	−3.0 [−6.0, +0.2]	0.564	0.464	+0.102 [+0.044, +0.146]	0.014	yes
family only	only bins	136	82.3%	−3.2 [−6.1, −0.2]	0.563	0.471	+0.108 [+0.049, +0.154]	0.014	yes
family only	only cumint	48	79.9%	−5.5 [−9.1, −2.0]	0.099	0.471	+0.104 [+0.040, +0.157]	0.029	yes
family only	only fft	187	80.3%	−5.2 [−8.6, −1.9]	0.064	0.477	+0.119 [+0.059, +0.176]	0.012	yes
family only	only stats	170	78.4%	−7.0 [−10.5, −3.6]	0.009	0.484	+0.123 [+0.055, +0.177]	0.014	yes
family only	only wavelet	85	50.0%	−35.5 [−40.6, −29.8]	<0.001	0.693	+0.336 [+0.244, +0.419]	<0.001	yes
family only	only xcorr	64	84.9%	−0.5 [−2.4, +1.3]	1.000	0.366	+0.004 [−0.032, +0.041]	1.000	no
family drop nomirror	without bins	555	82.6%	−0.2 [−1.9, +1.5]	1.000	0.403	−0.006 [−0.016, +0.008]	1.000	no
family drop nomirror	without cumint	643	83.4%	+0.7 [−1.0, +2.6]	1.000	0.405	−0.003 [−0.020, +0.017]	1.000	no
family drop nomirror	without fft	504	82.8%	+0.0 [−1.6, +1.7]	1.000	0.400	−0.008 [−0.019, +0.001]	0.797	no
family drop nomirror	without global	690	82.6%	−0.2 [−0.5, +0.0]	1.000	0.410	+0.000 [−0.000, +0.000]	1.000	no
family drop nomirror	without stats	521	84.8%	+2.0 [+0.9, +3.4]	0.097	0.413	+0.002 [−0.008, +0.012]	1.000	no
family drop nomirror	without wavelet	606	83.9%	+1.2 [+0.3, +2.2]	0.500	0.406	−0.005 [−0.009, −0.001]	0.163	no
family drop nomirror	without xcorr	627	79.9%	−2.8 [−5.8, −0.2]	0.520	0.490	+0.084 [+0.049, +0.113]	0.004	yes
family only nomirror	only bins	136	79.6%	−3.2 [−6.0, −0.5]	0.500	0.479	+0.071 [+0.035, +0.106]	0.015	yes
family only nomirror	only cumint	48	80.1%	−2.7 [−4.9, −0.5]	0.421	0.493	+0.081 [+0.036, +0.122]	0.022	yes
family only nomirror	only fft	187	78.3%	−4.5 [−8.7, −0.9]	0.342	0.530	+0.127 [+0.076, +0.184]	0.002	yes
family only nomirror	only global	1	49.7%	−33.1 [−39.2, −27.0]	<0.001	0.694	+0.292 [+0.221, +0.364]	<0.001	yes
family only nomirror	only stats	170	74.1%	−8.7 [−11.8, −5.7]	<0.001	0.532	+0.129 [+0.083, +0.168]	0.001	yes
family only nomirror	only wavelet	85	50.3%	−32.4 [−38.8, −26.1]	<0.001	0.694	+0.292 [+0.221, +0.364]	<0.001	yes
family only nomirror	only xcorr	64	84.8%	+2.0 [−2.1, +6.2]	1.000	0.377	−0.029 [−0.090, +0.041]	1.000	no

Even families and an abstaining ablation. The `wavelet` family alone with augmentation has no mirror-sensitive feature, so it returns $P(\text{right}) = 0.5$ for every trial. It abstains on every trial: it has no answered trial and no forced choice, and its 50.0 % is the half credit of the abstentions, not evidence about the family. Without mirroring the even families are tested properly: alone, `wavelet` reaches 50.3 % [47.0, 53.2] and `global` 49.7 % [47.8, 51.5], both at chance (the CIs include one half).

Decision. Keep every family (the pre-specified default). The families overlap, `xcorr` carries the most information that the others do not, and the redundancy costs little in a penalised model. Its odd correlations pair a chiral channel with a mirror-even one.

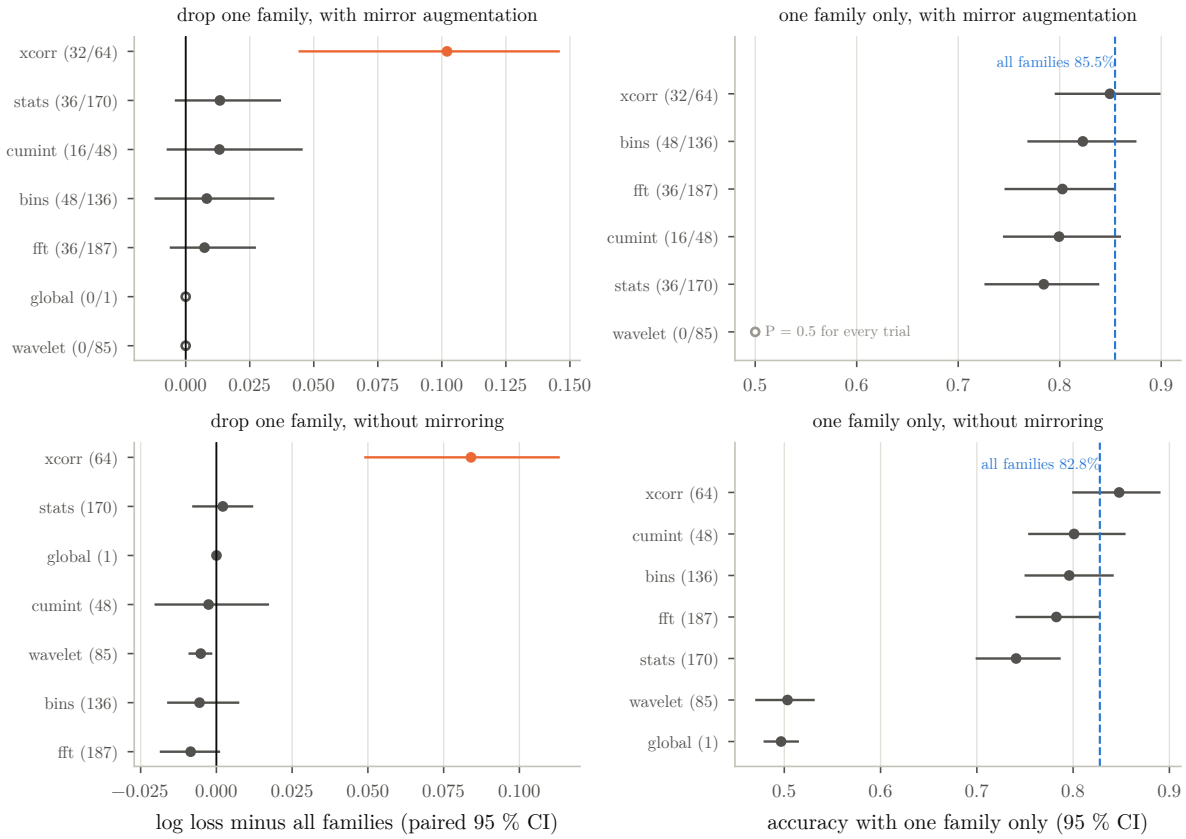


Figure 60: Feature-family ablation of the L2 LR with mirror augmentation (top; numbers: mirror-sensitive / all features of the family) and without mirroring (bottom). Left: log-loss change when one family is dropped (paired 95 % CI; orange: Holm-adjusted $p < 0.05$ within the design's family; open: identical within solver precision). Right: accuracy of each family alone; dashed: all families. Only dropping `xcorr` hurts; the `wavelet` family alone abstains under augmentation.

D.5. Mirror handling

Options. (i) None. (ii) Augmentation: every training trial also enters mirrored, through the exact feature map, with the opposite label [22]. (iii) The antisymmetric wrapper, which forces $P(\text{right})$ of a mirrored trial to be one minus that of the original:

$$p(\mathbf{x}) = \tfrac{1}{2}\big(f(\mathbf{x}) + 1 - f(m(\mathbf{x}))\big), \tag{5}$$

where f is the fitted model and $\mathbf{x}$ the feature vector. (iv) Both. The LR was run with all four, LGBM with none and augmentation. The comparisons belong to the data-treatment family of 12 (LR) and to a family of 1 comparison (LGBM).

Evidence. Figure 61. Without mirroring, the LR changes the log loss by +0.045 [+0.007, +0.080] (Holm $p = 0.242$) and the accuracy by −2.7 [−5.1, −0.2] points (Holm $p = 0.653$): not different after the adjustment. The wrapper alone does not differ either (log loss +0.008 [−0.005, +0.021], Holm $p = 1.000$). Augmentation plus the wrapper is identical to augmentation alone

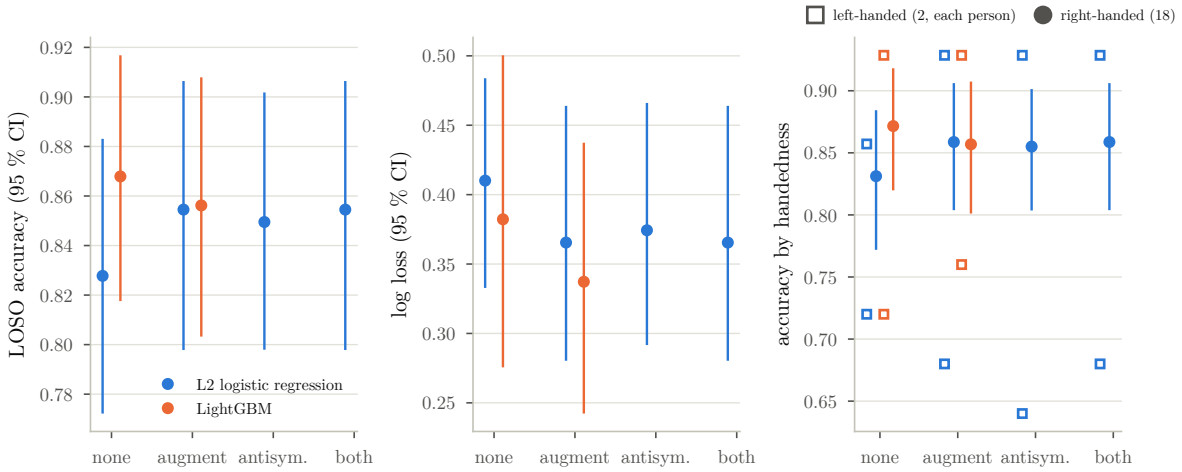


Figure 61: Mirror handling: none, augmentation, the antisymmetric wrapper (Eq. 5) and both, for the L2 LR (blue) and LGBM (orange; none and augmentation only). Left and middle: accuracy and log loss with 95 % CIs. Right: accuracy of the 18 right-handed persons (with CI) and of each of the 2 left-handed persons (open squares, no CI). Augmentation is kept.

(largest $|\Delta P(\text{right})|$ 1.8e-04; 0 trials change hand): an augmented linear model is already antisymmetric. For LGBM, dropping the augmentation worsens the log loss by +0.046 [+0.021, +0.075] (Holm $p = 0.002$), with an accuracy change of +1.2 [−0.6, +2.9] points (Holm $p = 0.599$).

Left-handers. Only 2 persons are left-handed, so they are shown individually. With augmentation, the LR scores 92.9 % on P11 and 68.0 % on P17; without mirroring, 85.7 % and 72.0 %. Two persons give no evidence either way. The 18 right-handers score 85.9 % [80.4, 90.6] with augmentation and 83.1 % [77.2, 88.4] without.

Decision. Keep augmentation (pre-specified). It builds in the symmetry that the physics predicts (Section 3), no option beats it, and LGBM is worse without it.

D.6. Learning curve

Design. Inside every outer fold, the default LR is trained on a random subset of 4, 8, 12 or 16 of the training persons, with 5 seeded repetitions per size; the held-out person is always scored (Figure 62, Table 54).

Result. The mean accuracy is 82.1 % (range 79.3 % to 84.4 %) with 4 training persons, 85.1 % (range 84.3 % to 86.0 %) with 16 and 85.5 % with all 19. The mean log loss falls from 0.458 to 0.391 and 0.365. From 16 persons to all, the accuracy changes by +0.3 points, within the range of the 16-person repetitions, while the log loss changes by −0.025, outside their range. The accuracy has flattened while the probabilities still sharpen: more persons would mainly improve the log loss. This reading is tentative: the full size is a single run, and the subsets share persons.

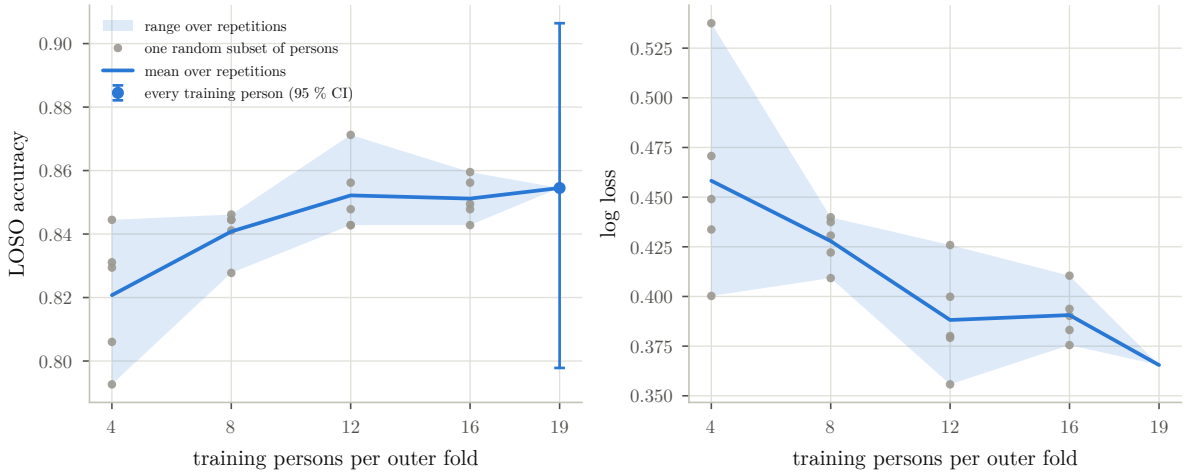


Figure 62: Learning curve of the default L2 LR: training persons subsampled inside every outer fold (5 seeded repetitions per size; band: range; line: mean); the held-out person is always scored. The right-most point uses every training person (one run, 95 % CI). Accuracy flattens; the log loss still falls.

Table 54: Lane m2: learning curve of the default L2 logistic regression (training persons subsampled inside every outer fold, seeded repetitions; min and max over the repetitions; the last row uses every training person, one run).

Training persons	Repetitions	Accuracy (mean)	Min	Max	Log loss (mean)	LL min	LL max	AUC
4	5	82.1%	79.3%	84.4%	0.458	0.400	0.538	0.886
8	5	84.1%	82.8%	84.6%	0.428	0.409	0.440	0.894
12	5	85.2%	84.3%	87.1%	0.388	0.356	0.426	0.912
16	5	85.1%	84.3%	86.0%	0.391	0.376	0.410	0.910
19	1	85.5%	85.5%	85.5%	0.365	0.365	0.365	0.920

D.7. What drives the predictions

Method. The importance of a feature group is its share of the mean absolute contribution to the logit on held-out persons, aggregated over the folds (Figure 63). For the LR the contribution is the coefficient times the scaled feature value; for LGBM it is the tree's additive contribution (TreeSHAP). A correlation feature is split equally between its two channels. These are associations on held-out persons, not causal effects.

Result. For the default LR, `xcorr` carries 40 %, `bins` 19 %, `stats` 17 %, `fft` 16 % and `cumint` 9 %; `wavelet` and `global` carry 0 % and 0 %, as augmentation implies. The leading channels are the yaw rate about gravity (18 %), the cumulative yaw (15 %), the device x acceleration (14 %) and the device z rotation rate (14 %). Odd features carry 89 %, swapped pairs 11 % and even features 0 %. For LGBM, `bins` (24 %) and `xcorr` (23 %) lead, the cumulative yaw is the top channel (21 %), and even features carry 27 %: trees can use them in interaction with odd ones. The yaw about gravity leads, as the physics predicts (Section 3; Appendix C).

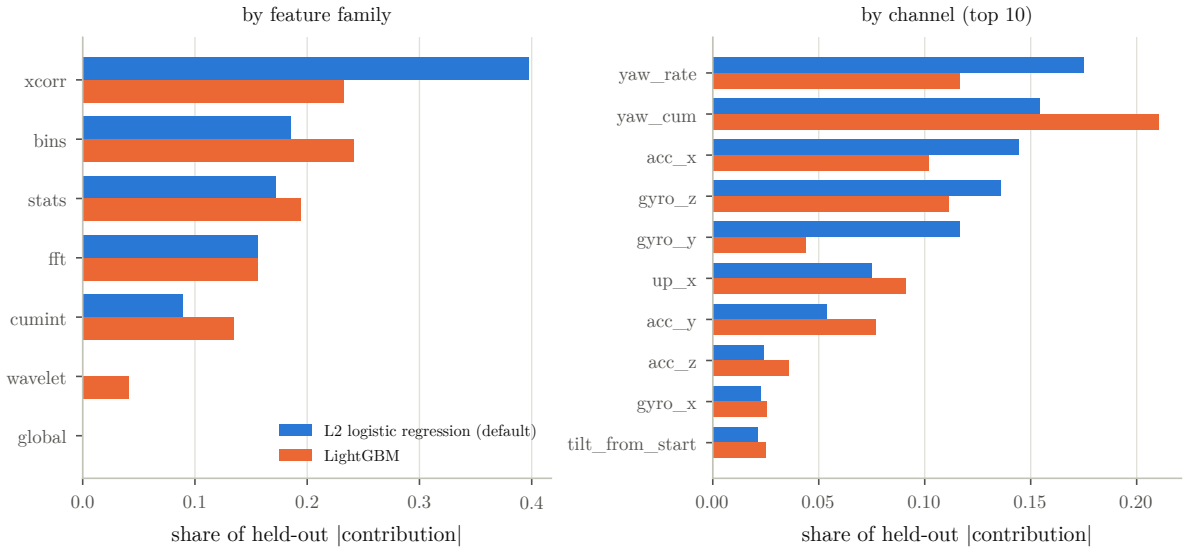


Figure 63: What drives the predictions on held-out persons: share of the mean absolute contribution to the logit (L2 LR: coefficient times scaled value; LGBM: TreeSHAP), by feature family (left) and by channel (right, top 10; a correlation pair is split equally between its channels). The yaw about gravity leads; the device-frame channels carry a large share.

Caveat: the device frame. The device-frame channels carry 60 % of the LR's contributions and 55 % of LGBM's. They depend on the grip. Lane final tests this by training on upright grips only (Appendix F): on the rotated grips the default LR falls to 51.3 % (−31.0 points, Holm $p < 0.001$), LGBM changes by −19.8 points (Holm $p = 0.014$) and the stack by −25.3 points (Holm $p = 0.013$), each Holm-adjusted within lane final's robustness family. Lane final therefore excludes these models as not robust.

D.8. Calibration

Method. The expected calibration error (ECE) uses 10 equal-mass bins. It is judged against its perfect-calibration null band: the ECE that outcomes drawn from Bernoulli(P(right)) would give with the same predictions (trials independent, so a lower bound on the noise). No percentile bootstrap CI of the ECE is used, because resampling inflates the ECE. The calibration slope and the CITL of a logistic recalibration come with person-bootstrap 95 % CIs; perfect calibration gives 1 and 0.

Detection. The nested selection has an ECE of 0.031 (null 97.5th percentile 0.050), a slope of 1.13 [0.84, 1.62] and a CITL of −0.12 [−0.45, +0.19]. The default LR has an ECE of 0.048 (null 97.5th percentile 0.052), a slope of 1.07 [0.79, 1.51] and a CITL of −0.03 [−0.28, +0.18]. LGBM has an ECE of 0.022 (null 97.5th percentile 0.049), a slope of 1.05 [0.80, 1.48] and a CITL of −0.12 [−0.46, +0.22]; the stack an ECE of 0.025 (null 97.5th percentile 0.046), a slope of 1.00 [0.75, 1.45] and a CITL of −0.11 [−0.45, +0.23] (Table 55; Figure 64). No model shows detectable miscalibration: every slope CI includes 1, every CITL CI includes 0 and every ECE lies within its null band.

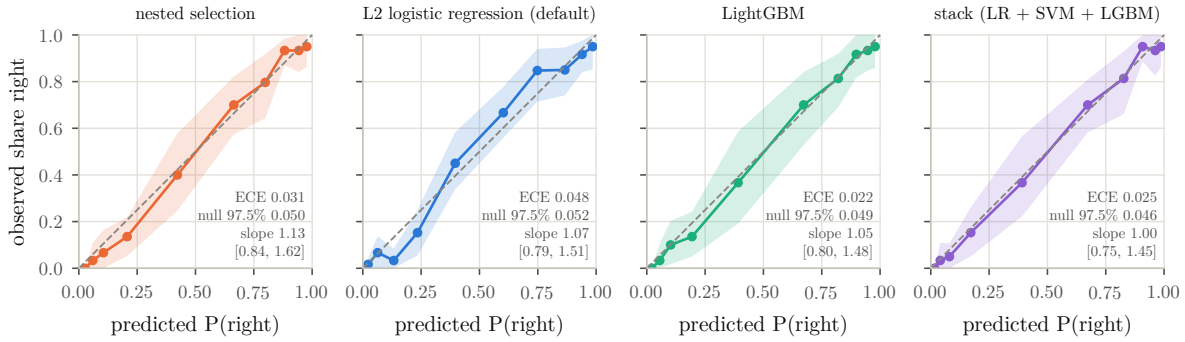


Figure 64: Reliability of the uncalibrated models on held-out persons (10 equal-mass bins; band: 95 % interval of the observed share; dashed: perfect calibration). Text: ECE with the 97.5th percentile of its perfect-calibration null, and the calibration slope with its 95 % CI. No model shows detectable miscalibration.

Remedies. Per model, 5 calibrators were tried: Platt scaling [21], temperature scaling [12], isotonic regression, beta calibration and a Platt map per platform. Each is fitted inside every outer fold on cross-fitted predictions of the training persons and applied to the held-out person. They are compared with the uncalibrated model by paired log-loss and Brier differences, Holm-adjusted over the 5 calibrators of a model (Figure 65). None lowers the log loss after the adjustment. The best for the selection is beta calibration (-0.006 [-0.021, $+0.008$], Holm $p = 1.000$), for the default LR beta calibration ($+0.000$ [-0.015, $+0.016$], Holm $p = 1.000$) and for LGBM temperature scaling (-0.004 [-0.016, $+0.007$], Holm $p = 1.000$). Decision: every model stays uncalibrated.

Platforms. For the selection, the Android persons (12) have a slope of 1.27 [0.88, 1.98], a CITL of -0.11 [-0.48, $+0.28$] and an ECE of 0.032 (null 97.5th percentile 0.063). The iOS persons (8) have a slope of 0.94 [0.57, 1.75], a CITL of -0.15 [-0.71, $+0.33$] and an ECE of 0.043 (null 97.5th percentile 0.079). A platform-specific Platt map against one global map changes the log loss by $+0.005$ [-0.009, $+0.020$] for the selection (Holm $p = 1.000$) and by $+0.013$ [$+0.002$, $+0.024$] for the default LR (Holm $p = 0.085$), adjusted over the 3 analysed models (Figure 66; Table 56). There is no evidence that the platforms need different maps, but the comparison is underpowered (8 iOS persons).

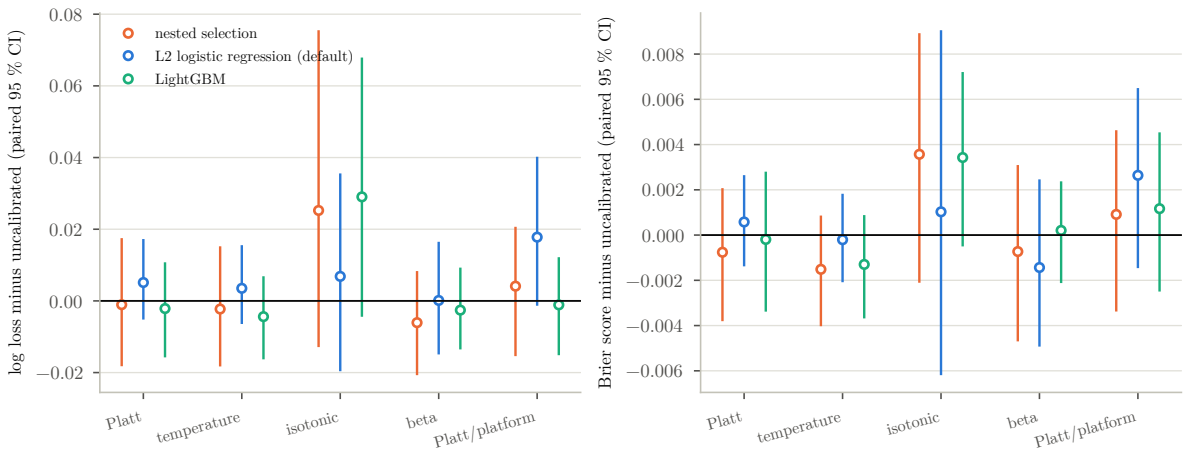


Figure 65: Post-hoc calibrators fitted inside every outer fold on cross-fitted predictions of the training persons: paired per-person differences against the uncalibrated model in log loss (left) and Brier score (right), 95 % CIs (below 0: better). Open markers: not significant after Holm over the 5 calibrators of a model, which is every case. Every model stays uncalibrated.

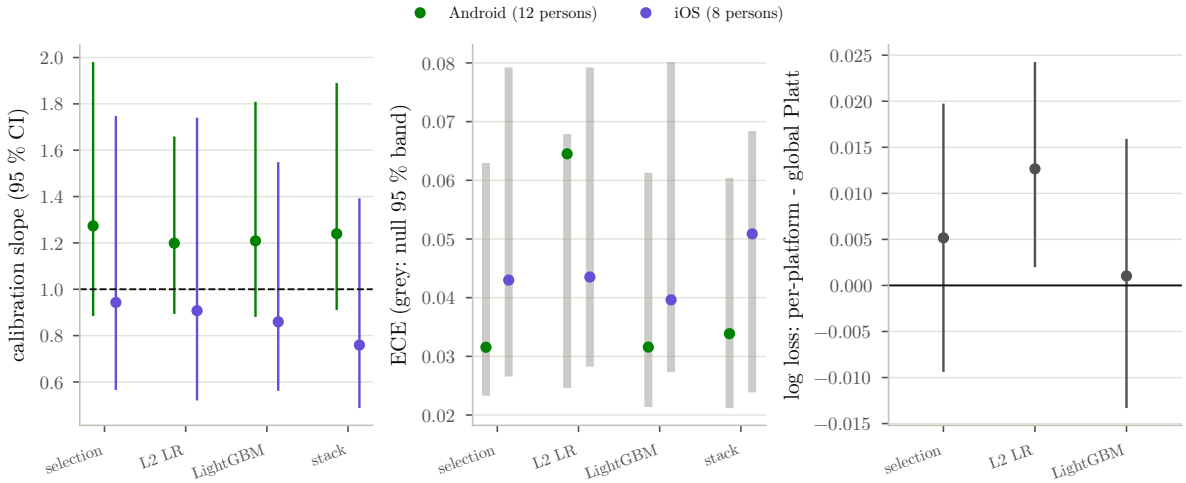


Figure 66: Calibration per platform (Android 12 persons, iOS 8): calibration slope with 95 % CI (left), ECE against its perfect-calibration null band (middle) and the paired log-loss difference of a per-platform Platt map against one global map (right; below 0: the per-platform map helps). No evidence for different maps; underpowered.

Table 55: Lane m2: calibration of the analysed models on held-out persons, uncalibrated and with each post-hoc calibrator fitted inside every outer fold on cross-fitted predictions of the training persons (inner person folds). ECE (10 equal-mass bins) is shown with the 97.5th percentile of the ECE that perfectly calibrated outcomes would give (y drawn from Bernoulli(p)); slope and calibration-in-the-large (CITL) of the logistic recalibration with person-bootstrap 95 % CIs (perfect: 1 and 0); log-loss and Brier differences paired against the uncalibrated model, p Holm-adjusted over the calibrators of a model.

Model	Calibrator	ECE	ECE null 97.5%	Slope [95% CI]	CITL [95% CI]	Brier	Log loss	Diff. log loss vs none	p (Holm)	Diff. Brier vs none
select	uncalibrated	0.031	0.050	1.13 [0.84, 1.62]	−0.12 [−0.45, +0.19]	0.099	0.334	–	–	–
select	Platt	0.028	0.046	0.97 [0.72, 1.40]	−0.04 [−0.42, +0.33]	0.098	0.333	−0.001 [−0.018, +0.018]	1.000	−0.0008 [−0.0038, +0.0021]
select	temperature	0.030	0.046	0.98 [0.73, 1.41]	−0.13 [−0.48, +0.22]	0.098	0.332	−0.002 [−0.018, +0.015]	1.000	−0.0015 [−0.0040, +0.0009]
select	isotonic	0.020	0.042	0.80 [0.56, 1.21]	−0.07 [−0.48, +0.37]	0.103	0.362	+0.025 [−0.013, +0.076]	1.000	+0.0036 [−0.0021, +0.0089]
select	beta	0.026	0.045	0.96 [0.71, 1.39]	−0.03 [−0.41, +0.37]	0.099	0.329	−0.006 [−0.021, +0.008]	1.000	−0.0007 [−0.0047, +0.0031]
select	Platt per platform	0.023	0.044	0.93 [0.71, 1.32]	−0.05 [−0.44, +0.32]	0.099	0.336	+0.004 [−0.015, +0.021]	1.000	+0.0009 [−0.0034, +0.0046]
lr_full	uncalibrated	0.048	0.052	1.07 [0.79, 1.51]	−0.03 [−0.28, +0.18]	0.110	0.365	–	–	–
lr_full	Platt	0.051	0.048	0.95 [0.70, 1.35]	+0.01 [−0.27, +0.26]	0.111	0.371	+0.005 [−0.005, +0.017]	1.000	+0.0006 [−0.0014, +0.0026]
lr_full	temperature	0.050	0.049	0.96 [0.70, 1.36]	−0.04 [−0.30, +0.20]	0.110	0.370	+0.004 [−0.006, +0.016]	1.000	−0.0002 [−0.0021, +0.0018]
lr_full	isotonic	0.043	0.045	0.87 [0.66, 1.17]	+0.03 [−0.26, +0.32]	0.111	0.374	+0.007 [−0.020, +0.036]	1.000	+0.0010 [−0.0062, +0.0091]
lr_full	beta	0.035	0.049	0.95 [0.70, 1.32]	+0.03 [−0.25, +0.30]	0.109	0.367	+0.000 [−0.015, +0.016]	1.000	−0.0014 [−0.0049, +0.0025]
lr_full	Platt per platform	0.058	0.048	0.90 [0.64, 1.30]	−0.00 [−0.30, +0.25]	0.112	0.382	+0.018 [−0.001, +0.040]	0.505	+0.0026 [−0.0015, +0.0065]
lgbm	uncalibrated	0.022	0.049	1.05 [0.80, 1.48]	−0.12 [−0.46, +0.22]	0.101	0.337	–	–	–
lgbm	Platt	0.022	0.048	1.02 [0.79, 1.44]	+0.01 [−0.36, +0.37]	0.101	0.334	−0.002 [−0.016, +0.011]	1.000	−0.0002 [−0.0034, +0.0028]
lgbm	temperature	0.027	0.047	1.04 [0.80, 1.46]	−0.12 [−0.46, +0.23]	0.100	0.332	−0.004 [−0.016, +0.007]	1.000	−0.0013 [−0.0037, +0.0009]
lgbm	isotonic	0.044	0.044	0.84 [0.60, 1.23]	−0.01 [−0.44, +0.44]	0.104	0.366	+0.029 [−0.004, +0.068]	0.789	+0.0034 [−0.0005, +0.0072]
lgbm	beta	0.019	0.048	1.01 [0.78, 1.41]	+0.01 [−0.36, +0.41]	0.102	0.334	−0.003 [−0.014, +0.009]	1.000	+0.0002 [−0.0021, +0.0024]
lgbm	Platt per platform	0.025	0.047	0.99 [0.78, 1.37]	−0.01 [−0.41, +0.37]	0.102	0.333	−0.001 [−0.015, +0.012]	1.000	+0.0012 [−0.0025, +0.0045]
ens_stack	uncalibrated	0.025	0.046	1.00 [0.75, 1.45]	−0.11 [−0.45, +0.23]	0.095	0.323	–	–	–

Table 56: Lane m2: calibration of the uncalibrated models per platform (statistics as in the calibration table, within the platform), and the paired log-loss difference of a platform-specific Platt map against one global Platt map (both fitted inside the outer folds; platform is known to production from the browser).

Model	Platform	Persons	Trials	ECE	ECE null 97.5%	Slope [95% CI]	CITL [95% CI]	Log loss	Brier	Diff. log loss
select	android	12	383	0.032	0.063	1.27 [0.88, 1.98]	−0.11 [−0.48, +0.28]	0.316	0.095	–
select	ios	8	215	0.043	0.079	0.94 [0.57, 1.75]	−0.15 [−0.71, +0.33]	0.366	0.106	–
select	per-platform vs global Platt	–	598	–	–	–	–	–	–	+0.005 [−0.009, +0.020]
lr_full	android	12	383	0.065	0.068	1.20 [0.89, 1.66]	+0.08 [−0.15, +0.33]	0.360	0.108	–
lr_full	ios	8	215	0.044	0.079	0.91 [0.52, 1.74]	−0.28 [−0.75, +0.16]	0.375	0.113	–
lr_full	per-platform vs global Platt	–	598	–	–	–	–	–	–	+0.013 [+0.002, +0.024]
lgbm	android	12	383	0.032	0.061	1.21 [0.88, 1.81]	−0.16 [−0.56, +0.28]	0.307	0.093	–
lgbm	ios	8	215	0.040	0.080	0.86 [0.56, 1.55]	−0.04 [−0.74, +0.56]	0.392	0.116	–
lgbm	per-platform vs global Platt	–	598	–	–	–	–	–	–	+0.001 [−0.013, +0.016]
ens_stack	android	12	383	0.034	0.060	1.24 [0.91, 1.89]	−0.09 [−0.41, +0.31]	0.294	0.088	–
ens_stack	ios	8	215	0.051	0.068	0.76 [0.49, 1.39]	−0.16 [−0.98, +0.51]	0.376	0.107	–

D.9. Breakdowns and persons

Strata. Conditions, handedness, flags and the predictor gates stratify the predictions only; they are never inputs. A stratum gets a CI only with at least 5 persons (Figure 67; Table 57). The strata are descriptive, not tests of the conditions. Android (12 persons) reaches 87.2 % [79.7, 92.9] and iOS (8) 85.1 % [75.8, 92.7]. The top-right grip reaches 78.9 % [68.4, 89.8], against 89.8 % [82.5, 96.1] for the top-left grip and 87.7 % [81.3, 92.9] upright. The lowest stratum with at least 5 persons is the twist variant: 66.7 % [33.3, 93.8] (15 trials, 8 persons). Trials that fail the predictor gates (28 trials of 14 persons) reach 92.9 % [84.8, 100.0], against 86.1 % [80.3, 91.3] for those that pass.

Small strata, person by person. The bent-start variant has 4 persons: P01 100.0 %, P09 0.0 %, P13 100.0 % and P15 100.0 %. The left-handers score 92.9 % (P11) and 72.0 % (P17). The trials flagged for a grip mismatch score 71.4 % (P05), 81.8 % (P09) and 100.0 % (P16); those flagged for possible clipping 71.4 % (P01) and 100.0 % (P20); the screen-rotated trials 100.0 % (P14).

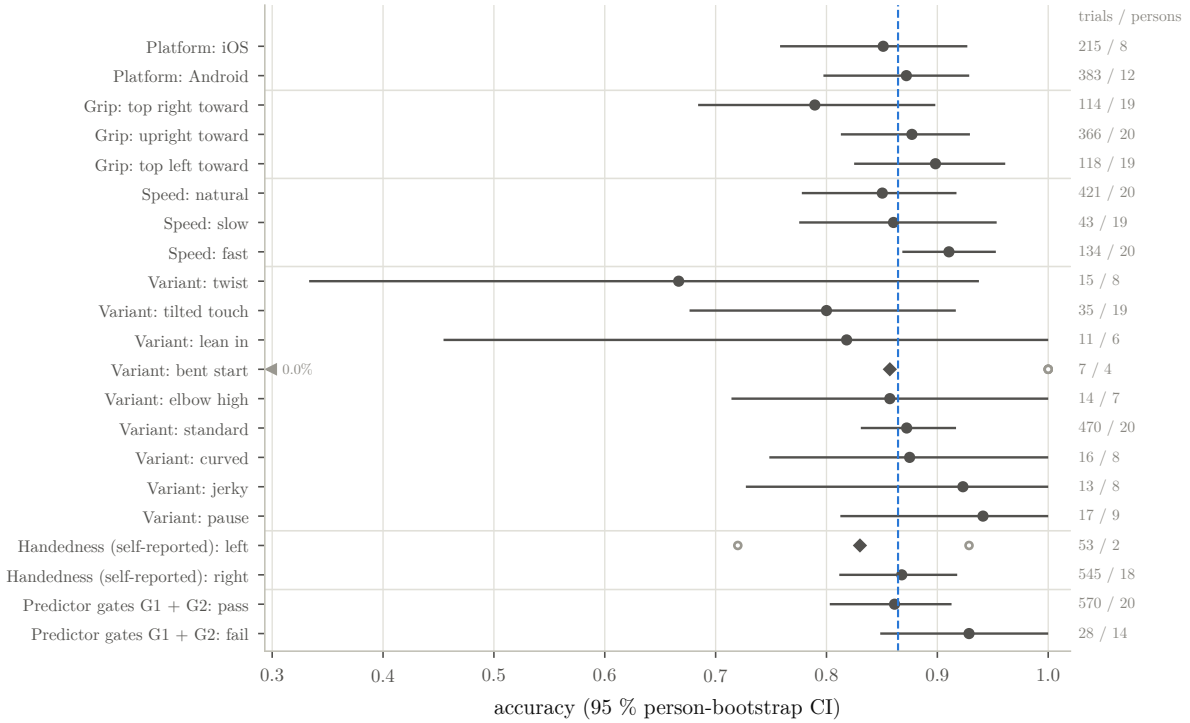


Figure 67: Accuracy of the nested selection by stratum (strata used after prediction only). 95 % CI only for strata with at least 5 persons; smaller strata show each person (open circles; an arrow marks a person below the axis) and the pooled value (diamond). Dashed: overall. Right: trials / persons.

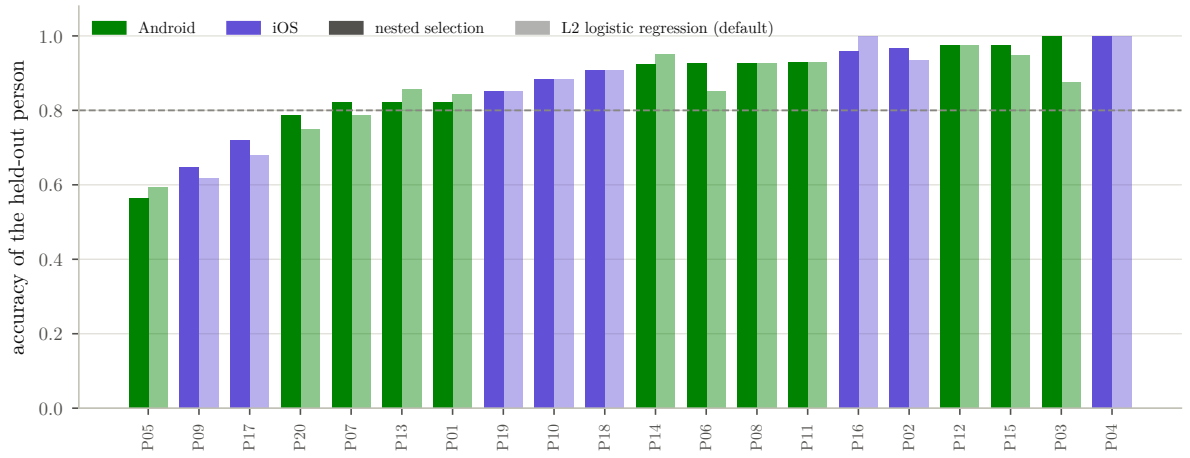


Figure 68: Accuracy of every held-out person, sorted by the nested selection (solid) with the default L2 LR (light); colour: platform; dashed: 80 %. 4 of 20 persons fall below 80 % under the nested selection.

Persons. Per-person accuracy, not the pooled mean, is what a new user experiences (Figure 68). Under the nested selection the median person scores 91.6 %, the lowest 56.2 % (P05), and 4 of 20 persons score below 80 %. The same person is lowest under the default LR (59.4 %, P05).

Table 57: Lane m2: accuracy of the headline model (m2_select) by stratum: conditions, handedness, flags and predictor gates stratify predictions only. 95 % person-bootstrap CI within the stratum, shown only with at least 5 persons.

Factor	Level	Trials	Persons	Accuracy	95% CI	Log loss
Platform	android	383	12	87.2%	[79.7, 92.9]	0.316
Platform	ios	215	8	85.1%	[75.8, 92.7]	0.366
Grip	top left toward	118	19	89.8%	[82.5, 96.1]	0.289
Grip	top right toward	114	19	78.9%	[68.4, 89.8]	0.436
Grip	upright toward	366	20	87.7%	[81.3, 92.9]	0.316
Speed	fast	134	20	91.0%	[86.8, 95.3]	0.249
Speed	natural	421	20	85.0%	[77.8, 91.7]	0.360
Speed	slow	43	19	86.0%	[77.5, 95.4]	0.340
Variant	bent start	7	4	85.7%	–	0.933
Variant	curved	16	8	87.5%	[74.8, 100.0]	0.386
Variant	elbow high	14	7	85.7%	[71.4, 100.0]	0.306
Variant	jerky	13	8	92.3%	[72.7, 100.0]	0.428
Variant	lean in	11	6	81.8%	[45.5, 100.0]	0.516
Variant	pause	17	9	94.1%	[81.2, 100.0]	0.236
Variant	standard	470	20	87.2%	[83.1, 91.7]	0.307
Variant	tilted touch	35	19	80.0%	[67.6, 91.7]	0.391
Variant	twist	15	8	66.7%	[33.3, 93.8]	0.615
Posture	sitting	438	15	86.3%	[78.5, 92.4]	0.352
Posture	standing	160	5	86.9%	[82.1, 92.3]	0.284
Start position	center	18	10	94.4%	[83.3, 100.0]	0.453
Start position	natural	563	20	86.5%	[80.7, 91.8]	0.329
Start position	shoulder	17	9	76.5%	[58.8, 93.8]	0.355
Handedness (self-reported)	left	53	2	83.0%	–	0.397
Handedness (self-reported)	right	545	18	86.8%	[81.2, 91.8]	0.328
Predictor gates G1 + G2	fail	28	14	92.9%	[84.8, 100.0]	0.283
Predictor gates G1 + G2	pass	570	20	86.1%	[80.3, 91.3]	0.336
Flag: grip mismatch	not flagged	575	20	86.6%	[81.1, 91.5]	0.325
Flag: grip mismatch	flagged	23	3	82.6%	–	0.543
Flag: segmentations disagree > 100 ms	not flagged	483	20	86.5%	[80.8, 91.9]	0.339
Flag: segmentations disagree > 100 ms	flagged	115	19	86.1%	[79.8, 92.0]	0.310
Flag: possible clipping	not flagged	590	20	86.6%	[81.1, 91.6]	0.334
Flag: possible clipping	flagged	8	2	75.0%	–	0.291
Flag: repeated samples dropped	not flagged	472	20	86.4%	[79.9, 92.1]	0.330
Flag: repeated samples dropped	flagged	126	20	86.5%	[80.3, 93.8]	0.348
Flag: false start	not flagged	470	20	86.8%	[80.9, 92.1]	0.322
Flag: false start	flagged	128	20	85.2%	[78.3, 92.4]	0.378
Flag: screen rotated	not flagged	596	20	86.4%	[80.9, 91.5]	0.334
Flag: screen rotated	flagged	2	1	100.0%	–	0.335

D.10. Several attempts and the displayed probability

Question. Production may combine k attempts of one person and hand. Two choices follow: which aggregate decides the hand, and which probability is shown. Here k held-out primary trials of the same person and hand stand in for k attempts. The analysis uses the 19 persons with at least 5 trials of each hand (Figure 69; Table 58).

Options. Four aggregates of the k attempts: the mean $P(\text{right})$, the mean logit, the sum of logits (attempts as independent evidence) and the tempered sum $\sum_i z_i/\tau_k$, where z_i is the

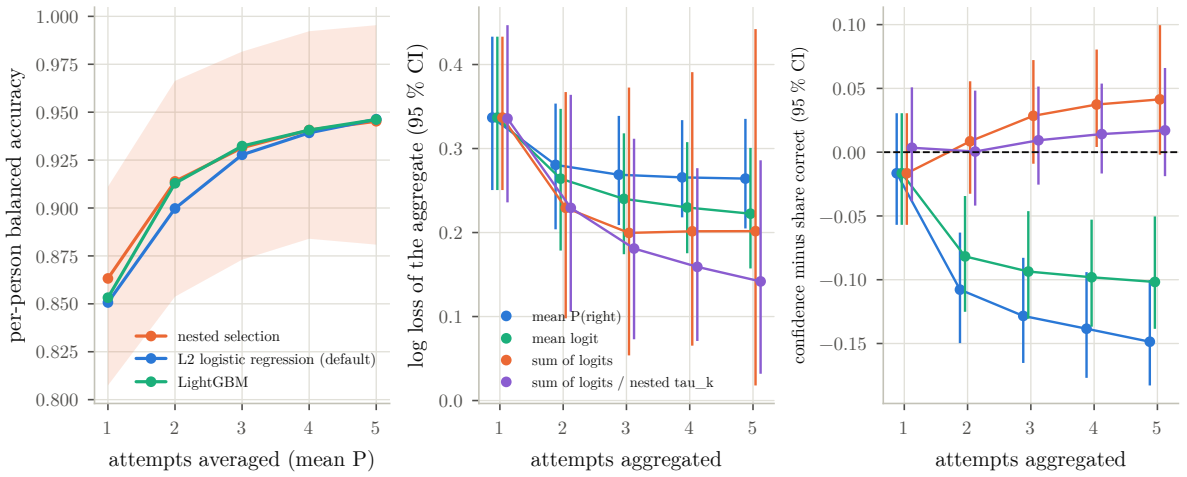


Figure 69: k attempts of one person and hand (19 persons). Left: per-person balanced accuracy of the mean $P(\text{right})$ (band: 95 % CI of the nested selection). Middle and right: log loss and confidence gap (above 0: over-confident) of four aggregates for the nested selection, with 95 % CIs. Decide with the mean $P(\text{right})$; display the tempered sum.

logit of attempt i. The scale τ_k is fitted inside each outer fold on cross-fitted predictions of the training persons, so the held-out person never informs its own τ_k.

Decision accuracy. Averaging $P(\text{right})$ raises the per-person balanced accuracy of the nested selection. It is 86.3 % [80.7, 91.1] with one attempt, 93.1 % [87.3, 98.2] with three and 94.5 % [88.1, 99.5] with five.

Displayed probability. At $k = 5$ the log loss is 0.264 [0.205, 0.335] for the mean $P(\text{right})$, 0.222 [0.157, 0.301] for the mean logit, 0.202 [0.018, 0.442] for the sum of logits and 0.142 [0.032, 0.286] for the tempered sum. The confidence gap (mean confidence minus share correct) is −0.149 [−0.183, −0.100], −0.102 [−0.138, −0.050], +0.041 [−0.002, +0.100] and +0.017 [−0.019, +0.066], respectively: the mean $P(\text{right})$ and the mean logit are under-confident, while the CIs of the two sums include 0. Over the three pairwise comparisons at $k = 5$ (Holm), the tempered sum has a lower log loss than the mean $P(\text{right})$ (−0.122 [−0.182, −0.048], Holm $p = 0.014$), but not than the plain sum (−0.060 [−0.172, +0.017], Holm $p = 0.531$). The median τ_5 is 1.83: five attempts carry about 2.7 independent attempts' worth of evidence.

Decision. Decide with the mean $P(\text{right})$ (the accuracy curve) and display the tempered sum, which has the lowest log loss. The aggregates were checked by the confidence gap only, without a slope, intercept or ECE null band, so this is weaker evidence than the single-attempt calibration above. Caveat: τ_k is fitted on inner predictions from models trained on fewer persons than the final ones. If those predictions are less confident, the tempered sum can be slightly over-confident on new persons. The production pipeline combines attempts with its own classifier and stopping rule (Appendix G).

Table 58: Lane m2: k attempts of one person and hand (m2_select): per-person balanced accuracy of the mean P(right) (95 % person-bootstrap CI), and the log loss and confidence gap (mean confidence minus share correct: above 0 over-confident, below 0 under-confident) of four aggregates; the tempered sum divides the sum of logits by tau_k fitted inside each outer fold on the training persons (median over folds shown).

k	Accuracy (mean P)	95% CI	LL mean P(right)	Gap mean P(right)	LL mean logit	Gap mean logit	LL sum of logits	Gap sum of logits	LL sum of logits / nested tau_k	Gap sum of logits / nested tau_k	tau_k (nested, median)
1	86.3%	[80.7, 91.1]	0.337	−0.017	0.337	−0.017	0.337	−0.017	0.336	+0.003	0.83
2	91.4%	[85.4, 96.6]	0.281	−0.108	0.264	−0.082	0.229	+0.009	0.229	+0.000	1.02
3	93.1%	[87.3, 98.2]	0.269	−0.128	0.240	−0.094	0.200	+0.029	0.181	+0.009	1.29
4	94.1%	[88.4, 99.2]	0.266	−0.138	0.230	−0.098	0.201	+0.037	0.159	+0.014	1.53
5	94.5%	[88.1, 99.5]	0.264	−0.149	0.222	−0.102	0.202	+0.041	0.142	+0.017	1.83

D.11. Caveats

- With 20 persons every CI is wide; differences of a point or two between pipelines are not resolved.
- Only the nested selection and the pre-specified default are headline-grade. The model comparison, the calibrators and the ablations are exploratory; every result records its number of configurations compared.
- The features use the oracle (app) segmentation. Production sees detected windows, and the approach features inherit any segmentation error there.
- The free-choice trials use one hand only: they give a right-hand recall, not a calibration or accuracy check over both hands.
- The experiments use 2000 person resamples, the derived intervals (strata, platforms, slopes) 1000 and the multi-attempt intervals 300.
- Open: does per-trial amplitude normalisation hold on detected windows?

E. Time-frequency and deep sequence models

Lane m3 tests whether models that learn their own representation of the motion beat engineered features on new persons, and whether they use the same evidence as the physics of Section 3: the hand-signed yaw about gravity early in the approach. The lane trained one-dimensional convolutional neural networks (1D CNN) and four other deep sequence architectures, two-dimensional (2D) CNNs and logistic regressions on time-frequency planes, and MiniRocket [20]. All models used the 598 primary trials of the 20 persons under leave-one-person-out (LOSO) evaluation (Appendix A). The first free-choice trial of each person was scored apart as the natural-use set. This appendix states every model and every data-treatment choice with its paired evidence, then the calibration, the attribution analysis, the breakdowns and the caveats. The network that became the final model comes from this lane; Appendix F describes how it was chosen among the candidates of all lanes, calibrated and exported.

E.1. Models, training and statistics

Model families. The lane compared four families (Table 59, Figure 70):

- deep sequence models on the time-normalised approach: a 1D CNN (the pre-specified default), a residual 1D network (ResNet1D), a reduced InceptionTime (InceptionTime-lite), a temporal convolutional network (TCN) and a gated recurrent unit network (GRU);
- MiniRocket: 84 fixed convolution kernels with dilations on a log grid and 4000 pooled features, followed by an L2-penalised logistic regression whose penalty was chosen on inner person-disjoint folds;
- time-frequency models: the short-time Fourier transform (STFT; Hann window of 16 samples, hop of 4) or the complex-Morlet continuous wavelet transform (CWT; 10 scales from 2 to 32 samples) of each channel, keeping the real and imaginary parts, either pooled into frequency bands and time blocks for a logistic regression or fed to a small 2D CNN;
- magnitude-only time-frequency models, kept only as a physics demonstration (Appendix E.3).

Input channels. The default input is the 8 gravity-frame channels: the magnitudes of $\mathbf{a}$ and $\boldsymbol{\omega}$, the vertical and horizontal parts of $\mathbf{a}$, the horizontal part of $\boldsymbol{\omega}$, the yaw rate $\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$, the cumulative yaw and the tilt from the start pose. Only the two yaw channels are chiral. Two richer sets were tried. "All 17" adds the 9 device-frame components of $\mathbf{a}$, $\boldsymbol{\omega}$ and $\hat{\mathbf{g}}$ in phone coordinates, whose values depend on the grip. "Gravity + body frame (12)" adds four channels of a heading-following body frame. With $\mathbf{n}$ the unit normal out of the screen, the frame is

$$\mathbf{h} = \frac{\mathbf{n} - (\mathbf{n} \cdot \hat{\mathbf{g}})\,\hat{\mathbf{g}}}{\|\mathbf{n} - (\mathbf{n} \cdot \hat{\mathbf{g}})\,\hat{\mathbf{g}}\|}, \qquad \mathbf{s} = \hat{\mathbf{g}} \times \mathbf{h}. \tag{6}$$

It gives the roll rate $\boldsymbol{\omega} \cdot \mathbf{h}$ and the lateral acceleration $\mathbf{a} \cdot \mathbf{s}$, which are chiral, and the pitch rate $\boldsymbol{\omega} \cdot \mathbf{s}$ and the forward acceleration $\mathbf{a} \cdot \mathbf{h}$, which are mirror-even. A rotation of the phone about its screen normal leaves all four unchanged, so the 12-channel set does not depend on the grip. The frame follows lane m1's definition (Appendix C); it is undefined only when the screen faces straight up or down, which did not occur in the recorded trials.

Training. Every model was fitted through the shared LOSO harness with its cap on training trials per person (Appendix A). Inside each outer fold the training persons were split into 4 person-disjoint parts. A deep model is a 4-member ensemble: member j trains on the other three parts and is early-stopped on part j (log loss on real, unaugmented trials; patience 6; 6 to 40 epochs). Its logits on part j give every training trial a cross-fitted logit, made by a member that never saw that trial's person. The calibrators and the attempt scale c_k (Appendix E.8) were fitted on these cross-fitted logits, never on the held-out person. The ensemble logit is the mean of the member logits. Each member fits its channel normaliser on its training trials and their mirror images, so every chiral channel is centred at exactly zero and normalising commutes with the mirror $m(\cdot)$. Training was augmented with the mirror image of every trial with the flipped label (Equation 1), a smooth monotone time warp of up to 0.3 of the window, magnitude scaling between 0.90 and 1.10, and jitter with a standard deviation of 0.03 [22]. The network heads keep coarse time position (8 time bins), because the early yaw reverses later in the approach. Every fit ran on one thread with deterministic algorithms, so a refit reproduces the saved predictions bit for bit.

Statistics. Every comparison of two models uses the same trials and one estimand: the mean per-person accuracy difference in percentage points, with a paired person-bootstrap 95 % confidence interval (CI) and an exact sign-flip p-value over persons. The p-values are Holm-adjusted within a named family of comparisons: `treatment` (16 comparisons), `treatment_body` (3), `grip` (9), `nested` (3), `equal_input` (3), `engineered` (18) and `physics_demo` (4). A difference is called better or worse only when the adjusted p-value is below 0.05 and the CI excludes 0; otherwise it is called not detectable. The per-person log-loss difference is tested in the same way. Appendix E.5 gives the calibration statistics.

Table 59: Lane m3 models, leave-one-person-out (95 % person-cluster bootstrap CI). Natural use: accuracy on the answered first free-choice trials (one per person, all right-handed: the recall of the right hand; a P(right) of exactly 0.5 is an abstention). $k = 3$: per-person balanced accuracy when the P(right) of 3 attempts are averaged. CPU min: training cost of one LOSO run (a nested choice: all its candidates). Configs: configurations compared on the outer folds before keeping this one (1 = pre-specified, a comparison or a check). Selection: how the configuration came to be; post hoc = designed after outer-fold results had been seen.

Model	Accuracy	95% CI	AUC	Log loss	ECE	Natural use	k = 3	CPU min	Configs	Selection
1D CNN	81.9%	[77.0, 86.9]	0.873	0.459	0.042	70%	89.0%	7.0	1	pre-specified
1D CNN, antisymmetric	81.1%	[76.2, 86.0]	0.876	0.455	0.052	85%	89.6%	6.3	1	comparison
GRU	79.6%	[73.6, 85.6]	0.864	0.472	0.035	75%	87.8%	20.1	1	pre-specified
InceptionTime-lite	80.8%	[75.6, 86.3]	0.871	0.461	0.048	80%	88.6%	18.4	1	pre-specified
ResNet1D	81.1%	[75.2, 86.7]	0.874	0.456	0.044	75%	88.7%	9.6	1	pre-specified
TCN	79.4%	[74.1, 85.0]	0.874	0.453	0.048	75%	88.0%	8.8	1	pre-specified
1D CNN, all 17 channels	88.3%	[84.6, 92.1]	0.945	0.308	0.032	90%	95.3%	9.6	1	comparison
1D CNN, mirror only	80.4%	[75.2, 85.7]	0.867	0.473	0.039	80%	88.3%	5.9	1	comparison
1D CNN, no augmentation	80.4%	[75.4, 85.4]	0.850	0.496	0.057	70%	87.8%	3.3	1	comparison
1D CNN, gravity + body frame (12)	89.1%	[84.8, 93.1]	0.951	0.290	0.029	85%	96.2%	7.8	16	best of 16, post hoc
1D CNN, chiral channels	82.3%	[77.3, 87.0]	0.881	0.439	0.038	80%	90.2%	8.5	1	comparison
1D CNN, fixed 3-s window	82.8%	[77.9, 88.0]	0.886	0.430	0.039	85%	89.2%	19.2	1	comparison
1D CNN, global-average head	78.9%	[73.4, 84.5]	0.869	0.460	0.044	75%	87.9%	8.4	1	comparison
1D CNN, no scaling	81.8%	[77.8, 85.9]	0.884	0.437	0.033	85%	89.6%	7.6	1	comparison
1D CNN, robust scaling	80.3%	[74.5, 85.7]	0.866	0.475	0.049	75%	87.9%	7.4	1	comparison
1D CNN, per-trial amplitude	83.1%	[77.5, 88.4]	0.876	0.441	0.044	85%	89.7%	6.8	1	comparison
1D CNN, body frame, antisymmetric	89.5%	[85.7, 92.7]	0.957	0.274	0.024	85%	97.2%	7.3	1	check, post hoc
1D CNN, body frame, fixed 3-s window	89.3%	[86.4, 92.2]	0.959	0.264	0.036	85%	97.1%	21.7	1	check, post hoc
1D CNN, body frame, per-trial amplitude	91.0%	[87.4, 94.2]	0.961	0.253	0.008	75%	97.3%	7.7	1	check, post hoc
1D CNN, all 17 channels, trained on upright grips only	76.8%	[72.3, 80.9]	0.786	0.927	0.150	90%	85.9%	6.9	1	check
1D CNN, gravity + body frame, trained on upright grips only	88.8%	[84.1, 93.3]	0.941	0.332	0.023	80%	95.4%	6.5	1	check, post hoc
deep, architecture chosen in every fold	80.8%	[74.7, 86.7]	0.871	0.467	0.044	85%	88.2%	70.2	1	nested
deep, configuration chosen in every fold	87.8%	[83.5, 91.7]	0.946	0.306	0.020	85%	95.4%	154.6	1	nested, post hoc
deep, grip-invariant configuration chosen in every fold	89.1%	[84.8, 93.1]	0.951	0.290	0.029	85%	96.2%	145.0	1	nested, post hoc
MiniRocket + logistic	79.8%	[76.1, 83.7]	0.865	0.477	0.049	75%	89.5%	3.8	1	pre-specified
MiniRocket, antisymmetric	81.4%	[77.3, 85.7]	0.881	0.446	0.047	70%	90.4%	2.1	1	comparison
MiniRocket, chiral channels	81.8%	[77.0, 86.5]	0.879	0.444	0.054	80%	89.7%	3.4	1	comparison
MiniRocket, per-trial amplitude	82.8%	[77.7, 87.6]	0.881	0.436	0.053	80%	90.1%	3.4	4	best of 4
2D CNN on STFT (signed)	82.8%	[78.4, 87.1]	0.871	0.479	0.073	80%	89.0%	10.6	3	best of 3
CWT signed + logistic	73.1%	[67.3, 78.9]	0.804	0.553	0.051	75%	82.3%	1.2	1	comparison
CWT magnitude + logistic	50.0%	[50.0, 50.0]	0.500	0.693	0.005	–	50.0%	0.1	1	demonstration
CWT magnitude + logistic, no mirror	50.8%	[47.0, 55.0]	0.515	0.697	0.048	15%	50.8%	0.5	1	demonstration
STFT signed + logistic	80.6%	[75.8, 85.7]	0.859	0.510	0.090	80%	88.3%	0.5	1	pre-specified
STFT magnitude + logistic	50.0%	[50.0, 50.0]	0.500	0.693	0.005	–	50.0%	0.1	1	demonstration
STFT magnitude + logistic, no mirror	50.8%	[48.0, 53.7]	0.505	0.696	0.054	5%	52.8%	0.4	1	demonstration

E.2. Selection: configurations tried and nested estimates

Each role's finalist is the configuration with the highest LOSO accuracy among the configurations of its role. The choice was made on the outer folds, so each finalist's own LOSO number is exploratory; Table 59 gives the number of configurations compared before each one was kept.

- Deep finalist: the 1D CNN on the gravity and body frames (12 channels), best of 16 deep configurations, shipped with Platt calibration. Its body-frame input was designed after the outer-fold results of the device-frame CNN had been seen (post hoc; Appendix E.10). Accuracy 89.3 % [85.0, 93.1], area under the receiver operating characteristic curve (AUC) 0.951 [0.912, 0.980], log loss 0.291 [0.196, 0.420].

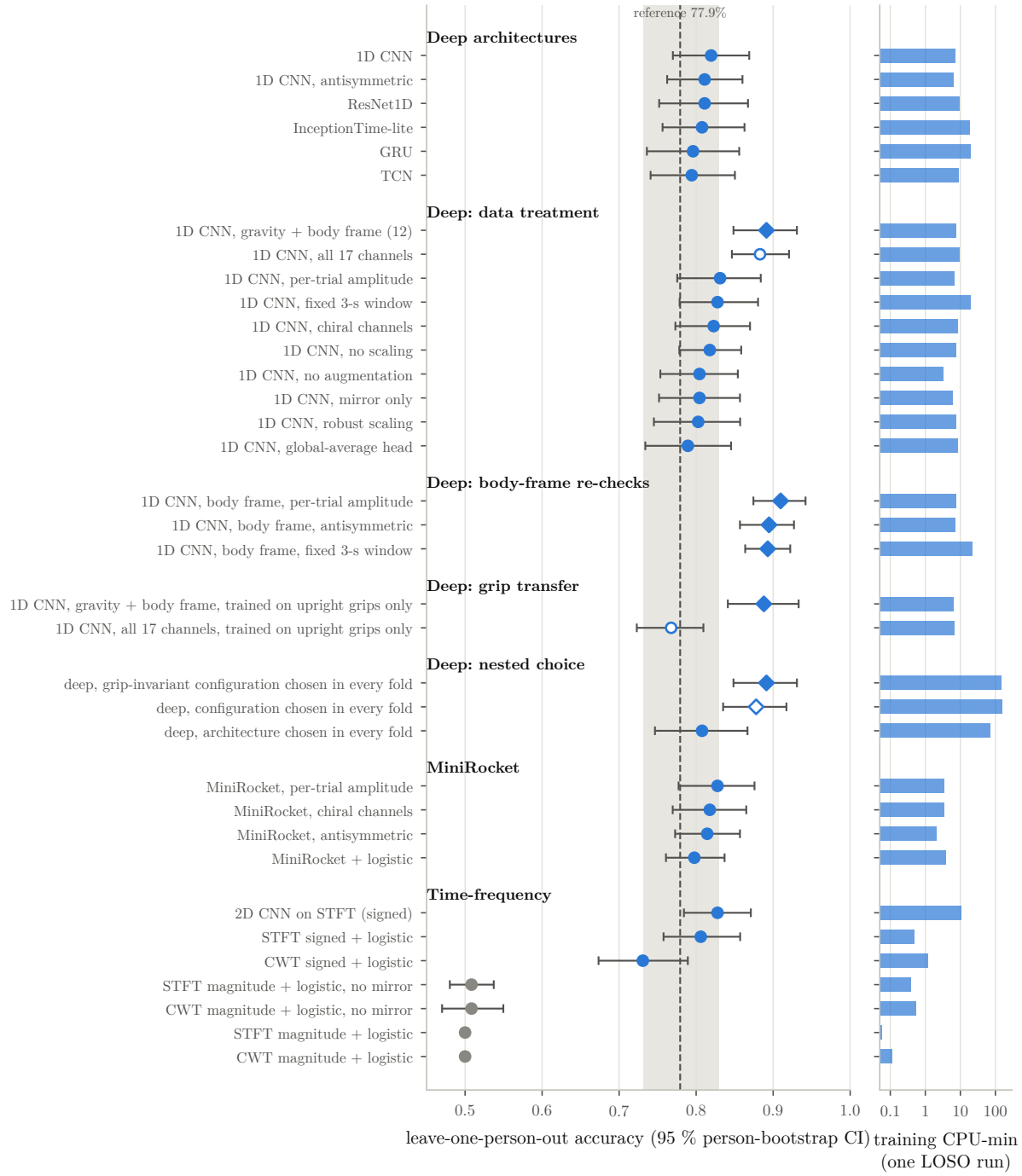


Figure 70: Every lane-m3 model under LOSO. Left: accuracy with its 95 % CI; dashed line and band: the reference yaw logistic regression (77.9 %). Right: training cost of one LOSO run in CPU-minutes (log scale; for a nested choice, all its candidates). Grey: magnitude-only demonstrations; open markers: grip-dependent device-frame input; diamonds: configurations designed after outer-fold results had been seen.

- MiniRocket finalist: per-trial amplitude scaling, best of 4: 82.3 % [77.4, 86.8].
- Time-frequency finalist: the 2D CNN on the signed STFT planes, best of 3: 83.1 % [79.2, 87.4].

A grip rule was declared before the body-frame results existed: a grip-dependent winner is replaced by the most accurate grip-invariant configuration unless their paired difference excludes 0. It did not act, because the most accurate deep configuration was already grip-invariant.

Three nested estimates repeat the deep choice inside every outer fold, by inner cross-validated log loss on the training persons only (Table 59; family `nested` in Table 60):

- among the 15 grip-invariant configurations, the shipping policy: 89.1 % [84.8, 93.1]. It chose the body-frame CNN in 20 of 20 folds, so its predictions are identical to that network's;
- among all 16 configurations: 87.8 % [83.5, 91.7]. It chose a grip-dependent configuration in 5 folds; the gap to the grip-invariant estimate comes from admitting device-frame inputs, not from the selection itself;
- among the 6 pre-specified gravity-frame configurations (the five architectures and the antisymmetric CNN): 80.8 % [74.7, 86.7], not detectably different from the default CNN (−0.8 points [−2.8, +1.5], Holm-adjusted $p = 0.501$).

The grip-invariant nested estimate is the lane's selection-aware deep number. Neither nested estimate removes the optimism of the post hoc body-frame design, because the body-frame candidate entered the candidate set after outer-fold results were known.

On natural use, the deep finalist's right-hand recall on the first free-choice trial of each person is 85 % [62, 97] (17 of 20; exact binomial CI [26]). These trials are all right-hand trials so far, so a constant 'right' answer would score 100 %. The model answered every one of them (coverage 100 %), so the answered and forced-choice recalls coincide.

E.3. Time-frequency inputs must keep the sign

Options. A time-frequency model can take the magnitude of each plane (spectrogram, scalogram) or the signed planes (real and imaginary parts).

Evidence. Under the mirror (Equation 1) a chiral channel only changes sign, $x \to -x$. Both transforms are linear, so the magnitude is unchanged: $|\mathrm{STFT}(-x)| = |\mathrm{STFT}(x)|$, and likewise for the CWT. Over every chiral channel of all primary trials, the largest magnitude difference between a trial and its mirror image is 0.0e+00 for the STFT and 0.0e+00 for the CWT, while the real parts change sign (Figure 71a–d). A magnitude-only input therefore sees a trial and its mirror image as one input. With mirror augmentation it receives every input twice with opposite labels, and it returns $P(\text{right}) = 0.5$, an abstention, on every trial. Its coverage is therefore zero (natural use: 0 % answered), its AUC is 0.500, and its accuracy with each abstention scored as half right, the expected forced-choice accuracy, is 50.0 %. Without mirror augmentation it can use only what is not mirror-symmetric between the hands, which does not transfer to new persons: AUC 0.505 [0.452, 0.558]. The signed STFT logistic regression reaches an AUC of 0.859 [0.812, 0.904] and the signed 2D CNN 0.871 [0.827, 0.913] (Figure 71e).

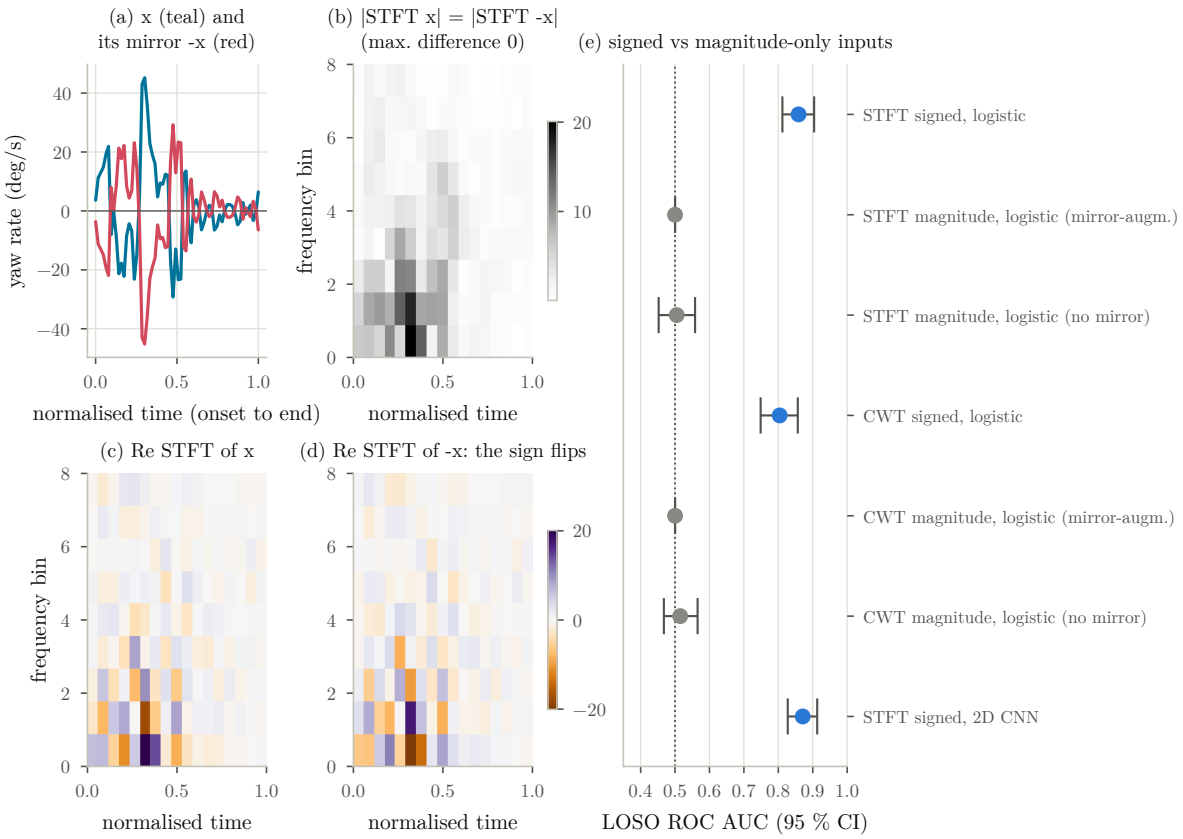


Figure 71: Why magnitude-only time-frequency inputs cannot see the hand. (a) Yaw rate of the median right-hand approach and its mirror image; (b) their STFT magnitudes are identical; (c, d) the real parts change sign. (e) LOSO AUC with its 95 % CI of signed and magnitude-only models: with mirror augmentation a magnitude-only model abstains on every trial, without it the model is at chance; the signed planes keep the hand.

Against its signed twin, the mirror-augmented magnitude model loses −31.4 points [−36.4, −26.6] of accuracy ($p < 0.001$, Holm over the `physics_demo` family); the CWT pairs behave alike (Table 60).

Decision and rationale. Every time-frequency model of the lane keeps the sign. The hand is carried by the sign of the chiral channels, and a magnitude discards it by construction.

E.4. Data treatment and model choices

Each option was paired against its family default on the same trials: the gravity-frame 1D CNN, MiniRocket on z-scored channels, or the signed STFT logistic regression (Figure 72, Table 60). An option replaces its default only when it is better after the Holm adjustment, and it is rejected when it is worse by the same rule; otherwise the default is kept. A log loss that is lower after adjustment marks a candidate improvement, not an adopted one. The finalists are a separate, exploratory choice (Appendix E.2), so a finalist can use an option whose comparison

with the default is not detectable. Upstream preprocessing (the 60 Hz grid, the low-pass filter, gap bridging and exclusions) was not varied in this lane; Appendix B covers it.

Channels. *Options:* the 8 gravity-frame channels (default), the 2 chiral channels alone, all 17 channels, and gravity + body frame (12). *Evidence:* the body frame adds +7.7 points [+4.4, +11.6] ($p < 0.001$, Holm, family `treatment`) and changes the per-person log loss by −0.174 [−0.223, −0.120]. All 17 channels add +6.1 points [+2.5, +9.8], also better after adjustment (Holm-adjusted $p = 0.047$). The 2 chiral channels alone are not detectably different from the 8 (+0.0 points [−1.8, +2.1]), so the even gravity-frame channels add nothing detectable to the yaw. *Decision:* gravity + body frame. *Rationale:* it is the most accurate set and it does not depend on the grip, whereas the device-frame channels of the 17-channel set carry evidence tied to phone coordinates that collapses on grips absent from training (Appendix E.10).

Window. *Options:* the time-normalised approach between the stored onset and end (default), or a fixed 3-s window in absolute time with a mask channel (augmented by a shift of up to 8 samples and a speed change of up to 15 % about the onset). *Evidence:* +1.3 points [−0.9, +3.5] on the gravity frame and −0.0 points [−2.2, +2.3] on the body frame, both not detectable. *Decision:* keep the time-normalised approach for model comparison. *Rationale:* no detectable difference. The fixed window needs no end bound, which matters when no cue marks the movement (Appendix G); cue-free bounds were not evaluated in this lane.

Normalisation. *Options:* a z-score per channel (default), no scaling, the median and interquartile range (IQR) winsorised at 5, and per-trial root-mean-square (RMS) amplitude. *Evidence:* on the gravity frame none differs detectably from the z-score (−0.0, −1.1 and +1.5 points). On the body frame, per-trial RMS gives +1.7 points [+0.4, +3.0] (unadjusted $p = 0.023$, Holm-adjusted 0.070, family `treatment_body`), and it changes the per-person log loss by −0.036 [−0.054, −0.017] (Holm-adjusted $p = 0.005$), a lower log loss after adjustment. This check is the most accurate model of the lane, 91.0 % [87.4, 94.2], but it was run post hoc as a re-check, not as a candidate. *Decision:* keep the z-score; per-trial RMS on the body frame is a candidate improvement for a future nested selection. *Rationale:* adopting it now would be one more choice made on these outer folds.

Augmentation. *Options:* the full set (mirror, time warp, scaling and jitter; default), mirror only, and none. *Evidence:* −0.8 points [−2.7, +1.1] without augmentation and −0.6 points [−2.9, +1.8] with the mirror only, both not detectable. *Decision:* keep the full set. *Rationale:* it costs no detectable accuracy, and the mirror term keeps the network close to mirror-equivariant, which the attribution analysis uses (Appendix E.6).

Network head. *Options:* a head that keeps 8 time bins (default) or global average pooling, which discards time position. *Evidence:* global pooling gives −2.8 points [−4.8, −0.7] (unadjusted $p = 0.018$, Holm-adjusted 0.229): not detectable after adjustment. *Decision:* keep the time-binned head. *Rationale:* the sign of the yaw reverses later in the approach (Section 3), so the position in time carries information; the unadjusted direction agrees.

Symmetry. *Options:* an unconstrained network trained with mirror augmentation (default), or an antisymmetric network with logit $z(x) = g(x) - g(m(x))$, which gives $P(\text{right} \mid m(x)) = 1 - P(\text{right} \mid x)$ exactly and makes mirror augmentation redundant [24, 27]. *Evidence:* −0.2 points [−2.1, +2.0] on the gravity frame; on the body frame +0.3 points [−1.2, +1.7] (Holm-adjusted $p = 1.000$), with a per-person log-loss change of −0.017 [−0.028, −0.005] (Holm-adjusted $p = 0.023$), a lower log loss after adjustment. *Decision:* keep the default; the antisymmetric body-frame network is a second candidate improvement. *Rationale:* as for per-trial RMS, adopting it now would be one more outer-fold choice.

MiniRocket. *Options:* z-scored channels (default), per-trial RMS amplitude, the chiral channels only, and antisymmetric features $\phi(x) - \phi(m(x))$. *Evidence:* +3.5 points [−0.6, +8.3], +2.5 points [−1.1, +6.1] and +1.6 points [−1.4, +4.2], none detectable. *Decision:* the default stays the comparison baseline; the MiniRocket finalist uses per-trial RMS as the most accurate of its configurations (exploratory).

Time-frequency transform and model. *Options:* the STFT or the CWT; a logistic regression on pooled bands or a 2D CNN on the planes. *Evidence:* the CWT is worse than the STFT by −9.2 points [−15.3, −4.3] (Holm-adjusted $p = 0.022$); the 2D CNN is not detectably better than the logistic regression (+1.9 points [−0.1, +3.7], Holm-adjusted $p = 0.877$). *Decision:* the STFT; the time-frequency finalist is the 2D CNN as the most accurate of its configurations (exploratory).

Re-checks on the body frame. The window, normalisation and symmetry checks were repeated on the body-frame CNN (family `treatment_body`; results above). The augmentation and head decisions were not repeated; the body-frame CNN inherits them from the gravity-frame CNN as an assumption.

Table 60: Data treatment and model choices of lane m3, each paired against its default on the same trials: mean per-person accuracy difference with its paired person-bootstrap 95 % CI, exact sign-flip p over persons and the p Holm-adjusted within the family. Verdict: better / worse only when the adjusted p is below 0.05 and the CI excludes 0; identical: the same P(right) on every trial.

Family	Comparison	Against	Accuracy	Diff.	95% CI	p	p (Holm)	Log-loss diff.	Verdict
treatment	antisymmetric g(x) - g(mirror x)	1D CNN	81.1%	−0.2%	[−2.1, 2.0]	0.890	1.000	−0.005	no detectable difference
treatment	fixed 3 s (absolute time, mask channel)	1D CNN	82.8%	+1.3%	[−0.9, 3.5]	0.275	1.000	−0.034	no detectable difference
treatment	all 17	1D CNN	88.3%	+6.1%	[2.5, 9.8]	0.003	0.047	−0.153	better
treatment	chiral only (2)	1D CNN	82.3%	+0.0%	[−1.8, 2.1]	0.976	1.000	−0.017	no detectable difference
treatment	gravity + body frame (12, grip-invariant)	1D CNN	89.1%	+7.7%	[4.4, 11.6]	0.000	0.000	−0.174	better
treatment_body	fixed 3 s (absolute time, mask channel)	1D CNN, gravity + body frame (12)	89.3%	−0.0%	[−2.2, 2.3]	0.989	1.000	−0.026	no detectable difference
treatment_body	per-trial RMS	1D CNN, gravity + body frame (12)	91.0%	+1.7%	[0.4, 3.0]	0.023	0.070	−0.036	no detectable difference
treatment_body	antisymmetric g(x) - g(mirror x)	1D CNN, gravity + body frame (12)	89.5%	+0.3%	[−1.2, 1.7]	0.732	1.000	−0.017	no detectable difference
treatment	none (raw units)	1D CNN	81.8%	−0.0%	[−1.5, 1.5]	0.977	1.000	−0.020	no detectable difference
treatment	median/IQR, winsorised at 5	1D CNN	80.3%	−1.1%	[−3.0, 0.9]	0.291	1.000	+0.018	no detectable difference
treatment	per-trial RMS	1D CNN	83.1%	+1.5%	[−0.3, 3.4]	0.157	1.000	−0.017	no detectable difference
treatment	none	1D CNN	80.4%	−0.8%	[−2.7, 1.1]	0.439	1.000	+0.035	no detectable difference
treatment	mirror only	1D CNN	80.4%	−0.6%	[−2.9, 1.8]	0.625	1.000	+0.011	no detectable difference
treatment	global average (no time position)	1D CNN	78.9%	−2.8%	[−4.8, −0.7]	0.018	0.229	+0.003	no detectable difference
treatment	phi(x) - phi(mirror x)	MiniRocket + logistic	81.4%	+1.6%	[−1.4, 4.2]	0.285	1.000	−0.032	no detectable difference
treatment	chiral only (2)	MiniRocket + logistic	81.8%	+2.5%	[−1.1, 6.1]	0.209	1.000	−0.035	no detectable difference
treatment	per-trial RMS	MiniRocket + logistic	82.8%	+3.5%	[−0.6, 8.3]	0.162	1.000	−0.041	no detectable difference
physics_demo	magnitude only, mirror-augmented	STFT signed + logistic	50.0%	−31.4%	[−36.4, −26.6]	0.000	0.000	+0.192	worse
physics_demo	magnitude only, no mirror augmentation	STFT signed + logistic	50.8%	−30.0%	[−35.4, −24.9]	0.000	0.000	+0.194	worse
treatment	CWT (Morlet)	STFT signed + logistic	73.1%	−9.2%	[−15.3, −4.3]	0.001	0.022	+0.051	worse
physics_demo	magnitude only, mirror-augmented	CWT signed + logistic	50.0%	−22.2%	[−27.7, −16.5]	0.000	0.000	+0.142	worse
physics_demo	magnitude only, no mirror augmentation	CWT signed + logistic	50.8%	−20.3%	[−27.3, −13.0]	0.000	0.000	+0.143	worse
treatment	2D CNN on the signed planes	STFT signed + logistic	82.8%	+1.9%	[−0.1, 3.7]	0.073	0.877	−0.033	no detectable difference
nested	deep, architecture chosen in every fold	1D CNN	80.8%	−0.8%	[−2.8, 1.5]	0.501	0.501	+0.007	no detectable difference
nested	deep, configuration chosen in every fold	1D CNN	87.8%	+6.3%	[3.5, 10.0]	0.000	0.000	−0.157	better
nested	deep, grip-invariant configuration chosen in every fold	1D CNN	89.1%	+7.7%	[4.4, 11.6]	0.000	0.000	−0.174	better

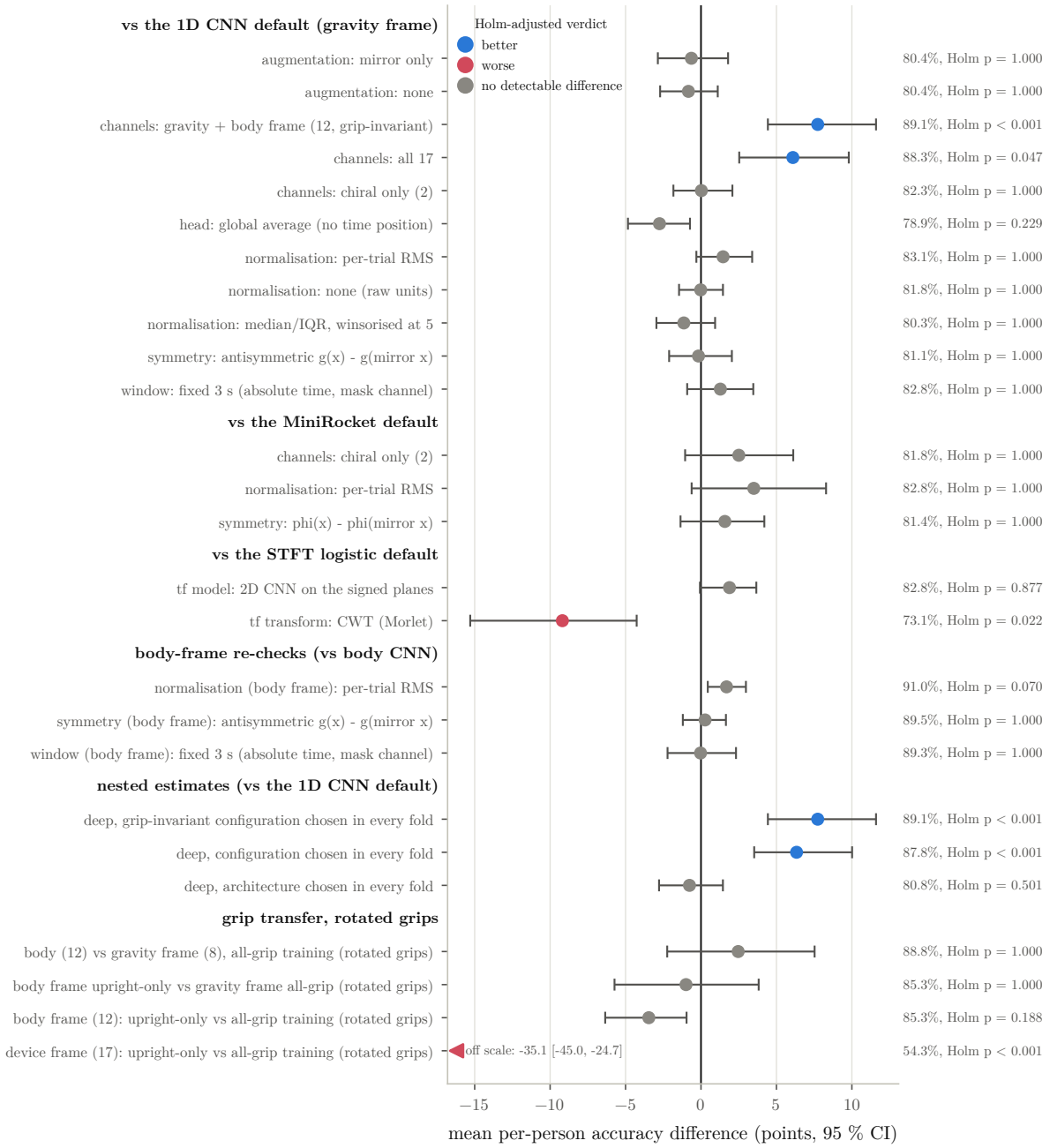


Figure 72: Data-treatment and model choices, each paired against its family default on the same trials: mean per-person accuracy difference with its paired person-bootstrap 95 % CI; right: the option's accuracy and its Holm-adjusted p within the family. Colour: the Holm-adjusted verdict. The off-scale row is the collapse of the device-frame CNN on grips absent from training (Appendix E.10). The figure supports the decisions of Appendix E.4.

E.5. Calibration and its remedies

The app shows $P(\text{right})$ as a percentage, so calibration on new persons is part of the result.

Detection. Reliability diagrams use 10 equal-mass bins with person-bootstrap bands, the bins held fixed. The expected calibration error (ECE) is judged against its distribution under perfect calibration (consistency resampling); its 95th percentile is the null band. The calibration slope and intercept carry person-bootstrap CIs; the Brier score [19] and the log loss complete the picture, overall and per platform (Figure 73, Tables 61 and 62).

Remedies. Platt scaling [21], temperature scaling [12], isotonic regression, beta calibration, and the calibrator chosen inside each fold. Each was fitted inside every outer fold on the training persons' cross-fitted logits and applied to the held-out person. The pre-specified remedy is Platt scaling for a model that is not antisymmetric. Each remedy was compared with the uncalibrated model by paired per-person log-loss and Brier differences, Holm-adjusted. The target of the analysis plan is an ECE of at most 0.05 and no reliability bin off the diagonal.

Which bins count as off. A bin whose trials all have the same hand has an observed share of exactly 0 or 1 and a person-bootstrap band of zero width; judged by that band alone, it is flagged for any prediction short of 0 or 1. A bin therefore counts as off only when its person-bootstrap band excludes the bin's mean prediction *and* its observed share lies outside the 95 % consistency band of a perfectly calibrated model with the same predictions. For the shipped deep finalist, the bootstrap band alone flags 2 bins, of which 2 are degenerate; the combined criterion flags 0. A perfectly calibrated model would show at least one bin off with probability 18 %.

Deep finalist. Uncalibrated, it already meets the target: ECE 0.029 against a null band of 0.038, slope 0.94 [0.67, 1.50], intercept −0.01 [−0.34, +0.28], and 0 bins off by the combined criterion. With Platt scaling (shipped), the ECE is 0.026 against a null band of 0.040, the slope 0.97 [0.69, 1.53] and the intercept +0.07 [−0.25, +0.41], with no bin off: the target is met. Platt scaling changes the per-person log loss by +0.0015 [−0.0060, +0.0073] (Holm-adjusted $p = 1.000$): it is harmless rather than needed. The calibrator chosen inside the fold kept the uncalibrated model in 20 of 20 folds, so its predictions are identical to the base model's.

MiniRocket finalist. Uncalibrated, it misses the target: ECE 0.053 against a null band of 0.056, slope 1.15 [0.93, 1.49], intercept −0.08 [−0.36, +0.16], and 2 bins off. With Platt scaling (shipped), the ECE is 0.049 against a null band of 0.053, the slope 0.99 [0.79, 1.30] and the intercept +0.00 [−0.28, +0.27], with 0 bins off: the target is met, narrowly.

Time-frequency finalist. Uncalibrated, its ECE of 0.073 lies above its null band of 0.051, with slope 0.81 [0.61, 1.16], intercept −0.09 [−0.34, +0.17] and 3 bins off. With Platt scaling (shipped), the ECE rises to 0.085 against a null band of 0.054, the slope is 0.89 [0.68, 1.28] and the intercept −0.01 [−0.25, +0.24], with 5 bins off: *the target is missed*, and the shipped remedy does not help. The slope and intercept are compatible with 1 and 0, so the miscalibration is not a global over- or under-confidence. The reliability curve is S-shaped (Figure 73, top right): the predictions are too low at the bottom, too high in the lower middle, too low in the upper

middle and too high at the top. A two-parameter monotone map such as Platt's cannot correct that shape. Isotonic regression lowers the ECE to 0.051, still above its null band of 0.050 (slope 0.82 [0.65, 1.07], intercept −0.06 [−0.31, +0.21]), without a detectable log-loss gain (−0.0176 [−0.0435, +0.0115], Holm-adjusted $p = 1.000$). It was not adopted, because choosing it here would be one more outer-fold selection.

Per platform. Table 62 repeats the statistics for the 12 Android and the 8 iOS persons. For the shipped deep finalist the ECE stays inside its null band on both platforms (Android 0.034 against 0.050; iOS 0.039 against 0.062). The slopes are 1.18 on Android and 0.75 on iOS; their CIs, in the table, are wide, above all for iOS, and both contain 1. The intercept was not computed per platform.

Decision. Ship the pre-specified Platt remedy for each role. The deep and MiniRocket finalists meet the target; the time-frequency finalist does not, and it is flagged as miscalibrated.

Table 61: Calibration of each role's best m3 model before and after the remedies, each fitted inside every outer fold on the cross-fitted logits of the training persons ("chosen in fold": the calibrator picked inside the fold). ECE (10 equal-mass bins) with the 95th percentile of its distribution under perfect calibration; calibration slope (1 = calibrated, < 1 over-confident) with its person-bootstrap CI; log-loss difference against the uncalibrated model per person, Holm-adjusted. Target: docs/09 M4 (ECE at most 0.05 and no reliability bin off by more than its CI).

Model	Calibration	Accuracy	ECE	ECE null 95%	Slope	Slope 95% CI	Brier	Log loss	LL diff. vs none	p (Holm)	Target
best deep sequence model	uncalibrated	89.1%	0.029	0.038	0.94	[0.67, 1.50]	0.083	0.290	–	–	met
best deep sequence model	Platt (pre-specified) (shipped)	89.3%	0.026	0.040	0.97	[0.69, 1.53]	0.083	0.291	+0.0015	1.000	met
best deep sequence model	temperature	89.1%	0.022	0.039	0.97	[0.69, 1.55]	0.083	0.290	+0.0013	1.000	met
best deep sequence model	isotonic	89.0%	0.015	0.037	0.83	[0.57, 1.32]	0.084	0.308	+0.0168	1.000	met
best deep sequence model	beta	89.1%	0.025	0.039	0.93	[0.65, 1.48]	0.084	0.300	+0.0094	1.000	met
best deep sequence model	chosen in fold	89.1%	0.029	0.038	0.94	[0.67, 1.50]	0.083	0.290	+0.0000	1.000	met
best MiniRocket	uncalibrated	82.8%	0.053	0.056	1.15	[0.93, 1.49]	0.137	0.436	–	–	missed
best MiniRocket	Platt (pre-specified) (shipped)	82.3%	0.049	0.053	0.99	[0.79, 1.30]	0.136	0.439	+0.0023	1.000	met
best MiniRocket	temperature	82.8%	0.050	0.053	1.01	[0.80, 1.32]	0.136	0.437	+0.0001	1.000	missed
best MiniRocket	isotonic	81.8%	0.028	0.051	0.90	[0.71, 1.16]	0.137	0.443	+0.0110	1.000	met
best MiniRocket	beta	81.1%	0.049	0.053	1.00	[0.80, 1.30]	0.137	0.438	+0.0016	1.000	met
best MiniRocket	chosen in fold	82.6%	0.055	0.056	1.08	[0.83, 1.44]	0.138	0.443	+0.0069	1.000	missed
best time-frequency model	uncalibrated	82.8%	0.073	0.051	0.81	[0.61, 1.16]	0.142	0.479	–	–	missed
best time-frequency model	Platt (pre-specified) (shipped)	83.1%	0.085	0.054	0.89	[0.68, 1.28]	0.142	0.473	−0.0054	1.000	missed
best time-frequency model	temperature	82.8%	0.080	0.054	0.89	[0.67, 1.27]	0.143	0.477	−0.0013	1.000	missed
best time-frequency model	isotonic	82.1%	0.051	0.050	0.82	[0.65, 1.07]	0.139	0.461	−0.0176	1.000	missed
best time-frequency model	beta	82.3%	0.072	0.054	0.88	[0.66, 1.26]	0.142	0.476	−0.0017	1.000	missed
best time-frequency model	chosen in fold	81.9%	0.073	0.051	0.79	[0.60, 1.08]	0.144	0.488	+0.0099	1.000	missed

Table 62: Calibration per platform, uncalibrated and shipped (leave-one-person-out; ECE with its perfect-calibration 95th percentile; slope with its person-bootstrap CI).

Model	Calibration	Platform	Trials	Persons	Accuracy	ECE	ECE null 95%	Slope	Slope 95% CI	Brier	Log loss
best deep sequence model	uncalibrated	android	383	12	90.1%	0.047	0.049	1.13	[0.86, 1.83]	0.077	0.259
best deep sequence model	uncalibrated	ios	215	8	87.4%	0.042	0.059	0.73	[0.41, 1.75]	0.094	0.346
best deep sequence model	Platt	android	383	12	90.6%	0.034	0.050	1.18	[0.92, 1.79]	0.077	0.258
best deep sequence model	Platt	ios	215	8	87.0%	0.039	0.062	0.75	[0.41, 1.84]	0.093	0.351
best MiniRocket	uncalibrated	android	383	12	83.3%	0.070	0.072	1.31	[1.05, 1.80]	0.134	0.432
best MiniRocket	uncalibrated	ios	215	8	81.9%	0.087	0.089	0.96	[0.64, 1.60]	0.140	0.444
best MiniRocket	Platt	android	383	12	82.8%	0.052	0.069	1.16	[0.92, 1.60]	0.134	0.430
best MiniRocket	Platt	ios	215	8	81.4%	0.072	0.082	0.81	[0.53, 1.37]	0.142	0.455
best time-frequency model	uncalibrated	android	383	12	83.0%	0.098	0.068	0.87	[0.61, 1.28]	0.142	0.483
best time-frequency model	uncalibrated	ios	215	8	82.3%	0.073	0.081	0.74	[0.48, 1.49]	0.141	0.471
best time-frequency model	Platt	android	383	12	82.2%	0.091	0.071	0.96	[0.67, 1.43]	0.144	0.481
best time-frequency model	Platt	ios	215	8	84.7%	0.066	0.085	0.82	[0.54, 1.64]	0.138	0.459

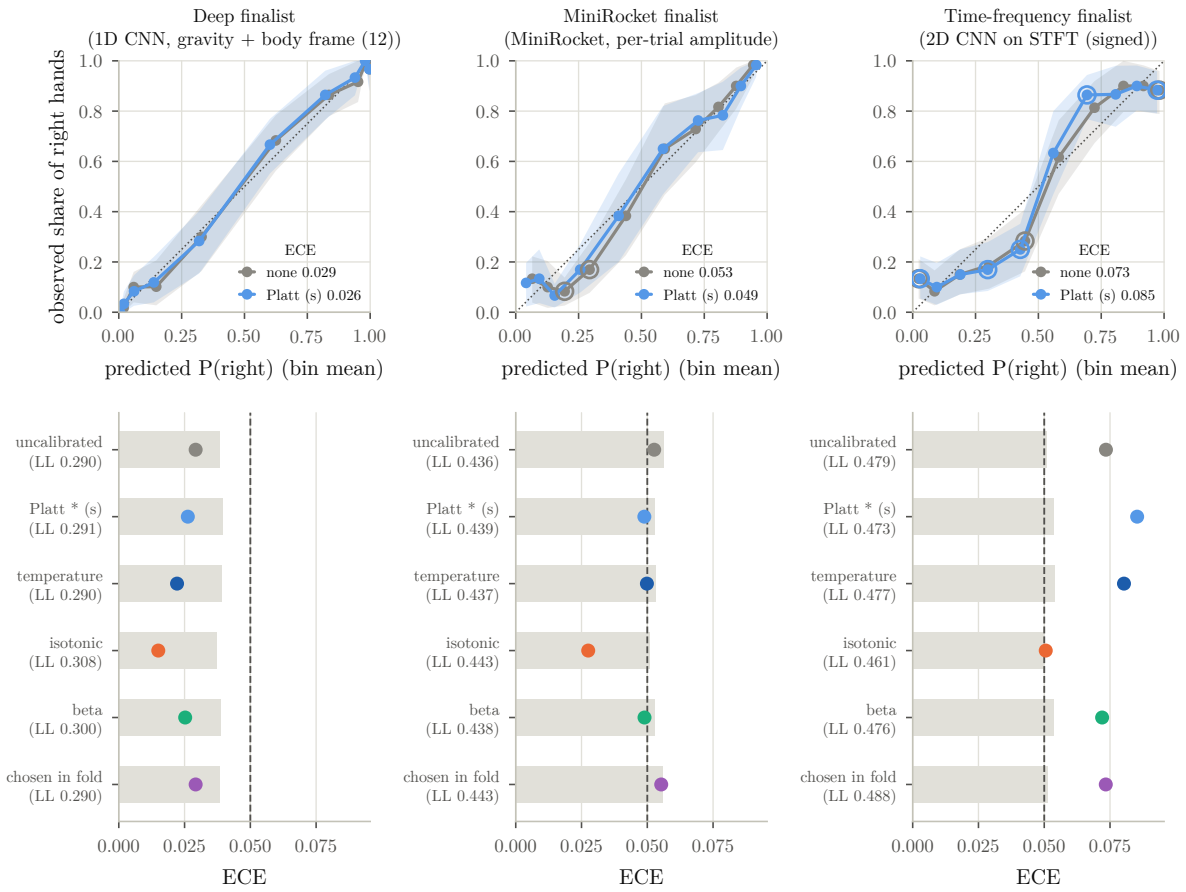


Figure 73: Calibration of each role's finalist under LOSO. Top: reliability with 10 equal-mass bins and person-bootstrap bands, uncalibrated (grey) and with the shipped Platt remedy; ringed points: bins off the diagonal by both criteria. Bottom: ECE of every remedy (points) against the 95th percentile of its distribution under perfect calibration (grey bars); dashed line: the target of 0.05; LL: log loss; ∗: pre-specified; (s): shipped. The deep and MiniRocket finalists meet the target; the time-frequency finalist misses it.

E.6. What the network uses: integrated gradients

Aim. To test whether the deep finalist relies on the evidence of the engineered features, early hand-signed motion, and to find the channels that carry it.

Method. Integrated gradients (IG) attribute each held-out trial's logit to its input samples. Each trial was attributed out of fold by its own fold's ensemble, with the normalised zero input as the baseline; that baseline is invariant under the mirror. The attributions are complete to a median relative error of 0.004, and the sign of the summed attribution agrees with the logit in 92 % of the trials. The calibrators are monotone maps of the logit, so they do not change the attribution.

Channels. The four chiral channels that do not depend on the grip (yaw rate, cumulative yaw, roll rate and lateral acceleration) carry 57 % of all |IG|, against 33 % if every channel weighed

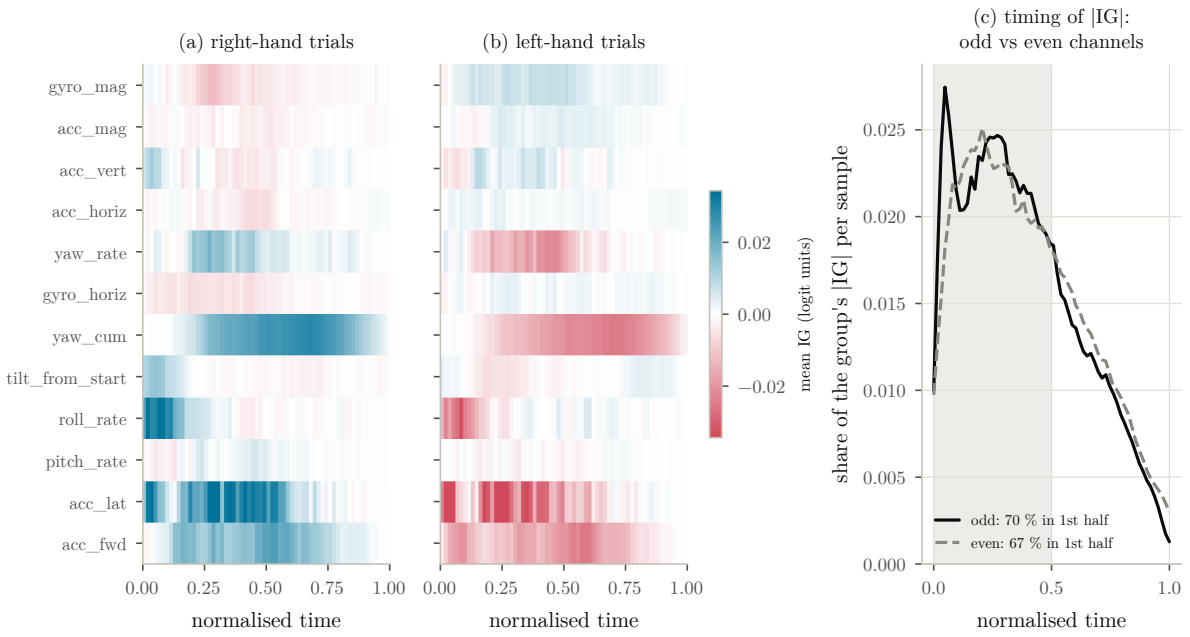


Figure 74: Integrated gradients of the deep finalist, out of fold. (a, b) Mean attribution per channel and time toward $P(\text{right})$ for right- and left-hand trials (teal: toward right; red: toward left). (c) Time profile of |IG| for the chiral, grip-invariant channels (solid) and the even channels (dashed); grey: the first half. The chiral channels carry 57 % of |IG| but are only slightly more front-loaded than the even ones.

the same (Figure 74). The lateral acceleration is the largest single channel (24 %), followed by the yaw rate (12 %), the roll rate (11 %) and the forward acceleration (11 %). The two yaw channels together carry 21 %; the two chiral body-frame channels carry 36 %.

Timing. 70 % of the chiral channels' |IG| falls in the first half of the time-normalised window, against 67 % for the even channels. The chiral channels are thus more front-loaded by only +3.3 points [+2.0, +4.6] (person bootstrap): most of the early concentration is the generic time profile of the attribution. The roll rate peaks at the start of the window (0.06 of its length), the lateral acceleration at 0.27 and the yaw rate at 0.43. The cumulative yaw peaks late (0.73), as an integral of the yaw rate should.

Interactions. In a mirror-equivariant model an even channel can push toward the true hand only through its interaction with the chiral channels. The forward acceleration does so: its attribution toward the true hand is +0.40 (sum of IG times the hand sign over the sum of |IG|; 1 means always toward it), against +0.19 for the yaw rate. The network therefore combines channels rather than reading a single hand-carrying one. The mirror consistency, the correlation between $\text{IG}(m(x))$ and $-\text{IG}(x)$, is 0.81 (1 for an exactly antisymmetric model).

Conclusion. The network agrees with the physics in using early, hand-signed motion, but its evidence is not the early yaw alone: it leans more on the body-relative roll and lateral motion.

E.7. Breakdown by platform and instructed condition

Platform and instructed conditions only stratify the results; they are never model inputs (Figure 75, Table 63). The shipped deep finalist reaches 90.6 % [86.8, 93.9] on Android (12 persons) and 87.0 % [77.3, 94.0] on iOS (8 persons). Fast approaches score 94.8 % [90.3, 98.5] and natural-speed approaches 87.4 % [82.0, 92.0]. Among the movement variants, only the twist has a CI entirely below the overall accuracy: 60.0 % [31.2, 86.7] on 15 trials of 8 persons. The levels were compared without a multiplicity adjustment, so this finding is exploratory. The bent-start variant has fewer than 5 persons and carries no CI (57.1 % on 7 trials of 4 persons).

Label-noise sensitivity. Two blocks of one person show the hand-swap signature found from the data (Appendix A); their labels were never changed. Without these 8 trials, the deep finalist scores 89.7 % [85.3, 93.4], against 89.3 % on all trials; on the swap trials themselves it scores 62.5 %.

Table 63: Shipped m3 deep finalist (m3_cal_deep_platt) by platform and instructed condition (conditions stratify the results only; they are never inputs; 95 % person-cluster bootstrap CI within each level, none for levels with fewer than 5 persons).

Condition	Level	Trials	Persons	Accuracy	95% CI	AUC
platform	android	383	12	90.6%	[86.8, 93.9]	0.961
platform	ios	215	8	87.0%	[77.3, 94.0]	0.935
speed	fast	134	20	94.8%	[90.3, 98.5]	0.986
speed	natural	421	20	87.4%	[82.0, 92.0]	0.938
speed	slow	43	19	90.7%	[76.7, 100.0]	0.961
orientation	top_left_toward	118	19	92.4%	[84.2, 99.1]	0.984
orientation	top_right_toward	114	19	86.8%	[78.3, 95.3]	0.947
orientation	upright_toward	366	20	89.1%	[84.7, 93.1]	0.947
variant	bent_start	7	4	57.1%	–	0.500
variant	curved	16	8	81.2%	[56.2, 100.0]	0.859
variant	elbow_high	14	7	85.7%	[64.3, 100.0]	0.959
variant	jerky	13	8	92.3%	[72.7, 100.0]	0.950
variant	lean_in	11	6	81.8%	[45.5, 100.0]	0.967
variant	pause	17	9	82.4%	[66.7, 94.4]	0.944
variant	standard	470	20	91.7%	[87.2, 95.5]	0.966
variant	tilted_touch	35	19	85.7%	[75.0, 94.7]	0.915
variant	twist	15	8	60.0%	[31.2, 86.7]	0.643
posture	sitting	438	15	88.8%	[83.1, 93.1]	0.951
posture	standing	160	5	90.6%	[84.7, 95.8]	0.955
start_position	center	18	10	94.4%	[83.3, 100.0]	0.889
start_position	natural	563	20	89.2%	[84.8, 93.0]	0.953
start_position	shoulder	17	9	88.2%	[72.2, 100.0]	0.986

E.8. Several attempts: aggregation

Options. Four ways to combine k attempts of one person and hand: the mean of P(right) (the harness definition), the mean logit, the sum of logits (attempts treated as independent

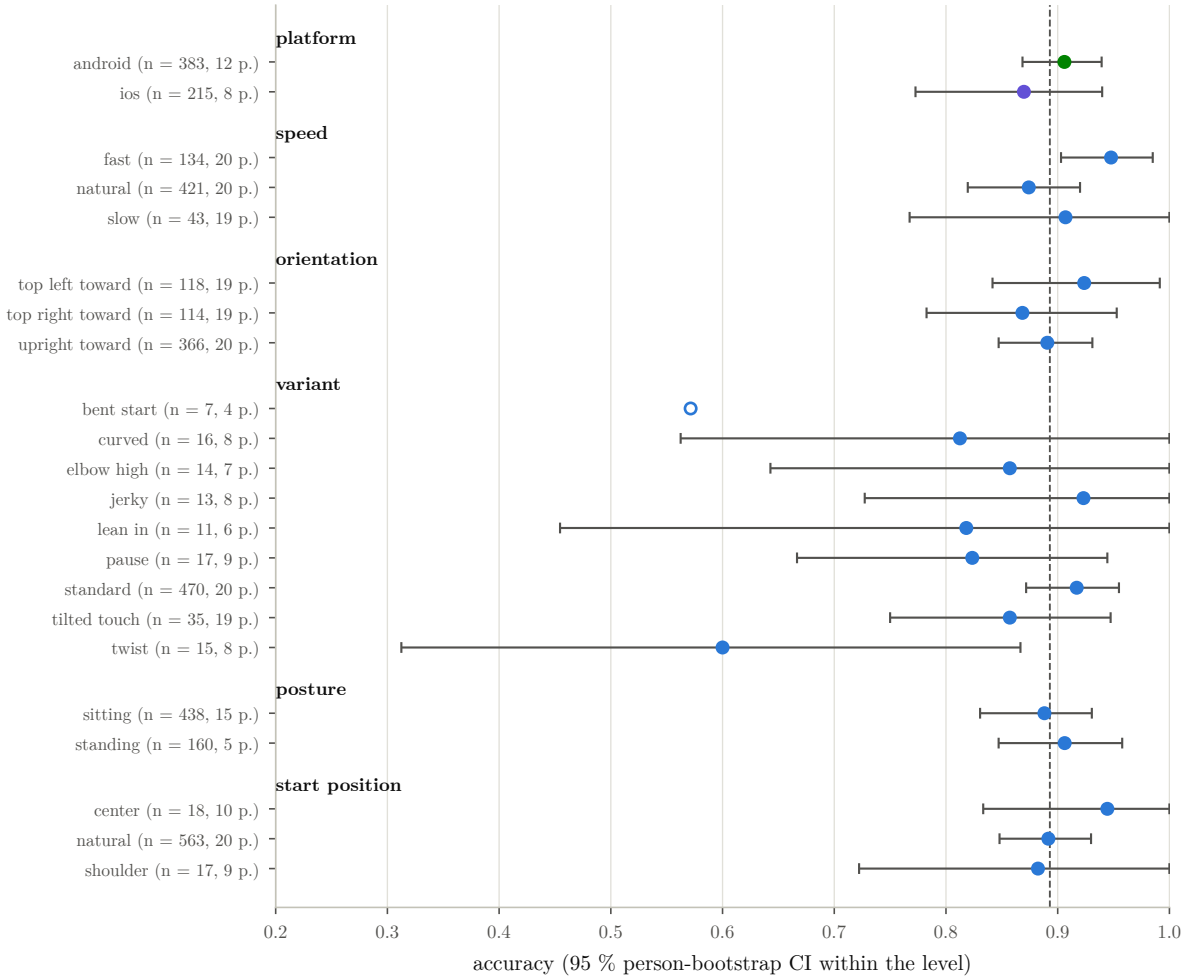


Figure 75: The shipped deep finalist by platform and instructed condition (n: trials; p.: persons; dashed line: overall accuracy). Levels with fewer than 5 persons carry no CI (open marker). The conditions only stratify the result, and the levels are compared without a multiplicity adjustment (exploratory).

evidence), and the sum of the raw logits scaled by c_k, fitted on the training persons' cross-fitted logits of each fold. The scale c_k plays the role of $1/\tau_k$ in the production rule (Section 6).

Evidence. The k-attempt curves use the 19 persons with at least 5 trials of each hand and report per-person balanced accuracy (Figure 76, Table 64). At $k = 3$ the deep finalist reaches 96.4 % [93.2, 98.9] with the mean of $P(\text{right})$ and 96.0 % [91.6, 99.1] with the c_k-scaled sum of logits. The aggregators differ in calibration. At $k = 3$ the over-confidence of the deep finalist (mean confidence minus accuracy) is -0.123 [-0.139, -0.105] for the mean of $P(\text{right})$, -0.068 [-0.087, -0.046] for the mean logit, $+0.013$ [-0.007, $+0.041$] for the sum of logits and $+0.007$ [-0.013, $+0.037$] for the c_k-scaled sum. Averaging $P(\text{right})$ is therefore under-confident. The median scale falls from 0.98 at $k = 1$ to 0.74 at $k = 3$ and 0.58 at $k = 5$: the attempts of one person are correlated, so their logits add less than independent evidence would.

Decision. The c_k-scaled sum of logits. It is the aggregator designed for correlated attempts,

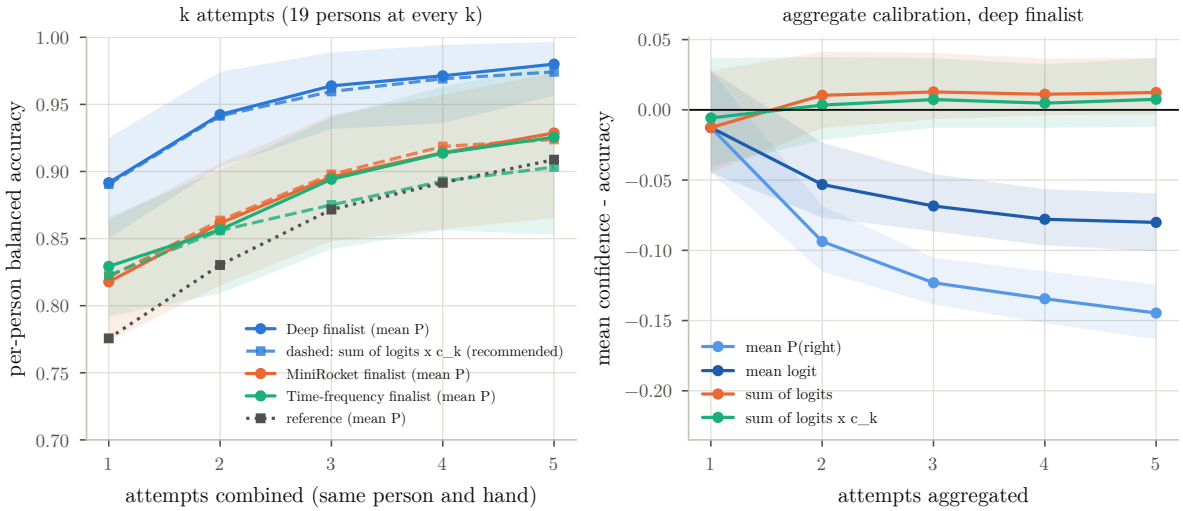


Figure 76: Combining k attempts of one person and hand (19 persons at every k). Left: per-person balanced accuracy of the finalists and the reference (solid: mean of $P(\text{right})$, with its 95 % person-bootstrap band; dashed: the c_k-scaled sum of logits). Right: over-confidence of the four aggregators for the deep finalist; the two sums stay calibrated and the two means are under-confident. The aggregator was chosen among four on these outer folds (exploratory).

and its over-confidence CI at $k = 3$ contains 0 for every role. It was chosen among the four aggregators on these outer folds, so the choice is exploratory. For the time-frequency finalist it gives 87.5 % [81.4, 92.6] at $k = 3$, against 89.4 % for the mean of $P(\text{right})$; the $k = 3$ column of Table 59 uses the mean of $P(\text{right})$.

Table 64: Aggregating $k = 3$ attempts of one person and hand (out of fold; over-confidence = mean confidence minus accuracy, person-bootstrap 95 % CI). Sum of logits treats the attempts as independent; c_k is fitted on the training persons of each fold. The recommended aggregator was chosen among 4 on these outer folds (exploratory).

Model	Single-trial calibration	Aggregator	Accuracy	Log loss	Over-conf.	95% CI
best deep sequence model	uncalibrated	mean of P(right)	96.2%	0.202	−0.118	[−0.136, −0.099]
best deep sequence model	uncalibrated	mean logit	96.0%	0.149	−0.062	[−0.082, −0.037]
best deep sequence model	uncalibrated	sum of logits (independence)	96.0%	0.124	+0.015	[−0.006, +0.045]
best deep sequence model	uncalibrated	sum of logits x c_k (fitted on training persons)	96.0%	0.124	+0.007	[−0.013, +0.037]
best deep sequence model	calibrated	mean of P(right)	96.4%	0.205	−0.123	[−0.139, −0.105]
best deep sequence model	calibrated	mean logit	96.1%	0.154	−0.068	[−0.087, −0.046]
best deep sequence model	calibrated	sum of logits (independence)	96.1%	0.120	+0.013	[−0.007, +0.041]
best deep sequence model	calibrated	sum of logits x c_k (fitted on training persons)	96.0%	0.124	+0.007	[−0.013, +0.037]
best MiniRocket	uncalibrated	mean of P(right)	90.1%	0.373	−0.174	[−0.213, −0.125]
best MiniRocket	uncalibrated	mean logit	89.8%	0.351	−0.148	[−0.189, −0.099]
best MiniRocket	uncalibrated	sum of logits (independence)	89.8%	0.257	+0.013	[−0.025, +0.059]
best MiniRocket	uncalibrated	sum of logits x c_k (fitted on training persons)	89.8%	0.267	+0.002	[−0.040, +0.051]
best MiniRocket	calibrated	mean of P(right)	89.9%	0.358	−0.155	[−0.193, −0.108]
best MiniRocket	calibrated	mean logit	89.4%	0.335	−0.122	[−0.164, −0.074]
best MiniRocket	calibrated	sum of logits (independence)	89.4%	0.277	+0.027	[−0.011, +0.074]
best MiniRocket	calibrated	sum of logits x c_k (fitted on training persons)	89.8%	0.267	+0.002	[−0.040, +0.051]
best time-frequency model	uncalibrated	mean of P(right)	88.9%	0.361	−0.144	[−0.194, −0.085]
best time-frequency model	uncalibrated	mean logit	87.5%	0.345	−0.089	[−0.142, −0.027]
best time-frequency model	uncalibrated	sum of logits (independence)	87.5%	0.354	+0.046	[−0.002, +0.102]
best time-frequency model	uncalibrated	sum of logits x c_k (fitted on training persons)	87.5%	0.328	+0.023	[−0.028, +0.083]
best time-frequency model	calibrated	mean of P(right)	89.3%	0.367	−0.157	[−0.200, −0.105]
best time-frequency model	calibrated	mean logit	87.9%	0.348	−0.107	[−0.154, −0.054]
best time-frequency model	calibrated	sum of logits (independence)	87.9%	0.317	+0.035	[−0.006, +0.083]
best time-frequency model	calibrated	sum of logits x c_k (fitted on training persons)	87.5%	0.328	+0.023	[−0.028, +0.083]

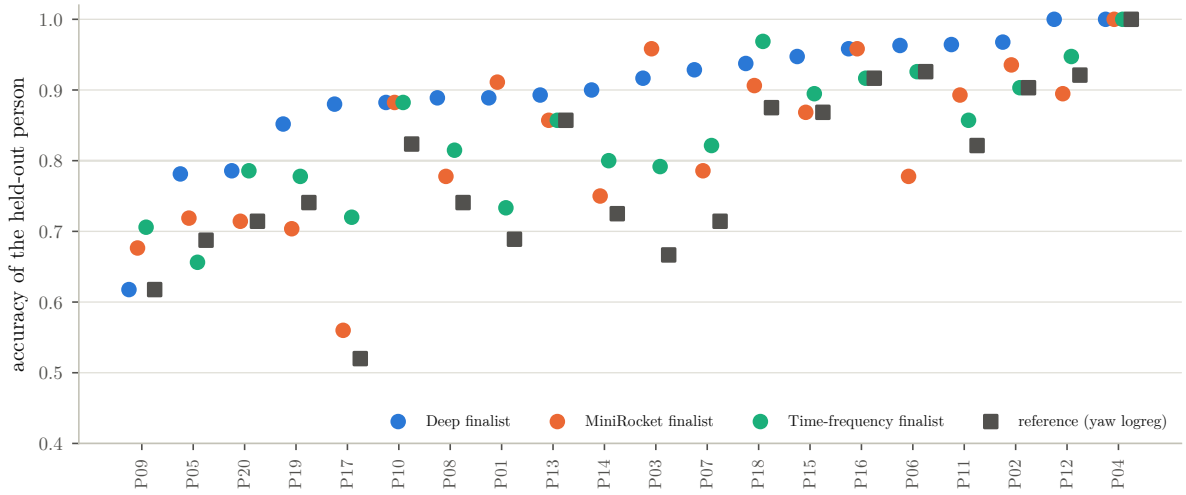


Figure 77: Accuracy of each of the 20 held-out persons for the three calibrated lane-m3 finalists and the reference yaw logistic regression, sorted by the deep finalist. It shows whether the deep and time-frequency models fail on the same persons as the engineered features.

E.9. Per person

Figure 77 shows the accuracy of each held-out person. No person scores lower with the deep finalist than with the reference; two persons score the same. The weakest person for the deep finalist, P09, is the second weakest for the reference, which scores the same on that person; the MiniRocket and time-frequency finalists do better there. The largest gains of the deep finalist over the reference are on persons for whom the reference is weak (P17, the reference's weakest person, then P03 and P07). The persons who are hard for the deep finalist (P09, P05, P20, P19) are also below the median for the reference, so the deep and engineered models fail partly on the same persons. These are readings of the figure on one fold per person, without a test.

E.10. Grip transfer: device frame against body frame

Detected. The device-frame channels add accuracy to the gravity-frame CNN (Appendix E.4), but their values depend on how the phone is held. The instructed grips were balanced across hands, so the gain is not a protocol artefact. Evidence tied to phone coordinates may still fail on a grip that training never saw, and production grips are free.

Test. A configuration was trained only on the trials whose grip, detected from the up vector before the onset, is upright, and it was tested on every grip; the instructed grip only stratifies the result. The grip detection is clean: 0 rotated-grip trials were detected as upright, and 6 upright trials were not detected as upright and were left out of that training. The statistic is the paired difference on the rotated grips (top left and top right; 232 trials of 19 persons), which upright-only training never saw (family `grip`, Holm).

Evidence (Figure 78, Table 65):

- device frame (all 17): trained on upright grips only, the CNN scores 54.3 % on the rotated grips against 88.8 % with all-grip training, a change of −35.1 points [−45.0, −24.7] ($p < 0.001$): it collapses toward chance;
- body frame (12): 85.3 % against 88.8 %, a change of −3.5 points [−6.3, −1.0] (unadjusted $p = 0.031$, Holm-adjusted 0.188): not detectable after adjustment. Its per-person log loss on those grips changes by +0.118 [+0.043, +0.218] (Holm-adjusted $p = 0.023$), a higher log loss after adjustment;
- on the rotated grips, the body-frame CNN trained on upright grips only is not detectably different from the gravity-frame CNN trained on all grips (−1.0 points [−5.7, +3.8]);
- with all-grip training, the body frame beats the gravity frame on the upright grips (+11.1 points [+7.5, +15.7], $p < 0.001$) but not detectably on the rotated grips (+2.5 points [−2.2, +7.5]).

Decision and rationale. Use the body frame. It avoids the collapse of the device frame on unseen grips. The frame itself is exactly invariant to a rotation about the screen normal, so the residual penalty on grips absent from training, suggested by the accuracy CI and shown by the log loss, reflects how people move with a rotated grip. Training data must therefore cover the grips expected in production. The check covers the three recorded grips, all with the screen toward the person; a screen-away grip reverses the facing axis **h** of the body frame and was never recorded.

Table 65: Grip transfer: upright-only against all-grip training and the body frame against the gravity frame, on all trials and on the rotated (never seen by upright-only training) and upright grips (instructed orientation, used only to stratify). Mean per-person difference, paired person-bootstrap 95 % CI, Holm-adjusted p.

Family	Comparison	Against	Accuracy	Diff.	95% CI	p	p (Holm)	Log-loss diff.	Verdict
grip	trained on sensor-detected upright trials only, tested on every grip	1D CNN, all 17 channels	76.8%	−10.5%	[−14.6, −6.2]	0.000	0.002	+0.599	worse
grip	trained on sensor-detected upright trials only, tested on every grip	1D CNN, gravity + body frame (12)	88.8%	−0.5%	[−2.4, 1.4]	0.650	1.000	+0.040	no detectable difference
grip	device frame (17): upright-only vs all-grip training (rotated grips)	1D CNN, all 17 channels	54.3%	−35.1%	[−45.0, −24.7]	0.000	0.000	+1.683	worse
grip	device frame (17): upright-only vs all-grip training (upright grips)	1D CNN, all 17 channels	91.0%	+3.5%	[−0.5, 7.6]	0.123	0.615	−0.012	no detectable difference
grip	body frame (12): upright-only vs all-grip training (rotated grips)	1D CNN, gravity + body frame (12)	85.3%	−3.5%	[−6.3, −1.0]	0.031	0.188	+0.118	no detectable difference
grip	body frame (12): upright-only vs all-grip training (upright grips)	1D CNN, gravity + body frame (12)	91.0%	+1.6%	[−0.6, 3.7]	0.185	0.738	−0.014	no detectable difference
grip	body (12) vs gravity frame (8), all-grip training (rotated grips)	1D CNN	88.8%	+2.5%	[−2.2, 7.5]	0.344	1.000	−0.124	no detectable difference
grip	body (12) vs gravity frame (8), all-grip training (upright grips)	1D CNN	89.3%	+11.1%	[7.5, 15.7]	0.000	0.000	−0.210	better
grip	body frame upright-only vs gravity frame all-grip (rotated grips)	1D CNN	85.3%	−1.0%	[−5.7, 3.8]	0.727	1.000	−0.006	no detectable difference

E.11. Against engineered features: input or model class

Two families of paired comparisons answer this question (Figure 79, Table 66).

Equal input. The family `equal_input` compares deep and linear models on matched information (Holm over its 3 comparisons). On the gravity frame alone, the CNN is not detectably better than lane m1's logistic regression on two yaw features (+2.8 points [+0.4, +5.5], Holm-adjusted $p = 0.087$). It is also not detectably different from lane m1's six-feature physics logistic regression (PHYS6: the yaw about gravity plus body-frame roll and lateral features; Appendix C), although the CNN has less input (−0.7 points [−2.8, +1.2]). With the body-frame channels, the CNN beats PHYS6 by +7.0 points [+3.3, +11.1] (Holm-adjusted $p = 0.006$). The deep advantage over the engineered linear model therefore needs the body-frame input: it comes from the input and the model class together, not from the model class alone.

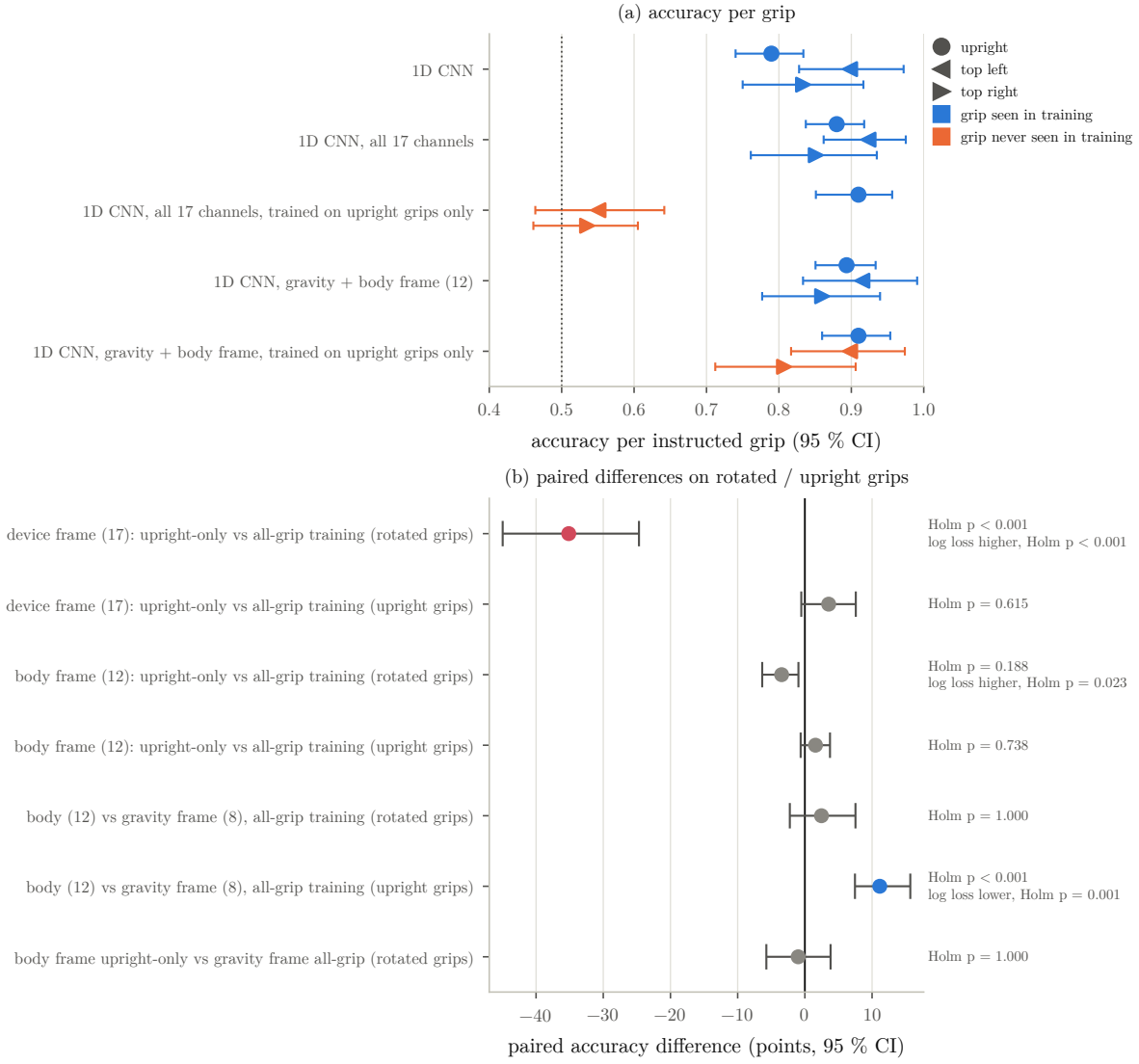


Figure 78: Grip transfer. (a) Accuracy per instructed grip with its 95 % CI; orange: grips that upright-only training never saw. (b) Paired per-person accuracy differences on the rotated and upright grips, Holm-adjusted within the grip family (right: the adjusted p and, where significant, the log-loss verdict). The device-frame CNN collapses on unseen grips; the body-frame CNN does not.

Against engineered-feature models. The family `engineered` (Holm over its 18 comparisons) pairs the shipped finalists and the three nested deep estimates with three models: the reference yaw logistic regression (77.9 %), lane m1's pre-specified physics logistic regression, and lane m2's best engineered-feature stack, itself an exploratory choice (Appendix D).

- The deep finalist is better than the reference (+11.1 points [+7.4, +15.2], $p < 0.001$) and than lane m1's physics logistic regression (+7.1 points [+3.4, +11.1], Holm-adjusted $p = 0.021$). It is not detectably different from lane m2's stack (+1.3 points [−1.5, +4.4], Holm-adjusted $p = 1.000$).
- The grip-invariant nested estimate shows the same pattern: +11.0 points against the reference ($p < 0.001$), +7.0 points against lane m1 (Holm-adjusted $p = 0.025$) and +1.2 points [−1.5, +4.2] against lane m2 (not detectable).
- The nested estimate over all configurations is better than the reference (+9.6 points, $p < 0.001$) and not detectably different from lane m1 (+5.6 points, Holm-adjusted $p = 0.051$) or lane m2 (−0.2 points).
- The architecture-only nested estimate on the gravity frame is worse than lane m2's stack (−7.3 points [−10.4, −4.5], $p < 0.001$).
- The MiniRocket and time-frequency finalists are not detectably worse than lane m2's stack after adjustment (−5.8 points [−9.7, −2.1], Holm-adjusted $p = 0.091$; −4.7 points [−8.1, −1.5], Holm-adjusted $p = 0.092$), but their per-person log loss is higher after the same adjustment (MiniRocket: +0.125 [+0.069, +0.177], Holm-adjusted $p = 0.009$; time-frequency: +0.147 [+0.095, +0.198], Holm-adjusted $p = 0.002$). Against the reference, the time-frequency finalist is better (+5.2 points, Holm-adjusted $p = 0.003$) and MiniRocket is not detectably better (+4.0 points, Holm-adjusted $p = 0.421$).

Conclusion. On these persons, representation learning does not beat the best engineered features: the body-frame CNN matches lane m2's stack and beats the simpler linear models. Its advantage over lane m1 rests on the body-frame input. It is also costlier: one LOSO run of the deep finalist takes 7.8 CPU-minutes, and the network needs a port to the browser with parity tests before it can run in the app (Appendix F).

Table 66: Lane-m3 models against linear and engineered-feature models on the same trials (mean per-person accuracy difference, paired person-bootstrap 95 % CI, sign-flip p, Holm-adjusted within the family). Equal input: deep and linear models on matched information; engineered: the shipped finalists and the nested deep estimates against the reference, lane m1 and lane m2.

Family	Comparison	Against	Accuracy	Diff.	95% CI	p	p (Holm)	Log-loss diff.	Verdict
equal_input	gravity frame only: CNN on the 8 gravity channels vs logistic on the yaw about gravity	m1 logistic on 2 yaw features	81.9%	+2.8%	[0.4, 5.5]	0.044	0.087	−0.053	no detectable difference
equal_input	CNN on the gravity frame vs logistic on yaw plus body-frame roll and lateral (the CNN has less input)	m1 physics logistic (PHYS6)	81.9%	−0.7%	[−2.8, 1.2]	0.543	0.543	+0.006	no detectable difference
equal_input	gravity + body frame: CNN on 12 channels vs logistic on yaw, roll and lateral features	m1 physics logistic (PHYS6)	89.1%	+7.0%	[3.3, 11.1]	0.002	0.006	−0.167	better
engineered	best deep sequence model vs reference (yaw logreg)	reference (yaw logreg)	89.3%	+11.1%	[7.4, 15.2]	0.000	0.000	−0.224	better
engineered	best deep sequence model vs m1 physics logistic (pre-specified)	m1 physics logistic (pre-specified)	89.3%	+7.1%	[3.4, 11.1]	0.002	0.021	−0.166	better
engineered	best deep sequence model vs m2 engineered-feature stack (best of lane m2)	m2 engineered-feature stack (best of lane m2)	89.3%	+1.3%	[−1.5, 4.4]	0.429	1.000	−0.036	no detectable difference
engineered	best MiniRocket vs reference (yaw logreg)	reference (yaw logreg)	82.3%	+4.0%	[0.4, 8.0]	0.053	0.421	−0.063	no detectable difference
engineered	best MiniRocket vs m1 physics logistic (pre-specified)	m1 physics logistic (pre-specified)	82.3%	+0.0%	[−2.6, 2.7]	0.997	1.000	−0.005	no detectable difference
engineered	best MiniRocket vs m2 engineered-feature stack (best of lane m2)	m2 engineered-feature stack (best of lane m2)	82.3%	−5.8%	[−9.7, −2.1]	0.009	0.091	+0.125	no detectable difference
engineered	best time-frequency model vs reference (yaw logreg)	reference (yaw logreg)	83.1%	+5.2%	[2.9, 7.5]	0.000	0.003	−0.042	better
engineered	best time-frequency model vs m1 physics logistic (pre-specified)	m1 physics logistic (pre-specified)	83.1%	+1.2%	[−1.7, 3.6]	0.411	1.000	+0.017	no detectable difference
engineered	best time-frequency model vs m2 engineered-feature stack (best of lane m2)	m2 engineered-feature stack (best of lane m2)	83.1%	−4.7%	[−8.1, −1.5]	0.010	0.092	+0.147	no detectable difference
engineered	deep, architecture chosen in every fold vs reference (yaw logreg)	reference (yaw logreg)	80.8%	+2.5%	[−0.6, 5.4]	0.120	0.843	−0.045	no detectable difference
engineered	deep, architecture chosen in every fold vs m1 physics logistic (pre-specified)	m1 physics logistic (pre-specified)	80.8%	−1.5%	[−3.9, 1.2]	0.282	1.000	+0.013	no detectable difference
engineered	deep, architecture chosen in every fold vs m2 engineered-feature stack (best of lane m2)	m2 engineered-feature stack (best of lane m2)	80.8%	−7.3%	[−10.4, −4.5]	0.000	0.000	+0.144	worse
engineered	deep, configuration chosen in every fold vs reference (yaw logreg)	reference (yaw logreg)	87.8%	+9.6%	[6.1, 13.7]	0.000	0.000	−0.208	better
engineered	deep, configuration chosen in every fold vs m1 physics logistic (pre-specified)	m1 physics logistic (pre-specified)	87.8%	+5.6%	[2.3, 9.3]	0.005	0.051	−0.150	no detectable difference
engineered	deep, configuration chosen in every fold vs m2 engineered-feature stack (best of lane m2)	m2 engineered-feature stack (best of lane m2)	87.8%	−0.2%	[−3.1, 3.0]	0.908	1.000	−0.020	no detectable difference
engineered	deep, grip-invariant configuration chosen in every fold vs reference (yaw logreg)	reference (yaw logreg)	89.1%	+11.0%	[7.3, 15.2]	0.000	0.000	−0.226	better
engineered	deep, grip-invariant configuration chosen in every fold vs m1 physics logistic (pre-specified)	m1 physics logistic (pre-specified)	89.1%	+7.0%	[3.3, 11.1]	0.002	0.025	−0.167	better
engineered	deep, grip-invariant configuration chosen in every fold vs m2 engineered-feature stack (best of lane m2)	m2 engineered-feature stack (best of lane m2)	89.1%	+1.2%	[−1.5, 4.2]	0.446	1.000	−0.037	no detectable difference

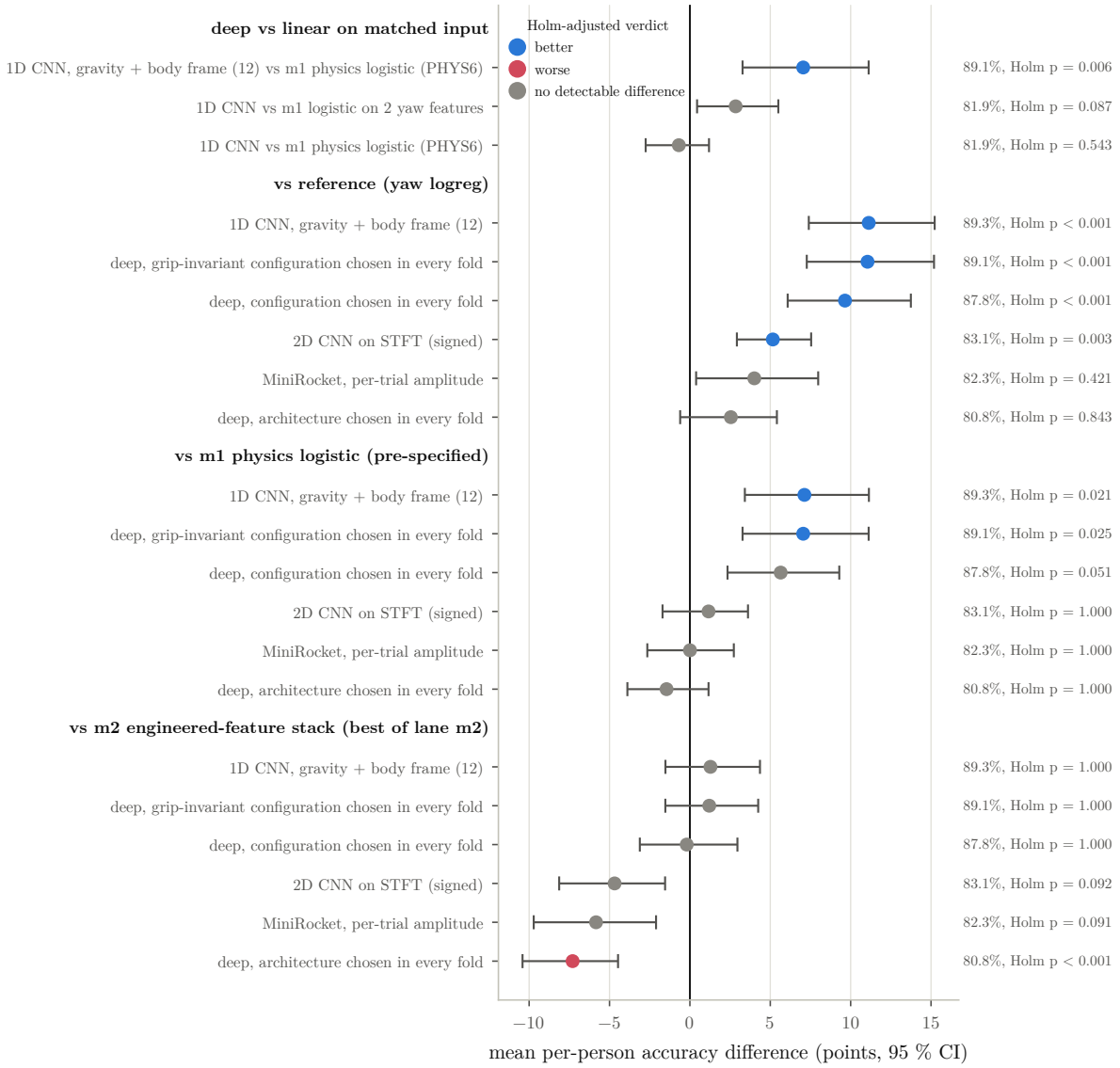


Figure 79: Lane-m3 models against linear and engineered-feature models on the same trials: mean per-person accuracy difference with its paired person-bootstrap 95 % CI; right: the lane-m3 model's accuracy and the Holm-adjusted p (families `equal_input` and `engineered`). Colour: the Holm-adjusted verdict. The deep advantage over the linear models needs the body-frame input, and no lane-m3 model is detectably better than lane m2's stack.

E.12. Caveats

- With 20 persons, most paired differences between configurations are not detectable after the adjustment, so the ranking of the architectures is not resolved.
- Outer-fold choices: each finalist was kept among its role's configurations; the body-frame input was designed after the grip-transfer result of the device-frame CNN had been seen, which no nested estimate removes; the aggregator recommendation is also an outer-fold choice.
- The natural-use set has one trial per person, all with the right hand: it measures right-hand recall only, with a wide exact CI, and a constant 'right' answer would score 100 %.
- The ECE null band treats trials as independent and ignores the clustering within persons, so it understates the noise: an ECE above the band is weaker evidence of miscalibration than it looks, and an ECE inside it is strong evidence of calibration.
- Deviations from a textbook inner cross-fit, kept because they are cheaper and checked by the calibration statistics: the calibrators are fitted on cross-fitted logits of single ensemble members but applied to the ensemble mean; each member's early-stopping epoch is chosen on the rows that also supply its cross-fitted logits; a refit on the whole fold is replaced by the cross-validated ensemble; the MiniRocket biases are fitted on all outer-training rows, inner-validation persons included (label-free, but transductive for the inner choices).
- The body-frame re-checks cover the window, normalisation and symmetry only; the other treatment decisions are inherited from the gravity-frame CNN.
- Every model was evaluated on the oracle windows; the cue-free segmentation of production (Appendix G) was not combined with these models in this lane.
- The body frame was given to the 1D CNN only; MiniRocket and the time-frequency models still use the gravity frame.
- Lane m2's engineered features include device-frame channels; whether its stack transfers to grips absent from training was not tested here.

F. Final-model selection, calibration and export

This appendix documents how the final model was chosen among the finalists of the model lanes, how its percentage $P(\text{right})$ was checked and calibrated, and how it was exported for the Stage 3 inference page. Every number comes from leave-one-person-out evaluation: each model is trained on the other persons and tested on the held-out person. Paired comparisons score two models on the same trials. Their statistic is the mean per-person difference, its 95 % CI comes from a paired person bootstrap, and its p-value from an exact sign-flip test over persons. Holm's adjustment runs within each named family of comparisons.

The final model is the body-frame 1D CNN of lane m3 (Appendix E). It is an ensemble of four 1D CNNs on 12 channels: 8 gravity-frame channels and 4 body-frame channels (roll rate, pitch rate, lateral and forward acceleration). Its pipeline z-scores the channels, augments the training trials by mirroring, time warping, scaling and jitter [22], and ends with a Platt

calibrator [21] fitted inside each fold. Its own LOSO numbers are exploratory: it was the best of 16 configurations in lane m3 and was then chosen here among 17 candidates. On the 598 primary trials of 20 persons it reaches an accuracy of 89.3 % [85.0, 93.1], an AUC of 0.951 [0.912, 0.980], a log loss of 0.291 [0.196, 0.420] and a Brier score [19] of 0.083 (all exploratory). The selection-aware estimate of the whole selection is an accuracy of 88.5 % [84.1, 92.3] (Appendix F.4).

F.1. The candidates and the context of every experiment

Options. The final model could have come from any of the several hundred experiments of the model lanes. Choosing among all of them on the outer folds that also report the winner's accuracy would make that accuracy optimistic. The lane therefore restricted the choice to a pre-specified list of 17 candidates:

- every trial-level finalist of lanes m1, m2, m3 and p1, read from their finalists files;
- the reference floor, a logistic regression on the cumulative yaw about gravity at 8 instants;
- three engineered models given the body-frame channels (Appendix F.5);
- one pre-specified ensemble, the mean logit of one finalist per lane (the body-frame CNN, the m2 logistic and the m1 PHYS6 logistic).

Lane m4, whose unit is a cue-free recording, and lane x1, the descriptive PD01 case study, provide no trial-level candidate.

Evidence. Figure 80 places every experiment of the lanes' registries on one plane of accuracy against log loss: 361 experiments, of which 127 were compared on the outer folds before being kept. Accuracy and log loss move together. The finalists sit at the upper-left edge of their lanes' clouds. The few lane m3 points further left are post hoc checks of the body-frame CNN, not finalists (Appendix E). Table 67 lists the candidates as their lanes deliver them, with their representation, data treatment, calibration, natural-use recall and number of outer-fold configurations. Its natural-use column is right-hand recall on the free-choice trials (all right-hand so far; a constant right answer would score 100 %). Of the 17 candidates, 7 were chosen among several configurations on the outer folds of their own lane, so their numbers are exploratory. Figure 5 in Section 5 plots the same comparison as a forest.

Reproduction. Before comparing the candidates, the lane rebuilt every finalist from its finalists file (module, feature builder, model factory and parameters) and re-ran it through the LOSO harness. 12 of 12 finalists reproduce their saved predictions on every primary and free-choice trial; the largest difference in $P(\text{right})$ is 0.0e+00. The m2 stack was not re-run because of its cost; its saved result is used as lane m2 saved it.

F.2. The selection rule and the leaders

Options. Three rules were considered: the highest LOSO accuracy, the lowest LOSO log loss, and a pre-specified rule that also weighs robustness and simplicity. Accuracy is a coarse score on 598 trials and ignores the percentage shown to the user. Log loss scores the percentage itself, but the lowest log loss among 17 candidates is optimistic, and its lead over its neighbours may be chance.

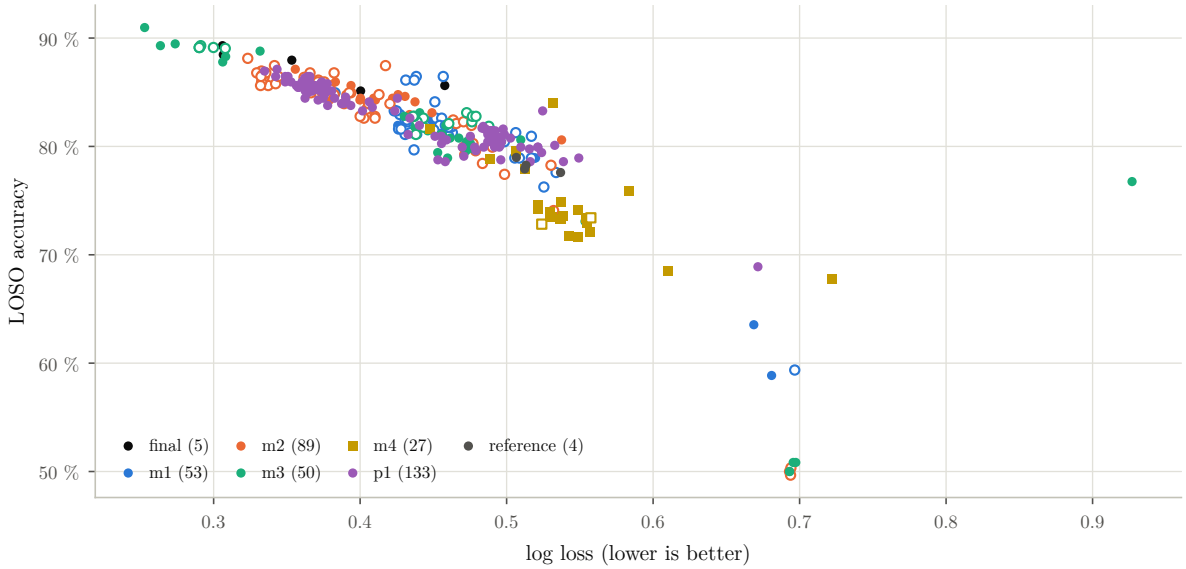


Figure 80: Every experiment of the model lanes' registries, the reference and lane final's own models (361 with a log loss below 1.2; the stress tests of lane final are left out): LOSO accuracy against log loss, coloured by lane. Open markers: compared on the outer folds before being kept (exploratory; 127). Squares: lane m4, whose unit is a cue-free recording, whose classifier is its own reference model (not the final model) and whose abstentions score one half; they are not comparable with the trial-level points. Points near 50 % and a log loss of 0.69 are ablations that cannot see the hand by design. The figure gives the context of the selection: the candidates come from the upper-left edge of their lanes.

Table 67: Candidates of the cross-lane selection as their lanes deliver them (leave-one-person-out, 95 % person-bootstrap CIs). ECE with the 95th percentile of its perfect-calibration null (*: P constant within each fold, so the pooled ECE mostly sorts the folds and is not comparable); after in-fold remedy: the production form (remedy chosen inside every fold); natural use: right-hand recall on the free-choice trials; configs: outer-fold choices in the candidate's lane.

model	lane	family	representation	data treatment	accuracy	AUC	log loss	ECE (null 95th pct.)	after in-fold remedy: log loss / ECE	natural use: right recall	k = 3	configs	nested	CPU s / LOSO
m3 1D CNN, body frame	m3	1D CNN ensemble (4 members)	12 channels: gravity frame + body frame	z-score, mirror / warp / scale / jitter augmentation, Platt in-fold	89.3 [85.0, 93.1]	0.951	0.291 [0.196, 0.426]	0.026 (0.039)	0.291 / 0.026	85 %	96.4	16	no	277
ensemble: mean logit of 3 lanes	final	mean logit of 3 finalists	m3 CNN + m2 logistic + m1 PHYS6	each component as in its lane	89.3 [85.3, 93.0]	0.946	0.306 [0.228, 0.402]	0.053 (0.046)	0.308 / 0.054	90 %	95.3	1	yes	289
m2 stack (LR + SVM + LGBM)	m2	stacked ensemble	691 features, 17 channels	components tuned in-fold, stacker on inner logits	88.1 [82.9, 92.9]	0.936	0.323 [0.224, 0.432]	0.025 (0.043)	0.323 / 0.025	90 %	93.7	12	yes	–
m2 LightGBM	m2	gradient boosting	691 features, 17 channels	mirror augmentation, Optuna on inner folds	85.6 [80.3, 90.8]	0.930	0.337 [0.242, 0.437]	0.022 (0.046)	0.337 / 0.048	90 %	93.2	12	yes	926
m2 logistic + body frame (21 ch.)	final	L2 logistic regression	m2 feature families on 17 + 4 body channels	z-score, mirror augmentation, C nested	88.0 [82.6, 93.3]	0.928	0.353 [0.243, 0.494]	0.057 (0.045)	0.353 / 0.057	85 %	94.1	1	yes	368
p1 LightGBM + calibrator	p1	gradient boosting (fixed parameters)	14 summary features x 17 channels	mirror augmentation, calibrator chosen in-fold	85.5 [81.0, 89.5]	0.921	0.358 [0.265, 0.463]	0.032 (0.043)	0.358 / 0.032	90 %	92.2	1	yes	109
m2 logistic (691 features)	m2	L2 logistic regression	691 features of 7 families, 17 channels	z-score, mirror augmentation, C nested	85.5 [79.8, 90.6]	0.920	0.365 [0.260, 0.464]	0.048 (0.048)	0.365 / 0.048	90 %	92.8	12	yes	68
m2 logistic on the CNN input (12 ch.)	final	L2 logistic regression	m2 feature families on the 8 gravity + 4 body channels	z-score, mirror augmentation, C nested	85.1 [80.1, 90.0]	0.908	0.400 [0.313, 0.508]	0.060 (0.050)	0.400 / 0.060	75 %	92.4	1	yes	92
m1 PHYS6 logistic	m1	antisymmetric logistic (6 features)	yaw, roll and lateral velocity at 50 / 100 % of the approach	treatment (scaling, winsorising, asinh) and C chosen in-fold	83.3 [79.2, 87.4]	0.890	0.423 [0.345, 0.506]	0.047 (0.052)	0.423 / 0.047	75 %	90.4	1	yes	140
m1 PHYS6 GAM	m1	additive spline logistic	PHYS6 features	knots and C chosen in-fold, mirror-symmetrised	81.6 [77.1, 85.9]	0.886	0.426 [0.351, 0.510]	0.029 (0.052)	0.429 / 0.026	75 %	89.8	6	yes	170
m3 MiniRocket	m3	MiniRocket + logistic	17 channels, 4000 random-kernel features	per-trial amplitude, C nested, Platt in-fold	82.3 [77.4, 86.8]	0.879	0.439 [0.372, 0.509]	0.049 (0.054)	0.439 / 0.049	80 %	89.6	4	yes	318
logistic, body-frame time courses	final	antisymmetric logistic (32 features)	8 time bins of yaw rate, cumulative yaw, roll rate, lateral acceleration	RMS scaling (no centring), no intercept, C nested	85.6 [80.7, 90.0]	0.896	0.458 [0.353, 0.567]	0.101 (0.052)	0.458 / 0.101	75 %	93.0	1	yes	37
p1 MiniRocket + ridge	p1	MiniRocket + ridge	gravity frame, about 2000 features	ridge alpha and Platt nested	80.6 [76.5, 85.1]	0.872	0.459 [0.397, 0.518]	0.055 (0.053)	0.459 / 0.055	80 %	88.4	1	yes	68
m3 2D CNN on STFT	m3	2D CNN ensemble	signed STFT planes of the gravity frame	z-score, augmentation, Platt in-fold	83.1 [79.2, 87.4]	0.871	0.473 [0.401, 0.548]	0.085 (0.054)	0.473 / 0.085	80 %	89.4	3	no	726
p1 logistic (chiral bins)	p1	logistic regression (16 features)	8 time bins of yaw rate and cumulative yaw	standardised, mirror augmentation, C = 1	81.4 [77.5, 85.4]	0.868	0.490 [0.421, 0.569]	0.069 (0.053)	0.498 / 0.085	85 %	89.7	1	no	6
reference: yaw logistic	reference	logistic regression (8 features)	cumulative yaw about gravity at 8 instants	mirror augmentation, C = 1	77.9 [73.1, 82.9]	0.855	0.512 [0.449, 0.579]	0.072 (0.057)	0.512 / 0.072	75 %	87.2	1	no	5
m1 rule: first-half yaw sign	m1	rule (2 parameters)	first-half yaw about gravity	direction and hit rate learned in-fold	78.9 [73.5, 83.8]	0.729	0.519 [0.453, 0.592]	0.097 (0.059) *	0.519 / 0.097	85 %	86.4	1	no	6

Decision. The lane used the pre-specified rule (`final_finalmodel.recommend`), in five steps.

1. *Production form.* Every candidate is scored as it would ship: its own pipeline plus a calibration remedy (none, temperature, Platt or beta) chosen inside every outer fold by grouped cross-validation on the training persons' inner cross-fitted predictions. The product is the calibrated percentage, so the comparison uses it.
2. *Leaders.* A leader is not detectably worse than the lowest-log-loss candidate in accuracy or in log loss (paired, Holm over the comparisons against that candidate). Two candidates whose log losses differ by less than 0.001 with no trial changing its answer are called identical.
3. *Robust leaders.* A robust leader shows no Holm-significant accuracy drop on grips absent from training, no natural-use shortfall against the person-matched expectation, and no

screen-away failure that the rest-pose check cannot repair (Appendices F.3, F.6 and F.12).

4. *Portability.* Among the robust leaders, the rule takes the one easiest to port to the browser (portability rank from 1, easiest, to 6).
5. *Ties.* Remaining ties go to the candidate with the fewest upstream outer-fold choices, then to the lowest log loss.

Evidence. Figure 81 and Table 68 compare every candidate with the lowest-log-loss candidate, the body-frame CNN. After Holm over the 16 comparisons against it, 11 candidates are worse in accuracy or in log loss. The 6 leaders are the CNN, the ensemble, the m2 stack (logistic regression, support vector machine and LightGBM), the m2 LightGBM, the m2 logistic on 691 features and the p1 LightGBM with its calibrator. The data cannot rank the leaders at 20 persons. Against the reference floor (Holm over 16 comparisons), 11 candidates are more accurate. The CNN gains +11.7 [+8.0, +15.6] points of accuracy over the reference (Holm $p < 0.001$) and changes the log loss by −0.223 (Holm $p = 0.008$).

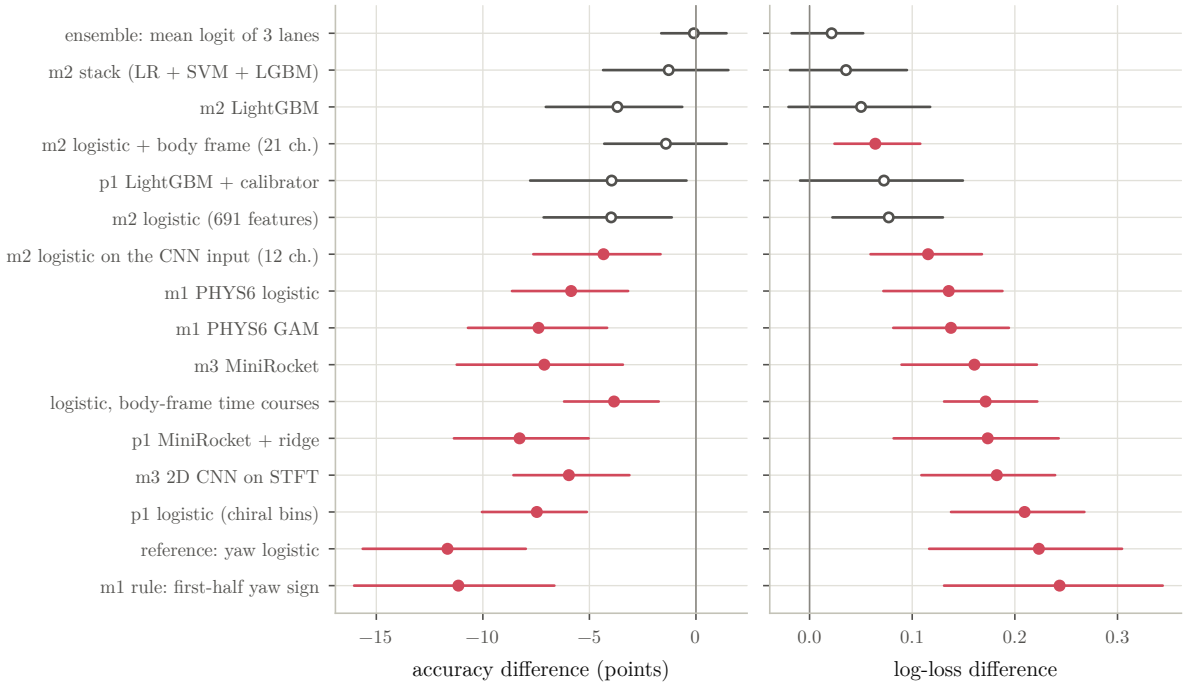


Figure 81: Every candidate against the lowest-log-loss candidate (the body-frame CNN) on the same trials, each in its production form: mean per-person difference in accuracy (left, points) and in log loss (right) with paired person-bootstrap 95 % CIs. Filled red: Holm-adjusted sign-flip p below 0.05 over the 16 comparisons. The candidates with open markers in both panels are the leaders, among which robustness and portability decide.

F.3. Robustness to unseen test conditions

Tests. The PI will hold the phone in ways the training data may not cover. The lane ran four tests on every candidate (Table 69).

1. *A grip absent from training:* train on the upright trials only and test on the sideways trials (top left and top right).

Table 68: Paired comparisons on the same trials: mean per-person differences with 95 % CIs; Holm within each family (against the best, against the reference).

model	vs best: accuracy	vs best: log loss	vs reference: accuracy	vs reference: log loss
reference: yaw logistic	−11.7 [−15.6, −8.0] (Holm < 0.001)	+0.223 [0.116, 0.304] (Holm 0.006)	(anchor)	(anchor)
m1 rule: first-half yaw sign	−11.1 [−16.0, −6.6] (Holm 0.002)	+0.244 [0.131, 0.344] (Holm 0.006)	+0.5 [−3.4, 4.2] (Holm 0.804)	+0.020 [−0.029, 0.075] (Holm 0.955)
m1 PHYS6 logistic	−5.9 [−8.6, −3.2] (Holm 0.008)	+0.136 [0.072, 0.188] (Holm 0.006)	+5.8 [2.9, 8.9] (Holm 0.007)	−0.088 [−0.143, −0.027] (Holm 0.097)
m1 PHYS6 GAM	−7.4 [−10.7, −4.2] (Holm 0.007)	+0.138 [0.082, 0.194] (Holm 0.003)	+4.3 [0.6, 8.3] (Holm 0.133)	−0.086 [−0.152, −0.013] (Holm 0.185)
m2 logistic (691 features)	−4.0 [−7.1, −1.1] (Holm 0.084)	+0.077 [0.022, 0.130] (Holm 0.051)	+7.7 [3.9, 11.3] (Holm 0.007)	−0.146 [−0.208, −0.073] (Holm 0.008)
m2 LightGBM	−3.7 [−7.0, −0.6] (Holm 0.186)	+0.050 [−0.021, 0.118] (Holm 0.554)	+8.0 [3.9, 12.1] (Holm 0.010)	−0.173 [−0.233, −0.112] (Holm 0.002)
m2 stack (LR + SVM + LGBM)	−1.3 [−4.4, 1.5] (Holm 1.000)	+0.036 [−0.019, 0.095] (Holm 0.554)	+10.4 [6.3, 14.8] (Holm 0.001)	−0.188 [−0.256, −0.111] (Holm 0.003)
m3 MiniRocket	−7.1 [−11.2, −3.4] (Holm 0.009)	+0.161 [0.090, 0.221] (Holm 0.005)	+4.5 [0.9, 8.6] (Holm 0.114)	−0.063 [−0.122, −0.001] (Holm 0.388)
m3 2D CNN on STFT	−6.0 [−8.6, −3.1] (Holm 0.008)	+0.182 [0.109, 0.239] (Holm 0.003)	+5.7 [3.5, 8.0] (Holm 0.002)	−0.041 [−0.079, 0.002] (Holm 0.388)
p1 LightGBM + calibrator	−4.0 [−7.8, −0.4] (Holm 0.186)	+0.072 [−0.009, 0.149] (Holm 0.421)	+7.7 [4.3, 11.0] (Holm 0.004)	−0.151 [−0.212, −0.090] (Holm 0.004)
p1 logistic (chiral bins)	−7.5 [−10.0, −5.1] (Holm < 0.001)	+0.209 [0.138, 0.268] (Holm < 0.001)	+4.2 [1.4, 7.4] (Holm 0.063)	−0.014 [−0.059, 0.039] (Holm 0.955)
p1 MiniRocket + ridge	−8.3 [−11.4, −5.0] (Holm 0.002)	+0.174 [0.082, 0.243] (Holm 0.007)	+3.4 [−0.4, 7.6] (Holm 0.245)	−0.050 [−0.099, −0.003] (Holm 0.388)
logistic, body-frame time courses	−3.8 [−6.2, −1.7] (Holm 0.029)	+0.172 [0.131, 0.222] (Holm < 0.001)	+7.8 [4.3, 11.3] (Holm 0.003)	−0.052 [−0.132, 0.050] (Holm 0.895)
m2 logistic on the CNN input (12 ch.)	−4.3 [−7.6, −1.7] (Holm 0.021)	+0.115 [0.059, 0.168] (Holm 0.007)	+7.3 [4.2, 10.6] (Holm 0.003)	−0.108 [−0.179, −0.031] (Holm 0.099)
m2 logistic + body frame (21 ch.)	−1.4 [−4.3, 1.4] (Holm 1.000)	+0.064 [0.024, 0.108] (Holm 0.034)	+10.3 [6.0, 14.8] (Holm 0.003)	−0.159 [−0.247, −0.046] (Holm 0.069)
ensemble: mean logit of 3 lanes	−0.1 [−1.6, 1.4] (Holm 1.000)	+0.021 [−0.017, 0.052] (Holm 0.554)	+11.6 [7.8, 15.6] (Holm < 0.001)	−0.202 [−0.265, −0.126] (Holm 0.002)
m3 1D CNN, body frame	(anchor)	(anchor)	+11.7 [8.0, 15.6] (Holm < 0.001)	−0.223 [−0.304, −0.116] (Holm 0.008)

2. *The screen held away:* test on screen-away twins synthesised from the recorded trials (Appendix F.12).
3. *The other platform:* train on the persons of one platform (Android or iOS) and test on the persons of the other.
4. *Label noise:* train without the blocks that curation flagged as possible wrong-hand or swapped recordings (Appendix A).

Each test gives a paired accuracy change on the same test trials, with Holm over the candidates within the test. Figure 7 in Section 5 plots the grip and screen-away tests for every candidate.

Evidence: grips. The grip test separates the leaders. The leaders built on lane m2's and lane p1's device-frame features lose accuracy on the sideways grips after Holm. The m2 logistic falls to 51.3 % (−31.0 points, Holm $p < 0.001$), the m2 LightGBM to 64.7 % (−19.8 points, Holm $p = 0.014$), the m2 stack to 61.6 % (−25.3 points, Holm $p = 0.013$) and the p1 LightGBM to 63.8 % (−21.6 points, Holm $p = 0.004$). The CNN keeps 85.3 % (−4.1 points, Holm $p = 0.188$). The ensemble keeps 79.3 % (−10.5 points, Holm $p = 0.312$): a large drop, but not significant after Holm. Over all candidates, 5 of 17 fail the grip test: the device-frame leaders above and the m2 logistic with the body-frame channels added (51.3 %, −36.8 points, Holm $p < 0.001$). Lane m3 reports the same grip test for its own models (Appendix E.10).

Evidence: platform and label noise. No candidate changes detectably when it is trained on the other platform or without the suspect blocks (Holm over the candidates; Table 69). Training on the other platform also means fewer training persons, so the CNN had a size-matched control: as many training persons, drawn at random from both platforms. Trained on the other platform only, the CNN reaches 86.5 %; the control reaches 89.3 %. The platform switch changes the accuracy by −2.3 [−4.7, +0.2] points (Holm $p = 0.209$), and the smaller training set by −0.4 [−2.8, +2.1] points (Holm $p = 0.789$). This Holm family holds the four paired accuracy comparisons of the final model: these two and the two screen-flip augmentation tests of Appendix F.12. Without the suspect blocks, the CNN changes by +1.1 points (Holm $p = 1.000$).

Decision. The grip test removes the device-frame leaders. The robust leaders in the sense of the rule (no Holm-significant drop) are the CNN and the ensemble; the CNN's own paired CI of the grip change still excludes zero (−4.1 [−7.0, −1.4] points, unadjusted $p = 0.016$). The platform and label-noise tests exclude no candidate.

Table 69: Robustness to unknown test conditions (accuracy in %, paired changes in points, Holm over the candidates within each test).

model	rotated grips, upright-only training	screen away	screen away + check	check cost (recorded)	other platform only	without suspect blocks
reference: yaw logistic	84.5 (+0.4, Holm 1.000)	77.9 (invariant)	77.9	+0.0	−1.4 (Holm 1.000)	−0.5 (Holm 1.000)
m1 rule: first-half yaw sign	84.1 (+0.0, Holm 1.000)	78.9 (invariant)	78.9	+0.0	+0.0 (Holm 1.000)	+0.0 (Holm 1.000)
m1 PHYS6 logistic	87.5 (−1.1, Holm 1.000)	51.5	82.4	−0.8	−1.3 (Holm 1.000)	+0.9 (Holm 1.000)
m1 PHYS6 GAM	86.2 (−0.7, Holm 1.000)	48.8	80.8	−0.8	−0.0 (Holm 1.000)	+1.2 (Holm 1.000)
m2 logistic (691 features)	51.3 (−31.0, Holm < 0.001)	63.0	85.3	−0.0	−4.2 (Holm 0.108)	−0.6 (Holm 1.000)
m2 LightGBM	64.7 (−19.8, Holm 0.014)	60.9	84.6	−1.0	+0.1 (Holm 1.000)	+1.4 (Holm 1.000)
m2 stack (LR + SVM + LGBM)	61.6 (−25.3, Holm 0.013)	60.0	87.3	−0.8	–	–
m3 1D CNN, body frame	85.3 (−4.1, Holm 0.188)	28.6	86.6	−2.7	−2.6 (Holm 0.170)	+1.1 (Holm 1.000)
m3 MiniRocket	81.0 (−3.8, Holm 0.422)	82.3 (invariant)	82.3	+0.0	−3.0 (Holm 1.000)	−0.6 (Holm 1.000)
m3 2D CNN on STFT	84.9 (−3.0, Holm 1.000)	83.1 (invariant)	83.1	+0.0	–	–
p1 LightGBM + calibrator	63.8 (−21.6, Holm 0.004)	59.2	84.4	−1.0	−4.7 (Holm 0.097) [p1]	−0.6 (Holm 1.000)
p1 logistic (chiral bins)	83.2 (−3.5, Holm 1.000)	81.4 (invariant)	81.4	+0.0	−1.3 (Holm 1.000) [p1]	−0.4 (Holm 1.000)
p1 MiniRocket + ridge	82.8 (−2.2, Holm 1.000)	80.6 (invariant)	80.6	+0.0	−3.4 (Holm 0.679)	−0.7 (Holm 1.000)
logistic, body-frame time courses	87.1 (−0.4, Holm 1.000)	47.3	84.6	−1.0	−2.3 (Holm 1.000)	+0.8 (Holm 1.000)
m2 logistic on the CNN input (12 ch.)	84.5 (−2.4, Holm 0.312)	60.2	84.6	−0.5	−2.0 (Holm 1.000)	+0.4 (Holm 1.000)
m2 logistic + body frame (21 ch.)	51.3 (−36.8, Holm < 0.001)	41.3	86.3	−1.6	−3.3 (Holm 1.000)	−0.2 (Holm 1.000)
ensemble: mean logit of 3 lanes	79.3 (−10.5, Holm 0.312)	43.0	88.5	−0.7	−2.4 (Holm 1.000)	−0.4 (Holm 1.000)

F.4. The decision and the selection-aware estimate

Decision. Table 70 applies the rule to every candidate. Of the 6 leaders, only the CNN and the ensemble are robust. Neither shows a natural-use shortfall, and the rest-pose check repairs the screen-away failure of both. The rule then takes the more portable of the two: the CNN (portability rank 5). The ensemble needs the CNN and two more models. The setting `FINAL_MODEL` of the lane agrees with the rule. Lane p1 kept the default data treatment in 11 of its 11 choices (Appendix B), so the final model uses the preprocessing of the cache unchanged.

Options for the headline number. The CNN's own LOSO accuracy is the best of 17 candidates on the folds that report it, so it is optimistic. A selection-aware estimate repeats the whole selection inside every outer fold (nested selection). In each outer fold, every candidate was cross-validated on the training persons only (5 inner person folds). The candidate with the lowest pooled inner log loss, the pre-specified criterion, was refitted on all training persons and predicted the held-out person.

Evidence. Table 71 shows how often each candidate was chosen. The CNN was chosen in 16 of 20 outer folds, the ensemble in 2, and the m2 LightGBM and the m2 stack in 1 and 1. The CNN also has the lowest mean inner log loss (0.304). The nested estimate is an accuracy of 88.5 % [84.1, 92.3], an AUC of 0.945 [0.906, 0.975] and a log loss of 0.307 [0.212, 0.432] (Table 5 and Figure 6 in Section 5). The naive pick, the CNN at 89.3 % (exploratory), is optimistic by +0.8 [−0.2, +2.3] points of accuracy and −0.015 [−0.034, −0.003] in log loss (pooled differences). As a mean per-person difference, the optimism in accuracy is +0.9 [−0.2, +2.4] points (sign-flip $p = 0.375$). The headline of the report is therefore the nested estimate.

Caveat. The nested selection removes the optimism of choosing among the 17 candidates. It does not remove the optimism of the lanes' own design choices, made on the same outer folds before the candidates reached this lane. Of the candidates, 7 were chosen among several configurations, and lane m3 designed the body-frame input post hoc.

F.5. Input against model class

Question. The CNN differs from the engineered models in two ways at once: its input (the body-frame channels) and its model class (a network on time courses). If the input carried the whole gain, a simpler model on the same channels would do. Lane m3 asked the same question

Table 70: Decision table: every candidate in its production form (accuracy, log loss, ECE before and after the in-fold remedy), robustness, portability and nested selection frequency (*: P constant within each fold, ECE not comparable).

model	accuracy	log loss	worse than best (Holm)	ECE before / after remedy	calibration after remedy	worst platform	worst variant (>= 5 persons)	grip transfer fails	screen away (raw / with check)	natural-use shortfall	bounds +-200 ms (points)	robust	port (1 easy - 6 hard)	nested picks	decision
m3 1D CNN, body frame	89.3	0.291	no	0.026 / 0.026	consistent	ios 87.0	twist 60	no	29 / 87	no	−0.8	yes	5	16	FINAL
ensemble: mean logit of 3 lanes	89.3	0.308	no	0.053 / 0.054	miscalibrated	ios 86.0	twist 60	no	43 / 88	no	–	yes	6	2	
m2 stack (LR + SVM + LGBM)	88.1	0.323	no	0.025 / 0.025	consistent	ios 86.5	twist 67	yes	60 / 87	no	–	no	4	1	
m2 LightGBM	85.6	0.337	no	0.022 / 0.018	consistent	ios 84.7	twist 67	yes	61 / 85	no	–	no	4	1	
m2 logistic + body frame (21 ch.)	88.0	0.353	yes	0.057 / 0.057	miscalibrated	ios 85.6	twist 67	yes	41 / 86	no	–	no	3	0	
p1 LightGBM + calibrator	85.5	0.358	no	0.032 / 0.032	consistent	ios 84.2	twist 67	yes	59 / 84	no	−2.0 [p1]	no	4	0	
m2 logistic (691 features)	85.5	0.365	no	0.048 / 0.048	consistent	ios 84.2	twist 47	yes	63 / 85	no	–	no	3	0	
m2 logistic on the CNN input (12 ch.)	85.1	0.400	yes	0.060 / 0.060	miscalibrated	ios 82.3	twist 60	no	60 / 85	no	–	yes	3	0	
m1 PHYS6 logistic	83.3	0.423	yes	0.047 / 0.047	consistent	ios 83.3	curved 62	no	52 / 82	no	–	yes	1	0	
m1 PHYS6 GAM	81.6	0.429	yes	0.028 / 0.026	consistent	android 81.5	curved 56	no	49 / 81	no	–	yes	2	0	
m3 MiniRocket	82.3	0.439	yes	0.049 / 0.049	consistent	ios 81.4	twist 60	no	82 / 82	no	–	yes	4	0	
logistic, body-frame time courses	85.6	0.458	yes	0.101 / 0.101	miscalibrated	ios 85.6	twist 67	no	47 / 85	no	–	yes	1	0	
p1 MiniRocket + ridge	80.6	0.459	yes	0.055 / 0.055	consistent	android 79.9	tilted_touch 60	no	81 / 81	no	–	yes	4	0	
m3 2D CNN on STFT	83.1	0.473	yes	0.085 / 0.085	miscalibrated	android 82.2	twist 60	no	83 / 83	no	–	yes	5	0	
p1 logistic (chiral bins)	81.6	0.498	yes	0.069 / 0.085	miscalibrated	ios 81.4	twist 60	no	81 / 81	no	−1.7 [p1]	yes	1	0	
reference: yaw logistic	77.4	0.512	yes	0.072 / 0.072	miscalibrated	android 77.8	twist 53	no	78 / 78	no	–	yes	1	0	
m1 rule: first-half yaw sign	78.9	0.519	yes	0.097 / 0.097 *	miscalibrated	ios 76.3	twist 60	no	79 / 79	no	–	yes	1	0	

Table 71: How often each candidate was chosen by the inner log loss across the outer folds.

candidate	folds chosen	mean inner log loss
m3 1D CNN, body frame	16	0.304
ensemble: mean logit of 3 lanes	2	0.314
m2 stack (LR + SVM + LGBM)	1	0.338
m2 LightGBM	1	0.349
m2 logistic + body frame (21 ch.)	0	0.354
p1 LightGBM + calibrator	0	0.370
m2 logistic (691 features)	0	0.385
m2 logistic on the CNN input (12 ch.)	0	0.410
m1 PHYS6 GAM	0	0.428
m1 PHYS6 logistic	0	0.430
logistic, body-frame time courses	0	0.440
m3 MiniRocket	0	0.445
p1 MiniRocket + ridge	0	0.449
m3 2D CNN on STFT	0	0.468
p1 logistic (chiral bins)	0	0.497
reference: yaw logistic	0	0.520
m1 rule: first-half yaw sign	0	0.522

with its own comparisons (Appendix E.11); this test gives the body-frame channels themselves to engineered models.

Options tested. The lane gave the body-frame channels to three engineered models and evaluated them like the other candidates (per-trial features, nested tuning of the regularisation):

- lane m2's feature families and logistic regression on the CNN's own 12 channels;
- lane m2's logistic regression on its 17 channels plus the 4 body-frame channels (21 channels);
- an antisymmetric logistic regression on 32 features: 8 time bins of the yaw rate, the cumulative yaw, the roll rate and the lateral acceleration, the mirror-odd (chiral) channels of the CNN's input.

Evidence. Figure 82 and Table 72 give six paired comparisons, with Holm over the six.

- On the same 12 channels, the CNN is more accurate than lane m2's features and logistic regression by +4.3 [+1.7, +7.6] points (Holm $p = 0.016$); the log loss changes by −0.115 [−0.168, −0.059] (Holm $p = 0.004$).

- Against the 32-feature logistic on the mirror-odd channels of its input, the CNN gains +3.8 [+1.7, +6.2] points (Holm $p = 0.021$); the log loss changes by −0.172 [−0.222, −0.131] (Holm $p < 0.001$).
- The two engineered models on the body-frame input do not differ detectably in accuracy: −0.5 [−3.3, +2.1] points (Holm $p = 0.757$).
- Adding the body-frame channels to lane m2's logistic changes its accuracy by +2.6 [+0.8, +4.5] points (Holm $p = 0.065$): not detectable after Holm.
- The 21-channel logistic does not differ detectably from the CNN in accuracy (−1.4 [−4.3, +1.4] points, Holm $p = 0.738$), but its log loss is higher by +0.064 [+0.024, +0.108] (Holm $p = 0.023$).
- The time-course logistic does not differ detectably from the m1 PHYS6 logistic, which summarises the same components in six features (+2.0 [−0.1, +4.1] points, Holm $p = 0.232$).

Conclusion. On the same input, the CNN is more accurate and has a lower log loss than both engineered models. Its gain needs more than the body-frame channels. The comparison is not symmetric in tuning effort: the CNN's configuration was the best of 16 in lane m3, while each engineered model ran once with nested tuning of its regularisation.

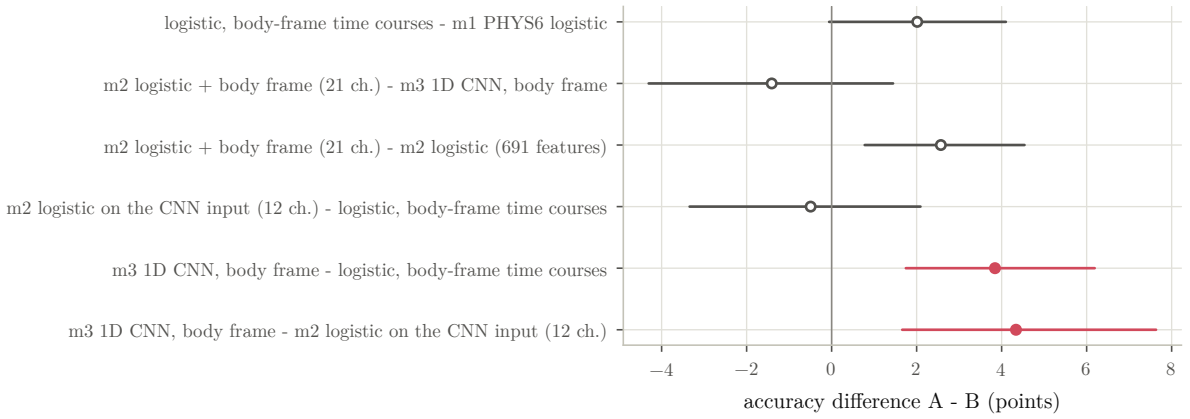


Figure 82: Input against model class: paired accuracy differences, model A minus model B, between models given the same channels (mean per person, paired person-bootstrap 95 % CI). Filled red: Holm-adjusted p below 0.05 over the six comparisons. On the same input the CNN is more accurate than both engineered models, so its gain needs more than the body-frame channels.

Table 72: Input against model class (task 2b): models given the same channels.

comparison (A - B)	isolates	accuracy	log loss
m3 1D CNN, body frame - m2 logistic on the CNN input (12 ch.)	model class on the same 12 channels (CNN vs m2 features + LR)	+4.3 [1.7, 7.6] (Holm 0.016)	−0.115 [−0.168, −0.059] (Holm 0.004)
m3 1D CNN, body frame - logistic, body-frame time courses	CNN vs a 32-feature logistic on the odd channels of its input	+3.8 [1.7, 6.2] (Holm 0.021)	−0.172 [−0.222, −0.131] (Holm < 0.001)
m2 logistic on the CNN input (12 ch.) - logistic, body-frame time courses	rich features vs time-binned odd channels (same frames)	−0.5 [−3.3, 2.1] (Holm 0.757)	−0.056 [−0.100, −0.014] (Holm 0.055)
m2 logistic + body frame (21 ch.) - m2 logistic (691 features)	adding the body-frame channels to lane m2's logistic	+2.6 [0.8, 4.5] (Holm 0.065)	−0.013 [−0.049, 0.034] (Holm 0.596)
m2 logistic + body frame (21 ch.) - m3 1D CNN, body frame	m2 logistic with 21 channels vs the CNN	−1.4 [−4.3, 1.4] (Holm 0.738)	+0.064 [0.024, 0.108] (Holm 0.023)
logistic, body-frame time courses - m1 PHYS6 logistic	time courses vs 6 physics features (same components)	+2.0 [−0.1, 4.1] (Holm 0.232)	+0.036 [−0.023, 0.101] (Holm 0.572)

F.6. Transfer to natural use

Why. A model could learn motion that only the instructed protocol produces. The free-choice trials test whether its evidence transfers to movements whose hand the person chose. Natural-

use accuracy is right-hand recall on the free-choice trials (all right-hand so far; a constant right answer would score 100 %), not a balanced accuracy.

Test. Raw recall on 20 trials compares models poorly, because persons differ in how readable they are. The lane therefore used a person-matched test. The expected correctness of each free-choice trial is the same person's right-hand accuracy on the instructed trials. The observed number of correct free-choice trials is compared with the Poisson-binomial distribution of these expectations (lower tail, Holm over the candidates).

Evidence. No candidate falls short of its expectation after Holm (Figure 83, Table 73). The CNN answers 17 of 20 free-choice trials correctly (right-hand recall 85.0 %; all right-hand so far, so a constant right answer would score 100 %), against 17.7 expected (Holm $p = 1.000$).

Which evidence transfers. An occlusion analysis asks which input components carry the CNN's evidence. The channels of one component are set to zero in the test window; the resulting change in the logit toward the reported hand is the component's contribution. On the free-choice trials, the yaw component points to the reported hand in 17 of 20 trials, the roll component in 16 and the lateral component in 11. Against the same persons' instructed trials of that hand (free choice minus instructed), the lateral contribution changes by -0.37 [-0.68, -0.04] (Holm over the three components, $p = 0.133$) and the yaw contribution by -0.19 [-0.67, $+0.26$] (Holm $p = 0.889$). The whole logit is $+1.70$ on the free-choice trials against $+2.74$ on the instructed ones (difference -1.05 [-1.85, -0.23], unadjusted sign-flip $p = 0.023$). So the CNN is less sure on uninstructed movements, and the lateral component points to the reported hand least often. For the body-frame time-course logistic, both the lateral contribution (-0.47 [-0.76, -0.21], Holm $p = 0.008$) and the yaw contribution (-0.46 [-0.81, -0.12], Holm $p = 0.034$) shrink after Holm.

Calibration on natural use. With one hand only, the mean level is the only calibration check. The mean $P(\text{right})$ on the free-choice trials is 74.1 %, against 82.8 % on the same persons' instructed right-hand trials (per-person difference -8.7 [-20.8, $+2.2$] points, sign-flip $p = 0.158$): no detectable shift.

Decision. The natural-use criterion of the rule excludes no leader. The lateral evidence transfers less than the yaw evidence, as lane m1 found for its lateral feature (Appendix C); leaning on yaw is the safer direction for Stage 3. The set holds one free-choice trial per person, so every interval here is wide.

F.7. Calibration: no remedy shipped

Why. The product is the percentage shown to the user. An over- or under-confident percentage misleads even when the decision is right.

Standard. Calibration is judged by three numbers together. The first is the ECE with its perfect-calibration null band: the ECE values that a perfectly calibrated model with the same predictions shows by chance, with outcomes simulated from the predicted probabilities. The other two are the calibration slope and intercept with person-bootstrap CIs; perfect calibration gives a slope of 1 and an intercept of 0. A model is called miscalibrated only when the slope CI excludes 1, the intercept CI excludes 0, or the ECE lies above its band. The null band treats trials as independent and ignores clustering by person. An ECE inside it is good evidence of

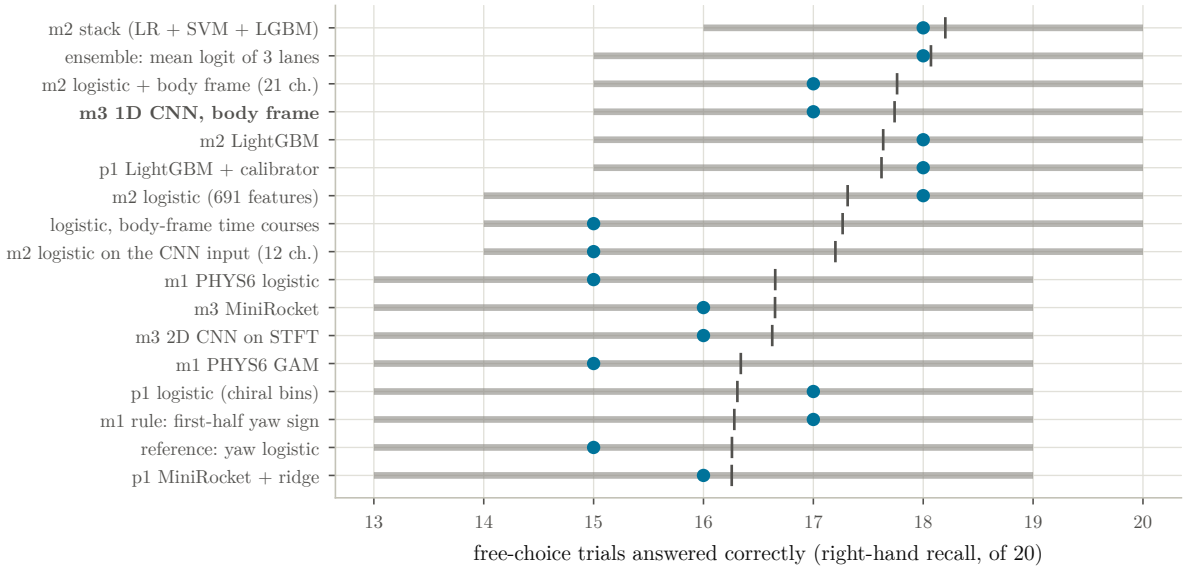


Figure 83: Person-matched natural-use test. Dots: free-choice trials answered correctly by each candidate, out of 20 (right-hand recall: every reported hand is the right one, so a constant right answer would score 100 %). Ticks: the count expected from the same persons' right-hand accuracy on the instructed trials; grey: its 95 % Poisson-binomial range. No candidate falls short after Holm over the candidates, so the natural-use criterion excludes none; the final model is in bold.

Table 73: Person-matched natural-use test: observed correct free-choice trials (right-hand recall) against the Poisson-binomial expectation from the same persons' instructed trials.

model	correct (right recall)	expected (person-matched)	p (lower tail)	Holm	shortfall
reference: yaw logistic	15 of 20	16.3 [13, 19]	0.309	1.000	no
m1 rule: first-half yaw sign	17 of 20	16.3 [13, 19]	0.761	1.000	no
m1 PHYS6 logistic	15 of 20	16.7 [13, 19]	0.220	1.000	no
m1 PHYS6 GAM	15 of 20	16.3 [13, 19]	0.287	1.000	no
m2 logistic (691 features)	18 of 20	17.3 [14, 20]	0.786	1.000	no
m2 LightGBM	18 of 20	17.6 [15, 20]	0.733	1.000	no
m2 stack (LR + SVM + LGBM)	18 of 20	18.2 [16, 20]	0.564	1.000	no
m3 1D CNN, body frame	17 of 20	17.7 [15, 20]	0.398	1.000	no
m3 MiniRocket	16 of 20	16.7 [13, 19]	0.440	1.000	no
m3 2D CNN on STFT	16 of 20	16.6 [13, 19]	0.446	1.000	no
p1 LightGBM + calibrator	18 of 20	17.6 [15, 20]	0.720	1.000	no
p1 logistic (chiral bins)	17 of 20	16.3 [13, 19]	0.760	1.000	no
p1 MiniRocket + ridge	16 of 20	16.3 [13, 19]	0.537	1.000	no
logistic, body-frame time courses	15 of 20	17.3 [14, 20]	0.114	1.000	no
m2 logistic on the CNN input (12 ch.)	15 of 20	17.2 [14, 20]	0.123	1.000	no
m2 logistic + body frame (21 ch.)	17 of 20	17.8 [15, 20]	0.392	1.000	no
ensemble: mean logit of 3 lanes	18 of 20	18.1 [15, 20]	0.602	1.000	no

calibration; one above it is weaker evidence of miscalibration than it looks. Table 67 gives the 95th percentile of the null, Table 74 the upper end of its central 95 % band, so the two thresholds of the CNN differ slightly.

Detection. Without an extra remedy, the CNN has an ECE of 0.026 (null band up to 0.042, $p = 0.470$), a slope of 0.97 [0.69, 1.50] and an intercept of +0.07 [−0.25, +0.39]: consistent with calibration.

Options. Four remedies were tried on top of the model's own in-fold Platt step: Platt scaling [21], temperature scaling [12], isotonic regression and beta calibration. Each was fitted in every outer fold on the training persons' inner cross-fitted predictions and applied to the held-out person.

Evidence. No remedy lowers the log loss (Figure 84a, Table 74). The paired log-loss differences against no remedy are +0.0026 [−0.0083, +0.0111] for Platt, +0.0020 [−0.0087, +0.0106] for temperature, +0.0071 [−0.0108, +0.0244] for isotonic and +0.0058 [−0.0128, +0.0200] for beta. After Holm over the four remedies, their p-values are 1.000, 1.000, 1.000 and 1.000.

Decision. A remedy ships only when its paired log-loss difference against none is negative with a Holm-adjusted p below 0.05; among several, the one with the fewest fitted parameters ships. None qualifies, so the model ships without an extra remedy.

Platforms. On Android (12 persons), the ECE is 0.034 (band up to 0.055) and the slope 1.18 [0.92, 1.76]. On iOS (8 persons), the ECE is 0.039 (band up to 0.067) and the slope 0.75 [0.41, 1.79]. On neither platform does the ECE lie above its band or the slope CI exclude 1; the lane saved no per-platform intercept, so the platform check rests on the ECE, the slope and the calibration-in-the-large difference below. Figure 8 in Section 5 shows the reliability curves, overall and per platform. The differences, Android minus iOS with persons resampled within each platform, are +0.43 [−0.62, +1.08] in slope ($p = 0.446$), +0.23 [−0.38, +0.93] in calibration-in-the-large ($p = 0.554$) and −0.093 [−0.422, +0.142] in log loss ($p = 0.620$). After Holm over these three, the smallest adjusted p is 1.000. With 8 and 12 persons, these platform intervals are wide.

Table 74: Calibration remedies of the final model, fitted on the training persons' inner cross-fitted predictions; Holm over the remedies.

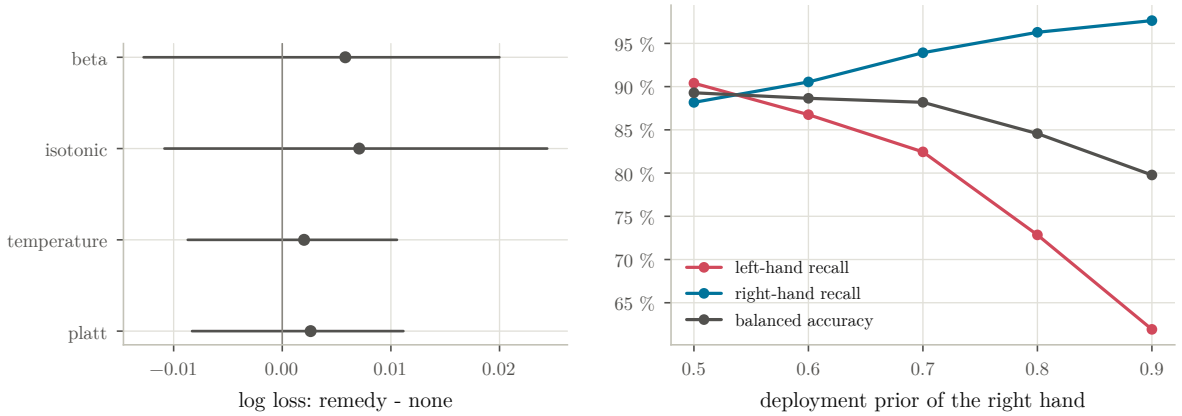
remedy	ECE (null 95 % band)	slope	intercept	log loss	Brier	log loss vs none	ECE vs none (basic CI)	Android ECE / slope	iOS ECE / slope	verdict
none (shipped)	0.026 (up to 0.042)	0.97 [0.69, 1.50]	+0.07 [−0.25, +0.39]	0.291	0.083	–	–	0.034 / 1.18	0.039 / 0.75	consistent with calibration
platt	0.040 (up to 0.045)	1.10 [0.77, 1.69]	+0.08 [−0.24, +0.40]	0.293	0.084	+0.0026 [−0.008, 0.011] (Holm 1.000)	+0.014 [0.007, 0.037]	0.054 / 1.34	0.042 / 0.84	consistent with calibration
temperature	0.034 (up to 0.045)	1.10 [0.77, 1.70]	+0.07 [−0.25, +0.40]	0.292	0.084	+0.0020 [−0.009, 0.011] (Holm 1.000)	+0.007 [−0.001, 0.023]	0.048 / 1.34	0.033 / 0.84	consistent with calibration
isotonic	0.021 (up to 0.042)	0.95 [0.71, 1.33]	+0.04 [−0.33, +0.30]	0.297	0.084	+0.0071 [−0.011, 0.024] (Holm 1.000)	−0.005 [−0.025, 0.009]	0.032 / 1.13	0.045 / 0.75	consistent with calibration
beta	0.041 (up to 0.044)	1.06 [0.75, 1.59]	+0.05 [−0.38, +0.32]	0.296	0.085	+0.0058 [−0.013, 0.020] (Holm 1.000)	+0.015 [0.006, 0.047]	0.056 / 1.26	0.050 / 0.84	consistent with calibration

F.8. The prior

The model assumes equal prior odds of the two hands, because its training data are balanced. A deployment prior π of the right hand would shift every logit z (`final_infer.apply_prior`):

$$z' = z + \log \frac{\pi}{1 - \pi}. \tag{7}$$

Figure 84b shows the cost on the primary trials. At a prior of 0.9, the right-hand recall rises to 97.6 %, while the left-hand recall falls from 90.4 % to 61.9 %.



(a) Calibration remedies of the final model: paired per-person log-loss difference against no remedy (95 % CI). None lowers the log loss (Holm over the four remedies), so none is shipped.

(b) Left- and right-hand recall and balanced accuracy on the primary trials when a deployment prior of the right hand shifts every logit (Equation 7). The shipped prior is 0.5.

Figure 84: The percentage shown to the user: (a) no calibration remedy is shipped; (b) the prior stays at 0.5.

Decision. The default stays at $\pi = 0.5$. The PI chooses the hand freely, and a right-leaning prior would buy natural-use recall with left-hand errors. The shift is exported and documented, so a later deployment can set it.

F.9. Several attempts: the displayed aggregate

Question. The page can combine k attempts of one person and hand. The aggregates below take nearly the same decision; they differ in the percentage they display.

Options. Four aggregates of the attempts' logits $z_1, \ldots, z_k$ were compared: the mean $P(\text{right})$, the mean logit, the sum of the logits, and the sum divided by a fitted scale τ_k:

$$P(\text{right})_k = \sigma\Big(\frac{1}{\tau_k}\sum_{i=1}^{k} z_i\Big), \tag{8}$$

where σ is the logistic function. A scale $\tau_k = 1$ gives the sum and $\tau_k = k$ the mean logit. The scale τ_k was fitted in every outer fold on the training persons' inner cross-fitted logits.

Evidence. Table 75 and Figure 9 (Section 5) evaluate the aggregates on k-subsets of the attempts of one person and hand (19 persons with enough trials of each hand, the same persons for every k). Calibration is judged here only in the large, by the mean confidence minus the accuracy: positive values are over-confident, and an aggregate is neither over- nor under-confident on average when the CI contains 0. The ECE with its null band, the slope and the intercept of the aggregates were not computed. At $k = 3$, all four aggregates reach the same accuracy, 96.3 %. The mean $P(\text{right})$ is under-confident (-0.122 [-0.138, -0.105]), and so is the mean logit (-0.070 [-0.089, -0.049]). The sum of the logits ($+0.011$ [-0.005, $+0.037$]) and the scaled sum (-0.008 [-0.024, $+0.016$]) are neither over- nor under-confident on average. The median τ_3 over the outer folds is 1.54. Against the mean $P(\text{right})$, the scaled sum

changes the log loss by −0.089 [−0.126, −0.028] (Holm over the two paired log-loss comparisons at $k = 3$, $p = 0.010$). At $k = 5$, the scaled sum stays so (+0.001 [−0.018, +0.028]) and reaches 97.3 %. At $k = 1$, the four aggregates are identical by construction. The harness's own $k = 3$ curve, which averages P(right) over the attempts, gives the CNN 96.4 % [93.2, 98.9] (Table 67).

Decision. The page shows $\sigma(\sum_i z_i/\tau_k)$, with τ_k exported in the model file, and never the mean P(right). The scaled sum is calibrated in the large and has the lower log loss; the mean P(right) understates the evidence of attempts that agree. These numbers use the oracle windows of instructed trials; how many attempts the cue-free pipeline accepts is a question for Appendix G.

Table 75: Multi-attempt aggregates of the final model (tau_k fitted on inner cross-fitted predictions of the training persons in every outer fold).

k	aggregate	accuracy	log loss	confidence - accuracy	ECE	calibrated
1	mean P(right)	89.2	0.287 [0.200, 0.414]	−0.011 [−0.042, 0.027]	0.026	yes
1	mean logit	89.2	0.287 [0.200, 0.414]	−0.011 [−0.042, 0.027]	0.026	yes
1	sum of logits	89.2	0.287 [0.200, 0.414]	−0.011 [−0.042, 0.027]	0.026	yes
1	sum / tau_k (nested)	89.2	0.287 [0.200, 0.414]	−0.011 [−0.042, 0.027]	0.026	yes
2	mean P(right)	94.2	0.219 [0.165, 0.282]	−0.096 [−0.114, −0.072]	0.092	no
2	mean logit	94.2	0.190 [0.124, 0.272]	−0.055 [−0.076, −0.027]	0.053	no
2	sum of logits	94.2	0.174 [0.071, 0.314]	+0.009 [−0.014, 0.038]	0.019	yes
2	sum / tau_k (nested)	94.2	0.169 [0.084, 0.276]	−0.013 [−0.035, 0.015]	0.017	yes
3	mean P(right)	96.3	0.204 [0.159, 0.252]	−0.122 [−0.138, −0.105]	0.124	no
3	mean logit	96.3	0.155 [0.101, 0.223]	−0.070 [−0.089, −0.049]	0.070	no
3	sum of logits	96.3	0.123 [0.034, 0.262]	+0.011 [−0.005, 0.037]	0.012	yes
3	sum / tau_k (nested)	96.3	0.115 [0.044, 0.220]	−0.008 [−0.024, 0.016]	0.012	yes
4	mean P(right)	96.8	0.204 [0.161, 0.258]	−0.134 [−0.150, −0.116]	0.134	no
4	mean logit	96.9	0.144 [0.094, 0.207]	−0.075 [−0.094, −0.054]	0.074	no
4	sum of logits	96.9	0.107 [0.022, 0.247]	+0.014 [−0.001, 0.039]	0.015	yes
4	sum / tau_k (nested)	96.9	0.094 [0.030, 0.194]	−0.004 [−0.020, 0.020]	0.006	yes
5	mean P(right)	97.5	0.201 [0.155, 0.251]	−0.142 [−0.160, −0.123]	0.143	no
5	mean logit	97.3	0.135 [0.088, 0.196]	−0.078 [−0.097, −0.056]	0.078	no
5	sum of logits	97.3	0.094 [0.012, 0.237]	+0.014 [−0.001, 0.040]	0.014	yes
5	sum / tau_k (nested)	97.3	0.092 [0.018, 0.216]	+0.001 [−0.018, 0.028]	0.007	yes

F.10. Breakdown by stratum and the learning curve

Strata. Table 76 stratifies the CNN's LOSO predictions; nothing is refitted. Figure 10 in Section 5 shows the same breakdown by person and by grip-by-speed cell. Strata with fewer than 5 persons get no CI; the table shows their persons' values instead. By platform, the accuracy is 90.6 % [86.6, 94.1] on Android (383 trials) and 87.0 % [77.0, 94.1] on iOS (215 trials). The weakest grip-by-speed cell with at least 5 persons is top right at natural speed: 86.5 % [76.7, 94.6] (74 trials). The weakest movement variant with at least 5 persons is the twist: 60.0 % [31.2, 87.5] (15 trials of 8 persons). The 2 left-handed persons are shown individually: P11 reaches 96.4 % and P17 88.0 %.

Persons and confident errors. The lowest person accuracy is 61.8 % (P09), and 3 persons fall below 80 %. Of the 64 errors, 24 are confident: the probability given to the true hand is below 0.2. P09 holds 7 of them, P20 5 and P05 4. By grip, 13 confident errors occur upright, 6 top right and 5 top left; by speed, 21 occur at natural speed and 3 at fast speed. These counts

follow the number of trials per condition (most trials are upright at natural speed); they locate the errors and do not test a difference. A person whom the model misreads consistently is not rescued by more attempts (Table 80).

Learning curve. The learning curve asks whether more participants would help. Inside every outer fold, the lane drew 3, 6, 9, 12 or 15 training persons at random, 5 times each (Table 77; Figure 11 in Section 5). The CNN rises from 84.0 % with 3 training persons to 89.3 % with all 19; the reference rises from 72.6 % to 77.9 %. An inverse power law fitted to these points extrapolates the CNN to 89.6 % [88.8, 90.8] at 40 training persons (asymptote 90.1 %), a gain of +0.5 [+0.1, +1.3] points; for the reference, the extrapolation is 78.6 % [77.4, 80.6], a gain of +0.6 [+0.0, +2.1] points. These intervals resample the repetitions at fixed persons, not the persons, so they understate the uncertainty; the fit also uses few sizes. The curve is an extrapolation, not a forecast. For the average accuracy, more participants look marginally worthwhile. They would still narrow every interval of this report and widen the coverage of movement styles, grips and natural use, where the robustness checks found the open risks.

Table 76: Final model by stratum (stratification only; a CI needs at least 5 persons).

factor	level	trials	persons	accuracy	95 % CI (or per person)
platform	android	383	12	90.6	[86.6, 94.1]
platform	ios	215	8	87.0	[77.0, 94.1]
grip x speed	top_left_toward, fast	41	19	95.1	[88.1, 100.0]
grip x speed	top_left_toward, natural	75	19	90.7	[80.5, 98.6]
grip x speed	top_left_toward, slow	2	1	100.0	persons: P01 100
grip x speed	top_right_toward, fast	38	19	86.8	[76.5, 95.0]
grip x speed	top_right_toward, natural	74	19	86.5	[76.7, 94.6]
grip x speed	top_right_toward, slow	2	1	100.0	persons: P01 100
grip x speed	upright_toward, fast	55	20	100.0	[100.0, 100.0]
grip x speed	upright_toward, natural	272	20	86.8	[80.6, 92.3]
grip x speed	upright_toward, slow	39	19	89.7	[76.3, 100.0]
variant	bent_start	7	4	57.1	persons: P01 50, P09 0, P13 100, P15 50
variant	curved	16	8	81.2	[56.2, 100.0]
variant	elbow_high	14	7	85.7	[64.3, 100.0]
variant	jerky	13	8	92.3	[72.7, 100.0]
variant	lean_in	11	6	81.8	[45.5, 100.0]
variant	pause	17	9	82.4	[66.7, 94.6]
variant	standard	470	20	91.7	[87.1, 95.4]
variant	tilted_touch	35	19	85.7	[73.5, 94.6]
variant	twist	15	8	60.0	[31.2, 87.5]
handedness	left	53	2	92.5	persons: P11 96, P17 88
handedness	right	545	18	89.0	[84.6, 92.7]

F.11. Tolerance to segmentation errors and a causal filter

Why. The CNN was trained and tested on oracle windows, the approach bounds the app stored. In production, a cue-free detector cuts the windows (Appendix G), and an online pipeline cannot use a zero-phase filter. The bound errors the model tolerates are therefore a requirement for the detector.

Test. The CNN, trained on oracle bounds, was tested on windows cut with errors. The windows come from lane p1's variant arrays, through the same canonical code path (Appendix B). The

Table 77: Learning curve: training persons drawn at random inside every outer fold.

model	training persons	repetitions	accuracy (mean, min-max)	log loss (mean)
m3 1D CNN, body frame	3	5	84.0 (81.9-85.3)	0.423
m3 1D CNN, body frame	6	5	87.1 (86.1-88.0)	0.361
m3 1D CNN, body frame	9	5	88.0 (87.3-88.6)	0.352
m3 1D CNN, body frame	12	5	88.5 (87.3-89.6)	0.318
m3 1D CNN, body frame	15	5	88.8 (88.0-89.3)	0.302
m3 1D CNN, body frame	19	1	89.3 (89.3-89.3)	0.291
reference: yaw logistic	3	5	72.6 (72.1-73.1)	0.709
reference: yaw logistic	6	5	75.5 (72.6-78.6)	0.553
reference: yaw logistic	9	5	77.2 (76.4-77.9)	0.535
reference: yaw logistic	12	5	77.4 (76.8-78.8)	0.532
reference: yaw logistic	15	5	77.5 (76.9-78.8)	0.520
reference: yaw logistic	19	1	77.9 (77.9-77.9)	0.512

variants are: both bounds jittered uniformly within ±50, ±100 or ±200 ms; the onset 100 ms late; and a causal 15-Hz low-pass filter instead of the zero-phase one.

Evidence. No variant changes the accuracy detectably (Figure 85, Table 78; Holm over the five variants). The changes are −0.4 [−1.1, +0.2] points for ±50 ms (Holm $p = 0.938$), −0.4 [−1.4, +0.4] points for ±100 ms (Holm $p = 0.938$), −0.8 [−2.1, +0.5] points for ±200 ms (Holm $p = 0.902$), −0.9 [−2.1, +0.3] points for the late onset (Holm $p = 0.771$) and +0.1 [−0.8, +0.9] points for the causal filter (Holm $p = 1.000$). The log loss rises slightly with the larger bound errors and the late onset (Table 78).

Requirement. Within the tested range, both bounds within ±200 ms of the oracle bounds and an onset at most 100 ms late, the model loses no detectable accuracy. The causal filter that an online pipeline needs costs nothing detectable. Larger errors were not tested.

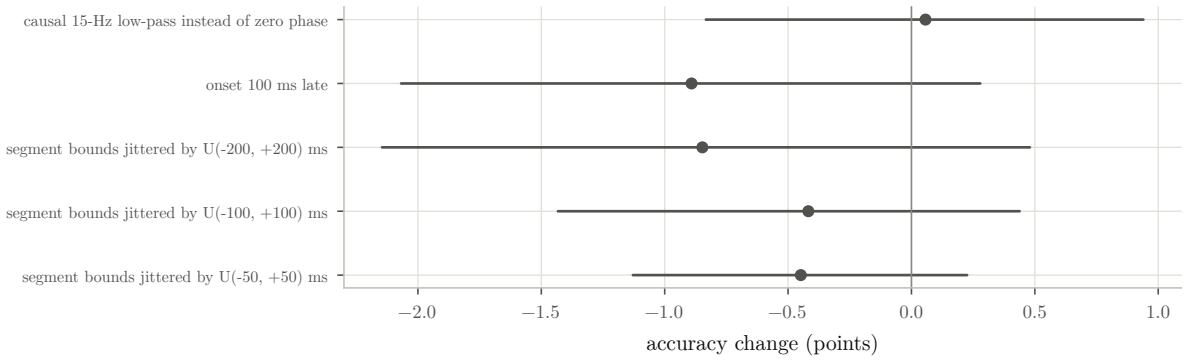


Figure 85: The final model, trained on oracle bounds, tested on windows with segmentation errors or a causal filter (lane p1's variant arrays): paired accuracy change with 95 % CIs. No variant is significant after Holm over the five; the tested range is the bound accuracy the production detector must reach.

Table 78: Final model on test windows with segmentation errors or a causal filter.

test-time variant	accuracy	change	log loss change
segment bounds jittered by U(−50, +50) ms	88.8	−0.4 [−1.1, 0.2] (Holm 0.938)	−0.001 [−0.005, 0.004]
segment bounds jittered by U(−100, +100) ms	88.8	−0.4 [−1.4, 0.4] (Holm 0.938)	+0.014 [0.005, 0.025]
segment bounds jittered by U(−200, +200) ms	88.3	−0.8 [−2.1, 0.5] (Holm 0.902)	+0.022 [0.009, 0.035]
onset 100 ms late	88.1	−0.9 [−2.1, 0.3] (Holm 0.771)	+0.021 [0.008, 0.037]
causal 15-Hz low-pass instead of zero phase	89.3	+0.1 [−0.8, 0.9] (Holm 1.000)	−0.002 [−0.008, 0.005]

F.12. The screen held away

Detection. No trial was recorded with the screen facing away from the face. The lane synthesised a screen-away twin of every recorded trial with the canonical flip of the channels (`standardize.screen_flip_channels`). On the twins, the CNN's accuracy falls to 28.6 % (paired change −60.7 [−67.8, −53.0] points). The body-frame roll and lateral channels change sign when the screen faces away, so the model answers the other hand, and with confidence. The failure is general: 11 of the 11 candidates whose inputs change under the flip lose accuracy after Holm over the candidates, and 5 fall below chance (Table 69). The inputs of the other candidates do not change under the flip.

Options. Three remedies were compared (Table 79).

1. *Check, then un-flip.* Lane m4's model-free rest-pose check flags the screen as away when the up component of the screen normal at rest is below its threshold (Appendix G); the flagged window is un-flipped before the model reads it.
2. *Check, then prompt.* The same check, followed by a prompt to turn the phone; the flagged attempt is an abstention.
3. *Augmentation.* Every training trial also enters as its screen-away twin with the same label; no check runs.

Evidence. The check flags 95.7 % of the twins and 4.3 % of the recorded trials. With the check and un-flipping, the CNN reaches 86.6 % on the twins (gain +58.1 [+50.0, +65.2] points). Its cost on the recorded trials is −2.7 [−7.4, +0.1] points, and it comes only from the recorded trials it flags (the screen tilted down at rest). With the prompt, the flagged attempts are abstentions; scoring each with half credit gives 49.2 % on the twins and 87.5 % on the recorded trials. Its coverage is the share of unflagged attempts, the complement of the flag rates above. Forced to answer the flagged attempts, it would score as option 1 if it un-flips them, or as the model as trained if it does not (28.6 % on the twins, 89.3 % on the recorded trials); its accuracy on the answered attempts alone is not among the saved results. Augmentation reaches 87.0 % on the twins, against 28.6 % for the model as trained (+57.8 [+50.4, +64.7] points, Holm $p < 0.001$). On the recorded trials it reaches 86.8 % against 89.3 % (−2.9 [−5.4, −0.4] points, Holm $p = 0.144$; log loss +0.055). This Holm family holds the four paired accuracy comparisons of the final model: these two and the two platform comparisons of Appendix F.3.

Decision. The two working remedies give nearly the same numbers: 86.6 % (check) against 87.0 % (augmentation) on the twins, and a log loss of 0.338 against 0.339 on the recorded trials; they were not tested against each other. Production keeps the model as trained and exports the model-free check with it. The check leaves the model unchanged, its cost on the recorded

trials comes only from the trials it flags, and on a flag the page can ask the user to turn the phone instead of un-flipping.

Caveats. The screen-away twins are synthesised, never recorded. The check reads gravity only and assumes a still phone before the movement. Unrecorded grips, such as landscape or the phone tilted forward, are untested.

Table 79: Screen-away remedies of the final model: accuracy and log loss on the recorded holds and accuracy on their screen-away twins (synthesised with standardize.screen_flip_channels); the rest-pose check is the model-free rule of lane m4.

remedy	recorded holds: accuracy	recorded holds: log loss	screen-away twins: accuracy
model as trained, no check	89.3	0.291	28.6
rest-pose check, then un-flip the window	86.6	0.338	86.6
rest-pose check, then prompt (abstain: half credit)	87.5	0.310	49.2
screen-flip augmentation of the training rows	86.8	0.339	87.0

F.13. The fallback model: the PHYS6 logistic

The simplest robust candidate is the m1 PHYS6 logistic (portability rank 1). It is an antisymmetric logistic regression on six physics features: the yaw, the roll and the lateral velocity at 50 and 100 % of the approach (Appendix C).

Evidence. In its production form, it is detectably worse than the CNN: −5.9 points of accuracy (Holm $p = 0.008$) and +0.136 in log loss (Holm $p = 0.006$; Holm over the comparisons against the best). On a grip absent from training the gap closes. With upright-only training, the PHYS6 logistic reaches 87.5 % on the sideways trials (change −1.1 points) and the CNN 85.3 %. The two models were not tested against each other on these trials. The CNN's own change there is −4.1 [−7.0, −1.4] points (unadjusted $p = 0.016$; log loss +0.124, unadjusted $p = 0.011$), which is not significant after Holm over the candidates ($p = 0.188$). The PHYS6 logistic also needs the rest-pose check: it reaches 51.5 % on the screen-away twins without it and 82.4 % with it. On natural use it answers 15 of 20 free-choice trials correctly against 16.7 expected (right-hand recall on the free-choice trials; all right-hand so far, so a constant right answer would score 100 %).

Decision. The CNN is the better choice when the PI uses the recorded grips, which the training data cover. The PHYS6 logistic is the fallback if a grip unlike any recorded one matters more than the accuracy on the recorded grips. It reads six features and is the easiest candidate to port.

F.14. Export and model card

Fit. The exported model was fitted on all 20 persons with the procedure of the outer folds. It uses the same per-person cap on training trials (Appendix A) and the inner person splits for the model's in-fold Platt calibrator. The scale τ_k and the (empty) remedy were fitted on the persons' cross-fitted predictions.

Format. The model is a plain JSON file, `final_model.json` (1585 kB, 46084 fitted numbers), in the export folder `analysis/reports/stage2/export/`. It holds the channel table and

the preprocessing parameters, the body-frame definition, the weights, the calibrator, τ_k, the rest-pose screen check (its threshold and flip signs), the prior (0.5), the decision threshold and the identity of the cached data it was fitted on. A numpy-only reference implementation (`analysis/stage2/lanes/final_infer.py`) is the specification for the Stage 3 JavaScript port.

Checks. The reference implementation reproduces the fitted pipeline on all 618 primary and free-choice trials; the largest difference in P(right) is 2.2e-16. The test `test_final_export.py` requires 10^{-9}. A float32 run is also compared with the fitted pipeline; the export folder keeps both comparisons in `final_model_check.json`. One prediction takes 20.05 ms in numpy. The lane's finalists file (`results/finalists_final.json`) rebuilds the shipped result by the same convention as the other lanes, with a largest difference of 0.0e+00.

Model card. Table 80 summarises the intended use, the inputs, the numbers above and the known failure modes. The failure modes are persons whom the model misreads consistently, unusual movements, a grip absent from training, a screen held away without the check, weaker lateral evidence on uninstructed movements, and windows from a detector that misses the bounds. The card's LOSO numbers are the exploratory ones of the shipped model; the selection-aware estimate is in Appendix F.4.

Table 80: Model card of the exported final model.

item	value
model	m3 1D CNN, body frame (1D CNN ensemble (4 members))
intended use	P(right) that the right hand brought the phone from arm's length to the nose, from the phone's motion sensors only (no demographic, no instructed condition)
inputs	canonical approach window: 17 channels x 64 samples; body-frame channels derived from it (roll_rate, pitch_rate, acc_lat, acc_fwd)
preprocessing	the cache pipeline (60 Hz, 15 Hz zero-phase low-pass, gyro-propagated gravity, onset to end (last-normalized to 64 samples)); README section 3
parameters	46984 fitted numbers; JSON 1585 kB; 20.05 ms per prediction in numpy
calibrator	none
accuracy (LOSO)	89.2 % [85.0, 93.1]
log loss	0.294 [0.196, 0.429]
AUC	0.951
calibration	ECE 0.026 (perfect-calibration band up to 0.042), slope 0.97
three attempts	96.3 % per person and hand with signal(3 runs of light-/ too_ft), confidence minus accuracy −0.000 [−0.024, +0.016]
one attempt	89.2 % [85.0, 93.1] (618 free-choice trials grouped by 77 from the same persons' instructed right-hand trials)
screen away	29 % without the rest-pose check, 87 % with it (check reported)
known failure modes	some persons are misread consistently (lowest P(0 62 %), and more attempts do not fix them; unusual movements (weakest variant: twist 66 %); a grip absent from the training data (−4.1 points on rotated grips when trained on upright ones); a screen held away without the rest-pose check (29 %); uninstructed movements carry less lateral evidence (it points to the reported hand in 11 of 20 free-choice trials); windows from a detector that misses the onset or end (final_variants)
prior	P(right) assumes equal prior odds of the two hands (balanced training data); final_infer.apply(..,prior shifts it
rule check	the decision rule recommends m3_deep; FINAL_MODEL = m3_deep

F.15. Left out for time and open issues

Not run. The lane left out 4 planned experiments for the compute budget: the platform and label-noise tests of the m2 stack and of the m3 2D CNN on short-time Fourier transform planes (`final_plat_m2_stack`, `final_noisy_m2_stack`, `final_plat_m3_tf` and `final_noisy_m3_tf`; the dashes in Table 69). All four are diagnostics of candidates that the decision already excludes. The command `python -m stage2.pipeline compute -lanes final` runs them.

A contradictory configuration count. For the m2 logistic on 691 features, lane m2's finalists file records 12 configurations compared on the outer folds, but its saved result records 1. This lane uses the count of the finalists file, the role in which the model became a candidate, so its numbers count as exploratory. The contradiction does not change the decision, because the model fails the grip test, but lane m2's records should be reconciled.

Open issues.

- Every number here uses oracle windows. The final model has not yet run inside the cue-free production simulation; the production numbers of Appendix G use lane m4's own reference classifier, not the final model.
- The natural-use set holds one free-choice trial per person, all right-hand.
- The screen-away twins are synthesised.

- The nested estimate does not remove the optimism of the lanes' own design choices.

G. The cue-free production pipeline

In production (the Stage 3 inference link) no GO cue marks the movement. The person holds the phone at arm's length and brings it to the nose once or a few times. The pipeline must find each approach in the continuous sensor stream, ignore the return and every other movement, decide when to stop, and output a calibrated $P(\text{right})$. Lane m4 built this pipeline and simulated it on the recordings; this appendix states each of its decisions with the options, the evidence and the rationale.

Which hand classifier. Every production number in this appendix uses lane m4's own reference hand classifier, yaw8: a mirror-augmented logistic regression on the cumulative yaw about gravity ($\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$) at 8 instants of the approach window. It does not use the final model, the body-frame 1D CNN of Appendix F. The hand classifier is a plug-in: the detector, the calibration and the stopping rule take any model that reads a window, and the final model inside this simulation is the first next step (Appendix G.13).

Unit, evaluation and scoring. The unit is one full recording of a primary trial: rest, approach, hold and return, without the app's stored event times. There are 598 recordings of 20 persons, evaluated leave-one-person-out; every fitted part (segmentation setting, segment classifier, hand model, calibrator, aggregation scale) is fitted on the training persons only. An abstention (no approach accepted, or a calibrated $P(\text{right})$ of exactly 0.5) is never a right-hand answer: it scores half a correct answer. Next to this half-credit accuracy, each result gives the coverage (the share answered), the accuracy on the answered recordings and the forced-choice accuracy (the best candidate always scored). In the tables of this appendix, 'answered' means that an approach was accepted; it still counts the calibrated outputs of exactly 0.5 (Appendix G.6), which are abstentions and score half. For the production pipeline, the share of recordings with a hand shown is therefore 76.4 % (457 of 598), right on 79.9 % of those; these two numbers are exact arithmetic on the saved counts, done by the report's asset step. Natural use is right-hand recall on the free-choice trials (all right-hand so far; a constant right answer would score 100 %).

Tests. Paired differences are means over persons with a person-bootstrap CI and an exact sign-flip test over persons. Holm's adjustment runs within named families: *trial* (the 26 comparisons of Table 88), *stress* (44 comparisons of the stress test), *sessions* (6 session comparisons), *calibrators* (each calibrator against none) and *grip* (11 comparisons of the grip check). The production pipeline was compared with 22 cue-free pipelines on the outer folds, so its own LOSO numbers are exploratory; the segmentation setting and the calibrator are chosen inside each fold.

Headline. With one touch, the production pipeline reaches a half-credit accuracy of 72.8 % [68.9, 76.6]. It accepts an approach in 82.9 % of the recordings (accuracy 77.5 % on those, a 0.5 output scoring half) and shows a hand for 76.4 %, right on 79.9 % of those; its forced-choice accuracy is 74.6 %; its AUC is 0.816 [0.777, 0.854]. With the stopping rule on synthetic multi-touch sessions it reaches 83.2 % [74.6, 90.8] after 3.66 accepted attempts on average (exploratory; Appendix G.9).

G.1. The baseline detector v0 and the evaluation contract

Options. A fixed rule built from the app's own thresholds, or a detector learned from labelled segments. The fixed rule, v0, comes first: it sets the baseline and the outcome categories that every later detector reports.

The rule. v0 takes the first sustained movement (smoothed $|\boldsymbol{\omega}| > 30$ deg/s for 100 ms) that follows at least 1 s of stillness ($|\boldsymbol{\omega}| < 20$ deg/s), and ends it at the next 250 ms of stillness. It reads the gyroscope only. Adding an accelerometer condition ($|\mathbf{a}| < 0.5$ m/s^2) leaves 90 % of the approaches with at least 1 s of stillness before them (median 1.8 s), against 94 % with the gyroscope alone (median 2.4 s).

Evidence. Figure 86 and Table 81 score v0's first detection against the oracle window, the approach bounds the app stored, by their intersection over union (IoU). The first detection is the approach (IoU ≥ 0.5) in 53 % of the primary recordings, part of it in 12 %, an earlier movement in 28 %, the return in 2 % and nothing in 5 %. The approach is among all of v0's detections in 72 % of the recordings. When v0 finds the approach, its onset is 48 ms late (median; 90th percentile of the absolute error 183 ms). With yaw8 on its first detection, v0 reaches 67.7 % [63.4, 72.1] (coverage 94.6 %, answered 68.7 %, forced choice 67.7 %).

Decision and rationale. v0 is the baseline, and its categories (approach, part, earlier movement, return, nothing, rejected) are the contract of every detector in this appendix. Its main failure is an earlier movement (the arm settling after the previous trial, a fidget), not the return.

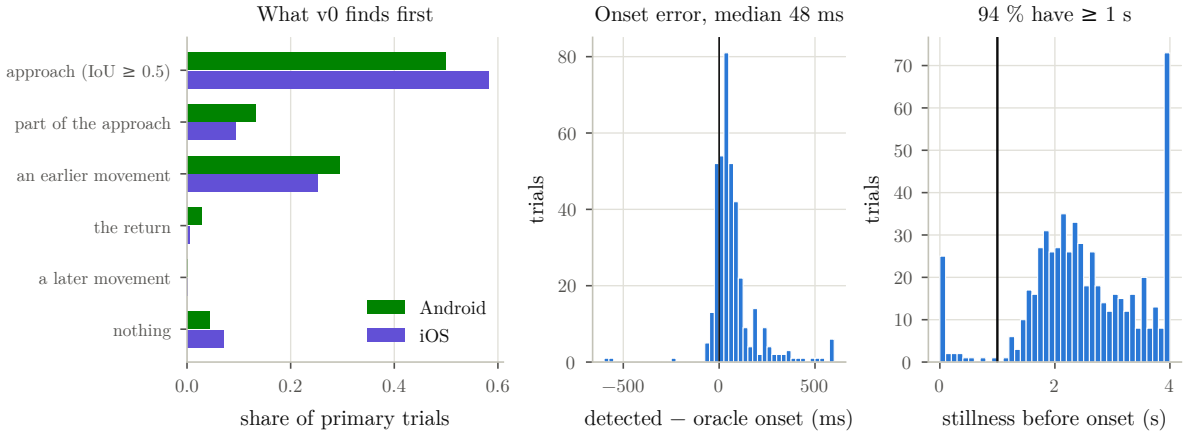


Figure 86: Baseline cue-free detector v0 on the primary recordings, by platform. Left: what v0 finds first (approach: IoU ≥ 0.5 with the oracle window). Middle: onset error when it finds the approach. Right: stillness before the approach in v0's own signal, which the collection app enforced; the shares are therefore upper bounds for production (Figure 87).

Table 81: Cue-free detector v0 on the recorded primary trials: what it finds first, whether the approach is among all its detections, and how late its onset is. The stillness before each approach was enforced by the collection protocol, so these shares are upper bounds for production (see the stress test).

Quantity	All	Android	iOS
Trials	598	383	215
First detection: the approach (IoU ≥ 0.5)	53%	50%	58%
First detection: part of the approach	12%	13%	9%
First detection: an earlier movement	28%	30%	25%
First detection: the return of the arm	2%	3%	0%
First detection: a later movement	0%	0%	0%
First detection: nothing	5%	4%	7%
The approach among all detections	72%	70%	77%
Onset lag when it finds the approach, median (ms)	48	50	33
Absolute onset error, 90th percentile (ms)	183	183	147

G.2. The stillness stress test: recorded shares are upper bounds

Problem. The collection app enforced the stillness before every approach: it waited for 400 ms of stillness, then a 400–900 ms foreperiod, then the reaction time. Production has no GO cue and no hold-still gate, so a user may start moving without holding still. Every share measured on the recordings as they are is therefore an upper bound.

Method. The stress test cuts the still run before each approach to a given length: the samples right after the preceding activity are removed and the rest of the recording moves back. Every pipeline then runs again on the shortened recording, fitted without the held-out person.

Evidence for v0. Figure 87 and Table 82 show that v0 needs its own 1 s rule. It finds the approach first in 53 % of the recordings as recorded and in 53 % with 1 s of stillness, but in 3 % with 0.75 s and 10 % with none. With a random stillness, uniform between 0 and 1.5 s, it finds the approach first in 24 %.

Evidence for the full pipelines. Figure 88 and Table 83 run the production pipeline, v0 and a detector whose stillness feature sees up to 3 s (context capped at 3 s) through the same cuts. The production pipeline scores 72.8 % as recorded, 73.1 % with 1 s, 71.5 % with 0.9 s, 67.6 % with 0.75 s, 61.1 % with 0.5 s and 56.4 % with none. With no stillness it answers only 33 % of the recordings and scores the return in 2 %; v0 then answers 60 %, scores the return in 23 % and falls to 46.2 %.

Options for the stillness feature. The stillness before a segment is a strong feature of the segment classifier (Appendix G.4). It can see the whole stillness (capped at 3 s) or only up to the instructed 1 s. Paired on the same recordings (robust detector minus the context capped at 3 s, both uncalibrated, family *stress*), the difference is -1.2 [-2.6, $+0.2$] points as recorded (Holm $p = 1.000$), $+7.9$ [$+4.3$, $+11.8$] points with 1 s (Holm $p = 0.013$), $+7.6$ [$+4.3$, $+10.9$] points with 0.9 s (Holm $p = 0.013$) and $+1.0$ [-0.2, $+2.2$] points with none (Holm $p = 1.000$). The production detector against v0 at 0.9 s: $+25.3$ [$+20.7$, $+30.1$] points (Holm $p < 0.001$).

Decision and rationale. The stillness feature is capped at the instructed 1 s (the robust feature set), and the inference page must ask for at least 1 s of stillness and check it, measured exactly as the feature is (the causal activity below the stillness threshold). The capped detector is better where production will operate, at the instructed 1 s; it is not better as recorded, where the long enforced stillness favours the uncapped one. Caveat: the advantage appears only at and just below its own cap.

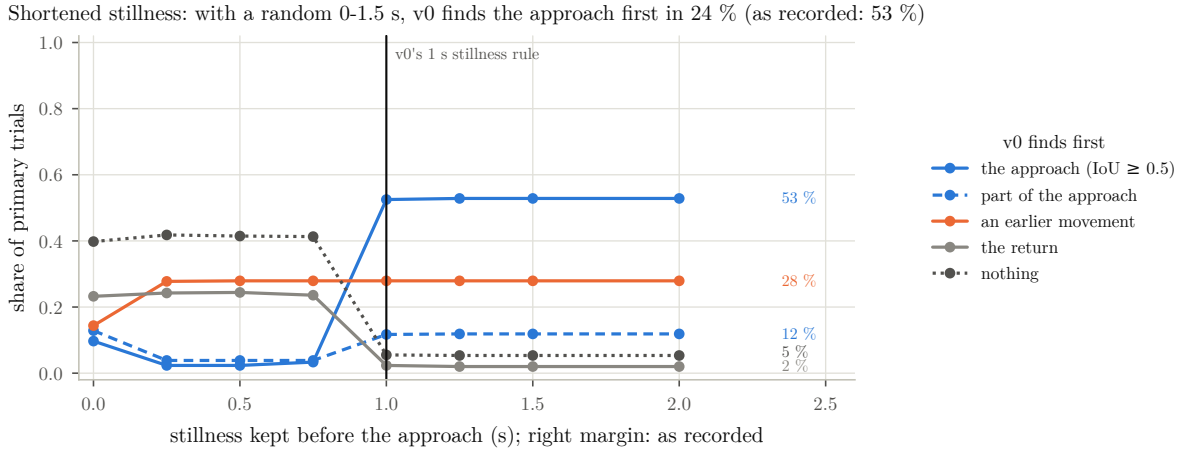


Figure 87: Stress test of v0: the stillness before each approach cut to the given length, then v0 run again (right margin: as recorded). Below its own 1 s rule, v0 can no longer separate the approach from what came before.

Table 82: Stress test of v0: the stillness before each approach is cut to the given length (recorded: unchanged; random: uniform per trial, seeded) and v0 is run again. Shares of what it finds first (the approach, part of it, an earlier movement, the return, nothing) and of trials whose approach is among all its detections (anywhere).

Stillness kept (s)	Approach	Part	Earlier	Return	Nothing	Anywhere
recorded	53%	12%	28%	2%	5%	72%
0	10%	13%	14%	23%	40%	13%
0.25	2%	4%	28%	24%	42%	3%
0.5	2%	4%	28%	24%	41%	4%
0.75	3%	4%	28%	24%	41%	5%
1	53%	12%	28%	2%	6%	72%
1.25	53%	12%	28%	2%	5%	72%
1.5	53%	12%	28%	2%	5%	72%
2	53%	12%	28%	2%	5%	72%
random	24%	8%	27%	16%	26%	32%

Table 83: Stillness stress test of single recordings: the stillness before each approach cut to the given seconds (recorded = as recorded), and the screen-away twin of every recording (the phone turned 180 deg about its long axis); each pipeline fitted without the person (an abstention (no approach found, P = 0.5) scores half a correct answer). Grip check: the inference page's rest-pose check (the screen-normal up component at rest before the first movement below 0: the page asks to turn the screen and does not answer).

Pipeline	Stillness kept (s)	Grip	Accuracy	CI low	CI high	Answered	Approach found	Return	Grip check fires	Acc. with check
production	recorded		72.8%	68.9%	76.6%	83%	67%	2%	4%	71.7%
robust_uncal	recorded		73.4%	68.7%	77.9%	83%	67%	2%	4%	72.4%
grip_uncal	recorded		68.5%	62.8%	74.0%	75%	56%	5%	4%	68.3%
uncapped	recorded		74.6%	70.0%	79.0%	86%	70%	2%	4%	73.5%
v0	recorded		67.7%	63.4%	72.1%	95%	53%	2%	4%	66.8%
production	0.0		56.4%	53.8%	59.2%	33%	18%	2%	3%	56.2%
robust_uncal	0.0		56.4%	53.6%	59.6%	33%	18%	2%	3%	56.2%
grip_uncal	0.0		52.5%	49.3%	56.1%	36%	15%	9%	3%	52.3%
uncapped	0.0		55.4%	52.7%	58.4%	31%	17%	2%	3%	55.1%
v0	0.0		46.2%	43.0%	49.4%	60%	10%	23%	3%	46.2%
production	0.5		61.1%	57.6%	64.7%	44%	28%	2%	4%	60.9%
robust_uncal	0.5		61.1%	57.2%	64.9%	44%	28%	2%	4%	61.0%
grip_uncal	0.5		52.7%	49.8%	55.6%	35%	12%	9%	4%	52.8%
uncapped	0.5		61.1%	57.0%	65.5%	42%	28%	2%	4%	61.0%
v0	0.5		45.5%	42.8%	48.6%	59%	2%	24%	4%	45.7%
production	0.75		67.6%	63.6%	71.7%	64%	47%	2%	4%	67.2%
robust_uncal	0.75		68.0%	63.6%	72.4%	64%	47%	2%	4%	67.6%
grip_uncal	0.75		59.4%	55.1%	63.6%	54%	30%	8%	4%	59.5%
uncapped	0.75		62.5%	57.9%	67.1%	50%	35%	2%	4%	62.2%
v0	0.75		45.7%	43.1%	48.8%	59%	3%	24%	4%	46.0%
production	0.9		71.5%	67.6%	75.2%	76%	60%	2%	4%	70.6%
robust_uncal	0.9		72.0%	67.5%	76.3%	76%	60%	2%	4%	71.2%
grip_uncal	0.9		65.6%	60.4%	70.3%	67%	46%	6%	4%	65.5%
uncapped	0.9		64.4%	60.0%	69.0%	56%	40%	2%	4%	64.1%
v0	0.9		46.2%	43.5%	49.2%	59%	4%	23%	4%	46.5%
production	1.0		73.1%	69.2%	76.8%	81%	65%	2%	4%	72.0%
robust_uncal	1.0		73.7%	69.2%	78.1%	81%	65%	2%	4%	72.7%
grip_uncal	1.0		68.1%	62.8%	73.1%	73%	54%	5%	4%	67.9%
uncapped	1.0		65.8%	61.6%	70.1%	58%	43%	2%	4%	65.3%
v0	1.0		67.5%	63.3%	71.8%	94%	53%	2%	4%	66.6%
production	1.25		73.2%	69.4%	76.7%	83%	67%	2%	4%	72.1%
robust_uncal	1.25		73.9%	69.5%	78.1%	83%	67%	2%	4%	72.9%
grip_uncal	1.25		68.5%	62.7%	74.1%	76%	56%	5%	4%	68.3%
uncapped	1.25		68.4%	63.9%	73.0%	65%	50%	2%	4%	67.8%
v0	1.25		67.9%	63.7%	72.2%	95%	53%	2%	4%	67.0%
production	1.5		73.2%	69.3%	76.8%	84%	67%	2%	4%	72.1%
robust_uncal	1.5		73.9%	69.3%	78.2%	84%	67%	2%	4%	72.9%
grip_uncal	1.5		68.5%	62.7%	74.1%	76%	56%	5%	4%	68.3%
uncapped	1.5		70.3%	65.8%	74.7%	71%	56%	2%	4%	69.6%
v0	1.5		67.9%	63.7%	72.2%	95%	53%	2%	4%	67.0%
production	2.0		73.0%	69.2%	76.6%	83%	67%	2%	4%	71.9%
robust_uncal	2.0		73.7%	69.2%	78.0%	83%	67%	2%	4%	72.7%
grip_uncal	2.0		68.6%	63.1%	74.0%	76%	57%	5%	4%	68.4%
uncapped	2.0		73.7%	69.2%	77.9%	80%	65%	2%	4%	72.7%
v0	2.0		67.7%	63.4%	72.1%	95%	53%	2%	4%	66.8%
production	recorded	screen away	42.5%	35.8%	49.6%	67%	17%	39%	96%	49.7%
robust_uncal	recorded	screen away	42.5%	35.6%	49.9%	67%	17%	39%	96%	49.7%
grip_uncal	recorded	screen away	68.5%	62.8%	74.0%	75%	56%	5%	96%	50.2%
uncapped	recorded	screen away	43.5%	37.3%	50.3%	68%	20%	36%	96%	49.8%
v0	recorded	screen away	67.7%	63.4%	72.1%	95%	53%	2%	96%	50.9%
production	1.0	screen away	41.9%	35.3%	49.0%	67%	16%	40%	96%	49.4%
robust_uncal	1.0	screen away	41.9%	35.1%	49.2%	67%	16%	40%	96%	49.4%
grip_uncal	1.0	screen away	68.1%	62.8%	73.1%	73%	54%	5%	96%	50.3%
uncapped	1.0	screen away	40.8%	35.5%	46.2%	61%	10%	37%	96%	49.5%
v0	1.0	screen away	67.5%	63.3%	71.8%	94%	53%	2%	96%	50.9%

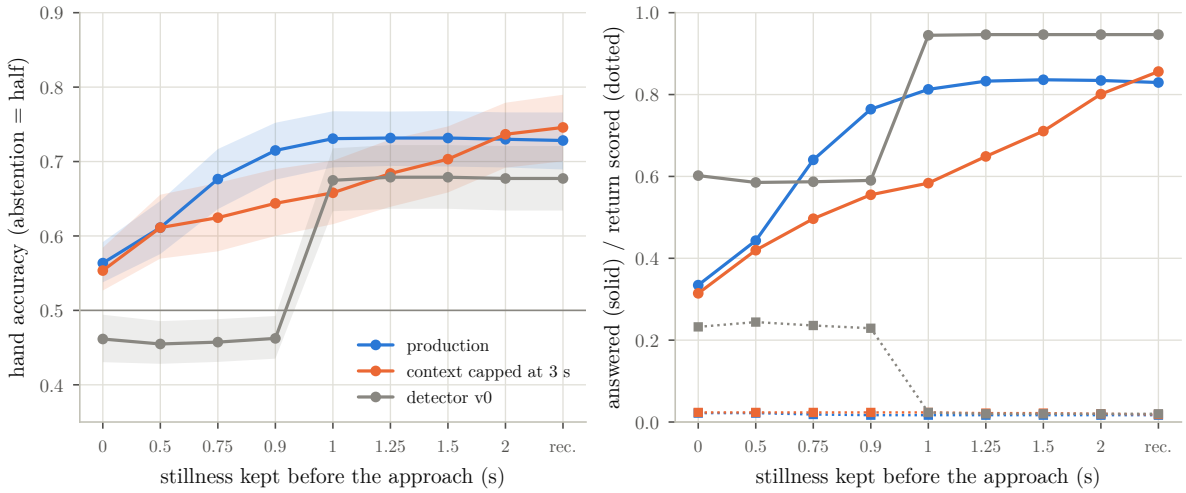


Figure 88: Stress test of the full pipelines. Left: half-credit accuracy (band: 95 % CI). Right: share answered (solid) and share where the scored segment is the return (dotted). The production detector sees at most the instructed 1 s of stillness; with 1 s it scores 73.1 % against 65.8 % for the context capped at 3 s. Supports capping the stillness feature at 1 s and asking for that stillness on the page.

G.3. The return: a time-reversed approach

Problem. After the hold, the arm returns to arm's length. The return retraces the approach backwards, so its chiral evidence is reversed.

Evidence. Figure 89 and Table 84 compare the class-mean yaw rate of the approach and of the return in the same 288 trials with a clean return window. The sign rule fitted on approaches (right if and only if the first-half yaw is positive) is right on 76 % of the approaches but on 17 % of the returns (18 % on the net yaw of the return); the AUC of the first-half yaw on the return is 0.11. A detector that scores the return therefore answers the other hand.

Decision. Returns are never pooled with approaches, and the detector must reject them: the segment classifier has a return class (Appendix G.4).

Option: the return as evidence. The return carries the hand too, with the opposite sign. Figure 90 tests a separate return head (the same hand model fitted on returns). On the stored windows of the trials with a clean return, the approach model scores 78.8 % [73.2, 84.1], the return head 81.6 % [75.7, 87.0] and the sum of their logits 84.0 % [77.6, 89.7]; the sum minus the approach is +5.2 [+1.7, +8.8] points (Holm $p = 0.747$, family *trial*). Cue-free, adding the return head when the segment after the chosen approach is classified as a return gives 75.9 % against 73.4 % (both uncalibrated, coverage unchanged; answered 81.2 %, forced choice 77.6 %); paired +2.5 [+0.8, +4.4] points (Holm $p = 0.203$). The price is time: the return's evidence arrives a median 2267 ms after the end of the approach (90th percentile 2768 ms).

Decision and rationale. The return head is not in production yet. No accuracy difference is shown after adjustment, and the summed logits need their own cross-fitted calibration. Between two touches the return happens anyway, so in a multi-touch session it would delay only the final answer; it is a next step.

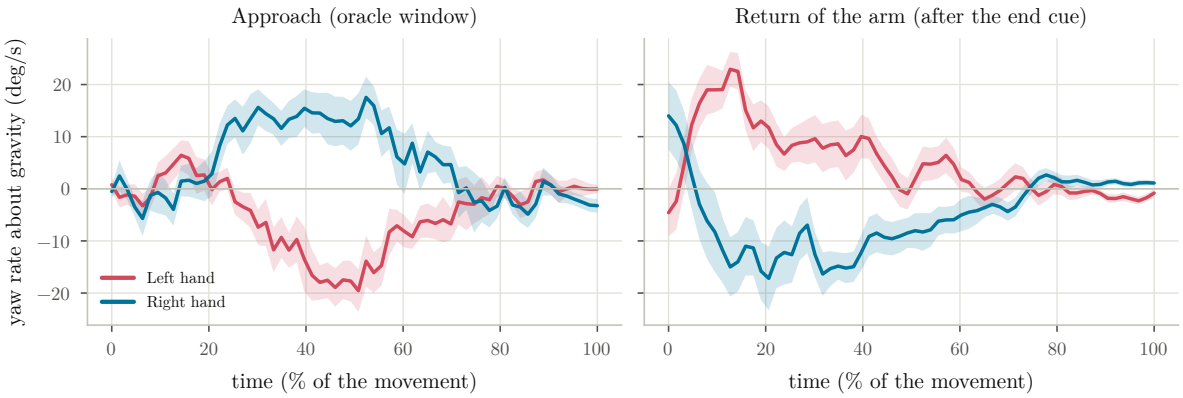


Figure 89: Class-mean yaw rate about gravity in the approach (left) and in the return (right) of the same 288 trials, by hand (band: 95 % CI). The return mirrors the approach in time, so a sign rule fitted on approaches answers the other hand on returns.

Table 84: What the approach rule says about the return of the arm (the cue-free risk).

Quantity	Value
Primary trials with a clean return window	288
Persons with a return window	20
Sign rule (right iff first-half yaw > 0) on the approach	76 %
The same rule on the return	17 %
The rule on the net yaw of the return	18 %
AUC of the first-half yaw on the return	0.11
AUC of the net yaw on the return	0.09
Return duration, median (s)	1.35

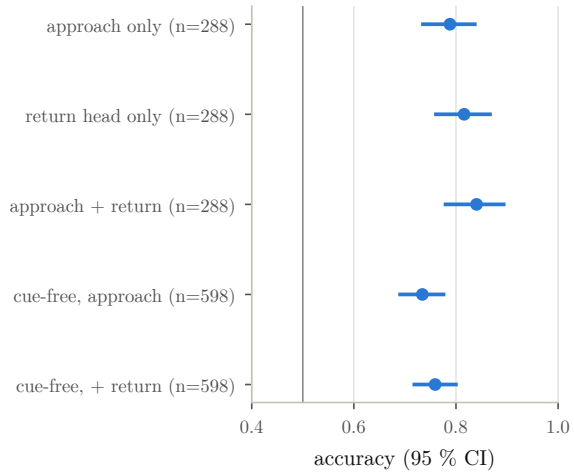


Figure 90: The return as evidence (half-credit accuracy, 95 % CI). Top three: stored windows of the 288 trials with a clean return. Bottom two: the cue-free production detector (uncalibrated) without and with the return head, on all 598 recordings.

G.4. The learned detector: candidate segments and the segment classifier

Candidate segments. A causal activity signal follows $|\boldsymbol{\omega}|$ through a one-pass second-order 5 Hz Butterworth low-pass; every bound is moved back by its group delay of 44 ms. Hysteresis cuts the stream into bouts: a bout becomes a movement once the activity exceeds 30 deg/s for 100 ms, and it ends when the activity stays below a lower threshold for a sustain time. Unlike v0, no stillness is required before a bout. Three settings of the lower threshold and the sustain time are compared inside each fold; 15 deg/s with 250 ms was chosen in 20 of 20 folds.

The segment classifier. A multinomial logistic regression (C = 1) labels every bout as an approach, a return or another movement, from hand-agnostic features scaled by their median and interquartile range and winsorised. Its labels come from the app's events of the training persons. Every feature is mirror-even (tested), so the segment classifier carries no hand information. The candidate with the highest P(approach) is scored; it is rejected (an abstention) when P(approach) is below 0.50. A segment is confirmed 0.5 s after its end, when its window is complete.

Evidence: what separates the approach from the return. Figure 91 shows the four strongest features. The displacement along the screen normal separates approaches from returns with an AUC of 0.89 (0.74 against other movements). The change of the up vector's screen-normal component has an AUC of 0.10, the reverse ordering. The vertical displacement reaches 0.74, and the stillness before the segment, capped at 1 s, reaches 0.83 against other movements (0.80 against returns). The screen-normal features are in the device frame, so they change sign when the screen faces away (Appendix G.10).

Options: five feature sets. *physics* (the movement features); *full* (physics plus the stillness before, capped at 3 s, and recording context); *nested* (physics or full chosen inside each fold: full in 20 folds); *robust* (physics plus the stillness before, capped at the instructed 1 s, and the stillness after); and *grip* (robust with the four screen-normal quantities signed by the side of the phone that faces up over the segment). Table 87 gives physics 71.7 %, full 74.6 %, robust 73.4 % and grip 68.5 %. Paired (family *trial*): robust minus nested −1.2 [−2.6, +0.2] points (Holm $p = 1.000$); grip minus robust −4.9 [−8.7, −1.8] points (Holm $p = 0.250$). The grip set is invariant to the screen-away grip by construction, but the side that faces up is a poor grip estimate on real recordings: its sign is negative for 15.8 % of the approach and 20.0 % of the return segments of screen-toward recordings.

Decision and rationale. The robust set is the production detector, an exploratory choice among 5 feature sets. It is the set that does not reward the enforced stillness of the protocol (Appendix G.2); the screen-away grip is handled by the instructions and the page's grip check instead of the grip set.

What the detector scores. Table 85 scores each detector's chosen segment against the stored events. The production detector scores the approach in 67 % of the recordings, an earlier movement in 5 % and the return in 2 %; it rejects 17 % instead of scoring a wrong segment. v0 scores the approach in 53 % and an earlier movement in 28 %. When the approach is found, the median bound errors (detected minus stored) are 22 ms at the onset and −77 ms at the end.

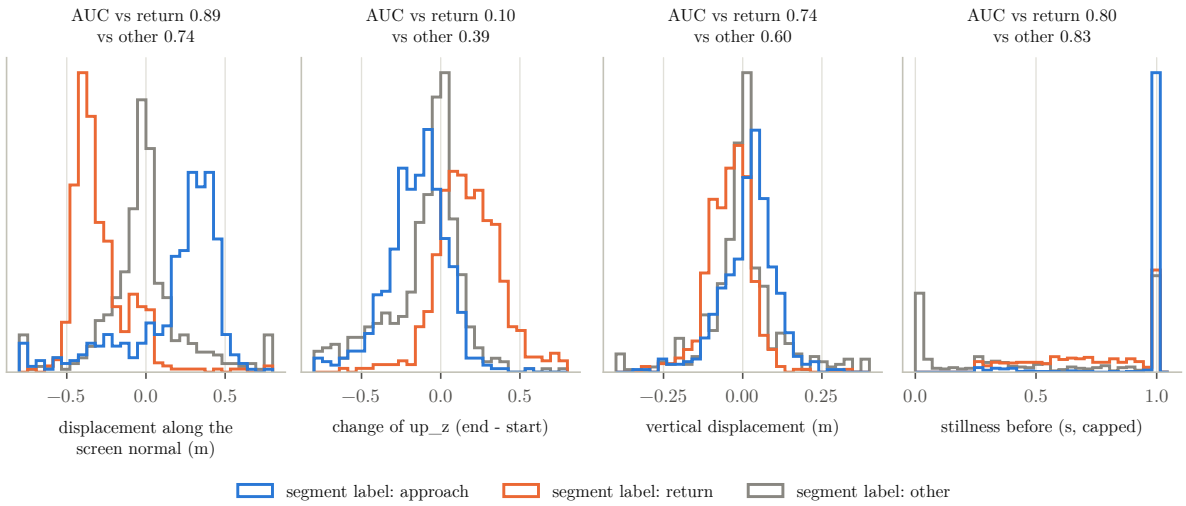


Figure 91: Hand-agnostic features of the candidate segments of the recordings, by their label from the app's events (densities; AUC against return and against other movements in the panel titles). They separate the approach from the return without any hand information and are the basis of the segment classifier.

Table 85: What the detector scores on each full primary recording, against the app's stored events (share of recordings); bound errors are medians over the recordings where the approach was found.

Detector	Approach	Part	Earlier	Return	Rejected	None	Onset err. (ms)	End err. (ms)
detector v0 (first movement)	53%	12%	28%	2%	0%	5%	48	−152
learned, physics features	59%	10%	6%	2%	23%	0%	22	−77
learned, physics + context	70%	10%	4%	2%	14%	0%	22	−78
learned, feature set nested	70%	10%	4%	2%	14%	0%	22	−78
learned, robust (production detector)	67%	10%	5%	2%	17%	0%	22	−77
learned, grip-normalised	56%	8%	6%	5%	24%	0%	22	−78
production (calibrator nested)	67%	10%	5%	2%	17%	0%	22	−77

G.5. Pipelines, references and data-treatment ablations

The ladder. Table 87 lists every cue-free pipeline and its references: the same hand model on the oracle window (no detection), v0, the five feature sets, the calibrators, the return head and the ablations. Table 86 adds, for the main rungs, what an abstention hides: coverage, accuracy on the answered recordings and forced-choice accuracy, on the primary recordings and on natural use.

Detection loss. Production minus the oracle window is −5.1 [−7.6, −2.7] points (Holm $p = 0.047$, family *trial*; Table 88): this is the price of finding the movement. Production minus v0 is +5.1 [+1.9, +8.3] points (Holm $p = 0.203$: no accuracy difference shown) and −0.198 [−0.276, −0.120] of log loss (Holm $p = 0.001$). The learned detector is kept because its probabilities are better and because v0 fails when the stillness is short (Appendix G.2).

Training windows of the hand model. Options: the oracle windows or the detector's windows. Detector minus oracle windows: −0.8 [−3.8, +1.8] points (Holm $p = 1.000$). Decision: oracle windows, the cached training arrays, so that any window model plugs in unchanged.

Table 86: Scoring of the main cue-free pipelines (yaw8 hand model; LOSO, 598 recordings of 20 persons). Accuracy: an abstention scores half credit. Answered: coverage. Forced: the best candidate always scored. Natural use: right-hand recall on the 20 free-choice trials (all right-hand so far; a constant right answer would score 100 %), as answered of 20, recall on the answered, forced choice and half credit.

	Primary recordings					Natural use (right-hand recall)			
Pipeline	Accuracy	95 % CI	Answered	Acc. answered	Forced	Answered	Answered	Forced	Half credit
oracle window	77.9 %	[73.1, 82.9]	100.0 %	77.9 %	77.9 %	20	75 %	75 %	75.0 %
v0 (first movement)	67.7 %	[63.4, 72.1]	94.6 %	68.7 %	67.7 %	16	69 %	65 %	65.0 %
learned, physics	71.7 %	[66.6, 76.3]	77.1 %	78.1 %	73.4 %	8	88 %	70 %	65.0 %
learned, full	74.6 %	[70.0, 79.0]	85.6 %	78.7 %	75.1 %	12	83 %	70 %	70.0 %
learned, robust (uncal.)	73.4 %	[68.7, 77.9]	82.9 %	78.2 %	74.9 %	11	82 %	70 %	67.5 %
learned, grip	68.5 %	[62.8, 74.0]	75.4 %	74.5 %	70.6 %	11	82 %	70 %	67.5 %
production (calibrated)	72.8 %	[68.9, 76.6]	82.9 %	77.5 %	74.6 %	11	86 %	72 %	70.0 %

Activity filter and stream low-pass. The activity signal runs causal (zero-phase minus causal: +0.8 [−0.4, +1.9] points, Holm $p = 1.000$), because it must run online. The stream itself is low-passed zero-phase over the buffer received so far, the code path of the training windows; a fully causal low-pass minus this is −1.7 [−3.5, +0.0] points (Holm $p = 1.000$). A replay re-standardised each raw record up to each decision time, as the page would: 100.0 % of the segments are found again, decisions change in 0.0 % of the recordings (statuses in 0.3 %), and the largest change of P(right) is 0.000.

Hand model. The second mirror-augmented hand model, chiral16, minus yaw8 is +1.5 [−1.2, +4.4] points cue-free (Holm $p = 1.000$) and +1.7 [−0.9, +4.3] points on the oracle window (Holm $p = 1.000$). Decision: yaw8 stays the pre-specified default; the hand model is pluggable.

Ablations. Table 89 varies one data treatment at a time against the uncalibrated production detector: feature scaling (z-score +0.5 [−0.1, +1.3], none −0.1 [−0.7, +0.6] points against median/IQR), the segment classifier's C nested in the folds (+0.2 [−0.4, +0.9] points) and the look-ahead (0.3 s +0.8 [+0.0, +1.8], 0.8 s −0.1 [−0.7, +0.5] points against 0.5 s). None shows an accuracy difference after Holm's adjustment (family *trial*). The pre-specified values stay: median/IQR scaling, C = 1, and a 0.5 s look-ahead, which adds directly to the latency.

P(approach) threshold. Re-thresholding the saved predictions on the outer folds (exploratory, outside the Holm family) trades coverage for accuracy: 0.30 gives 74.3 % (answers 90 %; paired +1.5 [+0.6, +2.5] points), 0.40 73.9 % (87 %), 0.50 72.8 % (83 %), 0.60 71.7 % (77 %) and 0.70 69.6 % (68 %). The pre-specified 0.5 stays; a threshold chosen inside the folds is a candidate for the next version.

Table 87: Lane m4: cue-free pipelines and their references (leave-one-person-out; 95 % person-cluster bootstrap CI). Accuracy: an abstention (no approach found, P = 0.5) scores half a correct answer. Answered: share of recordings with an approach found. Natural use (one first free-choice trial per person, all right hands): answered / scored, accuracy on the answered ones and half-credit accuracy. Configs: n_configs_tried recorded with the result.

Pipeline	Trials	Accuracy	95% CI	AUC	Log loss	ECE	Answered	Nat. answered	Nat. acc. answered	Nat. acc.	Configs
oracle window (no detection)	598	77.9%	[73.1, 82.9]	0.855	0.512	0.072	100%	20/20	75%	75.0%	1
oracle window, chiral16	598	79.6%	[75.5, 84.2]	0.859	0.506	0.086	100%	20/20	75%	75.0%	1
detector v0 (first movement)	598	67.7%	[63.4, 72.1]	0.732	0.722	0.093	95%	16/20	69%	65.0%	1
learned, physics features	598	71.7%	[66.6, 76.3]	0.806	0.549	0.054	77%	8/20	88%	65.0%	1
learned, physics + context	598	74.6%	[70.0, 79.0]	0.835	0.521	0.069	86%	12/20	83%	70.0%	1
learned, feature set nested	598	74.6%	[70.0, 79.0]	0.835	0.521	0.069	86%	12/20	83%	70.0%	1
learned, robust (production detector)	598	73.4%	[68.7, 77.9]	0.819	0.557	0.067	83%	11/20	82%	67.5%	5
learned, grip-normalised	598	68.5%	[62.8, 74.0]	0.762	0.610	0.047	75%	11/20	82%	67.5%	1
hand model on detector windows	598	73.7%	[70.0, 77.3]	0.822	0.530	0.075	86%	12/20	83%	70.0%	1
production detector, chiral16 (uncal.)	598	74.9%	[71.4, 78.6]	0.831	0.537	0.073	83%	11/20	82%	67.5%	1
calibrated: temperature	598	73.4%	[68.7, 77.9]	0.820	0.554	0.070	83%	11/20	82%	67.5%	1
calibrated: Platt	598	72.9%	[68.8, 77.0]	0.816	0.555	0.065	83%	11/20	82%	67.5%	1
calibrated: isotonic	598	73.5%	[69.1, 77.7]	0.811	0.531	0.052	83%	11/20	82%	67.5%	1
calibrated: beta	598	72.1%	[67.5, 76.7]	0.811	0.557	0.066	83%	11/20	82%	67.5%	1
calibrated: symmetrised isotonic	598	72.8%	[68.9, 76.6]	0.816	0.524	0.048	83%	11/20	86%	70.0%	1
production (calibrator nested)	598	72.8%	[68.9, 76.6]	0.816	0.524	0.048	83%	11/20	86%	70.0%	22
stored windows, approach	288	78.8%	[73.2, 84.1]	0.859	0.489	0.049	100%	–	–	–	1
stored windows, return head	288	81.6%	[75.7, 87.0]	0.888	0.447	0.065	100%	–	–	–	1
stored windows, approach + return	288	84.0%	[77.6, 89.7]	0.895	0.532	0.061	100%	–	–	–	1
production detector + return head (uncal.)	598	75.9%	[71.5, 80.3]	0.844	0.583	0.052	83%	11/20	82%	67.5%	1
production detector, zero-phase activity (uncal.)	598	74.2%	[69.4, 78.6]	0.823	0.549	0.068	83%	12/20	83%	70.0%	1
production detector, causal low-pass (uncal.)	598	71.7%	[67.4, 75.8]	0.815	0.543	0.069	83%	11/20	82%	67.5%	1
ablation: z-scored segment features	598	73.9%	[69.2, 78.4]	0.828	0.530	0.068	83%	11/20	82%	67.5%	1
ablation: unscaled segment features	598	73.3%	[68.8, 77.6]	0.822	0.537	0.067	83%	11/20	82%	67.5%	1
ablation: segment-classifier C nested	598	73.6%	[69.2, 77.9]	0.824	0.538	0.074	82%	11/20	82%	67.5%	1
ablation: look-ahead 0.3 s	598	74.2%	[69.9, 78.4]	0.832	0.521	0.068	83%	12/20	83%	70.0%	1
ablation: look-ahead 0.8 s	598	73.3%	[68.8, 77.7]	0.822	0.537	0.071	83%	11/20	82%	67.5%	1

Table 88: Paired comparisons on the same recordings (A - B; half-credit accuracy; person-bootstrap 95 % CI; exact sign-flip test over persons; Holm-adjusted over the 26 comparisons of this table, one family).

Comparison	A	B	Acc. A	Acc. B	Difference	CI low	CI high	p	Holm p	Log-loss diff.	Identical
prod_vs_oracle	production (calibrator nested)	oracle window (no detection)	72.8%	77.9%	−5.1%	−7.6%	−2.7%	0.002	0.047	+0.012	
prod_vs_v0	production (calibrator nested)	detector v0 (first movement)	72.8%	67.7%	+5.1%	+1.9%	+8.3%	0.008	0.203	−0.198	
full_vs_physics	learned, physics + context	learned, physics features	74.6%	71.7%	+2.9%	+0.7%	+5.8%	0.023	0.517	−0.028	
nested_vs_physics	learned, feature set nested	learned, physics features	74.6%	71.7%	+2.9%	+0.7%	+5.8%	0.023	0.517	−0.028	
robust_vs_nested	learned, robust (production detector)	learned, feature set nested	73.4%	74.6%	−1.2%	−2.6%	+0.2%	0.077	1.000	+0.036	
robust_vs_physics	learned, robust (production detector)	learned, physics features	73.4%	71.7%	+1.8%	−0.6%	+4.3%	0.126	1.000	+0.009	
grip_vs_robust	learned, grip-normalised	learned, robust (production detector)	68.5%	73.4%	−4.9%	−8.7%	−1.8%	0.011	0.250	+0.053	
detwin_vs_oraclewin	hand model on detector windows	learned, feature set nested	73.7%	74.6%	−0.8%	−3.8%	+1.8%	0.555	1.000	+0.008	
chiral16_vs_yaw8	production detector, chiral16 (uncal.)	learned, robust (production detector)	74.9%	73.4%	+1.5%	−1.2%	+4.4%	0.450	1.000	−0.021	
oracle_chiral16_vs_yaw8	oracle window, chiral16	oracle window (no detection)	79.6%	77.9%	+1.7%	−0.9%	+4.3%	0.200	1.000	−0.006	
return_vs_prod	production detector + return head (uncal.)	learned, robust (production detector)	75.9%	73.4%	+2.5%	+0.8%	+4.4%	0.008	0.203	+0.026	
zp_activity_vs_prod	production detector, zero-phase activity (uncal.)	learned, robust (production detector)	74.2%	73.4%	+0.8%	−0.4%	+1.9%	0.282	1.000	−0.009	
causal_lowpass_vs_prod	production detector, causal low-pass (uncal.)	learned, robust (production detector)	71.7%	73.4%	−1.7%	−3.5%	+0.0%	0.085	1.000	−0.015	
ret_sum_vs_app	stored windows, approach + return	stored windows, approach	84.0%	78.8%	+5.2%	+1.7%	+8.8%	0.037	0.747	+0.043	
ret_head_vs_app	stored windows, return head	stored windows, approach	81.6%	78.8%	+2.8%	−1.7%	+6.9%	0.294	1.000	−0.041	
abl_scale_z_vs_base	ablation: z-scored segment features	learned, robust (production detector)	73.9%	73.4%	+0.5%	−0.1%	+1.3%	0.172	1.000	−0.028	
abl_scale_none_vs_base	ablation: unscaled segment features	learned, robust (production detector)	73.3%	73.4%	−0.1%	−0.7%	+0.6%	0.934	1.000	−0.021	
abl_segc_nested_vs_base	ablation: segment-classifier C nested	learned, robust (production detector)	73.6%	73.4%	+0.2%	−0.4%	+0.9%	0.578	1.000	−0.019	
abl_look0p3_vs_base	ablation: look-ahead 0.3 s	learned, robust (production detector)	74.2%	73.4%	+0.8%	+0.0%	+1.8%	0.055	1.000	−0.036	
abl_look0p8_vs_base	ablation: look-ahead 0.8 s	learned, robust (production detector)	73.3%	73.4%	−0.1%	−0.7%	+0.5%	0.852	1.000	−0.020	
cal_temperature_vs_none	calibrated: temperature	learned, robust (production detector)	73.4%	73.4%	+0.0%	+0.0%	+0.0%	1.000	1.000	−0.004	
cal_platt_vs_none	calibrated: Platt	learned, robust (production detector)	72.9%	73.4%	−0.5%	−2.3%	+1.3%	0.616	1.000	−0.002	
cal_isotonic_vs_none	calibrated: isotonic	learned, robust (production detector)	73.5%	73.4%	+0.1%	−1.7%	+1.9%	0.890	1.000	−0.027	
cal_beta_vs_none	calibrated: beta	learned, robust (production detector)	72.1%	73.4%	−1.3%	−3.8%	+0.9%	0.383	1.000	−0.001	
cal_isotonic_sym_vs_none	calibrated: symmetrised isotonic	learned, robust (production detector)	72.8%	73.4%	−0.6%	−2.1%	+0.9%	0.594	1.000	−0.033	
cal_nested_vs_none	production (calibrator nested)	learned, robust (production detector)	72.8%	73.4%	−0.6%	−2.1%	+0.9%	0.594	1.000	−0.033	

Table 89: Data-treatment ablations of the production detector (uncalibrated; paired with it on the same recordings; an abstention (no approach found, P = 0.5) scores half a correct answer): feature scaling, the segment classifier's C, the look-ahead after a segment, and the P(approach) threshold (re-thresholded saved predictions of the production pipeline, paired with it; outside the Holm family).

Ablation	Accuracy	Base	Difference	CI	Holm p	Log-loss diff.	Log-loss CI
ablation: z-scored segment features	73.9%	73.4%	+0.5%	[−0.1, 1.3]	1.000	−0.028	[−0.070, −0.001]
ablation: unscaled segment features	73.3%	73.4%	−0.1%	[−0.7, 0.6]	1.000	−0.021	[−0.059, 0.003]
ablation: segment-classifier C nested	73.6%	73.4%	+0.2%	[−0.4, 0.9]	1.000	−0.019	[−0.059, 0.001]
ablation: look-ahead 0.3 s	74.2%	73.4%	+0.8%	[0.0, 1.8]	1.000	−0.036	[−0.075, −0.007]
ablation: look-ahead 0.8 s	73.3%	73.4%	−0.1%	[−0.7, 0.5]	1.000	−0.020	[−0.061, 0.005]
P(approach) threshold 0.30	74.3%	72.8%	+1.5%	[0.6, 2.5]	–	−0.006	[−0.019, 0.010]
P(approach) threshold 0.40	73.9%	72.8%	+1.1%	[0.3, 1.8]	–	−0.006	[−0.016, 0.004]
P(approach) threshold 0.60	71.7%	72.8%	−1.1%	[−2.0, −0.2]	–	+0.005	[−0.007, 0.016]
P(approach) threshold 0.70	69.6%	72.8%	−3.3%	[−5.1, −1.5]	–	+0.017	[−0.004, 0.038]

G.6. Calibration of one touch

Problem and detection. The product is the percentage shown to the user, so it must be calibrated. Uncalibrated, the pipeline's expected calibration error is 0.067, above the range a perfectly calibrated $P(\text{right})$ with the same values would show (null band 0.021–0.061, $p = 0.006$); its calibration slope is 0.79 [0.57, 1.10] and its intercept −0.11 [−0.25, 0.03] (Figure 92, left).

Options. Every calibrator is fitted on cross-fitted predictions of the training persons (person-grouped inner folds) and applied to the held-out person: none, temperature scaling [12], Platt scaling [21], isotonic regression, beta calibration and a symmetrised isotonic regression. The hand model is mirror-augmented, so a mirror-image recording gets exactly $1 - P(\text{right})$; a calibrator keeps this only if it is antisymmetric, $P(-z) = 1 - P(z)$ for a logit z. Platt, plain isotonic and beta are not: their largest deviation $|P(z) + P(-z) - 1|$ is 0.109, 0.477 and 0.144.

Evidence. Table 90 and Figure 92 (right) give each calibrator's paired log-loss difference to none (family *calibrators*, Holm). The log-loss difference to none is −0.027 [−0.054, −0.002] for plain isotonic (Holm $p = 0.305$) and −0.033 [−0.060, −0.009] for the symmetrised isotonic (Holm $p = 0.132$); temperature and Platt change little. No calibrator shows a log-loss difference after adjustment; the choice rests on the calibration diagnostics below.

Decision and rationale. The calibrator is chosen inside each fold among the antisymmetric ones (none, temperature, symmetrised isotonic) by person-grouped cross-validation of the cross-fitted logits; the symmetrised isotonic won in 20 of 20 folds. The production pipeline's ECE is 0.048 (null band 0.020–0.060), its slope 0.94 [0.75, 1.18] and its intercept −0.13 [−0.29, 0.03]: consistent with calibration. Against the uncalibrated pipeline its log loss changes by −0.033 [−0.060, −0.009] (Holm $p = 0.132$), its Brier score [19] by −0.0025 [−0.0066, +0.0016] and its accuracy by −0.6 [−2.1, +0.9] points. The flat centre of the symmetrised isotonic map returns exactly 0.5 for 39 of the 496 answered recordings: these show no hand and score half credit. Per platform, Android has ECE 0.067 (null band 0.024–0.075) and slope 1.07 [0.88, 1.28]; iOS has ECE 0.051 (null band 0.031–0.101) and slope 0.80 [0.54, 1.24] (per-platform intercepts were not saved).

Table 90: Calibrators of the cue-free production detector, each fitted on cross-fitted predictions of the training persons and applied to the held-out person; nested = chosen inside each fold among the antisymmetric ones (the production calibrator). Mirror dev.: largest |P(z) + P(-z) - 1| over the folds (0: a mirror-image recording gets exactly 1 - P). d log loss: paired difference to none (person bootstrap; Holm over the calibrators). Null band: 95 % range of the ECE of a perfectly calibrated P with the same values (persons resampled, outcomes redrawn). Slope: calibration slope of y on logit(P) (1 = calibrated).

Calibrator	Antisym.	Mirror dev.	Accuracy	Log loss	d log loss	d CI	Holm p	Brier	ECE	Null band	Slope	Slope CI
none	yes	0.000	73.4%	0.557	–	–	–	0.174	0.067	[0.021, 0.061]	0.79	[0.573, 1.103]
temperature	yes	0.000	73.4%	0.554	−0.004	[−0.016, 0.008]	1.000	0.177	0.070	[0.023, 0.063]	0.97	[0.688, 1.353]
platt	no	0.109	72.9%	0.555	−0.002	[−0.016, 0.010]	1.000	0.177	0.065	[0.021, 0.063]	0.94	[0.668, 1.311]
isotonic	no	0.477	73.5%	0.531	−0.027	[−0.054, −0.002]	0.305	0.173	0.052	[0.020, 0.060]	0.88	[0.705, 1.102]
beta	no	0.144	72.1%	0.557	−0.001	[−0.022, 0.016]	1.000	0.178	0.066	[0.022, 0.064]	0.92	[0.681, 1.251]
isotonic_sym	yes	0.000	72.8%	0.524	−0.033	[−0.060, −0.009]	0.132	0.171	0.048	[0.020, 0.060]	0.94	[0.752, 1.183]
nested	yes	0.000	72.8%	0.524	−0.033	[−0.060, −0.009]	0.132	0.171	0.048	[0.020, 0.060]	0.94	[0.752, 1.183]

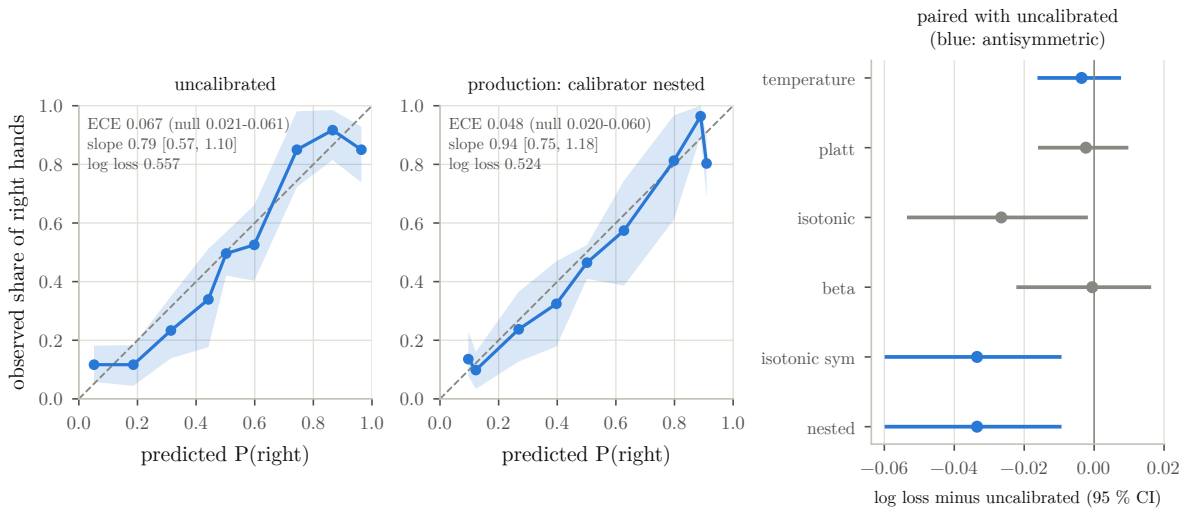
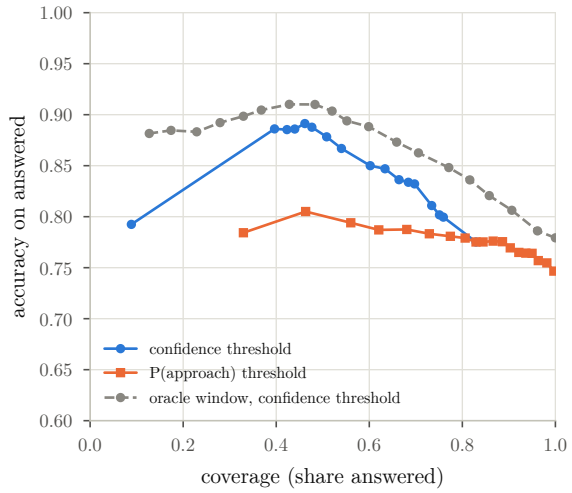


Figure 92: Calibration of one touch (10 equal-mass bins; band: 95 % CI of the observed share; text: ECE with its perfect-calibration null band, slope with its CI). Left: uncalibrated, not consistent with calibration. Middle: production, the calibrator chosen inside each fold (symmetrised isotonic in 20 of 20 folds), consistent. Right: paired log-loss difference of each calibrator to none (blue: antisymmetric).

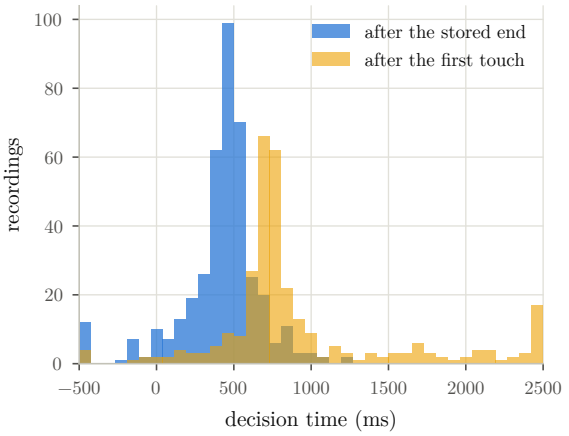
G.7. The single-touch abstain rule and the decision latency

Abstain rule. A single-touch page can ask the user to repeat instead of answering. Two rules were compared (Figure 93a): answer only when $\max(P(\text{right}), 1 - P(\text{right})) \geq c$, or only when the best candidate's P(approach) reaches a threshold. The confidence rule buys accuracy; the P(approach) rule barely does. With $c = 0.70$ the pipeline answers 60 % of the recordings and is right on 85.0 % of those; its forced-choice accuracy stays 74.6 %. The same rule on the oracle windows lies above the cue-free curve at every coverage: part of the gap is detection. The threshold is a product choice read off this curve, not tuned inside the folds; multi-touch sessions use the stopping rule instead (Appendix G.9).

Decision latency. The end of an approach is decided at the first stillness of the sustain time at the face (the deceleration to rest; the contact transient is not needed), plus the 0.5 s look-ahead. Figure 93b shows the result: the decision comes a median 451 ms after the stored end of the approach (90th percentile 666 ms) and 754 ms after the first touch of the screen on the nose (90th percentile 2002 ms). A shorter look-ahead would cut the latency without an accuracy difference shown (Appendix G.5).



(a) Accuracy on the answered recordings against coverage, for a confidence threshold (blue), a P(approach) threshold (orange) and the confidence threshold on the oracle windows (grey).



(b) Decision time of the online pipeline after the stored end of the approach (blue) and after the first touch (yellow); recordings whose approach was found; values beyond the axis drawn at its edge.

Figure 93: The single-touch abstain rule (a) and the latency of the end-of-approach decision (b), production pipeline.

G.8. Several touches: synthetic sessions and the k curve

Method. The recordings contain one approach each, so multi-touch sessions are simulated. Each synthetic session concatenates 8 recordings of one person and hand, with the stillness at every junction cut to a given length; there are 134 sessions per setting. Each fold's fitted pipeline runs over the held-out person's sessions online. Every bout that the segment classifier accepts as an approach is an attempt, and the attempts' calibrated logits z_i are combined:

$$P_k = \sigma(s_k\,\bar{z}_k) = \sigma\left(\frac{1}{\tau_k}\sum_{i=1}^{k} z_i\right), \qquad \tau_k = \frac{k}{s_k}, \tag{9}$$

where σ is the logistic function and $\bar{z}_k$ the mean of the first k logits.

Evidence: the k curve. Figure 94 (left) and Table 91 give the half-credit accuracy of the sign of the mean logit of up to k accepted attempts. With 1 s at each junction it is 78.0 % [68.5, 86.5] for $k = 1$, 71.3 % for $k = 2$, 76.5 % [69.2, 83.7] for $k = 3$ and 84.0 % [77.3, 90.6] for $k = 5$; 97 % of the sessions reach 5 accepted attempts. Keeping only the attempts that are true approaches (the attainable curve) gives 81.3 %, 88.4 % [82.0, 94.5] and 92.9 % for $k = 1$, 3 and 5. With no stillness at the junctions the curve stays near chance (50.4 % at $k = 3$); with 0.5 s it reaches 69.4 % and with 2 s 75.4 %.

Cause: false attempts. Only 77 % of the 1075 accepted attempts are (part of) an approach. The others are the arm settling before the approach (10 % of the attempts, 112), returns (10 %, 108) and later movements (3 %, 28). One attempt is right 80 % of the time when it is the approach, 76 % when it is part of it, 46 % when it is the arm settling, 45 % when it is a return and 34 % for later movements. The false attempts cancel the gain of more touches; rejecting them is the first next step.

Option: the attempt rule. The pairing rule treats the bout after an accepted approach as its return and skips it; the plain rule (no pairing) takes every accepted bout. Paired on the same sessions with the stopping rule (family *sessions*), no pairing minus pairing is −1.1 [−5.3, +3.0] points (Holm $p = 1.000$, 9 discordant sessions). No pairing recalls 77 % of the true approaches against 66 % with pairing, at the same precision (77 % and 77 %). Decision: no pairing, because the pairing rule discards a true approach whenever an arm-settling movement before it was accepted. For $k = 3$ the pairing rule scores 79.5 % [72.8, 86.4]; the attempt rule is one of the exploratory session choices (Appendix G.9).

Not a cue-free number. The harness's $k = 3$ of the same pipeline, 85.2 % [79.8, 90.2], averages three independent recordings of one person and hand, each with its best candidate. It is not a cue-free session and is not comparable with the oracle-window $k = 3$ of other lanes.

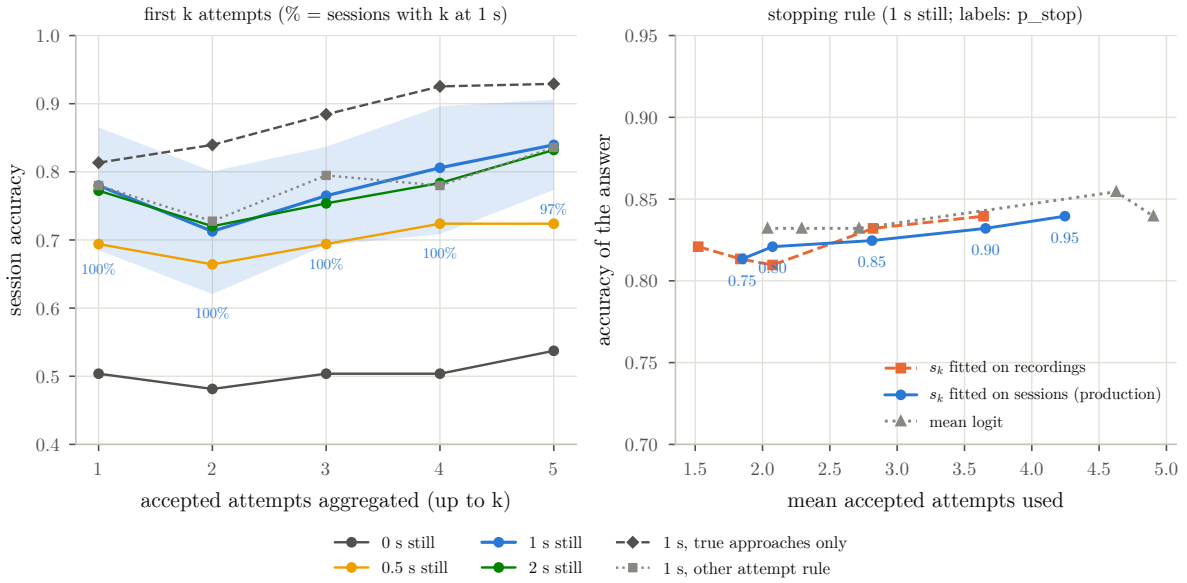


Figure 94: Synthetic multi-attempt sessions (134 per stillness setting). Left: half-credit accuracy of up to k accepted attempts, by junction stillness (band: 95 % CI at 1 s; labels: sessions with at least k attempts); dark dashed: true approaches only; grey dotted: the pairing rule. Right: the stopping rule at 1 s, one point per p_{stop}, for three aggregations. The gap between the 1 s curve and the true-approach curve is the cost of false attempts.

Table 91: Synthetic multi-attempt sessions (8 recordings of one person and hand each): half-credit accuracy when the first k accepted attempts are aggregated (the sign of their mean calibrated logit; fewer when the session has fewer: "up to k"), by the stillness left at each junction (production attempt rule: no pairing); then the pairing rule and the attainable curve when only the true approaches are kept, and the share of sessions with at least k accepted attempts.

Attempts (up to)	0 s	0.5 s	1 s	2 s	1 s, pairing rule	1 s, true approaches only	Sessions with k
1	50.4%	69.4%	78.0%	77.2%	78.0%	81.3%	100%
2	48.1%	66.4%	71.3%	72.0%	72.8%	84.0%	100%
3	50.4%	69.4%	76.5%	75.4%	79.5%	88.4%	100%
4	50.4%	72.4%	80.6%	78.4%	78.0%	92.5%	100%
5	53.7%	72.4%	84.0%	83.2%	83.6%	92.9%	97%

G.9. The stopping rule and the calibration of several touches

The rule. After each accepted attempt, P_k of Equation (9) is updated; the pipeline stops when P_k leaves $[1 - p_{\text{stop}}, p_{\text{stop}}]$ or after 5 attempts. $p_{\text{stop}} = 0.90$ is a product choice fixed before the sessions were evaluated.

Problem: the scale of several touches. Accepted attempts are not independent evidence: some are not approaches, and the attempts of one session share the person's bias. Three aggregations were compared (Figure 95, Table 92): the mean logit ($s_k = 1$, nothing fitted); s_k fitted on random recordings of the training persons (median $s_3 = 2.44$), which assumes neither problem; and $s_k = s_1 k^b$ fitted on cross-fitted synthetic sessions of the training persons (median $b = 0.69$, $s_3 = 1.79$). The sum of the logits (as if independent) is shown for $k = 3$ only.

Evidence at the stop (1 s junctions, $p_{\text{stop}} = 0.90$). Fitted on recordings: accuracy 83.2 % with 2.82 attempts, ECE 0.116 (null band 0.036–0.095), slope 0.81 [0.57, 1.23], intercept −0.74: not consistent with calibration. Fitted on sessions: accuracy 83.2 % with 3.66 attempts, ECE 0.107 (null band 0.039–0.108, at its upper edge), slope 1.03 [0.73, 1.81], intercept −0.70: consistent with calibration. Mean logit: accuracy 85.4 % with 4.63 attempts, ECE 0.177 (null band 0.053–0.143), slope 1.81 [1.17, 3.30], intercept −0.45: not consistent. The intercepts' CIs were not saved. Paired on the same sessions (family *sessions*), recording-fitted minus session-fitted is +0.0 [−3.0, +3.0] points of accuracy and +0.066 [−0.010, +0.147] of log loss (Holm $p = 0.598$); session-fitted minus mean logit is −2.2 [−6.2, +0.8] points and −0.052 [−0.158, +0.070] of log loss (Holm $p = 1.000$). No paired difference is shown.

Decision and rationale. A declared rule picks the aggregation: among those consistent with calibration, the one with the fewest fitted parameters. It picks s_k fitted on sessions. This choice is post hoc and exploratory: the session-fitted scale was added after the held-out sessions had shown the problem, and 6 configurations were compared (attempt rule × aggregation).

Result. With 1 s junctions the stopping rule reaches 83.2 % [74.6, 90.8] with 3.66 accepted attempts and 3.49 true touches by the decision on average; AUC 0.902, log loss 0.412. Every session answers (0 % without an attempt); 60 % stop on the threshold and are right 90.1 % [79.6, 98.8] of the time while claiming 93.6 % on average; the other 40 % stop at 5 attempts. With 0.5 s junctions it falls to 71.6 % [59.6, 82.8] and with none to 54.5 % [43.6, 64.9]. Per platform: Android 82.7 % (ECE 0.094, null band 0.054–0.139, slope 1.10; 84 sessions of 12 persons), iOS 84.0 % (ECE 0.099, null band 0.067–0.175, slope 0.95; 50 sessions of 8 persons); the CIs of these slopes and the intercepts were not saved, so these are descriptive.

Exploratory readings of the same sessions. Table 93 sets touches against certainty: $p_{\text{stop}} = 0.75$ gives 81.3 % with 1.85 attempts, 0.85 gives 82.5 % with 2.81, 0.95 gives 84.0 % with 4.25. An abstain rule after 5 attempts (answer only when $\max(P, 1 - P) \geq p_{\text{answer}}$) answers 93 % of the sessions with 86.3 % right at $p_{\text{answer}} = 0.6$, 84 % with 88.4 % at 0.7 and 67 % with 90.0 % at 0.8 (accuracy on the answered sessions).

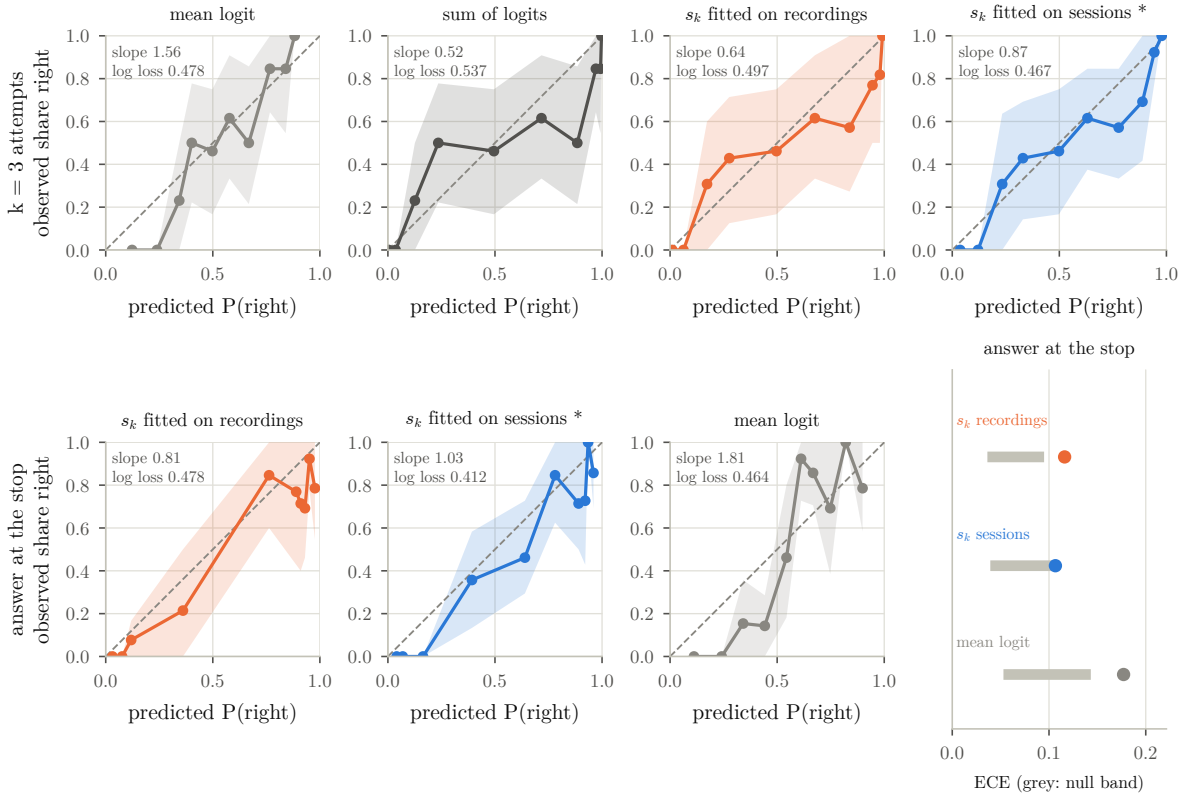


Figure 95: Calibration of several touches on the synthetic sessions (1 s junctions, production attempt rule; equal-mass bins, 95 % bands). Top: the first three accepted attempts under four aggregations. Bottom: the answer of the stopping rule ($p_{\text{stop}} = 0.90$), and its ECE against the perfect-calibration null band (right; * production). Only s_k fitted on sessions is consistent with calibration at the stop, at the edge of its band.

Table 92: Calibration of several attempts on the synthetic sessions (1 s of stillness at each junction, production attempt rule): three accepted attempts (k=3) and the answer of the stopping rule (p_stop 0.90) under each aggregation of the calibrated logits. Calibrated: ECE inside its perfect-calibration null band and slope CI containing 1; the production aggregation is the calibrated one with the fewest fitted parameters (declared rule).

Aggregation	Params	k=3 log loss	k=3 slope	Stop accuracy	Attempts	Stop log loss	Stop ECE	Null band	Slope	Slope CI	Calibrated
mean logit	0	0.478	1.56	85.4%	4.63	0.464	0.177	[0.053, 0.143]	1.81	[1.170, 3.301]	no
sum of logits	–	0.537	0.52	–	–	–	–		–		
s_k fitted on recordings	5	0.497	0.64	83.2%	2.82	0.478	0.116	[0.036, 0.095]	0.81	[0.574, 1.230]	no
s_k fitted on sessions (production)	2	0.467	0.87	83.2%	3.66	0.412	0.107	[0.039, 0.108]	1.03	[0.727, 1.809]	yes

Table 93: Sequential stopping rule on the synthetic sessions (production attempt rule; attempts aggregated with s_k fitted on sessions): stop when the aggregated P(right) leaves [1 - p_stop, p_stop], else answer after 5 accepted attempts or at the end of the session. Attempts: accepted attempts used; Touches: true approaches completed by the decision. Confident stops: share of sessions that stopped on the threshold, and their accuracy (an abstention (no approach found, P = 0.5) scores half a correct answer).

Stillness (s)	p_stop	Accuracy	CI low	CI high	Attempts	Touches	Confident stops	Acc. confident	ECE
0	0.75	52.6%	42.7%	62.0%	1.74	3.51	87%	49.1%	0.317
0	0.80	53.4%	43.7%	62.7%	1.90	3.80	81%	50.0%	0.314
0	0.85	53.7%	43.1%	63.4%	2.34	4.51	56%	52.0%	0.327
0	0.90	54.5%	43.6%	64.9%	2.87	5.34	31%	54.8%	0.300
0	0.95	53.7%	43.0%	64.1%	3.13	5.73	19%	52.0%	0.272
0.5	0.75	74.6%	65.2%	83.8%	1.56	2.64	92%	75.6%	0.110
0.5	0.80	73.9%	63.3%	83.6%	1.73	2.86	90%	75.0%	0.107
0.5	0.85	72.4%	61.3%	82.6%	2.37	3.72	74%	76.8%	0.122
0.5	0.90	71.6%	59.6%	82.8%	3.08	4.80	58%	80.8%	0.139
0.5	0.95	72.4%	60.9%	83.1%	3.66	5.54	38%	88.2%	0.123
1	0.75	81.3%	73.1%	88.2%	1.85	1.74	94%	81.7%	0.126
1	0.80	82.1%	73.7%	89.3%	2.07	1.91	87%	81.9%	0.104
1	0.85	82.5%	75.6%	89.1%	2.81	2.63	75%	84.0%	0.102
1	0.90	83.2%	74.6%	90.8%	3.66	3.49	60%	90.1%	0.107
1	0.95	84.0%	77.3%	90.6%	4.25	4.06	44%	93.2%	0.077
2	0.75	80.6%	72.2%	87.6%	1.88	1.72	91%	81.1%	0.092
2	0.80	82.1%	73.7%	89.3%	2.13	1.92	86%	82.6%	0.102
2	0.85	82.5%	75.8%	88.8%	2.87	2.62	75%	84.0%	0.086
2	0.90	82.5%	73.8%	90.3%	3.74	3.45	58%	89.7%	0.091
2	0.95	83.2%	76.4%	90.2%	4.30	4.02	40%	92.6%	0.076

G.10. Screen away and the rest-pose grip check

Problem and detection. Every held-out recording was also scored as its screen-away twin: the phone turned 180 degrees about its long axis, so the device x and z axes change sign. The gravity-frame channels and the yaw8 input do not change, but the strongest segment features are device-frame quantities along the screen normal (Appendix G.4). Screen away, the production pipeline falls to 42.5 % [35.8, 49.6]; it scores the return in 39 % of the twins, and its answers are right 38.8 % of the time (Figure 96). A confident wrong percentage is the worst outcome for the product.

Options. (1) A check of the chosen segment's own start pose; (2) the grip-normalised feature set, invariant by construction; (3) a model-free check of the rest pose on the inference page: while the phone rests at arm's length before the first movement, the screen-normal component of the up vector must be at least 0.00; otherwise the page asks to turn the screen towards the face and does not answer (scored as an abstention).

Evidence. (1) The segment check stopped 44 % of the twins and left their accuracy at 42.7 %; the answers it let through were right 18.5 % of the time. When the detector takes the return for the approach, the chosen segment starts at the nose, so its start pose says nothing about the grip. (2) The grip set scores 68.5 % screen away and 68.5 % as recorded, but grip minus robust on the recordings is −4.9 [−8.7, −1.8] points (Holm $p = 0.250$, family *trial*). (3) The rest-pose check stops 95.7 % of the twins. The share answered wrongly falls from 38.8 % to 2.0 % (−36.8 [−44.0, −29.8] points, Holm $p < 0.001$, family *grip*); the twins' half-credit accuracy moves to 49.7 % [48.6, 50.6], because a stopped recording scores as an abstention. On the

screen-toward recordings it stops 4.3 %: coverage goes from 83 % to 79 %, the share answered wrongly from 15.4 % to 14.5 %, and the accuracy from 72.8 % to 71.7 % (paired −1.1 [−3.0, +0.1] points, Holm $p = 1.000$, family *stress*).

Decision and rationale. Production keeps the robust features behind the rest-pose grip check, and the instructions ask for the screen facing the user, tilted slightly upwards. A user asked to turn the phone then gets the screen-toward accuracy. The first remedy was replaced; the grip set was rejected for its cost on real grips. Caveats: the check reads gravity only, so a phone held near vertical or tilted forward is also asked to adjust; it was evaluated on synthetic twins, not on real screen-away recordings; only the recorded grips (upright, top left and top right, all screen toward) were tested.

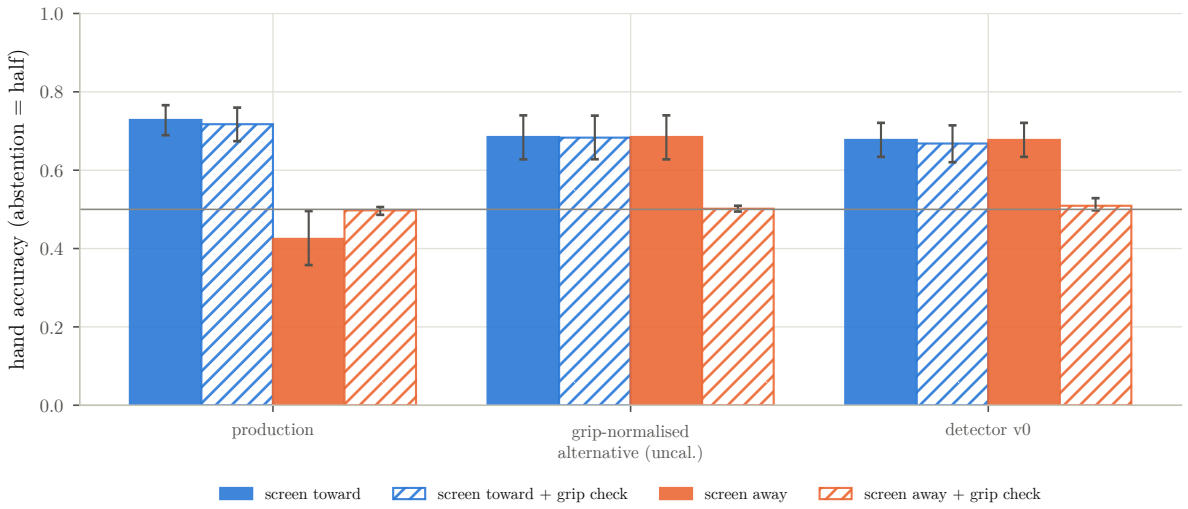


Figure 96: Every held-out recording scored as recorded (screen toward) and as its screen-away twin, by the pipeline of its fold (half-credit accuracy, 95 % CI). Hatched: with the rest-pose grip check, which stops 95.7 % of the twins and 4.3 % of the recordings. Supports the grip check over the grip-normalised features.

G.11. Platform and condition

Table 94 stratifies the production pipeline by platform and by instructed condition; the conditions stratify only and never enter a model. The platforms score alike: Android 72.5 % [67.7, 76.9] on 383 recordings, iOS 73.5 % [66.2, 80.2] on 215. Fast movements score higher than natural and slow ones, and the sideways grips higher than the upright grip; the slow, tilted-touch, twist and curved variants score lowest, with wide intervals. The bent-start variant has fewer than 5 persons and no CI. These strata are descriptive: no comparison between them was tested here (lane sec tests speed and grip; Appendix H).

Table 94: Cue-free production pipeline by platform and by instructed condition (conditions stratify only; person-bootstrap 95 % CI, none for strata with fewer than 5 persons; an abstention (no approach found, P = 0.5) scores half a correct answer).

Stratum	Level	Trials	Persons	Accuracy	CI low	CI high	Answered	ECE	Log loss
platform	android	383	12	72.5%	67.7%	76.9%	81%	0.067	0.512
platform	ios	215	8	73.5%	66.2%	80.2%	87%	0.051	0.545
speed	fast	134	20	83.2%	78.3%	88.7%	87%	0.150	0.371
speed	natural	421	20	70.5%	66.3%	74.4%	83%	0.045	0.564
speed	slow	43	19	62.8%	48.9%	77.0%	70%	0.177	0.611
orientation	top_left_toward	118	19	80.9%	74.0%	87.9%	88%	0.114	0.427
orientation	top_right_toward	114	19	79.8%	71.3%	87.3%	85%	0.059	0.432
orientation	upright_toward	366	20	68.0%	62.3%	73.5%	81%	0.075	0.584
posture	sitting	438	15	72.3%	67.4%	76.9%	85%	0.069	0.539
posture	standing	160	5	74.4%	69.5%	80.1%	78%	0.040	0.484
start_position	center	18	10	66.7%	52.5%	81.6%	72%	0.160	0.577
start_position	natural	563	20	73.4%	69.2%	77.5%	83%	0.055	0.516
start_position	shoulder	17	9	61.8%	35.3%	86.1%	82%	0.315	0.742
variant	bent_start	7	4	42.9%	–	–	100%	0.335	0.840
variant	curved	16	8	59.4%	37.5%	81.2%	69%	0.311	0.998
variant	elbow_high	14	7	78.6%	57.1%	92.9%	86%	0.176	0.448
variant	jerky	13	8	61.5%	31.8%	83.3%	85%	0.231	0.606
variant	lean_in	11	6	72.7%	59.1%	86.4%	45%	0.211	0.524
variant	pause	17	9	58.8%	38.2%	79.4%	65%	0.190	0.724
variant	standard	470	20	76.2%	72.2%	80.3%	84%	0.057	0.473
variant	tilted_touch	35	19	55.7%	41.2%	69.4%	86%	0.183	0.737
variant	twist	15	8	56.7%	34.4%	79.2%	87%	0.382	0.729

G.12. A failure mode: natural movements are rejected

Problem. The first free-choice movement of each person is the most production-like trial in the data: unpractised and chosen freely. The production detector rejects 9 of these 20 natural-use recordings (45 %), against 17 % of the primary recordings. From each person's own primary rejection rate, 3.2 rejections are expected (Poisson-binomial $p = 0.0015$; Fisher's exact test, natural against primary, $p = 0.0041$; post hoc tests, unadjusted).

What the pipeline achieves on natural use. It answers 11 of 20 (55 %; 1 of them at exactly 0.5), with a right-hand recall of 86 % on the answered trials (90.0 % on the 10 with a hand shown), 72 % in forced choice and 70.0 % [57.5, 82.5] with half credit; the oracle window gives 75.0 %. All free-choice trials are right-hand so far, so this is right-hand recall: a constant right answer would score 100 %.

Cause (post hoc). A candidate segment is the stored approach in 70 % of the natural recordings (84 % primary), and that candidate's P(approach) is below the threshold in 36 % of them (17 % primary). The natural approaches move less along the screen normal: median 0.03 m against 0.28 m, with 57 % of them below the primary 25th percentile; their median P(approach) is 0.59 against 0.83. The grip check fires on 5 % of the natural and 4 % of the primary recordings, so the grip does not explain the rejections.

Consequence. A first, unpractised movement is rejected as "not an approach" more often than a practised one. In production the page then asks for another touch; whether practised cue-free touches behave like the primary trials needs real cue-free streams.

G.13. Online operation, caveats and next steps

What runs on the phone. (1) Every standardised 60 Hz sample updates the causal activity. (2) Before the first touch, the rest-pose grip check runs. (3) Bouts are cut by hysteresis; a bout is a movement after 100 ms above 30 deg/s. (4) An approach ends at the first stillness at the face and is confirmed 0.5 s later; its window is cut by the same code as the training windows. (5) The segment classifier drops returns and other movements; every accepted approach adds its calibrated logit. (6) The stopping rule of Appendix G.9 decides when to answer. The same segmentation code runs online and offline; the replay of Appendix G.5 checks it on re-standardised raw records. The page must also check the instructed 1 s of stillness before it accepts a touch and say "hold still" otherwise.

Caveats.

- The production numbers use yaw8, lane m4's reference hand classifier, not the final model.
- The recordings come from a protocol with a GO cue and an enforced stillness; the stress test and the synthetic sessions approximate production but are spliced from those recordings, and the session-fitted scale is learned from such sessions too.
- Exploratory choices on the outer folds: the detector feature set (5 sets), the hand model, the attempt rule and the aggregation (6 session configurations); the production result records 22 cue-free pipelines. The segmentation setting and the calibrator are nested.
- Mild in-fold optimism, not leakage of the held-out person: the recording-fitted scale and the calibrator are fitted on the same cross-fitted logits, and the session-fitted scale uses the outer calibrator on the same training persons.
- Untested: iOS sample stalls during an approach; a shorter hold at the nose (the stillness after a segment is a feature); real screen-away and landscape grips.
- Session-level intervals are wide (134 sessions of 20 persons); natural use has one right-hand trial per person.

Next steps.

- Run the final model inside the production simulation (the hand model is a plug-in) and report its cue-free and multi-touch numbers.
- Reject the arm settling before an approach and returns, for example with a sequence model over the bouts (rest, approach, hold, return); the true-approach curve of Figure 94 is its ceiling.
- Add the return head to the stopping rule, with its own cross-fitted calibration.
- Augment the training streams with short stillness and screen-away grips; choose the P(approach) threshold inside the folds.
- Record a rehearsal session of real cue-free streams, with several touches per person, and re-test the natural-use rejections on it.

H. Secondary analyses in full

This appendix reports the secondary tasks of the protocol: the instructed speed (fast, natural, slow), the instructed grip (upright, top left, top right) and the dominant hand. It ends with a test of whether demographics would help the hand prediction. Speed, grip and dominance are

Table 95: Decisions of the secondary analyses: the options considered, the choice and where its evidence is.

Decision	Options	Choice	Evidence
Speed windows	the app's stored (oracle) windows; the app's detector re-run with one parameter set (label-blind); a fixed window from the onset	label-blind; oracle numbers kept as a label-informed, optimistic reference	H.2, Fig. 97
Common detector setting	the defaults with the longest movement cap; the slow trials' stricter end criterion	the defaults with the longest cap	H.2, Tab. 98
Manipulation check	all trials; standard-variant trials; with or without the session position	standard variant, both windows, position-adjusted as sensitivity	H.3, Tab. 99
Speed headline	LR on the duration; LR on 16 mirror-invariant kinematic features; LightGBM on every channel	the pre-specified LR on the kinematic features	H.4, Tab. 96
Speed metrics	accuracy; balanced accuracy; macro-F1	balanced accuracy (primary) and macro-F1, with per-class recall, precision and F1	H.1
Speed training trials	every primary trial; standard variant only	every primary trial	H.4
Speed vs the protocol	ignore; grip-only reference and grip strata	grip-only reference and grip strata	H.5, Tab. 103
Grip model	up vector before the onset; 12 orientation features; LightGBM on every channel	the pre-specified 12-feature LR (identical to the up vector alone)	H.8, Tab. 108
Grip labels	correct the probable errors; keep as instructed	keep as instructed; explain the errors	H.8, Fig. 102
Dominance inputs	every channel; mirror-invariant features only	mirror-invariant only (cannot encode which hand moved)	H.9
Dominance reference	AUC 0.5; within-person label-permutation null	the permutation null	H.9, Tab. 113
Person-level handedness	a model; individual values	individual values (2 left-handed persons)	H.9
Demographics	add to the hand model; leave out	leave out	H.10, Fig. 104
Calibration of these models	assess; not assess	not assessed (none feeds the displayed P(right))	H.1

targets here. They never enter the hand model as inputs, and no model of this appendix feeds the displayed P(right). Table 95 lists every decision with its options, its choice and its evidence; the subsections give the detail. The main-text summary is Section 7. Abbreviations used in this appendix: leave-one-person-out (LOSO), logistic regression (LR), gradient-boosted trees (LightGBM), area under the ROC curve (AUC), confidence interval (CI), standard deviation (SD), interquartile range (IQR), ordinary least squares (OLS), spectral arc length (SPARC).

H.1. Targets, design and evaluation

Targets. The 598 primary trials of the 20 persons carry an instructed speed (fast 134, natural 421, slow 43) and an instructed grip (upright 366, top left 118, top right 114). Of these, 470 are of the standard movement variant. The other variants (pause, jerky) were made at natural speed only; 0 fast or slow trials belong to them. Slow trials are few: a median of 2 per person, at most 8, and 1 person has none.

Evaluation. Every model was evaluated LOSO on the same person splits as the hand models, with the same cap on trials per person [7]. Hyperparameters were tuned inside each outer fold on inner folds of the training persons, on the class-balanced log loss, with class-balanced weights. Each model is pre-specified: one model family whose penalty or tree parameters are tuned by nesting, so no configuration was selected on the outer folds unless stated. Brackets are 95 % CIs from 2000 person resamples.

Metrics. The primary metric is the balanced accuracy (chance 33.3 % for three classes), reported with the macro-F1 and with the recall, precision and F1 of every class. Plain accuracy would reward answering natural, the class of 421 of the 598 trials. The price of the class-balanced weights is a low precision for the rare slow class (Appendix H.4). This is a metric choice, not a tested remedy.

Paired comparisons. Two models are compared on the trials both scored (Tables 96 and 97). The pooled balanced-accuracy difference is tested with an exact sign-flip test on the per-person contributions, the pooled macro-F1 difference with a person-swap test, and the mean per-person log-loss and accuracy differences with a sign-flip test over persons. Each metric forms its own Holm family over the 10 comparisons of the two tables. A pair in which no trial changes class and the log-loss difference is below 0.001 is called identical and is not tested.

No calibration claims. None of these models feeds the displayed percentage, which is the $P(\text{right})$ of the hand model. Their log losses are reported with the other metrics; their calibration was not assessed.

Table 96: Paired comparisons on the trials both models scored (a model of a subset of trials on that subset): the pooled balanced-accuracy difference (the primary metric; exact sign-flip test on the per-person contributions) and the pooled macro-F1 difference (person-swap test), A - B, each with a person-bootstrap 95% CI, Holm-adjusted within each metric over the comparisons listed. Identical: no trial changes class and the log-loss difference is below the solver precision.

A	B	trials	bal. acc. A - B	Holm p	macro-F1 A - B	Holm p (F1)	verdict
sec_speed_lr_common	sec_speed_dur_common	598	+20.3% [+12.7%, +27.7%]	< 0.001	+0.103 [+0.033, +0.177]	0.083	
sec_speed_gbm_common	sec_speed_lr_common	598	−4.7% [−9.9%, +0.6%]	1.000	+0.062 [+0.009, +0.117]	0.169	
sec_speed_lr_common	sec_speed_grip	598	+2.0% [−7.6%, +10.8%]	1.000	+0.160 [+0.108, +0.211]	< 0.001	
sec_speed_lr_std	sec_speed_lr_common	470	−0.7% [−2.8%, +1.1%]	1.000	+0.003 [−0.018, +0.024]	1.000	
sec_speed_lr_common	sec_speed_lr	598	−17.8% [−22.7%, −12.5%]	< 0.001	−0.183 [−0.240, −0.123]	< 0.001	
sec_speed_gbm_common	sec_speed_gbm	598	−26.5% [−30.7%, −22.5%]	< 0.001	−0.253 [−0.295, −0.222]	< 0.001	
sec_speed_ang_common	sec_speed_ang	598	−18.1% [−23.0%, −12.5%]	< 0.001	−0.185 [−0.243, −0.126]	< 0.001	
sec_speed_gbm	sec_speed_lr	598	+4.0% [−0.4%, +9.6%]	0.814	+0.133 [+0.081, +0.189]	< 0.001	
sec_grip_lr	sec_grip_up	598	+0.0% [+0.0%, +0.0%]	1.000	+0.000 [+0.000, +0.000]	1.000	identical
sec_grip_gbm	sec_grip_lr	598	−0.6% [−1.7%, +0.0%]	1.000	−0.006 [−0.018, +0.000]	1.000	

Table 97: The same comparisons on the mean per-person log-loss and accuracy differences (A - B; person-bootstrap 95% CIs; exact sign-flip p over persons, Holm within each metric); changed: trials whose predicted class differs.

A	B	log loss A - B	Holm p (log loss)	accuracy A - B	Holm p (accuracy)	changed
sec_speed_lr_common	sec_speed_dur_common	−0.104 [−0.144, −0.066]	< 0.001	+5.6% [−6.3%, +16.1%]	1.000	432
sec_speed_gbm_common	sec_speed_lr_common	−0.198 [−0.262, −0.135]	< 0.001	+20.3% [+14.6%, +26.4%]	< 0.001	266
sec_speed_lr_common	sec_speed_grip	−0.070 [−0.126, −0.016]	0.093	+21.3% [+15.5%, +27.1%]	< 0.001	364
sec_speed_lr_std	sec_speed_lr_common	+0.014 [−0.009, +0.039]	0.842	+1.2% [−1.9%, +4.5%]	1.000	56
sec_speed_lr_common	sec_speed_lr	+0.242 [+0.176, +0.301]	< 0.001	−18.0% [−24.8%, −11.8%]	< 0.001	178
sec_speed_gbm_common	sec_speed_gbm	+0.224 [+0.137, +0.317]	< 0.001	−8.5% [−11.1%, −5.9%]	< 0.001	118
sec_speed_ang_common	sec_speed_ang	+0.252 [+0.177, +0.315]	< 0.001	−18.7% [−25.9%, −11.8%]	< 0.001	222
sec_speed_gbm	sec_speed_lr	−0.180 [−0.227, −0.137]	< 0.001	+10.8% [+7.6%, +14.2%]	< 0.001	168
sec_grip_lr	sec_grip_up	+0.000 [−0.012, +0.012]	1.000	+0.0% [+0.0%, +0.0%]	1.000	0
sec_grip_gbm	sec_grip_lr	+0.017 [−0.039, +0.089]	1.000	−0.3% [−0.9%, +0.0%]	1.000	2

H.2. Speed: label-blind windows

The problem. The app resolved its online segmentation parameters per instructed condition (Appendix A.6). By default an approach ends after 250 ms below 30 deg/s and 0.6 m/s^2, with a movement cap of 6000 ms. A slow trial ends only after 450 ms below 10 deg/s and 0.25 m/s^2, with a cap of 12000 ms; pause and jerky trials also wait longer for their end. The stored (oracle) window of a slow trial is therefore chosen by its label. Its duration, and every feature computed over it, carries a criterion that a sensor-only system does not have.

Options. (i) The oracle windows. (ii) The app's own detector re-run on every stored stream with one parameter set for every trial: the label-blind windows. (iii) A fixed window from the label-blind onset; it was not run in this version. For (ii), two common settings were considered: the defaults with the longest movement cap of any condition (12000 ms), or the slow trials' stricter end criterion for every trial.

Evidence. With the stored parameters, the re-run detector reproduces the stored bounds of 598 of 598 trials exactly. With the common default setting it keeps the bounds of 98.3 % of the standard natural trials and 94.0 % of the standard fast ones, but of 0 of the 43 slow trials (Table 98; Figure 97a). The median slow approach shrinks from 5.52 s to 1.63 s, as short as a natural one (1.60 s). It now ends after a median 54 % of its rotation (IQR 28 % to 89 %). The

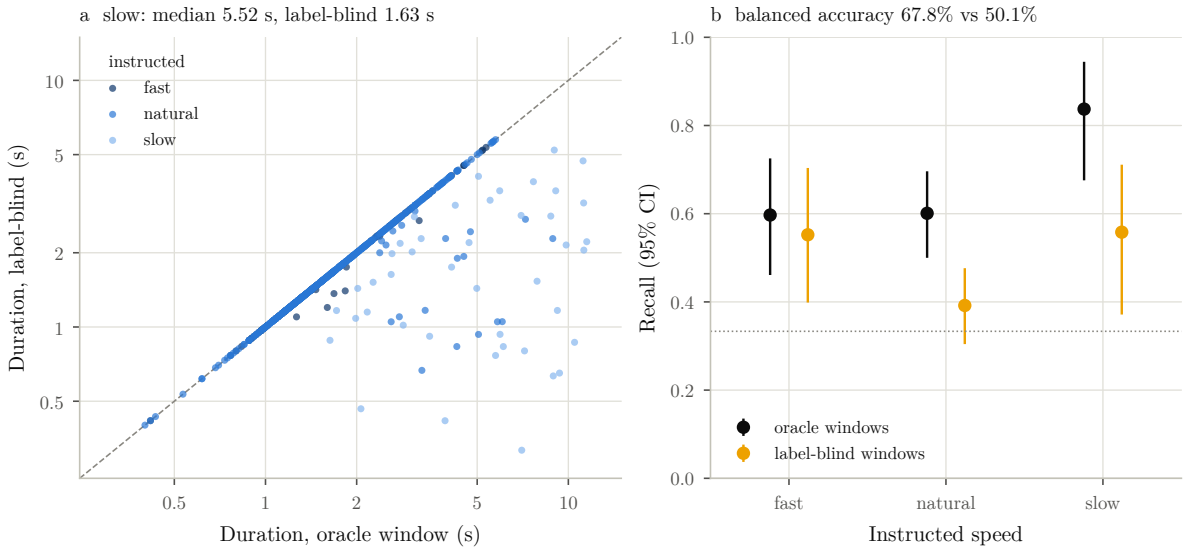


Figure 97: The approach window depended on the instructed speed. (a) Per trial, the approach duration of the app's stored (oracle) window against that of the app's detector re-run with one parameter set for every trial (label-blind; dashed: equal). The median slow approach shrinks from 5.52 s to 1.63 s; fast and natural trials barely change. (b) Recall per instructed speed of the headline LR fitted on each kind of window (LOSO; dotted: chance). The oracle-window speed results are optimistic, because the segmentation used the label.

stricter criterion is no usable common choice: applied to every trial, it ends the approach before the arm starts back in only 14.2 % of the standard fast and 17.7 % of the standard natural trials. The common default setting ends all 598 primary trials and 20 of the 20 free-choice trials. Every trial was re-standardised on the label-blind bounds through the cache's own code path. On the oracle bounds this path reproduces the cached arrays of all 598 trials (largest difference 0.0e+00).

Consequence. The same pre-specified LR falls from a balanced accuracy of 67.8 % [60.7, 73.7] on the oracle windows to 50.1 % [42.9, 56.4] on the label-blind windows (paired difference −17.8 [−22.7, −12.5] points, $p < 0.001$, Holm over the balanced-accuracy family). Its slow recall falls from 83.7 % to 55.8 % (Figure 97b). The same paired difference (label-blind minus oracle) is −26.5 [−30.7, −22.5] points for LightGBM and −18.1 [−23.0, −12.5] points for an LR on the 8 duration and angular features (both $p < 0.001$, Holm).

Decision. The label-blind windows carry every sensor-only speed result, and the label-blind LR is the speed headline. The oracle-window numbers are kept as a label-informed, optimistic reference. They are not an upper bound: the label-blind numbers belong to the app's default end rule, which truncates a slow reach at its first slow phase, and a detector built for slow movements could recover more. No segmentation parameter depends on the hand, so the hand models are unaffected.

Table 98: The approach window depends on the instructed speed. The app resolved its online segmentation parameters per condition (a slow trial ends only after a stricter and longer stillness, with a longer movement cap; a few trials also differ in their base parameters). Re-running the app's own detector on every stored stream with ONE parameter set (the defaults, with the longest movement cap of any condition) leaves fast and natural trials almost unchanged and ends a slow movement at its first slow phase. Durations: medians over trials; ratio: median per-trial ratio of the label-blind to the oracle duration. Strict end before return: the share of trials the slow trials' stricter end criterion, applied to every trial, ends before the arm starts back (why it is not the common set).

speed	trials	same bounds	duration, oracle (s)	duration, label-blind (s)	median ratio	strict end before return
fast	134	126/134	1.42	1.40	1.00	14.2%
natural	421	401/421	1.63	1.60	1.00	18.8%
slow	43	0/43	5.52	1.63	0.41	100.0%

H.3. Speed: the manipulation check

Method. The check asks whether the persons moved differently when told to. Per person, the median approach duration and peak angular speed were taken per instructed speed, on the 470 standard-variant trials (the variants occur at natural speed only). Across persons the analysis gives the median per-person ratio of the expected-higher to the expected-lower speed (1 = no difference) with its CI, the number of persons in the instructed order, and a Wilcoxon signed-rank test, Holm-adjusted over the four contrasts of each set of trials (Table 99; Figure 98).

Fast vs natural. The fast instruction raised the peak angular speed in 18 of 20 persons (ratio 1.28 [1.12, 1.54], $p < 0.001$, Holm). It shortened the reach in only 14 of 20 persons (natural/fast duration ratio 1.12 [1.00, 1.23], Holm p 0.195). The persons followed the fast instruction at the peak; its effect on the duration of the whole reach is weak and uncertain.

Order and practice. The protocol ran the fast trials before the natural ones for 20 of 20 persons, and natural reaches shortened over the session (median within-person Spearman ρ of the log duration with the position −0.24, negative for 14 of 19 persons, Wilcoxon p 0.005). The unadjusted contrast therefore mixes instruction and practice. Two sensitivity analyses form their own Holm family with the adjusted peak contrast. A per-person OLS of the log duration on speed and position gives natural longer than fast for 16 of 20 persons (ratio 1.10 [1.04, 1.20], Holm p 0.027). Keeping only the natural trials made before the person's last fast trial gives 13 of 20 persons (ratio 1.14 [0.96, 1.27], Holm p 0.083). The direction is the instructed one in each analysis; the size stays uncertain. Adjusted for position, the peak angular speed is higher for fast in 19 of 20 persons ($p < 0.001$, Holm).

Slow vs natural. On the oracle windows, slow reaches are longer than natural ones for 19 of 19 persons (ratio 2.67 [2.08, 3.27], $p < 0.001$, Holm). On the label-blind windows only 9 of 19 are (ratio 0.98 [0.74, 1.21], Holm p 0.709): most of the oracle contrast is the stricter end criterion (Appendix H.2). At the peak, natural reaches are faster than slow ones for 13 of 19 persons on the oracle windows (Holm p 0.490) and for 14 of 19 on the label-blind windows (Holm p 0.292).

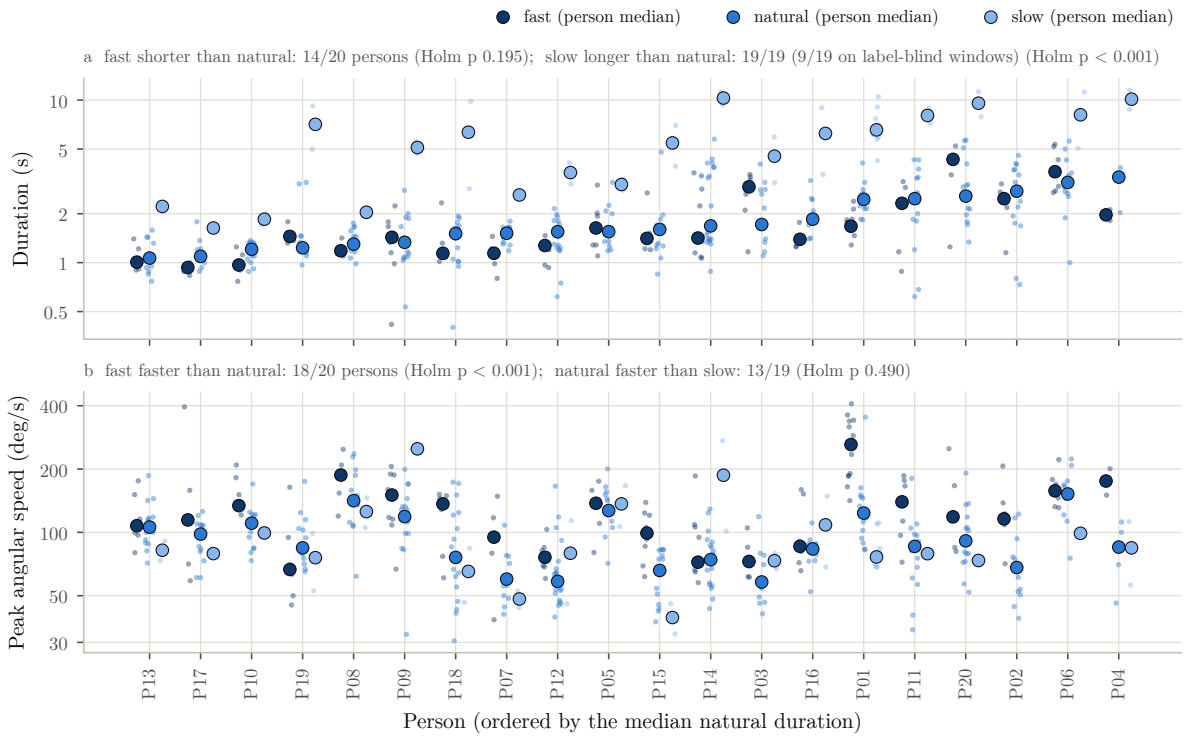


Figure 98: Manipulation check of the speed instruction on the oracle windows: every standard-variant trial (small dots) and the per-person median (large dots) of (a) the approach duration and (b) the peak angular speed per instructed speed, for each of the 20 persons. Panel titles: persons whose medians follow the instruction (Wilcoxon over persons, Holm). Fast reaches were faster at the peak for 18 of 20 persons; their shortening is uncertain and confounded with the session order. The slow durations are partly the app's stricter end criterion.

Decision. The check uses the standard-variant trials, both kinds of window, and the position-adjusted contrasts as a sensitivity analysis. Over every primary trial, variants included, fast reaches are shorter than natural ones for only 12 of 20 persons (Holm p 0.307).

Table 99: Manipulation check of the speed instruction: per person the median of each measure per instructed speed; the median per-person ratio of the higher-expected to the lower-expected speed (1 = no difference), the persons whose medians follow the instruction, a Wilcoxon signed-rank test over persons, Holm-adjusted within each set of trials (the position-adjusted fast vs natural contrasts form their own family). Median ratio with its person-bootstrap 95% CI. Oracle windows: the app's stored segment, whose parameters depend on the instructed speed; label-blind windows: the app's detector re-run with one parameter set for every trial. Position-adjusted: per person an OLS of the log measure on speed and the relative position in the session, or only the natural trials made before the person's last fast trial.

trials	measure	contrast	persons	as instructed	median ratio	Wilcoxon p	Holm p
standard variant	duration	natural vs fast	20	14	1.12 [1.00, 1.23]	0.097	0.195
standard variant	duration	slow vs natural	19	19	2.67 [2.08, 3.27]	< 0.001	< 0.001
standard variant	peak angular speed	natural vs slow	19	13	1.11 [0.93, 1.24]	0.490	0.490
standard variant	peak angular speed	fast vs natural	20	18	1.28 [1.12, 1.54]	< 0.001	< 0.001
standard variant, position (OLS)	duration	natural vs fast	20	16	1.10 [1.04, 1.20]	0.014	0.027
standard variant, position (OLS)	peak angular speed	fast vs natural	20	19	1.29 [1.19, 1.44]	< 0.001	< 0.001
standard variant, natural trials up to the person's last fast trial	duration	natural vs fast	20	13	1.14 [0.96, 1.27]	0.083	0.083
all primary	duration	natural vs fast	20	12	1.16 [0.94, 1.26]	0.154	0.307
all primary	duration	slow vs natural	19	19	2.67 [2.05, 3.51]	< 0.001	< 0.001
all primary	peak angular speed	natural vs slow	19	13	1.13 [0.92, 1.31]	0.275	0.307
all primary	peak angular speed	fast vs natural	20	16	1.28 [1.10, 1.47]	< 0.001	< 0.001
all primary, position (OLS)	duration	natural vs fast	20	16	1.14 [1.03, 1.32]	0.006	0.013
all primary, position (OLS)	peak angular speed	fast vs natural	20	19	1.26 [1.16, 1.40]	< 0.001	< 0.001
all primary, natural trials up to the person's last fast trial	duration	natural vs fast	20	13	1.17 [0.97, 1.26]	0.076	0.076
standard variant, label-independent windows	duration	natural vs fast	20	14	1.12 [1.00, 1.23]	0.097	0.292
standard variant, label-independent windows	duration	slow vs natural	19	9	0.98 [0.74, 1.21]	0.709	0.709
standard variant, label-independent windows	peak angular speed	natural vs slow	19	14	1.13 [1.01, 1.29]	0.123	0.292
standard variant, label-independent windows	peak angular speed	fast vs natural	20	18	1.28 [1.12, 1.54]	< 0.001	< 0.001
standard variant, label-independent windows, position (OLS)	duration	natural vs fast	20	16	1.10 [1.04, 1.20]	0.009	0.019
standard variant, label-independent windows, position (OLS)	peak angular speed	fast vs natural	20	19	1.29 [1.19, 1.44]	< 0.001	< 0.001
standard variant, label-independent windows, natural trials up to the person's last fast trial	duration	natural vs fast	20	13	1.14 [0.96, 1.27]	0.053	0.053

H.4. Speed: models and paired comparisons

Options. Four models were fitted on each kind of window (Table 100): a multinomial L2 LR on the log duration alone (the baseline); the same LR on the 16 mirror-invariant kinematic features of Table 107 (the pre-specified headline); LightGBM on summary statistics of every channel; and an LR on the up vector before the onset (the grip, a nuisance reference; Appendix H.5). Every feature is a function of one trial; scalers and models were fitted inside the folds [25].

Headline. On the label-blind windows the headline LR reaches a balanced accuracy of 50.1 % [42.9, 56.4] and a macro-F1 of 0.391 [0.340, 0.441] (chance 33.3 %). Recall, precision and F1 are 55.2 %, 36.6 % and 0.440 for fast; 39.2 %, 77.8 % and 0.521 for natural; and 55.8 %, 13.0 % and 0.211 for slow (Table 101). The class-balanced weights make it call slow 184 times for 43 slow trials. The largest confusion is between natural and slow: 32.3 % of the natural trials are called slow and 25.6 % of the slow trials natural (Table 102; Figure 15).

Kinematics vs duration. The duration alone reaches 29.8 % [24.2, 34.9]. The kinematic features add +20.3 [+12.7, +27.7] points of balanced accuracy ($p < 0.001$, Holm) and +0.103 [+0.033, +0.177] of macro-F1 (Holm p 0.083). Decision: the kinematic features, not the duration alone.

LR vs LightGBM. On the label-blind windows LightGBM reaches 45.4 % [39.4, 50.6] (macro-F1 0.453 [0.390, 0.501]). Its difference from the headline is −4.7 [−9.9, +0.6] points in balanced accuracy (Holm p 1.000) and +0.062 [+0.009, +0.117] in macro-F1 (Holm p 0.169); neither is significant. Its log-loss difference is −0.198 [−0.262, −0.135] and its accuracy difference +20.3 [+14.6, +26.4] points (both $p < 0.001$, Holm; Table 97). Plain accuracy weights the

classes by their size, and LightGBM recalls the largest class far more often (natural recall 74.3 % against 39.2 %) and slow far less often (14.0 % against 55.8 %). On the oracle windows LightGBM's macro-F1 difference is +0.133 [+0.081, +0.189] ($p < 0.001$, Holm), while its balanced accuracy differs by +4.0 [−0.4, +9.6] points (Holm p 0.814). Decision: the LR stays the headline because it was pre-specified and is interpretable, not because it scores best.

Training trials. Trained on the standard-variant trials only, the LR reaches 48.9 % [41.5, 55.2] on those trials. Against the all-trials model on the same 470 trials the difference is −0.7 [−2.8, +1.1] points (Holm p 1.000). Decision: train on every primary trial.

Free-choice trials. Each of the 20 free-choice trials was scored by its person's fold model on its label-blind window. The model calls 3 of them natural, 2 fast and 15 slow. This natural-speed recall of 15.0 % concerns free-choice trials only, all made at natural speed, where a constant natural answer would score 100 %; it is not a balanced measure.

Table 100: Instructed speed from sensor data (fast, natural, slow): leave-one-person-out, hyperparameters tuned on inner folds of the training persons, person-bootstrap 95% CIs. Label-blind windows: the app's detector re-run with one parameter set for every trial (the sensor-only estimates); oracle windows: the app's stored segment, whose end criterion depended on the instructed speed (label-informed, optimistic). Chance balanced accuracy is 33.3%.

experiment	windows	model	n	bal. acc.	macro-F1	log loss
sec_speed_dur_common	label-blind	LR, log duration only (1)	598	29.8% [24.2%, 34.9%]	0.288 [0.235, 0.330]	1.100
sec_speed_lr_common	label-blind	LR, mirror-invariant kinematics (16)	598	50.1% [42.9%, 56.4%]	0.391 [0.340, 0.441]	0.995
sec_speed_gbm_common	label-blind	LightGBM, summary statistics of every channel (271)	598	45.4% [39.4%, 50.6%]	0.453 [0.390, 0.501]	0.791
sec_speed_grip	label-blind	LR, up vector before the onset (the grip) (3)	598	48.0% [43.8%, 52.9%]	0.231 [0.202, 0.263]	1.070
sec_speed_lr_std	label-blind	LR, mirror-invariant kinematics (16)	470	48.9% [41.5%, 55.2%]	0.416 [0.366, 0.463]	1.007
sec_speed_dur	oracle	LR, log duration only (1)	598	56.2% [48.0%, 62.4%]	0.389 [0.314, 0.462]	0.924
sec_speed_lr	oracle	LR, mirror-invariant kinematics (16)	598	67.8% [60.7%, 73.7%]	0.574 [0.499, 0.643]	0.755
sec_speed_gbm	oracle	LightGBM, summary statistics of every channel (271)	598	71.9% [65.7%, 77.3%]	0.706 [0.651, 0.754]	0.579
sec_speed_ang	oracle	LR, duration and angular kinematics (8)	598	65.2% [57.9%, 71.2%]	0.546 [0.475, 0.616]	0.798
sec_speed_ang_common	label-blind	LR, duration and angular kinematics (8)	598	47.1% [40.5%, 52.8%]	0.361 [0.309, 0.408]	1.052

Table 101: Per-class recall, precision and F1 of the speed models (leave-one-person-out); n: trials of the class, predicted: trials predicted as it.

experiment	class	n	predicted	recall	precision	F1
sec_speed_dur_common	fast	134	116	23.1%	26.7%	0.248
sec_speed_dur_common	natural	421	244	43.0%	74.2%	0.544
sec_speed_dur_common	slow	43	238	23.3%	4.2%	0.071
sec_speed_lr_common	fast	134	202	55.2%	36.6%	0.440
sec_speed_lr_common	natural	421	212	39.2%	77.8%	0.521
sec_speed_lr_common	slow	43	184	55.8%	13.0%	0.211
sec_speed_gbm_common	fast	134	153	47.8%	41.8%	0.446
sec_speed_gbm_common	natural	421	415	74.3%	75.4%	0.749
sec_speed_gbm_common	slow	43	30	14.0%	20.0%	0.164
sec_speed_grip	fast	134	232	58.2%	33.6%	0.426
sec_speed_grip	natural	421	27	4.5%	70.4%	0.085
sec_speed_grip	slow	43	339	81.4%	10.3%	0.183
sec_speed_lr_std	fast	134	167	56.0%	44.9%	0.498
sec_speed_lr_std	natural	293	161	39.6%	72.0%	0.511
sec_speed_lr_std	slow	43	142	51.2%	15.5%	0.238
sec_speed_dur	fast	134	313	64.2%	27.5%	0.385
sec_speed_dur	natural	421	153	27.6%	75.8%	0.404
sec_speed_dur	slow	43	132	76.7%	25.0%	0.377
sec_speed_lr	fast	134	207	59.7%	38.6%	0.469
sec_speed_lr	natural	421	304	60.1%	83.2%	0.698
sec_speed_lr	slow	43	87	83.7%	41.4%	0.554
sec_speed_gbm	fast	134	148	46.3%	41.9%	0.440
sec_speed_gbm	natural	421	404	78.6%	81.9%	0.802
sec_speed_gbm	slow	43	46	90.7%	84.8%	0.876
sec_speed_ang	fast	134	201	53.7%	35.8%	0.430
sec_speed_ang	natural	421	305	58.2%	80.3%	0.675
sec_speed_ang	slow	43	92	83.7%	39.1%	0.533
sec_speed_ang_common	fast	134	194	50.0%	34.5%	0.409
sec_speed_ang_common	natural	421	194	35.4%	76.8%	0.485
sec_speed_ang_common	slow	43	210	55.8%	11.4%	0.190

Table 102: Confusion of the instructed speed: counts and row shares (rows = instructed speed); label-blind (*_common) and oracle windows.

experiment	instructed	n	predicted fast	predicted natural	predicted slow
sec_speed_dur_common	fast	134	31 (23.1%)	44 (32.8%)	59 (44.0%)
sec_speed_dur_common	natural	421	71 (16.9%)	181 (43.0%)	169 (40.1%)
sec_speed_dur_common	slow	43	14 (32.6%)	19 (44.2%)	10 (23.3%)
sec_speed_lr_common	fast	134	74 (55.2%)	36 (26.9%)	24 (17.9%)
sec_speed_lr_common	natural	421	120 (28.5%)	165 (39.2%)	136 (32.3%)
sec_speed_lr_common	slow	43	8 (18.6%)	11 (25.6%)	24 (55.8%)
sec_speed_gbm_common	fast	134	64 (47.8%)	69 (51.5%)	1 (0.7%)
sec_speed_gbm_common	natural	421	85 (20.2%)	313 (74.3%)	23 (5.5%)
sec_speed_gbm_common	slow	43	4 (9.3%)	33 (76.7%)	6 (14.0%)
sec_speed_dur	fast	134	86 (64.2%)	28 (20.9%)	20 (14.9%)
sec_speed_dur	natural	421	226 (53.7%)	116 (27.6%)	79 (18.8%)
sec_speed_dur	slow	43	1 (2.3%)	9 (20.9%)	33 (76.7%)
sec_speed_lr	fast	134	80 (59.7%)	46 (34.3%)	8 (6.0%)
sec_speed_lr	natural	421	125 (29.7%)	253 (60.1%)	43 (10.2%)
sec_speed_lr	slow	43	2 (4.7%)	5 (11.6%)	36 (83.7%)
sec_speed_gbm	fast	134	62 (46.3%)	70 (52.2%)	2 (1.5%)
sec_speed_gbm	natural	421	85 (20.2%)	331 (78.6%)	5 (1.2%)
sec_speed_gbm	slow	43	1 (2.3%)	3 (7.0%)	39 (90.7%)

H.5. Speed and the protocol: grip and order

The confound. The protocol confounds the instructed speed with the grip and with the position in the session (Table 103). Slow trials were upright in 39 of 43. Among standard-variant trials, 59.0 % of the fast, 50.9 % of the natural and 9.3 % of the slow trials were held sideways. Fast trials were more often sideways than natural ones for 16 of 20 persons. The median relative position in the session (0 = first, 1 = last) was 0.14 for slow, 0.37 for fast and 0.59 for natural trials; slow came before natural for 19 of 19 persons. A model can therefore predict the speed from the grip or from practice instead of from the movement.

Grip-only reference. An LR on the up vector before the onset alone predicts the instructed speed with a balanced accuracy of 48.0 % [43.8, 52.9] (slow recall 81.4 %). The headline does not differ from it significantly in balanced accuracy (+2.0 [−7.6, +10.8] points, Holm p 1.000) but beats it in macro-F1 (+0.160 [+0.108, +0.211], $p < 0.001$, Holm). The mirror-invariant features themselves carry some grip information: they separate sideways from upright trials with a balanced accuracy of 57.4 % [52.8, 62.3] (chance 50.0 %).

Within grip strata. The models were re-scored within the upright and within the sideways trials after prediction (Tables 104 and 105). A speed enters a stratum only with trials of at least 5 persons, so slow is left out of the sideways stratum. Within the upright trials the headline reaches 49.0 % over three classes and LightGBM 43.5 % (difference −5.5 [−13.1, +1.3] points, Holm p 0.389). Within the sideways trials (fast and natural) they reach 47.7 % and 61.1 % (+13.3 [+3.7, +23.7] points, Holm p 0.088). LightGBM, which sees the device-frame channels, predicts fast for 16.1 % of the upright and 40.5 % of the sideways trials, close to the instructed shares (15.0 % and 34.1 %). The headline predicts fast for 30.9 % and 38.4 %.

Decision. The grip-only model is reported as a nuisance reference, and the model comparisons are repeated within grip strata. The speed accuracies may still partly reflect the protocol's grip and order structure.

Table 103: The protocol confounds the instructed speed with the grip and with the position in the session: trials per speed and instructed grip (all variants), the share of standard-variant trials held sideways (top left or top right), and the median relative position of the trials in the person's sequence (0 = first, 1 = last). Descriptive.

instructed speed	trials	upright	top left	top right	sideways share (standard)	median position
fast	134	55	41	38	59.0%	0.37
natural	421	272	75	74	50.9%	0.59
slow	43	39	2	2	9.3%	0.14

Table 104: The speed models within grip strata, after prediction (instructed grip: upright; sideways = top left or top right). A speed enters a stratum only with trials of at least 5 persons (slow trials are almost all upright). Predicted / true fast: the share of the stratum's trials predicted fast and instructed fast.

experiment	grip stratum	speeds	n	bal. acc.	macro-F1	accuracy	predicted fast / true fast
sec_speed_lr_common	upright	fast, natural, slow	366	49.0%	0.378	42.1%	30.9% / 15.0%
sec_speed_lr_common	sideways	fast, natural	228	47.7%	0.341	46.1%	38.4% / 34.1%
sec_speed_gbm_common	upright	fast, natural, slow	366	43.5%	0.435	65.6%	16.1% / 15.0%
sec_speed_gbm_common	sideways	fast, natural	228	61.1%	0.404	62.7%	40.5% / 34.1%
sec_speed_lr	upright	fast, natural, slow	366	67.0%	0.559	61.5%	30.3% / 15.0%
sec_speed_lr	sideways	fast, natural	228	61.0%	0.409	61.4%	41.4% / 34.1%
sec_speed_gbm	upright	fast, natural, slow	366	71.2%	0.704	77.9%	17.2% / 15.0%
sec_speed_gbm	sideways	fast, natural	228	59.9%	0.399	62.7%	36.6% / 34.1%

Table 105: LightGBM vs the logistic regression within grip strata (paired on the same trials; person-bootstrap 95% CIs; Holm within each metric over the stratum comparisons).

A	B	stratum	trials	bal. acc. A - B	Holm p	macro-F1 A - B	Holm p (F1)
sec_speed_gbm_common	sec_speed_lr_common	upright	366	−5.5% [−13.1%, +1.3%]	0.389	+0.057 [−0.025, +0.129]	0.134
sec_speed_gbm_common	sec_speed_lr_common	sideways	228	+13.3% [+3.7%, +23.7%]	0.088	+0.063 [+0.025, +0.303]	0.093
sec_speed_gbm	sec_speed_lr	upright	366	+4.2% [−0.4%, +9.8%]	0.389	+0.145 [+0.087, +0.204]	< 0.001
sec_speed_gbm	sec_speed_lr	sideways	228	−1.1% [−6.5%, +4.3%]	0.737	−0.009 [−0.042, +0.204]	0.594

H.6. Speed: which features carry it

Model-free effects. Per person, the medians of each kinematic feature were compared between two instructed speeds on the standard-variant trials, in units of the feature's SD over all trials (Table 106; Figure 99a). A Wilcoxon test over persons is Holm-adjusted over every feature and both contrasts (16 features, two contrasts). On the label-blind windows, fast differs from natural most in peak linear speed (+0.78 SD, positive for 17 of 20 persons, Holm p 0.003), mean angular speed (+0.62 SD, 19 of 20) and mean linear acceleration (+0.62 SD, 17 of 20). In all, 7 features differ significantly between fast and natural, and 0 between slow and natural. The largest slow effect is in peak linear speed (−0.51 SD, Holm p 0.793). On the oracle windows the counts are 8 and 10, led by the slow trials' duration (+1.73 SD): the oracle slow contrasts are largely the label-dependent end criterion.

Model coefficients. The headline LR's largest coefficients (Figure 99b) are tilt path (−0.56 log-odds per SD) and angular path length (+0.53) for fast vs natural, and angular path length (−0.87) and tilt path (+0.75) for slow vs natural. These coefficients are conditional on the other, correlated features and vary across folds, so the model-free effects are the better guide to what carries the speed. Table 107 defines every feature and its mirror parity; smoothness follows the SPARC and jerk measures of the literature [28, 29].

Table 106: Which kinematic features carry the instructed speed (standard-variant trials, no model): per person the difference of the medians between two instructed speeds in units of the feature's SD over all trials; across persons the median, IQR, the persons with a positive difference and a Wilcoxon test, Holm-adjusted over every feature and contrast.

windows	feature	contrast	median difference (SD)	IQR	persons positive	Holm p
label-blind	duration	fast - natural	−0.23	[−0.49, +0.12]	6/20	1.000
label-blind	duration	slow - natural	−0.04	[−0.68, +0.42]	9/19	1.000
label-blind	peak angular speed	fast - natural	+0.46	[+0.17, +0.85]	18/20	0.003
label-blind	peak angular speed	slow - natural	−0.21	[−0.43, +0.08]	5/19	1.000
label-blind	mean angular speed	fast - natural	+0.62	[+0.25, +0.86]	19/20	0.003
label-blind	mean angular speed	slow - natural	−0.20	[−0.52, +0.03]	6/19	1.000
label-blind	time of the angular-speed peak	fast - natural	−0.08	[−0.42, +0.14]	8/20	1.000
label-blind	time of the angular-speed peak	slow - natural	−0.06	[−0.38, +0.63]	8/19	1.000
label-blind	angular path length	fast - natural	+0.17	[+0.03, +0.44]	16/20	0.018
label-blind	angular path length	slow - natural	−0.12	[−0.39, +0.16]	7/19	1.000
label-blind	SPARC	fast - natural	+0.13	[−0.08, +0.33]	14/20	1.000
label-blind	SPARC	slow - natural	−0.40	[−0.74, −0.11]	4/19	0.843
label-blind	log dimensionless jerk	fast - natural	+0.28	[−0.17, +0.50]	14/20	1.000
label-blind	log dimensionless jerk	slow - natural	−0.22	[−0.70, +0.30]	8/19	1.000
label-blind	angular-speed peaks	fast - natural	−0.16	[−0.32, +0.04]	5/20	1.000
label-blind	angular-speed peaks	slow - natural	+0.48	[+0.00, +0.96]	13/19	1.000
label-blind	net tilt	fast - natural	−0.11	[−0.20, +0.07]	8/20	1.000
label-blind	net tilt	slow - natural	−0.09	[−0.30, +0.20]	6/19	1.000
label-blind	absolute net yaw	fast - natural	−0.02	[−0.17, +0.24]	10/20	1.000
label-blind	absolute net yaw	slow - natural	−0.13	[−0.38, +0.01]	5/19	0.744
label-blind	tilt path	fast - natural	+0.08	[−0.02, +0.39]	13/20	0.367
label-blind	tilt path	slow - natural	−0.06	[−0.30, +0.33]	7/19	1.000
label-blind	yaw path	fast - natural	+0.30	[+0.06, +0.64]	17/20	0.016
label-blind	yaw path	slow - natural	−0.19	[−0.63, +0.08]	7/19	1.000
label-blind	peak linear acceleration	fast - natural	+0.48	[+0.14, +0.86]	19/20	< 0.001
label-blind	peak linear acceleration	slow - natural	−0.12	[−0.37, +0.29]	8/19	1.000
label-blind	mean linear acceleration	fast - natural	+0.62	[+0.23, +1.24]	17/20	0.003
label-blind	mean linear acceleration	slow - natural	−0.11	[−0.35, +0.15]	8/19	1.000
label-blind	linear path length	fast - natural	+0.10	[−0.01, +0.32]	14/20	0.140
label-blind	linear path length	slow - natural	−0.13	[−0.61, +0.26]	6/19	1.000
label-blind	peak linear speed	fast - natural	+0.78	[+0.33, +1.22]	17/20	0.003
label-blind	peak linear speed	slow - natural	−0.51	[−0.87, −0.01]	5/19	0.793
oracle	duration	fast - natural	−0.20	[−0.42, +0.10]	6/20	0.973
oracle	duration	slow - natural	+1.73	[+1.23, +2.15]	19/19	< 0.001
oracle	peak angular speed	fast - natural	+0.46	[+0.17, +0.84]	18/20	0.002
oracle	peak angular speed	slow - natural	−0.20	[−0.33, +0.23]	6/19	1.000
oracle	mean angular speed	fast - natural	+0.62	[+0.25, +0.86]	19/20	0.002
oracle	mean angular speed	slow - natural	−0.79	[−1.11, −0.67]	2/19	< 0.001
oracle	time of the angular-speed peak	fast - natural	−0.07	[−0.43, +0.20]	8/20	1.000
oracle	time of the angular-speed peak	slow - natural	−0.60	[−0.76, −0.00]	5/19	0.100
oracle	angular path length	fast - natural	+0.15	[+0.03, +0.39]	16/20	0.009
oracle	angular path length	slow - natural	+0.81	[+0.07, +1.47]	17/19	0.002
oracle	SPARC	fast - natural	+0.10	[−0.06, +0.25]	14/20	1.000
oracle	SPARC	slow - natural	−0.98	[−2.45, −0.47]	0/19	< 0.001
oracle	log dimensionless jerk	fast - natural	+0.25	[−0.15, +0.44]	14/20	0.701
oracle	log dimensionless jerk	slow - natural	−1.48	[−2.02, −1.28]	0/19	< 0.001
oracle	angular-speed peaks	fast - natural	−0.12	[−0.24, +0.03]	5/20	0.991
oracle	angular-speed peaks	slow - natural	+1.41	[+0.70, +2.81]	18/19	0.003
oracle	net tilt	fast - natural	−0.11	[−0.20, +0.07]	8/20	0.991
oracle	net tilt	slow - natural	−0.02	[−0.16, +0.04]	8/19	1.000
oracle	absolute net yaw	fast - natural	−0.06	[−0.17, +0.24]	9/20	1.000
oracle	absolute net yaw	slow - natural	−0.18	[−0.42, −0.03]	4/19	0.100
oracle	tilt path	fast - natural	+0.07	[−0.02, +0.35]	13/20	0.184
oracle	tilt path	slow - natural	+0.71	[+0.08, +1.33]	16/19	0.005
oracle	yaw path	fast - natural	+0.27	[+0.06, +0.57]	17/20	0.008
oracle	yaw path	slow - natural	+0.77	[+0.11, +1.52]	15/19	0.008
oracle	peak linear acceleration	fast - natural	+0.48	[+0.14, +0.87]	19/20	< 0.001
oracle	peak linear acceleration	slow - natural	−0.14	[−0.41, +0.36]	8/19	1.000
oracle	mean linear acceleration	fast - natural	+0.65	[+0.24, +1.29]	17/20	0.002
oracle	mean linear acceleration	slow - natural	−0.49	[−0.73, −0.26]	1/19	< 0.001
oracle	linear path length	fast - natural	+0.05	[+0.00, +0.18]	15/20	0.047
oracle	linear path length	slow - natural	+1.13	[+0.11, +2.45]	16/19	0.005
oracle	peak linear speed	fast - natural	+0.80	[+0.33, +1.24]	17/20	0.002
oracle	peak linear speed	slow - natural	−0.22	[−0.52, +0.09]	7/19	1.000

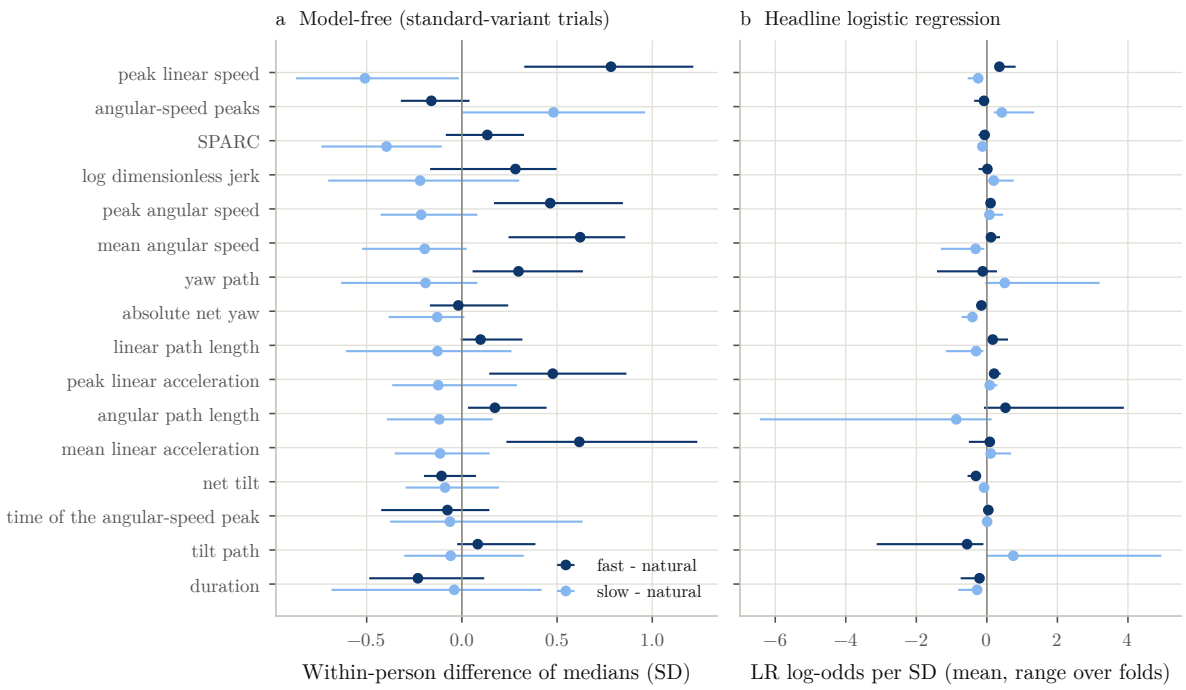


Figure 99: Which kinematic features carry the instructed speed, on label-blind windows. (a) Per person, the difference of the medians between two instructed speeds in units of the feature's SD (median and interquartile range over persons; standard-variant trials). (b) The log-odds contrasts of the headline LR per SD of each feature (mean and range over the outer folds; conditional on the other features). Fast differs from natural most in peak linear speed and mean angular speed; no slow effect survives Holm.

Table 107: The interpretable features of lane sec. Every feature of the even set is identical for a trial and its left/right mirror image (unit-tested); the grip set is device-frame and changes under the mirror.

set	feature	unit	source	definition
even	duration_log	log s	oracle bounds	natural log of the approach duration
even	ang_speed_peak	deg/s	60 Hz stream	peak angular speed |w|
even	ang_speed_mean	deg/s	60 Hz stream	mean angular speed |w|
even	ang_peak_time	fraction	60 Hz stream	time of the |w| peak as a fraction of the approach
even	ang_path	deg	60 Hz stream	angular path length: integral of |w| (total rotation)
even	ang_sparc	1	60 Hz stream	SPARC smoothness of |w|(t) (spectral arc length; closer to 0 = smoother)
even	ang_ldlj	1	60 Hz stream	log dimensionless jerk of |w|(t) (closer to 0 = smoother)
even	ang_n_peaks	count	60 Hz stream	local maxima of |w| with prominence >= 10 % of its peak
even	tilt_net	deg	approach	angle between the up vector at the onset and at the end
even	yaw_net_abs	deg	approach	absolute net rotation about gravity |yaw_cum(end)| (the sign would carry the hand)
even	tilt_path	deg	approach	integral of the tilting rate |w - (w.u)u|
even	yaw_path	deg	approach	integral of the absolute yaw rate |w.u|
even	acc_peak	m/s^2	approach	peak linear acceleration |a|
even	acc_mean	m/s^2	approach	mean linear acceleration |a|
even	lin_path	m	approach	linear path length of the phone: integral of |v|, v by strapdown integration with zero velocity at both bounds
even	lin_speed_peak	m/s	approach	peak linear speed |v| (same integration)
grip	up_before_x	1	fixed window	up vector, device x, mean over the 0.5 s hold before the onset
grip	up_before_y	1	fixed window	up vector, device y, mean over the 0.5 s hold before the onset
grip	up_before_z	1	fixed window	up vector, device z, mean over the 0.5 s hold before the onset
grip	up_onset_x	1	approach	up vector, device x, mean of the first 3 approach samples
grip	up_onset_y	1	approach	up vector, device y, mean of the first 3 approach samples
grip	up_onset_z	1	approach	up vector, device z, mean of the first 3 approach samples
grip	up_end_x	1	approach	up vector, device x, mean of the last 3 approach samples
grip	up_end_y	1	approach	up vector, device y, mean of the last 3 approach samples
grip	up_end_z	1	approach	up vector, device z, mean of the last 3 approach samples
grip	gyro_mean_x	deg/s	approach	mean angular velocity about device x over the approach
grip	gyro_mean_y	deg/s	approach	mean angular velocity about device y over the approach
grip	gyro_mean_z	deg/s	approach	mean angular velocity about device z over the approach

H.7. Speed: who follows the instruction

Figure 100 is a description, not evidence. Per person it plots the natural/fast duration ratio on the label-blind windows against the headline model's two-class fast-vs-natural balanced accuracy on the same standard trials (the class decided between fast and natural only; chance 0.5). The two are correlated (Spearman ρ +0.60, p 0.005, 20 persons, one exploratory test). The mean two-class balanced accuracy is 71.4 % for the 14 persons whose fast reaches were shorter and 52.2 % for the other 6. The analysis is partly circular, because the duration it stratifies by is an input of the model it scores, and it was added after seeing the results.

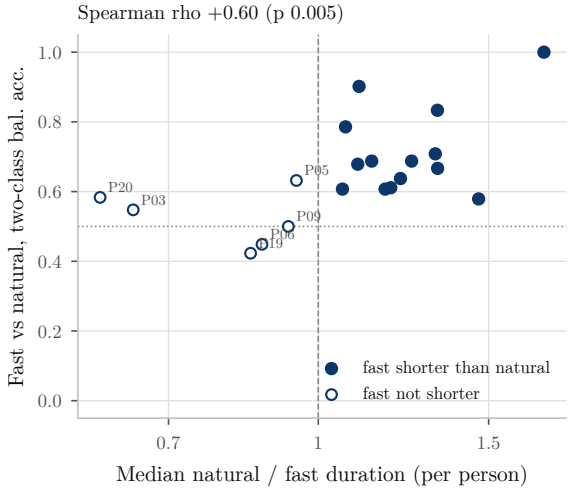


Figure 100: Description, not evidence: per person, the median natural/fast duration ratio (standard-variant trials, label-blind windows; right of the dashed line: fast reaches shorter) against the headline model's two-class fast-vs-natural balanced accuracy on the same trials (dotted: chance). Open circles: the persons whose fast reaches were not shorter. Partly circular, because the duration is an input of the model.

H.8. Grip: the up vector and the errors

Physics. The grip is the orientation of the phone in the hand, so it shows in where gravity points in the device frame during the still hold before the onset (Figure 102). The median unit up vector $\hat{\mathbf{g}}$ (device x, y, z) is (+0.00, +0.98, +0.18) for upright, with the top edge up; (+0.98, −0.01, +0.19) for top left, with the right edge up; and (−0.98, −0.00, +0.17) for top right, with the left edge up. The device-frame channels carry the grip directly. The gravity-frame channels do not encode the orientation, but the grip changes how the arm moves (Appendix H.5).

Models. Three nested models were fitted on the hold before the onset (Tables 108 and 109; Figure 101): an LR on the up vector alone, the pre-specified headline LR on the 12 device-frame orientation features, and LightGBM on the summary statistics of every channel. The headline reaches a balanced accuracy of 95.2 % [89.7, 100.0] against the instruction (chance 33.3 %), with recalls of 99.5 % for upright, 94.1 % for top left and 92.1 % for top right. The up-vector model makes the same predictions: 0 trials change class and the log-loss difference is +0.000, so the two models are identical and are not tested. LightGBM reaches 94.6 % [88.3, 100.0]; its difference from the headline is −0.6 [−1.7, +0.0] points (Holm p 1.000; Table 96). Decision: the headline stays the pre-specified 12-feature LR; the up vector before the onset carries all of its information.

Errors. The headline misclassifies 18 of 598 trials, all from 3 persons (Tables 110 and 111): P05 (top left held as top right in 3 of 5, top right held as top left in 3 of 6), P09 (top left held as top right in 3 of 8, top left held as upright in 1 of 8, top right held as upright in 1 of 5, upright held as top left in 2 of 21) and P16 (top right held as top left in 5 of 5). Of the errors, 14 are swaps between top left and top right. In 18 of the 18 errors the up vector lies closer to the median of the predicted grip than to that of the instructed one (ringed in Figure 102). A sign error is unlikely: the gravity sign was inferred from the instructed orientation for 0 trials, and 54 of the 56 upright trials of these persons are recognised. These trials were most likely held in the predicted grip.

Consistency checks. On the 575 trials that neither the app nor the gravity rule flagged, the headline's balanced accuracy is 100.0 %. Its prediction equals the app's own start grip for 589 of 589 trials (Table 112). Both references read the same up vector as the model, so they agree by construction: they are consistency checks, not an independent confirmation. No video or instruction log records the grip actually held.

Decision. The grip labels are kept as instructed; the errors are explained by physics, not corrected. Any result stratified by instructed grip, in every lane, mis-assigns these 18 trials.

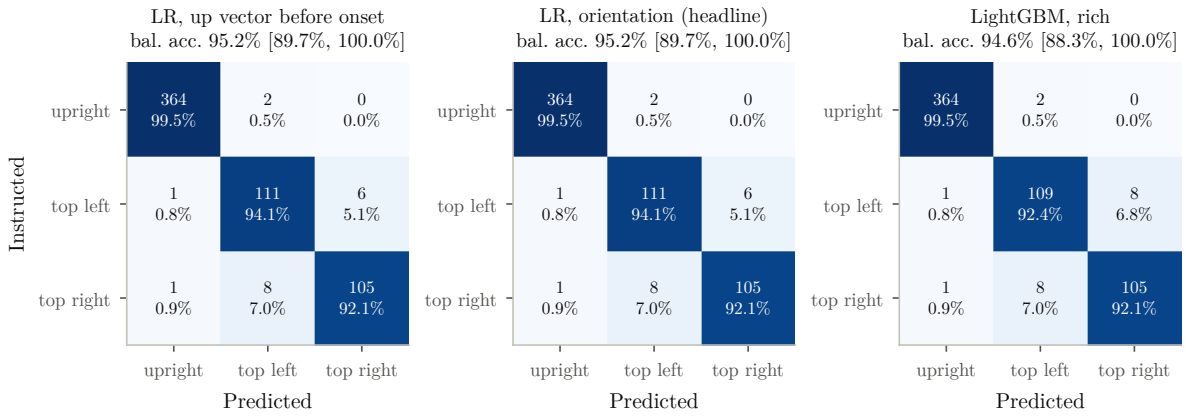


Figure 101: Confusion matrices of the instructed grip (LOSO; counts and row shares; rows: instructed grip) for the up vector before the onset alone, the headline orientation LR and LightGBM, with their balanced accuracies. The first two are identical. The headline's 18 errors include 14 swaps between top left and top right.

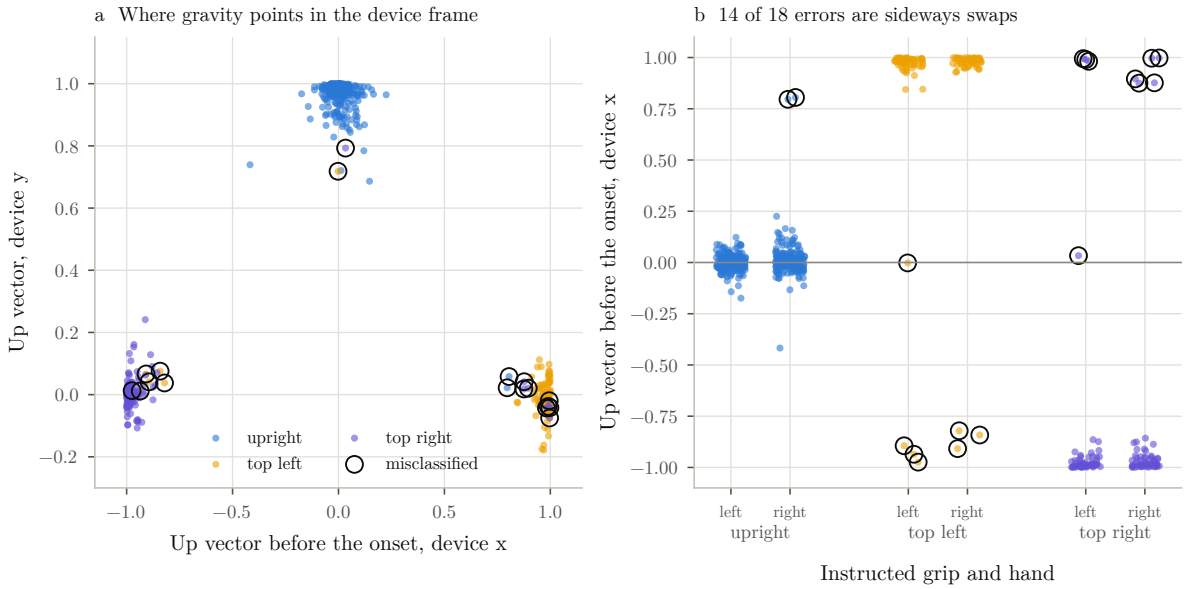


Figure 102: Why the grip is recognised and where it fails: the unit up vector in the device frame during the still hold before the onset, per trial, coloured by the instructed grip; (a) x against y, (b) x per instructed grip and hand. Upright puts the top edge ($+y$) up, top left the right edge ($+x$), top right the left edge ($-x$). Ringed: the trials the headline misclassifies; each of the 18 lies in the cluster of the predicted grip.

Table 108: Instructed grip from sensor data (upright, top left, top right): leave-one-person-out, nested tuning, person-bootstrap 95% CIs, scored against the instruction. Chance balanced accuracy is 33.3%.

experiment	windows	model	n	bal. acc.	macro-F1	log loss
sec_grip_up	oracle	LR, up vector before the onset (3)	598	95.2% [89.7%, 100.0%]	0.952 [0.897, 1.000]	0.158
sec_grip_lr	oracle	LR, device-frame orientation (12)	598	95.2% [89.7%, 100.0%]	0.952 [0.897, 1.000]	0.157
sec_grip_gbm	oracle	LightGBM, summary statistics of every channel (271)	598	94.6% [88.3%, 100.0%]	0.946 [0.883, 1.000]	0.171

Table 109: Per-class recall, precision and F1 of the grip models (against the instruction).

experiment	class	n	predicted	recall	precision	F1
sec_grip_up	upright	366	366	99.5%	99.5%	0.995
sec_grip_up	top left	118	121	94.1%	91.7%	0.929
sec_grip_up	top right	114	111	92.1%	94.6%	0.933
sec_grip_lr	upright	366	366	99.5%	99.5%	0.995
sec_grip_lr	top left	118	121	94.1%	91.7%	0.929
sec_grip_lr	top right	114	111	92.1%	94.6%	0.933
sec_grip_gbm	upright	366	366	99.5%	99.5%	0.995
sec_grip_gbm	top left	118	119	92.4%	91.6%	0.920
sec_grip_gbm	top right	114	113	92.1%	92.9%	0.925

Table 110: Confusion of the instructed grip: counts and row shares (rows = instructed grip).

experiment	instructed	n	predicted upright	predicted top left	predicted top right
sec_grip_up	upright	366	364 (99.5%)	2 (0.5%)	0 (0.0%)
sec_grip_up	top left	118	1 (0.8%)	111 (94.1%)	6 (5.1%)
sec_grip_up	top right	114	1 (0.9%)	8 (7.0%)	105 (92.1%)
sec_grip_lr	upright	366	364 (99.5%)	2 (0.5%)	0 (0.0%)
sec_grip_lr	top left	118	1 (0.8%)	111 (94.1%)	6 (5.1%)
sec_grip_lr	top right	114	1 (0.9%)	8 (7.0%)	105 (92.1%)
sec_grip_gbm	upright	366	364 (99.5%)	2 (0.5%)	0 (0.0%)
sec_grip_gbm	top left	118	1 (0.8%)	109 (92.4%)	8 (6.8%)
sec_grip_gbm	top right	114	1 (0.9%)	8 (7.0%)	105 (92.1%)

Table 111: The trials the headline grip model misclassifies: the instructed and predicted grip, the grip the app itself detected at the start and the grip-mismatch flag (both used after prediction only), and the up vector before the onset in the device frame.

trial_key	person_id	hand	instructed	predicted	app grip at start	flagged	up_before_x	up_before_y
P05-s1-t12a1	P05	left	top left	top right	top right	1.00	−0.94	+0.01
P05-s1-t13a1	P05	left	top left	top right	top right	1.00	−0.97	+0.01
P05-s1-t14a1	P05	left	top left	top right	top right	1.00	−0.89	+0.04
P05-s1-t18a1	P05	right	top right	top left	top left	1.00	+0.90	+0.02
P05-s1-t19a1	P05	right	top right	top left	top left	1.00	+0.88	+0.04
P05-s1-t20a1	P05	right	top right	top left	top left	1.00	+0.87	+0.02
P09-s1-t09a2	P09	left	top left	upright	unknown	1.00	−0.00	+0.72
P09-s2-t01a2	P09	right	upright	top left	unknown	1.00	+0.80	+0.02
P09-s2-t02a3	P09	right	upright	top left	unknown	1.00	+0.81	+0.06
P09-s2-t10a1	P09	left	top right	upright	unknown	1.00	+0.03	+0.79
P09-s2-t15a1	P09	right	top left	top right	top right	1.00	−0.82	+0.04
P09-s2-t16a2	P09	right	top left	top right	top right	1.00	−0.91	+0.07
P09-s2-t17a1	P09	right	top left	top right	top right	1.00	−0.84	+0.08
P16-s1-t15a1	P16	left	top right	top left	top left	1.00	+0.98	−0.04
P16-s1-t16a1	P16	left	top right	top left	top left	1.00	+0.99	−0.04
P16-s1-t17a1	P16	left	top right	top left	top left	1.00	+0.99	−0.02
P16-s1-t19a1	P16	right	top right	top left	top left	1.00	+1.00	−0.04
P16-s1-t20a1	P16	right	top right	top left	top left	1.00	+1.00	−0.08

Table 112: The grip models scored against three references (balanced accuracy, counts): the instruction on every trial; the instruction on the trials neither the app nor the gravity rule flagged as held in another grip (confirmed); and the grip the app itself detected at the start, on the trials where it could tell. The app reads the same physical quantity, so this separates wrong labels from sensor errors; it is not an independent sensor.

experiment	vs instruction, all trials	vs instruction, confirmed trials	vs the app's start grip
sec_grip_up	95.2% (18 errors / 598)	100.0% (0 errors / 575)	100.0% (589 agree / 589)
sec_grip_lr	95.2% (18 errors / 598)	100.0% (0 errors / 575)	100.0% (589 agree / 589)
sec_grip_gbm	94.6% (20 errors / 598)	99.4% (2 errors / 575)	99.4% (587 agree / 589)

H.9. Handedness and the dominant hand

Question. Can a LOSO model tell a dominant-hand trial from mirror-invariant kinematics alone? These inputs are identical for a trial and its mirror image (Section 3, Eq. (1)): every feature was bit-identical for 598 of 598 trials and their mirror images (largest difference 0). They cannot encode which hand moved, only how the dominant hand moves [16, 17]. The dominant hand is the right hand for the 18 right-handed persons and the left hand for the 2 left-handed ones, so dominance is confounded with the right hand for the majority. Every person has at least 4 dominant and 4 non-dominant trials. The first block was left-handed in 13 sessions and right-handed in 11 (exact binomial p 0.839).

Reference. The pre-specified reference is a label-permutation null, not an AUC of 0.5. The labels were shuffled within each person 200 times and the whole nested pipeline was re-run each time, which keeps each person's class balance.

Result. The pre-specified LR reaches a pooled AUC of 0.491 [0.453, 0.533], against a null mean of 0.495 (95th percentile 0.540; one-sided permutation p 0.582; Table 113; Figure 16). The mean within-person AUC is 0.495 [0.445, 0.548] (null mean 0.501, p 0.642); it is above 0.5 for 10 of 20 persons. The accuracy is 49.2 % [44.9, 53.4] and the log loss 0.695, against 0.693 for a constant 0.5. The nested tuning chose the smallest penalty parameter of the grid (0.001, the strongest regularisation) in 20 of 20 outer folds, and the predicted probability spans only 0.42 to 0.55. No dominant-hand signal is detected; the CI still allows a pooled AUC up to 0.533.

Exploratory second model. LightGBM (2 model classes tried in all, no permutation baseline) reaches a pooled AUC of 0.536 [0.497, 0.577], above the LR's null mean but close to its 95th percentile. Against the LR on the same trials, the per-person accuracy differs by +2.3 [−1.5, +6.4] points (sign-flip p 0.282) and the log loss by +0.013 [−0.006, +0.031] (p 0.213; one exploratory comparison, unadjusted). A weak signal that only a non-linear model finds cannot be excluded.

Per person. Table 114 gives each person's within-person AUC and accuracy. The left-handed persons are shown individually, without a CI: P11 0.653 and P17 0.237, against a right-handed median of 0.503 (range 0.305 to 0.742).

Within-person comparison. Per person, the medians of six kinematic features were compared between the dominant and the non-dominant hand (Table 115; Figure 103; Wilcoxon over persons, Holm over the 6 features). No feature differs detectably: 0 have a Holm p below 0.05, and none does when matched on speed, grip and variant. The CIs bound the differences still compatible with the data: the duration changes by −5.1 % [−12.1, +8.6] (dominant higher for 8 of 20 persons), the peak angular speed by −2.6 % [−5.6, +10.5], the angular path length by −11.1 % [−19.1, +1.2] (6 of 20 higher; Holm p 0.990) and the linear path length by +1.1 % [−7.6, +11.9]. SPARC differs by +0.024 [−0.017, +0.120] and the log dimensionless jerk by +0.089 [−0.288, +0.516]. The left-handed persons are shown individually (diamonds); their duration differences are −23.5 % (P11) and +19.7 % (P17).

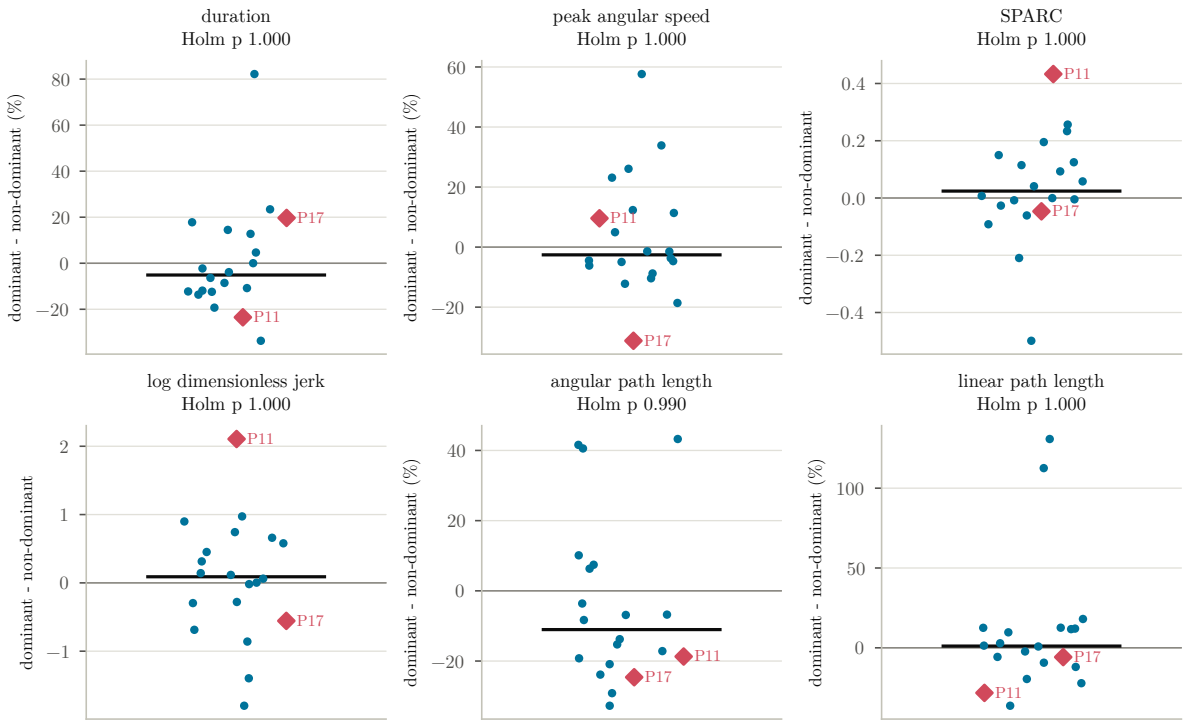


Figure 103: Dominant vs non-dominant movements within each person: per person, the difference of the medians (dominant minus non-dominant; a percent change for the duration, the peak angular speed and the path lengths); the bar is the median over persons. Wilcoxon over persons, Holm over the 6 features. Diamonds: the 2 left-handed persons, shown individually. No feature differs detectably; the CIs still allow moderate differences (text).

Concordance. Table 116 counts the persons by self-reported handedness, usual phone hand and the hand of the first free-choice trial; it is descriptive only. Of the 18 persons who name one usual phone hand, 18 name the right hand, and 20 of the 20 first free-choice trials used the right hand, which a constant would also give. The only contrast comes from the 2 left-handed persons: 2 of them chose the right hand for the free trial and 2 usually hold the phone in the right hand. Whether handedness or the usual phone hand goes with the free choice cannot be told apart with so few left-handed persons and no left-hand first free choice [30]. Of the 2 free-choice trials of repeat visits, 1 used the left hand.

Decision. No person-level handedness model is trained or evaluated: the 2 left-handed persons are fewer than the 5 persons a stratum needs for a CI, so their values are shown individually. Self-reported handedness is the only dominance measure; no handedness questionnaire was given.

Table 113: Can a leave-one-person-out model tell a dominant-hand trial from mirror-invariant kinematics alone? Nested models; person-bootstrap 95% CIs; label-permutation baseline of the pre-specified model (200 permutations: labels shuffled within each person, the whole nested pipeline re-run; one-sided p).

model	persons	pooled AUC	permutation p	null AUC mean [q95]	within-person AUC	permutation p (within)	accuracy
logistic regression (pre-specified)	20	0.491 [0.453, 0.533]	0.582	0.495 [0.540]	0.495 [0.445, 0.548]	0.642	49.2% [44.9%, 53.4%]
LightGBM (exploratory)	20	0.536 [0.497, 0.577]	not run				52.3% [49.4%, 55.4%]

Table 114: Dominant-hand model per person (leave-one-person-out): the AUC of its predictions within the person and its accuracy; the left-handed persons are shown individually (too few for a confidence interval).

person_id	handedness	n	dominant trials	within-person AUC	accuracy
P01	right	45	22	0.336	44.4%
P02	right	31	14	0.647	64.5%
P03	right	24	12	0.479	45.8%
P04	right	8	4	0.500	62.5%
P05	right	32	16	0.586	50.0%
P06	right	27	14	0.626	55.6%
P07	right	28	14	0.474	50.0%
P08	right	27	14	0.742	55.6%
P09	right	34	16	0.469	47.1%
P10	right	34	16	0.569	50.0%
P11	left	28	14	0.653	67.9%
P12	right	38	19	0.507	42.1%
P13	right	28	14	0.418	50.0%
P14	right	40	20	0.410	40.0%
P15	right	38	19	0.305	28.9%
P16	right	24	11	0.510	45.8%
P17	left	25	13	0.237	44.0%
P18	right	32	17	0.357	40.6%
P19	right	27	14	0.505	63.0%
P20	right	28	14	0.571	60.7%

Table 115: Dominant vs non-dominant movements within each person: per person the difference of the medians (a percent change for the duration, the peak angular speed and the path lengths); across persons the median with a person-bootstrap 95% CI, a Wilcoxon signed-rank test, Holm over the features. Matched: the same difference inside cells of equal instructed speed, grip and variant.

feature	scale	dominant - non-dominant	dominant higher	Wilcoxon p	Holm p	matched: difference	matched: Holm p
duration	percent change	−5.1% [−12.1%, +8.6%]	8/20	0.596	1.000	−4.3%	1.000
peak angular speed	percent change	−2.6% [−5.6%, +10.5%]	8/20	0.869	1.000	−6.7%	1.000
SPARC	difference	+0.024 [−0.017, +0.120]	11/20	0.261	1.000	+0.084	1.000
log dimensionless jerk	difference	+0.089 [−0.288, +0.516]	12/20	0.596	1.000	+0.195	1.000
angular path length	percent change	−11.1% [−19.1%, +1.2%]	6/20	0.165	0.990	−8.4%	1.000
linear path length	percent change	+1.1% [−7.6%, +11.9%]	11/20	0.898	1.000	−2.9%	1.000

Table 116: Self-reported handedness, usual phone hand (profile) and the hand of the first free-choice trial: counts of persons (descriptive only).

handedness	usual phone hand	free-choice hand	persons
left	right	right	2
right	both	right	2
right	right	right	16

H.10. Demographics add-on test

This test is a quick first pass and has not been audited. Production does not know a user's age, sex, height, handedness or usual phone hand; the test asks whether these would help if it did. Nothing from it enters the production model.

Design. Refitting the final model with covariate inputs was too slow for this pass. Instead, a second-stage L2 LR was fitted on the final model's logit (Appendix F), with and without the person covariates: age, sex, height (with a missing indicator; 3 heights are missing and were median-imputed inside the fold), self-reported handedness and usual phone hand. In each outer fold the stage was fitted on the final model's inner cross-fitted logits of the training persons and applied to the held-out person's logit from the saved LOSO predictions; it never uses other persons' outer-fold predictions. Its penalty was tuned on grouped inner folds. The variants are the sensor-only stage (the comparator), the stage with the covariates as main effects, the stage with covariates and covariate × logit interactions, the covariates alone (the floor), and two fixed rules for the base-rate note: answer the usual phone hand, or always answer right. Paired differences against the sensor-only stage are mean per-person differences with paired CIs and sign-flip p-values, Holm-adjusted over the two add-on variants within each metric.

Cohort. The 20 persons include 2 left-handed persons, 2 who use both hands for the phone and 0 who usually use the left; 12 are female and the ages span 19 to 29 years. Interactions such as 'left-handed persons move differently' are therefore barely identified.

Results. Table 117 and Figure 104 give the results. The sensor-only stage is neutral against the final model as shipped (accuracy +0.2 points, log loss +0.006; unadjusted sign-flip p 1.000 and 0.624) and reaches an accuracy of 89.5 % [85.1, 93.4]. Adding the covariates gives 86.5 % [81.8, 91.1]: a paired difference of −3.2 [−6.3, −0.7] points (Holm p 0.062) and a log-loss difference of +0.034 [−0.003, +0.073] (Holm p 0.098). The covariates change 28 primary answers; 1 person gains and 7 lose. With the interactions the accuracy is 87.5 % [83.1, 91.5] (difference −3.0 [−5.8, −0.6] points, Holm p 0.062), and the log loss rises by +0.056 [+0.014, +0.105] (Holm p 0.031, significant). The interactions change 24 answers; 2 persons gain and 8 lose. The covariates-only floor sits at chance: accuracy 50.3 % [49.5, 51.2], balanced accuracy 50.0 %, AUC 0.493. This is expected, since every person did both hands: a person-level covariate can help only through interactions, and the interaction variant does not help.

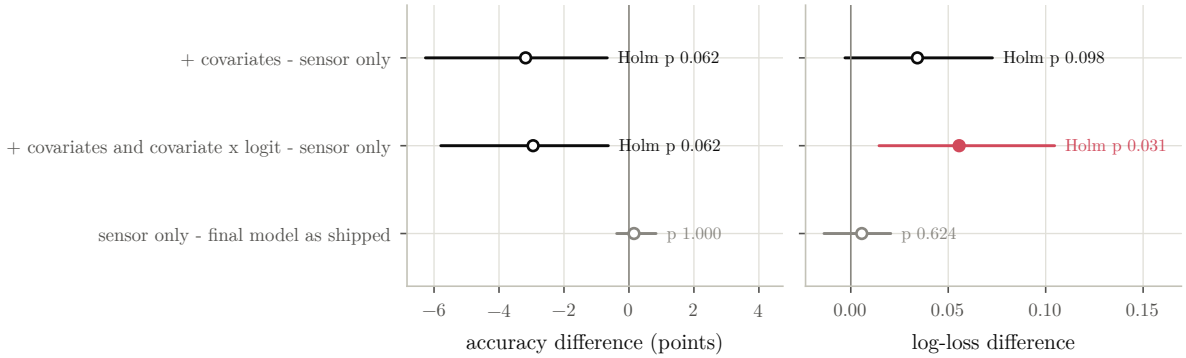


Figure 104: Demographics add-on test (quick first pass, not audited): the covariates added to a second stage on the final model's logit, against the same stage without them, on the same LOSO folds. Mean per-person differences with paired 95 % CIs and Holm-adjusted sign-flip p over the two add-on variants within each metric (filled red: below 0.05). Grey: the sensor-only stage against the shipped model (unadjusted). Negative accuracy and positive log-loss differences mean the covariates hurt.

Exploratory base. The absolute levels inherit the optimism of the final model as shipped, which is exploratory: the best of 16 configurations in lane m3, then chosen among 17 candidates. Its nested, selection-aware accuracy is 88.5 % [84.1, 92.3] (Appendix F). The paired differences share the base, so its optimism largely cancels in them. The stage's penalty is chosen on the same inner logits it is fitted on, a second-order effect that is not removed.

Natural use and the base rate. On natural use the measure is the right-hand recall on the free-choice trials (all right-hand so far; a constant right answer would score 100 %). The shipped model and the sensor-only stage reach 85.0 % and 85.0 %, the two add-ons 80.0 % and 70.0 %. The fixed rules show why this measure cannot judge an add-on (Figure 105). Always answering right scores 100.0 % and the phone-hand rule 95.0 %, yet both sit at chance on the primary set (balanced accuracy 50.0 % and 49.9 %). The phone-hand rule abstains (P(right) exactly 0.5) for the 2 persons who use both hands. On natural use it answers 18 of the 20 trials, all correctly; its score gives each of the 2 abstentions half credit, and it has no forced choice for them. The floor's 20.0 % is not a signal: its P(right) never moves more than 0.022 from 0.5 and sits just below it, because 49.5 % of the primary trials are right-hand trials.

Decision. Demographics are not added, even as a hypothetical: no add-on variant improves accuracy or log loss on the same folds. With 2 left-handed persons and 0 who usually hold the phone in the left hand, this is a null result on a small, mostly right-handed sample, not evidence of no effect. Not done in this pass: a refit of the network with covariate inputs, a replication on a cheaper base model, a calibration analysis of the add-ons, and an audit.

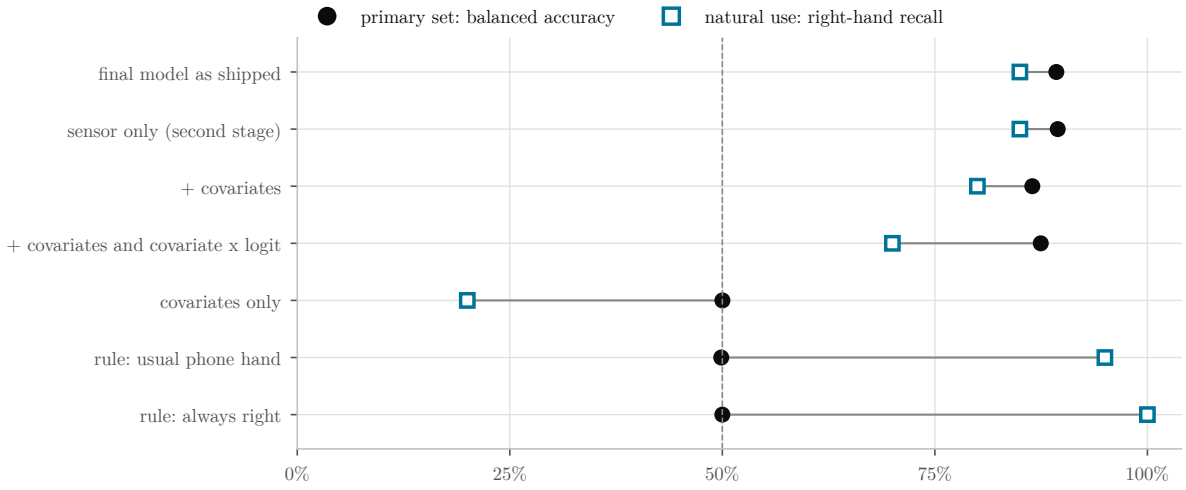


Figure 105: The base-rate effect: balanced accuracy on the primary set (both hands for every person) and right-hand recall on the 20 free-choice trials (all right-hand so far; a constant right answer would score 100 %). The fixed rules look excellent on natural use and sit at chance on the primary set, so natural use cannot judge an add-on; the add-on test is judged on the primary set.

Table 117: The demographics add-on test (leave-one-person-out; person-bootstrap 95% CIs). Second stage on the final model's logit, fitted on inner cross-fitted logits of the training persons; d = add-on minus the sensor-only second stage on the same folds (mean per-person accuracy and log-loss differences, paired person bootstrap), Holm across the add-on variants within each metric. Natural use: right-hand recall on free-choice trials (every one is a right-hand trial), not a balanced accuracy. Rules are fixed (no fitting) and shown for the base-rate effect only.

model	accuracy	bal. acc.	log loss	AUC	natural use	d accuracy (points)	d log loss
final logit, second stage (sensor only)	89.5% [85.1%, 93.4%]	89.5%	0.296 [0.214, 0.402]	0.951	85.0% (17 of 20)		
+ covariates (main effects)	86.5% [81.8%, 91.1%]	86.5%	0.327 [0.250, 0.408]	0.938	80.0% (16 of 20)	−3.2 [−6.3, −0.7], Holm p 0.062	+0.034 [−0.003, +0.073], Holm p 0.098
+ covariates and covariate x logit	87.5% [83.1%, 91.5%]	87.5%	0.335 [0.268, 0.417]	0.940	70.0% (14 of 20)	−3.0 [−5.8, −0.6], Holm p 0.062	+0.056 [+0.014, +0.105], Holm p 0.031
covariates only (floor)	50.3% [49.5%, 51.2%]	50.0%	0.693 [0.693, 0.694]	0.493	20.0% (4 of 20)		
rule: usual phone hand	49.4% [48.6%, 50.3%]	49.9%	0.830 [0.807, 0.849]	0.497	95.0% (18 of 20, 2 ties)		
rule: always right	49.5% [48.6%, 50.3%]	50.0%	0.842 [0.833, 0.852]	0.500	100.0% (20 of 20)		
final model as shipped	89.3% [85.0%, 93.1%]	89.3%	0.291 [0.196, 0.420]	0.951	85.0% (17 of 20)		

H.11. Caveats and open questions

- None of these models runs in production. A production speed or grip estimate would take its bounds from the cue-free detector (Appendix G), whose end detection meets the same difficulty with slow reaches; its sensitivity to end-detection errors would have to be measured.
- The label-blind speed estimate belongs to one detector setting, the app's defaults. A fixed window from the onset or a detector tuned for slow movements would give other numbers; neither was run.
- The linear path length integrates the acceleration with zero velocity at both bounds. The drift that the linear correction cannot remove grows with the duration.
- Slow trials are few (a median of 2 per person), so per-person slow statistics rest on very few movements.
- Added after seeing the results (exploratory; p-values not corrected for that): the per-person

description of Figure 100, the LightGBM dominance model, the grip consistency checks, the segmentation check, the grip-only reference, the grip strata and the position-adjusted contrasts.
- The null dominance result does not test whether the hand model's evidence lies only in the chiral channels; that remains a hypothesis.
- Open: a person-relative speed measure, normalised by the person's own natural movements; and a dominance-related movement style that transfers to left-handed persons, which needs more left-handed persons.

I. An exploratory case study: PD01

This appendix describes how one participant with Parkinson's disease, PD01, performed the nose-touch task, compared with the 20 modelling persons (the reference persons). It is descriptive and exploratory. PD01 never enters a model or a model comparison: no model was trained, tuned or scored on PD01's trials, and this appendix quotes no model output for PD01. Nothing here is a diagnostic claim.

The main finding is that PD01 is *unusual but not unique*. PD01 lies beyond every reference person in 9 of 42 pre-specified characteristics. A typical reference person, judged the same way against the other reference persons, lies beyond all of them in 4.0, and 2 of the 20 reach PD01's count (Appendix I.5). 5 pre-specified differences survive every check that could overturn them; none survives the Benjamini–Hochberg adjustment over the pre-specified characteristics (Appendix I.6).

I.1. The question, the participant and the quality-control caveats

Question. The case study asks what PD01 does, when performing this task, that the reference persons do not. The comparison covers preparation and holding still, movement time and speed, amplitude, smoothness, band power of the angular velocity in the hold, the reach and at the nose, change across repeats, speed scaling, task behaviour, and the hand signature itself, read without a classifier.

The participant. PD01 is in their fifties, decades older than every modelling person (reference ages 19 to 29 years). PD01 performed the task remotely and unsupervised, outside the group sessions, seated, with protocol version 2.0.0 and the app's street visit (Table 2). PD01 made 2 short sessions of 4.2 and 2.6 minutes: the first ended early and the second was left in progress (reference sessions: 4.3 to 21.1 minutes, median 8.4). PD01 therefore reached only protocol tiers 0, 1, the start of a visit. No clinical score and no medication state were recorded.

The trials. Of 10 attempts, 7 were accepted and analysable (Table 118). The app rejected 3: 1 for no movement within the time limit after the GO cue and 2 for still moving at the time limit. The accepted trials are 2 free-choice trials, both with the right hand (reported), and 5 instructed left-hand trials with the upright grip, screen toward the face (1 slow, 3 natural, 1 fast). PD01's accepted trials carried 9 false starts in total.

Same code path. PD01's raw records live outside the main data snapshot. They were cached by the same loader, standardisation, flags, gates and exclusion cascade as the main

Table 118: The accepted, analysable trials of PD01 (7 of 10 attempts; set: primary = instructed, natural_use / natural_use_repeat = first / later free-choice trial; clip flag: raw |a| above 3.5 g). Rejected attempts: failed: no movement within the time limit after GO (1); failed: still moving at the time limit (2).

trial	set	hand	label source	grip	speed	false starts	clip flag
PD01-s1-t00a2	natural_use	right	reported	free	natural	1	
PD01-s1-t01a1	primary	left	instructed	upright_toward	natural	0	yes
PD01-s1-t02a2	primary	left	instructed	upright_toward	fast	1	yes
PD01-s1-t03a1	primary	left	instructed	upright_toward	natural	0	
PD01-s2-t00a1	natural_use_repeat	right	reported	free	natural	1	yes
PD01-s2-t01a2	primary	left	instructed	upright_toward	slow	4	yes
PD01-s2-t02a1	primary	left	instructed	upright_toward	natural	2	

data (Appendix A.7), in a separate, git-ignored cache. A test checks that the builder code and parameters are identical, and that none of PD01's identifiers reaches the main curation or cache.

Quality-control caveats. Six properties of the recording limit every statement below.

- *Remote and unsupervised.* Nobody watched the visit; posture and grip are known only from the app's prompts and PD01's answers, and arm support was not recorded.
- *Clip flag.* The flag `clip_candidate` marks a raw specific force above 3.5 g or an angular velocity above 950 deg/s, where a sensor may have saturated. It is set on 4 of PD01's 7 accepted trials (accelerometer 4, gyroscope 0) and on 1 rejected attempt, against 8 of 598 reference primary trials (2 persons). PD01's acceleration magnitudes are therefore lower bounds, and every comparison is repeated without the flagged trials.
- *Rotation-rate mapping assumed, not verified.* In both of PD01's sessions the gyroscope axes use the assumed mapping of the W3C specification [1], not one verified by the app's gyro-gravity closure as in 22 of the 24 reference sessions (Appendix A.2). The closure still favours it: 0.80 deg against 4.58 deg for the next mapping in the first session, 0.78 deg against 2.62 deg in the second. The hand signature and the gravity-frame characteristics depend on this mapping.
- *Few trials.* PD01's values are means of 3 to 5 trials; the speed ratios rest on 1 slow and 1 fast trial.
- *Confounds.* Age, the phone (its sampling, noise floor and accelerometer range), phone mass, arm length, a seated posture with unknown arm support (reference: 18 sitting and 6 standing sessions) and the hand all differ from the reference. Whether the instructed hand matched PD01's dominant or usual phone hand is not balanced against the reference. Nothing here separates the disease from these factors.
- *The first screen touch* is the first touch event after GO. It is recorded in 100 % of PD01's left-hand upright natural trials and in 59 % of the reference trials, so the touch characteristics compare 14 reference persons. What touched the screen (face, finger or grip) is unknown.

I.2. Windows, characteristics and phenomena

Options. PD01 could have been scored by the hand models of this report, or described by kinematic characteristics. Scoring was ruled out from the start: one case cannot support a

performance claim, and a score invites a diagnostic reading. The lane therefore compares characteristics, each with a one-line kinematic rationale.

Windows. Each trial gives three windows. The *hold* is the 1.0 s before the movement onset; it lies inside the app's verified still window in 99.7 % of the reference trials, so it is a postural hold. The *reach* is the stored approach (the oracle window; Appendix A.6). *At the nose* is the first 0.6 s after the approach ends, kept only when the return of the arm starts later; it exists in 88.1 % of the reference trials. Two design limits follow. A hold that breaks the app's still gate restarts as a false start, so large hold movements never reach the hold window. The approach ends only after the phone is still, so the nose window describes only trials that became still (2 of PD01's rejections were still moving at the time limit).

Spectral resolution. A window of T seconds resolves $1/T$ Hz: 1.00 Hz in the hold, 1.67 Hz at the nose and about 0.59 Hz in a median reach of 1.68 s. The Hann window spreads a tone over $\pm 2/T$ Hz: a 6.5-Hz tone of RMS 7.07 deg/s puts 3.5 deg/s into the 7–12 Hz band of the hold window and 4.1 deg/s into that of the nose window. The 3–7 / 7–12 Hz split of the short windows is therefore coarse.

Characteristics. The lane compares 55 characteristics: 42 pre-specified and 13 post hoc (marked * in every figure and table). They include:

- the band RMS (root mean square) of the angular-velocity vector in 3–7 Hz (the band of parkinsonian rest tremor) and 7–12 Hz (the band of physiological and action tremor), the spectral peak in 3–12 Hz, and the 3–12 Hz power relative to 0.5–3 and 12–15 Hz in decibels. Band power is called a tremor only if the spectrum shows a peak, and nowhere in this appendix is it called one;
- the reach as recorded and in a movement band (gyroscope low-passed at 3 Hz), which separates the voluntary reach from faster components; smoothness by the log dimensionless jerk and the spectral arc length (SPARC) [28]; speed peaks and sub-movements; hesitations (at least 100 ms below the app's still threshold of 20 deg/s);
- the first screen touch: its time, its share of the approach, the rotation after it and the linear acceleration within 100 ms of it;
- the hand signature: the first-half and net yaw about gravity, mirror-normalised (Eq. 1) so that positive is the direction typical of the hand used. The hand label sets only this sign.

Phenomena. Characteristics that measure one thing count as one finding: for example the net rotation and the tilt change, or the time and the share of the approach of one touch. The 17 phenomena are declared by definition in the code, not by PD01's values.

Post hoc characteristics. The first-touch characteristics, the active movement time, the settling time, the active-time ratio and the early yaw peak were defined after PD01's traces were inspected. The components of the hold power ratio, the 7–12 Hz power per peak speed and the share of trials with the typical net-yaw sign were added after the audit, to test alternative explanations. They are hypotheses, not tests, and carry no q-value.

I.3. Matched comparisons and statistics

Strata. Each characteristic is judged in one pre-declared stratum. The reach uses the left-hand upright natural-speed trials; the hold, the nose and the hand signature use all left-hand upright

trials; the across-repeat measures use the first k left-hand upright natural reaches in visit order, with $k = 3$, the number PD01 made; the speed scaling uses the slow and fast trials. Table 125 gives every per-trial characteristic in every other stratum.

Statistics. A person's value is the mean of their trials in the stratum. PD01's value x is compared with the reference persons' values $r_1, \ldots, r_n$ by the mid-rank percentile, the robust z and the two-sided rank p:

$$z = \frac{x - \mathrm{med}(r)}{1.4826\ \mathrm{MAD}(r)}, \qquad p = \min\Big(1,\ \frac{2\min\{\#(r_j \geq x) + 1,\ \#(r_j \leq x) + 1\}}{n + 1}\Big), \tag{10}$$

where MAD is the median absolute deviation. The robust z uses the characteristic's declared scale: log for strictly positive, right-skewed quantities, linear otherwise. With $n = 20$, the smallest possible rank p is $2/(n + 1) = 0.095$: a rank test cannot reach $p < 0.05$ for one characteristic. The Crawford–Howell single-case t-test runs alongside, on the declared scale; it assumes normally distributed reference values. With fewer than 5 reference persons, no statistic is computed. This appendix gives no confidence intervals: with one case, every comparison is a rank among persons.

Multiplicity. The comparison family is the 42 pre-specified characteristics, adjusted by Benjamini–Hochberg; the adjusted p is called q. No pre-specified characteristic reaches $q < 0.05$ by either test: the smallest q is 0.50 for the rank test and 0.18 for the Crawford–Howell test (smallest p 0.008, peak linear acceleration). The smallest post hoc Crawford–Howell p ($p < 0.001$ for the early yaw peak on the linear scale, 0.058 on the log scale) is not evidence: the characteristic was chosen after looking at PD01, and its p depends on the scale. The rank p of the count of differences (Appendix I.5) and of the Mahalanobis distance (Appendix I.10) are single, unadjusted tests of the whole profile, outside the characteristic family.

I.4. Robustness: the checks that can overturn a difference

Options. Five sensitivity analyses were run (Table 122). Three recompute values and can overturn a "beyond every reference person" verdict: *first k*, each reference person judged on their first k matched trials, with k PD01's count (equal averaging noise and practice); *same protocol*, only reference trials of protocol 2.0.0 (1 reference session with 17 trials used the pilot protocol); and *no clip*, PD01 and the reference without the clip-flagged trials. Two only drop reference persons: *same posture* and *same platform*. A value beyond all 20 persons is beyond every subset of them, so these two can never overturn the verdict; they are marked (s).

Decision. A "beyond all" verdict is *robust* when it holds in every check that recomputes values and passes a size check: $|z| \geq 2$ on the declared scale whichever reference person is left out. A check that cannot be run, because too few of PD01's trials remain, withholds the label.

Rationale. The audit of the lane showed that an earlier definition, which counted the subset checks, called every "beyond all" verdict robust by construction. The size check guards against a value that is only just beyond the reference extreme.

I.5. The null reference of the count: unusual, not unique

Question. PD01 lies beyond every reference person in 9 pre-specified characteristics (7 phenomena). The question is how many such characteristics a young reference person reaches by chance. If PD01 were exchangeable with the reference persons, 4.5 would be expected.

Method. Each reference person was compared with the other 19, like for like: on the mean of their first k trials, as noisy as PD01's. PD01 was compared with all 20. PD01's rank among the 21 persons gives a rank p; it is conservative, since PD01 faces one more person.

Result. Reference persons lie beyond all others in a median of 4.0 pre-specified characteristics, with a maximum of 9 (Figure 106, bottom; Table 119). 2 reference persons (P09, P14) reach PD01's count, rank $p = 0.143$. Counted in phenomena, 1 reference person reaches PD01's 7, rank $p = 0.095$. Against any 19 of the reference persons, PD01's count is 9 to 10. PD01 is thus among the most unusual persons of the group, but not alone.

Other versions. When everyone is judged on all their trials, the reference median is 3.5 and the maximum 8 (rank $p = 0.048$). This version is not like for like: PD01's values are means of few trials, and noisier means reach the extremes more often. With the post hoc characteristics, PD01 reaches 14 (10 phenomena) against a reference maximum of 13 (rank $p = 0.048$), but those characteristics were chosen after looking at PD01, which inflates PD01's count. The like-for-like count of pre-specified characteristics is the fair comparison.

Table 119: The null reference of the count of characteristics (n) and phenomena (phen) in which a person lies beyond every reference person: each reference person against the other 19, PD01 (last row) against all 20. Like for like: each reference person judged on the mean of their first k trials, as many as PD01 made; main: everyone on all trials; all: the post hoc characteristics included (chosen after looking at PD01).

Person	pre, lfl: n	phen	pre, main: n	phen	all, lfl: n	phen	all, main: n	phen
P01	7	5	6	4	10	6	9	5
P02	6	4	6	4	6	4	6	4
P03	5	3	5	3	6	3	6	3
P04	2	2	2	2	2	2	2	2
P05	3	2	3	2	6	3	6	3
P06	8	5	8	5	13	7	13	7
P07	0	0	0	0	1	1	1	1
P08	6	5	6	5	6	5	6	5
P09	9	7	5	5	10	7	6	5
P10	2	1	1	1	3	2	1	1
P11	5	3	5	3	6	4	6	4
P12	3	2	2	2	4	3	2	2
P13	3	2	3	2	3	2	3	2
P14	9	5	4	3	12	5	5	4
P15	1	1	1	1	1	1	1	1
P16	3	3	3	3	4	3	4	3
P17	6	6	6	6	8	8	8	8
P18	1	1	1	1	2	2	2	2
P19	6	4	6	4	7	4	7	4
P20	3	3	3	3	6	5	6	5
PD01	9	7	9	7	14	10	14	10

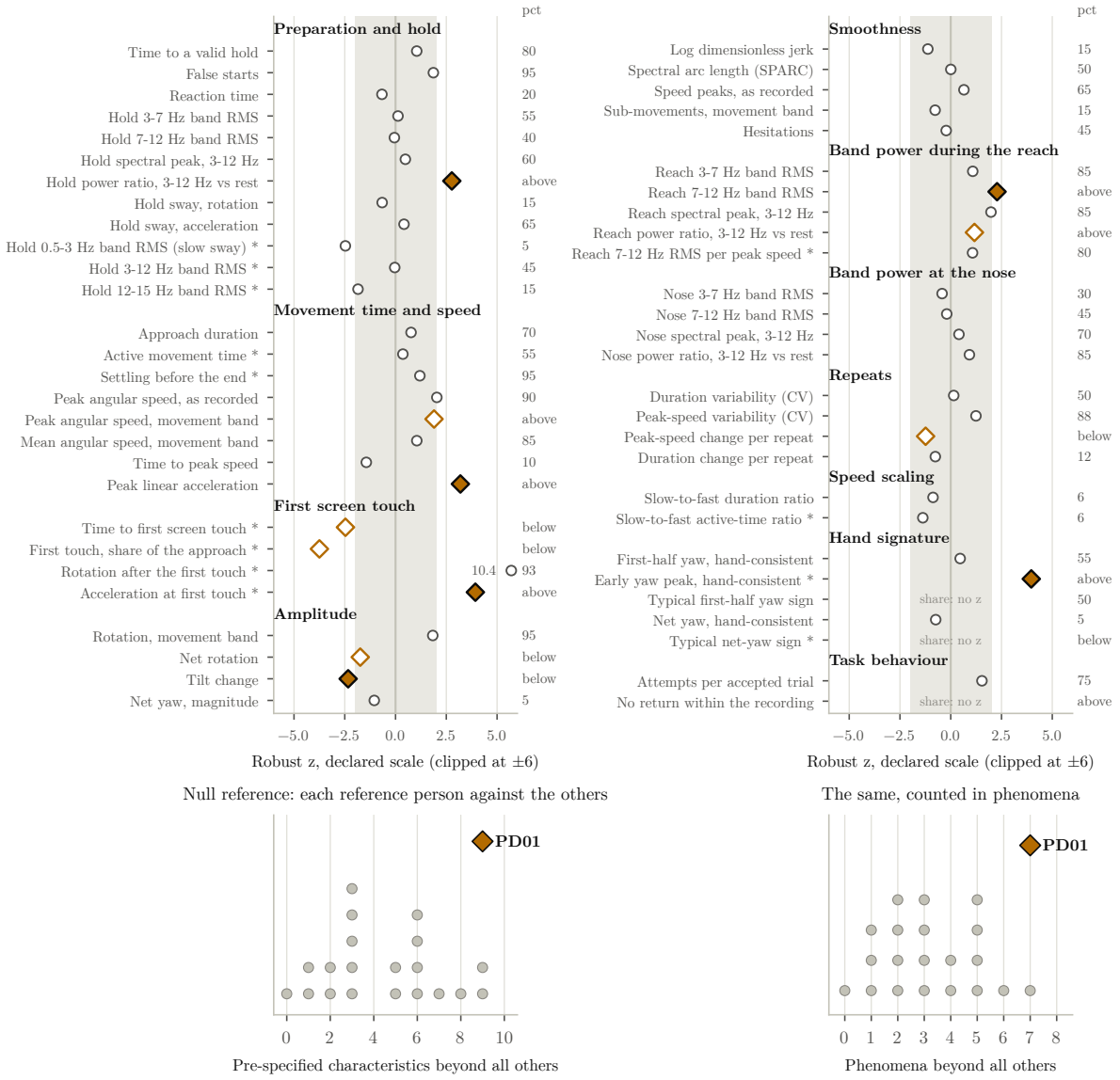


Figure 106: Overview of PD01 against the 20 modelling persons. Top: robust z of every characteristic on its declared scale (grey band: $|z| < 2$) and PD01's percentile; diamonds mark characteristics beyond every reference person (filled: robust; open: not robust). Bottom: the null reference of the count, the number of pre-specified characteristics (left) and phenomena (right) in which each reference person lies beyond all others, on as many trials as PD01 made. 2 reference persons reach PD01's count: unusual, not unique.

I.6. Robust differences

5 pre-specified characteristics in 5 phenomena, and 2 post hoc ones, are beyond every reference person and robust (Table 120, Table 122). Each is listed with its checks; Table 121 collects the checks of alternative explanations. Figure 110 shows the reach characteristics, and Figure 107 shows one of PD01's reaches next to a typical reference reach.

Hold power ratio: less slow sway, not tremor. In the hold before the movement, PD01's power ratio of 3–12 Hz to the rest is 8.6 dB, against a reference median of 2.3 dB (range −1.2 to 6.6; $z = +2.8$; 4 of 5 trials beyond the reference range). It holds on the first k trials, with the same protocol and without clip-flagged trials (7.1 dB against a reference extreme of 6.8), and its smallest leave-one-out z is +2.5. The ratio is high because its denominator is small. PD01's 0.5–3 Hz power is at percentile 5 and the 12–15 Hz power at percentile 15, while the 3–12 Hz power is typical (percentile 45). PD01's hold spectrum lies at 0.1 to 0.3 times the reference median in 0.5–3 Hz (Figure 109, left), and the hold 3–7 and 7–12 Hz band RMS sit at percentiles 55 and 40 (Figure 108). The finding is therefore less slow sway with typical tremor-band power. A supported forearm while seated, the phone's mass or its noise floor could explain it as well as the person.

Peak linear acceleration: a lower bound, and one event with the first touch. PD01's peak linear acceleration in the reach is 15.9 m/s^2, against a reference median of 4.2 and a maximum of 10.3 m/s^2 ($z = +3.2$ on the log scale; 3 of 3 trials beyond). Without the clip-flagged trials it is 12.0 m/s^2 against a reference extreme of 10.3: still beyond, but only just. The magnitudes are lower bounds, since the accelerometer may have saturated. The first touch falls within 100 ms of the largest acceleration in 6 of PD01's 7 touched trials, against 27 % of 424 reference touched trials. The acceleration at the first touch (post hoc) is 14.0 m/s^2 against a median of 3.2 (14 reference persons; without clip-flagged trials 9.2 against 9.0). The two characteristics measure one event and count as one finding: a large acceleration that coincides with an early screen touch.

Tilt change. The phone's tilt changes by 2.9 deg over PD01's reach, against a reference median of 9.3 and a minimum of 3.2 deg ($z = -2.3$ on the log scale; leave-one-out $z = -2.1$). The net rotation measures the same change (Spearman $\rho = +0.94$ across reference persons). It is also below every reference person (3.9 against a minimum of 5.1 deg) but not robust: it fails the same-protocol check (reference extreme 2.6 deg) and the size check. PD01 thus ends the approach with a smaller net change of the phone's orientation than any reference person.

Reach 7–12 Hz band power. PD01's 7–12 Hz band RMS during the reach is 44.6 deg/s, against a reference median of 15.1 and a maximum of 29.5 deg/s ($z = +2.3$ on the log scale; all 3 trials beyond). Four checks address two alternative explanations, a faster reach and the first touch (Table 121):

- *speed-matched*: all 21 reference reaches (11 persons) with a voluntary peak speed in PD01's range, 63 to 165 deg/s, stay below 31.7 deg/s; PD01's 3 reaches give 37.7 to 55.4;
- *time-domain band-pass*: 33.5 deg/s over the whole reach (reference maximum 20.4); 25.9 before the first touch (maximum 22.0, 14 persons); 33.0 without the touch (maximum 21.6). The band power is not only the contact;

- *spectral peak*: PD01's mean reach spectrum lies 3.3 to 14.9 times the reference median across 7–12 Hz, and its 3–12 Hz peak rises 13.5 dB above its surroundings (reference persons: median 6.2, maximum 11.3 dB; Figure 109, middle);
- *per peak speed* (post hoc): relative to the faster reach, the 7–12 Hz RMS is at percentile 80, inside the reference range.

The checks disagree on one point: matched on speed, PD01's band power is beyond every reference reach; scaled by speed, it is not extreme. One person without a clinical examination cannot separate an action tremor from the dynamics of a fast reach that stops at the face. It is therefore reported as band power, not as tremor.

No return of the arm within the recording. In all 5 of PD01's left-hand upright trials the arm did not start its return before the recording stopped, against a reference median share of 0.27 and a maximum of 0.83. It is a share of trials, not a time: the recording stops a fixed time after the end cue, so the time at the nose of a phone kept there is censored.

Not robust. The remaining characteristics beyond every reference person fail at least one check (Table 122). The peak angular speed in the movement band (113 against a maximum of 110 deg/s) falls to 88 deg/s without the clip-flagged trials and fails the size check. The time to the first screen touch (post hoc; 0.68 s against a minimum of 0.83 s) fails the size check, and its share of the approach (post hoc) fails without the clip-flagged trials. After the touch, PD01 still performs a share of 0.55 of the voluntary rotation (reference median 0.09, percentile 93): the touch comes during an ongoing movement. The reach power ratio fails the size check, and the net rotation fails as described above; the peak-speed change per repeat and the share of trials with the typical net-yaw sign are discussed in Appendices I.8 and I.9. Across the reference persons, some findings of different phenomena go together: the peak linear acceleration with the time to the first touch ($\rho = -0.74$) and with the peak angular speed ($\rho = +0.73$). They may reflect one behaviour, for example a faster reach with a harder arrival.

Table 120: Key characteristics of PD01 against the 20 modelling persons (exploratory; the full table is in the appendix). z: robust z on the declared scale; Trials beyond: PD01's trials beyond the reference range; q: Benjamini-Hochberg over the pre-specified characteristics; Robust: see the sensitivity table. *: post hoc.

Characteristic	n	PD01	Reference median [IQR]	Pct	z	Trials beyond	q	Robust
False starts (per trial)	5	1.40	0.38 [0.00, 0.62]	95	+1.9	2/5	0.67	
Reaction time (s)	5	0.65	0.90 [0.67, 1.17]	20	−0.7	1/5	0.74	
Hold 3-7 Hz band RMS (deg/s)	5	2.68	2.52 [1.78, 3.31]	55	+0.1	0/5	1.00	
Approach duration (s)	3	2.44	1.73 [1.44, 2.63]	70	+0.8	0/3	0.90	
Settling before the end (s) *	3	0.69	0.37 [0.23, 0.45]	95	+1.2	1/3	–	
Peak angular speed, movement band (deg/s)	3	113	52 [42, 77]	above all	+1.9	2/3	0.50	no
Time to first screen touch (s) *	3	0.68	1.24 [1.05, 1.41]	below all	−2.5	3/3	–	no
Acceleration at first touch (m/s^2) *	3	14.0	3.2 [2.4, 3.9]	above all	+3.9	2/3	–	yes
Rotation, movement band (deg)	3	88	50 [32, 58]	95	+1.8	0/3	0.67	
Log dimensionless jerk	3	−16.6	−14.5 [−16.0, −13.4]	15	−1.1	0/3	0.73	
Spectral arc length (SPARC)	3	−2.52	−2.52 [−3.09, −2.21]	50	+0.0	0/3	1.00	
Reach 3-7 Hz band RMS (deg/s)	3	29.1	19.2 [14.2, 24.0]	85	+1.1	0/3	0.73	
Reach 7-12 Hz band RMS (deg/s)	3	44.6	15.1 [11.7, 20.7]	above all	+2.3	3/3	0.50	yes
Nose 3-7 Hz band RMS (deg/s)	5	2.70	3.14 [2.60, 4.26]	30	−0.4	1/5	0.90	
Duration variability (CV)	3	0.18	0.17 [0.12, 0.24]	50	+0.1	–	1.00	
Slow-to-fast duration ratio	2	1.28	2.25 [1.87, 4.55]	6	−0.9	–	0.67	
First-half yaw, hand-consistent (deg)	5	5.0	3.1 [1.3, 6.9]	55	+0.5	1/5	1.00	

Strata: L-up = left hand, upright grip, standard movement, natural start (nat / slow / fast / all speeds); up nat, either = upright natural reaches of either hand; free = first free-choice trial (natural-use set); all = every accepted trial.

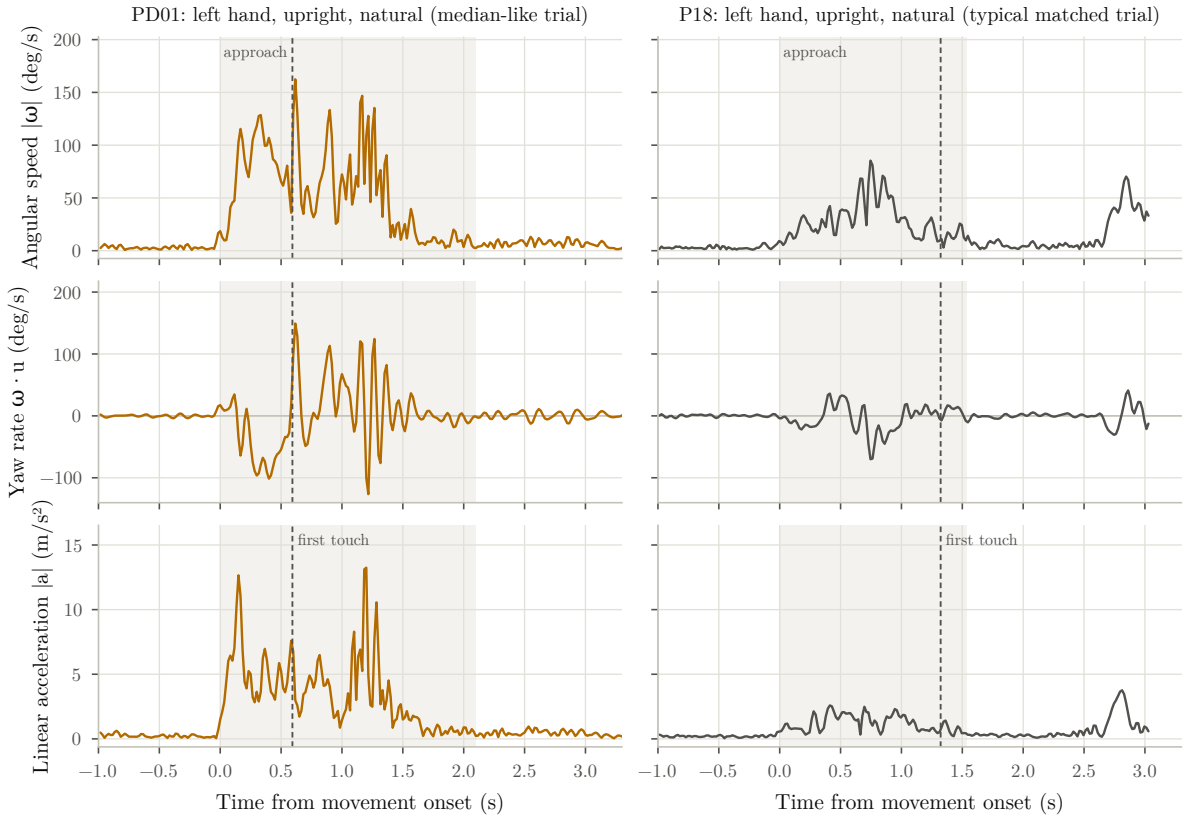


Figure 107: One left-hand upright natural reach of PD01 (left; the trial closest to PD01's medians) and the reference trial closest to the reference medians (right): angular speed, yaw rate about gravity and linear acceleration from the movement onset (shaded: the approach; dashed: the first screen touch). PD01's touch comes early, at a peak of the linear acceleration, during a fast, oscillating reach.

Table 121: Checks of alternative explanations for PD01's differences (left-hand upright natural reaches unless stated; *: post hoc). Every value is a numbers entry.

Check	PD01	Reference	Result
Reach 7-12 Hz RMS against speed-matched reference reaches (deg/s)	37.7-55.4	max 31.7 (21 trials, 11 persons)	3 of 3 case reaches above all
Reach 7-12 Hz band-pass RMS, whole reach (deg/s)	33.5	median 13.2, max 20.4 (20 persons)	0 persons as high
Reach 7-12 Hz band-pass RMS, before the first touch (deg/s)	25.9	median 13.7, max 22.0 (14 persons)	0 persons as high
Reach 7-12 Hz band-pass RMS, without the first touch (deg/s)	33.0	median 14.9, max 21.6 (14 persons)	0 persons as high
Prominence of the reach's 3-12 Hz spectral peak (dB, mean over trials)	13.5	median 6.2, max 11.3 (20 persons)	100th percentile
Reach 7-12 Hz RMS per peak speed *	0.45	median 0.26, max 0.55	at the 80th percentile
First touch within 100 ms of the largest acceleration	6 of 7 trials	27% of 424 touched trials	
Rotation after the first touch (share of the voluntary rotation) *	0.55	median 0.09, max 0.59	at the 93rd percentile
Early yaw peak to first touch (s)	median 0.13 (0.02-0.34)	median 0.43 (touched trials)	
Net yaw against the typical direction (share of trials)	6 of 7	median 35%, max 68%	0 persons as high
Band-power ratio of the hold: components (percentiles)	0.5-3 Hz 5, 3-12 Hz 45, 12-15 Hz 15		
Band-power ratio of the reach: components (percentiles)	0.5-3 Hz 80, 3-12 Hz 95, 12-15 Hz 95		
Band-power ratio of the nose: components (percentiles)	0.5-3 Hz 15, 3-12 Hz 25, 12-15 Hz 45		
6.5-Hz tone (RMS 7.07 deg/s) in the hold window	3.5 deg/s RMS in 7-12 Hz	resolution 1.00 Hz	
6.5-Hz tone (RMS 7.07 deg/s) in the nose window	4.1 deg/s RMS in 7-12 Hz	resolution 1.67 Hz	

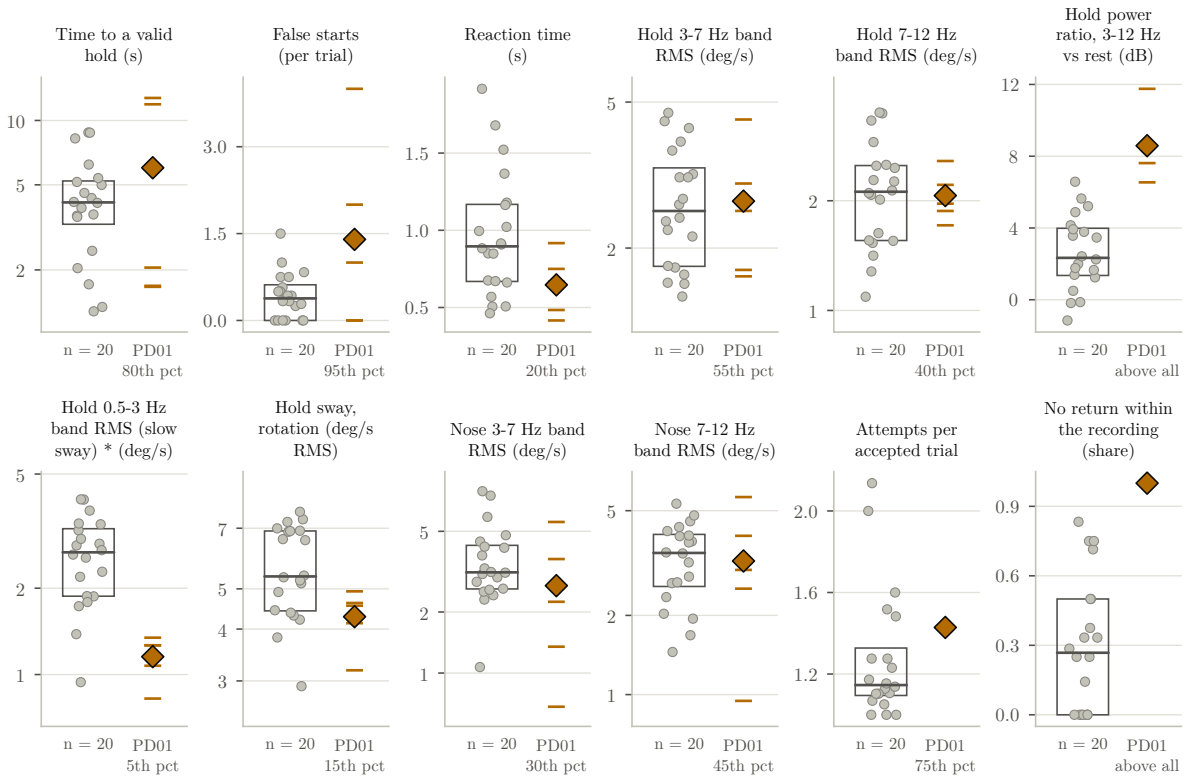


Figure 108: Preparation, hold, the phone at the nose and task behaviour. Grey dots: the 20 modelling persons (the mean of their left-hand upright trials), with median and interquartile box; diamond: PD01, ticks: PD01's trials; under each panel, PD01's percentile. The hold power ratio is above every reference person because of less slow sway (0.5–3 Hz), while the hold and nose 3–7 and 7–12 Hz band RMS are typical.

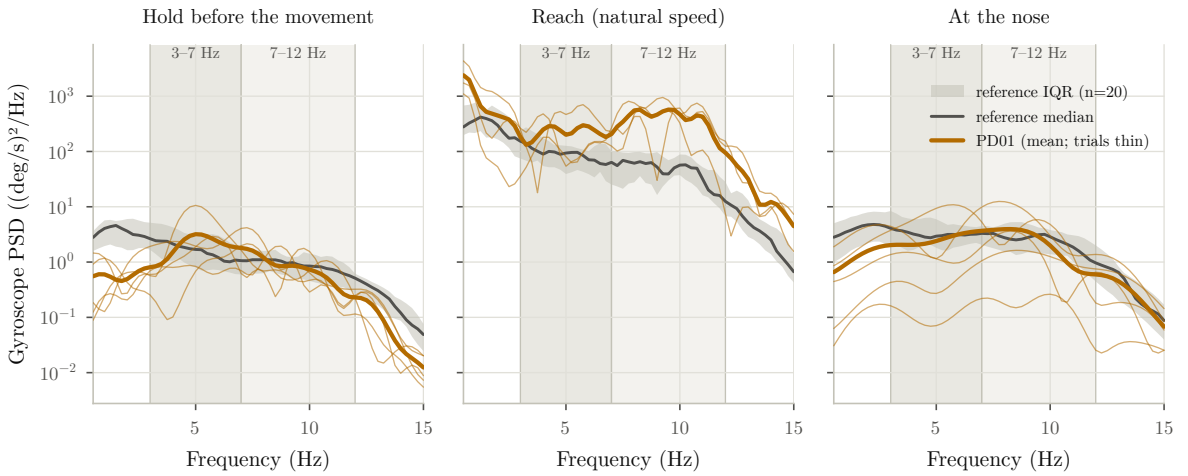


Figure 109: Power spectra of the angular velocity in the hold (1.0 s, 1.00-Hz resolution), the reach (natural speed) and at the nose (0.6 s, 1.67-Hz resolution): the reference persons' median and interquartile band, PD01's mean (thick) and trials (thin). PD01's spectrum departs from the reference in the reach (7–12 Hz) and, below 3 Hz, in the hold; at the nose it is typical.

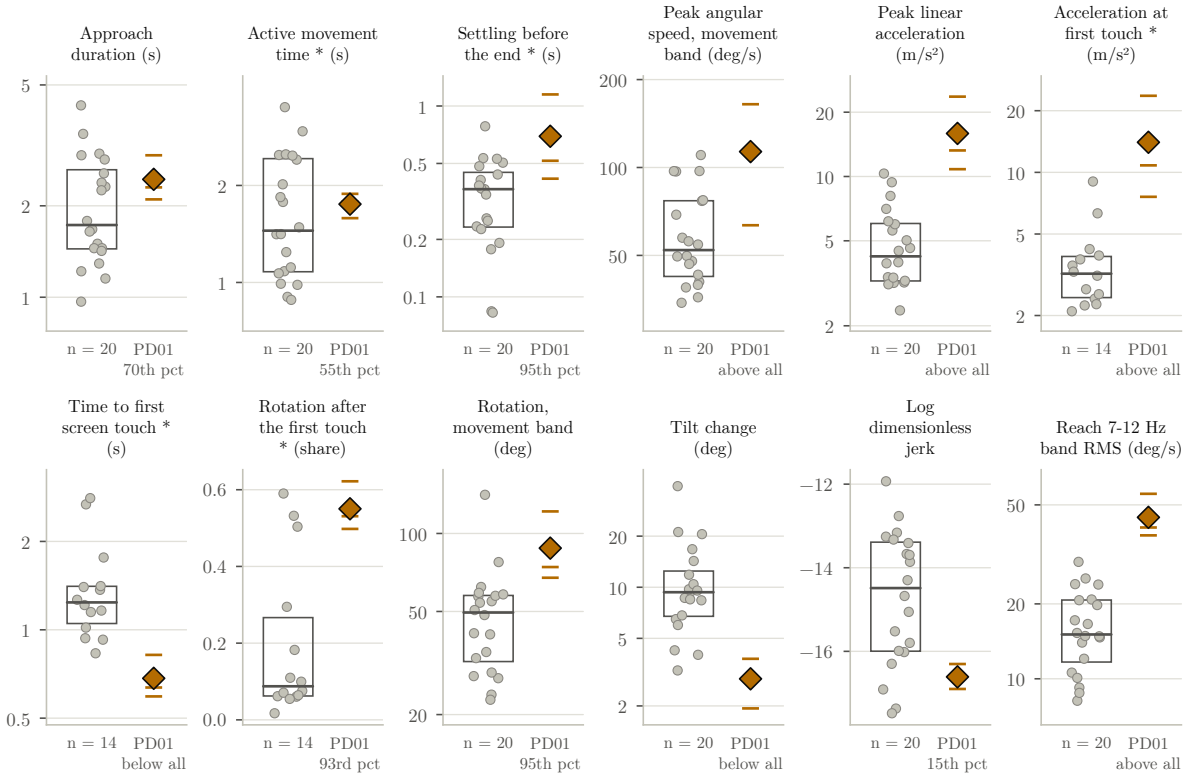


Figure 110: The reach (left hand, upright grip, natural speed): speed, first screen touch, amplitude, smoothness and band power, drawn as in Figure 108; the first-touch panels show the 14 reference persons who touched the screen. Accelerations of clip-flagged trials are lower bounds. Robust: peak linear acceleration, acceleration at the first touch, tilt change and reach 7–12 Hz band RMS; approach duration and smoothness are typical.

Table 122: Sensitivity of PD01's percentile (beyond: outside every reference person). Main: each person's mean over all matched trials; first k: each reference person on their first k matched trials, k = PD01's count; same protocol: reference trials of PD01's protocol version only; no clip: every person without clip-flagged trials; same posture / same platform: reference persons of PD01's posture / platform. (s): a pure subset of persons whose values did not change, which cannot overturn a beyond-all verdict. Size: |z| >= 2 on the declared scale with each reference person left out. Robust: beyond in the main analysis and in every check that recomputed values, and the size check passed. –: not computed, either not applicable (first k for person-level characteristics, no clip for the attempts) or too few case trials left, which withholds "robust". *: post hoc.

Characteristic	Main	First k	Same protocol	No clip	Same posture	Same platform	Size	Robust
Time to a valid hold (s)	80	75	80	85	80 (s)	75 (s)		
False starts (per trial)	95	95	95	92	beyond (s)	92 (s)		
Reaction time (s)	20	20	20	35	20 (s)	25 (s)		
Hold 3-7 Hz band RMS (deg/s)	55	60	55	30	47 (s)	67 (s)		
Hold 7-12 Hz band RMS (deg/s)	40	45	40	40	33 (s)	50 (s)		
Hold spectral peak, 3-12 Hz (Hz)	60	58	60	60	60 (s)	58 (s)		
Hold power ratio, 3-12 Hz vs rest (dB)	above	beyond	beyond	beyond	beyond (s)	beyond (s)	pass	yes
Hold sway, rotation (deg/s RMS)	15	15	15	30	13 (s)	25 (s)		
Hold sway, acceleration (m/s^2 RMS)	65	65	60	75	60 (s)	67 (s)		
Hold 0.5-3 Hz band RMS (slow sway) (deg/s) *	5	5	5	5	beyond (s)	8 (s)		
Hold 3-12 Hz band RMS (deg/s) *	45	45	45	40	33 (s)	58 (s)		
Hold 12-15 Hz band RMS (deg/s) *	15	20	15	beyond	7 (s)	25 (s)		
Approach duration (s)	70	70	70	55	73 (s)	75 (s)		
Active movement time (s) *	55	55	60	55	67 (s)	50 (s)		
Settling before the end (s) *	95	90	95	75	beyond (s)	92 (s)		
Peak angular speed, as recorded (deg/s)	90	90	90	80	93 (s)	83 (s)		
Peak angular speed, movement band (deg/s)	above	beyond	beyond	80	beyond (s)	beyond (s)	fail	no
Mean angular speed, movement band (deg/s)	85	90	80	70	80 (s)	92 (s)		
Time to peak speed (share of reach)	10	10	10	5	7 (s)	17 (s)		
Peak linear acceleration (m/s^2)	above	beyond	beyond	beyond	beyond (s)	beyond (s)	pass	yes
Time to first screen touch (s) *	below	beyond	beyond	beyond	beyond (s)	beyond (s)	fail	no
First touch, share of the approach (share) *	below	beyond	beyond	7	beyond (s)	beyond (s)	pass	no
Rotation after the first touch (share) *	93	93	93	86	beyond (s)	88 (s)		
Acceleration at first touch (m/s^2) *	above	beyond	beyond	beyond	beyond (s)	beyond (s)	pass	yes
Rotation, movement band (deg)	95	95	95	90	beyond (s)	92 (s)		
Net rotation (deg)	below	beyond	5	beyond	beyond (s)	beyond (s)	fail	no
Tilt change (deg)	below	beyond	beyond	beyond	beyond (s)	beyond (s)	pass	yes
Net yaw, magnitude (deg)	5	10	10	beyond	7 (s)	beyond (s)		
Log dimensionless jerk	15	20	15	15	13 (s)	8 (s)		
Spectral arc length (SPARC)	50	50	50	40	40 (s)	58 (s)		
Speed peaks, as recorded (count)	65	65	65	65	60 (s)	67 (s)		
Sub-movements, movement band (count)	15	15	15	15 (s)	20 (s)	8 (s)		
Hesitations (count)	45	42	45	50	53 (s)	42 (s)		
Reach 3-7 Hz band RMS (deg/s)	85	85	85	80	87 (s)	83 (s)		
Reach 7-12 Hz band RMS (deg/s)	above	beyond	beyond	beyond	beyond (s)	beyond (s)	pass	yes
Reach spectral peak, 3-12 Hz (Hz)	85	85	85	beyond	80 (s)	83 (s)		
Reach power ratio, 3-12 Hz vs rest (dB)	above	beyond	beyond	beyond	beyond (s)	beyond (s)	fail	no
Reach 7-12 Hz RMS per peak speed (ratio) *	80	80	80	95	73 (s)	83 (s)		
Nose 3-7 Hz band RMS (deg/s)	30	25	30	65	27 (s)	42 (s)		
Nose 7-12 Hz band RMS (deg/s)	45	45	45	55	40 (s)	67 (s)		
Nose spectral peak, 3-12 Hz (Hz)	70	70	70	30	60 (s)	67 (s)		
Nose power ratio, 3-12 Hz vs rest (dB)	85	90	85	95	87 (s)	92 (s)		
Duration variability (CV)	50	–	43 (s)	–	50 (s)	–		
Peak-speed variability (CV)	88	–	86 (s)	–	83 (s)	–		
Peak-speed change per repeat (%)	below	–	beyond (s)	–	beyond (s)	–	fail	no
Duration change per repeat (%)	12	–	14 (s)	–	17 (s)	–		
Slow-to-fast duration ratio	6	–	6	–	8 (s)	8 (s)		
Slow-to-fast active-time ratio *	6	–	6	–	8 (s)	8 (s)		
First-half yaw, hand-consistent (deg)	55	55	55	beyond	60 (s)	67 (s)		
Early yaw peak, hand-consistent (deg) *	above	beyond	beyond	beyond	beyond (s)	beyond (s)	pass	yes
Typical first-half yaw sign (share)	50	50	55	82	53 (s)	58 (s)		
Net yaw, hand-consistent (deg)	5	10	5	10	7 (s)	8 (s)		
Typical net-yaw sign (share) *	below	2	beyond	beyond	beyond (s)	beyond (s)		no
Attempts per accepted trial	75	–	75	–	73 (s)	83 (s)		
No return within the recording (share)	above	beyond	beyond	beyond	beyond (s)	beyond (s)		yes

I.7. What does not stand out

PD01 lies inside the reference interquartile range in 16 characteristics. Several of them were expected to differ in a person with Parkinson's disease: the hold 3–7 Hz band RMS (percentile 55), the nose 3–7 Hz band RMS (percentile 30), the SPARC (percentile 50), the hesitations (percentile 45), the approach duration (percentile 70) and the hand-consistent first-half yaw (percentile 55). Outside the interquartile range, though not beyond every reference person, are the log dimensionless jerk (percentile 15), the reaction time (percentile 20) and the false starts (1.40 per trial against a median of 0.38, percentile 95).

Two design limits bound the typical tremor-band power in the hold and at the nose. The still gate before GO keeps large hold movements out of the hold window, and the nose window exists only for trials that became still. PD01's false starts are recorded without their cause.

I.8. Speed scaling and repeats

Speed scaling. Asked to move slowly, PD01 slows down less than the reference persons do (Figure 111). PD01's slow-to-fast duration ratio is 1.28, against a reference median of 2.25 (interquartile range 1.87 to 4.55; percentile 6; 18 reference persons). The post hoc active-time ratio gives the same picture (0.72 against 1.83). Both ratios rest on 1 slow and 1 fast trial, so a single trial can move them.

Repeats. PD01's 3 left-hand upright natural reaches come from 2 sessions; 8 reference persons made at least as many (Figure 112). PD01's voluntary peak speed changes by −38 per cent per repeat, against a reference median of +6 (range −24 to +42). It is below every reference person but not robust: it fails the size check, and the check without clip-flagged trials cannot be run. The duration change per repeat is at percentile 12, and the variability of the duration and of the peak speed at percentiles 50 and 88. With 3 repeats across 2 sessions, fatigue, practice and the break between sessions cannot be told apart; a trend can only be suggested. The reach 7–12 Hz band RMS stays above every reference person in each repeat.

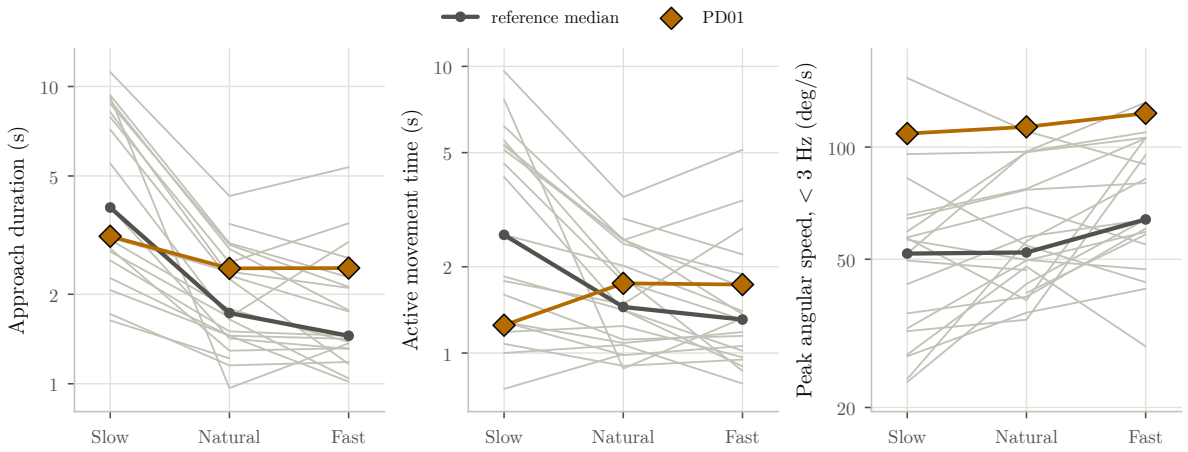


Figure 111: Instructed speed (left hand, upright grip): approach duration, active movement time and voluntary peak angular speed at slow, natural and fast speed. Grey lines: each modelling person's means; dark line: their median; diamonds: PD01 (1 slow, 3 natural and 1 fast trial). PD01 changes speed less than most reference persons, from one trial per extreme speed.

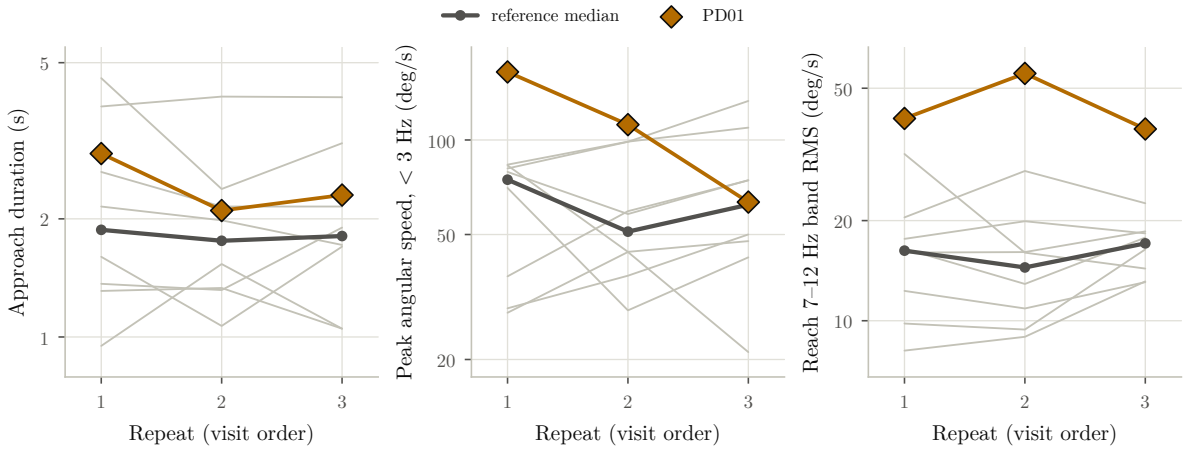


Figure 112: The first 3 left-hand upright natural reaches in visit order: grey lines, the 8 modelling persons with at least as many; dark line, their median; diamonds, PD01 (reaches from 2 sessions). PD01's peak speed falls across repeats (not robust); the reach 7–12 Hz band RMS stays high throughout.

I.9. The hand signature without a classifier

The hand signature is the turn of the phone about gravity during the approach (Section 3). It is read here directly, from the cumulative yaw, mirror-normalised so that positive is the direction typical of the hand used (Figure 113). No classifier is applied and no $P(\text{right})$ is computed for PD01.

The early turn is typical in direction and large. PD01's hand-consistent first-half yaw is 5.0 deg, at percentile 55. The early yaw peak (post hoc, defined after PD01's traces were inspected) is 26.3 deg, against a reference median of 7.4 and a maximum of 13.4 deg ($z = +4.0$; all 5 trials beyond); it passes every check. It falls 0.13 s (range 0.02 to 0.34) from the first touch, against 0.43 s in reference touched trials: the early turn and the early touch go together.

The net turn comes back. The hand-consistent net yaw is reversed, against the typical direction, in 6 of PD01's 7 accepted trials (4 below −2 deg). Reference persons reverse a median of 35 % and at most 68 % of their trials. PD01's mean net yaw (−1.2 deg) is at percentile 5 in its stratum, and below every reference person in the first free-choice trial (right hand, a single trial; Table 125). The share of trials with the typical net-yaw sign (post hoc) is below every reference person but fails the first-k check.

Reading. For PD01, the hand evidence lies in the early part of the approach: the early turn has the typical direction, and the net turn does not. Why the yaw turns back after the early peak, and whether this holds with the dominant hand instructed and with more repeats, are open questions. This appendix does not test whether the classifiers of this report would generalise to such movements.

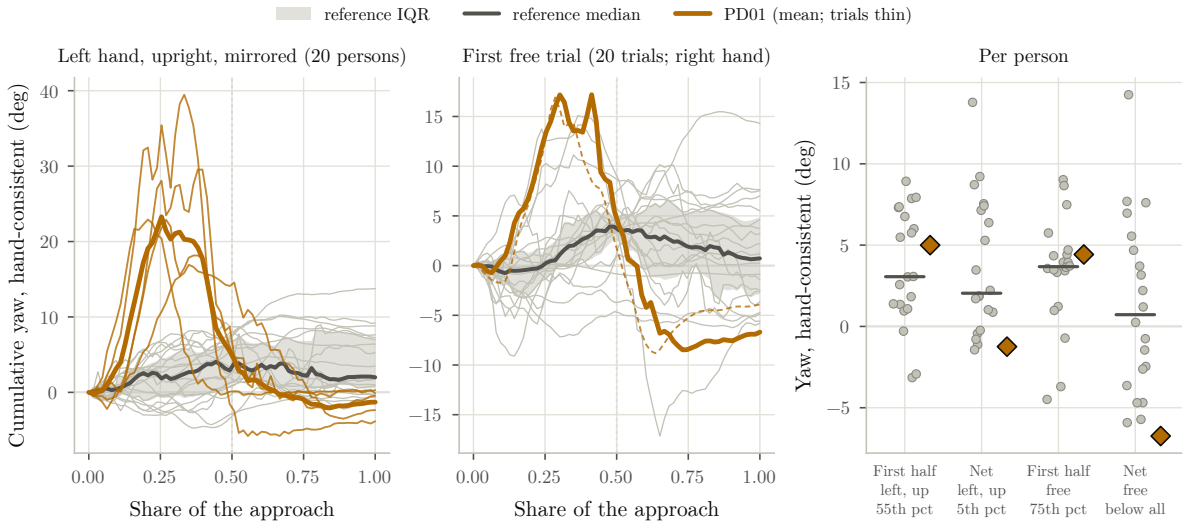


Figure 113: The hand signature without a classifier: cumulative yaw about gravity over the time-normalised approach, mirror-normalised so that positive is the direction typical of the hand used (dashed: half of the approach, where the first-half yaw is read). Left: left-hand upright trials (grey: each modelling person's mean; band: interquartile range); middle: the first free-choice trial of each modelling person (20 right-hand trials) and PD01's (solid; PD01's repeat-visit free trial dashed); right: first-half and net yaw per person, with PD01's percentile. PD01's early turn is typical in direction and large; the net turn comes back.

I.10. The multivariate view

Method. 10 pre-specified key characteristics (reaction time; hold 3–7 Hz band RMS; approach duration; peak angular speed and rotation in the movement band; log dimensionless jerk; SPARC; reach 3–7 and 7–12 Hz band RMS; hand-consistent first-half yaw) were taken on their declared scales (6 on the log scale) and robust-standardised against the reference persons. A principal component analysis (PCA) was fitted on the reference persons, with PD01 projected. The Mahalanobis distance used a Ledoit–Wolf shrinkage covariance; each reference person was left out of the standardisation and covariance of their own distance. The analysis was repeated on linear scales and with each characteristic left out.

Result. PD01's first principal component is 2.6, inside the reference range of −2.7 to 4.7 (the first two components explain 36 % and 31 % of the variance; Figure 114). PD01's Mahalanobis distance is 3.93; 4 of the 20 reference persons are at least as far (rank $p = 0.238$; the smallest possible is 0.048). With one characteristic left out, 1 to 9 are as far (Table 123). On linear scales the distance is 5.35, with 1 reference person as far (rank $p = 0.095$); without the reach 7–12 Hz band RMS, 5 are. The linear result depends on that one right-skewed characteristic. Taken together, PD01's profile is unusual but within the spread of the young reference persons.

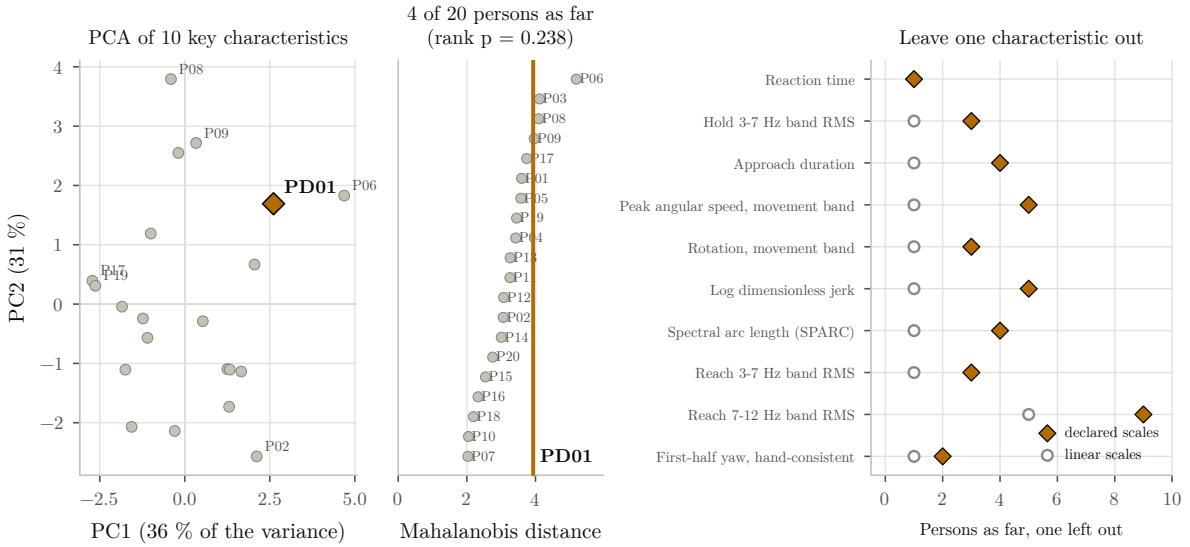


Figure 114: Multivariate view of the 10 pre-specified key characteristics, robust-standardised against the 20 modelling persons. Left: the first two principal components (fitted on the reference persons; PD01 projected). Middle: Mahalanobis distances (Ledoit–Wolf covariance, each reference person left out of their own standardisation); the line is PD01, and 4 reference persons are at least as far. Right: reference persons at least as far as PD01 with each characteristic left out, on declared and on linear scales.

Table 123: The Mahalanobis test with one key characteristic left out (declared scales; linear scales alongside): PD01's distance and the number of the 20 reference persons at least as far. All characteristics: 4 as far on the declared scales, 1 on linear scales.

Left out	Distance	As far	p	Distance (lin)	As far (lin)
Reaction time	3.89	1	0.095	5.25	1
Hold 3-7 Hz band RMS	3.85	3	0.190	5.27	1
Approach duration	3.83	4	0.238	5.30	1
Peak angular speed, movement band	3.34	5	0.286	4.70	1
Rotation, movement band	4.04	3	0.190	5.72	1
Log dimensionless jerk	3.65	5	0.286	5.11	1
Spectral arc length (SPARC)	3.76	4	0.238	5.19	1
Reach 3-7 Hz band RMS	3.90	3	0.190	5.31	1
Reach 7-12 Hz band RMS	3.23	9	0.476	3.60	5
First-half yaw, hand-consistent	3.93	2	0.143	5.26	1

I.11. The full characteristic table

Table 124 lists all 55 characteristics in their matched strata, with PD01's value, the reference median and interquartile range, the percentile, the robust z, PD01's trials beyond the reference range, the rank p, the Benjamini–Hochberg q over the 42 pre-specified characteristics, the Crawford–Howell p and the robustness verdict. No q is below 0.05. Table 125 gives PD01's percentile for every per-trial characteristic in every stratum.

Table 124: PD01 against the 20 modelling persons, one row per characteristic in its matched comparison (exploratory). Reference: median [IQR] of the person-level values; Pct: mid-rank percentile of PD01 (above / below all: beyond every reference person); z: robust z on the declared scale (log for positive, right-skewed characteristics; none for shares of binary outcomes); Trials beyond: PD01's trials beyond the reference persons' range; p (rank): two-sided, floor $2/(n+1) = 0.095$; q: Benjamini-Hochberg over the 42 pre-specified characteristics; p (C-H): Crawford-Howell test on the declared scale; Robust: a beyond-all verdict that survives every check that can fail (Table 122). *: post hoc, no q.

Characteristic	Stratum	n	PD01	Reference median [IQR]	n ref	Pct	z	Trials beyond	p (rank)	q	p (C-H)	Robust
Time to a valid hold (s)	L-up all	5	6.0	4.1 [3.3, 5.2]	20	80	+1.0	2/5	0.476	0.74	0.450	
False starts (per trial)	L-up all	5	1.40	0.38 [0.00, 0.62]	20	95	+1.9	2/5	0.190	0.67	0.028	
Reaction time (s)	L-up all	5	0.65	0.90 [0.67, 1.17]	20	20	−0.7	1/5	0.476	0.74	0.452	
Hold 3-7 Hz band RMS (deg/s)	L-up all	5	2.68	2.52 [1.78, 3.31]	20	55	+0.1	0/5	0.952	1.00	0.909	
Hold 7-12 Hz band RMS (deg/s)	L-up all	5	2.07	2.12 [1.56, 2.50]	20	40	−0.1	0/5	0.857	1.00	0.997	
Hold spectral peak, 3-12 Hz (Hz)	L-up all	5	7	6 [5, 7]	20	60	+0.5	0/5	0.857	1.00	0.746	
Hold power ratio, 3-12 Hz vs rest (dB)	L-up all	5	8.6	2.3 [1.4, 4.0]	20	above all	+2.8	4/5	0.095	0.50	0.012	yes
Hold sway, rotation (deg/s RMS)	L-up all	5	4.28	5.36 [4.42, 6.89]	20	15	−0.7	0/5	0.381	0.73	0.374	
Hold sway, acceleration (m/s^2 RMS)	L-up all	5	0.330	0.316 [0.299, 0.359]	20	65	+0.4	0/5	0.762	0.97	0.994	
Hold 0.5-3 Hz band RMS (slow sway) (deg/s) *	L-up all	5	1.15	2.67 [1.88, 3.22]	20	5	−2.5	1/5	0.190	–	0.063	
Hold 3-12 Hz band RMS (deg/s) *	L-up all	5	3.46	3.52 [2.66, 4.72]	20	45	−0.0	0/5	0.952	–	0.967	
Hold 12-15 Hz band RMS (deg/s) *	L-up all	5	0.49	0.80 [0.57, 0.93]	20	15	−1.8	2/5	0.381	–	0.271	
Approach duration (s)	L-up nat	3	2.44	1.73 [1.44, 2.63]	20	70	+0.8	0/3	0.667	0.90	0.567	
Active movement time (s) *	L-up nat	3	1.75	1.45 [1.08, 2.42]	20	55	+0.4	0/3	0.952	–	0.817	
Settling before the end (s) *	L-up nat	3	0.69	0.37 [0.23, 0.45]	20	95	+1.2	1/3	0.190	–	0.186	
Peak angular speed, as recorded (deg/s)	L-up nat	3	164	90 [79, 120]	20	90	+2.0	1/3	0.286	0.73	0.157	
Peak angular speed, movement band (deg/s)	L-up nat	3	113	52 [42, 77]	20	above all	+1.9	2/3	0.095	0.50	0.084	no
Mean angular speed, movement band (deg/s)	L-up nat	3	36	23 [19, 33]	20	85	+1.0	1/3	0.381	0.73	0.245	
Time to peak speed (share of reach)	L-up nat	3	0.26	0.44 [0.34, 0.52]	20	10	−1.4	0/3	0.286	0.73	0.210	
Peak linear acceleration (m/s^2)	L-up nat	3	15.9	4.2 [3.2, 6.0]	20	above all	+3.2	3/3	0.095	0.50	0.008	yes
Time to first screen touch (s) *	L-up nat	3	0.68	1.24 [1.05, 1.41]	14	below all	−2.5	3/3	0.133	–	0.102	no
First touch, share of the approach (share) *	L-up nat	3	0.29	0.76 [0.64, 0.84]	14	below all	−3.7	2/3	0.133	–	0.061	no
Rotation after the first touch (share) *	L-up nat	3	0.55	0.09 [0.06, 0.27]	14	93	+10.4	1/3	0.267	–	0.111	
Acceleration at first touch (m/s^2) *	L-up nat	3	14.0	3.2 [2.4, 3.9]	14	above all	+3.9	2/3	0.133	–	0.005	yes
Rotation, movement band (deg)	L-up nat	3	88	50 [32, 58]	20	95	+1.8	0/3	0.190	0.67	0.162	
Net rotation (deg)	L-up nat	3	3.9	10.3 [7.0, 14.4]	20	below all	−1.7	2/3	0.095	0.50	0.081	no
Tilt change (deg)	L-up nat	3	2.9	9.3 [6.7, 12.5]	20	below all	−2.3	2/3	0.095	0.50	0.068	yes
Net yaw, magnitude (deg)	L-up nat	3	2.1	4.6 [2.9, 7.6]	20	5	−1.0	1/3	0.190	0.67	0.226	
Log dimensionless jerk	L-up nat	3	−16.6	−14.5 [−16.0, −13.4]	20	15	−1.1	0/3	0.381	0.73	0.271	
Spectral arc length (SPARC)	L-up nat	3	−2.52	−2.52 [−3.09, −2.21]	20	50	+0.0	0/3	1.000	1.00	0.837	
Speed peaks, as recorded (count)	L-up nat	3	15.7	10.8 [8.4, 17.3]	20	65	+0.6	0/3	0.762	0.97	0.542	
Sub-movements, movement band (count)	L-up nat	3	2.0	3.5 [2.2, 6.1]	20	15	−0.8	0/3	0.476	0.74	0.363	
Hesitations (count)	L-up nat	3	0.33	0.50 [0.00, 1.54]	20	45	−0.2	0/3	0.952	1.00	0.611	
Reach 3-7 Hz band RMS (deg/s)	L-up nat	3	29.1	19.2 [14.2, 24.0]	20	85	+1.1	0/3	0.381	0.73	0.345	
Reach 7-12 Hz band RMS (deg/s)	L-up nat	3	44.6	15.1 [11.7, 20.7]	20	above all	+2.3	3/3	0.095	0.50	0.013	yes
Reach spectral peak, 3-12 Hz (Hz)	L-up nat	3	7.7	5.3 [4.7, 6.3]	20	85	+2.0	1/3	0.381	0.73	0.213	
Reach power ratio, 3-12 Hz vs rest (dB)	L-up nat	3	3.3	0.3 [−2.0, 1.5]	20	above all	+1.2	2/3	0.095	0.50	0.125	no
Reach 7-12 Hz RMS per peak speed (ratio) *	L-up nat	3	0.45	0.26 [0.21, 0.39]	20	80	+1.1	1/3	0.476	–	0.287	
Nose 3-7 Hz band RMS (deg/s)	L-up all	5	2.70	3.14 [2.60, 4.26]	20	30	−0.4	1/5	0.667	0.90	0.624	
Nose 7-12 Hz band RMS (deg/s)	L-up all	5	3.22	3.45 [2.58, 4.06]	20	45	−0.2	1/5	0.952	1.00	0.944	
Nose spectral peak, 3-12 Hz (Hz)	L-up all	5	8	8 [7, 8]	20	70	+0.4	1/5	0.667	0.90	0.656	
Nose power ratio, 3-12 Hz vs rest (dB)	L-up all	5	7.0	4.8 [3.2, 6.0]	20	85	+0.9	1/5	0.381	0.73	0.279	
Duration variability (CV)	L-up nat	3	0.18	0.17 [0.12, 0.24]	8	50	+0.1	–	1.000	1.00	0.826	
Peak-speed variability (CV)	L-up nat	3	0.45	0.30 [0.23, 0.37]	8	88	+1.2	–	0.444	0.74	0.316	
Peak-speed change per repeat (%)	L-up nat	3	−38	+6 [−16, +29]	8	below all	−1.2	–	0.222	0.67	0.148	no
Duration change per repeat (%)	L-up nat	3	−11	−3 [−11, +4]	8	12	−0.8	–	0.444	0.74	0.485	
Slow-to-fast duration ratio	L-up all	2	1.28	2.25 [1.87, 4.55]	18	6	−0.9	–	0.211	0.67	0.220	
Slow-to-fast active-time ratio *	L-up all	2	0.72	1.83 [1.24, 3.61]	18	6	−1.4	–	0.211	–	0.133	
First-half yaw, hand-consistent (deg)	L-up all	5	5.0	3.1 [1.3, 6.9]	20	55	+0.5	1/5	0.952	1.00	0.713	
Early yaw peak, hand-consistent (deg) *	L-up all	5	26.3	7.4 [3.4, 9.0]	20	above all	+4.0	5/5	0.095	–	<0.001	yes
Typical first-half yaw sign (share)	L-up all	5	0.80	0.79 [0.55, 1.00]	20	50	–	0/5	1.000	1.00	0.805	
Net yaw, hand-consistent (deg)	L-up all	5	−1.2	2.0 [0.6, 7.2]	20	5	−0.7	2/5	0.190	0.67	0.264	
Typical net-yaw sign (share) *	L-up all	5	0.20	0.75 [0.50, 1.00]	20	below all	–	4/5	0.095	–	0.093	no
Attempts per accepted trial	all	10	1.43	1.15 [1.09, 1.33]	20	75	+1.5	–	0.571	0.86	0.550	
No return within the recording (share)	L-up all	5	1.00	0.27 [0.00, 0.50]	20	above all	–	5/5	0.095	0.50	0.030	yes

Strata: L-up = left hand, upright grip, standard movement, natural start (nat / slow / fast / all speeds); up nat, either = upright natural reaches of either hand; free = first free-choice trial (natural-use set); all = every accepted trial.

Table 125: PD01's percentile for every per-trial characteristic in each matched stratum (above / below all: beyond every reference person). Case trials per stratum: L-up nat 3, L-up slow 1, L-up fast 1, L-up all 5, free 1, all 7. *: post hoc.

Characteristic	L-up nat	L-up slow	L-up fast	L-up all	free	all
Time to a valid hold	65	95	11	80	25	65
False starts	85	95	76	95	78	above all
Reaction time	30	11	68	20	40	35
Hold 3-7 Hz band RMS	45	84	21	55	70	50
Hold 7-12 Hz band RMS	70	37	53	40	20	35
Hold spectral peak, 3-12 Hz	65	53	79	60	35	65
Hold power ratio, 3-12 Hz vs rest	above all	above all	95	above all	above all	above all
Hold sway, rotation	35	26	5	15	10	below all
Hold sway, acceleration	60	63	21	65	35	55
Hold 0.5-3 Hz band RMS (slow sway) *	10	16	5	5	below all	below all
Hold 3-12 Hz band RMS *	55	74	21	45	50	30
Hold 12-15 Hz band RMS *	10	37	21	15	50	15
Approach duration	70	47	79	55	60	55
Active movement time *	55	21	74	40	50	40
Settling before the end *	95	68	above all	85	80	90
Peak angular speed, as recorded	90	95	above all	above all	above all	above all
Peak angular speed, movement band	above all	95	95	above all	above all	above all
Mean angular speed, movement band	85	89	68	80	above all	95
Time to peak speed	10	37	58	20	50	5
Peak linear acceleration	above all	above all	95	above all	above all	above all
Time to first screen touch *	below all	below all	8	below all	below all	below all
First touch, share of the approach *	below all	12	8	12	9	below all
Rotation after the first touch *	93	62	92	82	82	84
Acceleration at first touch *	above all	above all	92	above all	above all	above all
Rotation, movement band	95	63	84	95	95	95
Net rotation	below all	16	53	5	35	5
Tilt change	below all	5	58	15	10	5
Net yaw, magnitude	5	5	below all	below all	80	5
Log dimensionless jerk	15	58	11	25	50	20
Spectral arc length (SPARC)	50	95	63	70	80	75
Speed peaks, as recorded	65	18	66	50	35	50
Sub-movements, movement band	15	5	11	8	20	5
Hesitations	45	24	26	25	30	20
Reach 3-7 Hz band RMS	85	68	95	85	above all	above all
Reach 7-12 Hz band RMS	above all	above all	above all	above all	above all	above all
Reach spectral peak, 3-12 Hz	85	above all	21	75	74	95
Reach power ratio, 3-12 Hz vs rest	above all	89	89	above all	63	above all
Reach 7-12 Hz RMS per peak speed *	80	58	74	80	89	above all
Nose 3-7 Hz band RMS	50	5	below all	30	5	5
Nose 7-12 Hz band RMS	55	16	18	45	10	30
Nose spectral peak, 3-12 Hz	40	84	79	70	30	80
Nose power ratio, 3-12 Hz vs rest	85	79	65	85	85	95
First-half yaw, hand-consistent	65	42	79	55	75	55
Early yaw peak, hand-consistent *	above all	above all	above all	above all	above all	95
Typical first-half yaw sign	25	71	63	50	58	60
Net yaw, hand-consistent	below all	53	42	5	below all	below all
Typical net-yaw sign *	below all	74	16	below all	22	below all
No return within the recording	92	87	87	above all	35	95

Strata: L-up = left hand, upright grip, standard movement, natural start (nat / slow / fast / all speeds); up nat, either = upright natural reaches of either hand; free = first free-choice trial (natural-use set); all = every accepted trial.

I.12. What a follow-up study would need

This single case generates hypotheses; it tests none. A follow-up study would need:

- several participants with Parkinson's disease and age-, sex- and device-matched controls, with clinical scores (for example the motor examination of the Movement Disorder Society Unified Parkinson's Disease Rating Scale, MDS-UPDRS, and its tremor items) and the medication state recorded at the visit;
- supervised visits, both hands instructed, at least five repeats per speed, arm support recorded, and a longer still hold whose gate is logged rather than enforced, to measure rest tremor;
- a small, pre-registered set of characteristics: the phenomena with a robust pre-specified difference here (hold steadiness, linear acceleration, net orientation change, reach band power, return of the arm) and, as hypotheses, the post hoc ones (the early hand-consistent yaw), analysed by a mixed model over trials instead of single-case ranks;
- video or a contact sensor to establish what touches the screen first, and a verified rotation-rate mapping.